Haagerup Symmetry and Conformal Field Theory

From Fusion Rules to Lattice Constructions and Continuum Tests

A textbook for beginning mathematics and physics PhD students

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What this book constructs

A topological symmetry line in two dimensions can be deformed without changing a correlation function, provided that it does not cross an operator insertion. For an ordinary finite-group symmetry, every such line has an inverse. The Haagerup example asks us to take a more general possibility seriously: two lines can fuse into a direct sum of lines. The simplest characteristic relation is

$$\rho^2 = \mathbf{1} + \rho + \alpha\rho + \alpha^2\rho, \qquad \alpha^3 = \mathbf{1}.$$

The coefficient of the identity is one, but the product is not the identity. There is no inverse line that undoes ρ. Nevertheless, the line has well-defined junctions, acts on states, and imposes precise constraints on local physics. Its quantum dimension is

$$d = \frac{3 + \sqrt{13}}{2}, \qquad d^2 = 1 + 3d.$$

Understanding how this small piece of algebra becomes a conformal field theory is the organizing question of the book.

There are several construction problems hidden in that question. One can construct a fusion category that describes line junctions. One can construct a chiral algebra whose representations have a prescribed braiding and fusion. One can combine left- and right-moving fields into a local theory. One can construct a microscopic Hamiltonian with the line symmetry and investigate its long-distance limit. Each is a substantial achievement, and each requires different data. A useful education in the subject consists partly in learning to pass between these descriptions and partly in learning when a proposed passage has not yet been justified.

The central example in this book begins with a positive-definite lattice and its vertex operators. Inside the lattice vertex algebra

$$B = V_{E_6 \oplus A_2} \subset V_{E_8}$$

one specifies six currents, a conformal vector, and a weight-two field u. Their generated vertex algebra U has central charge eight. Associativity of its products comes from the ambient lattice theory. The difficult questions are whether this generated algebra is proper, whether it is rational, what its representations are, and whether all the positivity and analytic conditions needed for a unitary full CFT hold. The research construction studied here answers these questions under an explicit set of finite lattice identities and stated structural and analytic inputs. We turn that argument into a sequence of definitions, calculations, and implications that a beginning graduate student can follow.

This is a textbook exposition of a research construction. It is not an independent verification of every large finite lattice computation, and it does not treat every claim in a research manuscript as an established classification theorem. Model-specific algebraic identities and structural facts are identified as *construction inputs*; deductions from them state their assumptions. Saved numerical findings are presented as *numerical records* with their finite scope. Many of the small examples are proved in full on the page. This distinction makes the book usable both for learning the mathematics and for locating where a further verification would strengthen the construction.

The two full theories and the chain

Even after the chiral algebra U has been identified, there is more than one natural full theory. The diagonal theory has Hilbert space

$$\mathcal{H}_{\text{diag}} = \bigoplus_i M_i \otimes \overline{M_i},$$

where M_i runs over the twelve simple U-modules. A selected holomorphic extension gives another theory with Hilbert space

$$\mathcal{H}_{E_8} = V_{E_8} \otimes \overline{V_{E_8}}.$$

The selected extension includes an algebra structure and a specified Haagerup action; an equality of unflavoured characters alone would not choose those data. Both theories have $c_L = c_R = 8$, conventionally reported as $c = 8$. The diagonal theory has scalar primaries beginning at scaling dimension $2/3$, whereas the holomorphic extension's nonidentity scalar primaries begin at dimension two. Thus they answer different physical questions despite their shared central charge.

The regular Haagerup chain is a third object. Its local Hilbert space is encoded by fusion paths, and its Hamiltonian is

$$H = -\sum_j P_j^{\rho},$$

with projectors built from the actual unitary $\mathcal{H}_3$ associator. Numerical studies motivate a long-distance theory with central charge near two. The completed research record used in this book does not identify that continuum theory, establish its exact central charge, or rule out a sufficiently long crossover to a gapped phase. We therefore teach both the evidence and the tests that remain necessary. A correct finite matrix calculation and a controlled continuum conclusion are different kinds of statement.

There is no contradiction between central charge eight in the chiral construction and central charge near two in the chain investigation. The modular representation category of the chiral algebra is the *center* of a Haagerup fusion category. Having Haagerup lines in a full theory is a different requirement. In particular, the vanishing chiral anomaly of a center does not constrain every full CFT carrying those lines to have central charge divisible by eight. This distinction will become concrete in the chapters on centers, modular transformations, and defects.

Prerequisites and a route through the book

We assume linear algebra, elementary group representations, complex analysis through residues and Laurent series, and the basic language of Hilbert spaces. Familiarity with either quantum mechanics or abstract algebra is helpful; neither a previous CFT course nor a previous course in tensor categories is required. Some later proofs use results from operator algebras and vertex-algebra representation theory that cannot be developed from first principles in a single textbook. Before each such use, we explain the relevant objects and state the particular implication being imported.

The first part builds a common language. Fusion spaces and changes of association give a concrete entrance to categories. Radial quantization and Ward identities give a concrete entrance to CFT. Lattice vertex operators then provide a model in which many abstract formulas become ordinary oscillator and exponential calculations. The Haagerup fusion ring, its center, and its modular data are introduced only after these examples have supplied a reason to care about them.

The second part follows the central-charge-eight construction. It starts with an explicit field in an explicit lattice. It then explains why finite low-weight calculations can support an

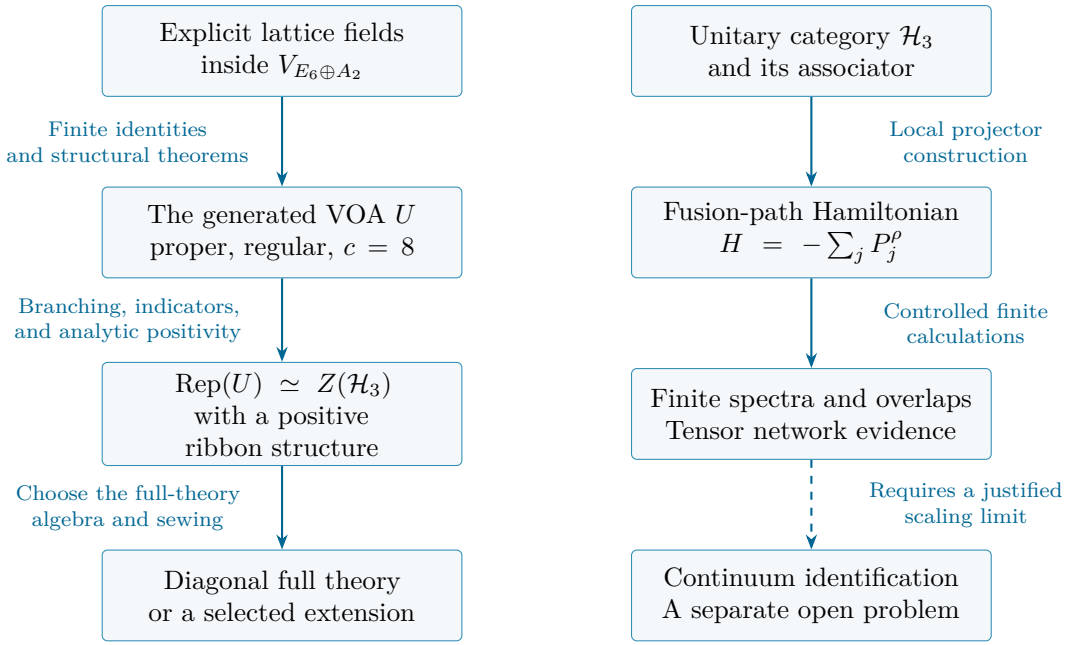


Figure 1: Two construction tracks studied in this book. The left track is the central-charge-eight research construction under its stated finite and structural inputs. The right track constructs the physical chain exactly but requires further work to identify its continuum limit. Sharing Haagerup data does not supply an arrow equating the two full theories.

all-weight strong-generation argument, why positive stress identities control representations, and how branching constrains the boundary fusion category. The passage from vacuum positivity to complete unitarity closes the structural argument. Characters then turn that structure into explicit graded state spaces. Their coefficients are consequences of the constructed algebra and its representation theory.

The third part turns the construction into observables. It derives the spectra of the two full theories, works out normalized three- and four-point functions, and computes defect and boundary quantities. The chapter on sewing explains exactly which data a torus partition function leaves undetermined. The fourth part examines cosets and stress splittings before returning to the physical chain. It develops finite-size scaling, overlap methods, tensor networks, and exact gapped comparison states. These tools explain what would count as an identification of the chain and why several attractive guesses remain conditional or fail specified tests.

The final mathematical part returns to the construction of the associators themselves. It explains how hyperbolic gamma functions solve finite coefficient equations, how exact algebra and positivity enter the noncyclic examples, and how planar diagrams reconstruct extended Haagerup. These comparisons clarify which methods generalize and which additional steps are still needed to obtain a CFT.

A reader mainly interested in physics can first study the CFT and lattice chapters, the explicit construction, characters, full theories, correlators, and the chain. The category chapters can be read alongside the first appearance of a defect or extension. A reader mainly interested in mathematics can first follow fusion categories, lattice vertex algebras, finite closure, rationality, branching, and unitarity, then use the physical chapters to see which information these structures actually predict. Both routes benefit from doing the small worked examples rather than treating them as illustrations.

A compact first tour can follow the Ising model across the book. Derive its null vector in Exercise 2.2, obtain its three chiral weights from the affine coset in Worked example 3.11, and construct its full four-point function in Section 17.2. Worked example 16.3 supplies its fixed and

free boundaries, while Worked example 20.2 derives its continuum spectrum from a periodic spin chain. Together these give a small complete comparison theory for the different stages of the Haagerup problem.

The following route begins with calculations a physics reader can carry through, then returns to the construction's existence proofs. The chapter references also make it possible to revisit a method when it is needed. Results imported from the construction retain the inputs and hypotheses stated there.

Read	Calculation or skill to acquire
Chapter 2; the Ising example in Chapter 20	Compute $\|L_{-2}\|0\rangle\|^2 = c/2$ and extract $v = 2J$, $c = 1/2$ from the exactly solved periodic Ising chain.
Chapters 1, 4, and 19	Turn an F-matrix into a local fusion projector and count the allowed periodic path states.
Chapters 3 and 7	Multiply two lattice exponentials, including their cocycle, and read a current level from the double pole.
Chapters 5–6 and 8–12	Track what finite closure, positive stresses, branching, and complete unitarity each establish about the constructed algebra.
Chapters 13–17	Remove Virasoro descendants from a character, count full scalar fields, and check the Ising four-point sewing example.
Chapter 18	Subtract commuting stresses and distinguish that calculation from constructing the reduced state space and its full CFT.
Chapters 20–22	Derive a stress estimate of c, normalize a wavefunction overlap, and obtain an MPS truncation from Schmidt coefficients.
Chapters 23–24	Compare a proven gapped parent with the physical Hamiltonian and state which spectral, OPE, and symmetry data a continuum identification still needs.

Chapters 25 and 26 can then be read for the exact arithmetic and reconstruction methods behind further associator examples.

How to use the examples and exercises

The clickable list after the contents gives the title and page of every worked example. It is useful for following one calculation across the different parts of the book or for choosing a short first reading.

The examples are chosen to expose mechanisms that recur in the hard construction. Diagonalizing a fusion matrix explains quantum dimensions and the exponential growth of a path Hilbert space. Multiplying two lattice exponentials explains both a current bracket and the origin of a cocycle sign. Subtracting descendants from a character explains why a large coefficient is not a count of new primary fields. A stress-tensor norm determines a finite-size estimator of c. A small vacuum algebra shows why a symmetric Hamiltonian can have several pure vacua. Each calculation should leave the reader with a method that can be used elsewhere.

Exercises include short checks, derivations that complete an argument, and problems that ask which hypothesis a tempting inference is missing. Worked solutions to every exercise are collected in Appendix B. The exercises do not require reproducing large research computations. Where saved numerical values are used, the point is to analyze what those values mean, not to launch a larger diagonalization.

We use a small set of conventions consistently. Vertex operators are expanded as

$$Y(a,z) = \sum_{n\in\mathbb{Z}} a_{(n)} z^{-n-1}, \qquad L_n = \omega_{(n+1)}.$$

Thus $a_{(n)}b$ has weight $\mathrm{wt}(a) + \mathrm{wt}(b) - n - 1$. For a full field, $\Delta = h + \bar{h}$ is its scaling dimension and $s_{\mathrm{spin}} = h - \bar{h}$ its spin. We reserve d for the noninvertible Haagerup quantum dimension and often write $s = 1 + d^2 = 2 + 3d$ in categorical formulas. Appendix A records other potentially confusing uses, including the difference between a central simple object labelled a and the invertible Haagerup object α.

References are deliberately selective. The main exposition should be readable without opening a paper or a program. A short source guide records the principal literature and the repository constructions behind the book, while a verification appendix describes the scope of exact computations and formal proofs. Its dependency ledger in Section C.2 collects the inputs and conclusions of the central-charge-eight proof. The purpose is to make the mathematics understandable and its logical dependencies inspectable.

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Part I

The language of the problem

Chapter 1

Fusion spaces and the meaning of an associator

The first obstacle in learning about Haagerup symmetry is that the word "symmetry" no longer means a group acting by invertible matrices. We will replace that familiar picture one piece at a time. This chapter requires linear algebra and the tensor product of vector spaces. Its main calculation is the Fibonacci associator: a small matrix whose entries are forced by consistency of fourfold fusion. The calculation will explain what an F-symbol contains, why its entries depend on choices of bases, and why a fusion table is only the beginning of a symmetry theory.

1.1 Three vector spaces that should be kept separate

A quantum system has a Hilbert space $\mathcal{H}$. A finite group G acts on it through unitary operators U_g satisfying $U_g U_h = U_{gh}$. In a two-dimensional field theory one can represent U_g by a line crossing the spatial circle. Moving this line without crossing an operator does not change a correlation function. Such a line is called *topological*. Putting two parallel lines together realizes their product.

For a more general topological line, putting two lines together can produce a direct sum of different line types:

$$a \otimes b \simeq \bigoplus_c N_{ab}^{\ c}\, c, \qquad N_{ab}^{\ c} \in \mathbb{Z}_{\geq 0}. \tag{1.1}$$

The labels a, b, c denote simple line types. "Simple" means that a line has no nontrivial direct-sum decomposition. The unit object $\mathbf{1}$ is the invisible line. An invertible line has an inverse under fusion; a noninvertible line does not.

There are now three different kinds of vector spaces:

1. The physical Hilbert space $\mathcal{H}$ contains states with energies, momenta, and possibly infinitely many descendants.

2. A fusion space
$$V_{ab}^{c} = \mathrm{Hom}(c, a \otimes b)$$
contains junctions at which a c line splits into a and b. Its dimension is the nonnegative integer $N_{ab}^{\ c}$.

3. A representation space of the algebra generated by closed line operators describes how a collection of states transforms. Its ordinary vector-space dimension is generally unrelated to the quantum dimension of a line.

These distinctions remain important even when all the fusion spaces have dimension zero or one. Multiplicity-free fusion makes individual junction spaces simple, but repeated fusion still creates larger spaces.

For instance, suppose a line τ obeys

$$\tau \otimes \tau \simeq \mathbf{1} \oplus \tau. \tag{1.2}$$

Then $\mathrm{Hom}(\tau, \tau^{\otimes 3})$ has dimension two. One can fuse the first pair to $\mathbf{1}$, or fuse it to τ, before reaching total charge τ. This is a two-dimensional space of junction networks; it is not an assertion that the physical theory has two states.

1.2 Semisimplicity, duality, and positivity

Definition 1.1. A fusion category over $\mathbb{C}$ is a $\mathbb{C}$-linear rigid monoidal category with a simple unit, finitely many simple isomorphism classes, finite-dimensional morphism spaces, and a decomposition of every object as a finite direct sum of simple objects.

Each word has a concrete role. Linearity permits superpositions of junctions. Semisimplicity means that every object decomposes into simples; the separate finiteness assumption limits the list of simple line types. The monoidal product describes putting lines together. Rigidity means that a line has a dual a^*, represented physically by reversing its orientation. Evaluation and coevaluation junctions can create and annihilate a line–antiline pair. Their zigzag identities say that a line bent into a cup and then a cap can be straightened. In formulas, choose a left dual X^* with maps

$$\mathrm{ev}_X : X^* \otimes X \to \mathbf{1}, \qquad \mathrm{coev}_X : \mathbf{1} \to X \otimes X^*.$$

With associators understood, the two straightening identities are

$$(1_X \otimes \mathrm{ev}_X)(\mathrm{coev}_X \otimes 1_X) = 1_X,$$
$$(\mathrm{ev}_X \otimes 1_{X^*})(1_{X^*} \otimes \mathrm{coev}_X) = 1_{X^*}.$$

For finite-dimensional vector spaces they say $v = \sum_i e_i e^i(v)$, using a basis and its dual. In a fusion category the same equations concern junction maps; they do not require the line X to be an ordinary vector space.

At the level of the fusion coefficients, rigidity implies

$$N_{ab}^{\mathbf{1}} = \delta_{b,a^*}. \tag{1.3}$$

More generally, moving a line through a cup gives Frobenius reciprocity, for example

$$\dim \mathrm{Hom}(c, a \otimes b) = \dim \mathrm{Hom}(a^* \otimes c, b).$$

The equality is an isomorphism of vector spaces obtained using actual evaluation and coevaluation maps. It is more than a coincidence between integers.

A unitary fusion category also supplies adjoints of morphisms and positive inner products. The associativity and unit isomorphisms are unitary, and the tensor product respects the adjoint. In suitable orthonormal fusion bases the recoupling matrices introduced below are therefore unitary. An abstract complex fusion category need not admit this structure.

A pivotal structure is a natural monoidal identification $j_X : X \to X^{**}$. Natural and monoidal mean that this identification respects morphisms and tensor products. It lets us close an endomorphism $f : X \to X$ without changing the orientation of its line. One closure gives

$$\mathrm{Tr}_{\mathrm{cat}}(f) = \mathrm{ev}_{X^*}\big((j_X f) \otimes 1_{X^*}\big)\,\mathrm{coev}_X\,.$$

Sphericity says that closing on the other side gives the same scalar. The quantum dimension is $d_X = \mathrm{Tr}_{\mathrm{cat}}(1_X)$. In a unitary fusion category the canonical spherical structure has positive dimensions d_a satisfying

$$d_a d_b = \sum_c N_{ab}^c d_c, \qquad d_{\mathbf{1}} = 1. \tag{1.4}$$

The unique positive solution is the Frobenius–Perron dimension function. We write $\mathrm{qdim}(X) = d_X$ for the spherical dimension and $\mathrm{FPdim}(X)$ for the positive fusion-ring dimension; they coincide in the canonical unitary structure. A dimension is the value of a closed line, not the number of basis states carried by an individual microscopic site. Irrational values cause no conflict with the integrality of Hilbert-space dimensions.

Worked example 1.2 (The golden ratio from a fusion matrix)**.** For (1.2), multiplication by τ on the basis $(\mathbf{1}, \tau)$ has matrix

$$N_\tau = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Its characteristic polynomial is $x^2 - x - 1$. The positive eigenvalue is

$$\varphi = \frac{1+\sqrt{5}}{2}, \qquad \varphi^2 = 1 + \varphi.$$

The vector $(1, \varphi)^{\mathsf{t}}$ is its positive eigenvector. Thus the positive dimension of τ is φ. The other root $-\varphi^{-1}$ is also a ring character, but it cannot be the positive dimension of a unitary line. Later we will encounter the Yang–Lee category, where a spherical dimension uses this other root. The same fusion ring supports distinct categorical structures.

This is an example of the Perron–Frobenius theorem. For a nonnegative matrix whose directed graph is strongly connected, the spectral radius $r = \max_\lambda |\lambda|$ is a simple eigenvalue with strictly positive left and right eigenvectors, unique up to scale. These are its *Perron eigenvectors*. If some power of the matrix has every entry strictly positive, the matrix is primitive and every other eigenvalue has modulus less than r. The Fibonacci fusion matrix above is primitive: its square has all entries positive.

Worked example 1.3 (A trace weights the multiplicity spaces)**.** For $X = \bigoplus_a a^{\oplus m_a}$, an endomorphism consists of matrices f_a on the multiplicity spaces. Closing each simple line contributes d_a, so

$$\mathrm{Tr}_{\mathrm{cat}}(f) = \sum_a d_a \,\mathrm{Tr}(f_a).$$

Thus on $\mathbf{1} \oplus \tau^{\oplus 2}$ in the Fibonacci category, an endomorphism (λ, B) with $B \in M_2(\mathbb{C})$ has trace $\lambda + \varphi\,\mathrm{Tr}(B)$. Its identity has trace $1 + 2\varphi$, not the unweighted sum of multiplicities 3. This weighting is the same one that will appear in modular link traces and twist moments.

1.3 Changing the order of fusion

There are two natural bases of $\mathrm{Hom}(d, a \otimes b \otimes c)$. The left basis first fuses a, b to an intermediate object e; the right basis first fuses b, c to f. Suppressing multiplicity labels for the moment, write the relation as

$$|((ab)e\,c); d\rangle = \sum_f [F_d^{abc}]_{ef} |(a\,(bc)f); d\rangle. \tag{1.5}$$

Only admissible channels occur. A left channel must satisfy $N_{ab}^e N_{ec}^d \neq 0$, and a right channel must satisfy $N_{bc}^f N_{af}^d \neq 0$. Associativity of the fusion ring guarantees equal dimensions for these two spaces. The associator tells us the actual change of basis.

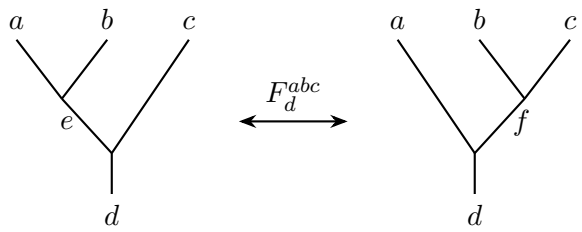


Figure 1.1: Two fusion-tree bases of the same junction space. The intermediate channel changes from e to f under recoupling.

When multiplicities are present, the entries of F have extra indices for bases of V_{ab}^{e} and V_{ec}^{d}, and similarly on the right. No new principle is involved, but one should never replace a multidimensional junction space by a scalar merely because the line labels are the same.

We take Hermitian inner products to be antilinear in the first argument and linear in the second. In a unitary category, orthonormal bases imply

$$FF^{\dagger} = F^{\dagger}F = I.$$

Here and below the displayed convention treats the left channel as a row and the right channel as a column. A reader using the inverse change of basis will transpose or conjugate some formulas. Such convention changes are harmless if carried through consistently.

Worked example 1.4 (A local projector from an F-matrix)**.** Suppose we want to project onto right intermediate channel f_0, while storing vectors in the left basis. Taking inner products in (1.5) gives

$$(P^{(f_0)})_{ee'} = \overline{F_{ef_0}} F_{e'f_0}. \tag{1.6}$$

It is an outer product, hence Hermitian and positive. Its square is itself because the corresponding column of F has norm one. Different channels give orthogonal projectors, and their sum is the identity. This small formula is the basic local ingredient of an anyon-chain Hamiltonian.

1.4 Gauge is a choice of junction bases

In a multiplicity-free category choose a nonzero basis vector t_{ab}^{c} in every nonzero fusion space. Replacing it by $t_{ab}^{c\prime} = u_{ab}^{c} t_{ab}^{c}$ changes the two fusion-tree bases by products of two factors. With convention (1.5),

$$[F_d^{abc}]'_{ef} = \frac{u_{ab}^{e} u_{ec}^{d}}{u_{bc}^{f} u_{af}^{d}} [F_d^{abc}]_{ef}. \tag{1.7}$$

For orthonormal bases the u's have absolute value one. In purely algebraic classifications any nonzero complex basis change is allowed. For higher multiplicities the scalars become invertible matrices.

An individual entry of F is therefore usually not an invariant. Statements comparing entries must specify a gauge or exhibit a cancellation of all gauge factors. A useful exception occurs when $a = b = c = d = e = f = \rho$. The numerator and denominator of (1.7) are then identical. Consequently

$$[F_{\rho}^{\rho\rho\rho}]_{\rho\rho} \tag{1.8}$$

is unchanged by trivalent-basis rescaling. We will use this elementary invariant to distinguish two Haagerup categories with exactly the same fusion table.

Gauge equivalence is also different from relabelling simple objects. A tensor equivalence may include a permutation preserving the fusion rules. When checking whether two categories are

equivalent, one must allow both operations. Conversely, a convenient common labelling does not identify their associators.

1.5 The pentagon is path independence

For four objects there are five binary parenthesizations. Moving between adjacent ones uses an associator. Two routes from $((a \otimes b) \otimes c) \otimes d$ to $a \otimes (b \otimes (c \otimes d))$ must agree. In terms of the associator $\mathfrak{a}_{a,b,c} : (a \otimes b) \otimes c \to a \otimes (b \otimes c)$, this is

$$\mathfrak{a}_{a,b,c\otimes d}\,\mathfrak{a}_{a\otimes b,c,d} = (1_a \otimes \mathfrak{a}_{b,c,d})\mathfrak{a}_{a,b\otimes c,d}(\mathfrak{a}_{a,b,c} \otimes 1_d). \tag{1.9}$$

Composition is read from right to left. In chosen fusion bases this becomes polynomial equations among the F-entries, with sums over intermediate channels. Those equations are the pentagon equations.

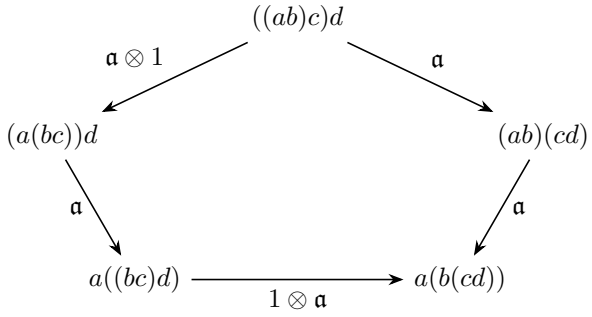


Figure 1.2: The pentagon compares a route of three reassociations with a route of two. Each edge is a linear map on the relevant fusion space; the two composite maps must agree.

Why are four objects enough? The coherence theorem for monoidal categories says that the pentagon, together with the unit axioms, guarantees that all reassociations of an arbitrary tensor product agree. One does not separately impose a new axiom for five, six, or one hundred objects. This is an early example of a theme that recurs in the CFT construction: a finite consistency condition can control arbitrarily complicated compositions, but only through a theorem explaining the passage.

Unitarity and the pentagon are independent demands. A random unitary matrix does not define a fusion category. Conversely, a solution of the pentagon over $\mathbb{C}$ does not automatically supply positive inner products making its recoupling matrices unitary.

1.6 A complete small pentagon calculation: Fibonacci

We now see the golden ratio appear a second time, from associativity rather than the dimension character. Use the two simple objects $\mathbf{1}, \tau$ with (1.2). Choose strict unit conventions and a real gauge in which the only nontrivial matrix is

$$F_\tau^{\tau\tau\tau} = \begin{pmatrix} u & v \\ v & -u \end{pmatrix}, \qquad u, v > 0, \tag{1.10}$$

in channels $(\mathbf{1}, \tau)$. Take the scalar $F_{\mathbf{1}}^{\tau\tau\tau} = 1$. The positive solution we obtain will be the standard unitary Fibonacci category.

The four-τ fusion space with total charge τ has dimension three. In the completely left-associated tree label its channels by

$$(e, f) = (\mathbf{1}, \tau), (\tau, \mathbf{1}), (\tau, \tau),$$

where e is the charge after two lines and f after three. To make every matrix reconstructible, name the four external lines a, b, c, d, all equal to τ. The five trees use the following intermediate labels:

tree, with total charge τ	ordered channel pair
$(((ab)_e c)_f d)_\tau$	(e, f)
$((a(bc)_g)_f d)_\tau$	(g, f)
$(a((bc)_g d)_h)_\tau$	(g, h)
$(a(b(cd)_k)_h)_\tau$	(k, h)
$((ab)_e(cd)_k)_\tau$	(e, k)

In every row, order the allowed pairs as $(\mathbf{1}, \tau), (\tau, \mathbf{1}), (\tau, \tau)$. The three-step route passes through the first four rows, and the two-step route passes from the first to the fifth and then the fourth. Along the three-step side the successive change-of-basis matrices are

$$A = \begin{pmatrix} u & 0 & v \\ 0 & 1 & 0 \\ v & 0 & -u \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & u & v \\ 0 & v & -u \end{pmatrix}, \qquad A.$$

Each block is simply (1.10) acting on the two channels it can mix; the isolated entry is a one-dimensional recoupling. Along the two-step side the matrices are B followed by

$$C = \begin{pmatrix} 0 & u & v \\ 1 & 0 & 0 \\ 0 & v & -u \end{pmatrix}.$$

The permutation visible in C comes from using the same final channel order as on the other route. Thus the pentagon is the ordinary matrix identity

$$ABA = BC. \tag{1.11}$$

Multiplication gives

$$ABA = \begin{pmatrix} u^2 - uv^2 & v^2 & uv + u^2 v \\ v^2 & u & -uv \\ uv + u^2 v & -uv & v^2 - u^3 \end{pmatrix}, \qquad BC = \begin{pmatrix} 0 & u & v \\ u & v^2 & -uv \\ v & -uv & u^2 \end{pmatrix}.$$

The $(2, 1)$ entry requires $v^2 = u$. Since $v \neq 0$, the $(1, 3)$ entry requires $u + u^2 = 1$. These two relations imply every other entry, including $u - u^3 = u^2$ in the bottom right. Positivity selects

$$u = \varphi^{-1}, \qquad v = \varphi^{-1/2}.$$

Therefore

$$F_\tau^{\tau\tau\tau} = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}. \tag{1.12}$$

Its square is the identity because $\varphi^{-2} + \varphi^{-1} = 1$. The total-charge-$\mathbf{1}$ four-τ pentagon reduces to this same identity. Pentagons containing the unit reduce to the strict unit conventions. This accounts for all cases in the two-object fusion rule.

This example illustrates the right order of reasoning. We specified the fusion spaces and a gauge ansatz, imposed associativity, selected the positive solution, and verified unitarity. Merely writing $\tau^2 = \mathbf{1} + \tau$ would not have supplied the matrix (1.12).

Worked example 1.5 (The Fibonacci projector)**.** Projecting two neighboring τ lines onto the vacuum channel, while keeping the left fusion basis of three lines with total τ, gives

$$P^{(\mathbf{1})} = \begin{pmatrix} \varphi^{-2} & \varphi^{-3/2} \\ \varphi^{-3/2} & \varphi^{-1} \end{pmatrix}.$$

Its trace is one and its determinant is zero. Its image is the span of $(\varphi^{-1}, \varphi^{-1/2})^{\mathsf{t}}$. The complementary projector is $P^{(\tau)} = I - P^{(\mathbf{1})}$. An interaction $-P^{(\mathbf{1})}$ energetically favors the vacuum fusion channel, but overlapping interactions need not commute because the two adjacent pairs are diagonal in different fusion bases.

1.7 What closed line operators remember

If Y_a is the operator obtained by wrapping a simple line around the spatial circle, then

$$Y_a Y_b = \sum_c N_{ab}^{\ c} Y_c, \qquad Y_a^\dagger = Y_{a^*}. \tag{1.13}$$

This representation remembers the fusion ring, but it does not remember all junctions or associators. In particular, knowing these matrices on an untwisted Hilbert space is generally insufficient to determine the transformation law of an operator product. Defect sectors and junctions enlarge the algebra to the tube algebra, studied in Chapter 5.

The vacuum transforms by the dimension character:

$$Y_a|0\rangle = d_a|0\rangle.$$

For an invertible line $d_a = 1$, so this agrees with the usual definition of an invariant state. A noninvertible invariant state instead has eigenvalue d_a. Requiring every line to have eigenvalue one would contradict fusion whenever a noninvertible object has more than one summand in its square.

1.8 Exercises

Exercise 1.1. Let $m_L = \dim \mathrm{Hom}(\mathbf{1}, \tau^{\otimes L})$ and $n_L = \dim \mathrm{Hom}(\tau, \tau^{\otimes L})$. Starting with $L = 0$, derive their recursion, list the first six pairs, and explain why their exponential growth rate is φ although every entry is an integer.

Solution on page 288.

Exercise 1.2. Derive (1.7) by rescaling the two trivalent vertices in each fusion tree. Verify explicitly that (1.8) is unchanged. Explain why a comparison of two other individual entries need not distinguish categories.

Solution on page 288.

Exercise 1.3. For an arbitrary unitary F-matrix, use (1.6) to prove $P^{(f)}P^{(g)} = \delta_{fg}P^{(f)}$ and $\sum_f P^{(f)} = I$. Keep the complex conjugates in the derivation.

Solution on page 288.

Exercise 1.4. Check every entry of (1.11) from $v^2 = u$ and $u^2 + u = 1$. Which step uses the assumption $v \neq 0$? Why does the negative root of $u^2 + u = 1$ fail the displayed positive, real gauge?

Solution on page 288.

Exercise 1.5. In the Fibonacci fusion ring solve the scalar equation $y^2 = 1 + y$. Interpret its two solutions as one-dimensional representations of the complexified fusion algebra. Explain why neither solution is an ordinary dimension of a vector space.

Solution on page 289.

Source note. The category and coherence language is standard. The Fibonacci calculation is included to make the recoupling convention explicit; the Haagerup specialization in Chapter 4 uses the transparent F-symbol presentation of Huang and Lin. No large table of coefficients is needed to follow the examples in this chapter.

Chapter 2

Conformal fields, states, and operator products

Conformal field theory organizes infinitely many local operators using symmetry. In two dimensions the organization is unusually powerful: local conformal symmetry separates into holomorphic and antiholomorphic parts, whose generators form two commuting Virasoro algebras. On the Euclidean plane $\bar{z}$ is the complex conjugate of z; treating the two parts separately is a way to organize the symmetry. This chapter develops the pieces of that structure needed later. The aim is to read an operator product expansion (OPE), compute a descendant norm, translate a conformal weight into a finite-circle energy, and recognize which information a character does and does not contain.

A *local field* $\phi(z, \bar{z})$ is an operator insertion at a point. Its measurable content is encoded in correlation functions such as $\langle \phi_1(z_1, \bar{z}_1)\phi_2(z_2, \bar{z}_2)\rangle$, the vacuum expectation value of the indicated insertions. Fields are generally operator-valued distributions, so coincident products need a prescription; separated-point correlators and the OPE provide one. Conformal transformations preserve angles locally. In two dimensions an orientation-preserving such transformation is locally a holomorphic change of coordinate with nonzero derivative. We first describe the physics of its action, and leave the formal algebraic definition of fields and their modules to Chapter 3.

2.1 From scale transformations to radial quantization

Write a Euclidean point as $z = x + iy$. A rotation and a dilation take z to $z' = \lambda e^{i\theta} z$, with $\lambda > 0$. For a field of definite scaling dimension and spin, our coordinate-change convention is

$$\phi'(z', \bar{z}') = \lambda^{-\Delta} e^{-i s_{\text{spin}} \theta} \phi(z, \bar{z}).$$

The *holomorphic and antiholomorphic conformal weights* $(h, \bar{h})$ are the two numbers satisfying

$$\Delta = h + \bar{h}, \qquad s_{\text{spin}} = h - \bar{h}. \tag{2.1}$$

The bar on $\bar{h}$ labels the antiholomorphic weight; it does not mean complex conjugation of h. For a bosonic local field, a full rotation must return the same field, so $s_{\text{spin}} \in \mathbb{Z}$. A scalar has $s_{\text{spin}} = 0$, a stronger condition than integer spin. A *chiral* field depends only on z in separated-point correlators; an *antichiral* field depends only on $\bar{z}$. A holomorphic field of weights $(1, 0)$ is called a chiral current. Its contour integral generates a symmetry. A field of weights $(1, 1)$ is instead a dimension-two scalar. These examples will prevent several otherwise easy mistakes when reading numerical spectra. The integer-spin requirement applies to full bosonic local fields. A chiral factor used to build one can have noninteger weight and need not itself define a single-valued local field of the full theory.

Radial quantization treats circles about the origin as equal-time slices. With an arbitrary reference length set to one, radial time is $t = \log|z|$; moving outwards is moving forwards in this time. The path integral over a disk sums field configurations in its interior with fixed boundary values; as a function of those boundary values it prepares a state on the boundary circle. An empty disk prepares the vacuum $|0\rangle$. A field inserted at the origin creates a state,

$$|\phi\rangle = \lim_{z,\bar{z}\to 0} \phi(z,\bar{z})|0\rangle.$$

Here the ket is a vector in the state space, whereas $\phi(z,\bar{z})$ is a field acting on states. The limit means the state prepared by the insertion; it does not assert that the field is a bounded operator. The *state–operator correspondence* identifies local fields with states in this way. We use it for theories with a normalizable vacuum and a discrete spectrum with finite degeneracies at fixed energy. More general theories require qualifications.

Write L_0 and $\bar{L}_0$ for the operators measuring the two weights:

$$L_0|\phi\rangle = h|\phi\rangle, \qquad \bar{L}_0|\phi\rangle = \bar{h}|\phi\rangle.$$

The generator of radial dilations is $L_0 + \bar{L}_0$; rotations are generated by $L_0 - \bar{L}_0$. The next section constructs these operators as members of two infinite families. Expanding a product of insertions inside a disk into states on its boundary will then give the OPE.

2.2 The stress tensor and its central term

The holomorphic stress tensor $T(z)$ generates infinitesimal conformal transformations. It has weight two. An *operator product expansion* expresses two nearby insertions as a sum of single insertions with coordinate-dependent coefficients. Such an equality is read inside correlators, with the other insertions farther away. The stress tensor's self-OPE is

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \tag{2.2}$$

The first term means $(c/2)(z-w)^{-4}$ times the identity field $\mathbf{1}$, whose insertion changes nothing. The derivative $\partial T(w)$ means $\partial_w T(w)$. The symbol $\sim$ retains the singular part. Regular terms exist and matter for composite fields, but they do not enter the contour commutator that we are about to compute. The number c is the chiral central charge. It measures the central extension of conformal symmetry and controls the vacuum's response to a change of geometry.

Expand the field in a Laurent series. Its coefficient operators, called *modes*, are

$$T(z) = \sum_{n\in\mathbb{Z}} L_n z^{-n-2}, \qquad L_n = \frac{1}{2\pi i}\oint z^{n+1}T(z)\,dz.$$

Each L_n is a linear operator on states. The contour is a small counterclockwise circle about the origin; the integral picks out the coefficient of z^{-1}. Radially ordered products place the operator at larger radius to the left. Subtracting the two radial orderings deforms the contours into one about the collision $z = w$. Evaluating that contour gives

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}. \tag{2.3}$$

Here $[A,B] = AB - BA$ and the Kronecker symbol $\delta_{m+n,0}$ is one if $m+n=0$ and zero otherwise. The scalar central term multiplies the identity operator on the state space; it commutes with all modes. These commutation relations define the *Virasoro algebra*. For example, the fourth-order pole contributes the third derivative of z^{m+1} divided by 3!. Its residue against w^{n+1} is

$c(m+1)m(m-1)\delta_{m+n,0}/12$, explaining both the cubic polynomial and its vanishing for $m = 0, \pm 1$. These three modes generate the ordinary global conformal algebra without a central term. It is the Lie algebra $\mathfrak{sl}_2$, generated here by L_{-1}, L_0, L_1; geometrically these generate translations, dilations, and special conformal transformations. In particular, $[L_0, L_n] = -nL_n$, so L_n shifts a weight by $-n$.

There is a commuting antiholomorphic copy with central charge $\bar{c}$. For the nonchiral theories studied in this book, $c = \bar{c}$. When we say that a theory has central charge eight, we mean $(c, \bar{c}) = (8, 8)$, not that their sum is eight.

Definition 2.1. A *Virasoro primary*, or *highest-weight state*, of weight h satisfies $L_0|\phi\rangle = h|\phi\rangle$ and $L_n|\phi\rangle = 0$ for every $n > 0$. Its field obeys

$$T(z)\phi(w) \sim \frac{h\phi(w)}{(z-w)^2} + \frac{\partial\phi(w)}{z-w}. \tag{2.4}$$

A quasiprimary state is annihilated by L_1 and has definite weight; it need not be annihilated by $L_2, L_3, \ldots$.

Thus a primary has no lower-weight state obtained by applying a positive Virasoro mode. Equivalently its field transforms under a local coordinate change as $\phi'(f(z)) = (f'(z))^{-h}\phi(z)$, with a local branch chosen for the power. A full primary obeys the corresponding conditions in both chiralities. Unless a larger algebra is specified, "primary" in this chapter always means Virasoro primary. The stress tensor is quasiprimary, but for $c \neq 0$ its fourth-order self-OPE pole prevents it from being primary.

Contour integration of (2.4) gives

$$[L_m, \phi(z)] = z^{m+1}\partial\phi(z) + (m+1)hz^m\phi(z).$$

In particular, L_{-1} differentiates a field. Acting with negative Virasoro modes generates *descendants*, the symmetry-related states above a primary. The descendant $L_{-n_1}\cdots L_{-n_r}|\phi\rangle$ has weight $h + n_1 + \cdots + n_r$, for positive integers n_i. The sum $n_1 + \cdots + n_r$ is its *level*, or grade. A representation, also called a module, of this algebra is a vector space on which the L_n act and satisfy (2.3). The *Verma module* is the universal highest-weight representation: start with one $|h\rangle$ and impose only the Virasoro relations and highest-weight conditions. Ordered words with $n_1 \geq \cdots \geq n_r \geq 1$ give its basis. A physical representation can be a quotient of this space by additional relations. Null states are crucial examples of such relations.

2.3 Unitarity and descendant norms

Unitarity means that the physical state space has a positive inner product compatible with time evolution and radial adjoints. For the Virasoro modes this compatibility reads $L_n^\dagger = L_{-n}$, where the dagger denotes the adjoint with respect to that inner product. Radial reflection exchanges the inside and outside of the unit circle. For a chiral primary of real weight h, it represents the bra by an insertion at infinity:

$$\langle\phi| = \lim_{R\to\infty} R^{2h}\langle 0|\phi^\dagger(R).$$

The conjugate chiral field $\phi^\dagger$ is still a field in the holomorphic theory; it is not the independent antiholomorphic field. We abbreviate this limit as $\phi^\dagger(\infty)$. Unit normalization means $\langle\phi^\dagger(\infty)\phi(0)\rangle = 1$. For a full primary on the positive real axis, the factor is $R^{2(h+\bar{h})}$.

If $|h\rangle$ is a unit primary, then

$$\|L_{-1}|h\rangle\|^2 = \langle h|[L_1, L_{-1}]|h\rangle = 2h.$$

Consequently $h \geq 0$. At $h = 0$ the translation descendant is null. In a positive semidefinite representation, a *null state* is a vector of norm zero and is orthogonal to every state. Such vectors are set equal to zero to obtain the physical Hilbert space. Before this quotient they can be nonzero formal vectors. In an indefinite inner product, zero norm alone does not imply orthogonality to every state; there one must specify the radical of the form, the subspace orthogonal to every vector, when taking a quotient. The vacuum obeys the stronger geometric condition $L_n|0\rangle = 0$ for $n \geq -1$.

Worked example 2.2 (The level-two Gram matrix)**.** At level two the Verma module has the ordered basis $v_1 = L_{-2}|h\rangle, v_2 = L_{-1}^2|h\rangle$. The *Gram matrix* records their pairwise inner products, $(G_2)_{ij} = \langle v_i|v_j\rangle$. It is

$$G_2(h,c) = \begin{pmatrix} 4h + c/2 & 6h \\ 6h & 4h(2h+1) \end{pmatrix}. \tag{2.5}$$

The first entry is $\langle h|[L_2, L_{-2}]|h\rangle$. For the off-diagonal entry, use $[L_2, L_{-1}] = 3L_1$ and $L_1 L_{-1}|h\rangle = 2h|h\rangle$ to obtain $6h$. Finally,

$$L_1 L_{-1}^2|h\rangle = 2(2h+1)L_{-1}|h\rangle,$$

which gives the bottom-right entry. Its determinant is

$$\det G_2 = 2h(16h^2 + 2(c-5)h + c).$$

For a coefficient column u, the corresponding state has squared norm $u^\dagger G_2 u$. A nonzero vector in the kernel of G_2 therefore describes a linear combination orthogonal to this whole level. Distinct levels are orthogonal because L_0 is self-adjoint, so a zero determinant detects a null relation. Positivity of every such Gram matrix at every level is much stronger than positivity of the primary norm.

For the vacuum, $L_{-1}|0\rangle = 0$, but

$$\|L_{-2}|0\rangle\|^2 = c/2. \tag{2.6}$$

The stress state is therefore $|T\rangle = L_{-2}|0\rangle$, and its unit-normalized version, when $c > 0$, is $\sqrt{2/c}\,|T\rangle$. This simple norm will later be the basis of a central-charge estimator for the chain.

Null vectors must be quotiented out to obtain the physical state space. Counting all words in the negative modes without doing this would overcount states. Conversely, an observed deficit in a character is not automatically a new null relation. A character is a graded state count, defined in Section 2.7; one first has to specify which highest-weight module is being counted.

2.4 What an operator product specifies

2.4.1 Choose the chiral algebra before naming its modules

The stress tensor need not be the only holomorphic field whose modes organize the states. A *chiral algebra* is a specified collection of mutually local holomorphic fields containing the identity and stress tensor and closed under derivatives and OPE products. Here mutual locality means that exchanging these algebra fields produces no nontrivial branch monodromy: their OPEs use integer powers, with the usual bosonic locality relations. Chapter 3 makes this precise using vertex operator algebras. The chiral algebra generated by T, whose modes form the Virasoro Lie algebra, is the minimal example. Adjoining currents or other holomorphic fields gives an *extended chiral algebra*.

A *module*, or representation, for a chosen chiral algebra is a vector space M on which all the modes of its fields act, obeying the same OPE identities as in the algebra. For example,

the modes $L_n : M \to M$ must satisfy the Virasoro commutators. If a current J belongs to the algebra, its modes must also act on M and satisfy their OPE relations with T, with each other, and with the other fields. Thus M consists of states together with specified operator actions. It is neither a single field nor merely a number of states.

A module is *irreducible*, also called *simple*, if it has no nonzero proper subspace preserved by every algebra mode. An irreducible representation type is often called a *sector*. The *vacuum module* contains $|0\rangle$ and the states obtained from it using the algebra fields. Other modules need not contain a vacuum. A Virasoro highest-weight module is the familiar example generated by the negative L_n from a primary, after any required quotient. A module for a larger algebra can contain many Virasoro primaries: its extra modes connect their Virasoro families. Consequently the number of irreducible modules depends on which chiral algebra we choose.

It is also useful to distinguish an algebra field from a general chiral field. The former acts within each module. A charged chiral field can instead carry a state from one module into another, and its products may involve fractional powers. Such fields are described by *intertwining operators*: field-valued maps between modules compatible with the algebra action. We will use their basic OPE meaning here; their precise definition comes in Chapter 3. A full local field combines holomorphic and antiholomorphic data so that its correlators have the required single-valuedness.

2.4.2 Coefficients, descendants, and fusion channels

If two local fields are close compared with every other separation, their product can be expanded in local fields at a common point. For chiral primaries the schematic form is

$$\phi_i(z)\phi_j(0) = \sum_k C_{ij}^{\ k} z^{h_k - h_i - h_j} \big(\phi_k(0) + \text{descendants}\big). \tag{2.7}$$

The descendant coefficients are constrained by conformal symmetry; the primary coefficients $C_{ij}^{\ k}$ contain dynamical information. The power of z balances the conformal weights. Each level-n descendant inside the parentheses comes with an additional z^n. For instance, if the exchanged primary has weight $h_k > 0$, the level-one term is

$$\phi_k(0) + z\,\frac{h_k + h_i - h_j}{2h_k}\partial\phi_k(0) + \cdots .$$

Its coefficient follows by applying L_1 and using $L_1 L_{-1}|h_k\rangle = 2h_k|h_k\rangle$. At higher levels one solves analogous linear equations using descendant Gram matrices, with null states removed. This is why symmetry fixes descendant coefficients once the primary coupling is chosen.

An *OPE channel* specifies the representation exchanged between the two insertions and the rest of the correlator. For a chosen chiral algebra, the allowed representation types form its *fusion rules*. A fusion coefficient counts independent compatible chiral couplings, just as a junction multiplicity did in Chapter 1; it does not count descendants. If more than one such coupling reaches the same output representation, one must attach an extra intertwiner label and choose a basis of that finite-dimensional coupling space. Equation (2.7) suppresses these labels and resolves fields into Virasoro families.

Choose an orthonormal primary basis, so that $\langle \phi_k^\dagger(\infty)\phi_l(0)\rangle = \delta_{kl}$. Then the three-point coefficient can be read as

$$C_{ij}^{\ k} = \langle \phi_k^\dagger(\infty)\phi_i(1)\phi_j(0)\rangle.$$

Conformal invariance then fixes all coordinate dependence:

$$\langle \phi_i(z_1)\phi_j(z_2)\phi_k(z_3)\rangle = \frac{C_{ijk}}{z_{12}^{h_i+h_j-h_k} z_{13}^{h_i+h_k-h_j} z_{23}^{h_j+h_k-h_i}}. \tag{2.8}$$

Here $z_{ij} = z_i - z_j$. For full fields there is an analogous antiholomorphic factor. The all-lower coefficient C_{ijk} uses a two-point pairing with the third field: if $g_{lk} = \langle \phi_l(\infty)\phi_k(0)\rangle$, then, with consistent field ordering and branches, $C_{ijk} = \sum_l C_{ij}^{\,l} g_{lk}$. For charged fields this pairing typically pairs opposite charges; lowering the index is not the same as ignoring a dagger. Locality and branch conventions must be respected when the chiral powers are not integers. A chiral contribution can be multivalued even when the full correlation function is single-valued. We define the individual contributions called conformal blocks in Section 2.8.

Field phases change individual complex coefficients. Under $\phi_i \mapsto e^{i\theta_i}\phi_i$, the coefficient $C_{ij}^{\,k}$ changes by $e^{i(\theta_i+\theta_j-\theta_k)}$. Norms do not fix these phases. Products of coefficients can cancel every phase and hence contain physical information. A real Hamiltonian or a positive Hilbert-space metric does not require every mixed coefficient in every charged basis to be real. The overlap chapter will use explicit invariants to separate phase conventions from measured relative phases.

Worked example 2.3 (The universal stress coefficient)**.** Let ϕ be a unit chiral primary of weight h and let $c > 0$. The contribution of the *Virasoro vacuum family*, consisting of the identity and its Virasoro descendants, to $\phi(z)\phi^\dagger(0)$ begins

$$\phi(z)\phi^\dagger(0)\Big|_{\text{Virasoro vacuum}} = z^{-2h}\left(\mathbf{1} + \frac{2h}{c}z^2 T(0) + \cdots\right). \tag{2.9}$$

To derive the coefficient, insert $T(w)$ far from the pair. A *Ward identity* expresses an insertion of a symmetry current through its action on the other fields. Summing the poles of (2.4) gives the stress-tensor Ward identity on the plane,

$$\left\langle T(w)\prod_i \phi_i(z_i)\right\rangle = \sum_i \left(\frac{h_i}{(w-z_i)^2} + \frac{1}{w-z_i}\partial_{z_i}\right)\left\langle \prod_i \phi_i(z_i)\right\rangle.$$

Apply it to the two-point function $(z_1 - z_2)^{-2h}$, and then set $z_1 = z, z_2 = 0$. The result is $hz^{2-2h}/[w^2(w-z)^2]$, whose leading large-w term is $hz^{2-2h}w^{-4}$. The proposed OPE instead gives $Az^{2-2h}\langle T(w)T(0)\rangle = A(c/2)z^{2-2h}w^{-4}$, so $A = 2h/c$. The coupling to the unit stress state is therefore $h\sqrt{2/c}$, whose square is $2h^2/c$. For $h = 2, c = 8$, this coefficient is one. Other families may also appear in the OPE. In particular a current can belong to the vacuum module of an extended algebra while being a separate Virasoro primary. Such a current contribution has not been included in the displayed Virasoro-vacuum projection.

2.5 Free bosons as a complete first laboratory

A chiral free boson can be normalized by $X(z)X(w) \sim -\log(z-w)$. This formula specifies its Gaussian two-point contraction, the part used when evaluating products by pairwise contractions. Because of the logarithm, X itself is an auxiliary chiral field with branch and zero-mode subtleties; its derivative is the single-valued current we use in the algebra. Set $J = i\partial X$. Differentiating the logarithm with the appropriate factors of i gives

$$J(z)J(w) \sim \frac{1}{(z-w)^2}.$$

Normal ordering, denoted by colons, subtracts the singular self-contraction before taking coincident points. For example,

$$:JJ:(w) = \lim_{z\to w}\left(J(z)J(w) - \frac{\mathbf{1}}{(z-w)^2}\right).$$

The limit is understood inside separated-point correlators. For these Gaussian fields, *Wick's rule* expresses a product of normal-ordered fields as the sum over all pairwise contractions between

them. The stress tensor $T = \frac{1}{2} : JJ :$ has central charge one: in $T(z)T(w)$ there are two double contractions, giving $\frac{1}{4} \cdot 2(z-w)^{-4} = \frac{1}{2}(z-w)^{-4}$. The single contractions give the required $2T$ and ∂T terms. Building a stress tensor quadratically from currents is called a *Sugawara construction*; here it is just this elementary free-boson calculation.

For a real charge p, the normal-ordered exponential $V_p =: e^{ipX} :$ is obtained by expanding the exponential and normal-ordering each term. Wick's rule resums the contractions between two exponentials into $\exp[-pq\langle X(z)X(w)\rangle] = (z-w)^{pq}$. It gives

$$J(z)V_p(w) \sim \frac{p}{z-w} V_p(w), \tag{2.10}$$

$$V_p(z)V_q(w) = (z-w)^{pq}\big(V_{p+q}(w) + \text{oscillator descendants}\big), \tag{2.11}$$

$$h(V_p) = p^2/2. \tag{2.12}$$

The exponent in the OPE agrees with $h(p+q) - h(p) - h(q) = pq$. This agreement is a useful elementary check on every later lattice-exponential formula.

The weight follows from the double contraction in $T(z)V_p(w)$: its double pole is $(p^2/2)V_p(w)$. The single contractions give $\partial V_p(w)$ in the simple pole, so V_p is a Virasoro primary.

2.5.1 Current modes and a concrete chiral module

Expand the current as

$$J(z) = \sum_{n\in\mathbb{Z}} J_n z^{-n-1}, \qquad J_n = \frac{1}{2\pi i}\oint z^n J(z)\, dz.$$

The JJ OPE gives $[J_m, J_n] = m\delta_{m+n,0}\,\mathrm{id}$. Wick's rule gives $T(z)J(w) \sim J(w)/(z-w)^2 + \partial J(w)/(z-w)$, so J is a weight-one primary and $[L_0, J_n] = -nJ_n$. Thus J_{-r} raises the weight by r for $r > 0$; it is a creation operator. The operator J_0, called the *zero mode* of the current, measures charge rather than raising or lowering weight.

Let $|p\rangle = V_p(0)|0\rangle$. Its defining conditions are

$$J_0|p\rangle = p|p\rangle, \qquad J_n|p\rangle = 0 \ (n>0), \qquad L_0|p\rangle = \frac{p^2}{2}|p\rangle.$$

The *Fock module* F_p is the vector space with basis

$$\prod_{r\geq 1} J_{-r}^{k_r}|p\rangle, \qquad k_r \in \mathbb{Z}_{\geq 0}, \quad \text{only finitely many } k_r \neq 0.$$

The negative modes act by adding factors, positive modes act through the commutators, and J_0 acts by p throughout. For example, $J_1 J_{-1}|p\rangle = |p\rangle$ and $J_1 J_{-2}|p\rangle = 0$. This is a concrete module for the chiral algebra generated by J and $T = \frac{1}{2} :JJ:$. Its basis state above has weight $p^2/2 + \sum_r r k_r$. The term *oscillator descendants* in the exponential OPE means precisely states formed by these current creation modes.

This example also distinguishes two kinds of descendant. The Sugawara formula gives $L_{-1}|p\rangle = pJ_{-1}|p\rangle$. At $p = 0$ the Virasoro translation descendant vanishes, but $J_{-1}|0\rangle$ does not. That state is the current itself, a new Virasoro primary inside the current algebra's vacuum module. We will count the states of F_p explicitly in Section 2.7.

2.5.2 Charge conservation and compactification

The current Ward identity is

$$\left\langle J(z)\prod_j V_{p_j}(z_j)\right\rangle = \sum_j \frac{p_j}{z-z_j}\left\langle \prod_j V_{p_j}(z_j)\right\rangle.$$

Integrating around all insertions computes their total J_0 charge. Since the vacuum at infinity has charge zero, a nonzero vacuum correlator requires $\sum_j p_j = 0$. For charges summing to zero,

$$\left\langle \prod_j V_{p_j}(z_j) \right\rangle = \prod_{i<j} (z_i - z_j)^{p_i p_j}; \qquad \sum_j p_j = 0. \tag{2.13}$$

Otherwise the correlator vanishes. With several bosons, replace $p_i p_j$ by the Euclidean inner product of charge vectors. A lattice theory also needs *cocycles*: consistent phase factors in multiplying its charged fields. They fix exchange signs that the bare exponentials alone do not supply; the next chapter explains their role.

Worked example 2.4 (A charged four-point function)**.** For $h = p^2/2$, place charges $(-p, p, -p, p)$ at $(\infty, 1, x, 0)$. Equation (2.13) gives

$$\langle V_{-p}(\infty) V_p(1) V_{-p}(x) V_p(0) \rangle = x^{-p^2} (1-x)^{-p^2}.$$

Near $x = 0$, expand $(1-x)^{-p^2} = 1 + p^2 x + \cdots$. The coefficient p^2 is the square of the coupling to the unit current J, as follows independently from its Ward identity. Indeed the full current-algebra OPE starts

$$V_p(z) V_{-p}(0) = z^{-p^2} \big(\mathbf{1} + pz J(0) + \cdots \big).$$

The current contribution explains the linear term in the four-point function; the Virasoro vacuum family alone has no level-one state. This is a small version of the charge-expansion calculation used for the defining Haagerup field u.

A *compact* full boson has a real field identified by $X_{\text{full}} \equiv X_{\text{full}} + 2\pi R$, with $R > 0$ its target circle's radius. This target circle differs from the spatial circle on which the quantum theory is placed. Its zero-mode wavefunctions have integer momentum label m, and a field configuration can wind n times around the target circle while going once around space. The pairs $(m, n) \in \mathbb{Z}^2$ form the momentum–winding lattice. In the radius convention used here, their left and right charges are

$$p_L = \frac{m}{R} + \frac{nR}{2}, \qquad p_R = \frac{m}{R} - \frac{nR}{2}, \qquad m, n \in \mathbb{Z},$$

with $h = p_L^2/2$, $\bar{h} = p_R^2/2$ for the ground state in each charge sector. Oscillator excitations add nonnegative integer levels to each weight. Consequently the ground-state dimension and spin are

$$\Delta = \frac{m^2}{R^2} + \frac{n^2 R^2}{4}, \qquad s_{\text{spin}} = mn \in \mathbb{Z}.$$

The radius changes dimensions while preserving the central charge. There are infinitely many charge sectors for the algebra generated only by the current. When R^2 is rational in this convention, an extension by suitable charged holomorphic fields can organize them into finitely many irreducible modules. Thus "rational radius" here refers to R^2, not R; the self-dual example $R = \sqrt{2}$ qualifies. Rationality is a property of a specified chiral algebra and its representations, not a synonym for having some rational-looking low dimensions.

2.6 From the plane to the cylinder

Set $z = e^{2\pi w/L}$, where $w = v t_E + i x$, t_E is Euclidean time, and x has spatial period L. The constant $v > 0$ converts time to length. Advancing x by L leaves z unchanged; advancing t_E moves radially outwards. This is why the map converts radial evolution on the plane into time evolution on a cylinder. Unlike a primary, the stress tensor acquires a geometry-dependent term

under a coordinate change. That term involves the following combination of derivatives, called the *Schwarzian derivative*:

$$T_{\text{cyl}}(w) = \left(\frac{dz}{dw}\right)^2 T(z) + \frac{c}{12}\{z,w\}, \qquad \{z,w\} = \frac{z'''}{z'} - \frac{3}{2}\left(\frac{z''}{z'}\right)^2.$$

Primes mean derivatives with respect to w. For this exponential map, $\{z,w\} = -\frac{1}{2}(2\pi/L)^2$. The cylinder Hamiltonian and momentum are therefore

$$H_{\text{cyl}} = \frac{2\pi v}{L}\left(L_0 + \bar{L}_0 - \frac{c+\bar{c}}{24}\right), \tag{2.14}$$

$$P_{\text{cyl}} = \frac{2\pi}{L}\left(L_0 - \bar{L}_0 - \frac{c-\bar{c}}{24}\right). \tag{2.15}$$

For equal chiral central charges and a vacuum of weight zero, $E_0 = -\pi v c/(6L)$. A lattice Hamiltonian adds an extensive nonuniversal energy $e_\infty L$, where e_∞ is the bulk ground-state energy per unit length. The negative $1/L$ vacuum term is the *Casimir energy*, the finite-size shift created by the change of geometry. Terms that vanish faster at large L can come from irrelevant perturbations, defined in Section 2.9. The lattice excitation gaps approach $2\pi v\Delta/L$ only when a CFT limit and the appropriate scaling regime have been reached.

For a state at left and right descendant levels $N, \bar{N}$ above weights $(h, \bar{h})$, replace Δ in that gap by $h + \bar{h} + N + \bar{N}$. For $c = \bar{c}$, its momentum relative to the vacuum is $2\pi(h - \bar{h} + N - \bar{N})/L$. Thus characters will count precisely the excitations whose circle energies these formulas describe. We use units with $\hbar = 1$.

The velocity v is a conversion between lattice time and continuum space; it is not fixed by conformal symmetry. An energy fit that extracts the coefficient of $1/L$ in the vacuum energy measures vc, not c separately. A second observable is needed to calibrate v. The distinction becomes crucial when comparing a theory of central charge exactly two with one near 2.025.

2.7 Characters and the torus

2.7.1 What a character counts

Recall from Section 2.4.1 that a chiral module M is a vector space carrying the modes of a specified chiral algebra. To define the character used here, suppose that L_0 is diagonalizable on M, has a lowest eigenvalue h, and gives a decomposition

$$M = \bigoplus_{n\geq 0} M_{h+n}, \qquad M_{h+n} = \{v \in M : L_0 v = (h+n)v\}, \qquad a_n = \dim M_{h+n} < \infty.$$

Some of these eigenspaces may be zero. This is the ordinary, positive-energy, integer-spaced grading appropriate to the untwisted modules in our examples. "Untwisted" means that the algebra fields return to themselves after going around the insertion, without an extra symmetry transformation. Twisted sectors can have different mode spacings. The integer n is the grade, and M_h is the *top space*, the lowest-weight eigenspace. Even an irreducible module can have $a_0 > 1$.

The *character* is the generating function for these graded dimensions, with the cylinder vacuum-energy shift included:

$$\chi_M(\tau) = \operatorname{Tr}_M q^{L_0 - c/24}, \qquad q = e^{2\pi i\tau}, \qquad \Im\tau > 0. \tag{2.16}$$

Here Tr_M is the sum of ordinary linear-algebra traces over the finite-dimensional graded pieces of M. On each of the a_n basis states at weight $h+n$, the traced operator acts by $q^{h+n-c/24}$. Consequently

$$\chi_M(\tau) = q^{h-c/24}\sum_{n\geq 0} a_n q^n.$$

The variable q records weight; it is not the field coordinate z. The condition $\Im\tau > 0$ makes $|q| < 1$. We may first regard the sum as a formal series. Analytic convergence also requires control of the growth of a_n; finite-dimensional grades alone do not prove it. The free-boson example below converges, as do the rational models used later under their stated hypotheses.

The integer a_0 counts lowest-weight states. A quantum dimension, in the sense introduced in Chapter 1, instead describes fusion and the value of a closed categorical line. When a category of modules is available these are still different notions of dimension. Likewise a character records multiplicities, but does not specify the matrices of every mode action or the OPE coefficients.

Worked example 2.5 (Counting a free-boson module)**.** For F_p the lowest weight is $h = p^2/2$ and $c = 1$. The first three grades have the following bases:

n	states	a_n
0	$\lvert p\rangle$	1
1	$J_{-1}\lvert p\rangle$	1
2	$J_{-2}\lvert p\rangle,\ J_{-1}^2\lvert p\rangle$	2.

At grade three the occupation patterns are 3, $2+1$, and $1+1+1$, giving three states. More generally, k_r copies of J_{-r} contribute grade rk_r. Summing each occupation number independently gives

$$\begin{aligned}\chi_{F_p}(\tau) &= q^{p^2/2-1/24}\prod_{r\geq 1}(1+q^r+q^{2r}+\cdots)\\ &= q^{p^2/2-1/24}\prod_{r\geq 1}(1-q^r)^{-1}\\ &= q^{p^2/2-1/24}(1+q+2q^2+3q^3+5q^4+\cdots).\end{aligned}$$

The grade-four count comes from the five partitions 4, $3+1$, $2+2$, $2+1+1$, and $1+1+1+1$. The product converges for $|q| < 1$ since the sum of the absolute values $\sum_{r\geq 1}|q|^r$ is finite.

The modules F_p and F_{-p} have the same character, although for $p \neq 0$ the current zero mode has different eigenvalues on them. Thus even a complete character need not determine the module. At $p = 0$ the q term counts the current state; it is present in the current algebra's vacuum module even though the Virasoro vacuum has no level-one descendant.

2.7.2 Pairing chiral sectors on a torus

A torus of complex structure τ is $\mathbb{C}/(\mathbb{Z}+\tau\mathbb{Z})$: identify points differing by an integer combination of the two periods 1 and τ. Taking $\Im\tau > 0$ fixes the orientation. It can be viewed as a spatial circle evolved through Euclidean time and glued back to itself, possibly after a spatial translation.

In the rational setting, finitely many irreducible chiral modules are available and the relevant state spaces decompose into direct sums of them. Complete reducibility, rather than finiteness alone, is part of the rationality assumptions made precise in Chapters 3 and 10. Suppose for simplicity that the left and right chiral algebras agree, with irreducible modules M_i. The full state space then decomposes as

$$\mathcal{H} = \bigoplus_{i,j} \mathbb{C}^{Z_{ij}} \otimes M_i \otimes \overline{M_j}.$$

This formula describes finite-energy states; the physical Hilbert space includes their Hilbert completion. The barred module denotes the corresponding antiholomorphic factor. The multiplicity Z_{ij} counts how many copies of the left–right pair occur and is therefore a nonnegative integer. Taking a trace on each tensor factor gives the full torus partition function

$$Z(\tau,\bar\tau) = \sum_{i,j} Z_{ij}\chi_i(\tau)\overline{\chi_j(\tau)}$$

where $\chi_i = \chi_{M_i}$. Different left and right algebras require two different character families, but the same trace argument. The index 0 labels the vacuum module. If it is the only sector with weight zero, a unique vacuum requires $Z_{00} = 1$. The simple choice $Z_{ij} = \delta_{ij}$ pairs each left sector with its corresponding right sector and gives $Z = \sum_i |\chi_i|^2$; it is the diagonal pairing in this convention.

Changing the oriented basis of the torus's two fundamental cycles generates $\tau \mapsto \tau + 1$ and $\tau \mapsto -1/\tau$. Neither cycle is contractible on the torus. The first transformation changes one basis cycle by adding the other; the second exchanges the cycles with an orientation sign. They generate the modular group acting on τ. A bosonic full theory with the anomaly cancelled between left and right must assign the same partition function to these equivalent descriptions of the torus. This *modular invariance* constrains the multiplicity matrix (Z_{ij}), but it does not give the three-point constants. Different local theories can have the same torus state count.

2.7.3 The thermal meaning of the trace

For a thermal circle of Euclidean-time circumference β, the inverse temperature in units with Boltzmann's constant one, take $\tau = iv\beta/L$. Then $Z = \mathrm{Tr}_{\mathcal{H}}\, e^{-\beta H_{\mathrm{cyl}}}$. Indeed multiplying $q^{L_0 - c/24}$ by its right-moving counterpart gives the Boltzmann factor from (2.14). More generally, writing $\tau = \tau_1 + i\tau_2$, the exponent in that full trace is

$$-2\pi\tau_2 \left(L_0 + \bar{L}_0 - \frac{c + \bar{c}}{24} \right) + 2\pi i \tau_1 \left(L_0 - \bar{L}_0 - \frac{c - \bar{c}}{24} \right).$$

Thus τ_2 weights energy and τ_1 inserts a spatial translation, or equivalently a phase determined by momentum.

The modular transformation $\tau \mapsto -1/\tau$ exchanges a short thermal cycle with a long one. In a unitary *compact* theory, meaning here a discrete finite-degeneracy spectrum and a convergent thermal trace, a unique normalizable vacuum controls the low-temperature limit. It therefore also determines the leading high-temperature growth through modular invariance. For equal chiral charges the result is $\log Z \sim \pi c L/(6v\beta)$ as $v\beta/L \to 0$; the exercise at the end of the chapter derives it from the vacuum term. In a nonunitary chiral theory whose smallest weight is $h_{\min}$, the analogous exponent involves the *effective central charge* $c_{\mathrm{eff}} = c - 24h_{\min}$, provided the modularly transformed trace has a nonzero leading contribution from that lowest-weight sector. For a continuum of states without a normalizable vacuum, one must analyze the spectral measure instead of inserting a vacuum term by hand. These qualifications are central to the later reduced-theory proposals.

2.8 Crossing and the meaning of a conformal block

2.8.1 A channel is a choice of intermediate states

Four distinct insertion points contain one complex variable that cannot be removed by a global conformal transformation. This *cross-ratio* can be written

$$x = \frac{z_{12} z_{34}}{z_{13} z_{24}}.$$

A Möbius transformation $z \mapsto (az + b)/(cz + d)$, with $ad - bc \neq 0$, puts the ordered points at $(\infty, 1, x, 0)$. Here a, b, c, d are coordinate-map coefficients, unrelated to the central charge. We use the chiral correlator

$$G(x) = \langle \phi_1(\infty)\phi_2(1)\phi_3(x)\phi_4(0) \rangle$$

with the infinity normalization already defined. Near $x = 0$ we expand the pair at $x, 0$; near $x = 1$ we expand the pair at $1, x$. These expansions, analytically continued with consistent branches,

must describe the same correlator. This is *crossing symmetry*. For full fields the corresponding function is $G(x, \bar{x})$.

A *conformal block* sums the intermediate states in one representation of the chosen symmetry algebra, with primary couplings factored out. To see what is being summed, choose at level N a basis $|p, N, a\rangle$ of descendants of an exchanged weight-p state, after removing null vectors. If G_N is its Gram matrix, the projector onto that level is

$$\Pi_{p,N} = \sum_{a,b} |p, N, a\rangle (G_N^{-1})_{ab} \langle p, N, b|.$$

Insert this projector between the pair of fields inside the radial circle and the pair outside. Each term is a product of two three-point matrix elements contracted with an inverse norm matrix. Summing over N gives the block. This explains why descendant norms, null quotients, and OPE coefficients enter the same calculation.

2.8.2 Global blocks and Virasoro blocks

The global algebra is generated by L_{-1}, L_0, L_1. Its highest-weight state need only be quasiprimary with respect to Virasoro, and its descendants are $L_{-1}^n|p\rangle$. For identical chiral external weights h and a unit-normalized exchanged state of weight $p > 0$, the squared norm at level n is $n!(2p)_n$. After the primary couplings are factored out, each three-point matrix element contributes $(p)_n$. Consequently the global block in our unreduced correlator is

$$x^{p-2h}\, {}_2F_1(p, p; 2p; x).$$

Here the series is explicitly

$${}_2F_1(p, p; 2p; x) = \sum_{n\geq 0} \frac{(p)_n^2}{(2p)_n\, n!} x^n, \qquad (p)_n = p(p+1)\cdots(p+n-1), \quad (p)_0 = 1.$$

It converges for $|x| < 1$ and sums L_{-1} descendants; continuation to other channels requires a choice of path. Multiplying $G(x)$ by x^{2h} defines a reduced correlator. Its corresponding block is $x^p\, {}_2F_1(p, p; 2p; x)$; the different prefactor is only a convention. The global identity block is x^{-2h} in the unreduced convention, since the identity has no nonzero L_{-1} descendants. One should not substitute $p = 0$ into the displayed fractions without treating this case separately. A Virasoro block also sums descendants involving $L_{-2}, L_{-3}, \ldots$, and depends on c. A block for an extended algebra sums still more states. These three expansions are related but are not interchangeable. In particular, the Virasoro vacuum block includes the stress state at level two, even though the global identity block has no such descendant.

Worked example 2.6 (An elementary global block)**.** At exchanged weight $p = 1$, $(1)_n = n!$ and $(2)_n = (n+1)!$. Thus the series coefficient is $1/(n+1)$ and

$$x^{1-2h} \sum_{n\geq 0} \frac{x^n}{n+1} = -x^{-2h} \log(1-x) = x^{1-2h} \left(1 + \frac{x}{2} + \frac{x^2}{3} + \cdots\right).$$

The logarithm here is the ordinary analytic function on the branch reached from $x = 0$. Its branch point at $x = 1$ shows why continuation between channels requires more than merely substituting a new real number into the original power series. This is a global block, not the complete four-point function or its Virasoro completion.

Reflection positivity means that reflecting an inner set of insertions through the quantization circle and taking their adjoints produces an outer set whose correlator with the inner set is a nonnegative squared norm. With several choices of inner state it produces a positive

semidefinite Gram matrix. In such a four-point channel, exchanged primaries carry nonnegative coefficients such as $|C|^2$ after orthonormalizing multiplicity spaces. Together with crossing, this positivity gives inequalities constraining allowed weights and OPE coefficients, a method called the *conformal bootstrap*. In a mixed charged channel, individual coefficients can instead be complex products of distinct OPE constants. Positivity must be applied to the correct matrix of correlators, not imposed on every coefficient separately.

When a rational chiral theory has the required tensor-category structure, changing the chiral fusion channel is encoded by an associator, of the kind studied in Chapter 1. Its finite-dimensional spaces label independent fusion couplings; each block separately sums an infinite family of descendant states. For the Haagerup construction the category of chiral modules and the category of symmetry lines are distinct categories related by the construction. Their associator matrices cannot simply be identified. Chapter 17 develops that distinction.

Crossing is a local consistency test, while modular invariance is a global torus consistency test. A construction must eventually satisfy both, as well as the other *sewing identities*: compatibility of correlators when surfaces are cut into simpler pieces and glued back together in different orders. Starting with an actual lattice vertex algebra supplies chiral associativity: the different orders of expanding its OPEs agree. It does not remove the work of identifying its representation category or the full two-dimensional theory built from it.

2.9 Perturbations and what dimension can tell us

Let S_* be the Euclidean action of a CFT. A *perturbation* changes it to

$$S = S_* + \lambda_{\text{phys}} \int d^2x\, \Phi(x),$$

where Φ is a scalar of dimension Δ. The action is dimensionless, the area element has units of length squared, and Φ has units of length to the power $-\Delta$. Consequently λ_{phys} has units of length$^{\Delta-2}$, or mass dimension $2-\Delta$. At an observation length ℓ, its dimensionless strength at leading order is $g(\ell) = \lambda_{\text{phys}}\ell^{2-\Delta}$.

The *renormalization group* tracks how the effective dimensionless couplings change when short-distance fluctuations are averaged out and the observation scale is increased. The derivative of a running coupling with respect to $\log \ell$ is its *beta function*. Including interaction corrections, and using the convention that ℓ increases toward long distances, it has the form

$$\frac{dg}{d\log \ell} = (2-\Delta)g + bg^2 + \cdots .$$

Thus $\Delta < 2$ is *relevant*: a small perturbation grows at long distances to first order. For $\Delta > 2$ it decreases and is *irrelevant*. When $\Delta = 2$ it is *marginal by dimension*; the linear term gives no answer about its eventual flow. The quadratic coefficient depends on normalized OPE data and on the coupling convention. For a single marginal coupling, keeping only that term and taking $g(\ell_0) = g_0$ gives

$$g(\ell) = \frac{g_0}{1 - bg_0 \log(\ell/\ell_0)}.$$

This approximation is valid while $|g|$ remains small. If $bg_0 < 0$ the coupling approaches zero logarithmically (marginally irrelevant); if $bg_0 > 0$ its magnitude grows (marginally relevant). If $b = 0$, higher terms can still produce running. An *exactly marginal* deformation has vanishing beta functions to all orders and gives a continuous family of CFTs. Several dimension-two fields can mix, and symmetry can permit or forbid their coefficients independently of their dimensions. Worked examples 14.2 and 14.3 derive a nonzero quadratic term from current OPEs and show how it produces slow logarithmic drift.

The *scalar gap* is the smallest positive scaling dimension of a nonidentity spin-zero primary in the specified spectrum. It is a dimension gap, not a nonzero energy gap in the infinite-volume Hamiltonian. The *symmetry-invariant scalar gap* restricts the same question to symmetry-preserving perturbations. For a group this means that every group element leaves the field unchanged. For a noninvertible line a, invariance means that its line operator acts on the field's state by the vacuum eigenvalue d_a, the quantum dimension, as explained in Chapter 1. A relevant scalar may exist but be forbidden in a symmetry-preserving Hamiltonian. Conversely, a gap of two in the invariant sector does not by itself prove a stable *conformal manifold*, a continuous family of inequivalent conformal theories parametrized by exactly marginal couplings. One must solve the coupled beta functions and identify deformations that merely relabel the same theory. We will compute these different gaps explicitly for the two full $c = 8$ theories.

2.10 Exercises

Exercise 2.1. Use the Virasoro commutator to show $\left\|L_{-1}^{n}|h\rangle\right\|^2 = n!(2h)_n$ for a unit primary. Here $(a)_n = a(a+1)\cdots(a+n-1)$. Derive the first two coefficients of the global block from these norms.

Solution on page 289.

Exercise 2.2. At $c = 1/2, h = 1/16$, find the null vector at level two from (2.5). Check directly that L_1 and L_2 annihilate it.

Solution on page 289.

Exercise 2.3. In the compact-boson convention of this chapter, determine the radius at which the $(m, n) = (1, 1)$ field is a holomorphic current. Show that the transformation $R \mapsto 2/R$, $m \leftrightarrow n$ preserves the spectrum.

Solution on page 289.

Exercise 2.4. Let T_1, T_2 be commuting stresses of positive central charges c_1, c_2, meaning that their mixed OPE has no singular part. Prove that $A = c_2 T_1 - c_1 T_2$ is a weight-two primary for $T = T_1 + T_2$, and compute its norm. Explain why A is not itself a stress tensor with this normalization.

Solution on page 289.

Exercise 2.5. For a unitary compact theory with $c = \bar{c}$, a discrete spectrum, a convergent thermal trace, and a unique normalizable vacuum, derive the leading high-temperature formula $\log Z \sim \pi c L/(6v\beta)$ from modular invariance and vacuum dominance at low temperature. Identify the step that fails when the vacuum is absent from the normalizable Hilbert space.

Solution on page 290.

Exercise 2.6. A character calculation finds thirty-six scalar primaries of dimension two invariant under a specified symmetry. State three additional kinds of information needed before asserting that they parametrize a thirty-six-dimensional conformal manifold.

Solution on page 290.

Chapter 3

Vertex algebras and the lattice laboratory

The preceding chapter described fields through their correlation functions and conformal transformations. A vertex algebra packages their operator products algebraically. A lattice vertex algebra is a particularly concrete example: its states are oscillator monomials attached to lattice momenta, and its products follow from one exponential formula. This is the setting in which the Haagerup construction becomes explicit.

We begin with the space of chiral states and explain how a field acts on that space. The distinction between a state, a field, and one coefficient of a field is essential: they are three different kinds of mathematical object. Once their types and notation are clear, the vertex-algebra axioms express the vacuum, translation, and locality properties familiar from CFT. We then build a lattice example in which the states and the operators can both be written explicitly.

3.1 The state–field map and the mode dictionary

3.1.1 States, formal fields, and coefficient operators

Let V be a complex vector space of chiral states. For the CFT-type examples below, it will be the algebraic direct sum of the finite energy spaces in the vacuum sector. An element of this direct sum has components in only finitely many energy spaces; a Hilbert-space completion is additional structure, introduced when we discuss unitarity. Write $\mathrm{End}_{\mathbb{C}}(V)$ for the space of linear maps $V \to V$. Its elements are operators on states.

For now z is a *formal variable*. A series in z means a specified list of coefficients, with algebraic rules for extracting, differentiating, and comparing them. It does not yet mean a function evaluated at a nonzero complex number. In particular, no convergence or boundedness of operators is assumed in the following definitions. We write

$$V[[z]] = \left\{ \sum_{j\geq 0} v_j z^j : v_j \in V \right\}, \qquad V((z)) = \bigcup_{N\geq 0} z^{-N} V[[z]].$$

Thus $V((z))$ consists of vector-valued Laurent series with only finitely many negative powers, but possibly infinitely many positive powers. Neither space is the original state space V.

A *field on* V is an operator-valued formal series

$$A(z) = \sum_{n\in\mathbb{Z}} A_n z^{-n-1}, \qquad A_n \in \mathrm{End}_{\mathbb{C}}(V),$$

such that, for every $b \in V$, one has $A_n b = 0$ for all sufficiently large n. This is *lower truncation*: after acting on b, the series $A(z)b$ belongs to $V((z))$. The bound on n can depend on b; there

need not be a single bound for the operators A_n on all of V. The indexing $-n-1$ is conventional. Large positive n correspond to large negative powers of z.

Denote the vector space of such fields by $\mathcal{F}(V)$. The *state–field map* is a linear map

$$Y : V \longrightarrow \mathcal{F}(V), \qquad a \longmapsto Y(a,z). \tag{3.1}$$

It is part of the vertex-algebra structure, just as multiplication is part of an associative algebra's structure. The vector space V alone does not determine it. Equivalently, regarding a field as its action on an input state, one can write $Y : V \to \mathrm{Hom}_{\mathbb{C}}(V, V((z)))$. The common notation $Y(a,z) \in \mathrm{End}_{\mathbb{C}}(V)[[z,z^{-1}]]$ uses the vector space of doubly infinite formal distributions. It allows powers in both directions; the separate lower-truncation condition is still required. It does not authorize arbitrary products of two series at the same variable. The normal product below gives the relevant prescription.

For a given state a, denote the coefficient *operator* of z^{-n-1} in its field by $a_{(n)}$, and call it the nth *mode* of a. Thus

$$Y(a,z) = \sum_{n\in\mathbb{Z}} a_{(n)} z^{-n-1}, \qquad a_{(n)} : V \longrightarrow V. \tag{3.2}$$

The symbol $a_{(n)}$ is neither a scalar coordinate of a nor another state. Applying it to a state produces a state: $a_{(n)}b \in V$. Linearity of Y implies that the product $(a,b) \mapsto a_{(n)}b$ is bilinear for every fixed integer n. There is a whole family of these products, indexed by n.

The types can be kept in view as follows:

Object	Where it lives or what it does
a, b	States in V
Y	Linear map from states to fields
$Y(a,z)$	Formal series of operators on V
$a_{(n)}$	One linear operator $V \to V$
$Y(a,z)b$	Vector-valued series in $V((z))$
$a_{(n)}b$	One state in V

For example, writing out a field's action makes the index shift visible:

$$Y(a,z)b = \cdots + (a_{(1)}b)z^{-2} + (a_{(0)}b)z^{-1} + a_{(-1)}b + (a_{(-2)}b)z + \cdots .$$

The negative-power tail on the left is finite for this particular b. The constant coefficient is $a_{(-1)}b$, whereas the residue is $a_{(0)}b$. In general,

$$a_{(n)}b = \mathrm{Res}_z\big(z^n Y(a,z)b\big),$$

where the *formal residue* means the coefficient of z^{-1}. It can be computed without choosing a contour or proving convergence.

3.1.2 The vacuum, translation, and locality axioms

To make these fields into a vertex algebra, specify a distinguished nonzero state $\mathbf{1} \in V$, the *vacuum*, and a linear operator $\mathsf{T} : V \to V$, the *translation operator*. The vacuum and translation identities are

$$Y(\mathbf{1},z) = \mathrm{id}, \qquad Y(a,z)\mathbf{1} = e^{z\mathsf{T}}a, \tag{3.3}$$

$$\mathsf{T}\mathbf{1} = 0, \qquad [\mathsf{T}, Y(a,z)] = Y(\mathsf{T}a,z) = \partial_z Y(a,z). \tag{3.4}$$

Here id is the identity *operator* on V, while $\mathbf{1}$ is a *state*. The exponential means the formal series $e^{z\mathsf{T}}a = \sum_{r\geq 0} z^r \mathsf{T}^r a/r!$. The commutator $[A, B] = AB - BA$ uses composition of linear operators; the derivative on the right is taken coefficient by coefficient. Translation therefore differentiates a field's position variable, as it does in the physical state–operator correspondence.

Worked example 3.1 (Recovering a state from its field)**.** Compare coefficients in $Y(a, z)\mathbf{1} = e^{z\mathsf{T}}a$. There are no negative powers on the right, so

$$a_{(n)}\mathbf{1} = 0 \quad (n \geq 0), \qquad a_{(-r-1)}\mathbf{1} = \frac{\mathsf{T}^r a}{r!} \quad (r \geq 0).$$

In particular $a_{(-1)}\mathbf{1} = a$ and $a_{(-2)}\mathbf{1} = \mathsf{T}a$. Thus the field determines its state: act on the vacuum and take the constant coefficient. If $Y(a, z) = 0$, this procedure gives $a = 0$, so Y is injective. It need not map onto every formal field on V.

For the vacuum field itself, $Y(\mathbf{1}, z) = \mathrm{id}$ has only one nonzero mode:

$$\mathbf{1}_{(-1)} = \mathrm{id}, \qquad \mathbf{1}_{(n)} = 0 \quad (n \neq -1).$$

This is why a vacuum state appearing as an OPE coefficient becomes an identity operator in the corresponding field formula.

Finally, locality says that for any a, b there is an integer $N \geq 0$ such that

$$(z - w)^N [Y(a, z), Y(b, w)] = 0$$

as an operator-valued formal distribution in two variables. The same N works on every input state for this pair of fields. Different expansions in z/w and w/z encode the two radial orderings; multiplication by $(z - w)^N$ removes the singularity that prevents these ordered expansions from agreeing. For example, the two formal expansions of a simple pole are

$$\left.\frac{1}{z - w}\right|_{|z|>|w|} = \sum_{r\geq 0} w^r z^{-r-1}, \qquad \left.\frac{1}{z - w}\right|_{|w|>|z|} = -\sum_{r\geq 0} z^r w^{-r-1}.$$

The domain labels recall their analytic geometric-series meanings; here they specify which expansion to use. The two series differ, but multiplying either by $z - w$ gives one. Locality is therefore compatible with a nonzero commutator of the unmultiplied fields.

These are the vacuum, translation, and locality requirements for the fields assigned by Y; an arbitrary assignment of formal fields to states need not satisfy them.

3.1.3 From state products to operator products

The connection to the OPE notation of Chapter 2 is

$$Y(a, z)Y(b, w) \sim \sum_{n\geq 0} \frac{Y(a_{(n)}b, w)}{(z - w)^{n+1}}. \tag{3.5}$$

The symbol $\sim$ retains the singular terms at $z = w$ in the chosen radial ordering. Each numerator on the right is a *field*; it is obtained by first forming the *state* $a_{(n)}b$ and then applying Y. The sum of singular terms is finite by lower truncation. The regular terms also have meaning: the first one is the field of $a_{(-1)}b$, the normal product discussed in the next section.

The combination of these axioms implies the Jacobi, or Borcherds, identity, which expresses compatibility of composing fields with taking their state products. One useful consequence is the mode commutator formula

$$[a_{(m)}, b_{(n)}] = \sum_{j\geq 0} \binom{m}{j} (a_{(j)}b)_{(m+n-j)}. \tag{3.6}$$

Here $(a_{(j)}b)_{(m+n-j)}$ means a mode of the field assigned to the state $a_{(j)}b$; it does not mean a product of two mode operators. For $j \geq 0$ the binomial coefficient is $\binom{m}{j} = m(m-1)\cdots(m-j+1)/j!$, with $\binom{m}{0} = 1$; this definition also applies to negative m. The sum is finite because the singular products $a_{(j)}b$ vanish for large j. Thus knowledge of the singular OPE determines the commutators of all modes. It does not determine an arbitrary abstract vertex algebra unless the generators and their relations are also controlled.

3.1.4 Conformal grading and the physics mode convention

A *conformal vector* is a distinguished state $\omega \in V$ whose field is the stress tensor. Write that field as

$$Y(\omega, z) = T(z) = \sum_{m\in\mathbb{Z}} L_m z^{-m-2}.$$

Comparing exponents with (3.2) defines $L_m = \omega_{(m+1)}$. To obtain a vertex operator algebra, or VOA, require these operators to satisfy the Virasoro relations, with $L_{-1} = \mathsf{T}$, and require L_0 to give an integer grading bounded below with finite-dimensional graded pieces. We mainly use VOAs of CFT type:

$$V = \bigoplus_{n\geq 0} V_n, \qquad V_0 = \mathbb{C}\mathbf{1}.$$

Here $V_n = \{v \in V : L_0 v = nv\}$, and a state in a single V_n is *homogeneous*, of weight $\mathrm{wt}(v) = n$. In particular ω has weight two. The grading compatibility identity

$$[L_0, Y(a,z)] = Y(L_0 a, z) + z\partial_z Y(a,z)$$

gives, for homogeneous a, $[L_0, a_{(n)}] = (\mathrm{wt}(a) - n - 1)a_{(n)}$. Acting on a homogeneous state b proves the weight rule

$$\mathrm{wt}(a_{(n)}b) = \mathrm{wt}(a) + \mathrm{wt}(b) - n - 1. \tag{3.7}$$

The displayed weight is the weight of a nonzero output; if the corresponding graded space is zero, the product must vanish. This rule provides a quick consistency check on every claimed mode identity. Two weight-two states have a weight-two first product $a_{(1)}b$ and a weight-three zeroth product $a_{(0)}b$.

Physics often writes a field of weight h as $\phi(z) = \sum_m \phi_m z^{-m-h}$. The dictionaries agree when $\phi_m = a_{(m+h-1)}$. In particular, a current of weight one has $J_m = J_{(m)}$, while a stress of weight two has $L_m = \omega_{(m+1)}$. For a weight-h state, the physics zero mode is $a_{(h-1)}$: it preserves conformal weight. The vertex-algebra zeroth product instead uses $a_{(0)}$, which changes weight by $h-1$. In particular, the stress has $\omega_{(0)} = L_{-1}$ and $\omega_{(1)} = L_0$. Keeping the convention explicit prevents a translation operator from being mistaken for an energy operator.

Worked example 3.2 (A current commutator from one OPE coefficient)**.** Let $J \in V_1$ be a current state and write $J_m = J_{(m)}$. Suppose $J_{(1)}J = \mathbf{1}$ and $J_{(j)}J = 0$ for $j \neq 1$, $j \geq 0$. Equation (3.6) gives

$$[J_m, J_n] = m\mathbf{1}_{(m+n-1)} = m\delta_{m+n,0}\,\mathrm{id},$$

because $Y(\mathbf{1}, z) = \mathrm{id}$ has only the mode $\mathbf{1}_{(-1)} = \mathrm{id}$. This is the Heisenberg algebra. For clarity, J here is a weight-one *state*, $J(z) = Y(J,z)$ is its field, and $J_m = J_{(m)}$ is an operator. The vacuum identities give $J_{-1}\mathbf{1} = J$ and $J_1\mathbf{1} = 0$, so the commutator also yields the state calculation

$$J_1 J = J_1 J_{-1}\mathbf{1} = [J_1, J_{-1}]\mathbf{1} + J_{-1}J_1\mathbf{1} = \mathbf{1}.$$

The OPE coefficient $J_{(1)}J = \mathbf{1}$ and the central term $[J_1, J_{-1}] = \mathrm{id}$ express the same structure at the state and operator levels. More generally, $J_m J_{-m}\mathbf{1} = m\mathbf{1}$ for every $m > 0$. Thus $Y(J,z)$ acting on $J_{-m}\mathbf{1}$ has a pole of order $m+1$. This gives a concrete example of lower truncation with no uniform pole bound over all input states.

3.2 Normal products, derivatives, and strong generation

The constant term in the state series $Y(a,z)b$ is $a_{(-1)}b$. Its corresponding field is the *normal product*:

$$Y(a_{(-1)}b,z) =: Y(a,z)Y(b,z) : .$$

To define the colons, split the first field into

$$Y(a,z)_- = \sum_{n<0} a_{(n)}z^{-n-1}, \qquad Y(a,z)_+ = \sum_{n\geq 0} a_{(n)}z^{-n-1},$$

and put

$$: Y(a,z)Y(b,z) := Y(a,z)_-Y(b,z) + Y(b,z)Y(a,z)_+.$$

This places the creation part on the left and the part that annihilates the vacuum on the right. Individual modes need not annihilate every other state. Lower truncation makes each coefficient of the displayed expression well defined on an input state, even though the fields themselves are infinite series. The equality with $Y(a_{(-1)}b,z)$ is a vertex-algebra identity, linking this operator prescription to the state product. In particular, $\mathsf{T}a = a_{(-2)}\mathbf{1}$. Differentiating the mode expansion gives

$$(\mathsf{T}a)_{(n)} = -n\, a_{(n-1)}. \tag{3.8}$$

Repeated negative modes can therefore be expressed using derivatives and normal products. They are not generally ordinary associative commutative products: changing parentheses or reversing factors adds terms governed by the singular OPE.

Skew symmetry is the identity

$$Y(a,z)b = e^{z\mathsf{T}}Y(b,-z)a. \tag{3.9}$$

For example, extracting the coefficient of z^{-n-1} gives

$$a_{(n)}b = \sum_{j\geq 0} \frac{(-1)^{n+j+1}}{j!}\mathsf{T}^j(b_{(n+j)}a).$$

The extra derivatives are precisely why a naive commutative algebra of OPE coefficients loses information.

Definition 3.3. A set $S \subset V$ *generates* V as a vertex algebra if all iterated modes of its elements, applied to the vacuum, span V. It *strongly generates* V if derivatives and iterated normal products of its elements span V.

Strong generation is the more restrictive statement. A set can generate a new low-weight state through singular products of higher-weight composites. Knowing a few low graded pieces does not exclude this possibility. For the Haagerup construction, proving that six chosen currents remain all of U_1 requires an all-weight argument. The finite-closure chapter explains how a controlled set of identities supplies that argument.

Worked example 3.4 (Why a weight-one conclusion follows from strong generation)**.** Suppose a CFT-type VOA is strongly generated by six weight-one currents and fields of weights at least two. Every derivative raises weight, and every normal product adds weights. Hence a weight-one state in their normal span can only be a linear combination of the six currents. Once strong generation is proved, the claim $\dim V_1 \leq 6$ is immediate. Without strong generation, the same count is not justified.

3.3 The Heisenberg Fock space

Let $\mathfrak{h}$ be an r-dimensional complex vector space with a nondegenerate symmetric form $(\ ,\ )$, obtained by complexifying a real Euclidean space. For each $a \in \mathfrak{h}$ and $n \in \mathbb{Z}$, introduce a mode operator $a(n)$, linear in a. These oscillators obey

$$[a(m), b(n)] = m(a,b)\delta_{m+n,0}\,\mathrm{id}.$$

The vacuum representation is generated from a vector $\mathbf{1}$ killed by $a(n)$ for $n \geq 0$. Negative modes commute with one another and act freely as creation operators. Thus its state space is the symmetric-algebra Fock space

$$M(1) = S(\mathfrak{h} \otimes t^{-1}\mathbb{C}[t^{-1}]).$$

Here $S(W)$ means the polynomial, or symmetric, algebra on a vector space W. The letter t labels oscillator modes and is distinct from the field variable z. The symbol $a \otimes t^{-n}$, $n > 0$, labels a creation mode $a(-n)$; a monomial represents a state $a_1(-n_1)\cdots a_s(-n_s)\mathbf{1}$. Positive modes act by commuting through these creation modes and then killing the vacuum. For example, $a(1)b(-1)\mathbf{1} = (a,b)\mathbf{1}$.

For $a \in \mathfrak{h}$, the associated weight-one state is $a(-1)\mathbf{1}$, and its field is

$$Y(a(-1)\mathbf{1}, z) = \sum_{n\in\mathbb{Z}} a(n) z^{-n-1}.$$

Thus $a(n)$ abbreviates $(a(-1)\mathbf{1})_{(n)}$; the vector $a \in \mathfrak{h}$, the state $a(-1)\mathbf{1} \in M(1)$, and its mode operators have now been related explicitly. An oscillator $a(-n)$ contributes weight n. For an orthonormal basis $u_1, \ldots, u_r$, the conformal vector is

$$\omega = \frac{1}{2}\sum_{j=1}^{r} u_j(-1)^2\mathbf{1},$$

and the central charge is r.

The character is the graded trace $\mathrm{ch}_{M(1)}(q) = \mathrm{Tr}_{M(1)}\, q^{L_0 - r/24}$. Its product formula follows from one geometric series for every oscillator species and frequency:

$$\mathrm{ch}_{M(1)}(q) = q^{-r/24}\prod_{n\geq 1}(1-q^n)^{-r} = \eta(\tau)^{-r}. \tag{3.10}$$

Here $q = e^{2\pi i\tau}$, with $\mathrm{Im}\,\tau > 0$ for an analytic character, and $\eta(\tau) = q^{1/24}\prod_{n\geq 1}(1-q^n)$ is the Dedekind eta function. One can also read the product coefficient by coefficient as a formal graded count. This q records energy; it is not the insertion coordinate z of a field. The Heisenberg algebra alone has a Fock module for every momentum in $\mathfrak{h}$; its natural representation theory is not finite. A positive even lattice supplies a mutually local discrete set of exponential fields and changes this situation substantially.

3.4 Even lattices and cocycles

A lattice L is a free abelian group of finite rank with a positive-definite integral form: $(\alpha, \beta) \in \mathbb{Z}$ for lattice vectors α, β. Thus, after choosing a lattice basis, each vector is an integer linear combination of basis vectors. Write $\alpha^2 = (\alpha, \alpha)$ and set $\mathfrak{h} = L \otimes_{\mathbb{Z}} \mathbb{C}$, so the Heisenberg space is the complexified lattice space. The lattice is *even* when $(\alpha, \alpha) \in 2\mathbb{Z}$ for every $\alpha \in L$. We will see how integrality and evenness enter the field construction after defining its states.

3.4.1 Momentum labels, states, and their fields

The Heisenberg vacuum space $M(1)$ describes oscillator excitations above a state of zero momentum. To include the momenta in L, we introduce one new *basis symbol* e^α for each $\alpha \in L$ and form the vector space

$$\mathbb{C}_\epsilon[L] = \bigoplus_{\alpha\in L} \mathbb{C}e^\alpha.$$

The subscript ϵ anticipates a multiplication to be defined below; it does not affect this vector-space definition. An element is a finite sum $\sum_\alpha c_\alpha e^\alpha$ with complex coefficients. The vector α belongs to the lattice, whereas e^α belongs to this new space of momentum labels. Here the notation names a basis vector; no power-series exponential of α is being taken.

Attach a copy of the oscillator space to each momentum label. The resulting vector space of the lattice VOA is

$$\begin{aligned} V_L &= M(1) \otimes \mathbb{C}_\epsilon[L] \\ &= \bigoplus_{\alpha\in L} (M(1) \otimes \mathbb{C}e^\alpha). \end{aligned} \tag{3.11}$$

The ground state in the momentum-α summand is $|\alpha\rangle = \mathbf{1} \otimes e^\alpha$, with no oscillator excitations. We conventionally abbreviate this state to e^α as well. The vacuum of V_L is $\mathbf{1} \otimes e^0$, also denoted $\mathbf{1}$. Thus the symbol e^0 labels the zero-momentum basis vector; its tensor product with the oscillator vacuum gives the full vacuum. An arbitrary lattice state is a finite linear combination of

$$a_1(-n_1)\cdots a_s(-n_s)e^\alpha, \qquad n_i > 0, \quad \alpha \in L.$$

The different summands have the same oscillator basis but different Heisenberg zero-mode actions. On the ground states,

$$a(n)e^\alpha = 0 \quad (n > 0), \qquad a(0)e^\alpha = (a, \alpha)e^\alpha, \qquad a \in \mathfrak{h}. \tag{3.12}$$

All zero modes commute with the creation modes, so every state in this summand has the same momentum α. Positive modes act on oscillator excitations through the Heisenberg commutators, just as in $M(1)$. These are the several-boson versions of the charged Fock modules F_p from Chapter 2.

The free-boson stress tensor determines the conformal grading. For an orthonormal basis $u_1, \ldots, u_r$ of $\mathfrak{h}$, its zero mode is

$$L_0 = \frac{1}{2}\sum_{j=1}^{r} u_j(0)^2 + \sum_{n>0}\sum_{j=1}^{r} u_j(-n)u_j(n).$$

Only finitely many terms act nontrivially on any fixed oscillator monomial. On e^α the second sum vanishes, and the first gives $\frac{1}{2}\sum_j (u_j, \alpha)^2 = \alpha^2/2$. Since a creation mode $a(-n)$ adds weight n, a nonzero oscillator monomial has

$$\mathrm{wt}(a_1(-n_1)\cdots a_s(-n_s)e^\alpha) = \frac{\alpha^2}{2} + n_1 + \cdots + n_s.$$

Now the role of evenness is visible: every momentum ground state has integer weight, so all conformal weights occurring in V_L are integers.

The exponential notation is motivated by the associated field $Y(e^\alpha, z)$, the lattice counterpart of the normal-ordered free-boson exponential from Chapter 2, customarily written $:\exp(i\alpha\cdot X(z)):$ in physics notation. Here $X = (X_1, \ldots, X_r)$ is a collection of independent chiral bosons with $X_j(z)X_k(w) \sim -\delta_{jk}\log(z-w)$, and $\alpha \cdot X = \sum_j (u_j, \alpha)X_j$. In the lattice theory this notation includes the momentum-shift and cocycle factors described below. The state e^α is the input to the state–field map; $Y(e^\alpha, z)$ is the resulting operator-valued field, which takes momentum β to momentum $\alpha + \beta$.

Worked example 3.5 (Momentum and oscillator excitations in rank one)**.** Let $L = \mathbb{Z}\alpha$ with $\alpha^2 = 2$. Its momentum labels are $e^{m\alpha}$ for $m \in \mathbb{Z}$, with ground-state weights m^2. The first examples are

state	momentum	weight
$\mathbf{1}$	0	0
$e^{\alpha},\ e^{-\alpha}$	$\alpha,\ -\alpha$	1
$\alpha(-1)\mathbf{1}$	0	1
$\alpha(-1)e^{\alpha}$	α	$2.$

Thus there are three independent weight-one states: two charged ground states and one neutral oscillator state. A superposition such as $e^{\alpha} + e^{-\alpha}$ is a sum of two different momentum states, whereas $e^{\alpha+(-\alpha)} = e^0$ labels the zero-momentum ground state. Keeping this distinction in mind prevents ordinary exponential manipulations from being applied to vector addition.

3.4.2 Multiplying momentum labels and choosing a cocycle

Adding charges suggests the ordinary group-algebra multiplication $e^{\alpha}e^{\beta} = e^{\alpha+\beta}$. The lattice construction allows a scalar factor in that rule:

$$e^{\alpha}e^{\beta} = \epsilon(\alpha,\beta)e^{\alpha+\beta}, \qquad \epsilon : L \times L \longrightarrow \mathbb{C}^{\times}.$$

Here $\mathbb{C}^{\times}$ denotes the nonzero complex numbers. We require $\epsilon(0,\alpha) = \epsilon(\alpha,0) = 1$, so e^0 is the identity for this multiplication. The resulting algebra is called a *twisted group algebra*. The concrete choice below takes values in $\{1,-1\}$.

This multiplication records how a charged field shifts the momentum label. For example, multiplication by e^{α} sends the label e^{β} to $\epsilon(\alpha,\beta)e^{\alpha+\beta}$. It supplies one ingredient of $Y(e^{\alpha},z)$; oscillator operators and powers of z supply the others. Consequently this auxiliary group-algebra product is different from the VOA product $e^{\alpha}_{(-1)}e^{\beta}$, which is a particular coefficient of the full field $Y(e^{\alpha},z)e^{\beta}$. Associativity of this multiplication requires the two-cocycle identity

$$\epsilon(\alpha,\beta)\epsilon(\alpha+\beta,\gamma) = \epsilon(\beta,\gamma)\epsilon(\alpha,\beta+\gamma).$$

The two sides are the coefficients of $e^{\alpha+\beta+\gamma}$ obtained by multiplying three basis labels in the two possible orders of association. A normalized function satisfying this identity is called a *two-cocycle*. Locality of the exponentials also requires

$$\frac{\epsilon(\alpha,\beta)}{\epsilon(\beta,\alpha)} = (-1)^{(\alpha,\beta)}. \tag{3.13}$$

This sign compensates for interchanging the factor $(z-w)^{(\alpha,\beta)}$ in the Gaussian OPE. The cocycle is therefore part of a consistent choice of field phases, not an optional decoration.

In the simply laced root bases used later, with Gram matrix G and diagonal entries two, write $a = (a_i)$ and $b = (b_i)$ for the integer coordinate columns of two lattice vectors. The entry G_{ij} is the pairing of the ith and jth basis vectors. One convenient cocycle is

$$\epsilon(a,b) = (-1)^{\sum_i a_i b_i + \sum_{i>j} a_i b_j G_{ij}}. \tag{3.14}$$

Its exponent is bilinear modulo two, so the cocycle identity follows by expanding both sides. The difference between the exponents for (a,b) and (b,a) is $a^{\mathsf{t}}Gb$ modulo two, proving (3.13). Moreover, $\epsilon(a,-a) = (-1)^{a^2/2}$ in this convention.

Rescaling the basis $e^{\alpha} \mapsto \lambda(\alpha)e^{\alpha}$, with $\lambda : L \to \mathbb{C}^{\times}$ and $\lambda(0) = 1$, changes ϵ by the coboundary $\lambda(\alpha)\lambda(\beta)/\lambda(\alpha+\beta)$. The condition at zero keeps the vacuum fixed. A rescaling of unit modulus is a rephasing. The resulting VOA is isomorphic. A calculation that compares two tables of coefficients must first align these gauges.

3.5 The exponential formula and a root calculation

The full vertex operator contains creation oscillators, annihilation oscillators, and the charge-shift operator introduced above. On a pure momentum state e^β, the positive oscillators vanish and the zero modes measure β, as in (3.12). Acting on a state without oscillator excitations, the exponential field reduces to the particularly useful formula

$$Y(e^\alpha, z)e^\beta = \epsilon(\alpha,\beta)z^{(\alpha,\beta)} \exp\left(\sum_{n>0} \frac{\alpha(-n)}{n} z^n\right) e^{\alpha+\beta}. \tag{3.15}$$

The prefactor $z^{(\alpha,\beta)}$ sets the first possible mode, and the exponential then generates the oscillator corrections. For an oscillator-excited input, the needed additional identity is

$$[a(m), Y(e^\alpha, z)] = (a,\alpha)z^m Y(e^\alpha, z).$$

For example, moving one creation mode past the field gives, for $r > 0$,

$$Y(e^\alpha, z)b(-r)e^\beta = \big(b(-r) - (b,\alpha)z^{-r}\big)Y(e^\alpha, z)e^\beta.$$

Repeated use of this identity and the Heisenberg relations reduces an arbitrary Fock monomial to the displayed pure-momentum formula. At a prescribed output weight, only finitely many terms of the exponential can contribute. This supplies an exact finite procedure for each mode product.

Worked example 3.6 (The affine $\mathfrak{sl}_2$ inside V_{A_1})**.** Let $L = \mathbb{Z}\alpha$ with $\alpha^2 = 2$. In the cocycle convention above, $\epsilon(\alpha,-\alpha) = -1$, so

$$Y(e^\alpha, z)e^{-\alpha} = -z^{-2}\left[\mathbf{1} + \alpha(-1)z + \frac{\alpha(-1)^2 + \alpha(-2)}{2} z^2 + \cdots\right].$$

Consequently

$$e^\alpha_{(1)}e^{-\alpha} = -\mathbf{1}, \qquad e^\alpha_{(0)}e^{-\alpha} = -\alpha(-1)\mathbf{1}.$$

Define the states $H = \alpha(-1)\mathbf{1}$, $E = e^\alpha$, and $F = -e^{-\alpha}$, and abbreviate their fields by $H(z) = Y(H,z)$, $E(z) = Y(E,z)$, and $F(z) = Y(F,z)$. Then

$$\begin{aligned} H(z)H(w) &\sim \frac{2}{(z-w)^2}, \\ H(z)E(w) &\sim \frac{2E(w)}{z-w}, \qquad & H(z)F(w) &\sim -\frac{2F(w)}{z-w}, \\ E(z)F(w) &\sim \frac{1}{(z-w)^2} + \frac{H(w)}{z-w}. \end{aligned}$$

These are the affine $\mathfrak{sl}_2$ relations at level one. The minus sign in F is forced by the chosen cocycle and ensures $[E_0, F_0] = H_0$.

At weight one, a positive even lattice contributes its rank from the oscillator states and one state for every lattice vector of squared norm two. Therefore

$$\dim(V_L)_1 = \operatorname{rank} L + \#\{\alpha \in L : \alpha^2 = 2\}. \tag{3.16}$$

For A_1 this is $1 + 2 = 3$. For E_8 it is $8 + 240 = 248$. This elementary count will make the properness of the six-current subalgebra a physically visible statement.

3.6 Modules, fusion, and intertwining fields

A representation of a Lie algebra tells us how its generators act on another vector space. A representation of a vertex algebra does the same for all field modes at once. A *V-module M* carries a map

$$Y_M : V \longrightarrow \mathrm{Hom}_{\mathbb{C}}(M, M((z))), \qquad v \longmapsto Y_M(v,z),$$

with mode expansion

$$Y_M(v,z) = \sum_{n\in\mathbb{Z}} v^M_{(n)} z^{-n-1}, \qquad v \in V,$$

whose coefficients $v^M_{(n)} : M \to M$ act on the module. These fields obey the vacuum identity $Y_M(\mathbf{1},z) = \mathrm{id}_M$, lower truncation, translation, and the same Jacobi or OPE compatibility identities as the fields of V. Here translation includes $Y_M(\mathsf{T}v,z) = \partial_z Y_M(v,z)$, where T acts on the input $v \in V$. In particular, the conformal vector acts by Virasoro operators on M. The module need not contain a vacuum state. Nor does its definition assign an endomorphism-valued field to every vector of M: the inputs of Y_M still belong to the vacuum algebra V.

A module is *simple* if it has no nonzero proper subspace invariant under all field modes. In the simple ordinary positive-energy modules used in the examples, L_0^M is diagonalizable and

$$M = \bigoplus_{N\ge 0} M_{h_M+N}, \qquad M_\lambda = \{u \in M : L_0^M u = \lambda u\},$$

with finite-dimensional eigenspaces and a lowest nonzero one M_{h_M}. This lowest space is the *top*; a state in M_{h_M+N} has *grade* N above it. Thus a module's top can contain several states even when the module is simple. The underlying module axioms alone do not impose this energy decomposition; the final section distinguishes the various grading assumptions.

A *module homomorphism* $f : M \to N$ is a linear map commuting with these actions:

$$f Y_M(v,z) = Y_N(v,z) f \qquad (v \in V).$$

This is the meaning of $\mathrm{Hom}_V(M,N)$. For simple ordinary modules with finite-dimensional energy eigenspaces, Schur's lemma makes it zero for inequivalent modules and one-dimensional for equivalent modules. For the scalar-endomorphism conclusion, restrict an endomorphism to a nonzero finite energy space and choose an eigenvalue; the kernel after subtracting that scalar is a nonzero submodule, so simplicity makes it the whole module.

A *chiral intertwining operator* is a different kind of object. Here we discuss ordinary modules, with diagonalizable conformal energy. An intertwiner of type $\binom{P}{M\ N}$ assigns to $u \in M$ a field

$$\mathcal{Y}(u,z) : N \longrightarrow P\{z\}.$$

For fixed u, its coefficient operators map N to P, and its action on a vector of N is a vector-valued formal series in $P\{z\}$. This notation permits the complex powers of z prescribed by the module weights; it does not denote evaluation of an analytic function. Generalized modules may require logarithmic intertwiners as well, which are outside these examples. The assignment is linear in u and satisfies lower truncation, the translation identity $\mathcal{Y}(L_{-1}u,z) = \partial_z \mathcal{Y}(u,z)$, and compatibility with the V-actions on all three modules. In physical language, moving a chiral-algebra field past this insertion must agree with first using its OPE with the inserted state. Thus a charged field can change the sector on which it acts, even though every ordinary V field preserves each V-module. Products of such charged fields supply the nonvacuum conformal blocks used later in the book.

The energy commutator is a useful part of this definition:

$$L_0^P \mathcal{Y}(u,z) - \mathcal{Y}(u,z) L_0^N = \mathcal{Y}(L_0^M u, z) + z\partial_z \mathcal{Y}(u,z).$$

If u, v have weights h_u, h_v, a term $z^r w$ in $\mathcal{Y}(u,z)v$ must therefore satisfy

$$r = h_w - h_u - h_v. \tag{3.17}$$

For top states in three modules, the possible top-to-top power is $h_P - h_M - h_N$. Its coefficient can vanish for particular choices of top components. The first nonzero term then comes from a higher grade of the same intermediate module; it need not signal a different fusion channel.

The *contragredient module* M' is the graded dual $\bigoplus_\lambda M_\lambda^*$ equipped with the dual vertex-algebra action. It uses the dual of each finite energy space, not the full algebraic dual of an infinite direct sum. The VOA is *self-contragredient* when $V' \cong V$ as V-modules. This is one of the hypotheses in the next theorem.

To identify these spaces with categorical fusion, we use the standard rational VOA tensor theorem under its usual hypotheses: simplicity, CFT type, self-contragredience, rationality, C_2-cofiniteness, and positive lowest weights for nonvacuum simple modules. The finiteness terms are explained at the end of this chapter. In this setting the ordinary modules form a semisimple rigid tensor category. Denote its fusion product by $\boxtimes_V$ to distinguish it from the ordinary tensor product of vector spaces. For simple modules,

$$M \boxtimes_V N \cong \bigoplus_P N_{MN}^P \, P, \qquad N_{MN}^P = \dim \mathcal{I}\binom{P}{M\ N} = \dim \mathrm{Hom}_V(M \boxtimes_V N, P), \tag{3.18}$$

where $\mathcal{I}\binom{P}{M\ N}$ is the space of chiral intertwiners of the indicated type. These are fusion maps $M \boxtimes_V N \to P$. Chapter 1 uses the oppositely oriented splitting space $\mathrm{Hom}(P, M \boxtimes_V N)$; composition pairs the two spaces nondegenerately. In the unitary setting, adjoints convert orthonormal fusion bases to splitting bases. The infinite tower of descendant states belongs inside each module P; the finite integer N_{MN}^P counts independent ways of coupling to that module. It does not count the dimension of $M \otimes_{\mathbb{C}} N$.

The contragredient M' supplies the categorical dual in this setting. A bar over a factor in a full-CFT Hilbert space instead indicates an antiholomorphic copy. The chosen right-moving pairing relates these notions; they are not the same notation for taking complex conjugates of coordinates. All twelve Haagerup center labels are self-dual, which will simplify the displayed diagonal pairings without eliminating the underlying duality convention.

3.7 Lattice modules and a rank-one example

The dual lattice is $L^* = \{x \in L \otimes \mathbb{R} : (x, L) \subset \mathbb{Z}\}$. For a positive even lattice, the simple modules are $V_{\lambda+L}$, indexed by $\lambda \in L^*/L$. Their conformal weight is the minimum of $x^2/2$ in the coset, and their twist is $e^{\pi i \lambda^2}$. Fusion is addition of cosets:

$$V_{\lambda+L} \boxtimes V_{\mu+L} \cong V_{\lambda+\mu+L}.$$

Every categorical dimension is one. The representation category is therefore pointed, even when some module tops have dimension greater than one because a coset has several shortest vectors. Concretely, $V_{\lambda+L}$ has the same oscillator monomials as V_L, but its momentum basis is labelled by $\lambda + L$. A vacuum-algebra field of momentum $\alpha \in L$ sends momentum $\beta \in \lambda + L$ to $\alpha + \beta$ in the same coset. This is how a charged lattice module can be preserved by every V_L field while containing no vacuum state of its own.

Worked example 3.7 (A discriminant group one can compute by hand)**.** Let $L = \mathbb{Z}\alpha$, $\alpha^2 = 2k$ with $k \in \mathbb{Z}_{>0}$. Then $L^* = \frac{1}{2k}\mathbb{Z}\alpha$, so $L^*/L \cong \mathbb{Z}/(2k)$. The module labelled by m has weight modulo one

$$q(m) = \frac{m^2}{4k} \pmod 1.$$

Its actual lowest weight uses the representative of m closest to zero modulo $2k$. The finite quadratic function q records twists, while the shortest-vector problem determines the integer part of the energy. The distinction will reappear for the Haagerup module whose twist exponent is $2/13$ but whose lowest weight is $15/13$.

The character of a lattice module is

$$\chi_{\lambda+L}(\tau) = \frac{\Theta_{\lambda+L}(\tau)}{\eta(\tau)^r}, \qquad \Theta_{\lambda+L} = \sum_{x\in\lambda+L} q^{x^2/2}. \tag{3.19}$$

The numerator counts momenta and the denominator counts oscillators. Poisson summation produces the modular S matrix

$$S_{\lambda\mu} = |L^*/L|^{-1/2} e^{-2\pi i(\lambda,\mu)}.$$

Thus the finite discriminant form is a direct bridge between a Euclidean lattice and a modular tensor category.

Worked example 3.8 (Poisson summation and the two sectors of the root lattice)**.** For $L = \mathbb{Z}\alpha$, $\alpha^2 = 2k$, write the theta series as

$$\Theta_m(\tau) = \sum_{\ell\in\mathbb{Z}} \exp\left(\frac{\pi i\tau}{2k}(2k\ell + m)^2\right), \qquad m \in \mathbb{Z}/(2k).$$

Poisson summation replaces a sum of Gaussian samples on a lattice by the Gaussian samples on its dual lattice. Applying the Gaussian Fourier transform to the arithmetic progression $m + 2k\mathbb{Z}$ gives

$$\Theta_m(-1/\tau) = \sqrt{\frac{-i\tau}{2k}} \sum_{r\in\mathbb{Z}} e^{\pi i r m/k} e^{\pi i \tau r^2/(2k)}.$$

Here $\operatorname{Im}\tau > 0$, and the analytic branch of $\sqrt{-i\tau}$ is chosen positive at $\tau = it$ for $t > 0$. Positive definiteness of the lattice makes these theta sums convergent in this domain. Split the integer r into its residue modulo $2k$ and an integer multiple of $2k$. Reindex the residue by its negative; $\Theta_{-n} = \Theta_n$ because reversing every momentum does not change its norm. The result is

$$\Theta_m(-1/\tau) = \sqrt{\frac{-i\tau}{2k}} \sum_{n=0}^{2k-1} e^{-\pi i m n/k} \Theta_n(\tau). \tag{3.20}$$

One can see both factors directly. The Fourier transform of the Gaussian supplies $\sqrt{-i\tau}$; the spacing $2k$ leaves the factor $(2k)^{-1/2}$. Splitting the dual sum into residue classes modulo $2k$ leaves the finite Fourier phase. The sign of that phase is the convention used here for S; reindexing $n \mapsto -n$ does not change these unflavoured theta series. Dividing by $\eta(-1/\tau) = \sqrt{-i\tau}\eta(\tau)$ cancels the continuous Gaussian factor. The modular transform of the character is therefore an ordinary finite Fourier transform.

At $k = 1$, there are two sectors, 0 and 1, and

$$S = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad T = e^{-\pi i/12}\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}.$$

The common phase in T comes from $c = 1$; the relative phase i comes from the nontrivial lowest weight $h = 1/4$. Their characters begin

$$\chi_0 = q^{-1/24}(1 + 3q + 4q^2 + \cdots),$$
$$\chi_1 = q^{5/24}(2 + 2q + 6q^2 + \cdots).$$

The two lowest states in the second sector are $e^{\alpha/2}$ and $e^{-\alpha/2}$. Nevertheless, $S_{10}/S_{00} = 1$, its quantum dimension. Its square fuses to the vacuum sector because adding its lattice coset to itself gives L.

As a further check, the normalized twist sum is $(1+i)/\sqrt{2} = e^{2\pi i/8}$, the phase appropriate to central charge one. Chapter 6 explains this relation in general. A nontrivial twist therefore need not mean a nontrivial quantum dimension, just as a nontrivial top multiplicity need not.

Worked example 3.9 (One fusion channel, two different leading powers)**.** Let $L = \mathbb{Z}\alpha$ with $\alpha^2 = 2$, and put $V = V_L$. The nontrivial simple module is $M_1 = V_{L+\alpha/2}$. Its top has basis

$$v_+ = e^{\alpha/2}, \qquad v_- = e^{-\alpha/2}, \qquad h_{M_1} = \frac{1}{4}.$$

Addition of lattice cosets gives $M_1 \boxtimes_V M_1 = V$, so there is exactly one vacuum-module intertwiner up to scale. This does not mean that the OPE of every pair of top components begins with the vacuum *state*.

Choose an intertwiner $\mathcal{Y}$ of type $\binom{V}{M_1\ M_1}$ and a branch for its fractional powers. The lattice exponential formula gives

$$\mathcal{Y}(v_+,z)v_- = \kappa_{+-} z^{-1/2}\left(\mathbf{1} + \frac{1}{2}\alpha(-1)\mathbf{1}\, z + O(z^2)\right), \tag{3.21}$$

$$\mathcal{Y}(v_+,z)v_+ = \kappa_{++} z^{1/2}\left(e^{\alpha} + \frac{1}{2}\alpha(-1)e^{\alpha}\, z + O(z^2)\right). \tag{3.22}$$

The nonzero phase coefficients κ_{+-}, κ_{++} depend on the chosen intertwiner, basis, and cocycle extension convention; they are not fixed merely by the cocycle on the original lattice L. With the usual unit exponential normalizations the common scale can be chosen so these coefficients have unit modulus. Their values are irrelevant to the leading-power distinction.

In the first product the charges add to zero and $(\alpha/2, -\alpha/2) = -1/2$. Its vacuum coefficient is nonzero. In the second product the charges add to α and $(\alpha/2, \alpha/2) = 1/2$. Charge conservation forbids a vacuum-state coefficient. The first output is instead the weight-one current e^{α} in the same vacuum module V. Equation (3.17) verifies both exponents:

$$0 - \frac{1}{4} - \frac{1}{4} = -\frac{1}{2}, \qquad 1 - \frac{1}{4} - \frac{1}{4} = \frac{1}{2}.$$

The gap of one is an intermediate-module grade. It does not change the fusion rule or create a second independent intertwiner.

For comparison, the inequivalent modules have no nonzero module homomorphism between them, whereas their charged field exists:

$$\mathrm{Hom}_V(M_1, V) = 0, \qquad \dim \mathcal{I}\binom{V}{M_1\ M_1} = 1.$$

A mode-preserving linear map between modules and a charged field joining three modules answer different representation-theoretic questions.

3.8 Affine currents, Sugawara stresses, and commutants

For a simple Lie algebra $\mathfrak{g}$ with invariant form normalized so long roots have squared length two, choose a basis t^a. Its associated current states J^a have weight one; write $J^a(z) = Y(J^a, z)$. The *level* k is the coefficient multiplying the invariant form in their double pole:

$$J^a(z)J^b(w) \sim \frac{k\kappa^{ab}}{(z-w)^2} + \frac{f^{ab}{}_c J^c(w)}{z-w}.$$

The structure constants and metric are defined by $[t^a, t^b] = f^{ab}{}_c t^c$ and $\kappa^{ab} = (t^a, t^b)$. The inverse metric κ_{ab} defines the dual basis $t_a = \kappa_{ab} t^b$ and the currents $J_a = \kappa_{ab} J^b$; repeated indices are summed. At positive integer level k, write $L_k(\mathfrak{g})$ for the simple integrable affine VOA. It is the simple quotient of the vacuum representation generated by these currents. Integrability requires the raising and lowering operators for affine simple roots to act locally nilpotently: repeated application to any fixed state eventually gives zero. For $\mathfrak{sl}_2$, one resulting relation is $(E_{(-1)})^{k+1}\mathbf{1} = 0$. At level one this is precisely the relation $E_{(-1)}E = 0$ visible in the root-lattice construction. Such relations are not imposed in the universal affine vacuum algebra. The Sugawara stress of the simple quotient is

$$T_{\mathfrak{g},k} = \frac{1}{2(k+h^\vee)} \sum_a :J^a J_a:, \qquad c_{\mathfrak{g},k} = \frac{k \dim \mathfrak{g}}{k+h^\vee}.$$

The dual Coxeter number $h^\vee$ is a constant of the simple Lie algebra: in this normalization the quadratic Casimir $\sum_a t^a t_a$ acts on the adjoint representation by $2h^\vee$. It supplies the correction from the simple poles when the current bilinears are contracted. For $\mathfrak{sl}_2$, $h^\vee = 2$ and $c = 3k/(k+2)$.

If an affine subalgebra A is embedded in V, its Sugawara stress need not equal the ambient stress. Suppose the currents are ambient weight-one primaries, their level is positive integral, and the embedding respects the ambient unitary grading. Under these coset-embedding hypotheses, the difference $\omega_C = \omega_V - \omega_A$ is a commuting stress of charge $c_V - c_A$. The *commutant*

$$\mathrm{Com}(A, V) = \{v \in V : a_{(n)}v = 0 \text{ for all } a \in A, n \geq 0\} \tag{3.23}$$

is the algebra of fields whose OPE with A has no singular part. Equivalently one can test a suitable generating set, with the appropriate closure justification.

This is a chiral construction. A commuting pair of stresses does not by itself imply a tensor-product decomposition of V. The ambient states can form a nontrivial extension of the tensor product, and can branch with multiplicities. Nor does a commutant automatically supply a modular-invariant full theory. Those are separate questions about representations and sewing.

An important example is the Cartan current in $L_k(\mathfrak{sl}_2)$, which generates a rank-one Heisenberg subalgebra $M(1)$. Its commutant is called the *parafermion VOA*:

$$K(\mathfrak{sl}_2, k) = \mathrm{Com}(M(1), L_k(\mathfrak{sl}_2)).$$

The Heisenberg stress has charge one, so this example supplies the subtraction requested in Exercise 3.5. Its states must satisfy the full commutant condition, not merely have zero Cartan charge.

Worked example 3.10 (The diagonal affine coset)**.** Embed $\mathfrak{sl}_2$ diagonally in level k and level one. The diagonal currents have level $k+1$, since their double poles add. The coset central charge is

$$\frac{3k}{k+2} + 1 - \frac{3(k+1)}{k+3} = 1 - \frac{6}{(k+2)(k+3)}.$$

At $k = 1$ this is $1/2$, the Ising central charge. The equality of central charges alone would not identify the coset, but the full Goddard–Kent–Olive (GKO) construction identifies this family with the unitary Virasoro minimal models. At $k = 39$ the charge is $286/287$, a value that will appear in the Haagerup stress analysis.

Worked example 3.11 (The three Ising weights and a branching grade)**.** Let $L_k(\lambda)$ denote the integrable affine $\mathfrak{sl}_2$ module of level k and highest weight $\lambda \in \{0, \ldots, k\}$. Its top has spin $j = \lambda/2$. In the invariant-form normalization above, the quadratic Casimir acts by $2j(j+1)$. Dividing by $2(k+2)$ in the Sugawara formula gives

$$h_k(\lambda) = \frac{j(j+1)}{k+2} = \frac{\lambda(\lambda+2)}{4(k+2)}.$$

Under the diagonal level-two algebra, write

$$L_1(\lambda) \otimes L_1(\mu) = \bigoplus_{\nu=0}^{2} L_2(\nu) \otimes B^{\nu}_{\lambda,\mu}.$$

The spaces B are coset modules. If their first diagonal highest vector occurs at numerator grade N, its coset weight is

$$h_C = h_1(\lambda) + h_1(\mu) + N - h_2(\nu). \tag{3.24}$$

Here N is an integer excitation above the numerator's own lowest energy, and cannot be dropped from the formula.

Only labels with $\lambda + \mu - \nu$ even are admissible: the diagonal $SU(2)$ center acts on the numerator by $(-1)^{\lambda+\mu}$ and on the denominator by $(-1)^{\nu}$. Affine current descendants preserve this parity. The diagonal GKO branching theorem supplies sufficiency and completeness in this example, and identifies

$$(\lambda, \mu; \nu) \sim (1 - \lambda, 1 - \mu; 2 - \nu).$$

This is an identification of coset modules, not of numerator modules. There are six admissible triples and three pairs, with no fixed triple. Representatives and their weights are

$(\lambda, \mu; \nu)$	N	h_C	Ising sector
$(0, 0; 0)$	0	0	$\mathbf{1}$
$(1, 0; 1)$	0	$\frac{1}{4} - \frac{3}{16} = \frac{1}{16}$	σ
$(1, 1; 0)$	0	$\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$	ϵ

The last two grade-zero branches are visible in the finite $\mathfrak{sl}_2$ products: doublet times singlet is a doublet, and doublet times doublet contains a singlet.

The equivalent label $(0, 0; 2)$ reveals the grade correction. Let J^a_1, J^a_2 be the two level-one currents and Ω the product vacuum. The three nonzero states

$$w^a = (J^a_{1,(-1)} - J^a_{2,(-1)})\Omega$$

occur at numerator grade one. For $J^b_D = J^b_1 + J^b_2$, the two equal central terms cancel, giving

$$J^b_{D,(n)} w^a = 0 \quad (n > 0), \qquad J^b_{D,(0)} w^a = f^{ba}{}_c w^c.$$

Thus the w^a form a diagonal affine spin-one top; its raising component is a highest vector with $\nu = 2$. There is no such top at grade zero, whose space is only the vacuum. Consequently

$$h_C = 0 + 0 + 1 - h_2(2) = 1 - \frac{1}{2} = \frac{1}{2},$$

as required by $(0, 0; 2) \sim (1, 1; 0)$. The states w^a carry diagonal charge; the coset sector is their multiplicity space. They are not themselves fields of the vacuum commutant.

The GKO theorem identifies these branching modules with the three irreducible Ising Virasoro modules and the commutant with the Ising chiral algebra. Central charge subtraction alone had not established that statement: it contained no branching selection, identification, or lowest-grade information.

3.9 Positive forms and a change of gauge

A lattice VOA has a canonical positive Hermitian form and an antilinear PCT involution θ. PCT denotes the conjugation entering the radial-adjoint identity; it preserves the vacuum and conformal vector. The invariant bilinear form often used in purely algebraic VOA calculations is

a different object. Taking complex conjugates of coordinates is not, by itself, the PCT operation in an arbitrary basis.

An automorphism of a VOA is an invertible linear map $g : V \to V$ with

$$g\mathbf{1} = \mathbf{1}, \qquad g\omega = \omega, \qquad gY(a,z)g^{-1} = Y(ga,z).$$

In a lattice VOA, scaling e^{α} by $\lambda(\alpha) \in \mathbb{C}^{\times}$ and fixing the oscillators gives such an automorphism when λ is a lattice character: $\lambda(\alpha+\beta) = \lambda(\alpha)\lambda(\beta)$. This preserves the chosen cocycle. The earlier arbitrary rescaling instead changes the cocycle presentation and supplies an isomorphism between the corresponding presentations.

Suppose such a VOA automorphism g changes the normalization of the exponential states. Transport the positive form and involution together:

$$\langle v, w\rangle_g = \langle g^{-1}v, g^{-1}w\rangle, \qquad \theta_g = g\theta g^{-1}. \tag{3.25}$$

The new form is positive because g is invertible. It may make a displayed basis nonorthonormal with unequal diagonal norms. If $v = \sum a_j v_j$ with orthogonal v_j, then $\|v\|_g^2 = \sum |a_j|^2 \, \|v_j\|_g^2$, not necessarily $\sum |a_j|^2$.

A subalgebra stable under the transported PCT inherits vacuum unitarity. One must check stability; it is not automatic for a complex-generated subalgebra. Even after vacuum unitarity is known, positivity on all modules and on fusion spaces remains a further problem. The complete-unitarity chapter develops this hierarchy.

3.10 Three finiteness notions that will recur

An *ordinary module* has diagonalizable L_0, finite-dimensional energy eigenspaces, and lower truncation in every integer-shift class: for each weight λ, the spaces of weights $\lambda - n$ vanish for sufficiently large integers n. A simple ordinary module in the positive-energy setting used here has spectrum contained in $h + \mathbb{Z}_{\geq 0}$ for one lowest weight h. Finite direct sums may involve several such translates; for example, a direct sum of an integral-weight module and a module of lowest weight $1/3$ has both integer and $1/3$-shifted energies. A weak module need not have these grading properties. A generalized module can contain nontrivial Jordan blocks for L_0, producing logarithmic behavior in correlation functions.

An *admissible module* is a weak module with a compatible nonnegative integer grading $M = \bigoplus_{n\geq 0} M(n)$: for homogeneous a,

$$a_{(m)}M(n) \subset M(n + \mathrm{wt}(a) - m - 1), \qquad M(j) = 0 \quad (j < 0).$$

This grading need not be the L_0 eigenspace grading, and its pieces are not required to be finite dimensional.

The C_2 subspace and quotient are

$$C_2(V) = \mathrm{span}\{a_{(-2)}b : a, b \in V\}, \qquad R_V = V/C_2(V). \tag{3.26}$$

Here $[a]$ denotes the coset $a + C_2(V)$. The quotient has a commutative associative product $[a][b] = [a_{(-1)}b]$ and a Poisson bracket $\{[a],[b]\} = [a_{(0)}b]$. A Poisson bracket is a bilinear antisymmetric operation satisfying the Jacobi identity and a derivation rule for the commutative product. The vertex-algebra identities make these operations independent of the representatives a, b. Derivatives vanish in the quotient because $\mathsf{T}a = a_{(-2)}\mathbf{1}$. The condition $\dim R_V < \infty$ is called C_2-cofiniteness. It controls a large amount of representation theory, but by itself does not exclude logarithmic nonsemisimple modules.

Rationality asserts complete reducibility of admissible modules. Regularity is the stronger-looking statement that every weak module is a direct sum of ordinary simple modules. The standard theorem used here is precise: for a VOA of CFT type, rationality together with C_2-cofiniteness implies regularity. This is the Abe–Buhl–Dong implication. Thus the strong rationality established in Chapter 10 supplies the weak-module decomposition needed in Chapter 12. We still distinguish the definitions, since C_2-cofiniteness or a finite list of ordinary modules alone would not supply that implication.

For the Haagerup algebra, the proof of finiteness is not a general principle that every subalgebra of a lattice VOA is rational. Heisenberg subalgebras already show why that would be false. Instead, an explicit positive identity among conformal vectors provides the needed control. Understanding why that identity works is one of the main tasks of Part II.

3.11 Exercises

Exercise 3.1. For a of weight three and b of weight four, list the possible nonzero singular products $a_{(n)}b$ in a CFT-type VOA and their weights. Which coefficient is forced to be a scalar multiple of the vacuum? Which coefficient is a candidate weight-two field?

Solution on page 290.

Exercise 3.2. Prove the cocycle and commutator identities for (3.14). Then show directly that multiplying ϵ by a coboundary leaves the commutator ratio unchanged.

Solution on page 290.

Exercise 3.3. For $L = \mathbb{Z}\alpha$, $\alpha^2 = 2$, compute the first three coefficients of $\Theta_L \prod_{n\geq 1}(1-q^n)^{-1}$ and of the corresponding series for $L + \alpha/2$. Explain why the second module has a two-dimensional top but categorical dimension one.

Solution on page 290.

Exercise 3.4. Show that the rank-one Heisenberg VOA has C_2 quotient $\mathbb{C}[x]$, with x represented by $J_{-1}\mathbf{1}$. Conclude that strong finite generation does not imply C_2-cofiniteness.

Solution on page 291.

Exercise 3.5. Derive the central charge of $K(\mathfrak{sl}_2, k)$ by removing the Cartan Heisenberg stress from $L_k(\mathfrak{sl}_2)$. Evaluate it at $k = 3$ and $k = 13$. Explain why this arithmetic alone does not identify a pair of unspecified weight-two fields with the two parafermion stresses.

Solution on page 291.

Exercise 3.6. Suppose g multiplies three orthonormal exponential states by $1/2, 1/\sqrt{40}, 1/\sqrt{40}$. Compute the transported squared norm of $v = e^a + \frac{\sqrt{13}+3i\sqrt{3}}{20}e^b + \frac{\sqrt{13}-3i\sqrt{3}}{20}e^c$. Explain why the answer differs from the sum of the three squared coefficient magnitudes.

Solution on page 291.

Chapter 4

The six Haagerup lines

We now turn the abstract language of fusion into a concrete algebra. The Haagerup example is small enough that its entire fusion ring can be understood by hand. Its associator already contains information that no group multiplication table can express. We will derive its quantum dimensions, all irreducible representations of its ordinary fusion algebra, an exact invariant-state projector, and a useful four-dimensional recoupling matrix. These calculations also fix the notation used throughout the construction.

4.1 A six-object fusion ring

Let α be an invertible object of order three and ρ a noninvertible object. The simple objects are

$$\mathbf{1}, \alpha, \alpha^2, \rho, \alpha\rho, \alpha^2\rho.$$

Their fusion rules are generated by

$$\alpha^3 = \mathbf{1}, \qquad \rho\alpha = \alpha^{-1}\rho, \qquad \rho^2 = \mathbf{1} + \rho + \alpha\rho + \alpha^2\rho. \tag{4.1}$$

Direct sums are written with plus signs in a fusion ring. Thus the last equality means an isomorphism with a four-term direct sum.

Put $A_g = \alpha^g$ and $R_g = \alpha^g\rho$, with $g \in \mathbb{Z}/3\mathbb{Z}$. Moving invertible objects past ρ gives the complete table in four formulas:

$$\begin{aligned} A_g A_h &= A_{g+h}, & A_g R_h &= R_{g+h}, \\ R_g A_h &= R_{g-h}, & R_g R_h &= A_{g-h} + R_0 + R_1 + R_2. \end{aligned} \tag{4.2}$$

For example,

$$R_1 R_2 = A_2 + R_0 + R_1 + R_2, \qquad R_2 R_1 = A_1 + R_0 + R_1 + R_2.$$

The ring is not commutative. This has an immediate categorical consequence: a category with these fusion rules cannot admit a braiding. A braiding would give an isomorphism $a \otimes b \simeq b \otimes a$ for every pair, making its Grothendieck ring commutative. The braided category relevant to the CFT is the *center* of the Haagerup category, not the six-object category itself.

The duality operation follows by looking for the unit in a product:

$$A_g^* = A_{-g}, \qquad R_g^* = R_g. \tag{4.3}$$

Every noninvertible simple is self-dual, even though the full ring is noncommutative.

Worked example 4.1 (An associativity check)**.** Using (4.2),

$$\begin{aligned}(R_gR_h)R_k &= R_{g-h+k} + \sum_x A_{x-k} + 3\sum_y R_y \\ &= R_{g-h+k} + \sum_x A_x + 3\sum_y R_y.\end{aligned}$$

Expanding $R_g(R_hR_k)$ gives the same expression. This verifies a representative family of associativity equations for the ring. It does not verify the pentagon for an associator. Ring associativity counts channels, while the pentagon compares linear maps between their bases.

4.2 The number $\sqrt{13}$ is forced by fusion

In a positive dimension function each invertible object has dimension one, and multiplication by α shows that all three R_g have the same dimension d. Applying the dimension function to (4.1) gives

$$d^2 = 1 + 3d.$$

Thus

$$d = \frac{3+\sqrt{13}}{2}, \qquad d_- := \frac{3-\sqrt{13}}{2} = -\frac{1}{d}. \tag{4.4}$$

Only d is positive. We will frequently use the abbreviations

$$s = 1 + d^2 = 2 + 3d = \frac{13+3\sqrt{13}}{2}, \qquad \mathcal{D} = 3s. \tag{4.5}$$

The global dimension of the six-object category is

$$\mathrm{Dim}(\mathcal{H}_3) = \sum_x d_x^2 = 3 + 3d^2 = 3s = \mathcal{D}. \tag{4.6}$$

The symbol $\mathcal{D}$ will also be the total quantum dimension of the twelve-object center. The center's global dimension is $\mathcal{D}^2$. This is why distinguishing a global dimension from its square root is essential.

The notation d_- should not suggest a second positive quantum dimension inside the same unitary category. It is another algebraic root and another scalar representation of the fusion ring. A physical unitary theory can contain states on which the closed-line operator Y_ρ has this negative eigenvalue; an operator eigenvalue need not be a categorical dimension.

4.3 One fusion ring, two unitary categories

There are two unitary Haagerup categories customarily denoted $\mathcal{H}_2$ and $\mathcal{H}_3$ with the six-object ring above. Their associators are inequivalent. They are Morita equivalent, and consequently their centers are braided equivalent, but these facts do not identify their microscopic anyon chains.

The regular ρ-projector chain investigated later in the book uses $\mathcal{H}_3$. The $\mathcal{H}_2$ presentation commonly used in algebraic Haagerup–Izumi calculations has a different invariant coefficient, compared below. The all-odd-order construction in Chapter 25 requires its own category and positivity analysis. A change of label on a computer program cannot convert one associator into the other.

Construction input 4.2 (The categorical Haagerup input)**.** We use the established unitary category $\mathcal{H}_3$ with fusion rule (4.1) and its transparent associator presentation. "Transparent"

here means a gauge in which admissible associators involving an external invertible simple object are identity maps. It is a statement about an associator gauge, not a claim that $\mathcal{H}_3$ is braided or that a line lies in a Müger center. The complete presentation satisfies the pentagon. We display and verify one important block below; that block alone is not a replacement for the complete pentagon data.

4.4 An explicit Haagerup recoupling matrix

Consider three ρ lines with total charge ρ. Every summand of ρ^2 is an admissible intermediate channel. Use the order

$$(\mathbf{1}, \rho, \alpha\rho, \alpha^2\rho).$$

Write $r = \sqrt{13}$ and define

$$x = \frac{2-r}{3}, \qquad y_- = \frac{5-r-\sqrt{6(1+r)}}{12}, \qquad y_+ = \frac{5-r+\sqrt{6(1+r)}}{12}. \tag{4.7}$$

In the convention of (1.5), the matrix is

$$F_\rho^{\rho\rho\rho} = \begin{pmatrix} d^{-1} & d^{-1/2} & d^{-1/2} & d^{-1/2} \\ d^{-1/2} & x & y_- & y_+ \\ d^{-1/2} & y_- & y_+ & x \\ d^{-1/2} & y_+ & x & y_- \end{pmatrix}. \tag{4.8}$$

Changing the sign of the second radical exchanges the roles of $\alpha\rho$ and $\alpha^2\rho$ in the corresponding full presentation. The displayed matrix is real and symmetric.

It is worthwhile to check its orthogonality entirely by hand. First,

$$x + y_- + y_+ = \frac{3-r}{2} = -\frac{1}{d}. \tag{4.9}$$

Next,

$$y_- y_+ = \frac{(5-r)^2 - 6(1+r)}{144} = \frac{2-r}{9} = \frac{x}{3}.$$

Consequently

$$xy_- + y_- y_+ + y_+ x = x\left(\frac{5-r}{6} + \frac{1}{3}\right) = \frac{(2-r)(7-r)}{18} = -\frac{1}{d}. \tag{4.10}$$

Squaring (4.9) and using (4.10) now gives

$$x^2 + y_-^2 + y_+^2 = d^{-2} + 2d^{-1} = 1 - d^{-1}. \tag{4.11}$$

The last equality is $d^{-2} + 3d^{-1} = 1$, which follows directly from the dimension equation.

The first row of (4.8) has squared norm $d^{-2} + 3d^{-1} = 1$. Its inner product with any other row is $d^{-1/2}(d^{-1} + x + y_- + y_+) = 0$. Every remaining row has squared norm $d^{-1} + x^2 + y_-^2 + y_+^2 = 1$. Finally, the inner product of any two distinct remaining rows is $d^{-1} + xy_- + y_- y_+ + y_+ x = 0$. Hence

$$(F_\rho^{\rho\rho\rho})^{\mathrm{t}} F_\rho^{\rho\rho\rho} = I. \tag{4.12}$$

This is an exact radical calculation; no decimal tolerance appears.

Worked example 4.3 (Two explicit local interactions)**.** The right-vacuum projector in this left basis is

$$P^{(\mathbf{1})} = \begin{pmatrix} d^{-2} & d^{-3/2} & d^{-3/2} & d^{-3/2} \\ d^{-3/2} & d^{-1} & d^{-1} & d^{-1} \\ d^{-3/2} & d^{-1} & d^{-1} & d^{-1} \\ d^{-3/2} & d^{-1} & d^{-1} & d^{-1} \end{pmatrix}.$$

The right-ρ projector is the outer product of

$$v_\rho = (d^{-1/2}, x, y_-, y_+)^{\mathsf{t}}$$

with itself. Both are rank one. Their product vanishes by (4.9). Thus $-P^{(\rho)}$ and $-P^{(\mathbf{1})}$ favor different fusion channels despite acting on the same four local path configurations. A periodic chain uses related blocks for every possible pair of exterior path labels, not only this block.

The entry x has an additional use. It is exactly $[F_\rho^{\rho\rho\rho}]_{\rho\rho}$, the gauge-invariant entry discussed in Chapter 1. For $\mathcal{H}_2$ the same entry is

$$x_{\mathcal{H}_2} = \frac{7-\sqrt{13}}{6}, \qquad x_{\mathcal{H}_3} = \frac{2-\sqrt{13}}{3}.$$

Their difference is

$$x_{\mathcal{H}_2} - x_{\mathcal{H}_3} = \frac{3+\sqrt{13}}{6} = \frac{d}{3} \neq 0. \tag{4.13}$$

They therefore cannot be related by a trivalent-basis gauge change with this distinguished object identified. Relabelling does not remove the distinction: for every R_g, the entry $[F_{R_g}^{R_g R_g R_g}]_{R_g R_g}$ is the same core entry $[F_\rho^{\rho\rho\rho}]_{\rho\rho}$ within each category. A fusion-rule permutation must take a noninvertible simple to a noninvertible simple, so it preserves this comparison. This concrete obstruction explains why a result proved for an $\mathcal{H}_2$ Hamiltonian cannot simply be relabelled as a result for the $\mathcal{H}_3$ chain.

4.5 All representations of the ordinary fusion algebra

The Grothendieck ring $K_0(\mathcal{H}_3)$ is the free abelian group on the six simple-object classes, with addition induced by direct sum and multiplication induced by fusion. Allowing complex coefficients gives the six-dimensional associative algebra $\mathcal{A} = K_0(\mathcal{H}_3) \otimes_{\mathbb{Z}} \mathbb{C}$. It has two one-dimensional representations:

$$\alpha \mapsto 1, \qquad \rho \mapsto d \quad \text{or} \quad \alpha \mapsto 1, \qquad \rho \mapsto d_-. \tag{4.14}$$

There is also a two-dimensional representation. Put $\omega = e^{2\pi i/3}$ and set

$$A = \begin{pmatrix} \omega & 0 \\ 0 & \omega^2 \end{pmatrix}, \qquad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{4.15}$$

Then $A^3 = I$, $RA = A^{-1}R$, and

$$I + R + AR + A^2R = I + (I + A + A^2)R = I = R^2.$$

Thus the fusion rules hold. The two eigenspaces of A are exchanged by R, so no one-dimensional subspace is invariant under both; this representation is irreducible.

Proposition 4.4. *There is an algebra isomorphism*

$$\mathcal{A} \simeq \mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}). \tag{4.16}$$

The two scalar summands are the dimension and negative-root characters, and the matrix summand is (4.15)*.*

Proof. The three representations define a homomorphism into the displayed algebra. The spectral projectors of A separate the two basis directions in the matrix summand, and multiplication by R gives its two off-diagonal matrix units. Thus that summand is generated. The element $E = (\mathbf{1} + \alpha + \alpha^2)/3$ vanishes in the matrix summand and equals one in the two scalar summands. The elements E and $E\rho$ separate those scalar summands because $d \neq d_-$. The map is therefore surjective. Its source and target both have dimension $6 = 1 + 1 + 4$, so it is an isomorphism. □

This is the entire representation theory of the ordinary closed-ring fusion algebra. It has three irreducible representations, not twelve. The twelve center sectors encountered later belong to the tube algebra, which includes defect sectors and junction operations.

4.6 An exact projector onto invariant states

Suppose the line operators Y_x on a Hilbert space obey the fusion rules and adjoint relations $Y_x^\dagger = Y_{x^*}$. Define

$$E = \frac{I + Y_\alpha + Y_{\alpha^2}}{3}. \tag{4.17}$$

This is the ordinary $\mathbb{Z}_3$-neutral projector. Since inversion permutes the group elements, E commutes with Y_ρ. On its image,

$$Y_\rho^2 = I + 3Y_\rho.$$

The two possible eigenvalues are therefore d and $d_- = -d^{-1}$. Lagrange interpolation produces their projectors:

$$P_+ = E\frac{Y_\rho - d_-}{d - d_-} = E\frac{I + dY_\rho}{1 + d^2}, \tag{4.18}$$

$$P_- = E - P_+. \tag{4.19}$$

The equality of the two expressions follows after multiplying numerator and denominator by d.

Expanding E yields a particularly useful formula:

$$P_+ = \frac{I + Y_\alpha + Y_{\alpha^2} + d(Y_\rho + Y_{\alpha\rho} + Y_{\alpha^2\rho})}{3s}. \tag{4.20}$$

It is the dimension-weighted average of all six line operators. For a finite group every dimension equals one and the analogous formula is the familiar group average.

Proposition 4.5. *P_+ is the orthogonal projector onto states v satisfying*

$$Y_x v = d_x v \qquad \text{for all six simple lines } x.$$

The projector P_- selects states neutral under α but not invariant under the full Haagerup symmetry.

Proof. On $\operatorname{im} E$, (4.18) is one at the eigenvalue d and zero at d_-. On $\ker E$ it vanishes. Since Y_ρ is self-adjoint and E is an orthogonal projector, this is an orthogonal spectral projector. Its image has $Y_\alpha = I$ and $Y_\rho = dI$, so $Y_{\alpha^g\rho} = dI$ as well. Conversely, a vector transforming by all these dimensions lies in $\operatorname{im} E$ with $Y_\rho = d$, hence is fixed by P_+. □

The charged central projector is $P_{\mathrm{ch}} = I - E$. These three projectors are pairwise orthogonal and sum to I. In the regular representation their ranks are $1, 1, 4$, respectively. In a physical Hilbert space each irreducible representation can occur many times, at different energies. A rank-one central block of the regular algebra therefore does not imply a unique invariant physical state.

There is a further locality caveat with a simple explanation. P_+ is built from lines wrapping the entire circle. Sandwiching a local operator O as P_+OP_+ can turn it into a nonlocal operator. A projector onto invariant *states* is not by itself a recipe for every local invariant Hamiltonian density.

4.7 Fusion paths and exact integer growth

A periodic anyon chain with physical object ρ has basis labels $x_0, x_1, \ldots, x_L = x_0$ satisfying $N_{x_j\rho}^{x_{j+1}} \neq 0$. For multiplicity-free fusion, each admissible path contributes one basis vector. In the order $(A_0, A_1, A_2, R_0, R_1, R_2)$, right multiplication by ρ has adjacency matrix

$$N_\rho^{\text{right}} = \begin{pmatrix} 0 & I_3 \\ I_3 & J_3 \end{pmatrix}, \tag{4.21}$$

where J_3 has every entry equal to one. Hence the number of length-L periodic paths is $\mathrm{Tr}((N_\rho^{\text{right}})^L)$.

On the two-dimensional subspace of vectors constant in each block, (4.21) reduces to $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 3 \end{smallmatrix}\right)$, with eigenvalues d, d_-. On the sum-zero subspace of each three-dimensional block, J_3 vanishes, leaving eigenvalues $+1$ and -1, each twice. Thus

$$\dim \mathcal{H}_L = d^L + d_-^L + 2(1 + (-1)^L). \tag{4.22}$$

The irrational powers combine to give an integer. Indeed, if $C_L = d^L + d_-^L$, then

$$C_0 = 2, \qquad C_1 = 3, \qquad C_L = 3C_{L-1} + C_{L-2}. \tag{4.23}$$

For $L = 1, 2, 3, 4, 5, 6$, the path counts are $3, 15, 36, 123, 393, 1302$.

This provides a useful resource estimate before constructing any Hamiltonian: the Hilbert space grows as d^L, with $d > 3$. It also demonstrates why dimensions of categorical symmetry sectors are not counts of low-energy primary fields. These are dimensions of entire finite lattice Hilbert spaces, including high-energy states.

4.8 Four quantities that are sometimes all called an index

The original Haagerup subfactor has Jones index

$$[M : N] = 1 + d = \frac{5 + \sqrt{13}}{2}. \tag{4.24}$$

Its relevant six-object even part is $\mathcal{H}_2$, with a connected algebra object $\mathbf{1} \oplus \rho$ of that dimension. The existence of this algebra structure is additional information beyond the fusion rule. In particular, one cannot infer the same algebra structure in $\mathcal{H}_3$ merely because its objects have the same dimensions.

The CFT construction will instead use a chain of chiral extensions

$$U \subset B = V_{E_6 \oplus A_2} \subset H = V_{E_8}.$$

On the proper branch established by its construction inputs, the extension dimensions are

Inclusion or category	Relevant dimension
Original Haagerup subfactor	$1 + d$
$U \subset B$	$s = 1 + d^2$
$B \subset H$	3
$U \subset H$	$3s$
$\mathcal{H}_3$	global dimension $3s$
$Z(\mathcal{H}_3)$	global dimension $(3s)^2$

The chiral extension entries will be justified later using quantum dimensions and algebra objects. The table is a notation guide, not an assumption that the original subfactor inclusion literally occurs inside this lattice VOA. Multiplying the two extension dimensions gives $3s$; replacing it by the original number $1 + d$ would confuse two different constructions.

4.9 Exercises

Exercise 4.1. Derive all four formulas in (4.2) from (4.1). Verify the duality rule (4.3), and give a two-line proof that the category cannot be braided.

Solution on page 291.

Exercise 4.2. Verify (4.10) directly from the radicals. Use only the three scalar identities (4.9), (4.10), and (4.11) to prove orthogonality of the full matrix. Explain why this calculation does not prove every pentagon of $\mathcal{H}_3$.

Solution on page 292.

Exercise 4.3. Using the fusion ring alone, expand the square of the numerator in (4.20) and prove $P_+^2 = P_+$. Show that $Y_\rho P_+ = dP_+$ without diagonalizing a matrix.

Solution on page 292.

Exercise 4.4. Compute the characteristic polynomial of (4.21). Use it to derive (4.22) and obtain the length-six path count.

Solution on page 292.

Exercise 4.5. Construct a unitary representation of the fusion algebra containing three copies of the positive scalar representation, two copies of the negative scalar representation, and four copies of the two-dimensional representation. What are the ranks of E, P_+, P_-, and P_{ch}? How many states are neutral under α, and how many are fully invariant?

Solution on page 292.

Source note. The explicit associators follow the transparent presentation of Huang and Lin. The elementary fusion-algebra decomposition and projector derivation also underlie the repository's symmetry-sector checks. Existing floating-point checks of the complete associator are useful implementation checks; the hand calculation here proves the displayed block identities exactly and claims no independent reconstruction of the entire category.

Chapter 5

Subfactors, centers, and the tube algebra

The six Haagerup lines are not braided, yet the chiral algebra in the construction has a braided category of twelve representations. This chapter explains the bridge. The main idea is that a charge must specify how it passes *every* symmetry line. This extra data is a half-braiding, and the objects carrying it form the Drinfeld center. An equivalent description uses an algebra of annular line networks, the tube algebra. Before treating Haagerup, we work out the complete example of $\mathbb{Z}_3$ symmetry, where all of these constructions reduce to finite Fourier transforms.

5.1 Why operator algebras enter the story

A von Neumann algebra is a unital adjoint-closed algebra of bounded operators on a Hilbert space, closed in the weak operator topology: convergence of every matrix element. It is a factor when its algebraic center consists only of scalar multiples of the identity. Such algebras arise naturally as local observable algebras in quantum field theory. A subfactor is an inclusion $N \subset M$ of factors. Finite index measures the size of M as an N-module, with the ordinary dimension replaced by a dimension compatible with the operator-algebra trace or standard representation. For an outer action of a finite group G, the fixed-point inclusion $M^G \subset M$ has index $|G|$. This recovers an integer in the group case, while general subfactors can have nonintegral index. Here an outer group action means that no nonidentity group element acts as $x \mapsto uxu^*$ for a unitary u inside M.

One should not try to obtain nonintegral index by counting an ordinary basis of M over N. The relevant module dimension is not the dimension of a finite-dimensional complex vector space. The same distinction already appeared when the positive dimension of a Fibonacci line was φ.

An inclusion has a basic bimodule, conventionally written ${}_N L^2(M)_M$. Fusing bimodules over the intermediate algebra produces more bimodules. Alternating this basic bimodule with its dual generates tensor categories of N–N and M–M bimodules, called the even parts. In a finite-depth situation there are finitely many simple types, giving fusion categories. Positivity comes from the Hilbert-space bimodule structure.

Write $X = {}_N L^2(M)_M$ for the basic bimodule. The *principal graph* records the decomposition process. Its even vertices are the simple N–N bimodules and its odd vertices the simple N–M bimodules generated by alternating X with its dual. An even vertex Z is joined to an odd vertex Y with multiplicity $\dim \mathrm{Hom}(Y, Z \otimes_N X)$. The root is $L^2(N)$; reversing N and M gives the dual principal graph. Finite depth means that these generated graphs have finitely many vertices.

The original Haagerup subfactor has index $(5+\sqrt{13})/2$. Its construction produces categorical data that are not explained by a finite group action. The category $\mathcal{H}_2$ and its Morita partners,

including $\mathcal{H}_3$, arise in this setting. For our purposes the key lesson is a method: a concrete operator-algebra inclusion can package an exotic fusion category in the same way a group fixed-point inclusion packages ordinary symmetry.

5.2 A Jones projection and the origin of a loop parameter

A short calculation makes the relation between index and diagrams more concrete. Work with a finite-index inclusion of type II_1 factors, which carry compatible normalized finite traces. Here $L^2(M)$ is the completion of M for $\langle x, y\rangle = \mathrm{tr}(x^*y)$, and left and right multiplication give its bimodule actions. There is a trace-preserving conditional expectation $E_N : M \to N$. It is an N-bimodule map: $E_N(n_1xn_2) = n_1E_N(x)n_2$. On $L^2(M)$ the map extending $x \mapsto E_N(x)$ is the orthogonal projection e_N onto $L^2(N)$.

For $x \in M$, viewed as left multiplication, apply the composite e_Nxe_N to a vector $y \in M$. The result is

$$E_N(xE_N(y)) = E_N(x)E_N(y).$$

Since M is dense in $L^2(M)$, this proves the operator relation

$$e_Nxe_N = E_N(x)e_N. \tag{5.1}$$

The algebra generated by M and e_N is the first basic construction. Iterating it produces a tower and Jones projections $e_1, e_2, \ldots$. For index $I = [M : N]$, their trace and local relations are

$$\mathrm{tr}(e_i) = I^{-1}, \qquad e_i^2 = e_i = e_i^\dagger, \qquad e_ie_{i\pm1}e_i = I^{-1}e_i, \tag{5.2}$$

while distant projections commute. These relations express the effect of creating a cup–cap pair, joining it to its neighbor, and simplifying the resulting contractible loop.

Set $\delta_{\mathrm{TL}} = \sqrt{I}$ and $U_i = \delta_{\mathrm{TL}}e_i$. Then

$$U_i^2 = \delta_{\mathrm{TL}}U_i, \qquad U_iU_{i\pm1}U_i = U_i. \tag{5.3}$$

These are the Temperley–Lieb relations. Closing a contractible strand contributes the scalar δ_{TL}. The loop parameter is the square root of the subfactor index, not the index itself.

Worked example 5.1 (The first even object at the Haagerup index)**.** For the original Haagerup inclusion, $I = 1 + d$, so $\delta_{\mathrm{TL}} = \sqrt{1+d}$. The fundamental shaded strand followed by its oppositely shaded dual has a unit summand. This composition returns to an even part: an N–M bimodule composes with an M–N bimodule. Removing that summand with the two-strand Jones–Wenzl projection $1 - e_1 = 1 - \delta_{\mathrm{TL}}^{-1}U_1$ leaves categorical dimension

$$\delta_{\mathrm{TL}}^2 - 1 = I - 1 = d.$$

In the relevant even part this is the object ρ. Thus the two numbers $\sqrt{1+d}$ and d have different diagrammatic origins: the former belongs to a shaded fundamental strand, the latter to the noninvertible even object. The Temperley–Lieb relations capture a useful subalgebra of diagram operations; they do not supply every Haagerup junction or its full associator.

5.3 Braiding, twists, and nondegeneracy

A braiding is a natural family of isomorphisms $c_{x,y} : x \otimes y \to y \otimes x$. Naturality says that exchanging two lines commutes with applying morphisms to them. The two *hexagon identities*

say that exchanging one line with a fused pair agrees with two successive exchanges. Suppressing associators, they read

$$c_{x,y\otimes z} = (1_y \otimes c_{x,z})(c_{x,y} \otimes 1_z),$$
$$c_{x\otimes y,z} = (c_{x,z} \otimes 1_y)(1_x \otimes c_{y,z}).$$

With parentheses restored, the associators turn each equality into a six-sided diagram. Thus the hexagons constrain braiding and associativity together. A commutative fusion table is necessary for a braiding, but does not supply these maps or identities.

In splitting bases $t_{ab}^{z,\mu} : z \to a \otimes b$, the exchange has a matrix called an R-symbol:

$$c_{a,b}t_{ab}^{z,\mu} = \sum_{\nu} [R_z^{ab}]_{\nu\mu} t_{ba}^{z,\nu}.$$

An F-matrix changes a fusion-tree basis; an R-matrix exchanges two neighboring factors. They are different operations, even when both happen to be scalars on a particular channel.

A ribbon structure includes compatible twists $\theta_x : x \to x$, dualities, and traces. For simple x the twist is a scalar. Its balancing identity is

$$\theta_{x\otimes y} = c_{y,x}c_{x,y}(\theta_x \otimes \theta_y).$$

Consequently double exchange on the simple output channel z is $\theta_z/(\theta_x\theta_y)$ times the identity. Double exchange is called *monodromy*; it need not be trivial even when the objects being exchanged are identical.

An object is *transparent* when its monodromy with every object is the identity. A ribbon fusion category is modular when the only transparent objects are direct sums of the unit; equivalently here, its normalized Hopf-link matrix is invertible. This is the nondegeneracy condition behind the modular matrices of the next chapter. A symmetric category, such as ordinary finite-group representations with the usual flip, instead makes every object transparent. Apart from the trivial category, it is not modular.

Topological lines in a two-dimensional spacetime do not automatically carry these exchange maps: moving intersecting lines through each other is extra data. The noncommutative Haagerup fusion ring forbids such a braiding on its six simple lines. The center construction below supplies a different category whose objects carry precisely the additional passing data.

5.4 An algebra object remembers how an extension multiplies

Suppose a larger chiral algebra H contains a smaller one U. Viewed as a U-module, H is a sum of irreducible modules. The sum alone does not specify H: its vertex operators also multiply these summands. Categorically, this multiplication is an algebra object.

Definition 5.2. An algebra object A in a monoidal category consists of maps

$$\mu : A \otimes A \longrightarrow A, \qquad \eta : \mathbf{1} \longrightarrow A$$

satisfying associativity, with the category's associator inserted, and the two unit axioms. It is connected if $\dim \mathrm{Hom}(\mathbf{1}, A) = 1$.

The associativity equation is

$$\mu(\mu \otimes 1_A) = \mu(1_A \otimes \mu)\mathfrak{a}_{A,A,A}.$$

Thus the existence of A depends on the F-symbols. The fact that the fusion product of an underlying object contains the right summands does not construct a multiplication satisfying this equation.

In a braided category, commutativity means $\mu c_{A,A} = \mu$. This is the categorical version of locality for a chiral extension. Separability means that multiplication has a splitting compatible with the left and right A actions. It is the property that keeps module categories semisimple in this setting. In a unitary category the corresponding positive structure is often called a *Q-system*: a compatible dagger Frobenius algebra, with a specialness normalization such as $\mu\mu^\dagger = \beta 1_A$ for a positive scalar β. The Frobenius condition says that the comultiplication $\Delta = \mu^\dagger : A \to A \otimes A$ respects both A actions. Suppressing associators, it reads

$$(\mu \otimes 1_A)(1_A \otimes \Delta) = \Delta\mu = (1_A \otimes \mu)(\Delta \otimes 1_A),$$

an equality of maps $A \otimes A \to A \otimes A$. Thus $\beta^{-1}\Delta$ is a bimodule splitting of multiplication. Normalization conventions vary, but the positivity and algebra compatibility are substantive.

Worked example 5.3 (The group algebra object)**.** In the category Vec_G of finite-dimensional G-graded vector spaces with trivial associator, the object $A = \bigoplus_{g\in G} g$ has a multiplication taking the $g \otimes h$ summand to gh. Associativity is exactly the group law. If the associator is instead twisted by a three-cocycle $\omega(g, h, k)$, scalar multiplication coefficients must satisfy an equation whose coboundary cancels ω. The same direct sum may therefore fail to admit the untwisted multiplication. An associator can obstruct an extension even when the fusion rules look unchanged.

Worked example 5.4 (The smallest associator obstruction to gauging)**.** Take the two labels $0, 1 \in \mathbb{Z}_2$, with fusion given by addition modulo two; write g for the nonidentity simple object. All fusion spaces are one dimensional. Define the associator scalar by

$$\omega(a, b, c) = (-1)^{abc}, \qquad a, b, c \in \{0, 1\}.$$

Thus its only nontrivial value is $\omega(1, 1, 1) = -1$. The pointed pentagon becomes the scalar identity

$$\omega(b, c, d)\omega(a, b + c, d)\omega(a, b, c) = \omega(a, b, c + d)\omega(a + b, c, d).$$

On the left the exponent is $bcd + abd + acd + abc$; on the right it is $abc + abd + acd + bcd$. They agree modulo two. The unit constraints hold because the phase is one whenever any argument is zero. We have therefore constructed a coherent unitary associator with the same fusion rule $g^2 = \mathbf{1}$ as the untwisted pointed category.

The sign is not removed by choosing different normalized binary junctions. If their phases are $u(a, b)$, with $u(0, a) = u(a, 0) = 1$, the gauge factor at $(1, 1, 1)$ is

$$\frac{u(1, 1)u(0, 1)}{u(1, 1)u(1, 0)} = 1.$$

The pentagon solutions called three-cocycles are identified under just these changes of junction basis; group cohomology packages that distinction. This example exhibits it without requiring the general cohomology machinery.

Now try to form $A = \mathbf{1} \oplus g$, with a unit-normalized multiplication and coefficient β on $g \otimes g \to \mathbf{1}$. Associativity restricted to $g \otimes g \otimes g$ says

$$\beta = -\beta.$$

The left-associated product contributes β, while the right-associated product includes the associator -1. Consequently $\beta = 0$ over $\mathbb{C}$. The resulting algebra has the square-zero ideal g and is not separable. Explicitly, its right regular module has quotient $\mathbf{1}$ on which g acts by zero. A splitting would have to embed that quotient as the unit summand of A, but right multiplication by g takes that summand to the nonzero g summand. Thus the quotient does not split as an A-module.

There is therefore no separable group-algebra condensation on this sum. When the two objects represent symmetry defects, this is an associator obstruction to gauging the whole $\mathbb{Z}_2$. No braiding was required for the obstruction. It concerns an algebra in the symmetry fusion category; a bosonic chiral extension in a braided representation category also requires the separate commutativity and locality conditions. For the Haagerup construction, the actual order-three subgroup is proved untwisted in Chapter 11; its group law alone would not have proved that fact.

For the original Haagerup subfactor, the connected algebra object $\mathbf{1} \oplus \rho$ in the relevant $\mathcal{H}_2$ category has dimension $1 + d$, agreeing with the subfactor index. Its multiplication is genuine additional data. The different associator of $\mathcal{H}_3$ prevents transferring that particular statement merely by copying the two summand labels.

5.5 Modules, local modules, and bimodules answer different questions

A right A-module is an object X with an action $r_X : X \otimes A \to X$ obeying the usual associative and unital identities. In a braided category, a commutative separable algebra allows relative tensor products over A, giving a category of all right modules. The braiding moves an A line to the side where it can act; one then identifies the two ways of applying that action. This is the categorical analogue of balancing a tensor product over an ordinary algebra. Our convention equips a right module with the left action $\lambda_X = r_X c_{X,A}^{-1} : A \otimes X \to X$ when forming relative tensor products; this fixes the braid direction in Theorem 6.3.

Not every module is *local*. A right module is local, or dyslectic, when moving A all the way around it before applying the action has no effect:

$$r_X c_{A,X} c_{X,A} = r_X. \tag{5.4}$$

The order of maps starts at $X \otimes A$. Local modules inherit a braiding and describe ordinary representations of a local chiral extension. Nonlocal modules still exist and can describe boundary or defect data, even when the extended chiral algebra is holomorphic.

There are therefore three related categories:

Category	Role in an extension
All right A-modules	Boundary or one-sided extension data; may be a nonbraided fusion category.
Local right A-modules	Representations of the extended local chiral algebra.
A–A bimodules	Defects compatible with the chosen extension description, with composition by relative tensor product.

For the selected holomorphic extension in this book, the local module category is Vec, while the category of all right modules is $\mathcal{H}_3$. The word “holomorphic” therefore does not mean that the extended theory has no interesting defects.

5.6 Morita equivalence changes the boundary description

For ordinary algebras, $M_n(\mathbb{C})$ and $\mathbb{C}$ have equivalent module categories: an $M_n(\mathbb{C})$-module is a sum of copies of the standard column module, and its multiplicity space is a vector space. The algebras are not isomorphic, but they present the same module theory. This is the elementary model of Morita equivalence.

For fusion categories the analogue of a representation is a *module category*. In a right-action convention it consists of a finite semisimple category $\mathcal{N}$, an action $(m, X) \mapsto m \triangleleft X$, and coherent

isomorphisms

$$(m \triangleleft X) \triangleleft Y \simeq m \triangleleft (X \otimes Y), \qquad m \triangleleft \mathbf{1} \simeq m.$$

Their coherence is the module version of the pentagon. A module endofunctor T also has coherent isomorphisms $T(m \triangleleft X) \simeq T(m) \triangleleft X$: it respects the action, rather than merely mapping objects to objects. Such functors form the *dual category*, with composition as tensor product. If $\mathcal{N}$ is indecomposable as a module category, this dual is a fusion category. For left module categories, the corresponding dual-category convention uses reverse composition.

For a small example, let untwisted $\mathrm{Vec}_{\mathbb{Z}_3}$ act on Vec by forgetting the grading. Each simple group label acts as the identity functor. A linear endofunctor is $T_W(v) = W \otimes v$; respecting the generator's action equips W with an operator U satisfying $U^3 = 1$. The three one-dimensional eigenvalue choices $1, \zeta_3, \zeta_3^2$, where $\zeta_3 = e^{2\pi i/3}$, are the three simple module endofunctors, and composition multiplies their characters. Thus the module category has one simple object, whereas its dual has three. Fusion categories related through such an indecomposable module category are Morita equivalent. Their simple objects and fusion rings can differ; their centers agree by the theorem below.

Theorem 5.5 (Morita invariance of the center)**.** *Morita equivalent fusion categories have braided equivalent Drinfeld centers. In the unitary setting the equivalence can be taken compatible with the unitary structures and canonical ribbon structures.*

We use this structural theorem as an input from tensor-category theory. Its physical interpretation is helpful: distinct boundary descriptions of the same three-dimensional topological theory can have the same bulk anyons. Those bulk anyons form the center. Thus $Z(\mathcal{H}_2) \simeq Z(\mathcal{H}_3)$ is entirely compatible with the inequivalence demonstrated by (4.13).

Morita equivalence does not itself give an isospectral map between arbitrarily chosen lattice Hamiltonians. To compare chains one would need a specified construction transporting their Hilbert spaces, local interactions, boundary conditions, and inner products. Center equivalence transports categorical bulk data, not an unspecified dynamical model.

5.7 A center object knows how to pass every line

Definition 5.6. An object of the Drinfeld center $Z(\mathcal{C})$ is a pair (z, σ) consisting of an object z of $\mathcal{C}$ and a natural family of isomorphisms

$$\sigma_x : z \otimes x \longrightarrow x \otimes z \qquad (x \in \mathcal{C}),$$

called a half-braiding. It is compatible with tensor products: passing z through $x \otimes y$ agrees with passing it through x and then through y, with the associators inserted.

In a strict notation the compatibility equation reads

$$\sigma_{x\otimes y} = (1_x \otimes \sigma_y)(\sigma_x \otimes 1_y). \tag{5.5}$$

Strict notation suppresses associators; it does not remove them from a calculation with actual F-matrices. Naturality means that the same motion commutes with every junction $x \to y$.

Two center objects can be braided: use the first object's half-braiding with the underlying object of the second. This makes $Z(\mathcal{C})$ braided even when $\mathcal{C}$ is not. With a spherical structure in the usual characteristic-zero setting, the center is modular. Its global dimension is

$$\mathrm{Dim}(Z(\mathcal{C})) = \mathrm{Dim}(\mathcal{C})^2. \tag{5.6}$$

The forgetful functor $F : Z(\mathcal{C}) \to \mathcal{C}$ discards the half-braiding. A simple center object can become a nonsimple object after forgetting; this is precisely what happens for most Haagerup center objects.

5.8 The complete $\mathbb{Z}_3$ example

Take $\mathcal{C} = \mathrm{Vec}_{\mathbb{Z}_3}$ with trivial associator. Its simple objects are $g \in \mathbb{Z}_3$, with fusion $g \otimes h = g + h$. Because the group is abelian, a half-braiding on simple object g with simple object h is a scalar σ_h. Equation (5.5) says $\sigma_{h+h'} = \sigma_h \sigma_{h'}$, so it is a character:

$$\sigma_h = \omega^{kh}, \qquad k \in \mathbb{Z}_3.$$

Here $\omega = e^{2\pi i/3}$. On an arbitrary graded object, the half-braiding with the generator acts on each graded multiplicity space by an operator of order three. It diagonalizes into the three characters, so every center object decomposes into these pairs (g, k). There are therefore nine simple center objects, labelled (g, k). The label g is a flux or twisted-sector label, and k is a charge. Their fusion and dimensions are

$$(g, k) \otimes (h, l) = (g + h, k + l), \qquad d_{(g,k)} = 1. \tag{5.7}$$

The twist, obtained by closing the half-braiding of the object with itself, is

$$\theta_{(g,k)} = \omega^{gk}. \tag{5.8}$$

The braiding is just as explicit. Passing (g, k) through (h, l) uses the first object's character on the second object's underlying flux, so its scalar is ω^{kh}. Passing back gives ω^{lg}, hence the monodromy is ω^{kh+lg}. This agrees with balancing, since

$$\frac{\theta_{(g+h,k+l)}}{\theta_{(g,k)}\theta_{(h,l)}} = \omega^{(g+h)(k+l)-gk-hl} = \omega^{kh+lg}.$$

It also proves nondegeneracy directly. A transparent (g, k) must have trivial monodromy with $(1, 0)$, which forces $k = 0$, and with $(0, 1)$, which forces $g = 0$. Only the unit is transparent.

To fix a torus convention that we will maintain in the next chapter, take

$$S_{(g,k),(h,l)} = \frac{1}{3}\omega^{-(kh+lg)}. \tag{5.9}$$

The minus sign corresponds to the inverse-monodromy orientation of the Hopf link and is compatible with the displayed positive twist. Using the opposite Hopf-link orientation requires the corresponding change in the modular generators. On the real, self-dual Haagerup data this orientation issue is invisible, which makes the present example a useful convention check.

Finite Fourier orthogonality immediately gives

$$S^2_{(g,k),(h,l)} = \delta_{h,-g}\delta_{l,-k}.$$

Thus S^2 is charge conjugation rather than the identity. Also

$$\sum_{g,k} \theta_{(g,k)} = \sum_g \sum_k \omega^{gk} = 3.$$

Dividing by total quantum dimension 3 gives Gauss–Milgram phase one. Direct finite Fourier multiplication gives $(S\Theta)^3 = S^2$, where Θ is the twist matrix. These statements will be generalized in Chapter 6.

5.9 The tube algebra in the same example

An annular diagram can carry an incoming defect label a, an outgoing label b, and a line c circling the annulus. Its junction belongs to

$$\mathrm{Hom}(c \otimes a, b \otimes c).$$

Taking the direct sum over simple labels gives the vector space

$$\mathrm{Tube}(\mathcal{C}) = \bigoplus_{a,b,c} \mathrm{Hom}(c \otimes a, b \otimes c). \tag{5.10}$$

Multiplication glues annuli. Resolving the two circling lines into simple channels uses fusion and the associator. This multiplication is the part that an ordinary list of fusion coefficients does not specify. In a unitary category, reflection gives a positive adjoint structure, so the tube algebra is a finite-dimensional C^*-algebra.

For $\mathrm{Vec}_{\mathbb{Z}_3}$, the morphism space is nonzero exactly when $a = b$. Let $e_{g,h}$ denote the annulus with defect label g and circling line h. Then

$$e_{g,h} e_{g',h'} = \delta_{g,g'} e_{g,h+h'}. \tag{5.11}$$

Thus the tube algebra is three copies of $\mathbb{C}[\mathbb{Z}_3]$, one for each flux. Its nine primitive central idempotents are

$$p_{g,k} = \frac{1}{3} \sum_{h=0}^{2} \omega^{-kh} e_{g,h}. \tag{5.12}$$

The identity $\sum_h \omega^{(k-l)h} = 3\delta_{kl}$ proves $p_{g,k} p_{g',l} = \delta_{gg'} \delta_{kl} p_{g,k}$. On the corresponding one-dimensional representation, $e_{g,h}$ acts by ω^{kh}. A Dehn twist is represented by $e_{g,g}$ and has eigenvalue ω^{kg}, reproducing (5.8).

This calculation exhibits the equivalence between simple tube representations and simple center objects without any large matrices. The ordinary untwisted sector has $g = 0$, so only three of the nine labels occur there. Looking only at operators U_h on that sector would reveal the three ordinary group characters, while omitting the six charges attached to nontrivial fluxes.

5.10 Changing the associator changes the center

Worked example 5.7 (Toric-code and double-semion twists)**.** Return to the two-object category of Example 5.4, with $\omega(a,b,c) = (-1)^{abc}$. All labels and sums below are in $\mathbb{Z}_2$. For a center object with underlying label g, write σ_x for its scalar half-braiding with x. Restoring the three associators in (5.5) gives

$$\sigma_{x+y} = \frac{\omega(x,g,y)}{\omega(g,x,y)\omega(x,y,g)} \sigma_x \sigma_y = (-1)^{gxy} \sigma_x \sigma_y.$$

Since $\sigma_0 = 1$, the choice $t = \sigma_1$ must satisfy $t^2 = (-1)^g$. There are two possibilities over each flux:

$$\mathbf{1} = (0,1), \qquad b = (0,-1), \qquad s = (1,i), \qquad \bar{s} = (1,-i).$$

The corresponding quadratic operator on any multiplicity space has distinct roots, so it diagonalizes into these choices. They are the four simple center objects.

The tensor-product half-braiding also contains associators. For (g,t) and (h,u) its value on the generator is multiplied by $\omega(g,h,1)\omega(1,g,h)/\omega(g,1,h) = (-1)^{gh}$. Consequently

$$(g,t) \otimes (h,u) = (g+h, (-1)^{gh} tu),$$

and hence

$$b^2 = s^2 = \bar{s}^{\,2} = \mathbf{1}, \qquad s\bar{s} = b, \qquad bs = \bar{s}, \qquad b\bar{s} = s.$$

The fusion ring is that of $\mathbb{Z}_2 \times \mathbb{Z}_2$. In particular all four simples are self-dual. The bar on $\bar{s}$ names the *anti-semion*, whose twist is the complex conjugate of the semion's; it does not denote the categorical dual of s.

Use the canonical positive spherical structure. A direct sign check is available in the left-dual convention of Chapter 1: take $\mathrm{coev}_g = 1$, $\mathrm{ev}_g = (-1)^g$, and pivotal map $j_g = (-1)^g$. Their trace gives dimension one. The Drinfeld bend is the map $g \to g^{**}$ made from a crossing, a cup, and a cap. Its associator and evaluation signs cancel here, giving scalar $u_g = \sigma_g$, so the ribbon twist is $u_g^{-1} j_g = \sigma_g$, using $t^2 = (-1)^g$. Thus

X	$\mathbf{1}$	b	s	$\bar{s}$
d_X	1	1	1	1
θ_X	1	1	i	$-i$

These are the double-semion twists. Balancing makes the monodromy of b with either semion equal to -1. Every nonidentity simple has a nontrivial monodromy with some object, so the category is nondegenerate.

For comparison, the untwisted center $Z(\mathrm{Vec}_{\mathbb{Z}_2})$ has labels (g, k) with ordinary componentwise fusion and twist $(-1)^{gk}$. Its simples $1, e, m, \epsilon$ therefore have twists $1, 1, 1, -1$, the toric-code pattern. Its fusion ring is again $\mathbb{Z}_2 \times \mathbb{Z}_2$, but no relabeling identifies these twists with $1, 1, i, -i$. Both centers have global dimension four and Gauss–Milgram phase one:

$$\frac{1+1+i-i}{2} = \frac{1+1+1-1}{2} = 1.$$

Fusion multiplicities, global dimension, and the chiral anomaly therefore leave information that the associator and center twists can distinguish. This is a small complete model of why actual Haagerup junction data are needed.

5.11 Haagerup forgetful multiplicities

The tube-sector decomposition calculated for $\mathbb{Z}_3$ also occurs for Haagerup symmetry, with a less trivial distribution of multiplicities. Use center labels

$$0, b, a, c_0, c_1, c_2, d_1, \ldots, d_6.$$

Here a is a center object; it is not the invertible object α of $\mathcal{H}_3$. Put $X = \rho \oplus \alpha\rho \oplus \alpha^2\rho$. The forgetful images are

$$\begin{aligned} &F(0) = \mathbf{1}, && F(b) = \mathbf{1} \oplus X, && F(a) = 2\mathbf{1} \oplus X, \\ &F(c_j) = \alpha \oplus \alpha^2 \oplus X, && F(d_j) = X. \end{aligned} \tag{5.13}$$

These are categorical center data. Their dimensions already give the complete list of central quantum dimensions:

$$1, \quad 1 + 3d = d^2, \quad 2 + 3d = s \quad \text{(four objects)}, \quad 3d \quad \text{(six objects)}.$$

For a simple defect x define the finite-dimensional space

$$W_x^\mu = \mathrm{Hom}_{\mathcal{H}_3}(F(\mu), x), \qquad n_{x\mu} = \dim W_x^\mu. \tag{5.14}$$

The multiplicity table, with identical columns grouped, is

x	0	b	a	each c_j	each d_j
$\mathbf{1}$	1	1	2	0	0
α	0	0	0	1	0
α^2	0	0	0	1	0
each $\alpha^g\rho$	0	1	1	1	1

The irreducible tube module $W^\mu = \bigoplus_x W_x^\mu$ therefore has ordinary vector-space dimension

$$1,\ 4,\ 5,\ 5,\ 5,\ 5,\ 3,\ 3,\ 3,\ 3,\ 3,\ 3. \tag{5.15}$$

These integers are not the central quantum dimensions.

Worked example 5.8 (The dimension of the Haagerup tube algebra)**.** Finite-dimensional C^*-algebra structure gives

$$\mathrm{Tube}(\mathcal{H}_3) \simeq \mathbb{C} \oplus M_4(\mathbb{C}) \oplus M_5(\mathbb{C})^{\oplus 4} \oplus M_3(\mathbb{C})^{\oplus 6} \tag{5.16}$$

as an algebra. Its dimension is

$$1^2 + 4^2 + 4 \cdot 5^2 + 6 \cdot 3^2 = 171.$$

This finite integer should be compared with neither $3s$ nor $(3s)^2$. Those are categorical global dimensions. The tube algebra is an ordinary algebra built from junction spaces and has an ordinary vector-space dimension.

The center equivalence gives its representations additional tensor and braiding structure. An abstract algebra isomorphism such as (5.16) by itself forgets that structure. Just as knowing $M_n(\mathbb{C})$ does not tell us which preferred matrices represent a Hamiltonian, knowing the matrix-block sizes does not reconstruct the categorical junction operations.

5.12 Finite tube factors versus physical multiplicity spaces

In a two-dimensional theory with $\mathcal{C}$ symmetry, let $\mathcal{H}_x$ be the Hilbert space on a circle intersected by a defect x. Its decomposition takes the form

$$\mathcal{H}_x \simeq \bigoplus_{\mu \in \mathrm{Irr}\, Z(\mathcal{C})} W_x^\mu \otimes \mathcal{V}_\mu. \tag{5.17}$$

Here $\mathrm{Irr}\, Z(\mathcal{C})$ denotes representatives of the simple isomorphism classes. The space W_x^μ is finite and categorical. The space $\mathcal{V}_\mu$ contains the dynamical states: in a CFT it carries energies, spins, and descendant towers. The tube algebra acts on the first factor, while the Hamiltonian commutes with that action and acts on the multiplicity space.

For Haagerup symmetry, the ordinary untwisted space is

$$\mathcal{H}_{\mathbf{1}} \simeq \mathcal{V}_0 \oplus \mathcal{V}_b \oplus (\mathbb{C}^2 \otimes \mathcal{V}_a). \tag{5.18}$$

Its three types agree with the two scalar representations and the doublet of Chapter 4. The factor of two is a real physical multiplicity in the full untwisted Hilbert space. A partition function defined for the multiplicity space $\mathcal{V}_a$ alone must not include it a second time.

For the selected chiral Haagerup action on the holomorphic E_8 full theory, the construction will identify

$$\mathcal{V}_\mu = M_\mu \otimes \overline{V_{E_8}}.$$

That is a dynamical realization of (5.17), not a consequence of the abstract finite fusion ring by itself.

5.13 Why the tube algebra is needed for operator products

For an ordinary group, the tensor product of representations is defined by $U_g \mapsto U_g \otimes U_g$. Trying the same rule for a noninvertible fusion algebra fails. If $Y_\rho = d$ on two one-dimensional invariant representations, the naive tensor-product rule would give $Y_\rho = d^2$. But d^2 is neither of the allowed scalar roots of $y^2 = 1 + 3y$. The putative tensor product is not even a representation of the fusion algebra.

The missing ingredient is the junction information specifying how lines act around two operators and how the two lassos are joined. The center, or equivalently the tube algebra equipped with its categorical structure, supplies that information. Local operators transform in center charges. If the symmetry is unbroken, a correlator with charges $\mu_1, \dots, \mu_n$ can be nonzero only if

$$\mathrm{Hom}_{Z(\mathcal{C})}(\mathbf{1}, \mu_1 \otimes \cdots \otimes \mu_n) \neq 0. \tag{5.19}$$

This is a selection rule, not a formula for a normalized three-point coefficient. Allowed tensor channels still carry dynamical amplitudes. Extracting such amplitudes from a lattice calculation requires identified operator families and coherent normalizations, not only closed-line eigenvalues.

5.14 Exercises

Exercise 5.1. Starting from (5.5), classify the half-braidings on each simple object of $\mathrm{Vec}_{\mathbb{Z}_3}$. Compute all twists and count how many are 1, ω, and ω^2.

Solution on page 293.

Exercise 5.2. Prove the projector multiplication rule following (5.12). Show that the sum of all nine projectors is the unit of the tube algebra. Which three projectors can act on the untwisted sector?

Solution on page 293.

Exercise 5.3. Verify the dimension 171 directly from (5.10) and the Haagerup fusion rules, using

$$\dim \mathrm{Hom}(ca, bc) = \sum_z N_{ca}^{\ z} N_{bc}^{\ z}.$$

First sum over the three invertible choices of c, then over the three noninvertible choices.

Solution on page 293.

Exercise 5.4. Use (5.13) to derive every central quantum dimension. Compare the quantum dimension and ordinary tube-module dimension of the object a. Why are these allowed to differ?

Solution on page 293.

Exercise 5.5. Explain how a holomorphic chiral extension can have a trivial category of local modules and a nontrivial category of all modules. Identify the extra equation a module must satisfy to be local, and explain why a direct-sum decomposition of the extension is insufficient to reconstruct its multiplication.

Solution on page 293.

Source note. The center–tube correspondence and the physical multiplicity-space interpretation are standard structural results. The treatment of symmetry charges follows the formulation of Lin, Okada, Seifnashri, and Tachikawa. The Haagerup forgetful table is supplied as established center data; the explicit decompositions and dimension calculations following it are derived on the page.

Chapter 6

The twelve-sector modular data

Modular data translate categorical information into formulas for characters, defects, and boundaries. They are powerful because a small pair of matrices constrains infinitely many states. They are also easy to overinterpret: modular matrices do not determine all operator products, and matching them does not in general prove an equivalence of braided categories. This chapter derives the main Haagerup formulas and works through a nontrivial Verlinde calculation. It ends by constructing the categorical algebra underlying the selected holomorphic extension.

6.1 Tori, twists, and the central-charge phase

A torus with complex parameter τ can be described using different bases of its two cycles. The basic changes are

$$\tau \longmapsto -1/\tau, \qquad \tau \longmapsto \tau + 1.$$

For a rational chiral algebra with irreducible modules M_i, its characters

$$\chi_i(\tau) = \mathrm{Tr}_{M_i}\, q^{L_0 - c/24}, \qquad q = e^{2\pi i \tau},$$

transform by matrices S and T_{ch}. If the lowest weight of M_i is h_i, then

$$(T_{\mathrm{ch}})_{ii} = e^{2\pi i (h_i - c/24)}. \tag{6.1}$$

The corresponding categorical twist is

$$\theta_i = e^{2\pi i h_i}, \qquad \Theta = \mathrm{diag}(\theta_i), \qquad T_{\mathrm{ch}} = e^{-2\pi i c/24}\Theta. \tag{6.2}$$

The phase multiplying Θ is not optional. Forgetting it confuses a topological Dehn twist with the action on a chiral character including its vacuum energy.

The normalized S-matrix can be obtained from a Hopf-link trace. We choose its orientation to agree with the finite Fourier convention (5.9). In a unitary category this is the normalized trace of inverse double braiding. Reversing all orientation conventions gives equivalent formulas after the appropriate changes to twists and modular generators. The Haagerup S-matrix below is real, so complex conjugation of its entries has no effect.

Let

$$\mathcal{D} = \left(\sum_i d_i^2 \right)^{1/2}$$

be the total quantum dimension of a unitary modular category. Normalization gives

$$S_{00} = \mathcal{D}^{-1}, \qquad S_{0i} = d_i/\mathcal{D}. \tag{6.3}$$

The vacuum row is therefore positive. Charge conjugation is the matrix $C_{ij} = \delta_{i,j^*}$, and

$$S^2 = C, \qquad (S\Theta)^3 = \xi C, \qquad \xi = \frac{1}{\mathcal{D}} \sum_i d_i^2 \theta_i. \tag{6.4}$$

For the representation category of a suitable unitary rational chiral theory, $\xi = e^{2\pi i c/8}$. Substituting (6.2) removes that phase and gives $(ST_{\mathrm{ch}})^3 = C$.

The twist determines h_i only modulo integers. Knowing $\theta = e^{4\pi i/13}$ permits lowest weights 2/13, 15/13, 28/13, and so on. Positivity, branching, and actual character coefficients are needed to decide which lift occurs.

6.2 The Haagerup labels and matrix blocks

Use the ordering

$$(0, b, a, c_0, c_1, c_2, d_1, d_2, d_3, d_4, d_5, d_6). \tag{6.5}$$

The first label is the vacuum. The d_j are names of sectors; their quantum dimensions will all be $3d$. Write $r = \sqrt{13}$ and set

$$A = \frac{1}{6} - \frac{1}{2r}, \qquad B = \frac{1}{6} + \frac{1}{2r}, \qquad u = \frac{1}{3}, \qquad v = \frac{1}{r}.$$

Let $\mathbf{1}_n$ denote the column of n ones. In blocks of sizes $1, 1, 4, 6$, the matrix is

$$S = \begin{pmatrix} A & B & u\mathbf{1}_4^{\mathsf{t}} & v\mathbf{1}_6^{\mathsf{t}} \\ B & A & u\mathbf{1}_4^{\mathsf{t}} & -v\mathbf{1}_6^{\mathsf{t}} \\ u\mathbf{1}_4 & u\mathbf{1}_4 & C_4 & 0 \\ v\mathbf{1}_6 & -v\mathbf{1}_6 & 0 & D_6 \end{pmatrix}, \tag{6.6}$$

where

$$C_4 = \frac{1}{3}\begin{pmatrix} 2 & -1 & -1 & -1 \\ -1 & 2 & -1 & -1 \\ -1 & -1 & -1 & 2 \\ -1 & -1 & 2 & -1 \end{pmatrix}, \qquad (D_6)_{ij} = -\frac{2}{r}\cos\frac{2\pi ij}{13}. \tag{6.7}$$

The twist matrix is

$$\Theta = \mathrm{diag}(1, 1, 1, 1, e^{2\pi i/3}, e^{4\pi i/3}, e^{2\pi i 6/13}, e^{2\pi i 11/13}, e^{2\pi i 2/13}, e^{2\pi i 5/13}, e^{2\pi i 7/13}, e^{2\pi i 8/13}). \tag{6.8}$$

Equivalently, $\theta_{d_j} = e^{2\pi i 6j^2/13}$ for $1 \le j \le 6$. The six exponents are the quadratic nonresidues modulo thirteen.

Construction input 6.1 (The Haagerup center data)**.** The standard unitary Haagerup center has the modular matrices (6.6)–(6.8) and the forgetful data of (5.13). In the CFT construction, identifying the actual module category with this center requires the separate structural and analytic arguments developed later. Writing down these matrices is not a substitute for those arguments.

In the bootstrap notation frequently used for these sectors, the same order is

$$(1, \pi_1, \pi_2, \sigma_1, \sigma_2, \sigma_3, \mu_5, \mu_2, \mu_1, \mu_3, \mu_6, \mu_4).$$

The permutation among the final six labels is fixed by their twists. Keeping equal numerical subscripts on d_j and μ_j would misidentify sectors. We retain (6.5) in calculations and state any translation when comparing conventions.

6.3 Checking normalization without a computer

The first entry can be written

$$A = \frac{\sqrt{13}-3}{6\sqrt{13}} = \frac{1}{3s} = \frac{1}{\mathcal{D}}.$$

Consequently the ratios of the vacuum-row entries to A are

$$(\mathrm{qdim}(M_i))_i = (1, d^2, s, s, s, s, 3d, 3d, 3d, 3d, 3d, 3d). \tag{6.9}$$

For example $u/A = s$, while $v/A = 3d$. The total squared dimension is indeed

$$\begin{aligned} 1 + d^4 + 4s^2 + 6(3d)^2 &= 1 + d^4 + 4(1+d^2)^2 + 54d^2 \\ &= 5 + 5d^4 + 62d^2 \\ &= 9(1+d^2)^2 = (3s)^2. \end{aligned}$$

The last step follows from $d^4 = 11d^2 - 1$, itself a consequence of $d^2 = 3d + 1$.

We can also verify $S^2 = I$ using finite Fourier sums. For $m \not\equiv 0 \pmod{13}$,

$$\sum_{j=1}^{6} \cos\frac{2\pi mj}{13} = -\frac{1}{2}. \tag{6.10}$$

It is half the sum over all nonzero residues, which equals -1. Using $2\cos x \cos y = \cos(x+y) + \cos(x-y)$ gives

$$\sum_{j=1}^{6} \cos\frac{2\pi ij}{13}\cos\frac{2\pi kj}{13} = \frac{13}{4}\delta_{ik} - \frac{1}{2} \quad (1 \le i, k \le 6). \tag{6.11}$$

Writing I_n for the identity matrix and J_n for the n-by-n matrix of ones, we obtain

$$D_6^2 = I_6 - \frac{2}{13}J_6, \qquad D_6\mathbf{1}_6 = v\mathbf{1}_6.$$

The first two columns of S contribute $2v^2J_6$, exactly the missing term in the bottom-right block of S^2.

For the middle block, let P fix its first two coordinates and exchange its last two. Then $C_4 = P - J_4/3$, so

$$C_4^2 = I_4 - \frac{2}{9}J_4, \qquad C_4\mathbf{1}_4 = -\frac{1}{3}\mathbf{1}_4.$$

The first two columns contribute $2u^2J_4$, again exactly the missing term. Mixed blocks vanish by these row sums and $A + B = u$, $A - B = -v$. Finally,

$$A^2 + B^2 + 4u^2 + 6v^2 = 1, \qquad 2AB + 4u^2 - 6v^2 = 0.$$

This verifies every block of $S^2 = I$. All twelve simples are therefore self-dual. This statement concerns the center labels; the six-object category still has $\alpha^* = \alpha^2$. For a simple object, ribbon duality gives $\theta_{x^*} = \theta_x$: the dual morphism is a linear transpose. Taking a Hermitian adjoint instead conjugates the scalar. Thus the self-dual label c_1 can have the nonreal twist $e^{2\pi i/3}$; categorical duality and right-moving complex conjugation are different operations.

6.4 A Gauss sum and central charge modulo eight

Let $\zeta = e^{2\pi i/13}$. The standard quadratic Gauss-sum evaluation gives

$$G_- := \sum_{j=1}^{6} \zeta^{6j^2} = \frac{-1-\sqrt{13}}{2}. \tag{6.12}$$

One way to see its form is to separate the twelve nonzero residues into squares and nonsquares. Their two sums add to -1, while their difference is the quadratic Gauss sum $\sqrt{13}$ with the standard positive choice for this prime. The coefficient 6 selects the nonsquares.

The Gauss–Milgram numerator is now

$$\begin{aligned} p^+ &= 1 + d^4 + s^2(1+1+\omega+\omega^2) + 9d^2 G_- \\ &= 1 + d^4 + s^2 + \frac{9d^2}{2}(-1-\sqrt{13}) = 3s = \mathcal{D}. \end{aligned} \tag{6.13}$$

Thus $\xi = 1$. This also follows abstractly because a Drinfeld center has trivial multiplicative central charge.

If a rational chiral algebra has representation category $Z(\mathcal{H}_3)$, this constrains its chiral central charge to $c \equiv 0 \pmod 8$. The construction in this book realizes the choice $c = 8$.

There is no contradiction with a proposed full CFT of central charge near two carrying Haagerup *defect symmetry*. The latter claim does not identify its chiral representation category with $Z(\mathcal{H}_3)$. A symmetry category acts on a full theory, while the modular category of chiral modules describes a chosen rational chiral algebra. These are different pieces of structure. Likewise, the holomorphic E_8 chiral algebra has a trivial representation category while its full theory can possess nontrivial defects preserving a smaller algebra U.

6.5 Verlinde diagonalizes fusion

The fusion matrices of a modular category commute and are simultaneously diagonalized by S. In our convention their structure constants are

$$N_{ij}{}^{k} = \sum_{\ell} \frac{S_{i\ell} S_{j\ell} \overline{S_{k\ell}}}{S_{0\ell}}. \tag{6.14}$$

The eigenvalue of the line i in sector ℓ is $S_{i\ell}/S_{0\ell}$. The vacuum eigenvalue is d_i, recovering the dimension character.

For Haagerup all entries are real, so the bar in (6.14) can be omitted. Although individual entries contain radicals and cosines, the sum gives nonnegative integers. That integrality is a strong consistency condition, but its validity alone would still not reconstruct an associator or prove that a candidate modular table comes from a CFT.

Worked example 6.2 (The identity coefficient)**.** For any two labels,

$$N_{ij}{}^{0} = \sum_{\ell} S_{i\ell} S_{j\ell} = (S^2)_{ij} = \delta_{ij}.$$

This agrees with self-duality of every Haagerup center simple. The vacuum fusion rule follows similarly from unitarity: $N_{0i}{}^{k} = \delta_{ik}$.

6.6 A full nontrivial fusion product from cosine sums

We compute $d_i \otimes d_i$. Here d_i is a center label, whereas the italic scalar d remains the quantum dimension of a noninvertible Haagerup line. Write $c_m(j) = \cos(2\pi mj/13)$.

For an output d_k, the first two columns of Verlinde contribute

$$v^3\left(\frac{1}{A} - \frac{1}{B}\right) = \frac{9}{13},$$

using $AB = 1/117$. The six final columns contribute $-(8/13)\sum_j c_i(j)^2 c_k(j)$. Thus

$$N_{d_i d_i}^{\ d_k} = \frac{9}{13} - \frac{8}{13}\sum_{j=1}^{6} c_i(j)^2 c_k(j). \tag{6.15}$$

Product-to-sum gives

$$c_i^2 c_k = \frac{1}{2}c_k + \frac{1}{4}c_{2i+k} + \frac{1}{4}c_{2i-k}.$$

Let $[m]$ denote the representative of $\{m, -m\}$ modulo thirteen in $\{1, \ldots, 6\}$. If $k \neq [2i]$, all three frequencies are nonzero modulo thirteen, so (6.10) makes the sum $-1/2$. Equation (6.15) then gives one. If $k = [2i]$, precisely one of $2i + k$ and $2i - k$ vanishes; its cosine sum is six instead of $-1/2$. The triple-product sum becomes $9/8$, giving zero. Therefore

$$N_{d_i d_i}^{\ d_k} = 1 - \delta_{k,[2i]}. \tag{6.16}$$

For any of a, c_0, c_1, c_2, only the first two columns contribute, and their sum is

$$uv^2\left(\frac{1}{A} + \frac{1}{B}\right) = 1.$$

The coefficient of b is zero: its first two columns contribute $11/13$, while the final block contributes $-\sum_j (D_6)_{ij}^2 = -11/13$. The vacuum coefficient is one. Putting everything together,

$$d_i^2 = \mathbf{1} + a + c_0 + c_1 + c_2 + \sum_{\substack{1 \le k \le 6 \\ k \neq [2i]}} d_k. \tag{6.17}$$

Here $\mathbf{1} = M_0$ denotes the vacuum object, so the first term has dimension one. The dimension check is

$$(3d)^2 = 1 + 4s + 5(3d) = 9 + 27d,$$

exactly as required by $d^2 = 1 + 3d$.

Another compact product is

$$b^2 = \mathbf{1} + b + a + c_0 + c_1 + c_2 + \sum_{j=1}^{6} d_j.$$

Its dimension identity is $d^4 = 1 + d^2 + 4s + 18d$. These examples show explicitly how finite Fourier cancellation turns algebraic and trigonometric entries into integer fusion rules.

6.7 The canonical extension algebra

Let $F : Z(\mathcal{H}_3) \to \mathcal{H}_3$ be the forgetful functor and I its right adjoint. The object $I(\mathbf{1})$ carries a canonical connected commutative separable algebra structure. Its summands follow from adjunction:

$$\mathrm{Hom}_{Z(\mathcal{H}_3)}(\mu, I(\mathbf{1})) \simeq \mathrm{Hom}_{\mathcal{H}_3}(F(\mu), \mathbf{1}).$$

Using the first row of the forgetful table gives

$$\mathcal{A} := I(\mathbf{1}) \simeq \mathbf{1} \oplus b \oplus a^{\oplus 2}. \tag{6.18}$$

Here $\mathbf{1} = M_0$ is the vacuum center object; the numerical label 0 remains its index in the modular matrices. Its quantum dimension is

$$\operatorname{qdim}(\mathcal{A}) = 1 + d^2 + 2s = 3s = \mathcal{D}. \tag{6.19}$$

Thus $(\operatorname{qdim}\mathcal{A})^2$ equals the global dimension of the center. Such an algebra is called Lagrangian.

For a connected commutative separable algebra in a nondegenerate braided category, the dimension formulas are

$$\begin{aligned} \operatorname{Dim}(\mathcal{M}_{\mathcal{A}}) &= \frac{\operatorname{Dim}(\mathcal{M})}{\operatorname{qdim}\mathcal{A}}, \\ \operatorname{Dim}(\mathcal{M}_{\mathcal{A}}^{\mathrm{loc}}) &= \frac{\operatorname{Dim}(\mathcal{M})}{(\operatorname{qdim}\mathcal{A})^2}. \end{aligned} \tag{6.20}$$

For $\mathcal{M} = Z(\mathcal{H}_3)$ these become $\mathcal{D}$ and one, respectively. More strongly, the canonical algebra has

$$Z(\mathcal{H}_3)_{\mathcal{A}} \simeq \mathcal{H}_3, \qquad Z(\mathcal{H}_3)_{\mathcal{A}}^{\mathrm{loc}} \simeq \mathrm{Vec}\,.$$

The first equivalence retains all modules; the second keeps only local ones.

6.7.1 Why a Lagrangian algebra recovers a center

Let A be a connected commutative separable algebra in the nondegenerate braided category $\mathcal{M}$. The tensor unit of the module category $\mathcal{M}_A$ is A itself. Its dimension there must be one even when its dimension in $\mathcal{M}$ is large. The normalization is

$$\operatorname{FPdim}_{\mathcal{M}_A}(M) = \frac{\operatorname{FPdim}_{\mathcal{M}}(M)}{\operatorname{FPdim}_{\mathcal{M}}(A)},$$

where the numerator forgets the module action. In our positive spherical setting these are also the categorical dimensions. For instance, in $Z(\mathrm{Vec}_{\mathbb{Z}_3})$ take $A = \bigoplus_k (0,k)$. The flux module $x_g = \bigoplus_k (g,k)$ has ambient dimension three and module dimension one. The three flux modules consequently have global dimension three, as the dimension formula predicts. Their actions and relative tensor products are worked out in Chapter 16.

There is also an actual functor behind reconstruction. The free module functor is $G_A : \mathcal{M} \to \mathcal{M}_A$, $G_A(X) = X \otimes A$. It is *central*: the ambient braiding supplies coherent half-braidings of $G_A(X)$ with every A-module. Equivalently, it lifts to a braided tensor functor

$$\widetilde{G}_A : \mathcal{M} \longrightarrow Z(\mathcal{M}_A).$$

This is the meaning of a central functor; it need not take values in a braided category before the lift. For a fusion category, $\operatorname{FPdim}(\mathcal{M})$ means the sum of the squares of its simple Frobenius–Perron dimensions. In the unitary setting it agrees with the global dimension $\operatorname{Dim}(\mathcal{M})$.

Theorem 6.3 (Lagrangian reconstruction)**.** *Let $\mathcal{M}$ be a nondegenerate braided fusion category and A a connected commutative separable algebra. If $\operatorname{FPdim}(A)^2 = \operatorname{FPdim}(\mathcal{M})$, then the central lift gives a braided equivalence $\mathcal{M} \simeq Z(\mathcal{M}_A)$, carrying A to the canonical algebra $I(\mathbf{1})$. Here I is the right adjoint of the forgetful functor $Z(\mathcal{M}_A) \to \mathcal{M}_A$.*

The structural input is that this lift is fully faithful: it induces an isomorphism on every morphism space. Once that is known, the dimension formulas explain why the Lagrangian condition makes it exhaustive:

$$\operatorname{FPdim} Z(\mathcal{M}_A) = \left(\frac{\operatorname{FPdim}\mathcal{M}}{\operatorname{FPdim} A}\right)^2 = \operatorname{FPdim}\mathcal{M}.$$

A fully faithful fusion subcategory with the full positive global dimension cannot omit any simple object. This general theorem is what turns a holomorphic extension H/U into the identification $\mathrm{Rep}(U) \simeq Z(\mathrm{Rep}(U)_H)$ later in the construction.

To apply the canonical Haagerup algebra $\mathcal{A}$ to U, one needs an actual unitary ribbon equivalence $\mathrm{Rep}(U) \simeq Z(\mathcal{H}_3)$ and the relevant extension theorem. Transporting $\mathcal{A}$ then gives a selected holomorphic extension. Matching S and Θ alone does not provide that transport. The original embedded extension in V_{E_8} may identify its boundary category only up to $\mathcal{H}_2$ or $\mathcal{H}_3$; selecting the canonical $\mathcal{H}_3$ algebra is a separate categorical choice.

6.8 A modular-invariant vector and the number 248

Let $m = (1, 1, 2, 0, \ldots, 0)^{\mathsf{t}}$ be the multiplicity vector of (6.18). Direct substitution in (6.6) gives

$$Sm = m, \qquad \Theta m = m. \tag{6.21}$$

For example the first component is $A + B + 2u = 1$, the a component is $2u + 2(2/3) = 2$, and every d_j component is $v - v = 0$. Therefore

$$Z_{\mathrm{ext}} = |\chi_0 + \chi_b + 2\chi_a|^2 \tag{6.22}$$

is modular invariant whenever the characters transform by these matrices. This is a consistency consequence of the extension algebra. Conversely, the same invariant expression written for formal character candidates would not construct local vertex operators or prove sewing.

For the actual $c = 8$ construction, the top spaces of M_b and M_a have weight one and dimensions 80 and 81, while $\dim U_1 = 6$. Restricting the selected extension to U gives

$$H'|_U \simeq U \oplus M_b \oplus 2M_a.$$

Consequently its weight-one space has dimension

$$6 + 80 + 2 \cdot 81 = 248. \tag{6.23}$$

The equality is a valuable branching check. Identifying the extension with V_{E_8} uses the holomorphic $c = 8$ classification and the extension's established hypotheses, not merely the familiar number 248.

For this $c = 8$ theory, $T_{\mathrm{ch}} = e^{-2\pi i/3}\Theta$. The module called d_3 has actual lowest weight $15/13$, although its twist exponent is $2/13$. This one example captures why a modular table cannot be read as a complete table of lowest energies.

6.9 Exercises

Exercise 6.1. Using (5.9), prove $S^2 = C$ by performing the two finite geometric sums. Explain why replacing C by the identity would be incorrect for this example.

Solution on page 294.

Exercise 6.2. Complete the block verification of $S^2 = I$ for (6.6), including the mixed 4×6 block. Derive $AB = 1/117$ and $A^2 + B^2 = 11/117$.

Solution on page 294.

Exercise 6.3. List the six squares and six nonsquares modulo thirteen. Verify that $6j^2$, for $1 \le j \le 6$, runs through all nonsquares. Use (6.12) to check every algebraic step of (6.13).

Solution on page 294.

Exercise 6.4. Write the product d_1^2 explicitly from (6.17). Which d_j is absent? Repeat for d_4^2, and check dimensions in both cases.

Solution on page 294.

Exercise 6.5. Prove (6.21) component by component. Show that the rank-one coefficient matrix $Z = mm^{\mathsf{t}}$ commutes with both S and Θ. State two additional kinds of information needed before interpreting this matrix as an actual local CFT.

Solution on page 295.

Exercise 6.6. Explain why $\mathrm{Rep}(U) \simeq Z(\mathcal{H}_3)$ at $c = 8$ and a conjectural full $c = 2$ theory with $\mathcal{H}_3$ symmetry are not inconsistent statements. Identify exactly which object carries the central-charge-modulo-eight constraint in the first statement.

Solution on page 295.

Part II

Constructing the central charge eight theory

Chapter 7

The lattice laboratory and the six-current embedding

An existence problem becomes much more manageable when the proposed fields already live in a known theory. Their products then have meanings before we try to recognize the algebra they generate. In this chapter we construct the laboratory for the Haagerup example: an eight-dimensional lattice theory, two commuting affine current algebras, and a particular field of weight two. The calculations require the state–field correspondence, Heisenberg oscillators, and the elementary representation theory of $\mathfrak{sl}_2$. Our aim is to see where the special numbers 13 and 39 enter, and to understand precisely what has, and has not, been constructed when the generators have been written down.

7.1 From a positive lattice to fields

Let L be a positive-definite even integral lattice. Integral means that $(\alpha, \beta) \in \mathbb{Z}$ for $\alpha, \beta \in L$; even means that $(\alpha, \alpha) \in 2\mathbb{Z}$. The vector space of its lattice vertex algebra is

$$V_L = S(\mathfrak{h} \otimes t^{-1}\mathbb{C}[t^{-1}]) \otimes \mathbb{C}_\epsilon[L], \qquad \mathfrak{h} = L \otimes_{\mathbb{Z}} \mathbb{C}. \tag{7.1}$$

Here S denotes the symmetric algebra of the negative oscillator modes, so the first factor is an oscillator Fock space. The second has one basis vector e^α for each lattice momentum, with multiplication $e^\alpha e^\beta = \epsilon(\alpha, \beta) e^{\alpha+\beta}$. The symbol ϵ records a sign convention. It is needed because the exponential fields have to be mutually local, including their signs.

A typical basis vector has the form

$$h_1(-n_1) \cdots h_r(-n_r) e^\alpha, \qquad n_i \geq 1,$$

and its conformal weight is

$$\frac{(\alpha, \alpha)}{2} + n_1 + \cdots + n_r. \tag{7.2}$$

Thus the weight is nonnegative and integral. Positive definiteness is doing two jobs: it bounds the possible momenta at fixed weight, and it makes the natural Hilbert-space form positive. Neither statement would remain automatic for an indefinite lattice.

For the root lattice used below, choose a simple-root basis $\alpha_0, \ldots, \alpha_7$ with Gram matrix G and $G_{ii} = 2$. We use the following concrete cocycle in this basis:

$$\epsilon(a, b) = (-1)^{\sum_i a_i b_i + \sum_{i>j} a_i b_j G_{ij}}. \tag{7.3}$$

Its exponent is bilinear modulo two, so associativity of the twisted group algebra follows immediately. The identities

$$\frac{\epsilon(a,b)}{\epsilon(b,a)} = (-1)^{(a,b)}, \qquad \epsilon(a,-a) = (-1)^{(a,a)/2}$$

are useful sign checks. They show explicitly how the lattice pairing controls locality. For a general even lattice basis, the diagonal contribution $\sum_i a_i b_i$ must be replaced by $\sum_i (G_{ii}/2) a_i b_i$; the root-basis formula is not a basis-independent prescription.

For a pure momentum state, the part of the vertex operator that acts on another pure momentum state is

$$Y(e^a, z)e^b = \epsilon(a,b) z^{(a,b)} \exp\left(\sum_{n>0} \frac{a(-n)}{n} z^n\right) e^{a+b}. \tag{7.4}$$

Oscillator commutators determine the products on all remaining states. This formula is not an asymptotic approximation. Expanding it to the required power of z computes an exact mode product.

Worked example 7.1 (Recovering an affine root triple)**.** Let $(a,a) = 2$ and set $H = a(-1)\mathbf{1}$, $E = e^a$, $F = -e^{-a}$. Since $\epsilon(a,-a) = -1$, the extra minus sign in F gives

$$Y(E,z)F = z^{-2}\left(\mathbf{1} + z a(-1)\mathbf{1} + O(z^2)\right).$$

Consequently $E_{(1)}F = \mathbf{1}$ and $E_{(0)}F = H$. The Heisenberg relations give $H_{(1)}H = 2\mathbf{1}$, $H_{(0)}E = 2E$, and $H_{(0)}F = -2F$. These are the level-one affine $\mathfrak{sl}_2$ relations. If we had written $F = e^{-a}$ without changing the cocycle convention, the displayed Lie bracket would have the wrong sign. A cocycle convention is part of a coordinate system, not a dispensable decoration.

The conformal vector is the rank-eight free-boson stress,

$$\omega_{V_L} = \frac{1}{2}\sum_{i,j} (G^{-1})_{ij} \alpha_i(-1)\alpha_j(-1)\mathbf{1}.$$

Its central charge is the rank, namely eight. This makes a lattice VOA an especially convenient place to construct a conformal subalgebra: we can include this actual vector among the generators and thereby fix the conformal structure from the outset.

7.2 The $E_6 \oplus A_2$ lattice and its E_8 extension

Take $G_{ii} = 2$ and put $G_{ij} = -1$ on the edges

$$(0,1), (1,2), (2,3), (3,4), (2,5), (6,7),$$

with all other off-diagonal entries zero. The first six basis vectors span E_6 and the last two span A_2. Write

$$B = V_{E_6 \oplus A_2}, \qquad H = V_{E_8}.$$

The root lattice $E_6 \oplus A_2$ is not unimodular: its determinant is $3 \cdot 3 = 9$. Both factors have discriminant group $\mathbb{Z}/3$, so the discriminant group of their direct sum has nine elements.

It is useful to understand the extension without memorizing an E_8 coordinate table. Choose a fundamental E_6 weight of squared norm $4/3$ and an A_2 weight of squared norm $2/3$. Their sum has squared norm two. Its discriminant class generates an isotropic order-three subgroup: the conformal weight of that class is

$$\frac{1}{2}\left(\frac{4}{3} + \frac{2}{3}\right) = 1 \equiv 0 \pmod 1.$$

Adjoining it produces an even overlattice of index three. Its determinant is $9/3^2 = 1$, so it is the positive even unimodular rank-eight lattice E_8. At the VOA level this gives

$$H = B \oplus J \oplus J^2, \tag{7.5}$$

where J is a *simple current*: a simple module invertible under fusion. Its order is three, meaning $J \boxtimes_B J \boxtimes_B J \cong B$. In Lie-theoretic notation the two new current spaces are $(27, 3)$ and its dual. Their dimensions are 81 each, and

$$78 + 8 + 81 + 81 = 248.$$

This elementary count will later become a remarkably strong branching constraint for the smaller algebra U.

Remark 7.2 (Three meanings of dimension). The numbers 86 and 248 count ordinary vector-space dimensions of weight-one spaces. The lattice index is three. Quantum dimensions of modules will enter later and are different numbers. Writing "dimension" without its object and meaning is particularly dangerous in this construction, since all three meanings occur in one diagram.

7.3 Two commuting copies of $\mathfrak{sl}_2$

Define

$$\ell = (-1, -2, -3, -2, -1, -2, 0, 0), \qquad h = (5, 8, 9, 8, 5, 0, 2, 2).$$

The first vector is a root and the second has squared norm 78. Let c_i denote the coordinates of h. The current triples are

$$(H_L, E_L, F_L) = (\ell(-1)\mathbf{1}, e^{\ell}, -e^{-\ell}), \tag{7.6}$$

$$(H_R, E_R, F_R) = \left(h(-1)\mathbf{1}, \sum_{c_i \neq 0} e^{\alpha_i}, -\sum_i c_i e^{-\alpha_i} \right). \tag{7.7}$$

The subscripts L, R label these two internal current factors. They do not mean the left-moving and right-moving halves of a full CFT. Both triples belong to a single chiral algebra.

Proposition 7.3 (The levels are one and thirty-nine)**.** *The currents in* (7.6) *and* (7.7) *generate commuting simple affine VOAs*

$$K = A_{1,1} \otimes A_{1,39}, \qquad A_{1,k} = L_k(\mathfrak{sl}_2).$$

Proof. The Gram matrix gives

$$(\ell, \ell) = 2, \quad (\ell, \alpha_i) = 0 \ (c_i \neq 0), \quad (h, \alpha_i) = 2 \ (c_i \neq 0), \quad (h, h) = 78.$$

The zero-mode brackets follow from the root calculation above. For example the terms with opposite simple roots in $E_{R(0)}F_R$ sum to $\sum_i c_i \alpha_i(-1)\mathbf{1} = H_R$. Distinct simple roots do not create additional singular terms in this product. Similarly,

$$E_{R(1)}F_R = \sum_i c_i \mathbf{1} = 39\mathbf{1}, \qquad H_{R(1)}H_R = 78\mathbf{1}.$$

The central term therefore has level 39. Orthogonality and the absence of the relevant mixed root sums give vanishing mixed singular products. The transported positive form constructed below makes these coordinate triples compact; equivalently, one may first conjugate to that compact gauge. The positive lattice representation is integrable; hence the generated affine factors are their simple positive-level quotients. □

There is a second way to understand the apparently unusual level. The right triple is the diagonal of principal $\mathfrak{sl}_2$ embeddings in A_5 and A_2. Their indices are 35 and 4. Levels add under a diagonal embedding, giving $35 + 4 = 39$. The coordinate calculation is preferable as a first proof because it also fixes every normalization used later.

Here is a short derivation of those embedding indices. In the principal $\mathfrak{sl}_2 \subset \mathfrak{sl}_n$, the defining n-dimensional representation is the irreducible spin $(n-1)/2$ representation. Its Cartan eigenvalues are $n-1, n-3, \ldots, 1-n$, so

$$\mathrm{Tr}_{\mathbb{C}^n} H^2 = \sum_{j=0}^{n-1} (n-1-2j)^2 = \frac{n(n^2-1)}{3}.$$

The standard $\mathfrak{sl}_2$ Cartan has squared norm two. The ratio of invariant forms, and therefore the level induced from level-one $\mathfrak{sl}_n$, is $n(n^2-1)/6$. At $n = 6$ and 3 this gives 35 and 4. The omitted branch node in Figure 7.1 makes the two principal factors visible in the chosen root basis.

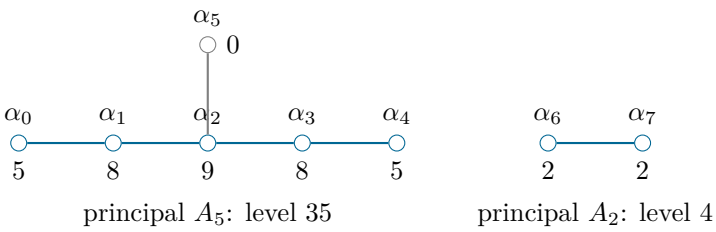


Figure 7.1: The $E_6 \oplus A_2$ root diagram in the chosen numbering. The integers below or beside the nodes are the coefficients c_i of the right Cartan $h = \sum_i c_i \alpha_i$. The nonzero nodes form $A_5 \oplus A_2$, whose principal triples combine diagonally to give level 39.

For a level-k triple the Sugawara vector is

$$\omega_k = \frac{1}{2(k+2)} \left(\frac{1}{2} H_{(-1)} H + E_{(-1)} F + F_{(-1)} E \right), \qquad c_k = \frac{3k}{k+2}. \tag{7.8}$$

Thus the two current factors account for

$$c_K = 1 + \frac{117}{41} = \frac{158}{41}.$$

They account for less than half of the ambient central charge eight. The complementary stress has charge 170/41. The desired algebra will contain more than its affine currents.

7.4 The exceptional field and the transported positive form

Put

$$a = (0,0,0,0,0,-1,1,1), \quad b = (0,0,0,1,1,-1,0,0), \quad c = (1,1,0,0,0,-1,0,0),$$

and define

$$u = e^a + \frac{\sqrt{13} + 3\sqrt{-3}}{20} e^b + \frac{\sqrt{13} - 3\sqrt{-3}}{20} e^c. \tag{7.9}$$

Each momentum has squared norm four. All three terms therefore have weight two. Their Cartan weights are $(1, 13)$, and substitution in the lattice product formula shows that the raising zero modes annihilate the displayed linear combination. Consequently u is a highest vector of current representation dimensions $(2, 14)$.

Worked example 7.4 (The three momenta are already highest vectors)**.** The Gram matrix makes the preceding assertions short calculations:

$$Ga = \begin{pmatrix} 0\\0\\1\\0\\0\\-2\\1\\1 \end{pmatrix}, \quad Gb = \begin{pmatrix} 0\\0\\0\\1\\1\\-2\\0\\0 \end{pmatrix}, \quad Gc = \begin{pmatrix} 1\\1\\0\\0\\0\\-2\\0\\0 \end{pmatrix}.$$

Taking coordinate dot products gives $a^2 = b^2 = c^2 = 4$, all three pairwise inner products equal to two, and

$$(\ell, a) = (\ell, b) = (\ell, c) = 1, \qquad (h, a) = (h, b) = (h, c) = 13.$$

Every displayed pairing with a simple root occurring in E_R is nonnegative; the only negative entry is at the omitted node 5. The exponential product then has no z^{-1} term. Thus $E_{R(0)}$ kills each exponential separately. The pairing with ℓ is also positive, so $E_{L(0)}$ kills each separately. Their linear span is already a three-dimensional space of highest vectors with the same charges. Highest-weight conditions alone therefore leave their relative coefficients undetermined. The special choice in (7.9) is selected by the further product identities.

The field is not chosen by demanding only its weight and Cartan charges. The relative coefficients matter in subsequent products. In a positive orthonormal momentum convention, the same vector is proportional to

$$e^a + \zeta e^b + \overline{\zeta} e^c, \qquad \zeta = \frac{\sqrt{13} + 3i\sqrt{3}}{\sqrt{40}}, \qquad |\zeta|^2 = \frac{13 + 27}{40} = 1.$$

There is no free continuous phase in the stated construction. The special phase is a concrete local datum, rather than a consequence of the fusion rule $d^2 = 1 + 3d$ alone.

Why are the coefficients in (7.9) not all of absolute value one? The current formulas use a convenient nonunitary torus gauge. Coordinate changes of this sort are legitimate, provided the positive form is transported with them.

Let $\widetilde{c} = (5, 8, 9, 8, 5, 1, 2, 2)$ and let g fix the oscillators and act on momenta by

$$g(e^\beta) = \prod_i \widetilde{c}_i^{-\beta_i/2} e^\beta.$$

If θ is the ordinary lattice PCT operation, use $\theta_g = g\theta g^{-1}$ and $\langle v, w\rangle_g = \langle g^{-1}v, g^{-1}w\rangle$. The form remains positive. In the explicit formula take the h_i in the real lattice Cartan and extend antilinearly to complex coefficients:

$$\theta_g(h_1(-n_1)\cdots h_r(-n_r)e^\beta) = (-1)^r \prod_i \widetilde{c}_i^{\beta_i} h_1(-n_1)\cdots h_r(-n_r)e^{-\beta}, \tag{7.10}$$

with antilinear action on scalar coefficients. The factors on opposite momenta cancel when the operation is applied twice.

Worked example 7.5 (A large normalization with a small meaning)**.** For the left root, $\prod_i \widetilde{c}_i^{\ell_i} = 8640^{-2}$. Hence $\theta_g(E_L) = -F_L/8640^2$. The root vectors have unequal coordinate norms even though they span the ordinary compact level-one current algebra. This is why a later compact diagonal current is $8640E_L + E_R$, paired with $F_L/8640 + F_R$, rather than the raw sum $E_L + E_R$. Positivity is a statement about the transported form, not about the apparent size of coordinates.

The positive real torus character extends from $E_6 \oplus A_2$ to the common real Cartan space, and hence to its E_8 overlattice. The entire tower can therefore be discussed in one positive convention.

Worked example 7.6 (The 28 states generated by one highest field)**.** We can now see an ordinary finite-dimensional representation inside the vertex algebra, including its positive normalization. Work with the compact zero-mode operators

$$e_L = 8640E_{L(0)}, \qquad f_L = \frac{F_{L(0)}}{8640}, \qquad e_R = E_{R(0)}, \qquad f_R = F_{R(0)}.$$

The transported form gives $e_L^\dagger = f_L$ and $e_R^\dagger = f_R$. The two Cartan zero modes are self-adjoint. For either internal label, $[e, f] = H_{(0)}$, $[H_{(0)}, e] = 2e$, and $[H_{(0)}, f] = -2f$; operators with different internal labels commute. These are two copies of the usual compact $\mathfrak{sl}_2$ representation problem. Both act on holomorphic fields: the subscripts L, R here do not denote physical chirality.

First normalize the highest vector u. Distinct lattice momenta are orthogonal, and the transported squared norms are

$$\|e^a\|_g^2 = 4, \qquad \|e^b\|_g^2 = 40, \qquad \|e^c\|_g^2 = 40.$$

The two nontrivial coefficients in (7.9) have squared modulus $(13+27)/400 = 1/10$. Consequently

$$\|u\|_g^2 = 4 + \frac{40}{10} + \frac{40}{10} = 12.$$

The already verified highest-vector conditions read

$$e_L u = e_R u = 0, \qquad H_{L(0)} u = u, \qquad H_{R(0)} u = 13u.$$

Recall the lowering calculation for one copy of $\mathfrak{sl}_2$. If $ev = 0$ and $H_{(0)}v = mv$, commuting e through q lowering operators gives

$$ef^q v = q(m - q + 1) f^{q-1} v.$$

For example, the coefficient is the sum $m + (m-2) + \cdots + (m - 2(q-1)) = q(m - q + 1)$. Taking the inner product and using $f^\dagger = e$ proves the recursion

$$\|f^q v\|^2 = q(m - q + 1)\|f^{q-1}v\|^2, \qquad \|f^q v\|^2 = q!\,\frac{m!}{(m-q)!}\|v\|^2 \quad (0 \le q \le m).$$

Every vector in this range is nonzero. At $q = m + 1$ the same recursion gives zero norm, so positivity gives $f^{m+1}v = 0$.

Apply this calculation to the two commuting copies with $m_L = 1$ and $m_R = 13$. Their joint lowered vectors are

$$u_{r,s} = f_L^r f_R^s u, \qquad r = 0, 1, \quad s = 0, 1, \ldots, 13.$$

Their squared norms and Cartan weights are

$$\|u_{r,s}\|_g^2 = 12\, s!\,\frac{13!}{(13-s)!},$$
$$(H_{L(0)}, H_{R(0)})u_{r,s} = (1 - 2r,\ 13 - 2s)u_{r,s}.$$

Here the first equality has no extra r factor because both nonzero vectors in a highest-weight-one representation have the same norm. The 28 pairs of eigenvalues are distinct. Self-adjointness therefore makes these vectors mutually orthogonal, and an orthonormal basis is

$$\widehat{u}_{r,s} = \sqrt{\frac{(13-s)!}{12\, s!\, 13!}}\, f_L^r f_R^s u. \tag{7.11}$$

In this basis the familiar angular-momentum coefficients appear:

$$f_L\widehat{u}_{0,s} = \widehat{u}_{1,s}, \qquad f_L\widehat{u}_{1,s} = 0,$$
$$f_R\widehat{u}_{r,s} = \sqrt{(s+1)(13-s)}\,\widehat{u}_{r,s+1}, \qquad e_R\widehat{u}_{r,s} = \sqrt{s(14-s)}\,\widehat{u}_{r,s-1}.$$

At an endpoint the corresponding coefficient is zero. The action of e_L reverses the nonzero f_L arrow. Reordering any word in the zero modes moves raising operators to the right, where they kill u, and reduces Cartan operators to their eigenvalues. Thus the cyclic zero-mode representation is exactly this 28-dimensional space, the irreducible tensor product of internal spins 1/2 and 13/2.

These lowering operations do not create Virasoro descendants. Each exponential in u is a primary of ambient conformal weight two. For a weight-one primary current J, the mode relation $[L_n, J_{(0)}] = 0$ holds for every n. Hence

$$L_0 u_{r,s} = 2u_{r,s}, \qquad L_n u_{r,s} = 0 \quad (n > 0).$$

All 28 vectors are weight-two Virasoro-primary components of one internal current multiplet. The affine levels $1, 39$, the internal highest weights $1, 13$, and the conformal weight 2 measure different things. This argument concerns current zero modes; it does not by itself assert annihilation by every positive affine mode. That stronger condition can be checked separately here. Each root $\pm\ell$ or $\pm\alpha_i$ occurring in the current triples pairs with a, b, c in $\{-1, 0, 1\}$. Their exponential OPEs therefore have no pole of order two or higher, so $J_{(n)}u = 0$ for $n \geq 1$ for each exponential current J. The positive Heisenberg modes also kill the pure exponentials. Together with $e_L u = e_R u = 0$, these identities prove that u is an affine highest vector. They use its actual lattice OPEs, beyond the two Cartan eigenvalues.

Finally, a three-dimensional space of available highest vectors is not a three-dimensional multiplet. It is a multiplicity space: choosing one highest vector selects one copy of the irreducible 28-dimensional representation. The particular line spanned by u is selected by its product identities. Cartan charges alone do not show that the PCT conjugate $\theta_g(u)$ belongs to this same cyclic copy; PCT stability remains a separate finite construction input.

7.5 What has actually been defined?

We now define the object of interest:

$$U = \langle K, \omega_B, u \rangle \subset B. \tag{7.12}$$

The angle brackets mean the smallest vertex subalgebra containing the listed states, closed under every mode. It is an actual algebra, not an abstract presentation whose Jacobi identities remain to be solved. It has the ambient stress because ω_B was explicitly included. The finite construction verifies PCT stability of the generated fields; the induced positive form then makes U a unitary VOA of CFT type.

At this point one has *not* proved $U \neq B$. This is the subtle question that motivates the next chapter. Iterated products of a few fields can eventually generate all ambient currents, even if the first several spaces look much smaller.

Construction input 7.7 (Initial generated spaces)**.** In the fixed lattice realization, closure of the original generators by the stated current and weight-two products constructs a PCT-stable space $S_2 \subset U_2$ of dimension 120. It consists of 22 affine vacuum states, two neutral stresses, and current multiplets

$$(1,5), (1,9), (1,9), (1,13), (2,2), (2,6), (2,8), (2,14).$$

The entries denote dimensions, so the multiplets contribute $5+9+9+13+4+12+16+28=96$ states. The generated grade-three space S_3 has dimension 956, and its lower normal subspace has dimension 634. These are exact finite vector-space statements. Until closure is established, they do not assert $S_i = U_i$.

There is a useful hand check of the number 22. A universal affine $\mathfrak{sl}_2$ vacuum module has three derivative states and six symmetric two-current states at weight two, hence nine. At level one the five-dimensional spin-two null multiplet is removed, leaving four. The level-39 factor still has nine at that grade. The mixed products contribute $3 \cdot 3 = 9$, giving $4+9+9=22$.

This calculation illustrates the correct division of labor. Small dimension checks explain why a finite record is plausible and teach its representation content. They do not replace the exact membership and product identities that identify the stated vectors with the algebra generated by (7.12).

7.6 The remaining construction as a chain of implications

The ambient lattice has solved the existence and associativity problem for the generated fields. The remaining claims have different inputs. The following table is a guide to the argument, not an additional assumption that all its conclusions hold.

Claim to establish	Input and role in the construction
The algebra is proper.	The finite mode identities and recursive closure theorem of Chapter 8 show that no extra currents appear at a later stage of generation. In particular, $U_1 \cong \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$ has dimension six.
All modules are controlled.	The parafermion and lattice cores of Chapter 9 give a positive conformal-vector identity. Chapter 10 uses it for C_2-cofiniteness and the splitting argument that yields rationality. The standard rationality–regularity implication then applies.
The boundary ring is the Haagerup ring.	Finite module tops, extension branching, and twist moments constrain the boundary category in Chapter 11. They determine its fusion ring and the modular-data target. Identifying its unitary tensor structure uses the later positivity step.
Every sector and fusion pairing is positive.	Chapter 12 compares actual modules and bounded field insertions with conformal-net sectors. Vacuum positivity is upgraded first to strong and then to complete unitarity. Unitary classification identifies the original boundary category as $\mathcal{H}_2$ or $\mathcal{H}_3$, and its center as their common Haagerup center.
The characters are determined.	Chapter 13 combines the resulting modular representation with the allowed leading powers and their coefficients. Uniqueness determines the vector of traces. This step counts states; it does not choose extension multiplication.
There are full local theories.	Chapters 14 and 17 apply rational sewing to the diagonal pairing and to a selected canonical $\mathcal{H}_3$ extension algebra. The selected holomorphic extension is isomorphic to V_{E_8}, with its specified boundary category.

This order also explains why replacing the defining field u by a plausible field of the same weight is not a harmless variation. The new field still generates a genuine subalgebra of the ambient lattice VOA. It need not satisfy the finite closure, core, or branching identities that drive the rest of the argument.

7.7 Exercises

Exercise 7.1. Use the cocycle formula to prove the commutator and opposite-momentum identities immediately following (7.3). Explain where evenness of the lattice is used.

Solution on page 295.

Exercise 7.2. Starting from (7.4), compute all singular terms in $Y(e^a, z)e^{-a}$ for $(a, a) = 4$. Identify the weights of their coefficients and compare with the mode weight rule.

Solution on page 295.

Exercise 7.3. For $\mathfrak{sl}_2$ at level k, the affine highest weight r has Sugawara weight $r(r{+}2)/(4(k{+}2))$. Compute the affine weight of u from its labels $(1, 13)$ at levels $(1, 39)$, and find its remaining weight under $\omega_B - \omega_K$.

Solution on page 295.

Exercise 7.4. Verify directly that $\theta_g^2 = 1$ and that the form transported by g is positive. Why would ordinary coefficient conjugation fail to give the correct adjoint of E_R in the displayed gauge?

Solution on page 296.

Exercise 7.5. Explain why an even overlattice obtained by adjoining an isotropic subgroup $A \subset L^*/L$ has determinant $\det(L)/|A|^2$. Apply the result to $E_6 \oplus A_2$.

Solution on page 296.

Chapter 8

How a finite calculation controls every conformal weight

The definition of U uses infinitely many possible iterated mode products. A calculation through weight four cannot, by itself, say what happens at weight twenty. The purpose of this chapter is to bridge that gap. We first prove an abstract recursive criterion, then explain how the particular finite spaces in the Haagerup construction satisfy its hypotheses. The central idea is that higher generators come with actual recipes in lower generators. Those recipes let Jacobi identities replace an unbounded search by a terminating induction.

The argument is useful far beyond this example. It also teaches how to read a computer-assisted proof: exact ranks, exact memberships, ambient dimensions, and recursive provenance support different logical steps. None can be substituted for another merely because the final numbers agree.

8.1 Generation and strong generation

For this chapter write $D = L_{-1}$ for translation and $:ab:= a_{(-1)}b$. We retain the mode convention $Y(a,z) = \sum_{n\in\mathbb{Z}} a_{(n)}z^{-n-1}$. The weight rule is

$$\mathrm{wt}(a_{(n)}b) = \mathrm{wt}(a) + \mathrm{wt}(b) - n - 1. \tag{8.1}$$

Products with $n \geq 0$ are the singular OPE coefficients. Products with negative n are normal products and their derivatives.

Definition 8.1. For a finite-dimensional homogeneous positive-weight space S, its *differential normal span* $\mathcal{N}(S)$ is the span of the vacuum and all expressions built from S using D and arbitrarily parenthesized normal products. The space S strongly generates a vertex algebra when this span is the whole algebra.

Allowing arbitrary parentheses avoids an unnecessary preliminary rewriting theorem. Normal multiplication in a VOA is neither literally commutative nor literally associative. Its defects are controlled by singular products and derivatives. If a preferred ordering is desired for a computer representation, its validity must be justified by those identities.

Vertex generation permits all modes at every step. Strong generation uses only normal products and derivatives after the strong generators have been named. The distinction is visible immediately in weight two. If a, b both have weight two, then $a_{(0)}b$ has weight three. It need not be a normal expression in a, b: the smallest normal product of them has weight four. Such a weight-three field may need to be added as a strong generator even though it is already a vertex product of the original generators.

Worked example 8.2 (A one-current test)**.** A Heisenberg current J satisfies $J_{(1)}J = \mathbf{1}$ and $J_{(0)}J = 0$, with all higher products zero. These singular products already lie in $\mathcal{N}(J)$. Wick reduction then shows that every singular product of two normal expressions in J lies there as well. Thus one current strongly generates its Heisenberg VOA. The point is the verified OPE closure, not merely the observation that the first few Fock-space dimensions agree with partition numbers.

8.2 The two identities that drive the proof

Translation obeys sesquilinearity, and zero modes are derivations:

$$[D, a_{(n)}] = -na_{(n-1)}, \qquad a_{(0)} : xy :=: (a_{(0)}x)y : + : x(a_{(0)}y) : .$$

The second identity is especially valuable because it has no correction term. Also $[D, a_{(0)}] = 0$.

For arbitrary modes the efficient language is the λ-bracket,

$$[a_\lambda b] = \sum_{n\geq 0} \frac{\lambda^n}{n!} a_{(n)}b.$$

The noncommutative Wick identity is

$$[a_\lambda : bc :] =: [a_\lambda b]c : + : b[a_\lambda c] : + \int_0^\lambda [[a_\lambda b]_\mu c]\, d\mu. \tag{8.2}$$

The integral is a formal polynomial integral, not an analytic operation. It accounts for the double contractions that a naive Leibniz rule would miss.

Lemma 8.3 (Filtered Wick reduction)**.** *Suppose singular products of generator pairs of total weight less than w belong to $\mathcal{N}(S)$. The same holds for normal expressions of total input weight less than w. At weight w, it holds whenever one input is a derivative or a normal product of positive-weight expressions.*

Proof. Induct on total input weight. Sesquilinearity removes a derivative and reduces that weight. Skew symmetry moves a composite to the right. Apply (8.2). In each of the first two terms one factor has been omitted from the bracket, so its total input weight is smaller. For the integral term, the coefficient $a_{(j)}b$ has weight $\mathrm{wt}(a) + \mathrm{wt}(b) - j - 1$. Pairing it with c gives total weight at most $\mathrm{wt}(a) + \mathrm{wt}(b) + \mathrm{wt}(c) - 1$, again smaller. If the coefficient is scalar its bracket vanishes. Normal multiplication and derivatives preserve the span by definition. This proves the induction without assuming that normal words are linearly independent. □

Notice what this lemma does not do. It reduces products containing composites, but it does not settle a new pair of primitive strong generators. We need one more source of structure for those pairs.

8.3 A recursive strong-generation theorem

Suppose we have finite spaces $S_d \subset V_d$ for $1 \leq d \leq h$, with $h \geq 2$. Let $K = S_1$, $A = S_2$, and $\mathcal{N} = \mathcal{N}(S_1 + \cdots + S_h)$. A subscript d on $\mathcal{N}$ denotes its homogeneous part of conformal weight d. Here K denotes the finite current space; in the Haagerup application it is the weight-one part of the affine algebra called K in the preceding chapter. A higher field has *recursive provenance* if, modulo lower normal expressions, it is a zero product of a weight-two field with a field one weight lower.

Theorem 8.4 (Finite recursive closure)**.** *Assume V is nonnegatively integer graded with $V_0 = \mathbb{C}\mathbf{1}$, and that the following identities hold in V:*

1. *For $k \in K$, $a \in A$, $s \in S_d$, and every $n \geq 0$, both $k_{(n)}s$ and $a_{(n)}s$ belong to $\mathcal{N}$.*
2. *For $3 \leq d \leq h$,*
$$S_d \subset A_{(0)}S_{d-1} + \mathcal{N}_d(S_1 + \cdots + S_{d-1}).$$

Then $\mathcal{N}$ is a vertex subalgebra. Hence the S_d strongly generate the vertex algebra that they generate.

Proof. First establish $a_{(0)}\mathcal{N} \subset \mathcal{N}$ for $a \in A$. This follows from the first hypothesis on generators, the zero-mode derivation identity, and commutation with D. It uses no unproved closure of other singular products.

Now order generator pairs (s, t) lexicographically by

$$(\mathrm{wt}(s) + \mathrm{wt}(t), \mathrm{wt}(s)).$$

We prove singular closure in that order. If the left weight is one or two, it is a hypothesis. Otherwise replace s by terms $a_{(0)}b$ and lower normal expressions, where $\mathrm{wt}(b) = \mathrm{wt}(s) - 1$. The normal terms are controlled by Lemma 8.3. Jacobi gives

$$(a_{(0)}b)_{(n)}t = a_{(0)}(b_{(n)}t) - b_{(n)}(a_{(0)}t). \tag{8.3}$$

In the first term the inner pair has lower total weight, and the outer $a_{(0)}$ preserves $\mathcal{N}$.

The first hypothesis gives a normal expression for $a_{(0)}t$. Every composite or derivative in that expression reduces the second term by filtered Wick. The only other possibility is a single undifferentiated generator of weight $\mathrm{wt}(t) + 1$. Pairing that generator with b has the original total weight but smaller left weight, so it is earlier in the lexicographic order. If $\mathrm{wt}(t) = h$ there is no such generator. The induction therefore terminates.

All generator singular products now close. Filtered Wick extends this to all normal expressions. For negative modes use

$$x_{(-r)}y = \frac{1}{(r-1)!} : (D^{r-1}x)y :, \qquad r \geq 1.$$

Thus every mode preserves $\mathcal{N}$, proving the assertion. □

The strength of the theorem is its finite range. The largest output weight in the first hypothesis is $h + 1$, arising from a zero product of weights two and h. Directly checking every pair of strong generators would reach weight $2h - 1$. More importantly, the theorem explains why those finitely many checks suffice in all weights.

Remark 8.5 (Why provenance is indispensable). Suppose an arbitrary extra vector were added at weight four because it made a character coefficient look correct. Its brackets with another weight-four vector would not be controlled by the theorem. The recipe $S_4 \subset A_{(0)}S_3+$ lower normals is what makes (8.3) available. A dimension is not a recipe.

8.4 The finite problem in the Haagerup algebra

Return to the actual $U \subset B$. Let K be its six known currents, and let S_2, S_3 be the constructed spaces of dimensions $120, 956$. Their lower normal space is

$$N_3 = K_{(-1)}S_2 + DS_2, \qquad \dim N_3 = 634,$$

and the exact provenance statement is

$$S_3 = N_3 + \operatorname{span}\{a_{(0)}b : a, b \in S_2\}.$$

Choose actual lifts of the quotient S_3/N_3 and an actual current triplet P_4 of highest weight $(0, 2)$. Here highest weights, rather than representation dimensions, are being used: $(0, 2)$ means the representation of dimensions $(1, 3)$.

Set

$$N_4 = K_{(-1)}S_3 + DS_3 + (S_2)_{(-1)}S_2, \qquad W_4 = N_4 + P_4, \tag{8.4}$$

$$N_5 = K_{(-1)}W_4 + DW_4 + (S_2)_{(-1)}S_3. \tag{8.5}$$

These are the full displayed spans; their exhaustion of the relevant normal expressions is justified below. A solver may use selected columns that lie in them, but those selected columns must not silently replace the full spaces in an algebraic argument.

Construction input 8.6 (Lower closure and provenance)**.** The actual spaces are stable under both current zero-mode algebras. All singular products of currents and weight-two fields with S_2 belong to the expected spaces in weights at most three. The triplet is constructed from an actual $a_{(0)}p$, with $a \in S_2$ and $p \in S_3$, by current projectors and lowering. It satisfies

$$P_4 \subset (S_2)_{(0)}S_3 + N_4, \qquad L_1P_4 \subset S_3, \quad L_2P_4 \subset S_2.$$

The stated inclusions are exact vector identities in the ambient lattice algebra and are available before the next two gates.

The two remaining central memberships are

$$(S_2)_{(0)}S_3 \subset W_4, \tag{8.6}$$

$$(S_2)_{(0)}P_4 \subset N_5. \tag{8.7}$$

We call them the weight-four and weight-five gates. The name refers to the output weight, not an empirical test that can be replaced by sampling a few products.

8.5 Raising operators turn seeds into a complete proof

In this representation-theoretic section, E_L, E_R, F_L, F_R abbreviate the zero-mode operators of the corresponding current states. Let $P = S_3/N_3$, of dimension 322. For a fixed highest field $a \in S_2$, define

$$f_a : P \longrightarrow B_4/W_4, \qquad [p] \longmapsto [a_{(0)}p].$$

Lower closure makes the map well defined. Because a is highest, its kernel is stable under the two commuting raising operators E_L, E_R.

Lemma 8.7 (Finite raising saturation)**.** *Suppose $C \subset \ker f_a$ and $P = E_LP + E_RP + C$. Then $f_a = 0$.*

Proof. On $P/\ker f_a$ the equation becomes $M = E_LM + E_RM$. The ideal generated by the two raising operators is nilpotent because the finite-dimensional representation has bounded weights. Iterating gives $M = (E_L, E_R)^rM = 0$ for sufficiently large r. □

Worked example 8.8 (One lowest vector per irreducible)**.** In an irreducible $\mathfrak{sl}_2$ representation of dimension m, the raising image has codimension one. Its missing direction is the lowest-weight vector. In a product of two irreducibles, $E_LP + E_RP$ again has codimension one: only the joint-lowest vector is missing. Therefore a direct sum of r irreducible constituents, counted with multiplicity, needs r such seeds. This explains the size of the Haagerup gate before any large matrix is written.

The exact current decomposition of P, in representation dimensions, is

$$\begin{aligned} P = {} & 3(1,3) + (1,5) + 5(1,7) + 2(1,9) + 3(1,11) + (1,13) \\ & + 2(1,15) + (1,19) + (2,2) + (2,4) + 2(2,6) + 2(2,8) \\ & + 2(2,10) + (2,12) + (2,14). \end{aligned}$$

There are 28 constituents and total dimension 322. Hence the raising image has dimension 294. The nine primitive tests are the eight listed nontrivial weight-two multiplets and one independent neutral stress; affine normal states and the ambient stress are already controlled. For each of these highest fields in the first input, exact memberships on a spanning set of the 28 missing directions prove the complete map vanishes. This is the conceptual content of the weight-four computation.

Some residual outputs have weight $(0,0)$ and require a little care. A vector being a highest class modulo W_4 does not mean its chosen ambient representative is literally a singlet. The following projection argument recovers the full membership.

Lemma 8.9 (Recovering a residual singlet)**.** *Let $v \in B_4$ have current weight $(0,0)$ and satisfy $E_L v, E_R v \in W_4$. Let Π be the polynomial projector onto joint singlets. If Πv lies in N_4, then $v \in W_4$.*

Proof. Each nonconstant factor in the projector contains FE, so acts as zero on the class of v modulo W_4. Thus $[\Pi v] = [v]$. The hypothesis makes the left side zero. □

There is no mysterious infinite representation theory in this projector. At ambient weight d, a momentum has norm at most $\sqrt{2d}$, so a Cartan eigenvalue obeys $m^2 \le 2d(h,h)$. At weight four the possible highest weights are at most 4 and 24 for the two current factors. On weight zero, FE acts in highest weight j by $j(j+2)/4$. Therefore

$$\Pi = \prod_{j=2,4} \left(1 - \frac{4F_L E_L}{j(j+2)}\right) \prod_{j=2,4,\dots,24} \left(1 - \frac{4F_R E_R}{j(j+2)}\right)$$

is an exact finite polynomial. Exact projected residuals complete the exceptional seed memberships.

8.6 Why the weight-five gate is a Jacobi problem

After the weight-four gate, Virasoro commutators recover the positive products that will be needed next. This step should be proved before any use of the weight-five gate, since its normal-space argument depends on precisely these products.

Lemma 8.10 (The intermediate positive-mode controls)**.** *The lower products, the direct $L_1 P_4, L_2 P_4$ controls, and the weight-four gate imply*

$$\begin{aligned} (S_2)_{(1)} S_3 &\subset S_3, & (S_2)_{(2)} S_3 &\subset S_2, & (S_2)_{(3)} S_3 &\subset K, \\ K_{(1)} W_4 &\subset S_3, & K_{(2)} W_4 &\subset S_2, & K_{(3)} W_4 &\subset K, \\ L_1 N_5 &\subset W_4, & L_2 N_5 &\subset S_3. \end{aligned}$$

The next mode in each of the first two lines is scalar; further modes vanish by weight.

Proof. Current products with S_3 follow first from its provenance. For example, if $k \in K$ and $a, b \in S_2$, then

$$k_{(1)}(a_{(0)}b) = (k_{(0)}a)_{(1)}b + (k_{(1)}a)_{(0)}b + a_{(0)}(k_{(1)}b).$$

Every right-hand term belongs to S_2 by lower closure. Higher current modes and normal inputs are handled by the same commutator and iterate identities.

For a general weight-two vector, the needed Virasoro commutators are

$$[L_1, a_{(n)}] = (2-n)a_{(n+1)} + (L_1 a)_{(n)},$$
$$[L_2, a_{(n)}] = (3-n)a_{(n+2)} + 3(L_1 a)_{(n+1)} + (L_2 a)_{(n)}.$$

These give $L_1 S_3 \subset S_2$, $L_2 S_3 \subset K$ and, on the three defining summands of N_4, $L_1 N_4 \subset S_3$, $L_2 N_4 \subset S_2$. For $L_2(a_{(-1)}b)$ the term $(L_2 a)_{(-1)}b$ must be retained: $L_2 a$ is scalar, but its (-1)-mode acts by that scalar. The direct controls on P_4 extend these inclusions to W_4 before the weight-five gate.

The primitive weight-two multiplets and the two neutral singlets are quasiprimary. Indeed L_1 is current-equivariant, and none of their current types occurs in K, whose types are $(3,1)$ and $(1,3)$. Thus $L_1 a = 0$ for these a. For $p \in S_3$, the weight-four gate now gives

$$2a_{(1)}p = L_1(a_{(0)}p) - a_{(0)}L_1 p \in S_3,$$
$$a_{(2)}p = L_1(a_{(1)}p) - a_{(1)}L_1 p \in S_2,$$
$$2a_{(3)}p = L_2(a_{(1)}p) - a_{(1)}L_2 p \in K.$$

In the last line $(L_2 a)_{(1)}p = 0$, because the positive modes of the vacuum are zero. This differs from the normal product term just discussed. The current-normal and derivative parts of S_2 use the general commutators and current iterates, so the first line of the lemma holds on all of S_2.

Next use $P_4 \subset (S_2)_{(0)}S_3 + N_4$ in the displayed current identity, now allowing $b \in S_3$. It gives $K_{(1)}P_4 \subset S_3$. Since a current is primary, $[L_1, k_{(1)}] = -k_{(2)}$ and $[L_2, k_{(1)}] = -k_{(3)}$. The established Virasoro controls therefore give the remaining current inclusions on W_4.

Finally apply L_1, L_2 to the three summands of N_5. The current and derivative summands use the controls already proved. For the normal input of weights two and three,

$$L_1(a_{(-1)}p) = 3a_{(0)}p + (L_1 a)_{(-1)}p + a_{(-1)}L_1 p,$$
$$L_2(a_{(-1)}p) = 4a_{(1)}p + 3(L_1 a)_{(0)}p + (L_2 a)_{(-1)}p + a_{(-1)}L_2 p.$$

The respective outputs lie in W_4 and S_3. Reordering a normal term uses only lower products that have already been established. This completes the proof. □

The point of spelling out this intermediate lemma is to make the logical order visible. We now possess the positive products needed to analyze all of the normal subspace N_5; no calculation at weight five has been used to establish its own hypotheses.

At this stage every normal word of weight five reduces to N_5. A leading derivative reduces to W_4; a normal product splits into weights $1+4$ or $2+3$, after reversing the factors if necessary. The reordering corrections are derivatives of $K_{(n)}W_4$ or $(S_2)_{(n)}S_3$, which have just been controlled. Higher negative modes reduce by translation:

$$x_{(-r)}v = \frac{D(x_{(-(r-1))}v) - x_{(-(r-1))}Dv}{r-1}, \qquad r \geq 2.$$

Thus no unproved S_3-by-S_3 singular closure is hidden in the following normal-space containments.

One obtains, before the final gate,

$$(S_2)_{(0)}N_4 \subset N_5, \qquad (N_3)_{(0)}S_3 \subset N_5.$$

The second inclusion follows from the iterate identity

$$(:xa:)_{(0)}p = \sum_{i\geq 0} x_{(-1-i)}(a_{(i)}p) + \sum_{i\geq 0} a_{(-1-i)}(x_{(i)}p),$$

for a current x. Every inner mode is already controlled, and grading makes the sums finite.

If $P_4 \subset N_4$, nothing remains. Otherwise $Q = W_4/N_4$ is the current triplet. Define

$$\beta(a,p) = [a_{(0)}p] \in Q, \qquad F(a,q) = [a_{(0)}\widetilde{q}] \in B_5/N_5.$$

The normal-space controls make F independent of the lift.

Lemma 8.11 (Jacobi exchanges the difficult input)**.** *If $a_{(0)}b \in N_3$, then*

$$F(a,\beta(b,p)) = F(b,\beta(a,p)). \tag{8.8}$$

Proof. The difference is the class of $(a_{(0)}b)_{(0)}p$, which lies in N_5 by the preceding inclusion. □

Here *charged* means a nontrivial representation of the retained current zero modes, while *neutral* means a current singlet. Write q_- for a lowest-weight vector of the triplet Q. The finite construction supplies two nonzero bottom witnesses $\beta(b,p) = cq_-$, $c \neq 0$. For each charged highest field a, one chooses such a witness for which $a_{(0)}b \in N_3$ and the weight of $a_{(0)}p$ is outside the three weights of Q. Then $\beta(a,p) = 0$, so (8.8) gives $F(a,q_-) = 0$. Raising this *proved zero identity* gives the other two triplet inputs; current covariance supplies the descendants of a.

The emphasized order matters. The weight exclusion holds at the chosen bottom input. It need not hold after that input is raised. What is raised is the zero identity already established by Jacobi, not an unverified assertion that all later intermediate weights are absent.

The remaining neutral family is reduced to charged brackets. Their current span exhausts P by the same nilpotent-ideal argument, now using lowering. Here is one explicit reduction. Let e be the retained neutral singlet and let $a_{\mathrm{ch}}, p_{\mathrm{bot}}$ be a charged bottom witness with $\beta(a_{\mathrm{ch}}, p_{\mathrm{bot}}) = cq_-$, $c \neq 0$. Charged spanning gives

$$e_{(0)}a_{\mathrm{ch}} = \sum_i t_i (x_i)_{(0)} y_i + n, \qquad n \in N_3,$$

with charged $x_i, y_i \in S_2$. Applying zero modes to p_{bot} gives

$$(e_{(0)}a_{\mathrm{ch}})_{(0)}p_{\mathrm{bot}} = \sum_i t_i [(x_i)_{(0)}, (y_i)_{(0)}]p_{\mathrm{bot}} + n_{(0)}p_{\mathrm{bot}} \in N_5.$$

The inner charged products belong to W_4 by the weight-four gate. The next charged products are controlled on both N_4 and P_4 by the normal-space results and the charged portion of the weight-five argument. A second Jacobi identity yields

$$e_{(0)}((a_{\mathrm{ch}})_{(0)}p_{\mathrm{bot}}) = (a_{\mathrm{ch}})_{(0)}(e_{(0)}p_{\mathrm{bot}}) + (e_{(0)}a_{\mathrm{ch}})_{(0)}p_{\mathrm{bot}} \in N_5.$$

The nonzero coefficient c proves the neutral bottom identity. The neutral zero mode commutes with current zero modes, so covariance gives the whole triplet. The other neutral stress follows by subtracting the retained stress and the known Sugawara stresses from the ambient stress.

Construction input 8.12 (The two completed gates)**.** For the actual finite spaces, the raising-seed memberships and their singlet residuals establish (8.6). The nonzero bottom witnesses, lower bracket memberships, and charged spanning identities described above establish (8.7). All equalities are over $\mathbb{Q}(\sqrt{13}, \sqrt{-3})$ in the fixed lattice basis. The gates are finite construction inputs; the recursive theorem is their all-conformal-weight consequence.

The construction manuscript's certificate ledger lists the completed lower-closure, actual-triplet, weight-four, charged-span, and weight-five records. The source guide gives the manuscript location; the finite identities remain inputs here, rather than calculations independently replayed by this textbook.

8.7 Properness is now an all-weight conclusion

We can now close the last positive modes explicitly. For a quasiprimary primitive $a \in S_2$ and $p \in P_4$, the weight-five gate and Lemma 8.10 give

$$\begin{aligned} 2a_{(1)}p &= L_1(a_{(0)}p) - a_{(0)}L_1p \in W_4, \\ a_{(2)}p &= L_1(a_{(1)}p) - a_{(1)}L_1p \in S_3, \\ 2a_{(3)}p &= L_2(a_{(1)}p) - a_{(1)}L_2p \in S_2, \\ a_{(4)}p &= L_2(a_{(2)}p) - a_{(2)}L_2p \in K. \end{aligned}$$

The next mode is scalar and further ones vanish. The current-normal and derivative portions again use the general identities in the bootstrap proof. Thus the zero-mode gates really do supply the all-singular-mode hypothesis of the recursive theorem.

Apply Theorem 8.4 with strong-generator spaces K, S_2, S_3, P_4. The gates, positive-mode bootstrap, and recursive recipes establish its hypotheses. Therefore the normal span is a vertex subalgebra. It contains the original generators of U, and every one of its generators is an actual U-vector, so it equals U.

A normal expression involving only positive-weight generators can have weight one only if it is one of the weight-one generators themselves. Thus

$$U_1 = K = \mathfrak{sl}_2 \oplus \mathfrak{sl}_2, \qquad \dim U_1 = 6 < 86 = \dim B_1. \tag{8.9}$$

In particular U is proper. The same normal-span argument recovers $U_2 = S_2$ and $U_3 = S_3$ from the lower identities, giving

$$\mathrm{ch}_U(\tau) = q^{-1/3}(1 + 6q + 120q^2 + 956q^3 + O(q^4)).$$

The statement is stronger than agreement of a finite character prefix: it rules out extra currents generated at any later stage.

8.8 What a finite-field calculation proves

Exact arithmetic over a small number field can still produce large rational expressions. Reduction modulo a suitable prime is often an efficient way to certify a nonzero determinant. The elementary principle is worth separating from more delicate uses.

Proposition 8.13 (A nonzero minor survives in characteristic zero)**.** *Let a matrix have entries in an integral domain of characteristic zero, and let a specialization to a finite field be defined on its entries and relevant denominators. If an $r \times r$ minor has nonzero image, that minor was nonzero before specialization. Hence the original rank is at least r.*

Proof. The determinant is a polynomial in the entries, so specialization commutes with it. A zero determinant could not have nonzero image. □

This proves a lower rank, not an upper rank. If the ambient space has independently known dimension 322, a rank-322 reduced matrix proves spanning in characteristic zero. Without that dimension bound, the same computation would not prove exhaustion. Similarly, a residual that is zero modulo one prime need not vanish in characteristic zero. Membership identities in this construction are checked as complete number-field equalities; finite fields select witnesses or certify nonzero minors. Example C.1 works through a two-by-two matrix whose columns become dependent at one admissible prime and stay independent at another.

For a reducible coefficient algebra, one must also distinguish one embedding from all algebraic conjugate components. Later irrational-axis arguments use units or all split fibers to control every component. This chapter's basic lesson is already sufficient: state the direction of every rank implication before interpreting a numerical-looking table as a theorem.

8.9 Exercises

Exercise 8.1. Prove the negative-mode identity in the final paragraph of the proof of Theorem 8.4. Why does it complete vertex closure once singular products and normal products are controlled?

Solution on page 296.

Exercise 8.2. In a tensor product of irreducible $\mathfrak{sl}_2$ representations of dimensions m, n, prove that the quotient by $E_L P + E_R P$ is one-dimensional. Give an explicit representative.

Solution on page 296.

Exercise 8.3. Check the constituent count and dimension of the displayed decomposition of P. Recover $\dim(E_L P + E_R P) = 294$ without a matrix calculation.

Solution on page 296.

Exercise 8.4. Construct an integer matrix whose rank drops modulo p, and explain why a reduced rank gives only a lower bound on the characteristic-zero rank. Construct an integer residual that vanishes modulo p but is not zero.

Solution on page 297.

Exercise 8.5. For a weight-two quasiprimary a, derive the L_1 commutator used in the positive-mode bootstrap. Then derive $2a_{(1)}p = L_1(a_{(0)}p) - a_{(0)}L_1 p$.

Solution on page 297.

Exercise 8.6. Explain, in the proof of recursive closure, why induction on total weight alone is insufficient and why the second lexicographic entry repairs it. Identify the unique type of term for which that second entry is needed.

Solution on page 297.

Chapter 9

Potts fields, parafermions, and positive stresses

The six currents explain only part of the energy in U. To control the rest we need familiar rational algebras embedded in the actual lattice realization. This chapter finds a three-state Potts factor, a level-thirteen parafermion factor, and two useful finite extensions. Their most important output is a positive linear expression for the ambient conformal vector. The next chapter will turn that expression into rationality. Here we work out the small calculations that make the expression tangible.

9.1 A parafermion is an affine algebra with its Cartan removed

The affine algebra $A_{1,k}$ contains a rank-one Heisenberg algebra generated by its Cartan current. Its commutant is the parafermion VOA $K(\mathfrak{sl}_2, k)$. Subtracting the Heisenberg stress gives

$$c_{\mathrm{PF}_k} = \frac{3k}{k+2} - 1 = \frac{2(k-1)}{k+2}. \tag{9.1}$$

The word "parafermion" does not mean that this vacuum algebra contains fractional-weight local generators. Its modules have fractional weights; the vacuum VOA is integrally graded.

Write $L(c,h)$ for the simple highest-weight Virasoro module of central charge c and lowest weight h. For $k=3$ the charge is $4/5$. The algebra is the chiral three-state Potts theory,

$$P = L(4/5, 0) \oplus L(4/5, 3).$$

Its extra weight-three field is essential. Finding only a Virasoro vector of charge $4/5$ does not establish that the Potts extension belongs to the generated algebra.

For $k=13$ the charge is $8/5$. It has a convenient presentation with generators of weights $2, 3, 4, 5$. The weight-four and weight-five fields can be obtained recursively from the weight-three field, but their singular products and decoupling relations are still needed to recognize the algebra. A matching central charge and a single cubic coupling would be insufficient.

9.1.1 Affine fusion and the six Potts modules

Recall that the integrable $SU(2)_k$ modules have highest-weight labels $\lambda = 0, 1, \dots, k$, internal top spin $\lambda/2$, and conformal weight $h_k(\lambda) = \lambda(\lambda+2)/(4(k+2))$. Their fusion is the level-truncated angular-momentum rule

$$[\lambda] \otimes [\mu] = \bigoplus_{\substack{\nu = |\lambda-\mu| \\ \nu \equiv \lambda+\mu \ (2)}}^{\min(\lambda+\mu,\, 2k-\lambda-\mu)} [\nu]. \tag{9.2}$$

The summation proceeds in steps of two. This is fusion of whole affine modules, not the ordinary tensor product of their finite current-algebra tops. For instance, at level three, $[2]\otimes[2]=[0]\oplus[2]$: the spin-two top channel that would occur in an ordinary spin-one tensor product is removed. The two labels $[0],[2]$ already have the Fibonacci fusion rule.

For PF_k, keep both the affine label and a Cartan charge: write its modules as $\mathsf{P}_{\lambda,m}$, where

$$\begin{gathered}0\le\lambda\le k,\qquad m\in\mathbb{Z}/(2k),\qquad m\equiv\lambda\pmod 2,\\(\lambda,m)\sim(k-\lambda,m+k).\end{gathered}\tag{9.3}$$

The parity condition follows because current modes change Cartan charge by even integers. Charge periodicity and the last field identification are part of the standard affine-to-parafermion branching theorem. Equivalently, decompose the affine module under the Cartan lattice with generator $\gamma^2=2k$:

$$L_k(\lambda)=\bigoplus_{\substack{m \bmod 2k\\ m\equiv\lambda\ (2)}} V_{\mathbb{Z}\gamma+m\gamma/(2k)}\otimes\mathsf{P}_{\lambda,m}.$$

The charge-m Heisenberg ground energy is $m^2/(4k)$. Consequently, if its first Cartan-highest vector occurs at affine grade N, its parafermion energy is

$$h_{\mathrm{PF}}=\frac{\lambda(\lambda+2)}{4(k+2)}+N-\frac{m^2}{4k}.\tag{9.4}$$

Dropping N gives a twist modulo one, not always a lowest weight. Fusion uses the range in (9.2) and adds the charge labels:

$$\mathsf{P}_{\lambda,m}\otimes\mathsf{P}_{\mu,n}=\bigoplus_{\nu}\mathsf{P}_{\nu,m+n}.$$

Their quantum dimensions depend only on the affine label:

$$d_{\lambda,m}=\frac{\sin((\lambda+1)\pi/(k+2))}{\sin(\pi/(k+2))}.$$

Worked example 9.1 (All six Potts sectors and their actual top weights)**.** At $k=3$, the identification in (9.3) lets us use the even affine labels $0,2$ and charges $0,2,-2$. Put $\psi=\mathsf{P}_{0,2}$ and $\tau=\mathsf{P}_{2,0}$. The complete table is

Module	(λ,m)	first grade N	lowest weight	dimension
$\mathbf{1}$	$(0,0)$	0	0	1
ψ	$(0,2)$	1	2/3	1
ψ^2	$(0,-2)$	1	2/3	1
τ	$(2,0)$	0	2/5	φ
$\tau\psi$	$(2,2)$	0	1/15	φ
$\tau\psi^2$	$(2,-2)$	0	1/15	φ

Here $\varphi=(1+\sqrt{5})/2$. For the charged vacuum branches, $E_{(-1)}\mathbf{1}$ and $F_{(-1)}\mathbf{1}$ are the first states of charges ±2; they have affine grade one and Heisenberg energy $4/12=1/3$. Their parafermion weight is therefore $2/3$. For affine label two, all three charges occur already in the spin-one top, at affine energy $2/5$. Subtraction gives $2/5$ at charge zero and $2/5-1/3=1/15$ at charges ±2.

The fusion formula gives $\psi^3=\mathbf{1}$, $\tau^2=\mathbf{1}\oplus\tau$, and the other products by multiplying these relations. The neutral Fibonacci subcategory and the pointed subcategory generated by ψ centralize one another: for example their double-braiding scalar is

$$\frac{\theta_{\tau\psi}}{\theta_\tau\theta_\psi}=e^{2\pi i(1/15-2/5-2/3)}=1.$$

The same holds for ψ^2. These two braided subcategories have trivial intersection and their products exhaust the six simples, giving

$$\mathrm{Rep}(P) \simeq \mathrm{Fib} \boxtimes \mathcal{C}(\mathbb{Z}/3, -r^2/3).$$

Here $\boxtimes$ between categories denotes their external tensor product: its simple objects are pairs $X \boxtimes Y$, and fusion and braiding act separately in the two factors. Dimensions and twists therefore multiply. This differs from $M \boxtimes_V N$, which fuses two modules inside the single category $\mathrm{Rep}(V)$. For a finite abelian group G, the notation $\mathcal{C}(G, q)$ denotes the pointed braided category with simples $g \in G$, fusion given by group addition, and twist $e^{2\pi i q(g)}$. The displayed quadratic form means that ψ^r has twist $e^{-2\pi i r^2/3}$; the Fibonacci object has twist $e^{4\pi i/5}$.

This table is the origin of the functions p_1 and p_τ used in Chapter 11: $p_\epsilon(m)$ is the lowest weight in the sector $\epsilon\psi^r$ with $r = m \bmod 3$. Thus a charge divisible by three gives 0 or $2/5$, while either nonzero residue gives $2/3$ or $1/15$. This charge-orbit label is r; the Cartan label of the parafermion notation above is $2r$.

9.1.2 What the seven even level-thirteen labels mean

For odd level k, the even affine labels form a fusion subcategory, denoted $SO(3)_k$ here. At $k = 13$ they are $2a$, $0 \le a \le 6$. Equation (9.2) becomes

$$a \otimes b = \bigoplus_{c=|a-b|}^{\min(a+b,\,13-a-b)} c, \qquad h_a = \frac{a(a+1)}{15}.$$

For example, $6 \otimes 6 = 0 \oplus 1$. This is why the symbol $SO(3)_{13}$ later contributes seven simple labels to the core, and hence fourteen after combining it with Fibonacci.

The oddness of thirteen also lets us represent every parafermion class uniquely by $(\lambda, m) = (2a, 2r)$ with $r \in \mathbb{Z}/13$. For the representative $-6 \le r \le 6$, its first affine grade is $\max(0, |r| - a)$. Charges with $|r| \le a$ are already in the finite spin-a top. At affine grade N, current descendants have finite spin at most $a + N$: each negative current mode has spin one and costs at least one grade. Hence $|r| \le a + N$; zero modes cannot evade this bound because they stay inside those finite spin representations. Applying $E_{(-1)}$ to the highest top, or $F_{(-1)}$ to the lowest top, attains the bound in this range; affine integrability allows these powers because $|r| - a \le 6 - a \le 13 - 2a$. The resulting vectors are killed by positive Cartan modes, as their commutators reduce to raising or lowering modes that kill the appropriate extreme top. Therefore

$$h_{\mathrm{PF}}(a, r) = \frac{a(a+1)}{15} - \frac{r^2}{13} + \max(0, |r| - a).$$

In particular the simple-current charge $a = 0, r = 1$ has actual weight $1 - 1/13 = 12/13$, not $-1/13$. This same integer-grade correction will matter when the fourteen core tops are minimized. The thirteen invertible labels $(0, 2r)$ have quadratic form $-r^2/13$ modulo one. These are the pointed parafermion charges used in the later order-thirteen extension.

9.2 The Potts stress in a scaled A_2 plane

With the three momenta a, b, c of (7.9), put $\beta_1 = b - a$, $\beta_2 = c - b$, and $\beta_3 = \beta_1 + \beta_2$. The Gram matrix of β_1, β_2 is

$$\begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}.$$

Thus they span $\sqrt{2}A_2$. To keep scalar notation separate from the Haagerup dimension, write $r = \sqrt{13}$ and $t = \sqrt{-3}$ in the next formulas. Define

$$p = \frac{\beta_1(-1)^2 + \beta_2(-1)^2 + \beta_1(-1)\beta_2(-1)}{15}\mathbf{1} + \sum_{\delta=\pm\beta_1,\pm\beta_2,\pm\beta_3} p_\delta e^\delta, \tag{9.5}$$

where

δ	p_δ	$p_{-\delta}$
β_1	$-(r+3t)/100$	$(-r+3t)/10$
β_2	$(7+3rt)/100$	$(7-3rt)/100$
β_3	$(-r+3t)/100$	$-(r+3t)/10$

The unequal opposite coefficients reflect the transported torus gauge. They do not indicate failure of positivity.

Worked example 9.2 (Computing the Potts central term by hand)**.** The inverse of the displayed Gram matrix shows that its plane stress is

$$\omega_{\text{plane}} = \frac{1}{6}(\beta_1(-1)^2 + \beta_2(-1)^2 + \beta_1(-1)\beta_2(-1))\mathbf{1}.$$

The Heisenberg part of p is therefore $(2/5)\omega_{\text{plane}}$. Since the plane has charge two, this part contributes $4/25$ to $p_{(3)}p$. Opposite momenta of squared norm four have cocycle sign $+1$. Moreover

$$p_{\beta_1}p_{-\beta_1} = \frac{r^2 - 9t^2}{1000} = \frac{1}{25}, \qquad p_{\beta_2}p_{-\beta_2} = \frac{49 - 9(rt)^2}{10000} = \frac{1}{25},$$

and the third pair gives the same result. Both orders of each pair occur, giving $6/25$. Thus

$$p_{(3)}p = \left(\frac{4}{25} + \frac{6}{25}\right)\mathbf{1} = \frac{2}{5}\mathbf{1},$$

which is the central term $c/2$ for $c = 4/5$.

There is an equally short reality check. The torus PCT factors for $\beta_1, \beta_2, \beta_3$ are $10, 1, 10$. Hence, for instance, $10\overline{p_{\beta_1}} = p_{-\beta_1}$, and the other pairs work in the same way. The Heisenberg quadratic is fixed by PCT.

To obtain the full Potts algebra set

$$A = \beta_1 - \beta_2, \quad C = \beta_1 + 2\beta_2, \quad E = -2\beta_1 - \beta_2, \quad X_\delta = 5p_\delta e^\delta,$$

and use the weight-three field

$$\begin{aligned} w_P = {} & :h_A h_C h_E: -9(:h_A(X_{\beta_3} + X_{-\beta_3}): \\ & + :h_C(X_{\beta_1} + X_{-\beta_1}): + :h_E(X_{\beta_2} + X_{-\beta_2}):), \end{aligned} \tag{9.6}$$

where $h_A = A(-1)\mathbf{1}$, and similarly for the other Heisenberg currents.

Construction input 9.3 (The actual Potts embedding)**.** The vectors in (9.5) and (9.6) belong to U, are stable under its PCT operation, and satisfy

$$p_{(1)}p = 2p, \quad p_{(2)}p = 0, \quad p_{(0)}p = Dp, \qquad p_{(1)}w_P = 3w_P, \quad p_{(n)}w_P = 0 \ (n \geq 2).$$

Together with the displayed standard scaled-A_2 realization, these fields generate the regular unitary Potts algebra P. Their intrinsic translation agrees with ambient translation on this subalgebra.

The hand checks of central term and PCT explain two parts of this input. They do not check every OPE coefficient or prove membership of w_P in the original generated algebra. Membership means an actual linear identity in mode words of the original generators, not merely that w_P occurs in the ambient lattice theory.

9.3 Recognizing the level-thirteen factor

Let z be the second embedded parafermion stress and let a_3 be its normalized weight-three primary. We use the subscript to distinguish this field from the lattice momentum a. Put

$$L = z, \quad b_4 = (a_3)_{(1)}a_3, \quad d_5 = (a_3)_{(1)}b_4, \quad c = \frac{8}{5}, \quad \lambda = \frac{14}{473}.$$

The self-OPE is especially instructive:

n	5	4	3	2	1	0
$(a_3)_{(n)}a_3$	$\frac{8}{15}\mathbf{1}$	0	$2L$	DL	b_4	$\frac{1}{2}Db_4 - \frac{1}{12}D^3L.$

(9.7)

The normalizations $c/3$ and $2L$ are standard for a spin-three primary. The last entry is constrained by skew symmetry, rather than an independent free parameter.

The next row contains the nonlinear information:

$$\begin{aligned}
(a_3)_{(n)}b_4 &= 0 \quad (n \geq 4),\\
(a_3)_{(3)}b_4 &= \frac{69283}{2365}a_3,\\
(a_3)_{(2)}b_4 &= \frac{8}{3}\big(5 - 2\lambda(2+c)\big)Da_3,\\
(a_3)_{(1)}b_4 &= d_5,\\
(a_3)_{(0)}b_4 &= \frac{2}{5}Dd_5 + \frac{32\lambda}{5} : LDa_3 : -\frac{48\lambda}{5} : (DL)a_3 : +\frac{2}{15}\big(-5 + 2\lambda(c-1)\big)D^3a_3.
\end{aligned} \tag{9.8}$$

The weight-six product $R = (a_3)_{(1)}d_5$ is not a new independent generator. It obeys the decoupling relation

$$\begin{aligned}
R = &-\frac{148744}{473} : Lb_4 : +\frac{566160}{473} : a_3a_3 : -\frac{4800}{121} : L(:LL:) :\\
&+\frac{3860}{11} : (D^2L)L : -\frac{33320}{121} : (DL)(DL) : -\frac{489817}{4730}D^2b_4 + \frac{2193061}{52030}D^4L.
\end{aligned} \tag{9.9}$$

This is where a finite presentation prevents an endless sequence of new primary generators.

For completeness, the remaining singular row is fixed by

$$\begin{aligned}
(a_3)_{(7)}d_5 &= c\big(55 - 16\lambda(2+c)\big)\mathbf{1}, \qquad (a_3)_{(6)}d_5 = 0,\\
(a_3)_{(5)}d_5 &= \frac{4}{3}\big(175 - 64\lambda(2+c)\big)L,\\
(a_3)_{(4)}d_5 &= \big(110 - 32\lambda(2+c)\big)DL,\\
(a_3)_{(3)}d_5 &= \big(95 - 16\lambda(11+4c)\big)b_4 + 128\lambda : LL :\\
&\quad + \big(10 + 8\lambda(c-1)\big)D^2L,\\
(a_3)_{(2)}d_5 &= 64\lambda : (DL)L : +\big(75/2 - 4\lambda(13+5c)\big)Db_4\\
&\quad + \frac{-25 + 8\lambda(9+5c)}{12}D^3L,\\
(a_3)_{(0)}d_5 &= \frac{1}{3}DR + \frac{448}{1419} : LDb_4 : -\frac{896}{1419} : (DL)b_4 :\\
&\quad - \frac{224}{1419} : (D^3L)L : -\frac{34019}{28380}D^3b_4 + \frac{9473}{170280}D^5L.
\end{aligned}$$

The weight-eight, seven, and six formulas have deliberately retained their normal ordering. Treating normal products as ordinary commutative polynomials can change the derivative terms.

Construction input 9.4 (Parafermion field identities)**.** Actual PCT-stable fields $z, a_3 \in U$ satisfy the stress and primary relations at charge $8/5$, with intrinsic translation D, and all singular rows (9.7)–(9.9) and the subsequent displayed row hold as full lattice-vector equalities. These are the corresponding defining rows of the simple $K(\mathfrak{sl}_2, 13)$ presentation.

Why do these rows identify the factor? An iterate identity gives

$$(b_4)_{(n)}v = (a_3)_{(1)}((a_3)_{(n)}v) - (a_3)_{(0)}((a_3)_{(n+1)}v) + (a_3)_{(n+1)}((a_3)_{(0)}v) - (a_3)_{(n)}((a_3)_{(1)}v).$$

Taking $v = b_4, d_5$ will recover the b_4, b_4 and b_4, d_5 rows, but that statement alone does not rule out circularity: the inner products can be normal polynomials, whose evaluation must be available first. We now give an order in which every required operation has already been justified.

For this explanation abbreviate a_3, b_4, d_5 by a, b, d. Commuting Virasoro modes through their definitions gives

$$L_m b = (2m+1)a_{(m+1)}a, \qquad L_m d = (2m+1)a_{(m+1)}b + a_{(1)}L_m b \quad (m \geq 1).$$

The supplied rows make these expressions linear in named fields and derivatives. Thus Virasoro creation operators can be moved outside a normal word using known commutators, and sesquilinearity removes derivatives. Three non-Virasoro factors would already have weight at least nine. The possible two-factor cores are therefore only

weight bound	cores besides individual fields
6	$:aa:$
7	$:aa:,\ :ab:$
8	$:aa:,\ :ab:,\ :bb:,\ :ad:$

Changing the order of a pair uses skew symmetry and its known bracket. No assumption that normal products commute is made.

Let A_r mean that every nonnegative mode of a can be evaluated on every normal polynomial of weight at most r, and let B_r mean the same assertion for b. The dependency order is

$$\begin{aligned} A_6 &\longrightarrow (b,b) \longrightarrow A_7 \longrightarrow (b,d) \\ &\longrightarrow B_6 \longrightarrow A_8 \longrightarrow B_7 \longrightarrow (d,d). \end{aligned}$$

To verify the first arrow, the only new core in A_6 is $:aa:$. Its Wick integral brackets a coefficient of $[a_\lambda a]$ with a. That coefficient is linear in the vacuum and derivatives of L, b, whose brackets with a are supplied. The displayed iterate then computes (b,b) because every inner $a_{(j)}b$ has weight at most six. The new core $:ab:$ in A_7 needs only this newly computed (b,b) row. The same iterate with $v = d$ then computes (b,d), since the inner a products have weight at most seven.

For B_6, the core $:aa:$ reduces by skew symmetry to A_6 on the coefficients of $[b_\lambda a]$; single-field inputs now use the known (b,b) and (b,d) rows. In A_8, the new core $:bb:$ uses B_6, and $:ad:$ uses the known brackets of L, b with d. Finally the new $:ab:$ core for B_7 uses B_6 on coefficients of $[b_\lambda a]$.

The final iterate is

$$d_{(n)}s = a_{(1)}(b_{(n)}s) - a_{(0)}(b_{(n+1)}s) + b_{(n+1)}(a_{(0)}s) - b_{(n)}(a_{(1)}s).$$

At $s = d$, its first two terms use A_8 and its last two use B_7. Hence it computes (d,d) without calling any unknown row. All generator brackets are fixed by the printed first rows, including their simple poles and the weight-six decoupling.

The uniqueness argument is also concrete. Given two simple graded algebras with this presentation, generate a subalgebra of their product by the diagonal pairs of the named fields.

Common polynomial closure makes its degree-zero space the single diagonal vacuum line. The sum of all proper graded ideals still misses that line, so there is a unique maximal proper graded ideal. Both coordinate projections have that kernel. Their simple quotients are therefore isomorphic. Neither freeness nor linear independence of normal monomials is needed: positive weights alone force the one-dimensional degree-zero space of the diagonal algebra.

The two targets here really satisfy the simplicity premise. The standard parafermion factor is simple. The actual generated subalgebra is PCT-stable with its positive inherited form, and its intrinsic translation and grading agree with the ambient ones on its states. It is consequently of CFT type and simple. One can see the last assertion from an orthogonal-ideal argument: a proper graded ideal misses the vacuum; its invariant orthogonal complement contains the vacuum and hence every state obtained by acting on the vacuum. That complement is the full algebra, so the proper ideal is zero. This argument identifies the actual embedded factor after the field identities have been supplied; it does not replace their verification.

9.4 The rank-two lattice–Potts core

Return to the lattice momenta a, b, c in (7.9); the temporary field abbreviations in the preceding calculation are no longer in use. Set $v = a + b + c$. Directly,

$$v = (1, 1, 0, 1, 1, -3, 1, 1), \qquad r_0 = v/3 = \ell/2 + h/6.$$

The vectors ℓ, h are orthogonal, with squared norms $2, 78$, so $r_0^2 = 8/3$. Let

$$L_0 = \mathbb{Z}\ell + \mathbb{Z}v, \qquad G_{L_0} = \begin{pmatrix} 2 & 3 \\ 3 & 24 \end{pmatrix}, \qquad \det L_0 = 39.$$

We write L_0 for the lattice here to distinguish it from the parafermion stress L in the preceding section. This lattice symbol is unrelated to the Virasoro energy operator, also conventionally written L_0; a lattice coset or lattice VOA always makes the meaning clear.

Construction input 9.5 (A charged lattice witness)**.** The actual fields satisfy

$$p_{(1)}u = \frac{2}{3}u, \qquad p_{(2)}u = p_{(3)}u = 0, \qquad u_{(-5)}(u_{(-3)}u) = -\frac{3}{5}e^v.$$

The Potts plane is orthogonal to the plane of L_0.

The last identity has output weight twelve, but its verification uses only the leading powers of three exponential products. It does not require enumerating an entire weight-twelve lattice space. This is a good example of why following the algebraic route of a computation can be much cheaper and clearer than building every ambient state.

Together with the left root currents and PCT, the witness puts V_{L_0} inside U. The field u lies in $V_{L_0+r_0} \otimes \psi$, where ψ is the order-three Potts simple current of weight $2/3$. Its total weight is

$$\frac{r_0^2}{2} + \frac{2}{3} = \frac{4}{3} + \frac{2}{3} = 2.$$

The generated algebra is the three-sector extension

$$\mathcal{T} = (V_{L_0} \otimes P) \oplus (V_{L_0+r_0} \otimes \psi) \oplus (V_{L_0+2r_0} \otimes \psi^2). \tag{9.10}$$

Its stress is

$$\omega_{\mathcal{T}} = \frac{1}{4}\ell(-1)^2\mathbf{1} + \frac{1}{156}h(-1)^2\mathbf{1} + p, \qquad c_{\mathcal{T}} = 2 + \frac{4}{5} = \frac{14}{5}.$$

The extension is finite, unitary, and by integer-weight simple currents; the standard extension theorem makes it strongly rational, in the sense defined in Chapter 10.

Worked example 9.6 (The discriminant arithmetic)**.** The class $g = h/39$ generates $L_0^*/L_0 \cong \mathbb{Z}/39$ and has quadratic form $q(g) = g^2/2 = 1/39$. Moreover $r_0 = 26g$ modulo L_0: indeed $r_0 - 26g = \ell/2 - h/2 = 2\ell - v \in L_0$. The order-three lattice class has quadratic value $26^2/39 = 1/3 \pmod 1$, canceled by the Potts simple-current weight $2/3$. This is the locality condition for the extension. The orthogonal remaining order-thirteen class $3g$ has $q(3g) = 3/13$. Consequently

$$\mathrm{Rep}(\mathcal{T}) \simeq \mathrm{Fib} \boxtimes \mathcal{C}(\mathbb{Z}/13, 3j^2/13).$$

The Fibonacci factor comes from the neutral Potts module data. Its nontrivial object has lowest weight $2/5$ and twist $e^{4\pi i/5}$, which specifies the braiding choice used here.

Sharing central charge with another familiar algebra is not recognition. For example $G_{2,1}$ also has charge $14/5$, but $\mathcal{T}$ has four currents and different representation data. Central charge is one coordinate of a theory, not its name.

9.5 An order-thirteen extension and fourteen useful modules

The factors $\mathcal{T}$ and PF_{13} commute, giving charge $22/5$. After choosing generators for their pointed parts, the combined pointed quadratic form is

$$q(j,k) = -\frac{j^2 + k^2}{13}, \qquad (j,k) \in (\mathbb{Z}/13)^2.$$

Replacing the old order-thirteen generator by twice itself converts the $3j^2/13$ description above into this form, since $3(2j)^2 \equiv -j^2 \pmod{13}$. Quadratic forms must be compared up to a change of generator.

The isotropic nonzero lines satisfy $k = \pm 5j$, because $5^2 = -1$ modulo thirteen. Actual weight-two vectors in U have simultaneous energies

$$h_{\mathcal{T}} = 14/13, \qquad h_{\mathrm{PF}_{13}} = 12/13, \qquad h_{\mathrm{complement}} = 0.$$

Their Cartan weights are $(\pm 1, \pm 5)$ and $(0, \pm 8)$. These witnesses select one isotropic line. The resulting extension, denoted A_L, has

$$\mathrm{Rep}(A_L) \simeq \mathrm{Fib} \boxtimes SO(3)_{13}. \tag{9.11}$$

There are fourteen simple modules: a Fibonacci label 1 or τ, and one of the seven even affine labels $0, 2, \ldots, 12$. Here $SO(3)_{13}$ carries the braiding inherited from this even-label subcategory of the standard $SU(2)_{13}$ category.

Here actual lowest energies distinguish labels that twists alone would not. In a connected unitary extension, an invertible constituent has multiplicity at most one, and supported invertibles form a subgroup. The nonzero witnessed charge generates its full order-thirteen line. Both isotropic lines cannot occur: their joint span contains nonintegral-weight charges. The line therefore specifies the extension without an unresolved fixed-point multiplicity.

9.6 The positive conformal-vector identity

Let ω_L, ω_R be the two affine stresses and let e, f be the two commuting current-commutant stresses already constructed in S_2. Their charges are

$$1, \quad \frac{117}{41}, \quad \frac{2400}{1271}, \quad \frac{70}{31},$$

and their sum is eight. These four vectors are mutually orthogonal for the invariant form; their squared norms are half their charges. They span the current-invariant part of the actual space S_2.

Average p and z over the compact current group. A group average is the orthogonal projection onto invariant vectors, so the projection coefficients are obtained by dividing overlaps by squared norms. The exact calculation in this finite-dimensional space gives

	ω_L	ω_R	e	f
$\overline{p}$	0	2/45	9/100	39/175
$\overline{z}$	0	14/195	13/20	13/175

where each row lists the coefficients in the four-axis basis.

Construction input 9.7 (Invariant projection data)**.** The four-axis invariance statement and the two projection rows above hold for the actual PCT-stable fields. Thus they are identities in the finite-dimensional space S_2, independently of whether one has already proved $S_2 = U_2$.

Proposition 9.8 (Positive stresses cover the ambient energy)**.** *The following identity holds in U:*

$$\omega_U = \omega_L + \frac{3403}{4563}\omega_R + \frac{25}{6}\overline{p} + \frac{25}{26}\overline{z}. \tag{9.12}$$

Proof. Check the four coordinates. The left coordinate is one. The right coordinate is

$$\frac{3403}{4563} + \frac{25}{6}\frac{2}{45} + \frac{25}{26}\frac{14}{195} = \frac{3403 + 845 + 315}{4563} = 1.$$

The remaining coordinates are

$$\frac{25}{6}\frac{9}{100} + \frac{25}{26}\frac{13}{20} = \frac{3}{8} + \frac{5}{8} = 1,$$

and

$$\frac{25}{6}\frac{39}{175} + \frac{25}{26}\frac{13}{175} = \frac{13}{14} + \frac{1}{14} = 1.$$

The four stresses sum to ω_U, proving the identity. □

The bars should not be discarded. An averaged stress is generally not itself a Virasoro vector, and the many conjugate Potts and parafermion copies need not commute with one another. What survives is positivity. In the finite-dimensional real PCT-fixed subspace, the Haar average belongs to the convex hull of the compact orbit. Carathéodory's theorem expresses it as a finite convex combination. Therefore

$$\omega_U = \sum_{i=1}^{N} t_i e_i, \qquad t_i > 0, \tag{9.13}$$

where each e_i is the conformal vector of an actual regular unitary subalgebra: one of the affine factors, or a compact conjugate of Potts or PF_{13}. Zero coefficients can be omitted.

This is not a tensor decomposition. It is a positive energy identity. The next chapter uses inequalities and traces that remain valid even when the different partial energy operators do not commute. That is why the averaged construction is sufficient.

9.7 Exercises

Exercise 9.1. Compute the central charges of PF_2, PF_3, and PF_{13} from (9.1). Identify which has the Ising charge.

Solution on page 297.

Exercise 9.2. Verify all three opposite-root products in the Potts central-term calculation, and check PCT reality for all three momentum pairs.

Solution on page 297.

Exercise 9.3. Use skew symmetry of the weight-three self-bracket to derive the simple pole $\frac{1}{2}Db_4 - \frac{1}{12}D^3L$ in (9.7).

Solution on page 298.

Exercise 9.4. Show that the two isotropic lines of $-(j^2+k^2)/13$ are exactly $k = \pm 5j$. Explain why adjoining both lines would violate integral vacuum grading.

Solution on page 298.

Exercise 9.5. Verify all four coefficients of (9.12). Explain why checking the norms of p and z alone would not prove this identity or establish either embedded factor.

Solution on page 298.

Exercise 9.6. Give a finite-dimensional example of two noncommuting positive self-adjoint matrices. Show that their positive linear combination still gives a valid expectation-value inequality. Which part of the next chapter's argument should therefore avoid simultaneous diagonalization?

Solution on page 298.

Chapter 10

Why the generated algebra is rational

The construction so far gives an infinite-dimensional unitary algebra inside a lattice theory. Rationality asks for a much stronger form of control: a finite and semisimple representation theory. A unitary vacuum representation does not automatically provide this. The positive conformal-vector identity from the previous chapter is the extra ingredient. We will use it first to bound a quotient of the vacuum algebra, then to constrain arbitrary modules, and finally to split extensions of the vacuum. A categorical theorem turns the last property into semisimplicity.

The proof is worth following even for readers primarily interested in physics. It explains why the later finite list of characters is a conclusion of the construction rather than an initial assumption.

10.1 The finiteness properties that must be distinguished

For a VOA V define

$$C_2(V) = \operatorname{span}\{a_{(-2)}b : a, b \in V\}.$$

The notation C_2 here has no relation to central charge two. The VOA is C_2-cofinite when $V/C_2(V)$ is finite dimensional. This quotient removes many derivatives and normal composites, so it measures a different kind of finiteness from $\dim V_n < \infty$. Finite-dimensional homogeneous spaces alone do not imply C_2-cofiniteness.

Worked example 10.1 (The Heisenberg warning)**.** A rank-one Heisenberg VOA has finite-dimensional weight spaces, but the classes of $J^n_{(-1)}\mathbf{1}$ survive in its C_2 quotient. The quotient is a polynomial algebra in the current class and is infinite dimensional. Correspondingly, there are continuously many Fock modules labeled by momentum. A simple positive vacuum representation is fully compatible with this nonrational behavior.

Worked example 10.2 (A five-dimensional quotient of the rank-one lattice theory)**.** Let $V = V_{A_1} = L_1(\mathfrak{sl}_2)$, with the weight-one currents E, F, H normalized as in Example 7.1. The VOA identities make

$$R = V/C_2(V), \qquad [a][b] = [a_{(-1)}b],$$

a commutative associative graded algebra. Derivatives vanish in R, so strong generation by E, F, H implies that their classes e, f, h generate it as an ordinary commutative algebra.

The lattice exponential formula supplies the decisive relation. For a root α of squared norm two, $Y(e^\alpha, z)e^\alpha$ starts at z^2. Its constant coefficient is therefore zero:

$$E_{(-1)}E = 0, \qquad e^2 = 0.$$

This is a relation in the actual simple lattice VOA. The universal affine vacuum algebra at level one has not yet imposed this integrability relation.

Zero modes preserve $C_2(V)$ and act as derivations on R. The derivation ∂_F induced by $F_{(0)}$ obeys

$$\partial_F(e) = -h, \qquad \partial_F(h) = 2f, \qquad \partial_F(f) = 0.$$

Apply it successively to $e^2 = 0$, $eh = 0$, $h^2 - 2ef = 0$, and $hf = 0$. The resulting equations are

$$-2eh = 0, \qquad -h^2 + 2ef = 0, \qquad 6hf = 0, \qquad 2f^2 = 0.$$

Consequently every class is a linear combination of

$$1, \quad e, \quad h, \quad f, \quad ef.$$

Every cubic monomial vanishes; for instance $h^3 = 2ehf = 0$. This proves an upper bound of five, but it does not yet prove that the last class survives.

Weights zero and one already give four independent classes, since $C_2(V)$ has no component in either weight. At weight two, the weight formula for $a_{(-2)}b$ gives exactly

$$C_2(V) \cap V_2 = DV_1, \qquad D = L_{-1}.$$

The conformal vector satisfies $L_1\omega = 0$, so $\langle Da, \omega\rangle = \langle a, L_1\omega\rangle = 0$ for $a \in V_1$. Its squared norm is $c/2 = 1/2$, hence $\omega \notin C_2(V)$. Finally the level-one Sugawara formula gives

$$[\omega] = \frac{1}{6}\left(\frac{h^2}{2} + 2ef\right) = \frac{ef}{2} = \frac{h^2}{4}.$$

Thus $ef \neq 0$, and the displayed spanning set is a basis:

$$\dim R = 5, \qquad \sum_{n\geq 0} (\dim R_n) t^n = 1 + 3t + t^2.$$

Adding the lattice exponential fields to the Heisenberg algebra has made the surviving current classes nilpotent and the quotient finite. This establishes C_2-cofiniteness; by itself it is not a proof of rationality. Nor is five a count of simple modules: this VOA has the two affine simples labelled $\ell = 0, 1$.

An ordinary module has diagonalizable L_0, finite-dimensional eigenspaces, and lower truncation in each integer translate of an eigenvalue. The simple ordinary modules used below have spectrum in one $h + \mathbb{Z}_{\geq 0}$. A grading-restricted generalized module allows finite Jordan blocks for L_0. Such modules have to be considered before rationality has been proved: excluding them at the definition stage would assume part of the conclusion.

For the present discussion, rationality means semisimplicity of the appropriate grading-restricted module theory; under the standard C_2-cofinite hypotheses this agrees with the usual rationality condition. A VOA is strongly rational when it is simple, of CFT type, self-contragredient, rational, and C_2-cofinite. Regularity of the component algebras below means that every weak module decomposes as an algebraic direct sum of ordinary simple modules. This stronger component property is known for the positive-integral affine, Potts, and parafermion algebras being used.

10.2 Compatible gradings and partial energies

Let $A \subset V$ be a vertex subalgebra with its own conformal vector e. We do not require $e = \omega_V$. The inclusion is *grading compatible* when a homogeneous state of A has the same intrinsic weight in A as its ambient weight in V. The two energy operators need not agree on all of V.

This distinction is familiar for a tensor product: $A \otimes \mathbf{1}$ has its own stress and its own energy, while the total energy also counts the other factor. The Haagerup construction is subtler because the many regular subalgebras in its positive identity need not commute. Nevertheless each inclusion is grading compatible, so a mode of an A-state changes ambient and intrinsic degrees by the same amount.

We use the following precise package of hypotheses.

Definition 10.3 (A positive regular stress family)**.** A simple unitary VOA V of CFT type has a positive regular stress family if there are finitely many PCT-stable, grading-compatible, regular unitary subalgebras, carrying the inherited positive form, $A_i \subset V$ with conformal vectors e_i such that

$$\omega_V = \sum_i t_i e_i, \qquad t_i > 0, \tag{10.1}$$

and each A_i has only nonnegative simple lowest weights, with the vacuum its unique simple module of lowest weight zero. We require each e_i to be an ambient weight-two quasiprimary fixed by PCT:

$$L_0^V e_i = 2e_i, \qquad L_1^V e_i = 0, \qquad \theta(e_i) = e_i.$$

The invariant-form identity then gives $(L_n^{e_i})^\dagger = L_{-n}^{e_i}$ on V.

The affine, Potts, and PF_{13} embeddings established in the preceding chapter give exactly such a family for U. Their PCT-real stresses are ambient quasiprimary vectors of weight two; the invariant-form adjoint identity therefore gives the displayed mode adjoints. In particular the partial energies are self-adjoint on each ambient grade. No properness assumption is needed here. The regularity argument would still work if the generated algebra had turned out to be the ambient B.

10.3 Bounding the C_2 quotient by energy

Regularity implies C_2-cofiniteness by Li's theorem; this is an available implication, not a further assumption about the embedded components. We also use the module-cofiniteness theorem for a regular C_2-cofinite algebra A: for each ordinary simple A-module M, the quotient

$$M/C_{2,A}(M), \qquad C_{2,A}(M) = \mathrm{span}\{a_{(-2)}m : a \in A,\ m \in M\},$$

is finite dimensional. The theorem includes modules, not only the vacuum algebra. Because A has finitely many simple modules, one can choose a uniform upper bound on the intrinsic energies that survive in these quotients.

Proposition 10.4 (Positive stresses imply C_2-cofiniteness)**.** *A VOA with a positive regular stress family is C_2-cofinite.*

Proof. For each A_i, choose a number D_i bounding the intrinsic energy of every quotient $M/C_{2,A_i}(M)$ over its finitely many simple modules. Regularity decomposes the weak A_i-module V into such modules. The component C_2 construction commutes with an algebraic direct sum: $C_{2,A_i}(\bigoplus_j M_j) = \bigoplus_j C_{2,A_i}(M_j)$. An isomorphism of component modules preserves their intrinsic conformal operator, so repeated copies do not introduce arbitrary intrinsic energy shifts. Thus the same D_i works even when a component simple occurs infinitely often in V. Therefore every $L_0^{e_i}$ eigenvector of energy greater than D_i belongs to $C_{2,A_i}(V)$, which is contained in $C_2(V)$.

The operator $L_0^{e_i}$ preserves each finite-dimensional ambient space V_n and is self-adjoint for its positive form. Take $v \in V_n \cap C_2(V)^\perp$. Its components along all intrinsic energies greater than D_i vanish, so

$$\langle L_0^{e_i} v, v\rangle \le D_i \|v\|^2.$$

Taking the zero mode of (10.1) yields

$$n\|v\|^2 = \sum_i t_i \langle L_0^{e_i} v, v\rangle \le \left(\sum_i t_i D_i\right) \|v\|^2.$$

Thus no nonzero v exists once $n > \sum_i t_i D_i$. Since $C_2(V)$ is ambient graded, these orthogonal complements represent the graded pieces of $V/C_2(V)$. Only finitely many finite-dimensional grades survive. □

Only one partial energy is diagonalized at a time. The proof would be invalid if it assumed a simultaneous eigenbasis for all $L_0^{e_i}$. It instead applies separate expectation-value bounds to the same vector and adds them using positive coefficients. This is the precise point of the positive-stress construction.

The proof is existential: it gives a finite cutoff from the component quotients but does not compute a sharp numerical cutoff for $U/C_2(U)$. That is enough for the structural conclusion. A tight cutoff would be useful for an explicit presentation of its Zhu algebra, the associative algebra that encodes degree-zero mode actions on lowest spaces. Such a presentation is not required to establish rationality by the route below.

10.4 Lowest spaces are complete component tops

Before applying positivity to arbitrary V-modules we must avoid an easy mistake. A partial conformal energy can be positive on the vacuum Hilbert space without a positive form having yet been supplied on every module. The argument below uses regular component representation theory and traces; it does not assume all V-modules are already unitary.

For a weak A-module M, its highest space is

$$\Omega_A(M) = \{v : a_{(n)}v = 0 \text{ for homogeneous } a \in A,\ n \ge \mathrm{wt}_A(a)\}.$$

These are precisely the modes that lower intrinsic degree. The degree-zero modes $o(a) = a_{(\mathrm{wt}(a)-1)}$ act on this space through the Zhu algebra. Regularity makes that finite-dimensional top algebra semisimple on the highest spaces under consideration.

Lemma 10.5 (Compatible top spaces)**.** *Let $A \subset V$ be regular and grading compatible. A finite space E in a weak V-module, stable under all A degree-zero modes and killed by all negative-degree A modes, is a direct sum of simple A top representations. In particular, L_0^A is diagonalizable on E with eigenvalues among the simple lowest weights of A.*

Proof. Compatibility places E in $\Omega_A(M)$. Decompose M as a weak A-module using regularity. The highest space of an ordinary simple module is its top: a vector of higher energy killed by every lowering mode would, after mode reordering, generate a nonzero submodule at its own energy and above. That submodule could not contain the original lowest space, contradicting simplicity. Thus $\Omega_A(M)$ is a sum of those tops. The finite invariant subspace E is consequently a sum of simple representations of the semisimple top algebra. □

The condition concerns lowering modes of *all* A-states. Checking only a few generator modes requires an additional argument that they imply the full highest-space condition.

Proposition 10.6 (Nonnegative lowest weights)**.** *Every simple ordinary V-module for a positive regular stress family has real nonnegative lowest weight. Its only simple module of lowest weight zero is the vacuum module V.*

Proof. Let $E = M_h$ be the finite lowest space. Degree-zero modes preserve it and lowering modes kill it. By Lemma 10.5, each $L_0^{e_i}$ is diagonalizable on E with nonnegative real spectrum. Taking traces gives

$$h \dim E = \sum_i t_i \operatorname{Tr}_E L_0^{e_i} \geq 0. \tag{10.2}$$

This proves the assertion about h even if the partial energies do not commute.

If $h = 0$, all nonnegative traces vanish. Each partial energy is diagonalizable, so it vanishes on E. The unique-zero-top hypothesis puts E in sums of vacuum tops for every A_i. Therefore $L_{-1}^{e_i} E = 0$, and the conformal-vector identity gives $L_{-1}^V E = 0$. The vacuum-like lemma below supplies a nonzero module map $V \to M$. Simplicity makes it an isomorphism. □

10.5 A translation-invariant vector is a vacuum vector

Lemma 10.7 (Vacuum-like vectors)**.** *If v in a weak V-module satisfies $L_{-1}v = 0$, then $a_{(n)}v = 0$ for every $a \in V$ and $n \geq 0$. The map*

$$s : V \longrightarrow M, \qquad s(a) = a_{(-1)}v$$

is a module homomorphism with $s(\mathbf{1}) = v$.

Proof. Lower truncation says that $a_{(n)}v = 0$ for sufficiently large n. Translation covariance gives

$$L_{-1}a_{(n+1)}v = -(n+1)a_{(n)}v.$$

Downward induction proves vanishing for every $n \geq 0$. Taking the appropriate constant coefficient in module associativity, or using the module Jacobi identity, now gives $b_{(m)}(a_{(-1)}v) = (b_{(m)}a)_{(-1)}v$. This is precisely the homomorphism assertion. □

This lemma explains why the translation mode, rather than only energy zero, appears in the projectivity proof. A generalized weight-zero vector could initially belong to a nontrivial Jordan extension. The positive component structure forces it to satisfy the full vacuum condition.

10.6 Splitting every extension onto the vacuum

In an abelian category, an object P is projective when every epimorphism onto P splits, equivalently when $\operatorname{Hom}(P, -)$ is exact. Think of a possible nonsplit extension

$$0 \longrightarrow X \longrightarrow E \longrightarrow V \longrightarrow 0.$$

Projectivity says that a vacuum copy can always be lifted back into E as an actual submodule.

Proposition 10.8 (The vacuum is projective)**.** *For a positive regular stress family, V is projective in the category of grading-restricted generalized modules.*

Proof. Let $\pi : E \twoheadrightarrow V$. The standard C_2-cofinite module theory gives finite length and makes its simple composition factors ordinary modules. The preceding lowest-weight result therefore implies that every composition factor has nonnegative energy. Generalized weight spaces are exact in these short exact sequences: module maps commute with L_0, and every vector has a finite decomposition into generalized eigenspaces. Induction along a composition series then shows that E has no negative generalized weights. Exactness also gives a surjection $E_{[0]} \twoheadrightarrow V_0$.

Consider the finite generalized weight-zero space $E_{[0]}$. It is killed by every negative-degree mode and is invariant under degree-zero modes. By compatible-top semisimplicity, each partial energy $L_0^{e_i}$ is diagonalizable there with nonnegative spectrum. But L_0^V is nilpotent on $E_{[0]}$, so

$$0 = \operatorname{Tr}_{E_{[0]}} L_0^V = \sum_i t_i \operatorname{Tr}_{E_{[0]}} L_0^{e_i}.$$

Every partial energy vanishes. As before, their vacuum-top translation operators vanish, so $L_{-1}^V E_{[0]} = 0$.

Since π respects generalized weights, choose $v \in E_{[0]}$ with $\pi(v) = \mathbf{1}$. Lemma 10.7 gives $s(a) = a_{(-1)}v$. Then $\pi s(a) = a_{(-1)}\mathbf{1} = a$, so s is a section of π. □

No Hermitian form on E was used. The positivity in the proof lies in the known nonnegative spectra of the regular component tops and the positive coefficients of the stress identity. This avoids assuming strong unitarity before it is proved by the later analytic argument.

10.7 From a projective unit to a semisimple category

At this stage standard C_2-cofinite VOA tensor theory provides a finite complex linear abelian braided category $\mathcal{C}$ of generalized modules. Its tensor product is right exact. Contragredient duality is a Grothendieck–Verdier duality with dualizing object V, which is the tensor unit. These are the categorical inputs to the next argument; we do not assume rigidity or semisimplicity already.

In this categorical section D denotes contragredient duality, so DX is the contragredient object; here D is not the translation operator. More precisely, this duality gives natural isomorphisms

$$\operatorname{Hom}(Y \otimes X, \mathbf{1}) \simeq \operatorname{Hom}(Y, DX).$$

At $Y = DX$, the identity of DX corresponds to the evaluation $DX \otimes X \to \mathbf{1}$, so this map is nonzero when $X \neq 0$. The assertion represents maps *into* the unit; it does not yet supply a coevaluation from the unit satisfying the snake identities or the exactness consequences of rigidity.

Worked example 10.9 (Right exactness already bounds growth)**.** Let A be a finite-dimensional algebra, and let $P = A$ be its left regular module. A module M of vector-space dimension n is a quotient of $P^{\oplus n}$: choose a vector-space basis $m_1, \dots, m_n$ and send the ith copy's unit to m_i. For any additive right-exact functor G between finite module categories, this gives a surjection $G(P)^{\oplus n} \twoheadrightarrow G(M)$, hence $\dim G(M) \leq n \dim G(P)$.

This estimate does not require G to preserve injections. For instance, with $A = \mathbb{C}[\varepsilon]/(\varepsilon^2)$, $G(M) = M/\varepsilon M$ is right exact. The inclusion $\varepsilon A \hookrightarrow A$ becomes the zero map $\mathbb{C} \to \mathbb{C}$, so G is not left exact. The dimension bound still holds. This is the modest amount of exactness used in the growth argument below.

We invoke the following rigidity theorem in its precise form: in a Cauchy-complete linear braided category with finite-dimensional Hom spaces and $\operatorname{End}(\mathbf{1}) = \mathbb{C}$, a nonzero indecomposable X is rigid if (i) $\mathbf{1}$ is a direct summand of $Y \otimes X$ for some Y, and (ii) $\dim \operatorname{End}(X^{\otimes n}) < n!$ for some n. These are called non-negligibility and moderate growth. The theorem does not assume tensor exactness. The finite abelian category at hand is Cauchy complete, meaning that every idempotent splits as a projection onto an actual direct summand. It also has finite-dimensional Hom spaces and finite decompositions into indecomposables, so the theorem can be applied one summand at a time.

Proposition 10.10 (Projective unit implies semisimplicity)**.** *Let $\mathcal{C}$ be finite complex linear abelian and braided, with simple unit, right-exact tensor product, and Grothendieck–Verdier duality with dualizing object $\mathbf{1}$. If $\mathbf{1}$ is projective, then $\mathcal{C}$ is rigid and semisimple.*

Proof. Realize the underlying finite abelian category as finite-dimensional modules over a finite-dimensional algebra. Let P correspond to its regular module and let $d_0(Z)$ denote the underlying vector-space dimension. Every Z is a quotient of $P^{\oplus d_0(Z)}$. For fixed X, right exactness gives

$$d_0(Z \otimes X) \le C_X d_0(Z), \qquad C_X = \max(1, d_0(P \otimes X)).$$

Iterating,

$$\dim \mathrm{End}(X^{\otimes n}) \le d_0(X^{\otimes n})^2 \le d_0(\mathbf{1})^2 C_X^{2n} < n!$$

for sufficiently large n. The last inequality is the elementary fact that factorial growth eventually dominates any fixed exponential.

For nonzero indecomposable X, duality gives a nonzero evaluation $DX \otimes X \to \mathbf{1}$. Since the unit is simple, it is surjective. Projectivity makes it split. The rigidity theorem therefore applies to X. Finite direct sums and Krull–Schmidt decompositions make all objects rigid.

Only now may we use that tensoring is exact. Rigidity identifies

$$\mathrm{Hom}(X, -) \cong \mathrm{Hom}(\mathbf{1}, - \otimes X^*).$$

The right side is exact because the unit is projective and tensoring by X^* is exact. Thus every object is projective. Every short exact sequence splits, which is semisimplicity. □

The order is important. Using tensor exactness to prove rigidity would create a circular proof. The growth bound above uses only right exactness, a property available before rigidity.

To connect this categorical conclusion with ordinary VOA rationality, we use one more standard consequence of C_2-cofiniteness: every finitely generated weak module belongs to the grading-restricted generalized category. Each vector of any weak module belongs to its cyclic submodule. Once that finite category is semisimple, these cyclic submodules are sums of ordinary simple modules. The whole weak module is therefore a sum of simple submodules, hence a direct sum of them by the usual semisimple-module argument. In this setting the conclusion actually gives regularity as well as rationality.

Theorem 10.11 (Rationality of the Haagerup candidate)**.** *Under the actual embedded-factor and positive-stress inputs of Chapter 9, the generated U is strongly rational.*

Proof. The lattice construction gives simplicity, CFT type, and a positive invariant form; the latter gives self-contragredience. The positive stress family gives C_2-cofiniteness by Proposition 10.4 and projectivity of the vacuum by Proposition 10.8. The standard finite VOA category and Proposition 10.10 give semisimplicity, hence rationality. □

This theorem does not identify the module category with the Haagerup center. It does not yet construct positive forms on all simple modules or on their intertwining-operator spaces. What it does provide is the finite semisimple setting in which the branching and character arguments of the next chapters become legitimate.

10.8 Exercises

Exercise 10.1. Explain why $C_2(V)$ is homogeneous for ambient weight. Show that the orthogonal complement in each V_n represents the grade-n quotient even if $C_2(V)$ is not closed in the Hilbert completion.

Solution on page 299.

Exercise 10.2. Prove that a diagonalizable operator with nonnegative real eigenvalues and zero trace is zero. Give a nondiagonalizable counterexample if diagonalizability is omitted.

Solution on page 299.

Exercise 10.3. In the vacuum-like lemma, write the downward induction explicitly for $a_{(2)}v$, $a_{(1)}v$, and $a_{(0)}v$. Why does the coefficient $n+1$ never obstruct the induction through $n=0$?

Solution on page 299.

Exercise 10.4. For the algebra $\mathbb{C}[\varepsilon]/(\varepsilon^2)$, exhibit a nonsplit short exact sequence of modules with one-dimensional endpoints. Compare the role of a projective simple module with the vacuum projectivity argument above.

Solution on page 299.

Exercise 10.5. Prove directly that $A^n < n!$ for all sufficiently large n, for any fixed $A > 0$. Use it to fill in the moderate-growth bound.

Solution on page 299.

Exercise 10.6. List the point in the proof at which each of these hypotheses is used: positive stress coefficients; component regularity; grading compatibility; the unique component vacuum at weight zero; braiding; right exactness. Which arguments would fail if a component had a second simple module of weight zero?

Solution on page 300.

Chapter 11

From finite tops to the Haagerup boundary ring

We now know that U is proper and strongly rational. These two facts still leave many possible representation categories. The next step uses information already visible near the bottom of the lattice theory: six retained currents, eighty complementary currents, and the two eighty-one-dimensional glue spaces. A universal bound on lowest module weights makes that small information exhaustive.

The chapter follows a chain of implications. Fourteen modules of a known core bound all U-module tops. That bound fixes the branching of B and H. The extension algebra then limits the boundary category to rank four or rank six. Character and indicator arguments select the Haagerup fusion ring. The final tensor-category identification still needs unitarity and classification, so we keep it separate from the ring calculation.

11.1 Lowest weights of the fourteen core modules

The regular subalgebra $A_L \subset U$ has category $\mathrm{Fib} \boxtimes SO(3)_{13}$. Label its fourteen simple modules by $x_a = (1, a)$ and $y_a = (\tau, a)$ for $0 \le a \le 6$; the actual even affine label is $2a$.

Their top spaces can be calculated from a small lattice minimization. The six Potts modules in Example 9.1 give, for an integer charge m,

$$p_1(m) = \begin{cases} 0 & 3 \mid m, \\ 2/3 & 3 \nmid m, \end{cases} \qquad p_\tau(m) = \begin{cases} 2/5 & 3 \mid m, \\ 1/15 & 3 \nmid m. \end{cases}$$

The actual lowest lattice–Potts weight in charge j is

$$h_{\mathcal{T}}(\epsilon, j) = \min_{m \equiv j \ (13)} \left(\frac{m^2}{156} + \frac{\mathbf{1}_{m \text{ odd}}}{4} + p_\epsilon(m) \right). \tag{11.1}$$

The parity term is the left level-one affine contribution. Even m gives a left singlet and odd m a doublet. The right Cartan charge is m.

For $-6 \le k \le 6$, the grade calculation of Section 9.1.2 gives the parafermion lowest weight

$$h_{\mathrm{PF}}(a, k) = \frac{a(a+1)}{15} - \frac{k^2}{13} + \max(0, |k| - a). \tag{11.2}$$

The last term is important. When the required Cartan charge lies outside the finite affine highest-weight representation, it first appears at a positive affine grade. A twist calculation modulo one would lose precisely this integer correction.

To find an A_L top, minimize the sum of (11.1) and (11.2) over the extension charges, with $k = -5j$ modulo thirteen. This is genuinely finite. For $h_{\mathcal{T}}$ in each charge class, taking $|m| \le 6$ supplies a candidate at energy at most $14/13$, whereas $|m| \ge 13$ already exceeds that bound from its quadratic term alone. The parafermion contribution is fixed within the charge class. Hence $-12 \le m \le 12$ contains every minimizer.

Worked example 11.1 (Two top weights)**.** For x_1, the neutral charge gives zero lattice–Potts energy and parafermion energy $1 \cdot 2/15 = 2/15$. To prove that it is the minimum, first consider $|k| \ge 2$. The parafermion contribution is

$$\frac{2}{15} - \frac{k^2}{13} + |k| - 1 > \frac{2}{15} \qquad (2 \le |k| \le 6),$$

as follows by checking the minimum of the concave quadratic on this integer interval at its endpoints. The remaining $k = \pm 1$ require $j = \pm 5$. Their lattice–Potts minimum is $14/13$, already larger than $2/15$. The neutral candidate therefore wins, and its top is one dimensional.

For y_3, choose $m = 2$, hence $j = 2$ and $k = 3$. The two parts are

$$\frac{4}{156} + \frac{1}{15} = \frac{6}{65}, \qquad \frac{12}{15} - \frac{9}{13} = \frac{7}{65}.$$

Their sum is $1/5$. The conjugate charge $m = -2$ is also a minimizer, giving a two-dimensional top. This multiplicity has a concrete charge origin; it is not its quantum dimension.

11.2 Energy traces and the universal $17/10$ bound

Let E be the lowest space of any simple ordinary U-module, of weight h and dimension D. The compatible-top lemma says that E is a direct sum of complete A_L top representations. Let x_a and y_a now also denote their nonnegative integer multiplicities. The meaning is unambiguous when they occur in sums of dimensions.

On a core top let L, P, Z be the traces of the left affine, Potts, and PF_{13} energy operators, and let $Q = \operatorname{Tr} H_R(0)^2$. Define the additive bookkeeping quantity

$$W = L + \frac{83}{6084}Q + \frac{25}{6}P + \frac{25}{26}Z. \tag{11.3}$$

Why is the coefficient of Q correct? In a complete right $\mathfrak{sl}_2$ representation of highest weight r,

$$\frac{\operatorname{Tr} H_R(0)^2}{\dim} = \frac{r(r+2)}{3}, \qquad L_0^{A_{1,39}} = \frac{r(r+2)}{164}.$$

Thus the Sugawara trace on the complete space E equals $3Q/164$. Multiplying by the positive-stress coefficient $3403/4563$ gives $83Q/6084$. An individual core top need not be a complete right $\mathfrak{sl}_2$ representation. Its table entry W is therefore not asserted to be an ambient-energy trace on that core top, or an energy eigenvalue times its dimension. Only after all core tops are assembled into the actual E does their sum satisfy $W_{\text{total}} = \operatorname{Tr}_E L_0 = hD$. This distinction is why an isolated row can have $W/\dim$ greater than the final universal bound.

The finite minimization gives the following complete table:

	dim	h_{A_L}	$4680W$		dim	h_{A_L}	$4680W$
x_0	1	0	0	y_0	1	2/5	7800
x_1	1	2/15	600	y_1	7	8/15	36960
x_2	5	2/5	10440	y_2	15	4/5	110880
x_3	9	4/5	56640	y_3	2	1/5	4080
x_4	2	1/3	5520	y_4	14	11/15	81480
x_5	14	1	148560	y_5	6	2/5	18240
x_6	6	4/5	54240	y_6	2	1/5	4920

This table is a finite arithmetic input obtained from the actual core realization. It can be reconstructed from the two minimization formulas and charge multiplicities; it does not assert any list of U-modules in advance.

It is instructive to recover the charges that are hidden by the three numerical columns. In the following table, an entry lists all minimizing absolute charge classes $|j|$; both signs are included when $j \neq 0$:

x_0	0	y_0	0
x_1	0	y_1	$0, 5$
x_2	$0, 3$	y_2	$0, 2, 3, 5$
x_3	$0, 2, 3, 6$	y_3	2
x_4	6	y_4	$1, 2, 4, 6$
x_5	$1, 4, 6$	y_5	$1, 4$
x_6	4	y_6	4

One must keep tied representatives of a charge class. For $\epsilon = 1$ the ties at $j = 1, 4, 5$ are respectively $m = 1, -12$, $m = 4, -9$, and $m = 5, -8$. For $\epsilon = \tau$, the ties at $j = 3, 5, 6$ are $m = 3, -10$, $m = 5, -8$, and $m = 6, -7$. Negative j gives the opposite charges. An odd m contributes a left doublet, so it contributes two states at that right charge.

For example, the $j = \pm 1$ ties of x_5 supply charges ± 1 with multiplicity two and ± 12 with multiplicity one. Its $j = \pm 4$ ties supply charges ± 4 with multiplicity one and ± 9 with multiplicity two. The $j = \pm 6$ contribution supplies one state at each of ± 6. The total is $2(2+1+1+2+1) = 14$, reproducing the dimension column.

Completeness of the right current representations imposes two further inequalities,

$$y_2 \geq x_5, \qquad y_4 \geq x_5 + x_6. \tag{11.4}$$

The minimizers just listed give, for the whole space E,

$$\mathrm{mult}(12) = x_5, \quad \mathrm{mult}(10) = y_2, \quad \mathrm{mult}(9) = 2(x_5 + x_6), \quad \mathrm{mult}(7) = 2y_4.$$

In a finite $\mathfrak{sl}_2$ representation the multiplicity of a nonnegative weight cannot increase as the weight increases by two. Comparing 10 with 12 and 7 with 9 gives exactly (11.4). The factors of two in the odd charges are the left doublets. A right-current representation cannot be assembled from isolated Cartan eigenlines while omitting its required lower-weight companions.

The following nonnegative certificate packages the entire bound:

$$\begin{aligned} 4680\left(\frac{17}{10}D - W_{\mathrm{total}}\right) &= 8460(y_2 - x_5) + 28716(y_4 - x_5 - x_6) \\ &\quad + 7956x_0 + 7356x_1 + 29340x_2 + 14964x_3 + 10392x_4 \\ &\quad + 22212x_6 + 156y_0 + 18732y_1 + 11832y_3 \\ &\quad + 1188y_4 + 29496y_5 + 10992y_6 \geq 0. \end{aligned} \tag{11.5}$$

Each coefficient can be checked directly from the table. For example, the y_0 coefficient is $4680(17/10) - 7800 = 156$. For x_5, the right side contributes $-8460 - 28716 = -37176$, agreeing with $4680(17/10) \cdot 14 - 148560$. This is a small linear-programming dual certificate written as an identity. The two representation constraints absorb the troublesome negative coefficient, and every remaining term is visibly nonnegative. No optimization program is needed to verify it.

Proposition 11.2 (Every simple top lies below weight two)**.** *Every simple ordinary U-module has lowest weight*

$$0 \leq h \leq \frac{17}{10} < 2.$$

Proof. The preceding chapter gives $h \geq 0$. Trace the positive-stress identity on E: compact averaging preserves trace, so $W_{\mathrm{total}} = hD$. Equations (11.4) and (11.5) give the upper bound. □

The bound need not be optimal. Its decisive feature is being less than two. That excludes a new integer-graded summand that first appears at weight two or later.

11.3 Why eighty currents determine the whole B branching

Since U is strongly rational, its restriction inside B is semisimple. Every summand has integer lowest weight, because B is integer graded. The universal bound therefore permits only weights zero and one. The weight-zero summand is the vacuum, occurring once.

There are $86 - 6 = 80$ remaining currents in B_1. The finite construction provides an irreducibility certificate for their zero-mode action. Its logic is elementary and valuable.

Lemma 11.3 (A cyclic one-dimensional eigenline)**.** *Let a finite-dimensional positive Hermitian space carry a star-closed algebra of operators containing commuting self-adjoint Cartan operators. If one joint Cartan eigenspace is one dimensional and a nonzero vector on that line is cyclic, the representation is irreducible.*

Proof. An invariant subspace has an invariant orthogonal complement because the algebra is star closed. Both are Cartan invariant. The one-dimensional joint eigenspace must lie entirely in one of them. Cyclicity makes that summand the entire space. □

Construction input 11.4 (Five irreducible finite tops)**.** On the eighty-dimensional current complement, the known U zero modes are star closed and a vector in the one-dimensional Cartan weight $(0, 10)$ line is cyclic. In each of the four mixed nontrivial $E_6 \oplus A_2$ lattice sectors of lowest weight one, the analogous star-closure, one-dimensional eigenline, and cyclicity tests prove irreducibility of its eighty-one-dimensional top.

There are four mixed sectors because both $\mathbb{Z}/3$ discriminant coordinates are nonzero. The E_8 extension chooses two of these B-module sectors as its glue sectors. The bound rules out any summand invisible in these tops. Hence

$$B\downarrow_U = U \oplus X, \qquad h_X = 1, \quad \dim X_1 = 80. \tag{11.6}$$

The two actual glue sectors in H/B restrict to simple modules Y, Y^* with 81-dimensional tops, giving

$$H\downarrow_U = \mathbf{1} \oplus X \oplus Y \oplus Y^*. \tag{11.7}$$

The notation Y^* means the contragredient. It does not yet say whether Y and Y^* are isomorphic.

11.4 The extension algebra remembers a boundary category

Let $\mathcal{C} = \mathrm{Rep}(U)$ and regard H as its commutative extension algebra. Because H is holomorphic, this algebra is Lagrangian: its local-module category is trivial and its dimension is the square root of the global dimension of $\mathcal{C}$. Let

$$\mathcal{F} = \mathcal{C}_H$$

be the category of *all* right H-modules. Local modules and all modules must not be confused: the former are trivial here, while the latter will be the desired nontrivial boundary category. Theorem 6.3 gives $\mathcal{C} \simeq Z(\mathcal{F})$.

Temporarily write $\sigma = \mathrm{FPdim}_{\mathcal{C}} B$. Extension dimension identities give

$$\mathrm{qdim}\, X = \sigma - 1, \qquad \mathrm{qdim}\, Y = \sigma, \qquad \mathrm{Dim}\, \mathcal{C} = 9\sigma^2, \quad \mathrm{Dim}\, \mathcal{F} = 3\sigma.$$

Here Dim means global dimension, the sum of squared simple quantum dimensions. Positivity of the nonvacuum conformal weights gives the positive spherical structure through character

asymptotics. At this stage pseudo-unitarity is sufficient; a categorical C^* structure is supplied later.

For center induction I and forgetful functor F, the canonical algebra is $I(\mathbf{1})$. The induction identity $FI(\mathbf{1}) = \bigoplus_z z \otimes z^*$, over the simple boundary objects z, implies

$$\operatorname{rank} \mathcal{F} = \dim \operatorname{End}_{\mathcal{C}}(H) = \begin{cases} 4 & Y \ncong Y^*, \\ 6 & Y \cong Y^*. \end{cases}$$

Indeed, the endomorphism dimension of a semisimple object is the sum of the squares of its multiplicities. Four distinct summands give $1+1+1+1=4$; identifying the two glue summands gives $1+1+2^2=6$.

Why should an automorphism of an extension say anything about a boundary particle? The canonical-algebra theorem gives a useful answer that does not yet require a unitary tensor category.

Lemma 11.5 (Extension automorphisms and invertible boundary objects)**.** *For a pseudo-unitary fusion category $\mathcal{F}$, the algebra automorphism group of $I(\mathbf{1}) \in Z(\mathcal{F})$ injects into the group of invertible objects of $\mathcal{F}$.*

Proof. The canonical algebra has a normalized special Frobenius structure with multiplication m, comultiplication Δ, and $m\Delta = \mathrm{id}$. Besides composition, its endomorphisms carry the convolution product $f * g = m(f \otimes g)\Delta$. The standard two-product description identifies composition with multiplication in $K_0(\mathcal{F}) \otimes \mathbb{C}$, while the primitive convolution idempotents are $d_z[z]$, with $d_z > 0$ the simple dimensions.

An algebra automorphism obeys $f * f = f$, and its composition inverse does too. Consequently their fusion-algebra expansions have coefficients either zero or d_z. Their fusion product is $[\mathbf{1}]$. Nonnegative fusion coefficients prevent cancellation, so every pair of objects in their supports must multiply only to copies of the unit. Rigidity implies that such a pair consists of inverse invertible simples. Both supports are therefore singletons, with dimension one. Composition respects the assignment, proving the group injection. □

The actual order-three automorphisms of H/B therefore supply three invertible objects in $\mathcal{F}$. More is known than their fusion group: their associator is untwisted. The discriminant form of B is

$$q(a,b) = \frac{2a^2 + b^2}{3}, \qquad (a,b) \in (\mathbb{Z}/3)^2.$$

Its diagonal and antidiagonal isotropic lines are complementary. Induction from the latter across the extension selected by the former gives a full tensor copy of $\mathrm{Vec}_{\mathbb{Z}/3}$ in $\mathcal{F}$. Knowing only that there are three invertible objects would not rule out a nontrivial associator class.

11.5 Excluding a sign extension and the rank-four fork

First, X cannot be invertible. Otherwise the extension $B = U \oplus X$ would have an order-two grading and a sign automorphism g with $B^g = U$. Its fixed current Lie algebra would be $\mathfrak{sl}_2 \oplus \mathfrak{sl}_2$, of rank two. But every finite-order automorphism of E_6 fixes a toral subalgebra of dimension at least four, and every such automorphism of A_2 fixes one of dimension at least one. The nonisomorphic ideals cannot be exchanged. The fixed rank is therefore at least five, a contradiction. This also excludes the possibility that all six boundary simples are invertible.

At rank four there are three invertibles and one noninvertible, with near-group ring

$$\alpha^3 = 1, \quad \alpha\rho = \rho\alpha = \rho, \qquad \rho^2 = 1 + \alpha + \alpha^2 + m\rho.$$

The relevant pseudo-unitary near-group theorem restricts m to $0, 2, 3, 6$. This is a specialized categorical theorem, not a consequence of associativity of the displayed ring alone.

The case $m = 0$ makes the extension B/U have dimension two, so X would be invertible. For $m = 2$, the categorical reduction gives a possibly twisted A_4 double and a holomorphic charge-eight extension whose A_4 fixed algebra is U. The holomorphic extension is E_8. An elementary current estimate then excludes it. The adjoint traces of nontrivial order-two and order-three automorphisms of E_8 are bounded below by -8 and -4, respectively. Since A_4 has three involutions and eight elements of order three,

$$\dim(E_8^{A_4}) \geq \frac{248 + 3(-8) + 8(-4)}{12} = 16 > 6.$$

The remaining two cases are excluded by the following specialized lemma from the construction's modular-character argument. We state its hypotheses because the conclusion is not a general restriction on arbitrary near-group categories.

Lemma 11.6 (Rank-four character obstruction)**.** *Suppose a strongly rational charge-eight VOA has positive nonvacuum lowest weights, an extension $B = V_{E_6 \oplus A_2}$ with branching $B = U \oplus X$, and the rank-four pseudo-unitary near-group center obtained above, with its untwisted order-three subgroup and the specified lattice extension. For multiplicities $m = 3, 6$, its current dimension must have one of the following forms:*

m	*necessary number of currents*
3	$15 + A + 10B$ *or* $14 + 42A$
6	$8 + A$ *or* $5 + 12P$

$A, B, P \in \mathbb{Z}_{\geq 0}$.

The letters A, B, P in this table are weighted dimensions of polar module tops, not the previously named vertex algebras.

The integers are nonnegative because they count actual top states. None of the displayed forms equals six. We explain the mechanism of the specialized lemma next; the assertion includes the finite theta-coefficient calculation rather than treating its entries as new lattice construction assumptions.

The mechanism deserves explanation even without reproducing a theta atlas. A *polar* character term is a negative power of q, equivalently a state of weight below $c/24$; the cusp is $q = 0$, or $\tau \to i\infty$. Center induction determines a finite modular representation for a suitable difference of branching characters. Multiplication by η^8 makes it a weight-four theta-type function. The allowed polar data, before that multiplication, are specified by the nonnegative module-top dimensions above. Once those are prescribed, principal-part cancellation determines the function; its vacuum coefficient yields the displayed restrictions. One keeps the factor of two between the vacuum character and the sum/difference branching coordinates. Dropping it changes the current obstruction. This is an application of modular functions to a finite categorical alternative, not an assumption that the desired character vector was known from the start.

11.5.1 A complete theta cancellation in one rank-four alternative

We work the first $m = 3$ orientation in detail. This demonstrates where the integers 15 and 10 come from, and lets us check the potentially dangerous vacuum factor of two.

The categorical projection in this orientation is indexed like the six integrable $G_{2,3}$ modules, whose finite-dimensional labels are $1, 7, 27, 77, 14, 64$, tensored with the three $A_{2,1}$ labels. For each pointed A_2 charge, retain the four-component difference vector

$$v = (\chi_1 - \chi_{64}, -\chi_{77}, -\chi_{14}, -\chi_7).$$

Here the labels describe the modular projection fixed by the rank-four induction argument; they do not assert that the unknown VOA is already an affine algebra. The known lattice extension gives $\chi_1 + \chi_{64} = \chi_B$ in the pointed vacuum charge. Since $\chi_1 = \chi_U$, a difference between two solutions of this extension problem obeys

$$\delta v_{0,0} = 2\,\delta\chi_U.$$

The lattice character is fixed, so the plus and minus coordinates cannot vary independently.

After multiplication by η^8, the projected modular representation is the even Weil representation on $(r,s) \in (\mathbb{Z}/7) \times (\mathbb{Z}/3)$, even in each coordinate. Representatives are $0 \le r \le 3$ and $s = 0, 1$. Its T phases are $\exp(2\pi i(r^2/7 + s^2/3))$. One convenient normalization of S is the tensor product of

$$S^{(7)}_{00} = -1/\sqrt{7}, \qquad S^{(7)}_{0r} = -2/\sqrt{7}, \qquad S^{(7)}_{r0} = -1/\sqrt{7},$$
$$S^{(7)}_{rs} = -2\cos(4\pi rs/7)/\sqrt{7} \quad (r,s > 0)$$

with the even order-three Fourier matrix $3^{-1/2}\left(\begin{smallmatrix} 1 & 2 \\ 1 & -1 \end{smallmatrix}\right)$. The unbalanced first row and column simply record that a nonzero even coordinate represents two finite-group elements.

We construct weight-four forms for this representation using the rank-four lattice $L_7 \oplus A_2$, where

$$\mathrm{Gram}(L_7) = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}.$$

Let u, v be its displayed basis and put $w = v - u/2$, so $u \perp w$, $u^2 = 2$, $w^2 = 7/2$. Write a vector in its real plane as $pu + tw$, and put

$$a = p^2, \qquad b = 7t^2/4, \qquad c = (y,y)/2 \quad (y \in A_2 \otimes \mathbb{R}).$$

These lowercase letters are temporary quadratic coordinates. Both $H_1 = a - b$ and $H_2 = a + b - c$ are harmonic homogeneous quadratics. The two coordinates in each plane contribute equal Laplacians, which cancel in these differences. Poisson summation therefore makes the coset theta sums

$$\Theta_{H,\lambda}(\tau) = \sum_{x \in L+\lambda} H(x) q^{(x,x)/2}$$

vector-valued modular forms of weight $4/2 + 2 = 4$.

Only three small coefficients are needed:

(r,s)	q-power	Θ_{H_1}	Θ_{H_2}
$(1,0)$	$1/7$	$-1/7$	$1/7$
$(3,0)$	$2/7$	$3/7$	$4/7$
$(0,0)$	1	2	-4

Each entry can be counted by hand. The first coset has shortest vector $2w/7$, giving $a = 0, b = 1/7$. In the second there are two shortest vectors with $a = 1/4, b = 1/28$, each of energy $2/7$; hence their total H_1 value is $3/7$ and their total H_2 value is $4/7$. In the vacuum coset, the two L_7 roots contribute 2 to both forms. The six A_2 roots contribute zero to H_1 and -6 to H_2.

The only allowed nonvacuum polar top dimensions are A at weight $2/7$ and B at weight $1/7$. Either can be zero if the actual module begins an integer grade higher. Use $G_{2,3} \otimes A_{2,1}$ as a reference solution of the same modular projection and lattice-extension problem. Its central charge is $3 \cdot 14/(3+4) + 2 = 8$, its currents number $14 + 8 = 22$, and its polar dimensions are $A = 7, B = 0$. The fundamental seven-dimensional G_2 representation has quadratic Casimir two in this normalization, so its Sugawara weight is $2/(3+4) = 2/7$. Thus the polar coefficients of $\eta^8 \delta v$ are $7 - A$ at $(3,0)$ and $-B$ at $(1,0)$.

Matching them against $\lambda\Theta_{H_1} + \mu\Theta_{H_2}$ gives two linear equations,

$$\frac{-\lambda+\mu}{7} = -B, \qquad \frac{3\lambda+4\mu}{7} = 7 - A.$$

Their solution is $\lambda = 7 - A + 4B$, $\mu = 7 - A - 3B$. After division by η^8, the unmatched remainder has no poles at the cusp. The finite-image modular representation makes it a holomorphic vector of modular functions on a compact finite cover, so it is constant. Its only possible T-fixed coordinate is $(r, s) = (0, 1)$, and the displayed Fourier S matrix does not preserve that line. Hence the constant is zero: the matched theta expression is the whole difference, not merely its first few coefficients.

Finally the vacuum coefficient of the theta expression is $2\lambda - 4\mu$. By the vacuum factor of two above it is also $2(\dim U_1 - 22)$; the leading vacuum terms canceled before multiplication by η^8, so no further eta coefficient enters this difference. We obtain

$$2(\dim U_1 - 22) = 2(7 - A + 4B) - 4(7 - A - 3B), \qquad \dim U_1 = 15 + A + 10B.$$

Since A, B are nonnegative integers, six currents are impossible. The other three alternatives in Lemma 11.6 are the corresponding specialized theta cancellations for the reverse orientation and the $m = 6$ projected representation. The complete calculation here explains the method and fixes its normalization; the remaining coefficient formulas are retained as the stated specialized lemma.

11.6 Deriving the six-object fusion ring

Rank four is excluded, so $H = \mathbf{1} \oplus X \oplus 2Y$. Its endomorphism algebra is

$$\mathrm{End}_{\mathcal{C}}(H) \cong \mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}).$$

The two-product description of the canonical center algebra identifies this composition algebra with the complexified boundary fusion algebra. In particular the fusion ring is noncommutative.

The group of three invertibles acts on the three noninvertibles. An order-three permutation is either trivial or transitive. The trivial case would split off the two nontrivial characters of $\mathbb{C}[\mathbb{Z}/3]$ as one-dimensional ideals, leaving an $M_2(\mathbb{C})$ ideal on which the dimension character is nonzero. A matrix algebra has no nonzero one-dimensional representation, so this is impossible.

Choose a noninvertible ρ and label the orbit by $\rho, \alpha\rho, \alpha^2\rho$. Duality conjugates the left and right actions of the invertible group, so the right action is transitive too. It commutes with the left action and hence is the same three-cycle or its inverse; the identity permutation has now been excluded. The same cycle would make the algebra generated by α, ρ commutative. Thus $\rho\alpha = \alpha^2\rho$. To see the self-duality explicitly, write $\rho^* = \alpha^k\rho$. Then $(\alpha^i\rho)^* = \alpha^{i+k}\rho$; applying duality twice gives $2k = 0$ modulo three. Hence $k = 0$, and all three noninvertibles are self-dual. Frobenius reciprocity gives exactly one invertible in ρ^2, namely the unit; commuting with α equates the remaining three coefficients. Therefore

$$\alpha^3 = 1, \quad \rho\alpha = \alpha^2\rho, \qquad \rho^2 = 1 + m(\rho + \alpha\rho + \alpha^2\rho), \quad m \geq 1. \tag{11.8}$$

There is still a multiplicity m to determine. Recognizing the shape of the Haagerup ring is not enough to set it to one.

11.7 Indicators: rotating a junction without braiding

Let $\mathcal{C}$ be a spherical fusion category over $\mathbb{C}$, with a fixed pivotal identification $j_X : X \to X^{**}$. A unitary fusion category has a canonical positive spherical structure. The theorem used here needs

the spherical structure, without assuming the complete unitarity proved later in Chapter 12. For $n \geq 1$, consider the finite-dimensional junction space $\mathrm{Hom}(\mathbf{1}, X^{\otimes n})$. Bend its first outgoing strand around the junction and return it as the last strand, using evaluation, coevaluation, and j_X^{-1}. With the required reassociations this defines the *pivotal cyclic rotation* $R_{X,n}$. It uses no crossing and requires no braiding on $\mathcal{C}$. In formulas, put $Y = X^{\otimes(n-1)}$ and let $f : \mathbf{1} \to X \otimes Y$. The two stages are

$$\widetilde{f} = (\mathrm{ev}_X \otimes 1_Y)(1_{X^*} \otimes f) : X^* \to Y,$$
$$R_{X,n}(f) = (\widetilde{f} \otimes j_X^{-1})\,\mathrm{coev}_{X^*} : \mathbf{1} \to Y \otimes X,$$

with the same associators understood as in the cup and cap identities. The nth Frobenius–Schur indicator is the ordinary linear trace

$$\nu_n(X) = \mathrm{Tr}_{\mathrm{Hom}(\mathbf{1},X^{\otimes n})}(R_{X,n}). \tag{11.9}$$

Pivotal coherence gives $R_{X,n}^n = 1$. For simple X, this immediately explains two values used below:

$$\nu_1(X) = \delta_{X,\mathbf{1}}, \qquad \nu_2(X) = \begin{cases} 0, & X \not\simeq X^*, \\ +1 \text{ or } -1, & X \simeq X^*. \end{cases}$$

Indeed, $\mathrm{Hom}(\mathbf{1}, X^{\otimes 2}) \simeq \mathrm{Hom}(X^*, X)$ is zero or one-dimensional, and its rotation squares to the identity. The sign measures the symmetry of the chosen self-duality under pivotal transpose; it is not fixed merely by the equation $X \simeq X^*$.

Worked example 11.7 (The group formula from an invariant projector)**.** Take $\mathcal{C} = \mathrm{Rep}(G)$ for a finite group, with its usual tensor product and positive spherical structure. For a representation $D : G \to \mathrm{GL}(V)$, the junction space is $(V^{\otimes n})^G$. Pivotal rotation is the cyclic permutation

$$C(v_1 \otimes \cdots \otimes v_n) = v_2 \otimes \cdots \otimes v_n \otimes v_1.$$

The projector onto invariants is $P = |G|^{-1} \sum_{g \in G} D(g)^{\otimes n}$, and C commutes with P. Therefore its trace on invariants equals $\mathrm{Tr}(CP)$ on the whole tensor power. For any matrix A, expansion in a tensor basis gives

$$\mathrm{Tr}(CA^{\otimes n}) = \sum_{i_1,\ldots,i_n} A_{i_1 i_2} A_{i_2 i_3} \cdots A_{i_n i_1} = \mathrm{Tr}(A^n).$$

Thus, writing $\chi_V(g) = \mathrm{Tr}\, D(g)$, we obtain the classical formula

$$\nu_n(V) = \frac{1}{|G|} \sum_{g \in G} \chi_V(g^n). \tag{11.10}$$

For $G = \mathbb{Z}_3 = \langle t \rangle$ and the one-dimensional character $\chi(t) = \zeta_3 = e^{2\pi i/3}$, this becomes

$$\nu_n(\chi) = \frac{1 + \zeta_3^n + \zeta_3^{2n}}{3} = \begin{cases} 1, & 3 \mid n, \\ 0, & 3 \nmid n. \end{cases}$$

In particular $\nu_3(\chi) = 1$, although $\nu_1(\chi) = \nu_2(\chi) = 0$. The invariant projector and the cyclic trace identity explain this without any modular data.

Worked example 11.8 (A unitary self-dual object with negative second indicator)**.** The quaternion group $Q_8 = \{1, -1, \pm i, \pm j, \pm k\}$ has a unitary irreducible representation V of dimension two, with character $\chi_V(1) = 2$, $\chi_V(-1) = -2$, and zero on the other six elements. It is self-dual: its character is real. The two central elements square to 1, while the other six square to -1. Thus

$$\nu_2(V) = \frac{2 \cdot 2 + 6 \cdot (-2)}{8} = -1.$$

The invariant bilinear pairing is alternating rather than symmetric. Its sign does not conflict with the positive Hermitian inner product that makes the representation unitary; these are different pairings. Consequently even a positive unitary setting does not allow us to replace the sign $\varepsilon = \nu_2(\rho)$ below by $+1$ without an argument.

The bridge to modular data is the following structural theorem. Let $F : Z(\mathcal{C}) \to \mathcal{C}$ be forgetful and I its two-sided adjoint. Equip the center with the ribbon twist induced by the same spherical structure. Then, for every object X and positive integer n,

$$\nu_n(X) = \frac{\mathrm{Tr}_{\mathrm{cat}}(\theta^n_{I(X)})}{\mathrm{Dim}(\mathcal{C})}, \qquad \mathrm{Dim}(\mathcal{C}) = \sum_{a \text{ simple}} d_a^2. \tag{11.11}$$

This is the center-induction theorem of Ng–Schauenburg. Here the numerator is a *categorical* trace: if $I(X) = \bigoplus_Z b_Z Z$, then it is $\sum_Z b_Z d_Z \theta_Z^n$. The twist is θ_Z, without the common central-charge phase of a VOA's modular T matrix.

Worked example 11.9 (Checking the normalization in the pointed center)**.** For $\mathcal{C} = \mathrm{Vec}_{\mathbb{Z}_3}$ with trivial associator and positive spherical structure, let α be the simple object of degree one. The center simples from Chapter 5 are (g, k), with $d_{(g,k)} = 1$, twist ζ_3^{gk}, and $F(g, k) = g$. Adjunction gives

$$I(\alpha) = (1, 0) \oplus (1, 1) \oplus (1, 2), \qquad \frac{\mathrm{Tr}_{\mathrm{cat}}(\theta^n_{I(\alpha)})}{3} = \frac{1 + \zeta_3^n + \zeta_3^{2n}}{3}.$$

Directly, $\mathrm{Hom}(\mathbf{1}, \alpha^{\otimes n})$ is zero unless $3 \mid n$; when nonzero it is one-dimensional and the untwisted pivotal rotation is the identity. This verifies both sides of (11.11). The fusion relation $\alpha^3 = \mathbf{1}$ alone would not prove $\nu_3(\alpha) = 1$: a nontrivial associator can give a nontrivial rotation phase on that same one-dimensional junction space.

For the rank-six fusion ring just derived, let d_m be the positive solution of $d_m^2 = 1 + 3md_m$. The simple dimensions are $1, 1, 1, d_m, d_m, d_m$. Put $s_m = 1 + d_m^2$. Its global dimension is $3s_m$, so the theorem says exactly

$$\mathrm{Tr}_{\mathrm{cat}}(\theta^n_{I(\rho)}) = 3s_m\nu_n(\rho).$$

The first moment is therefore zero, and the second is $3s_m\varepsilon$ with $\varepsilon = \nu_2(\rho) \in \{1, -1\}$. There is no extra dimension factor d_m on the right. On the left, a residual induced summand Z_j of multiplicity n_j, dimension $3d_m n_j$, and twist θ_j contributes $3d_m n_j^2 \theta_j^n$. This is the source of both the normalization and the squared multiplicities in the following argument.

11.8 Twist moments force multiplicity one

Put $d_m = (3m + \sqrt{9m^2 + 4})/2$, so $d_m^2 = 1 + 3md_m$, and put $s_m = 1 + d_m^2$. We use the positive spherical structure already obtained above from character asymptotics. No categorical adjoint is needed for the following positive-dimension trace argument.

First derive the part of center induction that contains invertibles. Write $R = \rho + \alpha\rho + \alpha^2\rho$. The underlying-object formula $FI(x) = \bigoplus_z z \otimes x \otimes z^*$ and the fusion ring give

$$FI(\alpha) = FI(\alpha^2) = 3\alpha + 3\alpha^2 + 3mR.$$

For example, the three invertible z contribute 3α to $FI(\alpha)$; the three self-dual noninvertibles each contribute $\alpha^2 + mR$. By adjunction,

$$\dim \mathrm{End}\, I(\alpha) = \dim \mathrm{End}\, I(\alpha^2) = \dim \mathrm{Hom}(I(\alpha), I(\alpha^2)) = 3.$$

An endomorphism dimension of three is a sum of squared nonnegative integral multiplicities, so each induction object is a multiplicity-free sum of three simples. Their mutual inner product

three makes them the same sum, say $C_0 + C_1 + C_2$. They are disjoint from $I(\mathbf{1})$. Adjunction gives one copy of each α, α^2 in $F(C_j)$. Centrality makes its three noninvertible multiplicities equal. Hence

$$F(C_j) = \alpha + \alpha^2 + b_j R, \qquad \sum_j b_j = 3m, \qquad d_{C_j} = 2 + 3b_j d_m.$$

The pointed subcategory has trivial associator in this construction, so $\nu_3(\alpha) = 1$. The third twist trace therefore says

$$\sum_j d_{C_j} \theta_{C_j}^3 = 3s_m = \sum_j d_{C_j}.$$

The twists have modulus one and the dimensions are positive. Equality in the triangle inequality forces each $\theta_{C_j}^3 = 1$. This does not yet say that all three cubic roots occur. The additional identity $\nu_1(\alpha) = 0$ gives

$$2\sum_j \theta_{C_j} + 3d_m \sum_j b_j \theta_{C_j} = 0.$$

The number d_m is a real quadratic irrational, while the twists now lie in $\mathbb{Q}(\zeta_3)$. These two quadratic fields intersect in $\mathbb{Q}$, so both sums vanish separately. Three cubic roots summing to zero must be $1, \zeta_3, \zeta_3^2$, once each. The weighted sum then forces $b_0 = b_1 = b_2 = m$. Thus all three C_j have dimension s_m.

Together with $I(\mathbf{1}) = \mathbf{1} + X + 2Y$, this supplies six central simples of dimensions $1, d_m^2$ and four copies of s_m, with twists $1, 1, 1, 1, \zeta_3, \zeta_3^2$. All central summands whose forgetful object contains an invertible have now been exhausted. Every remaining central simple Z_j has twist θ_j and forgets to $n_j R$, with n_j a positive integer; centrality again equates its three noninvertible multiplicities. The global-dimension identity and the first two Frobenius–Schur indicator identities give

$$\sum_j n_j^2 = 4m^2 + 2, \tag{11.12}$$

$$\sum_j n_j^2 \theta_j = -\frac{m^2 + m\sqrt{9m^2+4}}{2}, \tag{11.13}$$

$$\sum_j n_j^2 \theta_j^2 = \frac{-m^2 + (2\varepsilon - m)\sqrt{9m^2+4}}{2}, \qquad \varepsilon \in \{1, -1\}. \tag{11.14}$$

The roots of unity in these sums are counted with multiplicity n_j^2. Distinctness is not assumed.

Here is how the normalizations enter. Denote the known nontrivial summands of $I(\mathbf{1})$ by X, Y, of dimensions d_m^2, s_m, and the three summands of $I(\alpha)$ by C_0, C_1, C_2, each of dimension s_m. Their twists are respectively $1, 1$ and $1, \zeta_3, \zeta_3^2$, as just derived using both the first and third indicators. The center's global dimension is the square of $3s_m$. Therefore

$$9s_m^2 = 1 + d_m^4 + 4s_m^2 + 9d_m^2 \sum_j n_j^2.$$

Using $d_m - d_m^{-1} = 3m$ reduces this to $\sum_j n_j^2 = 4m^2 + 2$.

Adjunction gives the induction decomposition

$$I(\rho) = mX + mY + m(C_0 + C_1 + C_2) + \sum_j n_j Z_j.$$

The first indicator of a nonunit simple is zero. Thus the first twist trace of this object is

$$0 = m(d_m^2 + s_m) + 3d_m \sum_j n_j^2 \theta_j,$$

because the three C_j contributions cancel. This yields $-(m/3)(2d_m + d_m^{-1})$ for the sum, which is the first moment in (11.13). The second twist trace equals $3s_m\varepsilon$, where $\varepsilon = \nu_2(\rho)$ is 1 or -1 for a self-dual simple. Again the C_j contributions cancel:

$$3s_m\varepsilon = m(d_m^2 + s_m) + 3d_m \sum_j n_j^2\theta_j^2.$$

Subtract the first equation and use $s_m/d_m = d_m + d_m^{-1} = \sqrt{9m^2+4}$ to obtain (11.14).

Here is the arithmetic force of the equations. Write $9m^2 + 4 = t^2D$ with D squarefree. The allowed squareclasses have strong restrictions. Since $(3m)^2 < 9m^2 + 4 < (3m+1)^2$, one has $D > 1$. Reduction modulo three shows that $3 \nmid t$ and $D \equiv 1 \pmod 3$. Also $7 \nmid D$: divisibility by seven would require $m^2 \equiv 5 \pmod 7$, which is impossible.

Here $\varphi(n)$ is Euler's totient, the number of residues modulo n that are coprime to n. Larson's root-sum bound is a theorem about expressing a quadratic irrationality as a sum of roots of unity; it is stronger than the triangle inequality. Applying it to two copies of the root multiset clears the half-integral coefficients in (11.13) and gives

$$4m^2 + 2 \geq \frac{mt}{2}\varphi(2D).$$

For precisely these squareclasses, except $D = 10$, elementary prime-factor estimates give $\varphi(2D)/\sqrt{D} \geq 12/\sqrt{13}$. For odd D, this ratio is the product of $(p-1)/\sqrt{p}$ over its distinct prime factors. A single allowed prime is at least thirteen; two distinct prime factors contribute at least $40/\sqrt{55}$, from five and eleven, which already exceeds $12/\sqrt{13}$. Every additional factor is greater than one. For even D, an additional factor $\sqrt{2}$ occurs. The smallest single odd prime is five, giving the exception $D = 10$; otherwise it is at least eleven, and two odd factors give an even larger bound. These restrictions matter: the inequality is false for arbitrary squarefree numbers such as two or five. Consequently

$$4m^2 + 2 \geq \frac{6m\sqrt{9m^2+4}}{\sqrt{13}}.$$

For $m > 1$ this contradicts

$$36m^2(9m^2+4) - 13(4m^2+2)^2 = 4(m^2-1)(29m^2+13) > 0.$$

For the exceptional case we need a little field arithmetic. A cyclotomic field $\mathbb{Q}(\zeta_N)$ is generated by $\zeta_N = e^{2\pi i/N}$. Its automorphisms send ζ_N to ζ_N^a with $\gcd(a, N) = 1$, preserving rational polynomial relations. Averaging over the automorphisms that fix a subfield K lands in K. For example, the two conjugates of ζ_3 have average $(\zeta_3 + \zeta_3^2)/2 = -1/2$. The conductor of an abelian number field is the smallest N for which it lies in $\mathbb{Q}(\zeta_N)$. A root of order exactly N is primitive. A Ramanujan sum adds primitive roots without weights; a quadratic Gauss sum weights roots by a quadratic character, which modulo an odd prime is 1 on nonzero squares, -1 on nonsquares, and 0 at zero. The weighted sums isolate the square-root part of the average below.

The exceptional squareclass cannot simply be ignored. For a sum of N roots of unity equal to $-a - b\sqrt{10}$, with nonnegative integers a, b, the conductor-forty orbit estimate gives $N \geq a + 8b$. Equality, when the squared-root sum also lies in $\mathbb{Q}(\sqrt{10})$, forces that second sum to be a.

Here is the orbit mechanism behind this specialized arithmetic lemma. Take a common cyclotomic field and form the union of all Galois images of the root multiset under automorphisms fixing $\mathbb{Q}(\sqrt{10})$. This gives an orbit-constant multiset without assuming that the original multiplicities were constant on orbits. The imprimitive quadratic Gauss-sum formulas separate two costs. Primitive orders not divisible by forty contribute rational Ramanujan sums. Orders

divisible by forty have zero rational part, with nonzero quadratic orbit sums $\pm\sqrt{10}$ costing at least eight roots each. Thus a negative rational contribution of size a costs at least a roots, independently of the $8b$ cost of the quadratic part. Equality allows only copies of -1 and the negative quadratic orbit of primitive fortieth roots. Squaring those eight roots runs once through the primitive twentieth roots, whose sum is $\mu(20) = 0$. Here μ is the Möbius function: it vanishes when its argument has a squared prime factor, and otherwise is $(-1)^r$ for r distinct prime factors. Squaring -1 contributes one. Finally, Galois averaging multiplies the original second sum by the group order precisely because that sum lies in $\mathbb{Q}(\sqrt{10})$. Dividing by that order gives a.

When $9m^2 + 4 = 10t^2$, reduction modulo two first makes m even; reduction modulo four then makes t even. The conductor-forty bound reads

$$4m^2 + 2 \geq \frac{m^2}{2} + 4mt.$$

For $m \geq 4$, $t/m = \sqrt{(9 + 4/m^2)/10} > \sqrt{9/10} > 15/16$, so the right side is greater than $17m^2/4 > 4m^2 + 2$, a contradiction. The only remaining positive possibility below four is $m = t = 2$, where the first-moment bound is an equality and requires second sum 2. But (11.14) gives -2 or $-2 - 4\sqrt{10}$. Thus $m = 1$.

Proposition 11.10 (The resulting dimensions and twists)**.** *The boundary fusion ring is the Haagerup ring, with*

$$d = \frac{3 + \sqrt{13}}{2}, \qquad s = 1 + d^2 = 2 + 3d.$$

Its center has twelve simple objects, with quantum dimensions

$$1, \quad d^2, \quad s \text{ four times}, \quad 3d \text{ six times}.$$

The six residual twists have exponents $2, 5, 6, 7, 8, 11$ *modulo thirteen, divided by thirteen.*

The constituent X is the unique dimension-d^2 central simple, later called b. Retain the name Y for the original embedded extension's dimension-s constituent. The later choice of a center equivalence and the canonical $\mathcal{H}_3$ extension uses a for that selected algebra's constituent. The two twist-one dimension-s simples, a and c_0, have the same unflavoured characters; the original Y cannot be identified with the selected a from branching counts alone. None of these central labels denotes the boundary invertible object α.

For $m = 1$, the first moment is $(-1 - \sqrt{13})/2$ and has exactly six roots available. Average each root over a cyclotomic Galois group fixing $\sqrt{13}$. The quadratic Gauss-sum estimate bounds the absolute value of its $\sqrt{13}$ coefficient by $1/12$, with equality only for primitive orders thirteen or twenty-six. Thus the required coefficient $-1/2$ exhausts all six roots, each at its negative extremum. This argument allows the fractional orbit multiplicities that averaging may produce. The corresponding six-root negative quadratic periods are, respectively,

$$\frac{-1 - \sqrt{13}}{2}, \qquad \frac{1 - \sqrt{13}}{2}.$$

Here the per-root estimate can be seen directly from the conductor. For a primitive root of order N, the average over the subgroup fixing $\sqrt{13}$ is rational unless 13 divides N. When it does, insert the quadratic character modulo thirteen into the ordinary cyclotomic average. The untwisted numerator is the Ramanujan sum $\mu(N)$, where μ is the Möbius function. The twisted numerator vanishes unless $N = 13m$, with m squarefree and coprime to thirteen: a redundant prime-power factor cancels by a geometric sum. In the remaining case the Chinese remainder theorem reduces the twisted numerator to $\pm\sqrt{13}$, using the quadratic Gauss sum modulo thirteen. Division by $\varphi(N) = 12\varphi(m)$ gives an irrational coefficient of absolute value $1/[12\varphi(m)]$. Equality with $1/12$ requires $m = 1$ or 2, giving exactly the two orders above. Their rational coefficients are

respectively $-1/12$ and $1/12$. This explains both the six-root budget and the sign test without assuming that the original multiset was itself a union of Galois orbits.

The entire root budget is already needed for the irrational coefficient, so there is no room for rational-only orbits or cancellation. The prescribed rational part therefore selects order thirteen. All original roots are among the six nonresidue roots. These are linearly independent over $\mathbb{Q}$: a relation would give a polynomial of degree at most twelve, with zero constant term, divisible by $\Phi_{13}(t) = 1 + t + \cdots + t^{12}$, hence the zero polynomial. Comparison with the quadratic period forces each root to occur once. Since every simple was repeated n_j^2 times, there are six such simples and each has $n_j = 1$. Multiplication by two interchanges residues and nonresidues, so the second moment is $(-1 + \sqrt{13})/2$, giving $\varepsilon = 1$. This is why conductor thirteen appears in the center even though the original lattice extension had order three.

11.9 A ring is not yet a unitary tensor category

The ring and twists determine the standard Haagerup modular-data target in this setting, but modular data alone do not identify a braided category in general. The later analytic argument proves complete unitarity, allowing the extension algebra to be treated as a commutative Q-system and its all-module category as a unitary fusion category. The unitary classification of the untwisted six-object ring then leaves two possibilities, $\mathcal{H}_2$ and $\mathcal{H}_3$.

These categories are Morita equivalent, so their centers are equivalent. Consequently the chiral representation category is the Haagerup center $Z(\mathcal{H}_3)$. The original embedded H/U all-module category is still only identified as $\mathcal{H}_2$ or $\mathcal{H}_3$. To obtain an extension with the specifically chosen $\mathcal{H}_3$ boundary category, one transports its canonical Lagrangian algebra through the center equivalence. The resulting holomorphic VOA is again isomorphic to V_{E_8}, but the embedding and its chosen boundary realization need not be the original ones.

Keeping these last distinctions visible prevents a common shortcut: "the numbers look Haagerup" is not a tensor-category proof, and "the center is Haagerup" does not select a particular Morita representative as the boundary category.

11.10 Exercises

Exercise 11.1. Using $m = 6$ and the neutral Fibonacci label, verify the top weight $1/3$ for x_4. Compute the two summands separately.

Solution on page 300.

Exercise 11.2. Check the coefficients of x_0, x_5, and y_4 in (11.5). Explain why a positive bound for each individual ratio $W/\dim$ would be weaker than using the current character inequalities.

Solution on page 300.

Exercise 11.3. If a semisimple object decomposes as $\bigoplus_i m_i X_i$ with pairwise nonisomorphic simples, prove $\dim \mathrm{End}(\bigoplus_i m_i X_i) = \sum_i m_i^2$. Apply this to both branching alternatives for H.

Solution on page 300.

Exercise 11.4. Show that the ring (11.8) has two one-dimensional representations and one two-dimensional representation over $\mathbb{C}$. Write the matrices in the two-dimensional representation and check all defining relations.

Solution on page 301.

Exercise 11.5. Verify the difference-of-squares factorization excluding $m > 1$ outside the exceptional squareclass. For $m = 2$, compute both possible second moments and compare them with the equality-case requirement.

Solution on page 301.

Exercise 11.6. Use $d^2 = 1 + 3d$ to check that the squared dimensions of the twelve central simples sum to $[3(1 + d^2)]^2$. Distinguish this global dimension from the total quantum dimension and from the rank twelve.

Solution on page 301.

Chapter 12

From positive states to a unitary theory

The lattice construction begins in a positive Hilbert space. Why, then, is a separate chapter on unitarity necessary? The answer is that the vacuum representation is only one part of a rational CFT. Other modules describe charged sectors, and intertwining operators describe their fusion. Positivity must be compatible with all of these structures. An abstract equivalence of fusion rules, or even of braided categories without specified positive structures, does not automatically establish this compatibility for the actual VOA operators.

This chapter explains the analytic route used in the construction. The model-specific inputs are the regularity and branching results of Part II and the twelve-module count. Standard comparison theorems for lattice VOAs, conformal nets, and annulus insertions supply the analytic framework. We prove the elementary domain lemmas in full and describe how the remaining comparison fits together. The account retains the research manuscript's finite-input qualifications.

12.1 Three levels of unitarity

Vacuum unitarity means that U itself has a positive invariant Hermitian form and the corresponding antilinear PCT involution. This follows here from PCT stability inside the positive lattice theory.

Strong unitarity means that every simple ordinary U-module admits a positive invariant Hermitian form. It is a statement about all sectors, including ones not explicitly visible as subspaces of the original lattice vacuum.

Complete unitarity additionally requires positivity of the canonical Hermitian forms on fusion, or intertwining-operator, spaces. It gives the VOA representation category its unitary modular tensor structure. Choosing positive norms separately on the simple modules does not prove this last condition: the associativity and braiding operators must also be compatible with the fusion pairings.

A finite-dimensional analogy is helpful. Giving each vector space in a diagram an inner product does not make every linear map in the diagram unitary. Nor can arbitrary independent changes of metric be assumed to respect a collection of tensor products and composition laws. Fusion positivity is the corresponding compatibility problem for chiral fields.

12.2 Why smeared fields lead to operator algebras

An individual field $Y(v, z)$ is usually unbounded and may not define an operator on every vector of the Hilbert completion. To obtain a well-controlled operator, smear it against a smooth

function on the circle:

$$Y(v,f)=\sum_n \widehat{f}_n\, v_{(n)},$$

with the indexing adjusted to the chosen circle convention. Polynomial energy bounds have the schematic form

$$\left\|v_{(n)}\xi\right\| \le C(1+|n|)^s\,\|(1+L_0)^r\xi\| \tag{12.1}$$

on finite-energy vectors. Smooth Fourier coefficients decay faster than any power, so these estimates permit smearing on a dense domain and then taking the closure.

A *conformal net* assigns to every interval I of the circle a von Neumann algebra $\mathcal{N}(I)$ of bounded observables. Inclusions of intervals give inclusions of algebras, disjoint intervals give commuting algebras, and conformal transformations transport the assignment. The vacuum is cyclic, and rotations have positive energy. A von Neumann algebra is an algebra of bounded operators closed under adjoints and weak operator limits.

For a strongly local unitary VOA, the closed smeared fields generate such a net. An unbounded operator can be affiliated with a von Neumann algebra through the bounded operators in its polar decomposition and its spectral projections. Strong locality is the analytic statement that the algebras associated with disjoint supports commute; formal OPE locality alone must not silently be substituted for it.

Lattice VOAs satisfy the required locality and energy properties, and the relevant unitary subalgebra inheritance theorems apply to $U \subset B \subset H = V_{E_8}$. Since all three share the ambient conformal vector, they also share the rotation generator on their vacuum spaces and central charge eight.

12.3 Sectors, statistical dimension, and complete rationality

A sector is a representation of the local observable algebras that is compatible with their inclusions. In the chiral net setting, localized transportable representations admit fusion, statistical dimensions, and braiding. These are the operator-algebraic counterparts of VOA modules and their tensor products. Localized means equivalent to the vacuum outside a chosen interval; transportable means that the charge can be relocated to any other interval by an equivalent representation.

For a finite-index sector realized by a localized endomorphism σ, the statistical dimension can be defined through the local inclusion

$$d(\sigma)^2 = [\,\mathcal{N}(I) : \sigma(\mathcal{N}(I))\,].$$

Here I contains the localization region and the index has the standard minimal-index normalization. Thus the statistical dimension is a square root of an operator-algebra index, rather than the number of vectors in the sector's Hilbert space.

A completely rational net satisfies suitable split and strong additivity conditions and has finite two-interval index. The split property expresses a controlled independence of observables in separated intervals. Strong additivity says that removing one point from an interval does not lose its generated observable algebra. Under these conditions, the sectors form a unitary modular tensor category, and the two-interval index is

$$\mu(\mathcal{N}) = \sum_i d_i^2.$$

A holomorphic net has only its vacuum sector and $\mu = 1$.

There is a second way to form the local algebras, using point insertions regularized by incoming and outgoing annuli. Its hypothesis is called *bounded localized vertex operators*. For

each allowed insertion z between annuli A, B, one requires some $0 < s < 1$ for which

$$v \otimes \xi \longmapsto BY(s^{L_0}v, z)A\xi$$

extends to a bounded operator on the Hilbert tensor product. This controls the inserted state as well as the vector on which the field acts. The local algebras generated by these insertions and by the no-insertion products BA must commute for disjoint intervals. The lattice theory has this property, and the unitary subalgebra inherits it. The annulus section below explains the geometry and its module counterpart.

The construction first obtains complete rationality through a bounded-insertion version of the subnet. Regularity of U and the lattice ambient theory imply finite index for this inclusion; finite index in a holomorphic completely rational net then implies complete rationality of the subnet. A vacuum-net comparison carries that property to the net generated by smeared fields.

At this stage only an isomorphism-invariant property has been transported. An abstract isomorphism between two nets does not yet identify their particular embedded extensions or their individual annulus insertion operators. Those more precise identifications are proved when needed below.

12.4 A domain lemma that prevents a hidden gap

Algebraic branching decomposes finite-energy states. To claim a decomposition of the actual represented net, we must show that the projections commute with the closures of the fields. Commutation on a domain is not always enough for unbounded operators. The right statement is the following.

A subspace D is a *core* for a closed operator T when it is dense in $D(T)$ for the graph norm $(\|\xi\|^2 + \|T\xi\|^2)^{1/2}$. Approximation in this norm controls both the vector and its image, which is exactly what the next proof needs.

Lemma 12.1 (Reduction from a common core)**.** *Let $T_0 : D \to \mathcal{K}$ be a densely defined closable operator, and let P be an orthogonal projection. Suppose $PD \subset D$ and $T_0 P\xi = PT_0\xi$ for $\xi \in D$. Then P reduces $T = \overline{T_0}$ and its adjoint. It also commutes with the von Neumann algebra generated by T.*

Proof. If $\xi \in D(T)$, choose $\xi_j \in D$ with $\xi_j \to \xi$ and $T_0\xi_j \to T\xi$. Then $P\xi_j \to P\xi$ and $T_0 P\xi_j = PT_0\xi_j \to PT\xi$. Closedness gives $P\xi \in D(T)$ and $TP\xi = PT\xi$. The same argument applies to $1 - P$, so T is the direct sum of its restrictions to $P\mathcal{K}$ and $(1-P)\mathcal{K}$. Its adjoint is the direct sum of their adjoints. The polar partial isometry and the spectral projections of $|T|$ are consequently block diagonal, proving the last assertion. □

The proof does not assume that D is a core for T^*. It does require domain invariance $PD \subset D$, which is an actual condition, not a consequence of a formal commutator written without domains.

Apply this lemma to the orthogonal projections in

$$H \downarrow_U = U \oplus X \oplus Y^{\oplus 2}.$$

The algebraic summands carry the inherited positive forms and energy bounds. Here the smeared field is, by construction, the closure of its restriction to finite-energy vectors. This agrees with closing its restriction to smooth energy vectors: let $Q_R = \mathbf{1}_{[0,R]}(L_0)$ be the spectral projection onto energies between zero and R. The energy bound $\|Y(v,f)\xi\| \le C_{v,f}\,\|(1+L_0)^r\xi\|$ implies $Y(v,f)Q_R\xi \to Y(v,f)\xi$ on the smooth domain. Thus finite-energy vectors are a core for these particular closures; no claim about an arbitrary closed extension is needed. The projections preserve finite-energy vectors and commute with the modes; hence they reduce the closed smeared

fields. Conversely, a net intertwiner commutes with rotations and, on finite-energy vectors, with the Fourier coefficients of the fields. It is therefore a U-module map. Algebraically inequivalent simple summands remain inequivalent net sectors, with the same multiplicities $1, 1, 2$.

12.5 Counting the net sectors without assuming a tensor comparison

First we must establish finite index for the *actual* smeared-field inclusion $\mathcal{N}_U \subset \mathcal{N}_H$. Complete rationality as an abstract property is not a substitute for this step. Write I' for the interior complementary interval. If $x \in \mathcal{N}_H(I) \cap \mathcal{N}_U(I)'$, locality also makes x commute with $\mathcal{N}_U(I')$. Strong additivity lets us remove the two endpoints of I from any test interval, so these two algebras generate the global represented subnet. Thus x commutes with its local diffeomorphisms and the common rotation generator L_0. The zero-energy space of H is one-dimensional, and consequently $x\Omega$ is a scalar multiple of Ω. The ambient vacuum is separating for $\mathcal{N}_H(I)$, meaning that $y\Omega = 0$ for y in this algebra implies $y = 0$. Indeed, locality makes such a y kill every vector $b\Omega$ with $b \in \mathcal{N}_H(I')$, and these vectors are dense by local vacuum cyclicity, the Reeh–Schlieder property of a conformal vacuum net. This is cyclicity for a nonempty interval, not merely for the algebra of the whole circle. Applying this to x minus its vacuum scalar proves local irreducibility.

The finite-index theorem for irreducible local extensions of a completely rational net now applies. Its hypotheses use the finite number of sectors and their finite statistical dimensions. The dual canonical object is the already established restriction $\mathbf{1} \oplus X \oplus 2Y$, and the ambient net is holomorphic. The corresponding extension algebra is therefore Lagrangian. Its endomorphism algebra is

$$\operatorname{End}(H \downarrow_U) \cong \mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}). \tag{12.2}$$

Its dimension is six and its multiplication is noncommutative. The center-induction correspondence from Chapter 11 identifies this dimension with the number of boundary simples and this composition algebra with their complex fusion algebra. There is one further bridge before repeating the algebraic boundary argument. If the rank-six boundary category were pointed, its global dimension would be six. The actual lattice-net inclusion has $[\mathcal{N}_H : \mathcal{N}_B] = 3$, so multiplicativity of index would give $[\mathcal{N}_B : \mathcal{N}_U] = 2$. A local index-two extension has a sign involution. The unitary VOA/net automorphism correspondence lifts that involution to B; compatibility with fixed points and injectivity of the subalgebra-to-subnet correspondence identify its fixed VOA with U. This contradicts the explicit no-sign calculation in Chapter 11.

The order-three lattice extension automorphisms give three invertible boundary objects. The intermediate discriminant form provides an untwisted pointed subcategory. The rank-six and indicator argument then gives the Haagerup–Izumi ring with multiplicity one.

The net boundary category is already a C^* fusion category, so its spherical structure is positive. Its center has the twelve dimensions

$$1,\ d^2,\ \underbrace{s, s, s, s}_{4},\ \underbrace{3d, 3d, 3d, 3d, 3d, 3d}_{6}, \qquad s = 1 + d^2.$$

Because the extension is holomorphic, the Lagrangian-center correspondence identifies this center with the subnet's sector category. Thus the net has twelve simple sectors.

The logical point is the independence of the count. We have not assigned the VOA's character S matrix to the net by a presumed tensor equivalence. We used the actual branching, extension, and positive boundary category to obtain the count on the net side. This leaves room to prove the comparison of modules next.

12.6 Reconstructing ordinary modules and exhausting them

The net-to-VOA reconstruction theorem produces a positive weak module from a net representation on a Hilbert space $\mathcal{K}$. In its bounded-energy version, one starts with

$$M = \bigcup_{R>0} \mathbf{1}_{[0,R]}(L_0)\mathcal{K},$$

without assuming in advance that the net sector has finite energy multiplicities. Henriques–Tener's Definition 9.4 defines the reconstructed smeared fields by closing their restrictions to M, so M is their common core. Theorem 9.5 identifies these closures with the represented net fields; Remark 9.8 extends this statement to the bounded-energy domain without the finite-multiplicity assumption. The closed-field comparison also identifies the Virasoro actions, so the conformal zero mode agrees with the rotation generator. The underlying vacuum reconstruction is the original strongly local U.

Why is an irreducible net sector an ordinary simple module rather than a larger weak module? Regularity of U decomposes M algebraically into ordinary simple modules. Choose one nonzero summand S. Its graded pieces are finite-dimensional. Let P be the orthogonal projection onto its Hilbert closure. Spectral decomposition gives

$$P\mathbf{1}_{[0,R]}(L_0)\mathcal{K} \subset \bigoplus_{h+n\le R} S_{h+n} \subset S. \tag{12.3}$$

Thus $PM \subset M$ before any field commutation is asserted. Invariance of the Hermitian form expresses adjoints of modes as finite combinations of modes of PCT-transformed states. Since all these modes preserve S, testing matrix elements against S shows that P commutes with the modes on M.

Smearing preserves this commutation as a Hilbert-space equality. Lemma 12.1 then makes P commute with the net. Irreducibility forces $P = 1$. Equation (12.3) now implies $M = S$, proving both ordinaryness and simplicity. Finite energy multiplicity has been established, not assumed for the net sector being reconstructed.

The comparison is fully faithful on finite direct sums. One useful detail concerns boundedness: two positive invariant forms on a simple ordinary module differ by a positive scalar. Indeed, compare the forms on each finite-dimensional energy space. The resulting grade-preserving operator commutes with all modes. An eigenspace meeting the lowest space is a nonzero submodule, hence the whole simple module. An algebraic isomorphism therefore has a unitary rescaling and extends to the Hilbert completions.

There are twelve inequivalent net simples and twelve algebraic simple U-modules. The fully faithful reconstruction sends the former injectively into the latter, so it exhausts them. Every simple algebraic module is therefore unitarizable: strong unitarity has been proved.

This is so far a linear $*$-comparison. Counting simples does not construct a tensorator, identify braidings, or prove fusion positivity. A tensorator consists of coherent isomorphisms comparing the image of a fusion product with the fusion product of the images. The next step supplies the additional analytic input.

12.7 Annuli regularize point insertions

A round annulus acts, up to the conventional central-charge scalar, as r^{L_0} for $0 < r < 1$. Its effect is to suppress high energy. For a nonround annulus the operator also contains conformal reparametrizations. Placing a field between an incoming and an outgoing annulus leads to a regularized insertion

$$X_v(B, z, A) = \overline{B_U Y_U(v, z) A_U}. \tag{12.4}$$

The overline denotes the bounded extension of the insertion form in the allowed geometric interior. It does not mean that an unsmeared field was everywhere defined beforehand.

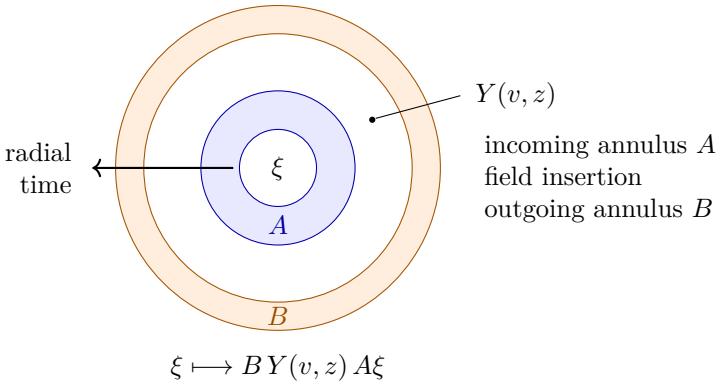


Figure 12.1: Radial ordering of a regularized insertion. The inner annulus prepares the input, the field lies in the intervening region, and the outer annulus regularizes the output. This round schematic illustrates the order BYA; actual geometric annuli need not be concentric, and boundedness requires the analytic insertion theorem rather than the picture alone.

Worked example 12.2 (Energy damping in one oscillator)**.** Let $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$ on an oscillator Hilbert space and $N|n\rangle = n|n\rangle$. The operator $a^\dagger$ is unbounded, but $a^\dagger r^N$ is bounded because

$$\left\|a^\dagger r^N\right\|^2 = \sup_{n\geq 0}(n+1)r^{2n} < \infty.$$

Exponential damping beats polynomial growth. The annulus argument is a geometric and many-field version of this mechanism. The inverse r^{-N} remains unbounded; density arguments must therefore be stated with their domains.

Lemma 12.3 (A dense domain for undoing an incoming annulus)**.** *Suppose A, C are bounded injective annulus operators with dense range, and $AC = \lambda r^{L_0}$ with $\lambda \neq 0$, $0 < r < 1$. On an ordinary module, $M^{\mathrm{fin}} \subset \operatorname{Ran} A$ and $A^{-1}M^{\mathrm{fin}} = CM^{\mathrm{fin}}$ is dense, where M^{fin} is the algebraic finite-energy space and the inverse is taken only on* $\operatorname{Ran} A$.

Proof. For a finite-energy vector v, $r^{-L_0}v$ is again a finite vector, so $v = AC\lambda^{-1}r^{-L_0}v$ lies in $\operatorname{Ran} A$. Injectivity of A then gives $A^{-1}M^{\mathrm{fin}} = CM^{\mathrm{fin}}$. The right side is dense because M^{fin} is dense and C is bounded with dense range. □

The chosen geometric annuli satisfy these hypotheses. The inner Jordan domain of an incoming annulus contains the origin. Place a sufficiently small round disc $r\mathbb{D}$ inside that domain and choose the intervening annulus C, with matching boundary parametrizations. Sewing A and C gives the round annulus, so its central lift yields $AC = \lambda r^{L_0}$. The geometric annulus theorem supplies injectivity and dense range for both factors. The outgoing adjoint admits the same construction. Thus the lemma verifies actual range and density conditions needed below.

When both test vectors become finite-energy after the incoming and outgoing annulus actions, their formal field matrix element is a finite Laurent polynomial. For an input energy p, output energy q, and field weight k, only the mode $n = p + k - q - 1$ contributes. This elementary grading observation allows analytic continuation of matrix coefficients through the insertion interior. Here is the geometric detail behind this continuation. Write

$$D_A = \{\xi : A_M\xi \in M^{\mathrm{fin}}\}, \qquad D_{B^*} = \{\eta : B_M^*\eta \in M^{\mathrm{fin}}\}.$$

Both are dense by the preceding lemma and its outgoing analogue. For one geometric representative of the pair, let $O_{\mathrm{in}}, O_{\mathrm{out}}$ be the incoming and outgoing comparison regions and put

$$\Omega_{\mathrm{ins}} = \mathrm{int}\big(\overline{O_{\mathrm{in}}} \cup \overline{O_{\mathrm{out}}}\big).$$

The geometric point-insertion theorem supplies a bounded operator $G_v(z)$ whose matrix coefficients are holomorphic on this domain, which lies in $\mathbb{C}^\times$. The finite-Laurent coefficient is holomorphic there as well. Every component of Ω_{ins} meets $O_{\mathrm{in}} \cup O_{\mathrm{out}}$: the part omitted from this union lies in two nowhere-dense boundaries and cannot contain a component, which is open. The identity theorem therefore extends each tested finite-Laurent coefficient throughout that component. Repeat for the geometric representatives; the prescribed insertion interior is the union of their allowed interiors. Boundedness and the dense test domains make these continuations unique on overlaps. The argument compares bounded extensions of sesquilinear forms; it does not assume that an unsmeared field is defined on every Hilbert-space vector.

12.8 Why paired lifts matter

Annulus and diffeomorphism actions involve central extensions. A full rotation acts on a simple ordinary module of lowest weight h by $Z = e^{2\pi i L_0} = e^{2\pi i h}$, while it acts trivially on the integer-graded vacuum VOA. Consequently two geometric lifts can agree in the vacuum and still differ by a nontrivial phase in a charged sector. Normalizing a vacuum scalar cannot erase that ambiguity.

The comparison used here chooses lifts for an incoming–outgoing pair after localizing both with one common diffeomorphism. Undoing that common lift gives actions A_M, B_M in each sector. More explicitly, choose a common localizer δ so that $\mathsf{U}_U(\delta)A_U$ and $B_U\mathsf{U}_U(\delta)^*$ belong to the interval algebra. With one lift $\widetilde{\delta}$, the actions are

$$A_M = \mathsf{U}_M(\widetilde{\delta})^* \pi_M(\mathsf{U}_U(\delta)A_U)\,,$$
$$B_M = \pi_M(B_U\mathsf{U}_U(\delta)^*)\,\mathsf{U}_M(\widetilde{\delta}).$$

Here U denotes diffeomorphism action, not the VOA U, and π_M is the net representation on the Hilbert completion of M. The geometric comparison matches the incoming annulus parametrization and its central normalization as part of this construction. Changing the common lift by Z^m changes the pair to

$$A_M \longmapsto Z^{-m}A_M, \qquad B_M \longmapsto B_M Z^m.$$

Every U-mode shifts energy by an integer, so it commutes with Z. The regularized insertion is unchanged. This cancellation uses the pair; it does not require a global choice of lift for each individual annulus.

The resulting specific comparison identity is

$$\pi_M\big(X_v(B,z,A)\big) = \overline{B_M Y_M(v,z)A_M}. \tag{12.5}$$

The geometric transport theorem proves it first in the incoming interior, the adjoint identity proves the outgoing version, and the finite-Laurent continuation proves it on the full allowed interior. Empty-interior pairs are covered by the no-insertion identity $\pi_M(BA) = B_M A_M$. On the vacuum space, locality and Haag duality then identify the concrete bounded-insertion and smeared-field nets. Write $\mathcal{A}_{\mathrm{bl}}$ for the bounded-insertion vacuum net. Haag duality says that the algebra of an interval is exactly the commutant of the complementary interval's algebra, in the vacuum representation. On the common vacuum Hilbert space the argument is the chain

$$\mathcal{A}_{\mathrm{bl}}(I) \subset \mathcal{N}_U(I) \subset \mathcal{A}_{\mathrm{bl}}(I')' = \mathcal{A}_{\mathrm{bl}}(I).$$

The first inclusion is the comparison just proved; the second uses locality and the first inclusion at I'; the equality uses Haag duality of the conformal vacuum net $\mathcal{A}_{\mathrm{bl}}$. Thus the conclusion identifies the actual algebras.

12.9 The final positivity implication and its scope

The fusion-positivity theorem applies to a simple regular unitary VOA with bounded localized vertex operators, provided every ordinary module is unitarizable and has the corresponding net representation. These hypotheses are now established for U: strong unitarity supplies the positive modules, and their representations satisfy (12.5). It gives positivity of the canonical fusion forms. Thus the construction concludes that U is completely unitary, under the preceding finite construction inputs and structural and analytic theorems.

Now the actual extension algebra can be treated as a positive Q-system. Its category of all modules is a unitary fusion category with the already determined Haagerup–Izumi ring and untwisted pointed subcategory. The unitary classification gives the two possibilities $\mathcal{H}_2$ and $\mathcal{H}_3$. Their Morita equivalence identifies their centers, so

$$\operatorname{Rep}(U) \simeq Z(\mathcal{H}_3)$$

as unitary ribbon categories. This conclusion uses the unitary classification and center correspondence, rather than asserting that a natural linear reconstruction functor has already been proved to preserve every tensor operation.

For the original embedded E_8/U extension, the boundary category may be $\mathcal{H}_2$ or $\mathcal{H}_3$. To obtain a specified $\mathcal{H}_3$ extension, transport its canonical Lagrangian Q-system along the center equivalence. The positive extension correspondence then gives a holomorphic strongly rational unitary VOA of central charge eight, which the central-charge-eight holomorphic VOA classification identifies with V_{E_8}. Isomorphism of the ambient holomorphic VOAs does not identify their embeddings of U.

The net-side diagonal full theory has a route using complete rationality and sector reconstruction before the final fusion-form comparison. The specified VOA extension uses the stronger complete-unitarity conclusion. Distinguishing these routes explains which claims survive if a particular analytic comparison is under review, and prevents one construction from being used circularly to prove its own hypotheses.

12.10 Exercises

Exercise 12.1. For $0 < r < 1$, bound $\sup_{n\geq 0}(n+1)r^{2n}$ explicitly using the corresponding function on $x \geq 0$. Show why the bound diverges as $r \to 1$.

Solution on page 301.

Exercise 12.2. Prove Lemma 12.1 for the complementary projection $1 - P$ and explain exactly where it is used to obtain a statement about T^*.

Solution on page 302.

Exercise 12.3. A fully faithful linear functor sends twelve simple objects into a semisimple category with twelve simples. Prove essential surjectivity on finite direct sums. Explain why the same argument does not construct a tensorator.

Solution on page 302.

Exercise 12.4. On a module with $h = 1/3$, compute the phase of a full rotation. Show that it is invisible on the vacuum and cancels in $BZ^mY(v,z)Z^{-m}A$ for $v \in U$. What grading property is needed?

Solution on page 302.

Exercise 12.5. If an extension restricts as $\mathbf{1} \oplus X \oplus Y \oplus Y^*$, compute its endomorphism algebra when $Y \not\cong Y^*$ and when $Y \cong Y^*$, assuming the named simple objects are otherwise distinct. Relate the dimensions to the rank-four/rank-six alternatives.

Solution on page 302.

Chapter 13

Characters and the spectrum they determine

A character is a remarkably efficient record of a chiral state space. It tells us how many states occur at every energy, but it suppresses their matrix elements and can even fail to distinguish inequivalent modules. In the present construction, characters become available after properness, regularity, and the modular data have been established. We will see how the six-current result selects the character vector and then turn its coefficients into physical primary counts.

13.1 Three different dimensions attached to a module

Let M_i be a simple ordinary module, with lowest weight h_i. Write

$$\chi_i(q) = q^{h_i - 1/3} A_i(q), \qquad A_i(q) = \sum_{n \geq 0} a_i(n) q^n. \tag{13.1}$$

The shift $1/3$ is $c/24$ at $c = 8$. The top dimension is $a_i(0)$, an ordinary positive integer. The categorical quantum dimension is $d_i = S_{0i}/S_{00}$, generally an irrational algebraic number. The number of Virasoro primary components at grade n is a third quantity, obtained after removing descendants. None of these three numbers can be substituted for another.

For example, the simple module b has top dimension 80 but quantum dimension d^2, where $d = (3 + \sqrt{13})/2$. The vacuum module has top dimension one, six states at weight one, and 120 states at weight two. Only 113 of the last states are Virasoro primaries. The remaining seven are the stress tensor and the derivatives of the six currents.

13.2 A hand calculation of the ambient lattice characters

The rank-eight oscillator factor begins

$$\prod_{m \geq 1} (1 - q^m)^{-8} = 1 + 8q + 44q^2 + 192q^3 + \cdots . \tag{13.2}$$

There are eight weight-one oscillators. At weight two there are eight single oscillators of frequency two and $\binom{9}{2} = 36$ symmetric pairs of frequency-one oscillators. At weight three the possibilities are one frequency-three oscillator, one frequency-two and one frequency-one oscillator, or three frequency-one oscillators. Their counts are $8 + 8 \cdot 8 + \binom{10}{3} = 192$.

The first theta coefficients of the root lattices are

$$\Theta_{E_6} = 1 + 72q + 270q^2 + 720q^3 + \cdots , \qquad \Theta_{A_2} = 1 + 6q + 6q^3 + \cdots .$$

Their product is $1 + 78q + 702q^2 + 2346q^3 + \cdots$. Multiplying by (13.2) gives

$$\chi_B = q^{-1/3}(1 + 86q + 1370q^2 + 11586q^3 + \cdots). \tag{13.3}$$

For instance, the weight-two coefficient is $702 + 78 \cdot 8 + 44 = 1370$. Momentum states and oscillators both contribute, which is why counting roots alone stops working after weight one.

For E_8, the theta series is the weight-four Eisenstein series

$$E_4(q) = 1 + 240 \sum_{n \geq 1} \sigma_3(n) q^n = 1 + 240q + 2160q^2 + 6720q^3 + \cdots ,$$

where $\sigma_3(n) = \sum_{m|n} m^3$. This follows from the modularity of an even unimodular lattice theta series and the one-dimensionality of weight-four modular forms for $SL_2(\mathbb{Z})$, together with the constant term one. Hence

$$\chi_{E_8} = \frac{E_4}{\eta^8} = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + \cdots). \tag{13.4}$$

These short computations already supply useful checks on branching. They do not identify the unknown subalgebra by themselves.

Worked example 13.1 (Why the ambient theory has 248 currents)**.** The first character coefficient can also be obtained without a modular form. In an orthonormal basis $e_1, \ldots, e_8$ of $\mathbb{R}^8$, use the standard lattice model

$$E_8 = D_8 \cup (D_8 + s), \qquad D_8 = \{n \in \mathbb{Z}^8 : \textstyle\sum_i n_i \in 2\mathbb{Z}\}, \qquad s = \tfrac{1}{2}(e_1 + \cdots + e_8).$$

A current comes from either one of the eight oscillator states or a momentum of squared norm two. The integer roots are $\pm e_i \pm e_j$ with $i < j$, giving $4\binom{8}{2} = 112$. A half-integer vector has squared norm at least $8/4 = 2$. Equality requires every coordinate to be $\pm 1/2$, and membership in $D_8 + s$ requires an even number of minus signs. Half of the 2^8 sign choices have that parity, giving 128 more roots. Thus there are 240 roots and $8 + 240 = 248$ currents.

The same model explains the $E_6 \oplus A_2$ count. Choose the A_2 simple roots $\beta = e_7 + e_8$ and $\gamma = e_6 - e_7$; their Gram matrix has diagonal two and off-diagonal minus one. A root orthogonal to both obeys $x_6 = x_7 = -x_8$. Among integer roots, this forces all nonzero coordinates into the first five positions, giving $4\binom{5}{2} = 40$. Among half-integer roots the last three signs are either $(+,+,-)$ or $(-,-,+)$. In each case exactly 16 choices of the first five signs give even total parity. Hence the orthogonal subsystem has $40 + 32 = 72$ roots.

To identify its type explicitly, take

$$e_1 - e_2, \quad e_2 - e_3, \quad e_3 - e_4, \quad e_4 - e_5, \quad e_4 + e_5,$$
$$\tfrac{1}{2}(-e_1 - e_2 - e_3 - e_4 - e_5 + e_6 + e_7 - e_8).$$

Their Gram matrix has diagonal entries two and the E_6 diagram: the third root meets the second, fourth, and fifth; the second meets the first, and the fifth meets the sixth. All other distinct pairings vanish. The six A_2 roots and the 72 perpendicular roots therefore contribute 78 momentum currents to B. Adding its eight oscillators gives $86 = (72 + 6) + (6 + 2)$, the dimensions of $\mathfrak{e}_6 \oplus \mathfrak{sl}_3$.

There remain $240 - 72 - 6 = 162$ roots outside this root subsystem. The product lattice has determinant $3 \cdot 3 = 9$, hence index $\sqrt{9} = 3$ in unimodular E_8. The remaining roots lie in the two nontrivial glue cosets. Negation exchanges those cosets, so each contributes 81 currents. This recovers the two glue-top dimensions used in the branching argument directly from lattice geometry. The orthonormal coordinates here are for counting; the explicit generators in Chapter 7 use their own simple-root coordinate basis.

Worked example 13.2 (Why a glue top is $27 \otimes 3$)**.** The preceding count gives 81 currents in each glue class. We can also see the finite representation carried by those currents, without a table of exceptional Lie-algebra representations. Keep $\beta = e_7 + e_8$ and $\gamma = e_6 - e_7$, and choose the A_2 fundamental weight

$$\nu = \frac{2\beta + \gamma}{3} = \frac{e_6 + e_7 + 2e_8}{3}, \qquad (\nu, \beta) = 1, \quad (\nu, \gamma) = 0, \quad \nu^2 = \frac{2}{3}.$$

An E_8 root r projects orthogonally to ν exactly when

$$r_6 = r_7, \qquad r_7 + r_8 = 1.$$

Among the integer roots, the solutions are $e_8 \pm e_i$ for $1 \le i \le 5$, together with $e_6 + e_7$: eleven roots. Among the half-integer roots, the last three coordinates must all be $+1/2$. An even number of the first five signs must be negative, giving 16 choices. There are therefore $11 + 16 = 27$ roots with this fixed A_2 projection.

Their perpendicular components $w = r - \nu$ are E_6 weights, all of squared length

$$w^2 = r^2 - \nu^2 = 2 - \frac{2}{3} = \frac{4}{3}.$$

Each has multiplicity one, since a lattice momentum supplies one state. For $r = e_6 + e_7$, the weight $w_0 = \frac{2}{3}(e_6 + e_7 - e_8)$ pairs to one with the sixth E_6 simple root in the preceding example and to zero with the other five. It is a fundamental highest weight.

Here is a direct check that these 27 weights belong to one representation. Write

$$\eta = \tfrac{1}{2}(-e_1 - e_2 - e_3 - e_4 - e_5 + e_6 + e_7 - e_8).$$

Reflection in η sends w_0 to the weight with first five coordinates $+1/2$ and last three $(1/6, 1/6, -1/6)$. The roots $\pm e_i \pm e_j$ in the first five coordinates generate permutations and even sign changes, producing all sixteen half-integer choices. Reflecting one with four minus signs in η reaches one of the ten remaining weights; the same permutations and sign changes reach the other nine. Thus all 27 form one Weyl orbit. The root zero modes connect their one-dimensional weight spaces. This is the irreducible *minuscule* 27: its weights form a single Weyl orbit, each with multiplicity one.

The A_2 fundamental representation has the three weights

$$\nu, \qquad \nu - \beta, \qquad \nu - \beta - \gamma.$$

Its Weyl transformations commute with those of E_6, so each of these three projections has the same 27 perpendicular weights. They give $27 \cdot 3 = 81$ roots in one glue class. Negation gives the dual representations in the other class. Including the two adjoint current spaces, the finite branching is

$$\mathfrak{e}_8\big|_{\mathfrak{e}_6 \oplus \mathfrak{sl}_3} \cong (78, 1) \oplus (1, 8) \oplus (27, 3) \oplus (\overline{27}, \overline{3}), \qquad 248 = 78 + 8 + 81 + 81.$$

The choice of which A_2 fundamental is called 3 fixes the conjugation labels. The two factors of a glue momentum have conformal weights $2/3$ and $1/3$, because lattice weight is half the squared length. Their sum is one, explaining why these are current tops without oscillator excitations.

This is branching under the finite current algebra $\mathfrak{e}_6 \oplus \mathfrak{sl}_3$. Restriction to the smaller algebra U still needs the irreducibility argument of Chapter 11. Neither this count nor equality of characters specifies the selected Haagerup extension algebra or its junctions.

13.3 Why the principal part can determine the whole vector

For a rational VOA, the characters form a vector-valued modular function. The T matrix contains the phase $e^{2\pi i(h_i - c/24)}$, not merely the categorical twist $e^{2\pi i h_i}$. The difference matters when deciding whether a constant vector is modular invariant.

The *principal part* is the sum of all negative powers of q in the character vector. In the Haagerup modular representation, strictly positive nonvacuum lowest weights leave only two possible polar terms:

$$q^{-1/3} \quad \text{in the vacuum coordinate,} \qquad \gamma q^{-7/39} \quad \text{in the } d_3 \text{ coordinate.}$$

The second exponent would arise from a lowest weight $2/13$. The integer $\gamma \geq 0$ is its top dimension if that weight occurs; $\gamma = 0$ means that this sector begins one or more grades later.

Proposition 13.3 (Uniqueness for the specified principal part)**.** *For the Haagerup character representation at $c = 8$, two character vectors with the same γ agree in every coefficient.*

Proof. The modular representation has finite image, a consequence of the congruence theorem for the rational VOA modular data used here. Its kernel Γ therefore has finite index in $SL_2(\mathbb{Z})$. Subtract the two vectors. Every component of the difference is holomorphic at infinity and, by its modular transformation law, at all cusps of Γ. A cusp is a Γ-orbit of rational boundary points or infinity. The quotient $\Gamma\backslash\mathfrak{H}$ of the upper half-plane becomes a compact Riemann surface after adding these finitely many cusps. Each component is holomorphic on $\mathfrak{H}$, and the absence of negative powers in its local q-expansion at every cusp means that it extends holomorphically over the added points. The maximum principle then makes it constant on this compact modular curve. The resulting vector must be fixed by both S and T_{ch}.

Only the coordinate with categorical twist $e^{2\pi i/3}$ is fixed by $T_{\mathrm{ch}} = e^{-2\pi i/3}\Theta$. But its S-image has a nonzero vacuum component, so no nonzero vector supported on that coordinate is also fixed by S. The constant difference is zero. □

Existence of the relevant vector-valued modular functions is a separate input. The Evans–Gannon construction supplies fundamental solutions $\Xi_{:,0}$ and $\Xi_{:,d_3}$ with the two allowed principal parts; the colon denotes the full twelve-component column. Their low coefficients give

$$\boldsymbol{\chi} = \Xi_{:,0} + \gamma \Xi_{:,d_3}, \qquad \dim U_1 = 6 + 13\gamma. \tag{13.5}$$

The independently proved six-current result forces $\gamma = 0$. Together with Proposition 13.3, this selects the entire character vector. The logic uses much more than matching a few coefficients: modular covariance, the allowed poles, absence of invariant constants, and a proved dimension of U_1 all enter.

This also clarifies the order of the construction. The first few vacuum dimensions $1, 6, 120, 956$ are outputs of finite lattice closure. The subsequent coefficient 6010 is supplied by the character theorem. Using that coefficient to prove the earlier closure statement would reverse the dependency.

13.4 The twelve characters at their first grades

The selected vector has the following initial data. Every row represents a simple extended-algebra module, and the grade is measured above its actual lowest weight.

Module	h_i	$a_i(0)$	$a_i(1)$	$a_i(2)$	$a_i(3)$
0	0	1	6	120	956
b	1	80	1250	10630	65042
a	1	81	1377	11583	71037
c_0	1	81	1377	11583	71037
c_1	1/3	3	243	2916	21870
c_2	2/3	27	594	5967	39852
d_1	6/13	7	292	3204	23010
d_2	11/13	42	777	7147	45367
d_3	15/13	119	1623	12996	76429
d_4	5/13	5	229	2738	19942
d_5	7/13	13	347	3804	26390
d_6	8/13	14	441	4445	30329

There are two instructive features. First, d_3 has twist exponent $2/13$ but lowest energy $15/13$. Twists know energies only modulo one. Second, a and c_0 are different simple modules with identical unflavoured characters. A trace of $q^{L_0-c/24}$ forgets enough structure to make this possible. Their defect eigenvalues and their categorical labels still distinguish them.

Worked example 13.4 (Branching checked grade by grade)**.** The restriction $B\downarrow_U = U \oplus b$ implies $\chi_B = \chi_0 + \chi_b$. Using the table gives

$$86 = 6 + 80, \qquad 1370 = 120 + 1250, \qquad 11586 = 956 + 10630,$$

in agreement with the independent lattice calculation. For the selected holomorphic extension,

$$\chi_{E_8} = \chi_0 + \chi_b + 2\chi_a. \tag{13.6}$$

The first three nontrivial checks are

$$\begin{aligned} 248 &= 6 + 80 + 2 \cdot 81, \\ 4124 &= 120 + 1250 + 2 \cdot 1377, \\ 34752 &= 956 + 10630 + 2 \cdot 11583. \end{aligned}$$

Replacing a by c_0 leaves the scalar character identity unchanged. Thus the identity verifies graded dimensions once branching is known; it cannot select an extension algebra or prove its multiplication.

13.5 Removing Virasoro descendants

Let $p(n)$ be the integer partition number. A positive-weight irreducible Virasoro Verma module has descendant generating function $\prod_{m\geq 1}(1-q^m)^{-1}$. At $c = 8$, every positive real highest weight has such an irreducible Verma module. Here is a useful check of this otherwise nontrivial statement.

Parametrize $c = 13 - 6(t + t^{-1})$. For $c = 8$, t lies on the unit circle with $t + t^{-1} = 5/6$ and nonzero imaginary part. The Kac determinant is the general-level analogue of the Virasoro Gram determinant computed in Chapter 2. Its possible singular weights are

$$h_{r,s} = \frac{1}{4}\big((r^2-1)t + (s^2-1)t^{-1} - 2rs + 2\big), \qquad r, s \in \mathbb{Z}_{>0}.$$

Reality requires $r = s$. Then $h_{r,r} = -7(r^2-1)/24 \leq 0$. No positive real highest weight is singular. At weight zero, the primitive vacuum null is $L_{-1}\mathbf{1}$, which has already been removed in the vacuum module.

To turn this irreducible-Verma calculation into a count of actual primary components, use the strong-unitarity conclusion proved in Chapter 12. In each finite-dimensional energy space, take the orthogonal complement of the descendants of lower-weight vectors. Repeating in increasing energy decomposes each module into Virasoro highest-weight summands. For the vacuum module, the needed positive form is already inherited from the lattice.

Put $P(q) = \prod_{m\geq 1}(1-q^m)$. Let $v_i(n)$ count Virasoro primary components of weight $h_i + n$, including the vacuum once. For nonvacuum modules,

$$\sum_{n\geq 0} v_i(n)q^n = P(q)A_i(q), \qquad i \neq 0. \tag{13.7}$$

For the vacuum module the correct formula is

$$\sum_{n\geq 0} v_0(n)q^n = P(q)A_0(q) + q. \tag{13.8}$$

To understand the correction, the vacuum descendants contribute $(1-q)/P(q)$ rather than $1/P(q)$. If $B_0(q)$ counts all vacuum module primaries, then $A_0 = (B_0 - q)/P$, hence $B_0 = PA_0 + q$.

Worked example 13.5 (The first interacting primary space)**.** Using $P(q) = 1 - q - q^2 + O(q^5)$ and $A_0 = 1 + 6q + 120q^2 + 956q^3 + 6010q^4 + \cdots$, one obtains

$$\sum_{n} v_0(n)q^n = 1 + 6q + 113q^2 + 830q^3 + 4934q^4 + \cdots .$$

At weight two the subtraction is $120 - 6 - 1 = 113$: six current derivatives and one stress. At weight three it is $956 - 120 - 6 = 830$. The generating-function method automatically accounts for descendants of descendants. For c_1, the first new grade-one primaries number $243 - 3 = 240$, since each of its three top states has one L_{-1} descendant.

The same calculation for E_8 gives 248 weight-one primaries and $4124 - 248 - 1 = 3875$ weight-two primaries. These are component counts, not counts of affine highest-weight multiplets. A whole multiplet can contribute many Virasoro primaries of the same weight.

13.6 Which deductions require more than a character

A character establishes a graded count once the module is known to exist. It does not give a preferred orthonormal basis, the action of the currents, the reduced matrix elements of an OPE, or a defect junction associator. In particular, the coefficient 113 predicts the dimension of a space on which a three-point tensor can be computed. It does not specify that tensor.

A finite truncation also has a precise coverage limit. If all characters are known through grade N and all lowest weights are nonnegative, then the diagonal full spectrum is complete through total dimension N: every contributing grade is at most N. The rectangular product of those truncated series contains some states at larger dimensions, but is generally incomplete there. Reporting the larger partial rectangle as a complete spectrum would be an error even if every retained coefficient were exact.

There is a useful division of labor. The lattice construction gives actual fields and products. The structural argument gives modules and modular data. The character theorem gives every unflavoured graded dimension. Descendant subtraction gives Virasoro-primary counts. Finally, a choice of full theory specifies how holomorphic and antiholomorphic pieces combine. We carry out that last step in Chapter 14.

13.7 Exercises

Exercise 13.1. Derive the coefficient 726 of q^4 in the rank-eight oscillator factor by classifying the partitions 4, $3+1$, $2+2$, $2+1+1$, and $1+1+1+1$. Use it and E_4 to obtain the weight-four coefficient of V_{E_8}.

Solution on page 302.

Exercise 13.2. Compute the first three primary multiplicities above the top of the b module and of the d_4 module. Reconstruct the corresponding character coefficients by convolving with partition numbers.

Solution on page 302.

Exercise 13.3. Explain why the categorical twist $e^{4\pi i/13}$ alone permits lowest weights $2/13, 15/13, 28/13, \ldots$. Identify exactly which steps in this chapter select $15/13$.

Solution on page 303.

Exercise 13.4. Two distinct simple modules have equal unflavoured characters. Explain why this does not violate linear independence of simple objects in the Grothendieck group. Give two types of observables that could distinguish the modules.

Solution on page 303.

Exercise 13.5. In Proposition 13.3, replace the modular representation by one containing a trivial summand. Explain why the same principal part need no longer determine a unique function. What extra normalization would be required?

Solution on page 303.

Part III

Local fields, defects, and observables

Chapter 14

Two full theories at central charge eight

A chiral algebra organizes left-moving fields. A full local CFT requires a compatible right-moving theory and a prescription for pairing their sectors. The Haagerup construction gives two especially natural choices. They have the same central charge and sharply different spectra, making them a useful example of why the phrase "the CFT of a chiral algebra" needs qualification.

Throughout this chapter we assume the construction inputs and the structural and positivity conclusions developed in Part II. The full-theory existence statements use the rational extension and diagonal constructions. The spectrum calculations themselves then follow from the character table by elementary counting.

14.1 The diagonal theory and a holomorphic extension

The diagonal theory has

$$\mathcal{H}_{\mathrm{diag}} = \bigoplus_i M_i \otimes \overline{M_i}, \qquad Z_{\mathrm{diag}} = \sum_i |\chi_i|^2. \tag{14.1}$$

The twelve labels i are the simple U-modules. They are self-dual in the present modular category, so charge conjugation does not change the displayed diagonal pairing. The diagonal Longo–Rehren construction applied to the completely rational net of Chapter 12 gives a unitary local full conformal net with this Hilbert space. This establishes an actual full theory; the modular-invariant sum alone would not supply the existence theorem.

The selected Haagerup extension uses the canonical Lagrangian algebra of $\mathcal{H}_3$ transported into $\mathrm{Rep}(U)$. Its chiral state space is

$$H' \downarrow_U = \mathbf{1} \oplus b \oplus a^{\oplus 2}, \qquad H' \cong V_{E_8}.$$

The positive extension is the E_8 lattice theory, so its tensor product with the antiholomorphic theory gives the full CFT

$$\mathcal{H}_{E_8} = H' \otimes \overline{H'}, \qquad Z_{E_8} = |\chi_0 + \chi_b + 2\chi_a|^2 = \left|\frac{E_4}{\eta^8}\right|^2. \tag{14.2}$$

This expression contains off-diagonal U-sector pairings. If $v = (1, 1, 2, 0, \ldots, 0)$ in the order $(0, b, a, c_0, \ldots)$, the coupling matrix is $Z_{ij} = v_i v_j$, whereas the diagonal theory has $Z_{ij} = \delta_{ij}$. For example, the diagonal $c_1 \otimes \overline{c_1}$ sector supplies nine scalars of dimension $2/3$, whereas its coefficient in the extension partition function is zero. Conversely, the extension contains four copies of

$a \otimes \bar{a}$: each chiral side has two independent copies of a. Changing the pairing changes actual state multiplicities, not just their names.

Both theories have a unique vacuum and $(c_L, c_R) = (8, 8)$. Holomorphicity of H' means that it has only one simple module as a chiral algebra. It does not mean that the full theory has only one state, one local field, or no interesting defects preserving a smaller algebra. The extension changes which fields are regarded as chiral-algebra generators.

14.2 Reading a full spectrum from chiral grades

In the diagonal theory, left grade n and right grade m in sector i give

$$\begin{aligned} (h, \bar{h}) &= (h_i + n, h_i + m), \\ \Delta &= 2h_i + n + m, \qquad s_{\text{spin}} = n - m, \\ d_{\text{states}} &= a_i(n)a_i(m), \qquad d_{\text{primary}} = v_i(n)v_i(m). \end{aligned} \tag{14.3}$$

The last line counts fields primary under both Virasoro algebras. A product of left and right extended-algebra top states is automatically Virasoro primary; higher grades need descendant subtraction first.

The scalar condition is $n = m$ in a diagonal sector. Consequently a positive grade raises a scalar dimension by at least two. Every relevant scalar, meaning $0 < \Delta < 2$, therefore comes from a chiral top with $h_i < 1$. Squaring the corresponding top dimensions gives the complete list:

Δ	2/3	10/13	12/13	14/13	16/13	4/3	22/13
Multiplicity	9	25	49	169	196	729	1764

Their total is 2941. The number looks large relative to twelve extended sectors because each top space contains multiple local components. A module with 42 chiral top states contributes $42^2 = 1764$ scalar components when paired diagonally.

Worked example 14.1 (A relevant spinning level)**.** The c_1 module has $h = 1/3$, three top states, and $243 - 3 = 240$ new Virasoro primaries at grade one. Pairing a grade-one primary with a top state gives

$$\Delta = \frac{1}{3} + 1 + \frac{1}{3} = \frac{5}{3}, \qquad s_{\text{spin}} = \pm 1,$$

with $240 \cdot 3 = 720$ components for each sign. These fields are relevant by dimension but cannot appear alone as rotationally invariant scalar perturbations.

At dimension two the scalar primaries come from the tops of b, a, c_0 and from current bilinears in the vacuum sector:

$$d_{\Delta=2,s=0} = 80^2 + 81^2 + 81^2 + 6^2 = 19558. \tag{14.4}$$

There are also 113 chiral weight-two primaries at each spin ± 2. Thus the total primary count at dimension two is 19784, while the state count is 19798. The difference is exactly two stress tensors and twelve current derivatives.

14.3 Why the extension has no relevant scalar

The holomorphic E_8 algebra is integer graded. The only state of weight zero is its vacuum, and the first nontrivial states are the 248 currents at weight one. A nonidentity full scalar must have equal positive left and right weights, so its dimension is at least two. The first scalar space is

$$(H')_1 \otimes \overline{(H')_1}, \qquad \dim = 248^2 = 61504.$$

Every such current bilinear is a Virasoro primary. Chiral currents paired with the opposite vacuum have dimension one and spin ± 1, so the ordinary primary gap is one even though the scalar gap is two.

The two theories can now be compared without ambiguity:

Quantity	Diagonal U	Selected E_8 theory
Currents per chirality	6	248
First nonidentity primary dimension	2/3	1
First nonidentity scalar dimension	2/3	2
Relevant scalar components	2941	0
Dimension-two scalar components	19558	61504

These are exact consequences of the selected characters and full pairings. They do not identify a lattice Hamiltonian flowing to either theory.

14.4 Which symmetry is being preserved?

In the diagonal theory, the twelve chiral-algebra-preserving Verlinde lines form the fusion category underlying $\mathrm{Rep}(U) = Z(\mathcal{H}_3)$. The fully invariant bulk space is $U \otimes \overline{U}$. Its first nonidentity scalars are the $6 \cdot 6 = 36$ current bilinears of dimension two.

The selected extension has a category of U-preserving bimodule defects equivalent to $\mathcal{H}_3 \boxtimes \mathcal{H}_3^{\mathrm{op}}$. Choosing the first chiral factor as the Haagerup symmetry gives invariant space $U \otimes \overline{H'}$. Its dimension-two scalar multiplicity is $6 \cdot 248 = 1488$. Requiring invariance under both independent factors gives $U \otimes \overline{U}$ and multiplicity 36. Both choices have invariant scalar gap two, but they are different symmetry requirements.

This is why a bootstrap comparison must name the symmetry action, not only the ordinary partition function. Two actions on an isomorphic full CFT can organize local fields differently. The defect chapter derives the invariant spaces from explicit line and tube data.

A dimension-two invariant scalar is only marginal by dimension. Whether a perturbation is exactly marginal depends on OPE coefficients, mixing, and higher-order compatibility. Commuting current constructions provide a standard source of exactly marginal directions, but the count of all current bilinears is not the dimension of a conformal manifold.

Worked example 14.2 (A dimension-two scalar that runs)**.** Consider a unitary $\mathrm{SU}(2)_k$ current algebra, with k a positive integer. Fix Cartesian current bases with

$$J^a(z)J^b(0) \sim \frac{k\delta^{ab}}{2z^2} + \frac{i\varepsilon^{abc}J^c(0)}{z}, \qquad a,b,c = 1,2,3.$$

Use the same $+i\varepsilon^{abc}$ convention for the independent antiholomorphic currents. Repeated Lie algebra indices are summed. The rescaled currents $j^a = \sqrt{2/k}\, J^a$ then satisfy

$$j^a(z)j^b(0) \sim \frac{\delta^{ab}}{z^2} + i\sqrt{\frac{2}{k}}\,\frac{\varepsilon^{abc}j^c(0)}{z}.$$

The isotropic current bilinear

$$\mathcal{O} = \frac{1}{\sqrt{3}}\sum_{a=1}^{3} j^a\bar{j}^a$$

has weights $(1,1)$ and is invariant under simultaneous $\mathrm{SU}(2)$ rotations of the two current triples. Its normalization follows directly from the double poles:

$$\langle \mathcal{O}(z,\bar{z})\mathcal{O}(0)\rangle = \frac{1}{3}\sum_{a,b}\frac{\delta^{ab}\delta^{ab}}{z^2\bar{z}^2} = \frac{1}{|z|^4}.$$

To find its self-OPE coefficient, multiply the two simple poles. The factor $i^2 = -1$ and the identity $\sum_{a,b} \varepsilon^{abc}\varepsilon^{abd} = 2\delta^{cd}$ give

$$-\frac{2}{3k|z|^2}\sum_{a,b,c,d}\varepsilon^{abc}\varepsilon^{abd}j^c\bar{j}^d = -\frac{4}{3k|z|^2}\sum_c j^c\bar{j}^c = -\frac{4}{\sqrt{3}\,k}\frac{\mathcal{O}}{|z|^2}.$$

Thus

$$\mathcal{O}(z,\bar{z})\mathcal{O}(0) = \frac{\mathbf{1}}{|z|^4} + C_{\mathcal{O}\mathcal{O}\mathcal{O}}\frac{\mathcal{O}(0)}{|z|^2} + \cdots, \qquad C_{\mathcal{O}\mathcal{O}\mathcal{O}} = -\frac{4}{\sqrt{3}\,k}. \tag{14.5}$$

Terms with one double pole and one simple pole vanish because $\delta^{ab}\varepsilon^{abc} = 0$. The omitted terms include spinning fields and higher-dimension fields. In this invariant current sector, the only dimension-two scalar generated by the displayed collision is $\mathcal{O}$ itself.

Now fix the action and scale conventions, since these determine the sign of the beta function. Use the Euclidean weight e^{-S_g} with

$$S_g = S_0 + g\int d^2x\,\mathcal{O}(x), \qquad d^2x = dx_1\,dx_2.$$

The velocity has been absorbed into the Euclidean time coordinate. There is no extra $1/(2\pi)$ in the definition of g. Let a be the short-distance cutoff and increase the coarse-graining length by $e^{d\ell}$. The second-order term in e^{-S_g} is proportional to $+g^2/2$. Its short-distance shell contribution is

$$\frac{g^2}{2}\int d^2y\int_{a<|r|<ae^{d\ell}} d^2r\,\frac{C_{\mathcal{O}\mathcal{O}\mathcal{O}}}{|r|^2}\mathcal{O}(y) = \pi C_{\mathcal{O}\mathcal{O}\mathcal{O}}g^2 d\ell\int d^2y\,\mathcal{O}(y),$$
$$\int_{a<|r|<ae^{d\ell}}\frac{d^2r}{|r|^2} = 2\pi\,d\ell.$$

The factor $1/2$ prevents double counting the two integrated insertions. Matching the term linear in $\mathcal{O}$ to $-(g+dg)\int\mathcal{O}$ therefore gives

$$\beta_{\text{length}}(g) := \frac{dg}{d\ell} = -\pi C_{\mathcal{O}\mathcal{O}\mathcal{O}}g^2 + O(g^3) = B_k g^2 + O(g^3), \qquad B_k = \frac{4\pi}{\sqrt{3}\,k} > 0. \tag{14.6}$$

There is no linear term because $\Delta_{\mathcal{O}} = 2$. The identity collision shifts the vacuum energy, while a circular shell averages the nonzero-spin terms to zero. A beta function defined using an inverse length μ has the opposite sign: $\mu\,dg/d\mu = -\beta_{\text{length}}(g)$.

Dropping the cubic and higher terms gives the leading running,

$$g(\ell) = \frac{g_0}{1 - B_k g_0 \ell}, \qquad \ell = \log(r/a).$$

For $g_0 < 0$, the coupling approaches zero from below and is *marginally irrelevant*. In the asymptotic regime,

$$g(r) = -\frac{1}{B_k\log(r/a)} + O\left(\frac{\log\log(r/a)}{\log^2(r/a)}\right),$$

where the first subleading terms also encode the initial coupling and higher-order beta function. A dimensionless observable with an expansion $Q(g) = Q_* + q_1 g + O(g^2)$ consequently approaches its limit as $1/\log r$ when $q_1 \neq 0$. If that linear coefficient vanishes, its leading drift can be smaller; one should not fit every observable with the same logarithm. For $g_0 > 0$, the coupling grows and is marginally relevant. The leading solution indicates loss of weak-coupling control at a length of order $a\exp[1/(B_k g_0)]$. Its pole does not determine a physical infrared endpoint.

Compare the unit-normalized Cartan scalar $\mathcal{O}_{\mathrm{C}} = j^3\bar{j}^3$. Since $\varepsilon^{33c} = 0$, its self-OPE has no $\mathcal{O}_{\mathrm{C}}/|z|^2$ term, so this quadratic beta-function obstruction is absent. Exact marginality requires

more than that one vanishing coefficient: the abelian current-current theorem, under its unitary commuting-current hypotheses, supplies the all-order local conformal family. The isotropic sum uses noncommuting currents and has the nonzero obstruction just computed. Both perturbing fields have dimension two at the starting CFT, but their renormalization-group behavior differs.

Worked example 14.3 (The same running produces two different logarithms)**.** For a concrete comparison model, take the standalone diagonal $\mathrm{SU}(2)_1$ WZW theory and perturb it by the preceding isotropic scalar with $g_0 < 0$. Its spin-$(1/2, 1/2)$ primary has $(h, \bar{h}) = (1/4, 1/4)$. Under the diagonal global SU(2) its four components split into a singlet and a triplet, labelled by $J = 0, 1$. Both have unperturbed scaling dimension $\Delta_0 = 1/2$. These are internal SU(2) representation spins; all four components have conformal spin $h - \bar{h} = 0$.

The current Ward identities turn the simple poles with a primary into the left and right spin generators t_L^a, t_R^a. Before specializing the level, the coefficient of that primary in $\mathcal{O}(z, \bar{z})\Phi_J(0)$ is

$$\mathcal{O}(z, \bar{z})\Phi_J(0) \sim \frac{q_J}{|z|^2}\Phi_J(0), \qquad q_J = \frac{2}{\sqrt{3}\,k}\, t_L \cdot t_R = \frac{J(J+1) - 3/2}{\sqrt{3}\,k}.$$

The last equality is ordinary angular-momentum addition: $2t_L \cdot t_R = (t_L + t_R)^2 - t_L^2 - t_R^2$, with $t_L^2 = t_R^2 = 3/4$. The singlet and triplet diagonalize the first-order mixing.

With the action convention above, first-order perturbation theory for $\langle \Phi_J(r)\Phi_J^\dagger(0)\rangle$ inserts $-g\int \mathcal{O}$. The two small neighborhoods of the external fields each contribute $2\pi q_J \log(r/a)$. Thus the correlator has relative correction $-4\pi g q_J \log(r/a)$. Comparing with $(r/a)^{-2(\Delta_0+\delta\Delta_J)} = (r/a)^{-2\Delta_0}\big[1 - 2\delta\Delta_J \log(r/a) + \cdots\big]$ gives

$$\delta\Delta_J = 2\pi g q_J + O(g^2).$$

Replacing g by its running value resums the leading logarithms:

$$\begin{aligned} G_J(r) &\propto r^{-2\Delta_0} \exp\left[-4\pi q_J \int_0^{\log(r/a)} g(\ell)\, d\ell\right] \\ &\propto r^{-2\Delta_0}\big[1 + B_k |g_0| \log(r/a)\big]^{4\pi q_J / B_k}. \end{aligned}$$

The amplitudes and the reference scale inside the logarithm depend on the normalization and initial coupling. The leading powers follow from the two coefficients already calculated:

$$\frac{4\pi q_J}{B_k} = J(J+1) - \frac{3}{2}.$$

At $k = 1$ this gives, to leading logarithmic accuracy,

$$G_0(r) \propto \frac{1}{r[\log(r/a)]^{3/2}}, \qquad G_1(r) \propto \frac{[\log(r/a)]^{1/2}}{r}.$$

Two fields with the same limiting dimension can therefore drift in opposite directions and carry different multiplicative logarithms. This is a canonical WZW comparison calculation. Applying it to a particular lattice chain would require identifying its continuum currents, perturbation, and coupling sign. The calculation neither identifies the Haagerup chain nor fixes a gapped endpoint, and it does not transport Haagerup defect junctions along a deformation.

14.5 Four explicit marginal directions and a radius example

We can say more than the general warning. The two Cartan currents in U commute and can be normalized as

$$j^1 = \frac{H_L}{\sqrt{2}}, \qquad j^2 = \frac{H_R}{\sqrt{78}}, \qquad j^a(z)j^b(0) \sim \frac{\delta^{ab}}{z^2}.$$

The subscripts on H_L, H_R still label the two internal affine factors, both holomorphic. Introduce an independent antiholomorphic copy $\bar{j}^b$. The four fields

$$\mathcal{O}_{ab} = j^a \bar{j}^b, \qquad a, b = 1, 2, \tag{14.7}$$

are scalar primaries of dimension two, with

$$\langle \mathcal{O}_{ab}(z, \bar{z}) \mathcal{O}_{cd}(0) \rangle = \frac{\delta^{ac}\delta^{bd}}{|z|^4}.$$

Thus their seed-point Zamolodchikov metric, the two-point metric on infinitesimal deformations in this normalization, is the identity matrix.

The abelian current-current deformation theorem supplies a simultaneous local exactly marginal family in these four directions. Its hypotheses involve mutually commuting left currents and mutually commuting right currents with the positive current two-point forms. The vanishing structure constants explain the absence of the usual quadratic current-current beta-function obstruction; the theorem controls the all-order statement. One should not infer that statement from the vanishing of a single three-point coefficient alone.

There is a useful geometric description. For r commuting currents on each side, collect a state's abelian charges in $p = (p_L, p_R) \in \mathbb{R}^{r,r}$. The charge zero-mode contributions are $p_L^2/2$ and $p_R^2/2$. A general state also has Heisenberg oscillator levels and neutral-sector weights; these pieces stay fixed in the charge-boost description. An infinitesimal boost

$$\delta p_L = \varepsilon p_R, \qquad \delta p_R = \varepsilon^{\mathsf{t}} p_L, \tag{14.8}$$

preserves $p_L^2 - p_R^2$ and therefore the spin contribution. It changes the total dimension by $\delta\Delta = 2p_L^{\mathsf{t}}\varepsilon p_R$. Independent rotations of left and right current bases do not change the theory's charge geometry. The associated local deformation space is consequently

$$O(r, r)/(O(r) \times O(r)), \qquad \dim = r^2.$$

To see the count, an infinitesimal $O(r, r)$ transformation has blocks

$$X = \begin{pmatrix} A & B \\ B^{\mathsf{t}} & D \end{pmatrix}, \qquad A^{\mathsf{t}} = -A, \quad D^{\mathsf{t}} = -D.$$

The antisymmetric blocks rotate the two current bases. After removing them, the arbitrary real r-by-r matrix B supplies r^2 directions. For the chosen two Cartans this gives four local directions. This description does not assert that U is a tensor product of two free bosons with a neutral algebra; the ambient theory can impose nontrivial charge-sector selection rules. Nor does it determine the global duality quotient.

Worked example 14.4 (An enhanced current leaves the chiral algebra)**.** In the compact-boson convention of Chapter 2, let $R = \sqrt{2}e^t$. The state with $(m, n) = (1, 1)$ has

$$p_L = \sqrt{2}\cosh t, \qquad p_R = -\sqrt{2}\sinh t,$$

so

$$h = \cosh^2 t, \qquad \bar{h} = \sinh^2 t, \qquad s_{\text{spin}} = 1, \qquad \Delta = \cosh(2t).$$

At $t = 0$ it is a holomorphic weight-one current. For $t \neq 0$ it is a nonchiral local field of the same integer spin, while the theory remains conformal with $c = 1$. This is how an exactly marginal deformation can remove an enhanced chiral symmetry without destroying conformal invariance. In charge space, the radius change is the boost $p_L(t) = \cosh t\, p_L(0) - \sinh t\, p_R(0)$ and its right-moving counterpart.

For the ordinary full $E_{8,1}$ theory, the eight Cartans on each side give the familiar 64 toroidal directions, locally $O(8,8)/(O(8)\times O(8))$. The special rational point has many more dimension-two current bilinears than these 64 unobstructed directions. The same distinction applies to the 36 bilinears in the diagonal U theory.

The four fields in (14.7) are categorical singlets at the rational point. Continuing a full system of topological lines along a deformation also requires transporting their junctions and checking locality with the deformed fields. Exact marginality by itself does not prove that the specified Haagerup category survives everywhere on the family. This is a question about the deformed symmetry data, separate from the existence of the ordinary conformal deformation.

14.6 Thermodynamics from the same spectrum

Let $t = \beta v/L$ and $b = 2\pi t$. For either full theory, define the dimensionless energy $E = \Delta - 2/3$, since $(c_L + c_R)/24 = 2/3$. Then

$$Z(b) = \sum_{\Delta,s} d_{\Delta,s} e^{-bE}, \qquad H = \frac{2\pi v}{L} E. \tag{14.9}$$

The vacuum energy is $-4\pi v/(3L)$. Once Z converges,

$$\langle E\rangle = -\partial_b \log Z, \qquad C_V = b^2\big(\langle E^2\rangle - \langle E\rangle^2\big), \qquad S = \log Z + b\langle E\rangle.$$

A finite spectral truncation gives a lower bound on Z because all degeneracies are nonnegative. To make it a quantitative approximation, one needs an upper bound on the omitted tail.

Proposition 14.5 (A modular tail bound)**.** *Assume a convergent positive modular-invariant partition function $Z(b) = Z(4\pi^2/b)$. Suppose every state with $\Delta \le D$ is known, where $D > 2/3$. Write $Z = L_D + H_D$ for the known and omitted sums, and put $E_D = D - 2/3$. If $b > 2\pi$ and $b' = 4\pi^2/b$, then*

$$0 \le H_D(b) \le \frac{L_D(b) - L_D(b')}{e^{(b-b')E_D} - 1}. \tag{14.10}$$

At $b = 2\pi$, provided the differentiated trace converges, $0 \le H_D(b) \le L_D'(b)/E_D$.

Proof. Every omitted state has $E > E_D > 0$, so $H_D(b) \le rH_D(b')$ with $r = e^{-(b-b')E_D} < 1$. Modular invariance gives $H_D(b') = H_D(b) + L_D(b) - L_D(b')$. Substitution and division by $1 - r$ prove the first claim. At the fixed point, differentiating the modular identity gives $Z'(2\pi) = 0$. Hence $L_D' = -H_D' \ge E_D H_D$. □

The estimate is an example of a global consistency relation controlling information beyond a finite table. It requires modular invariance and positivity as hypotheses; finite agreement of a few coefficients cannot establish either one. For moment tails choose $0 < b_0 < b$ and use

$$\sum_{\Delta>D,s} d_{\Delta,s} E^k e^{-bE} \le H_D(b_0) \sup_{E\ge E_D} E^k e^{-(b-b_0)E}. \tag{14.11}$$

For $k > 0$, the supremum is at E_D if $(b - b_0)E_D \ge k$, otherwise at $k/(b - b_0)$. Thus one can control the energy and heat capacity as well as the partition function.

Worked example 14.6 (A universal value at the modular fixed point)**.** At $b = 2\pi$, modular invariance gives $\langle E\rangle = 0$. Therefore

$$\langle\Delta\rangle = \frac{2}{3}, \qquad S = \log Z$$

at $t = 1$ in either theory. The same mean dimension does not imply the same distribution, entropy, or heat capacity. The two spectra give different values of these other observables.

14.7 What has and has not been reconstructed

We now know exact state counts, Virasoro-primary multiplicities, several symmetry-resolved gaps, and a way to obtain controlled thermal observables. These are much more than a central charge. They remain less than a complete list of local OPE constants. The next chapter computes actual chiral correlators in the lattice realization, including a normalized four-point function that can be expanded by hand. The sewing chapter will explain the additional data needed for nonvacuum bulk correlators of the diagonal theory.

14.8 Exercises

Exercise 14.1. Reproduce the seven relevant scalar multiplicities and their sum 2941. Then compute the spinning multiplicities at dimensions 23/13 and 25/13 using the d_4 and d_1 rows.

Solution on page 303.

Exercise 14.2. Show that the full E_8 theory has 69752 states at dimension two. Separate them into scalar primaries, chiral primaries, current derivatives, and stresses.

Solution on page 303.

Exercise 14.3. Derive (14.11) and evaluate its supremum for $k = 1, 2$. Explain why a reliable heat-capacity error bound needs more than an error bound on Z alone.

Solution on page 303.

Exercise 14.4. Suppose a proposed branching prescription gives $Z = 6|\chi_0|^2 + \cdots$ with all coefficients nonnegative and all omitted sectors of positive weight. Explain the vacuum problem. Why does dividing the expression by six not automatically produce a new local CFT?

Solution on page 304.

Exercise 14.5. Explain the three different meanings of an invariant scalar count 36, 1488, or 61504 in this chapter. Which of these numbers uses no symmetry restriction at all?

Solution on page 304.

Exercise 14.6. Let $\eta = \mathrm{diag}(I_r, -I_r)$ and $X = \left(\begin{smallmatrix} 0 & \varepsilon \\ \varepsilon^{\mathsf{t}} & 0 \end{smallmatrix}\right)$. Verify $X^{\mathsf{t}}\eta + \eta X = 0$ and derive (14.8). For $r = 1$, recover the radius example and show explicitly why a chiral current can acquire $\bar{h} > 0$ without losing integer spin.

Solution on page 304.

Chapter 15

Worked operator products and correlation functions

Characters count states. Correlation functions tell us how those states interact. The lattice realization is particularly valuable because it supplies actual fields, positive norms, and exact mode products. This chapter develops several calculations small enough to reproduce by hand and explains how symmetry organizes the larger weight-two three-point tensor.

The correlators below are chiral. Their adjoints are chiral radial adjoints, not additional antiholomorphic fields. Products with antiholomorphic counterparts give full vacuum-sector observables. Generic nonvacuum bulk correlators of the diagonal theory require the additional sewing data discussed in Chapter 17.

15.1 Normalize before comparing coefficients

For orthogonal states v_i with positive norms $N_i = \langle v_i, v_i\rangle$, write $\widehat{v}_i = v_i/\sqrt{N_i}$, taking the positive square root. For three weight-two primaries, the coefficient of the weight-two field in their singular OPE is

$$C_{ij}{}^{k} = \frac{\langle v_k, (v_i)_{(1)} v_j\rangle}{\sqrt{N_i N_j N_k}}. \tag{15.1}$$

The Hermitian form is antilinear in the first slot. Thus this coefficient is the correlator with $\widehat{v}_k^{\dagger}$ at infinity. Exact algebraic numerators and positive square roots of norm products keep both the magnitude and phase of the answer.

The six currents split into affine levels one and thirty-nine. For their Cartans,

$$H_L(z)H_L(0) \sim 2z^{-2}, \qquad H_R(z)H_R(0) \sim 78z^{-2}, \qquad H_L(z)H_R(0) \sim 0.$$

The defining field u has weight two, charges $(1, 13)$, and transported norm squared twelve. Consequently $\widehat{u} = u/\sqrt{12}$ has unit two-point coefficient. The squared couplings to the unit Cartan currents are

$$\frac{1^2}{2} = \frac{1}{2}, \qquad \frac{13^2}{78} = \frac{13}{6}. \tag{15.2}$$

The unit total stress is $T/2$ at $c = 8$, and its coupling to $\widehat{u}, \widehat{u}^{\dagger}$ is one by the Ward identity. These numbers provide independent checks on the first terms of the four-point function.

15.2 Fifteen Gaussian terms give a complete four-point function

In the positive unitary exponential basis, the normalized field is

$$\widehat{u} = \frac{1}{\sqrt{3}}\left(e^{a_1} + \zeta e^{a_2} + \bar{\zeta} e^{a_3}\right), \qquad \zeta = \frac{\sqrt{13} + 3i\sqrt{3}}{\sqrt{40}}, \quad |\zeta| = 1. \tag{15.3}$$

Its three momenta satisfy

$$a_i^2 = 4, \qquad (a_i, a_j) = 2 \quad (i \neq j).$$

These inner products, the cocycle convention, and charge conservation are enough for the calculation. The raw torus-gauge coefficients in the construction chapter are different because their exponential basis has squared norms $4, 40, 40$.

Consider

$$G_0(x) = \langle \widehat{u}^\dagger(\infty)\widehat{u}(1)\widehat{u}^\dagger(x)\widehat{u}(0)\rangle.$$

Expanding the four sums gives $3^4 = 81$ assignments. The Gram matrix of the three momenta has eigenvalues $8, 2, 2$, so they are linearly independent. Charge conservation therefore says exactly that the two positive charges and the two negative charges have the same multiset. There are three assignments using one momentum four times and twelve using two distinct momenta. In the fixed adjoint and cocycle convention, each surviving term has coefficient $1/9$.

The fifteen terms fall into three groups. The three identical-charge assignments contribute $1/[3x^4(1-x)^4]$. The two possible pairing patterns for distinct charges contribute six terms each, giving

$$G_0(x) = \frac{1}{3x^4(1-x)^4} + \frac{2}{3x^2(1-x)^4} + \frac{2}{3x^4(1-x)^2}.$$

Putting these over a common denominator yields

$$G_0(x) = \frac{3 - 4x + 4x^2}{3x^4(1-x)^4}. \tag{15.4}$$

This is the full meromorphic chiral correlator, not a truncated series or a fitted ansatz.

The equal-charge ordering similarly gives

$$G_+(x) = \langle \widehat{u}^\dagger(\infty)\widehat{u}^\dagger(1)\widehat{u}(x)\widehat{u}(0)\rangle = \frac{x^2(4 - 4x + 3x^2)}{3(1-x)^4}. \tag{15.5}$$

The identities $G_0(1-x) = G_0(x)$ and $G_+(x) = x^{-4}G_0(1/x)$ are exact crossing checks. At $x = 1/2$, for instance, $G_0 = 512/3$ and $G_+ = 11/3$.

Worked example 15.1 (The first equal-charge exchanged state)**.** The sum of two distinct momenta has squared norm twelve, so its exponential has weight six. Equal momenta instead give weight eight. Hence the leading equal-charge OPE has weight six and starts at $x^{6-2-2} = x^2$. The three distinct sums each occur twice in the product of (15.3); their coefficients have magnitude $2/3$. The resulting state has squared norm $3(2/3)^2 = 4/3$. It is a weight-six primary, and its normalized leading coupling can be chosen as $2/\sqrt{3}$. This reproduces the leading coefficient of (15.5).

15.3 Extracting global blocks by a triangular subtraction

For external weight two, expand the neutral channel in global conformal blocks:

$$x^4 G_0(x) = A_0 + \sum_{n \geq 1} A_n x^n {}_2F_1(n, n; 2n; x). \tag{15.6}$$

Each block begins with x^n, so its coefficient can be found by subtracting all lower blocks from the power series. The first terms of the reduced correlator are

$$x^4 G_0 = 1 + \frac{8}{3}x + 6x^2 + 12x^3 + \frac{65}{3}x^4 + \cdots .$$

Thus $A_0 = 1$, $A_1 = 8/3$. The weight-one global block is $x+x^2/2+x^3/3+x^4/4+\cdots$. It contributes $4/3$ at order x^2, leaving $A_2 = 6 - 4/3 = 14/3$. At the next order the known contributions are $8/9 + 14/3$, leaving $A_3 = 58/9$. Continuing one more step gives

$$(A_0, A_1, A_2, A_3, A_4) = \left(1, \frac{8}{3}, \frac{14}{3}, \frac{58}{9}, \frac{107}{15}\right). \tag{15.7}$$

The first coefficient equals $1/2 + 13/6$, exactly the two Cartan exchanges in (15.2). At weight two, the total stress contributes one. The other $11/3$ is the sum of squared couplings to the weight-two Virasoro primaries. A global coefficient is a sum over orthonormal quasiprimaries; it is not generally one OPE coefficient or one Virasoro-block coefficient. This distinction is essential when comparing a four-point expansion with a table of individual couplings.

15.4 Resolving the weight-two space

The 113 Virasoro primaries consist of 96 fields in eight current multiplets and seventeen additional fields. The eight types are written as dimensions of the two internal left and right $\mathfrak{sl}_2$ representations. Both current algebras act on holomorphic fields; these labels do not denote physical chirality. The types are

$$(1,5), (1,9), (1,9), (1,13), (2,2), (2,6), (2,8), (2,14).$$

Their dimensions sum to $5 + 9 + 9 + 13 + 4 + 12 + 16 + 28 = 96$. The extra fields comprise an internal right spin-two multiplet $(1,5)$, a mixed-current multiplet $(3,3)$, and three stress differences. At internal left affine level one the candidate $(5,1)$ multiplet is null, so it does not add another five states.

There are four mutually commuting stresses (T_L, T_R, T_e, T_f) with central charges

$$\left(1, \frac{117}{41}, \frac{2400}{1271}, \frac{70}{31}\right). \tag{15.8}$$

Their sum is the total stress. A combination $A = \sum_i a_i T_i$ is a total-Virasoro primary precisely when $\sum_i a_i c_i = 0$: this cancels the fourth-order pole in its OPE with the total stress, while the remaining poles give weight two and translation. Since the stress space has dimension four, there are three independent such differences.

Worked example 15.2 (A normalized three-point coefficient from two stresses)**.** Take two commuting stresses and set $A = c_2 T_1 - c_1 T_2$. Their orthogonal norms give $N_A = c_1 c_2 (c_1 + c_2)/2$. The first product is $A_{(1)}A = 2c_2^2 T_1 + 2c_1^2 T_2$, so

$$\langle A, A_{(1)}A \rangle = 2N_A(c_2 - c_1).$$

Consequently

$$|C_{AAA}|^2 = \frac{8(c_2 - c_1)^2}{c_1 c_2 (c_1 + c_2)}. \tag{15.9}$$

At $c_1 = 1, c_2 = 117/41$, this is $23104/9243$. The calculation checks a nontrivial entry of the large lattice coupling tensor using only the Virasoro OPE.

Current zero modes act as derivations. If P projects orthogonally onto the primary space and F is a lowering zero mode, then

$$P\big((Fv_i)_{(1)}v_j\big) = FP\big((v_i)_{(1)}v_j\big) - P\big((v_i)_{(1)}Fv_j\big). \tag{15.10}$$

The projection commutes with the current action because current zero modes commute with the total Virasoro algebra and the compact form acts unitarily. Thus a set of highest-weight seed products determines all lowered components. In a conventional finite $\mathfrak{sl}_2$ basis $v_p = F^p v_0$ of dimension n, $Ev_p = p(n-p)v_{p-1}$. Raw torus-gauge norms must still be transported consistently when using this recursion.

The recorded exact calculation fills the complete 113^3 tensor, with 18365 nonzero ordered entries. The important mathematical content is the specified primary basis, exact seed products, equivariant completion, and normalized positive form. A sparse table is complete only when its unlisted entries have been proved zero in that defined space. This tensor concerns weight-two vacuum-module fields; it is not a claim to have computed every nonvacuum bulk OPE.

15.5 Spin-three fields and a null-channel test

The Potts and level-thirteen parafermion factors contain a canonical weight-three field a with intrinsic central charge c_0. To avoid confusion, c_0 here denotes a number, not the central simple-object label used elsewhere. Set

$$N = \frac{c_0}{3}, \quad b = a_{(1)}a, \quad A = 31 - 16\lambda(c_0+2), \quad B = NA.$$

The parameter pairs are $(c_0,\lambda) = (4/5, 4/13)$ for Potts and $(8/5, 14/473)$ for the parafermion factor. In the canonical positive normalization, the singular OPE is

$$a(z)a(0) \sim \frac{N}{z^6} + \frac{2T_0}{z^4} + \frac{\partial T_0}{z^3} + \frac{b}{z^2} + \frac{\frac{1}{2}\partial b - \frac{1}{12}\partial^3 T_0}{z}, \tag{15.11}$$

with $a_{(3)}b = Aa$ and $\|b\|^2 = B$.

One can reconstruct the whole four-point function from these poles. Define the unnormalized function $F(x) = \langle a(\infty)a(1)a(x)a(0)\rangle$. The Ward identity gives $\langle a(\infty)a(1)T_0(0)\rangle = 3N$. Its first and third derivatives at the last insertion give $6N$ and $72N$. The corresponding b coefficient and first derivative are B and $4B$. Therefore the poles at zero have coefficients $N^2, 2c_0, 2c_0, B, 2B-2c_0$ at orders $6,4,3,2,1$. Crossing determines the poles at one, and the identity channel fixes the constant at infinity. Hence

$$\begin{aligned} F(x) = {} & N^2\big[1 + x^{-6} + (1-x)^{-6}\big] \\ & + 2c_0\big[x^{-4} + (1-x)^{-4} + x^{-3} + (1-x)^{-3}\big] \\ & + B\big[x^{-2} + (1-x)^{-2}\big] + (2B - 2c_0)\big[x^{-1} + (1-x)^{-1}\big]. \end{aligned} \tag{15.12}$$

Subtracting this rational function from the actual correlator leaves a holomorphic function on the sphere vanishing at infinity, so the difference is zero. Divide by N^2 for unit external fields.

The weight-four quasiprimary descendant of the intrinsic stress is

$$\Lambda =: T_0T_0 : -\frac{3}{10}\partial^2 T_0, \qquad \|\Lambda\|^2 = \frac{c_0(5c_0+22)}{10}.$$

Removing it and the derivative from b gives $W_4 = b - \frac{32}{5c_0+22}\Lambda - \frac{3}{10}\partial^2 T_0$. This subtraction can be checked entirely with Virasoro Gram products. Put $Q = 5c_0 + 22$. Then

$$\big\|\partial^2 T_0\big\|^2 = 20c_0, \qquad \langle \Lambda, \partial^2 T_0\rangle = 0, \qquad \|\Lambda\|^2 = \frac{c_0 Q}{10}.$$

Since W_4 is orthogonal to these descendants,

$$\|W_4\|^2 = B - \left(\frac{3}{10}\right)^2 20c_0 - \left(\frac{32}{Q}\right)^2 \frac{c_0 Q}{10} = B - \frac{9c_0}{5} - \frac{512c_0}{5Q}.$$

The squared coupling to the new intrinsic weight-four primary is $\|W_4\|^2/N^2$. Substituting $B = (c_0/3)[31 - 16\lambda(c_0+2)]$ and simplifying gives

$$\gamma_4^2 = \frac{48(c_0+2)\{8-\lambda(5c_0+22)\}}{c_0(5c_0+22)} = \begin{cases} 0, & \text{Potts,} \\ 60552/2365, & \text{level-thirteen parafermion.} \end{cases} \tag{15.13}$$

For the parafermion point, $Q = 30$ and $8 - \lambda Q = 3364/473$, while $48(c_0+2)/(c_0 Q) = 18/5$. Their product is the displayed $60552/2365$. At the Potts point, $Q = 26$ and $\lambda Q = 8$, so the same formula vanishes exactly. In the Potts case this channel is null. In the parafermion case it is a positive new channel. The test is with respect to the intrinsic Virasoro algebra of the factor; a field's decomposition under the total $c = 8$ Virasoro algebra is a different question.

15.6 A nonvacuum block with fractional powers

The lightest U-module c_1 can be seen by restricting either fundamental A_2 lattice sector. Choose its three weights

$$\beta_0 = (2/3, 1/3), \quad \beta_1 = (-1/3, 1/3), \quad \beta_2 = (-1/3, -2/3), \qquad (\beta_i, \beta_j) = \delta_{ij} - \frac{1}{3}.$$

Each exponential has weight $1/3$. Paired lattice intertwiners with unit vacuum coefficient give, on $0 < x < 1$,

$$\mathcal{F}_{ij}(x) = \langle V_{-\beta_i}(\infty) V_{\beta_i}(1) V_{-\beta_j}(x) V_{\beta_j}(0)\rangle = x^{-2/3}(1-x)^{-(\beta_i,\beta_j)}. \tag{15.14}$$

The diagonal blocks are $[x(1-x)]^{-2/3}$ and the off-diagonal ones are $x^{-2/3}(1-x)^{1/3}$. Analytic continuation requires a path because these are chiral blocks with fractional powers.

The intermediate lattice vacuum restricts to $U \oplus b$. Here b again denotes the simple module in the modular table, rather than the preceding weight-four field $a_{(1)}a$. The internal right Cartan H_R charges are $q_i = (2, 0, -2)$. These form an internal spin-one triplet, whose chiral conformal weight is $1/3$. The first reduced-block coefficient splits as

$$(\beta_i, \beta_j) = \frac{q_i q_j}{78} + \left[(\beta_i, \beta_j) - \frac{q_i q_j}{78}\right].$$

The first term is U-current exchange and the second is exchange through the b top. At $i = j = 0$ these are $2/39$ and $8/13$, whose sum is $2/3$. The matrix of the second terms has eigenvalues $0, 1, 35/39$, consistent with positivity.

Restriction of a lattice intertwiner is a genuine U intertwiner. Thus these are actual nonvacuum chiral data. They are selected linear combinations of channels determined by the ambient extension. They do not form a complete channel basis and do not by themselves determine the diagonal theory's lightest-scalar bulk four-point function. Multiplying one selected block by its complex conjugate would be a proposed sewing choice, which still needs justification.

15.7 Exercises

Exercise 15.1. Enumerate the three groups of surviving charge assignments in the neutral u four-point function and derive their powers of x and $1 - x$ from pairwise momentum inner products.

Solution on page 304.

Exercise 15.2. Complete the triangular subtraction leading to $A_4 = 107/15$. Use the leading power in the charged channel to determine its first global-block coefficient.

Solution on page 305.

Exercise 15.3. Derive (15.9) without using a four-point function. What happens when $c_1 = c_2$, and why does that not make the field A null?

Solution on page 305.

Exercise 15.4. Verify the Potts zero in (15.13). Derive the coefficient of the simple pole in (15.12) directly from the last term of (15.11).

Solution on page 305.

Exercise 15.5. Write the 3×3 matrix $K_{ij} = \delta_{ij} - 1/3 - q_i q_j/78$ explicitly. Find its eigenvectors and eigenvalues without a numerical diagonalization. Explain why an off-diagonal negative entry is compatible with positivity of K.

Solution on page 305.

Chapter 16

Defect lines, twisted sectors, and boundaries

A topological line can be wrapped around an operator, inserted in a thermal trace, or drawn along the time direction to define a twisted Hilbert space. A conformal boundary can be viewed either as a state in a closed-channel Hilbert space or as an endpoint condition for an open-channel Hamiltonian. Modular transformations relate these descriptions. The Haagerup examples make all of these relations explicit and show why six lines, twelve central sectors, and thirty-six bimodule defects can coexist without contradiction.

16.1 The eigenvalue of a contractible line

For an ordinary symmetry element, acting on the vacuum gives one. For a general simple topological line a, a contractible vacuum loop evaluates to its quantum dimension d_a. Therefore the appropriate invariance condition on a local state is

$$\widehat{L}_a|\psi\rangle = d_a|\psi\rangle \quad \text{for every symmetry line } a. \tag{16.1}$$

Demanding eigenvalue one for a noninvertible line would generally discard the vacuum itself.

In a rational diagonal theory, the chiral-algebra-preserving Verlinde line labelled by a acts on the bulk sector $M_i \otimes \overline{M_i}$ by

$$\lambda_a(i) = \frac{S_{ai}}{S_{0i}}. \tag{16.2}$$

At $i = 0$ this is d_a. Fusion of line operators follows from the Verlinde formula because the columns of S diagonalize all fusion matrices simultaneously.

For the Haagerup center all labels are self-dual and S is real. The Verlinde coefficients are

$$N_{ab}^{\,c} = \sum_i \frac{S_{ai} S_{bi} S_{ci}}{S_{0i}}. \tag{16.3}$$

They are nonnegative integers, even though individual line eigenvalues need not be positive. The value of a trace with a line inserted is therefore not a state degeneracy.

Worked example 16.1 (An invariant projector from modular orthogonality)**.** Let $\mathcal{D} = 1/S_{00} = 3(1 + d^2)$. On the diagonal bulk space,

$$\Pi_{\mathrm{inv}} = \frac{1}{\mathcal{D}^2} \sum_a d_a \widehat{L}_a.$$

Its eigenvalue on sector i is

$$\frac{1}{\mathcal{D}^2 S_{00} S_{0i}} \sum_a S_{a0} S_{ai} = \frac{\delta_{0i}}{\mathcal{D}^2 S_{00} S_{0i}} = \delta_{0i}.$$

Thus the invariant space is exactly $U \otimes \overline{U}$. Equivalently, if all line eigenvalues in sector i equalled their vacuum values, column i of S would be proportional to the vacuum column, forcing $i = 0$.

16.2 Twining and twisting are different traces

Insert the line operator in the ordinary Hilbert-space trace:

$$Z^{[a]}(\tau, \bar{\tau}) = \sum_i \lambda_a(i) |\chi_i(\tau)|^2. \tag{16.4}$$

This is a *twined* partition function. Its coefficients can be negative or algebraic because they are traces of a symmetry operator on states. Exchange the temporal and spatial cycles of the torus. The line now intersects the spatial circle, and the corresponding *twisted* Hilbert space has partition function

$$Z_{[a]}(\tau, \bar{\tau}) = \sum_{i,j} N_{ai}^{\ j} \chi_i(\tau) \overline{\chi_j(\tau)}. \tag{16.5}$$

To prove this relation, transform each character in (16.4) by S. The coefficient of $\chi_i \overline{\chi_j}$ is $\sum_k S_{ak} S_{ik} S_{jk} / S_{0k}$, exactly the Verlinde sum. Here every coefficient is a nonnegative integer because a twisted Hilbert space counts states.

The ground dimension in a twisted sector need not equal its first scalar dimension. A term $M_i \otimes \overline{M_j}$ has scalar states only if $h_i - h_j \in \mathbb{Z}$ and the grades can compensate that difference. For the present twelve lowest weights, the only unequal values differing by an integer are zero and one. Thus the lowest scalar count can be obtained from the top dimensions and the six vacuum currents without expanding many character grades.

Worked example 16.2 (A light twisted ground state need not be a scalar)**.** The line c_1 has twisted ground dimension $1/3$, with six components from the left/right placements of its three top states against the vacuum. Their spins are $\pm 1/3$. Its first scalar dimension is instead $2/3$, with multiplicity $2 \cdot 3^2 = 18$. We can derive the factor two from the modular matrix instead of reading a fusion table. Put $A = S_{00}$, $B = S_{0b}$, so that $AB = 1/117$ and $A + B = 1/3$. The nonzero entries in the c_1 row are the first two $1/3$ entries and a middle block consisting of one $2/3$ and three $-1/3$ entries. Verlinde gives

$$N_{c_1 c_1}^{\ c_1} = \frac{1}{27}\left(\frac{1}{A} + \frac{1}{B}\right) + 3\left[\left(\frac{2}{3}\right)^3 + 3\left(-\frac{1}{3}\right)^3\right] = \frac{13}{9} + \frac{5}{9} = 2.$$

Pairing the three left top states with the three right top states then gives eighteen. Fractional spin is permitted for states attached to the nontrivial defect; it would not describe an ordinary bosonic local field in the untwisted theory.

For orientation, some exact twisted gaps are

Line	Ground Δ	Multiplicity	First scalar Δ	Multiplicity
0	0	1	2/3	9
b	2/3	18	2/3	18
a, c_0	2/3	9	2/3	9
c_1	1/3	6	2/3	18
c_2	2/3	63	2/3	9
d_1	6/13	14	2/3	9
d_2, d_3	2/3	9	2/3	9
d_4	5/13	10	2/3	9
d_5	7/13	26	2/3	9
d_6	8/13	28	2/3	9

The vacuum is excluded from the phrase "first scalar" in the identity row. The common scalar gap does not make the twisted spaces equivalent; their ground states, spins, and multiplicities already distinguish them.

16.3 Cardy boundaries and open–closed duality

A conformal boundary preserving U glues the holomorphic and antiholomorphic symmetry. For the stress tensor, the boundary state obeys

$$(L_n - \bar{L}_{-n})|a\rangle = 0.$$

The opposite mode number comes from the reversal of orientation at the boundary. For each compatible sector there is an Ishibashi state $|i\rangle\!\rangle$, a formal solution of the gluing equations. It is not a normalizable vector before a positive Euclidean propagation length is inserted. It pairs matching left and right basis states in the module. With the standard normalization, the regularized inner product is

$$\langle\!\langle i|\tilde{q}^{(L_0+\bar{L}_0-c/12)/2}|j\rangle\!\rangle = \delta_{ij}\chi_i(\tilde{q}).$$

Each matching basis pair contributes one power $\tilde{q}^{h_i+n-c/24}$, which explains the character.

The annulus admits two quantizations (Figure 16.1). On an interval of length ℓ, with Euclidean time period β, the open-channel Hamiltonian is $H_{\text{open}} = (\pi v/\ell)(L_0 - c/24)$ and $q = e^{-\pi v\beta/\ell}$. Quantizing across the interval instead gives propagation between boundary states on a circle, with $\tilde{q} = e^{-4\pi\ell/(v\beta)}$. These two nomes are related by the modular S transformation. A consistent boundary condition must make the closed-channel overlap into a trace with nonnegative integer open-channel multiplicities.

The factors of two follow from the geometry. Unfolding the open interval gives one chiral circle of length 2ℓ, hence the coefficient $\pi v/\ell$ in H_{open}. In the closed channel, the spatial circumference is $v\beta$. Propagation over distance ℓ therefore inserts

$$\exp\left[-\frac{2\pi\ell}{v\beta}(L_0 + \bar{L}_0 - c/12)\right],$$

which is exactly the Ishibashi regulator above with the stated $\tilde{q}$. This also explains why the open trace is a sum of chiral characters rather than absolute squares of characters.

The diagonal theory's elementary Cardy boundaries are

$$|a\rangle = \sum_i \frac{S_{ai}}{\sqrt{S_{0i}}}|i\rangle\!\rangle. \tag{16.6}$$

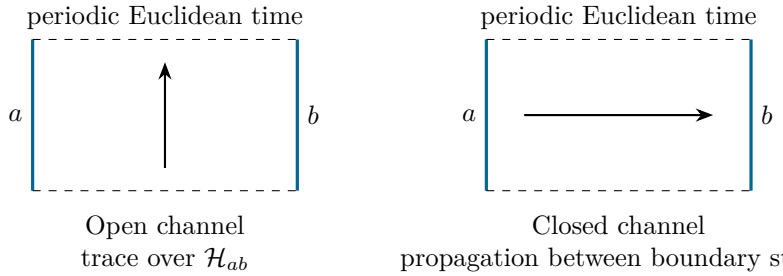


Figure 16.1: The same annulus viewed in two directions. The dashed horizontal edges are identified. The solid vertical edges carry boundary conditions a, b. The open channel propagates states along periodic time; the closed channel propagates from one boundary state to the other.

Their closed-channel overlap contains coefficients $S_{ai}S_{bi}/S_{0i}$. Applying a modular transformation gives the open-channel annulus

$$Z_{ab}^{\text{open}}(q) = \sum_i N_{ab}^{\ i}\chi_i(q). \tag{16.7}$$

The usual conjugate boundary label has disappeared only because the twelve center labels are self-dual. It will be necessary again for the six Haagerup boundary labels.

The vacuum overlap is the boundary g factor:

$$g_a = \frac{S_{a0}}{\sqrt{S_{00}}} = \frac{d_a}{\sqrt{\mathcal{D}}}, \qquad \log g_a \text{ is the boundary entropy.} \tag{16.8}$$

It is not an integer ground-state degeneracy. In the Haagerup diagonal theory its four possible values occur in the ratios $1 : d^2 : s : 3d$.

Worked example 16.3 (Fixed and free Ising boundaries)**.** For the Ising labels $(\mathbf{1}, \epsilon, \sigma)$,

$$S = \frac{1}{2}\begin{pmatrix} 1 & 1 & \sqrt{2} \\ 1 & 1 & -\sqrt{2} \\ \sqrt{2} & -\sqrt{2} & 0 \end{pmatrix}.$$

The fusion rules are $\epsilon^2 = \mathbf{1}$, $\epsilon\sigma = \sigma$, and $\sigma^2 = \mathbf{1} + \epsilon$. The boundaries labelled $\mathbf{1}, \epsilon$ are the two fixed-spin boundaries, and σ is the free boundary. Cardy's formula gives

$$Z_{++} = \chi_{\mathbf{1}}, \qquad Z_{+-} = \chi_\epsilon, \qquad Z_{ff} = \chi_{\mathbf{1}} + \chi_\epsilon, \qquad Z_{+f} = \chi_\sigma.$$

For instance, the free boundary has one nonidentity relevant boundary primary of weight $1/2$, while a boundary-changing field between a fixed and a free boundary has weight $1/16$. The vacuum row gives $g_+ = g_- = 1/\sqrt{2}$ and $g_f = 1$. The noninteger value $1/\sqrt{2}$ is entirely consistent because g is a vacuum overlap, not an open-channel ground multiplicity.

Worked example 16.4 (The vacuum and b boundary annuli)**.** The vacuum boundary has $Z_{00}^{\text{open}} = \chi_0$. Its first positive boundary weight is one, with six currents. For b, the fusion rule

$$b^2 = \mathbf{1} + b + a + c_0 + c_1 + c_2 + \sum_{j=1}^{6} d_j$$

puts three weight-$1/3$ states in Z_{bb}^{open}, coming from the top of c_1. They are relevant boundary perturbations. One must not remove the entire vacuum representation when asking for a first positive boundary energy: doing so would discard the currents at the vacuum boundary.

16.4 A nine-object example of condensation

Before applying the extension dictionary to Haagerup symmetry, consider the pointed center $Z(\mathrm{Vec}_{\mathbb{Z}_3})$ from Chapter 5. Its simples are (g,k) with $g,k\in\mathbb{Z}_3$, fusion adds both labels, and the double-braiding phase of (g,k) with (h,ℓ) is $\omega^{kh+\ell g}$, where $\omega=e^{2\pi i/3}$. Take the algebra

$$A=(0,0)\oplus(0,1)\oplus(0,2),$$

whose multiplication adds the second label. This algebra condenses the three pure charges. The construction is an ordinary finite Fourier problem because every component is one-dimensional and the relevant associator is trivial.

For each g, define the right module

$$x_g=(g,0)\oplus(g,1)\oplus(g,2).$$

The right action of $(0,\ell)$ sends the k component to the $k+\ell$ component. These three modules are simple as A-modules: a submodule containing any one component must contain all three. Every simple right module has one of these fixed fluxes, so the list is complete. The balanced tensor product identifies the three equivalent charge placements and gives

$$x_g\otimes_A x_h=x_{g+h}.$$

Thus the category of all modules is $\mathrm{Vec}_{\mathbb{Z}_3}$.

Locality imposes an extra requirement on the action. The double braiding with every condensed component $(0,\ell)$ has phase $\omega^{\ell g}$. Since the action is nonzero on all components, it is unchanged by this phase only when $g=0$. Hence the local module category has just its unit $x_0=A$.

There are more bimodules. One can twist the left action by any character $\ell\mapsto\omega^{t\ell}$ of $\mathbb{Z}_3$, independently for each x_g. The two labels (g,t) give nine simple bimodules. This small calculation displays the three different answers to three different questions: one local simple, three all-module simples, and nine bimodule simples. A holomorphic extension can therefore have a trivial local representation category and still carry nontrivial boundary and defect categories preserving the smaller chiral algebra.

16.5 Where the six Haagerup lines appear in the extension

The twelve Verlinde lines just discussed act on diagonal sectors $M_i\otimes\overline{M_i}$. In the extension construction below, the same twelve central labels index tube multiplicity spaces with dynamical sectors $M_i\otimes\overline{H'}$. The shared categorical labels do not identify these Hilbert spaces or their partition functions.

Let A be the selected canonical extension algebra and $\mathcal{C}=\mathrm{Rep}(U)_A\simeq\mathcal{H}_3$ the category of all right A-modules. Local A-modules form the trivial category because the extension is holomorphic. The nonlocal modules remain essential: discarding them would discard the six Haagerup objects.

The full E_8 theory has the corresponding U-preserving bimodule defects $\mathcal{C}\boxtimes\mathcal{C}^{\mathrm{op}}$, with 36 simple ordered pairs. This counts the rational defects preserving the specified U structure; it is not a classification of all defects preserving only the stress tensor. Here $\mathcal{C}^{\mathrm{op}}$ reverses tensor order. The displayed equivalence is the canonical-algebra bimodule theorem. The following calculation illustrates why one factor reverses: let $T_{x,y}(m)=x\otimes m\otimes y$. Then

$$T_{x,y}T_{x',y'}(m)=(x\otimes x')\otimes m\otimes(y'\otimes y).$$

Thus ordinary composition keeps the left order and reverses the right order. There is a further convention when comparing directly with right A-modules: a bimodule B acts by $T_B(M) = M \otimes_A B$, and

$$T_B \circ T_C \simeq T_{C\otimes_A B}.$$

This correspondence reverses composition. Passing to the opposite endofunctor category and exchanging the two factors recovers the displayed $\mathcal{C}\boxtimes\mathcal{C}^{\mathrm{op}}$ bimodule category. The $T_{x,y}$ calculation alone should therefore not be read as an identification with unchanged tensor conventions. Choosing the first factor gives six lines acting on the holomorphic side and leaving the antiholomorphic H' factor untouched. These defects preserve the original stress, because U and H' have the same conformal vector. They need not preserve every enlarged E_8 current. This is compatible with the trivial local module category of the maximal holomorphic algebra.

Let $F : Z(\mathcal{C}) \to \mathcal{C}$ forget the half-braiding, and abbreviate $X = \rho \oplus \alpha\rho \oplus \alpha^2\rho$. The forgetful multiplicities are

$$\begin{aligned} &F(0) = \mathbf{1}, && F(b) = \mathbf{1} \oplus X, && F(a) = 2\mathbf{1} \oplus X, \\ &F(c_j) = \alpha \oplus \alpha^2 \oplus X, && F(d_j) = X. \end{aligned} \tag{16.9}$$

Write $n_{x\mu} = \dim \mathrm{Hom}_{\mathcal{C}}(F(M_\mu), x)$. Free–forgetful adjunction proves that restriction of boundary object x to U is

$$Rx \cong \bigoplus_\mu n_{x\mu} M_\mu.$$

Under the rational defect construction for this selected extension, the first-factor twisted Hilbert space is

$$\mathcal{H}_x \cong \bigoplus_\mu W_x^\mu \otimes \mathcal{V}_\mu, \qquad W_x^\mu = \mathrm{Hom}(F(M_\mu), x), \quad \mathcal{V}_\mu = M_\mu \otimes \overline{H'}. \tag{16.10}$$

The finite tube factor W_x^μ and the infinite-dimensional CFT multiplicity space $\mathcal{V}_\mu$ have different roles. For instance $\dim W_1^a = 2$; the corresponding untwisted projector image contains two copies of $\mathcal{V}_a$. At left grade k and right grade ℓ in a specified μ summand, the number of states is $n_{x\mu} a_\mu(k) a_{E_8}(\ell)$. Thus $n_{1a} = 2$ and $a_a(0) = 81$ give 162 holomorphic weight-one states against the antiholomorphic vacuum. The small tube matrix acts on the two-copy factor, not on a two-state physical Hilbert space.

The twelve sector functions and the six twisted traces are

$$Z_\mu^{\mathrm{tube}} = \chi_\mu \overline{\chi_{E_8}}, \qquad Z_x = \sum_\mu n_{x\mu} Z_\mu^{\mathrm{tube}}. \tag{16.11}$$

The left and right central-charge phases cancel, so the twelve functions transform by (S, Θ) rather than (S, T_{ch}). For $x = 1$, the formula reduces to the ordinary E_8 partition function.

16.6 An explicit noninvertible action on untwisted states

On the multiplicity spaces of $H' = U \oplus b \oplus (\mathbb{C}^2 \otimes a)$, choose bases so that

	$\widehat{\alpha}$	$\widehat{\rho}$
U	1	d
b	1	$-d^{-1}$
$\mathbb{C}^2 \otimes a$	$\begin{pmatrix} \omega & 0 \\ 0 & \omega^2 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$\omega = e^{2\pi i/3}$. (16.12)

The last matrices satisfy $\alpha\rho = \rho\alpha^{-1}$. Their three noninvertible matrices sum to zero, so $\rho^2 = 1$ on that block agrees with $\rho^2 = 1 + \rho + \alpha\rho + \alpha^2\rho$. On the neutral blocks the same rule is the quadratic equation with roots $d, -1/d$.

Taking traces gives

$$Z^{[\alpha]} = (\chi_0 + \chi_b - \chi_a)\overline{\chi_{E_8}},$$
$$Z^{[\rho]} = (d\chi_0 - d^{-1}\chi_b)\overline{\chi_{E_8}}. \tag{16.13}$$

The absence of χ_a in the second line is the zero trace of the off-diagonal 2×2 matrix. Squaring the holomorphic factors in absolute value would describe a different paired left/right insertion, not this selected first-factor action.

Worked example 16.5 (Recovering twisted spectra from the line action)**.** Write the holomorphic coefficient vectors of the two twined traces in the center ordering as

$$t_\alpha = (1, 1, -1, 0, \dots, 0), \qquad t_\rho = (d, -d^{-1}, 0, \dots, 0).$$

Multiplication by the explicit S matrix gives

$$St_\alpha = (0, 0, 0, 1, 1, 1, 0, \dots, 0), \qquad St_\rho = (0, 1, 1, 1, 1, 1, 1, \dots, 1).$$

For the second vector, the vacuum entry vanishes because $S_{0b} = d^2 S_{00}$. The middle entries are $(d - d^{-1})/3 = 1$, and the six final entries are $(d + d^{-1})/\sqrt{13} = 1$. The b entry is one as well, using $d^2 - d^{-2} = 3\sqrt{13}$. These are exactly the forgetful multiplicities of α and ρ in (16.9). Thus the small line matrices, modular transformation, and actual restriction give three compatible descriptions of the same twisted traces.

Every antiholomorphic E_8 weight is integral. Hence the c_1, c_2, d_j tube multiplicity spaces contain no scalar states at any grade: their left weights are nonintegral modulo one. The first invariant scalars are instead in $U_1 \otimes \overline{H_1'}$, with dimension two and multiplicity $6 \cdot 248 = 1488$. This derives the invariant gap quoted earlier.

16.7 Six boundaries of the extended theory

The same module category gives six elementary boundaries preserving U. For boundary labels $x, y \in \mathcal{C}$,

$$Z_{xy}^{E_8,\mathrm{open}} = \sum_{z,\mu} N_{x^* y}^{\ z}\, n_{z\mu} \chi_\mu. \tag{16.14}$$

Now the dual is essential: $\alpha^* = \alpha^2$, whereas the three noninvertible simples are self-dual. At the level of morphisms, duality and the central half-braiding give

$$\mathrm{Hom}_{\mathcal{C}}(F(M_\mu) \otimes x, y) \cong \mathrm{Hom}_{\mathcal{C}}(F(M_\mu), x^* \otimes y).$$

Free–forgetful adjunction then identifies this space with the M_μ multiplicity in the restriction of $x^* \otimes_A y$. This derives the dual in the annulus formula from the boundary junction rather than guessing it from a commutative fusion table. The identity multiplicity is consequently $\dim \mathrm{Hom}(x, y) = \delta_{xy}$ for elementary boundaries. For example, the α self-annulus uses $\alpha^* \alpha = \mathbf{1}$; omitting the dual would incorrectly use α^2 and remove its boundary identity. Restriction multiplies quantum dimensions by $\dim A = \mathcal{D}$. Taking the closed-channel vacuum coefficient gives $S_{00} \dim R(x^* \otimes_A y) = d_x d_y$. Reflection positivity selects positive overlaps, so

$$(g_1, g_\alpha, g_{\alpha^2}, g_\rho, g_{\alpha\rho}, g_{\alpha^2\rho}) = (1, 1, 1, d, d, d).$$

Worked example 16.6 (A noninvertible boundary self-spectrum)**.** For $x = \rho$, use $\rho^2 = 1 + \rho + \alpha\rho + \alpha^2\rho$. The three noninvertible objects have identical restriction characters. Thus

$$Z_{\rho\rho}^{E_8,\mathrm{open}} = \chi_0 + 4\chi_b + 5\chi_a + 3(\chi_{c_0} + \chi_{c_1} + \chi_{c_2}) + 3\sum_{j=1}^{6} \chi_{d_j}.$$

The identity occurs once and the first positive weight is $1/3$, with $3 \cdot 3 = 9$ states. By contrast, the invertible boundary self-annuli equal χ_{E_8} and start at positive weight one with 248 currents. The different boundary entropies and relevant boundary spectra make the categorical distinction observable.

16.8 Exercises

Exercise 16.1. Check directly that the invariant projector defined above is idempotent and Hermitian in the diagonal theory. Why is the denominator $\mathcal{D}^2$ rather than $\mathcal{D}$?

Solution on page 305.

Exercise 16.2. Using the matrices in (16.12), verify every Haagerup fusion relation and derive both traces in (16.13).

Solution on page 305.

Exercise 16.3. Compute $R\alpha$, its lowest conformal weight, and its lowest top multiplicity. Explain why this twisted sector can contain fractional spin without violating locality of the untwisted full theory.

Solution on page 306.

Exercise 16.4. Compute the number of weight-$1/3$ boundary fields between two distinct noninvertible boundaries of the extension. Compare it with the self-boundary count nine.

Solution on page 306.

Exercise 16.5. Explain what is counted by 6, 12, and 36 in this chapter, and identify which category is braided. Why would substituting local extension modules for all extension modules erase the Haagerup boundary category?

Solution on page 306.

Chapter 17

What it means to complete a local CFT

A modular-invariant partition function is a powerful constraint, but it is still a count of states. A local theory must also assign correlation functions consistently when surfaces are cut and sewn. This chapter explains that additional structure through a completely worked Ising example, then returns to the Haagerup construction and the reduced-theory proposals.

17.1 Cutting a surface introduces a channel basis

Cut a four-punctured sphere along a circle separating two insertions from the other two. Insert a complete set of intermediate states on that circle. In a rational chiral theory this produces a finite sum of conformal blocks, each block already summing infinitely many descendants in an intermediate module. If the relevant intertwiner space has dimension greater than one, the channel also carries a multiplicity index.

Let $\mathcal{F}_p^{(s)}(z)$ be a chosen basis of holomorphic blocks for one cut. A full correlator takes the form

$$G(z,\bar{z}) = \sum_{p,q} D_{pq}\, \mathcal{F}_p^{(s)}(z)\overline{\mathcal{F}_q^{(s)}(z)}. \tag{17.1}$$

Here p, q include multiplicity labels, and D contains products of normalized bulk three-point coefficients. It is not fixed by the torus matrix Z_{ij} alone.

Changing the cut changes the block basis by a fusing matrix. Analytically continuing an insertion around another changes it by a braiding or monodromy matrix. The pairing D must transform consistently so that the full correlator is single-valued and independent of the chosen cut. In a reflection-positive channel, the two groups of insertions are related by reflection and adjoint. If one group prepares states Ψ_a, the reflected pairing is a Gram matrix:

$$K_{ab} = \langle \Psi_a, \Psi_b \rangle, \qquad \sum_{a,b} \overline{c_a} K_{ab} c_b = \Big\| \sum_b c_b \Psi_b \Big\|^2 \geq 0.$$

This explains the positive semidefinite coefficient pairing in such a channel, with separated insertions and positive Euclidean propagation understood. A basis change can move phases between blocks and coefficients, but cannot remove a failure of these compatibility conditions.

The categorical associator and braiding record these changes of chiral channel. They constrain a local completion, while the choice of bulk algebra and positive pairing specifies how the left and right chiral data are combined. For U conformal blocks, these fusing matrices belong to $\mathrm{Rep}(U) \simeq Z(\mathcal{H}_3)$. The associators for junctions of the six selected symmetry lines belong to $\mathcal{H}_3$. The center construction relates the two categories, but their F matrices are different objects. In

particular the four-by-four ρ recoupling block is not a conformal-block matrix for arbitrary U primaries.

17.2 A complete two-channel example from the Ising model

The Ising spin field σ has chiral weight $1/16$, and $\sigma \times \sigma = \mathbf{1} + \epsilon$, with $h_\epsilon = 1/2$. We normalize the full fields by

$$\langle \sigma(z,\bar z)\sigma(0)\rangle = |z|^{-1/4}, \qquad \langle \epsilon(z,\bar z)\epsilon(0)\rangle = |z|^{-2}.$$

The full spin and energy fields have weights $(1/16, 1/16)$ and $(1/2, 1/2)$ respectively. We calculate $G_\sigma = \langle \sigma(\infty)\sigma(1)\sigma(z,\bar z)\sigma(0)\rangle$, where $\sigma(\infty) = \lim_{|w|\to\infty} |w|^{1/4}\sigma(w,\bar w)$. A useful pair of chiral blocks is

$$\begin{aligned} \mathcal{F}_1(z) &= \frac{[z(1-z)]^{-1/8}}{\sqrt{2}}\sqrt{1+\sqrt{1-z}}, \\ \mathcal{F}_\epsilon(z) &= \frac{[z(1-z)]^{-1/8}}{\sqrt{2}}\sqrt{1-\sqrt{1-z}}. \end{aligned} \tag{17.2}$$

Choose positive square roots on $0 < z < 1$ and then specify paths for analytic continuation. Near zero,

$$\mathcal{F}_1(z) = z^{-1/8}(1 + O(z^2)), \qquad \mathcal{F}_\epsilon(z) = \frac{1}{2}z^{3/8}(1 + O(z)).$$

The exponent difference is the exchanged weight $1/2$.

These blocks can be obtained from the level-two null vector of σ. The Gram-matrix exercise gave $(L_{-2} - \frac{4}{3}L_{-1}^2)|\sigma\rangle = 0$. Insert this null field in a correlator and move the stress contour onto the other insertions. The Ward identity replaces the L_{-2} insertion by $\sum_{i\neq z}[h_\sigma/(z-z_i)^2 + (z-z_i)^{-1}\partial_{z_i}]$ acting on the correlator. Use the three global conformal Ward identities to fix the other insertions at $0, 1, \infty$. Their derivatives become $\partial_0 f = (z-1)f' + 2h_\sigma f$ and $\partial_1 f = -zf' - 2h_\sigma f$ before substituting the fixed positions. For $f(z) = \langle \sigma(\infty)\sigma(1)\sigma(z)\sigma(0)\rangle$ the resulting differential equation is

$$f''(z) + \frac{3(1-2z)}{4z(1-z)}f'(z) - \frac{3}{64z^2(1-z)^2}f(z) = 0. \tag{17.3}$$

Its indicial equation at zero is $r(r-1) + 3r/4 - 3/64 = 0$, with roots $-1/8$ and $3/8$. They are precisely $h_{\text{exchange}} - 2h_\sigma$ for the vacuum and energy channels. Substitution verifies the two radical solutions above. A second-order equation has a two-dimensional local solution space away from its singularities, so these independent solutions exhaust the chiral blocks for this null-vector problem.

Under exchange of the s and t cuts,

$$\begin{pmatrix} \mathcal{F}_1(1-z) \\ \mathcal{F}_\epsilon(1-z) \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} \mathcal{F}_1(z) \\ \mathcal{F}_\epsilon(z) \end{pmatrix}. \tag{17.4}$$

The identity follows by squaring the half-angle radical relations and checking their signs on $0 < z < 1$. The matrix is real orthogonal. It does not preserve either individual channel: a vacuum block in one cut contains both channels in the other.

Suppose the full correlator has pairing matrix D. Monodromy around zero gives phases $e^{-\pi i/4}$ and $e^{3\pi i/4}$ to the two holomorphic blocks. The phases differ by a sign, so single-valuedness removes off-diagonal entries of D. Write $D = \operatorname{diag}(a, b)$. Crossing requires $F^{\mathrm{t}}DF = D$, and its off-diagonal entry is $(a-b)/2$. Thus $a = b$. Unit normalization of the vacuum channel sets $a = 1$, yielding

$$G_\sigma(z,\bar z) = |\mathcal{F}_1(z)|^2 + |\mathcal{F}_\epsilon(z)|^2. \tag{17.5}$$

This is a local, crossing-symmetric positive pairing. In the $z \to 0$ expansion, the energy channel has relative coefficient $1/4$, so the normalized full coefficient obeys $|C_{\sigma\sigma\epsilon}| = 1/2$.

The final answer also has a branch-independent form:

$$G_\sigma(z,\bar{z}) = \frac{\sqrt{1+|z|+|1-z|}}{\sqrt{2}\,|z(1-z)|^{1/4}}. \tag{17.6}$$

Indeed, for $w = \sqrt{1-z}$ one has $(|1+w|+|1-w|)^2 = 2(1+|1-z|+|z|)$. The expression makes single-valuedness explicit and verifies both crossing generators:

$$G_\sigma(1-z,1-\bar{z}) = G_\sigma(z,\bar{z}), \qquad G_\sigma(z,\bar{z}) = |z|^{-1/4} G_\sigma(1/z,1/\bar{z}).$$

The prefactor in the second equation is fixed by the external dimension; crossing is not invariance under every fractional linear transformation without such factors.

Channel mixing explains why a single chiral block usually does not give a full correlator by taking its absolute square. Here $|\mathcal{F}_1|^2$ alone fails crossing, even though it is nonnegative on the real interval and has the correct leading vacuum singularity. Those two properties are insufficient.

17.3 Algebra objects organize rational sewing

In the rational framework, a suitable symmetric special Frobenius algebra object A in the representation category of an existing rational chiral theory organizes the oriented full-CFT sewing data. The chiral theory and its analytic conformal blocks are inputs; an abstract modular category alone does not construct them. Multiplication specifies how algebra lines meet, associativity makes different decompositions agree, and the Frobenius relation relates multiplication to a compatible comultiplication. In the unitary Q-system normalization this comultiplication is the adjoint of multiplication. The positive Q-system and unitary chiral theory provide the adjoint and Hilbert-space pairings; the purely algebraic Frobenius identities alone do not assert reflection positivity. The resulting bulk state space is described by the full center of A.

The TFT construction proves mapping-class invariance and bulk and boundary factorization for these categorical correlators on oriented world sheets. Their interpretation as analytic amplitudes on Riemann surfaces also uses the chiral conformal-block theory, including its sewing and convergence theorems. The annulus comparison in Chapter 12 establishes the positivity and VOA–net statements used here; it is not itself a proof of every higher-genus analytic sewing theorem. The diagonal local conformal net already exists by the construction in Chapter 14.

For the Cardy or diagonal construction, $A = \mathbf{1}$. Its bulk object is the canonical sum

$$\bigoplus_i M_i \boxtimes M_i^*$$

in $\mathrm{Rep}(U) \boxtimes \mathrm{Rep}(U)^{\mathrm{rev}}$, with a multiplication built from the chiral fusion and duality data. The torus trace is the diagonal partition function, with the right-moving contragredient convention understood. All twelve Haagerup labels, and all Ising labels used here, are self-dual, so the distinction between diagonal and charge-conjugate labelings does not change these examples. For a local chiral extension, a connected commutative separable twist-free algebra in $\mathrm{Rep}(U)$, with the positive structure and grading conditions of the extension theorem, supplies additional chiral fields; the selected Haagerup Lagrangian algebra gives the holomorphic extension studied above.

The underlying object is only part of the algebra. A decomposition such as $A = \mathbf{1} \oplus b \oplus 2a$ does not state its multiplication maps or their associativity. Equal characters of a and c_0 make this particularly visible: a scalar trace can conceal an extension choice that the junction data distinguish.

Morita-equivalent algebra objects give equivalent full-CFT descriptions in the appropriate rational construction, while their boundary presentations can look different. This is one reason module categories, rather than only fusion rings, are natural physical objects. The associator determines whether a candidate multiplication is actually associative; a list of integer fusion multiplicities cannot replace it.

17.4 Existence and explicit OPE tables are different achievements

Existence of the diagonal full theory does not require printing every coefficient in a basis of its bulk primary fields. Conversely, a long table of low three-point coefficients does not by itself prove all sewing identities in all sectors and genera. These are complementary forms of information.

For the Haagerup $c = 8$ construction, the lattice vacuum algebra supplies explicit chiral OPEs, the complete weight-two primary tensor, and several exact four-point functions. The structural and analytic arguments supply the representation category, unitarity, and full-theory existence under their stated inputs. The selected lattice intertwiners in the lightest module give additional nonvacuum chiral data.

What is not supplied by those particular calculations is a complete basis of all nonvacuum intertwiner spaces together with the numerical or algebraic bulk sewing tensor for every diagonal sector. In particular the nine simple lattice blocks of (15.14) are not a declaration of the complete lightest-scalar bulk four-point function. They remember the containing B extension and its chosen intertwiners.

For full vacuum-sector fields that factor as $\Phi(z, \bar{z}) = v(z)\bar{w}(\bar{z})$, correlators do factor into the corresponding chiral and antiholomorphic correlators. This gives many explicit local observables in both full theories. The factorization follows from the specified tensor-product sector, not from a general rule that every nonvacuum bulk field is the product of any two selected chiral blocks.

17.5 A coset is a chiral operation

Suppose A and $C = \mathrm{Com}(A, V)$ commute inside a VOA V, with complementary conformal vectors $\omega_V = \omega_A + \omega_C$. When an ambient module has a semisimple restriction with compatible ordinary gradings, it can branch as

$$V_i \downarrow_{A\otimes C} = \bigoplus_\lambda A_\lambda \otimes B_{i\lambda}, \qquad \chi_i^V = \sum_\lambda \chi_\lambda^A b_{i\lambda}. \tag{17.7}$$

Here $B_{i\lambda}$ is a multiplicity space carrying C fields, and $b_{i\lambda}$ is its graded trace when that trace is defined. The equality of characters is a branching identity. Deleting the A characters from a full partition function is not a general sewing theorem.

Several issues can arise. A multiplicity space might fail to be irreducible or ordinary for C. Infinitely many branching labels can produce a divergent reduced thermal trace. Distinct ambient sectors can produce repeated copies of the reduced vacuum. A putative left/right pairing can fail modular invariance or crossing. Finally, an ambient defect need not descend with coherent junctions. These are separate questions and can fail independently.

The exact reverse-commutant example in Chapter 18 gives a memorable finite warning: a naive branching sum produces $6Z_L$, because multiplicities contribute $1^2 + 1^2 + 2^2 = 6$ vacuum copies. The lattice CFT Z_L may exist, but obtaining it requires a new, justified choice of completion. Dividing a trace by six does not construct its multiplication or its defect action.

For an irrational stress splitting, the challenge can be analytic as well as categorical. A formal diagonal pairing of branching spaces with one vacuum may still have infinitely many states below

some energy and an infinite thermal trace. Conversely, projecting to a single chiral vacuum character can violate modular invariance. A continuous modular density with no normalizable vacuum describes yet another possibility and has different effective-central-charge behavior.

17.6 Defect descent requires junctions

A line preserving a candidate stress tensor preserves its local conformal transformations. This is necessary for a topological symmetry of the reduced theory. It is not sufficient to identify the reduced defect category with $\mathcal{H}_3$. One needs images of the lines and of their junction spaces, compatible tensor-product maps, and the same coherent associativity relations. In categorical language, one needs an appropriate monoidal transport, not only a list of operators commuting with L_0.

For example, operators may satisfy the fusion-algebra relations, including $\rho^2 = 1+\rho+\alpha\rho+\alpha^2\rho$, on a selected vector space while carrying no information about the F symbols of four line junctions. The ordinary Haagerup fusion algebra has only two one-dimensional representations and one two-dimensional representation. It cannot encode the full tensor product of local charge sectors by itself. The tube algebra and its half-braiding data supply more information, but their transport must still be tied to the actual reduced fields and sewing.

This distinction is constructive rather than merely restrictive. In the rational $c = 6$ commutant discussed later, the extension correspondence transports an algebra object and its multiplication. That supplies junction data, which is stronger than a numerical agreement of line eigenvalues. For the irrational near-two axis, an analogous full transport has not been completed in the research record.

17.7 How a proposed identification should be assessed

A convincing identification of a microscopic continuum theory should specify the actual Hilbert space and vacuum, a convergent torus trace, normalized local fields and their OPEs, crossing and sewing, and the intended symmetry action including junctions. One rarely computes all of these from a lattice simulation. Instead, an exact construction can supply them and numerical data can test whether the lattice approaches that construction.

Matching central charge is the first coarse test. Matching dimensions, spins, multiplicities, and symmetry sectors is much stronger. Matching normalized OPE ratios and phase invariants is stronger again. Every comparison needs control of finite-size, finite-entanglement, and operator-normalization errors. A mismatch can exclude a specified candidate under explicit limiting assumptions, but an unexplained mismatch need not classify all possible CFTs at the same central charge.

This is the perspective we will use for the physical Haagerup chain. It makes the existing evidence informative without turning a suggestive decimal or a chiral stress vector into a completed bulk construction.

17.8 Exercises

Exercise 17.1. Verify (17.4) on the interval $0 < z < 1$ by elementary radical identities. Use it to show directly that $|\mathcal{F}_1|^2$ alone fails crossing.

Solution on page 306.

Exercise 17.2. Starting with a general Hermitian 2×2 matrix D, use the two Ising monodromy phases and the fusing matrix to determine all local crossing-invariant pairings. Impose positivity

and vacuum normalization at the end.

Solution on page 307.

Exercise 17.3. Explain the difference between an equality of characters, an isomorphism of underlying modules, an isomorphism of algebra objects, and an equivalence of full local CFTs. Give one piece of data forgotten at each step toward the character.

Solution on page 307.

Exercise 17.4. A proposed reduced theory has one vacuum but infinitely many states with energies bounded by a fixed constant. Prove that its thermal trace diverges at every positive inverse temperature. Why does positivity make this argument especially direct?

Solution on page 307.

Exercise 17.5. An ambient Haagerup line action commutes with two complementary stress tensors. List the extra structures needed to turn its restriction into a specified Haagerup symmetry of one reduced full theory. Which of these are visible in a torus trace?

Solution on page 307.

Part IV

Reduced theories and the physical Haagerup chain

Chapter 18

Cosets, stress splitting, and the near-two frontier

A central-charge-eight construction naturally invites a smaller theory: can part of its stress tensor be removed while preserving Haagerup symmetry? This chapter explains both a successful charge-six reduction and the obstacles in the charge-two direction. It also constructs an irrational stress of charge $2.025004294\ldots$ and follows the precise distance between that chiral construction and a local full CFT.

The distinction is structural. Finding a Virasoro vector is an OPE problem inside one chiral algebra. A full reduced theory additionally needs a state space, a modular pairing, convergent positive OPEs, and coherent defect junctions. None of those later requirements can be inferred from a central charge being close to two.

18.1 A commuting stress is not automatically a tensor factor

Let V be unitary, with stress ω and translation operator $D = L^V_{-1}$. Write $L^e_n = e_{(n+1)}$ for the modes of a PCT-real weight-two vector e. Here PCT-real means fixed by the antilinear involution of Chapter 3, with the positive form transported in the chosen coordinate gauge.

Definition 18.1 (Semiconformal Virasoro subalgebra)**.** The vector e is semiconformal if the vertex subalgebra $\mathrm{Vir}(e) \subset V$ has e as its conformal vector and

$$L^e_n|_{\mathrm{Vir}(e)} = L^V_n|_{\mathrm{Vir}(e)} \quad (n \geq 0), \qquad L^e_{-1}|_{\mathrm{Vir}(e)} = D|_{\mathrm{Vir}(e)}.$$

These are equalities on the generated subalgebra; they are not equalities on all of V. The last equality specifies compatibility with the inherited translation operator.

This definition explains why $f = \omega - e$ is a commuting stress. For every $a \in \mathrm{Vir}(e)$ and $m \geq 0$, the restrictions give $f_{(m)}a = (L^V_{m-1} - L^e_{m-1})a = 0$. Skew symmetry then gives $a_{(m)}f = 0$ as well. Thus e and f have no singular mixed OPE. Subtracting their contributions from the ambient Virasoro OPE shows that f is a Virasoro vector of charge $c_V - c_e$; its translation on $\mathrm{Vir}(f)$ is again the inherited D. In particular,

$$e(z)e(w) \sim \frac{c_e/2}{(z-w)^4} + \frac{2e(w)}{(z-w)^2} + \frac{\partial e(w)}{z-w}, \qquad e(z)f(w) \text{ is regular},$$

and $c_f = c_V - c_e$.

The commutant

$$\mathrm{Com}(A, V) = \{v \in V : a_{(n)}v = 0 \text{ for all } a \in A,\ n \geq 0\}$$

collects fields whose OPEs with A are regular. In the unitary setting it is often an energy-zero space for a complementary stress. This gives an actual chiral subalgebra with a definite positive form. It does not imply that V is the tensor product of the two commutants. Nonvacuum paired modules may be needed to reconstruct V.

Worked example 18.2 (A lattice extension with two commuting stresses)**.** The plane lattice L_0 of Chapter 9 and its orthogonal rank-six lattice N give commuting stress tensors inside V_{E_8}. Their central charges add to eight. Nevertheless the E_8 lattice contains glue cosets beyond $L_0 \oplus N$, so V_{E_8} is a finite extension of $V_{L_0} \otimes V_N$, not that tensor product itself. The extra paired sectors are exactly what a naive stress subtraction would overlook.

There is also a categorical obstruction to an actual rational $c = 2$ times $c = 6$ factorization of U. Its simple quantum dimensions are $1, d^2$, four copies of $s = 1 + d^2$, and six copies of $3d$. The unit is its only invertible simple, and

$$(3d)^2 > s.$$

If $\mathrm{Rep}(U) = \mathcal{D} \boxtimes \mathcal{E}$ with both factors nontrivial, neither could be pointed. Choose nonunit simples X, Y from the two factors. Each has dimension at least $3d$, whereas the simple $X \boxtimes Y$ would exceed the largest allowed dimension s. Thus no such categorical factorization exists. If one VOA tensor factor were holomorphic, its central charge would be a multiple of eight, incompatible with charges two or six. This excludes a tensor product, while allowing proper conformal extensions and commutants.

18.2 The concrete charge-six commutant

Recall

$$L_0 = \mathbb{Z}\ell + \mathbb{Z}v, \qquad G_{L_0} = \begin{pmatrix} 2 & 3 \\ 3 & 24 \end{pmatrix}, \qquad \det L_0 = 39.$$

Let P be its real plane in the E_8 lattice Λ. Since 39 is squarefree, the displayed lattice is saturated: $L_0 = \Lambda \cap P$. Indeed an integral overlattice of index r would have determinant $39/r^2$, forcing $r = 1$.

Let $N = \Lambda \cap P^{\perp}$. Its rank is six and its determinant is also 39. An explicit description is

$$N = (A_2 \oplus A_2 \oplus A_1 \oplus \langle 78 \rangle) + \mathbb{Z}(\omega_1, \omega_2, \alpha/2, \gamma_{78}/6), \tag{18.1}$$

Here ω_i is a fundamental weight of the ith A_2 factor, α generates A_1, and γ_{78} generates the rank-one lattice $\langle 78 \rangle$. Their squared norms are $\omega_i^2 = 2/3$, $\alpha^2 = 2$, and $\gamma_{78}^2 = 78$. The glue vector has squared norm

$$\frac{2}{3} + \frac{2}{3} + \frac{1}{2} + \frac{78}{36} = 4.$$

It has order six modulo the direct-sum lattice. Its index-six extension has determinant $3 \cdot 3 \cdot 2 \cdot 78/6^2 = 39$, as required.

Define

$$C_6 = \mathrm{Com}(V_{L_0}, U). \tag{18.2}$$

Positive lattice energy gives an explicit ambient realization,

$$C_6 = U \cap V_N.$$

To see it, a state commuting with the plane Heisenberg algebra has neither plane oscillator excitations nor plane momentum. Such states are exactly V_N. Conversely, commuting with the full plane lattice is no stronger on this subspace. Its stress is the perpendicular free-boson stress, of charge six.

The pair is mutual:

$$\mathrm{Com}(C_6, U) = V_{L_0}.$$

The key is that C_6 contains its full perpendicular stress. A vector commuting with that stress has zero perpendicular energy, so has no perpendicular momenta or oscillators. Saturation identifies the remaining lattice states with precisely V_{L_0}. This is an all-grade argument, not a low-weight count.

The regular lattice-coset theorem now applies: a strongly rational VOA with this positive-definite lattice double commutant has a strongly rational Heisenberg/lattice coset. Thus C_6 is strongly rational. This uses a specific theorem with verified lattice hypotheses; rationality of arbitrary cosets is not being assumed.

18.3 Why the Haagerup extension data survive at charge six

Let $P_{39} = \mathrm{Rep}(V_{L_0})$. It is the nondegenerate pointed category on $\mathbb{Z}/39$ with quadratic form $q(n) = n^2/39$. The actual core A_L contains every one of these lattice labels. The mirror theorem for the rational mutual commutants therefore identifies their branching subcategory inside $\mathrm{Rep}(C_6)$ with P_{39}^{rev}. The superscript rev reverses the braiding and twist; reversing tensor order is instead denoted op below. Nondegeneracy splits off this pointed factor:

$$\mathrm{Rep}(C_6) = P_{39}^{\mathrm{rev}} \boxtimes \mathcal{D}.$$

The extension U of $V_{L_0} \otimes C_6$ is the diagonal algebra on the two pointed factors, tensored with the unit in $\mathcal{D}$. Condensing that diagonal algebra leaves $\mathcal{D}$, while the extension correspondence says that it leaves $\mathrm{Rep}(U)$. Hence

$$\mathrm{Rep}(C_6) \simeq P_{39}^{\mathrm{rev}} \boxtimes \mathrm{Rep}(U). \tag{18.3}$$

In the Haagerup construction its rank is $39 \cdot 12 = 468$. The reverse pointed factor has chiral central charge six modulo eight, consistent with the actual charge-six stress.

Let A denote the actual algebra of the original E_8/U extension. Transitivity of extension algebras identifies the E_8 algebra over $V_{L_0} \otimes C_6$ as

$$\mathrm{Diag}(P_{39}) \boxtimes A.$$

Taking the V_{L_0} vacuum multiplicity leaves $\mathbf{1} \boxtimes A$. The same vacuum multiplicity in V_{E_8} is V_N, so the actual extension V_N/C_6 realizes this same algebra, including its multiplication. Put $\mathcal{F} = \mathrm{Rep}(U)_A$, the category of all right A-modules. The algebra A is Lagrangian in $\mathrm{Rep}(U)$, and the canonical-algebra reconstruction therefore gives

$$\begin{aligned} {}_{\mathbf{1}\boxtimes A}\,\mathrm{Rep}(C_6)_{\mathbf{1}\boxtimes A} &\simeq P_{39}^{\mathrm{rev}} \boxtimes {}_A\,\mathrm{Rep}(U)_A \\ &\simeq P_{39}^{\mathrm{rev}} \boxtimes \mathcal{F} \boxtimes \mathcal{F}^{\mathrm{op}}. \end{aligned} \tag{18.4}$$

Here $\mathcal{F}^{\mathrm{op}}$ reverses tensor order. The total algebra $\mathbf{1} \boxtimes A$ is not Lagrangian in $\mathrm{Rep}(C_6)$: its local modules retain the pointed factor P_{39}^{rev}.

For this original embedded extension, the result of Chapter 12 is $\mathcal{F} \in \{\mathcal{H}_2, \mathcal{H}_3\}$. No step here selects between them. In particular, A in this section continues to mean the original extension algebra, not the separately selected canonical $\mathcal{H}_3$ algebra of Chapter 16. The transported multiplication preserves whichever of these boundary categories the original embedding has.

The charge-six full state space can now be written explicitly. Let $\mathcal{H}_{\lambda+N}$ be the Hilbert completion of the lattice module $V_{\lambda+N}$, where $\lambda \in N^*/N$. The ordinary diagonal lattice theory has

$$\mathcal{H}_6 = \bigoplus_{\lambda \in N^*/N} \mathcal{H}_{\lambda+N} \otimes \overline{\mathcal{H}_{\lambda+N}}, \qquad c_L = c_R = 6. \tag{18.5}$$

The bar denotes the conjugate antiholomorphic sector. There are $|N^*/N| = \det N = 39$ chiral sector labels, and its torus amplitude is

$$Z_6(\tau,\bar\tau) = \sum_{\lambda\in N^*/N} \left| \frac{\Theta_{\lambda+N}(\tau)}{\eta(\tau)^6} \right|^2 .$$

Positive definiteness of N gives a unique zero-energy vector, and the standard lattice construction supplies the positive local OPE and sewing. This is an actual compact full theory, rather than only a proposed central charge or branching function.

To interpret (18.4) as physical defects on (18.5), one uses the positive rational extension and bimodule-to-defect correspondence of Chapters 12 and 16 for the actual inclusion $C_6 \subset V_N$. Its analytic premise is compatibility of the unitary sector realization with this extension and its intertwiners. Regularity of C_6 and the lattice bounded-insertion theorem already give a finite-index rational subnet; that fact alone should not be substituted for the required sector comparison. With this analytic correspondence in place, the unit pointed spectator in (18.4) transports the original $\mathcal{F} \boxtimes \mathcal{F}^{\mathrm{op}}$ defects, including their junctions, to the charge-six theory. These defects preserve C_6. The simple defects required to preserve the larger lattice algebra V_N pointwise remain pointed. This distinction also explains why retaining the smaller preserved algebra is essential here.

The multiplication is the decisive extra information. Restricting the matrix representing a line on a few states would not prove that its junctions still associate correctly. Here the extension correspondence transports the algebra that encodes those junctions.

18.4 The reverse coset gives six vacua if labels are retained

The same calculation explains why deleting the charge-six factor does not automatically give the desired charge-two theory. The reverse commutant in E_8 is exactly V_{L_0}. Write the original algebra as

$$A = \mathbf{1} \oplus X \oplus 2Y.$$

For the C_6 labels indexed by a lattice charge n and an extension label a, the complementary branching functions are $b_{n,a} = m_a \chi_n^{L_0}$ with multiplicities $m_a = 1, 1, 2$. An unreduced diagonal sum gives

$$\sum_{n,a} |b_{n,a}|^2 = (1^2 + 1^2 + 2^2) \sum_n |\chi_n^{L_0}|^2 = 6 Z_{L_0}. \tag{18.6}$$

Its vacuum coefficient is six. Identifying the repeated branching spaces instead gives the ordinary unique-vacuum torus theory, but then its rational category is the pointed P_{39}, not the Haagerup center. The two tasks—removing repeated vacua and retaining the noninvertible junction category—are independent.

An ordinary rational full theory preserving the lattice chiral algebra V_{L_0} has a weakly integral defect category: every simple defect has integer squared Frobenius–Perron dimension. Here weak integrality means that the global Frobenius–Perron dimension is an integer; the assertion about its simple dimensions is a theorem for such fusion categories. A unitary strong tensor action of $\mathcal{H}_3$ would preserve the simplicity of ρ by Proposition 18.8 below. Its dimension has irrational square, $d^2 = (11 + 3\sqrt{13})/2$, so this rational lattice interpretation cannot carry the desired lattice-preserving Haagerup action. A new current-breaking action would be a separate construction, not a consequence of (18.6).

18.5 What ordinary invariant cosets cannot remove

There are several useful obstructions with carefully bounded scope. The retained current Lie algebra in the accepted charge-eight realization is $U_1 = \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$, generating the affine subalgebra $A_{1,1} \otimes A_{1,39}$. The currents of any transparent compact denominator lie in this retained Lie algebra. Positivity of its complementary Sugawara stress gives

$$c_{\text{denominator}} \leq \frac{158}{41}, \qquad c_{\text{coset}} \geq 8 - \frac{158}{41} = \frac{170}{41} > 2.$$

Allowing each original Haagerup line to transmit the denominator by one group automorphism does not evade this bound. If a group label g_x is assigned to each simple line and every fusion summand z of $x \otimes y$ obeys $g_z = g_x g_y$, then ρ^2 containing both 1 and ρ forces $g_\rho^2 = 1 = g_\rho$. Thus $g_\rho = 1$, and the other summands force $g_\alpha = 1$. All labels are trivial.

The statement assumes one automorphism per simple line and compatible transmission through fusion junctions. General noninvertible maps that mix or annihilate fields need not have this form. Generalized quotients that produce new lines after gauging are also outside the argument.

The standard A_2 and E_6 stresses are not in U. A particularly short diagnostic is to apply the A_2 stress operator $L_0^{A_2}$ to H_R: it selects the A_2 Cartan part of h, which is not in the two-dimensional retained Cartan span. If the stress belonged to U, closure would put that extra current in U_1, contradicting the six-current theorem.

Finally $C_{6,1} = 0$ and its exact weight-two algebra has dimension seven. Its polynomial Virasoro-vector equations have no solutions of charges one or two, even over $\mathbb{C}$. Complementation inside its charge-six stress also excludes charges five and four. Thus a further ordinary charge-four conformal coset of this particular C_6 cannot produce charge two. These are exclusions of specified mechanisms, not a classification of all possible Haagerup CFTs at charge two.

18.6 A nearby rational stress from diagonal affine currents

The compact diagonal of $A_{1,1} \otimes A_{1,39}$ has level forty. This is the diagonal affine construction used in Example 3.11: the current double poles add the levels, and the coset stress subtracts the diagonal Sugawara stress from the two numerator stresses. In the coordinate gauge of Chapter 7, its triple is

$$(H_L + H_R,\ 8640E_L + E_R,\ F_L/8640 + F_R).$$

The reciprocal rescaling makes this a PCT-stable compact diagonal. Its GKO stress

$$e_0 = \omega_L + \omega_R - \omega_{A_{1,40}}$$

has charge

$$1 + \frac{117}{41} - \frac{120}{42} = \frac{286}{287} = 1 - \frac{6}{41 \cdot 42}.$$

It is the unitary minimal-model stress $M(41, 42)$. Add the diagonal Cartan stress, of charge one, to obtain the exact near-two value

$$\frac{573}{287} = 2 - \frac{1}{287}.$$

The vector $u_0 = (\ell + h)/2 = v - \ell$ has norm twenty and its exponential has the minimum compatible affine energy ten. Its GKO weight is therefore zero. The rational algebra

$$M(41, 42) \otimes V_{\langle 20 \rangle}$$

is actually embedded with this stress.

This is an existence statement for a rational chiral algebra near charge two, not for a Haagerup full theory. Its rational categorical dimension field is ramified only at $2, 3, 7, 41$, while the Haagerup dimension requires $\sqrt{13}$. Within the ordinary finite rational framework preserving this algebra, that is an obstruction. Merely preserving its stress would be a weaker premise.

Worked example 18.3 (A discriminant detects the missing Haagerup dimension)**.** Here is the arithmetic behind that obstruction. For a number field K, the field trace $\mathrm{Tr}_{K/\mathbb{Q}}$ sums the images under its complex embeddings. If $b_1, \dots, b_r$ is an integral basis of its ring of integers, its discriminant is $\det(\mathrm{Tr}_{K/\mathbb{Q}}(b_i b_j))$. A rational prime is *ramified* precisely when it divides this field discriminant. We use two standard facts: a ramified prime of a subfield also ramifies in the larger field, and the only primes that can ramify in $\mathbb{Q}(\zeta_N)$ divide N.

For $\mathbb{Q}(\sqrt{13})$, an integral basis is $1, \xi$ with $\xi = (1+\sqrt{13})/2$ and $\xi^2 = \xi + 3$. Consequently

$$\operatorname{disc} \mathbb{Q}(\sqrt{13}) = \det \begin{pmatrix} 2 & 1 \\ 1 & 7 \end{pmatrix} = 13.$$

This is a discriminant of a number field, not the Gram determinant of an E_6 or A_2 root lattice.

The minimal-model quantum dimensions above belong to $K = \mathbb{Q}(\zeta_{82}, \zeta_{84}) = \mathbb{Q}(\zeta_{3444})$; the norm-twenty lattice contributes only dimension-one simples. Since $3444 = 2^2 \cdot 3 \cdot 7 \cdot 41$, the prime thirteen does not ramify in K. Hence K cannot contain $\mathbb{Q}(\sqrt{13})$, or the number $d = 1 + \xi$.

For any simple separable Frobenius sewing algebra A in the stated representation category $\mathcal{C}$, with A simple as an A–A bimodule, the dimension of a defect bimodule X is

$$\mathrm{FPdim}_{{}_A\mathcal{C}_A}(X) = \frac{\mathrm{FPdim}_{\mathcal{C}}(X)}{\mathrm{FPdim}_{\mathcal{C}}(A)} \in K.$$

Here numerator and denominator refer to underlying objects; direct sums also have dimension in K. Thus neither a simple nor a reducible defect preserving this rational algebra can have Haagerup dimension d. No approximation to c enters the argument. The same arithmetic will filter the ordinary W_3 family in Section 24.8.3.

18.7 An exactly solvable four-dimensional stress algebra

Now consider the commutant of the whole diagonal $A_{1,40}$ inside U. Its weight-two space is four dimensional. Besides e_0 above, it contains the two stresses e_1, e_2 of charges $2400/1271$ and $70/31$, and a PCT-real diagonal singlet x from the current representation $(2, 2)$. Normalize $\langle x, x\rangle = 1$.

Use the Griess product $a * b = a_{(1)}b$. Put

$$f_i = \langle e_i, e_i\rangle, \qquad f = \left(\frac{143}{287}, \frac{1200}{1271}, \frac{35}{31}\right), \qquad \lambda = \left(\frac{11}{41}, \frac{192}{1271}, \frac{49}{31}\right).$$

The complete multiplication table is

$$e_i * e_j = 2\delta_{ij} e_i, \quad e_i * x = \lambda_i x, \quad x * x = \sum_i \frac{\lambda_i}{f_i} e_i. \tag{18.7}$$

The last coefficients are forced by invariance of the form: $\langle x * x, e_i\rangle = \langle x, x * e_i\rangle = \lambda_i$. Representation parity excludes an x component. Exact ambient OPE identities verify that no omitted weight-two component remains.

For $e = \sum_i a_i e_i + tx$, the Virasoro idempotent equation $e * e = 2e$ becomes

$$a_i(1 - a_i) = q_i t^2, \qquad t(\lambda \cdot a - 1) = 0, \qquad q = \left(\frac{7}{26}, \frac{2}{25}, \frac{7}{10}\right). \tag{18.8}$$

Its central charge is $c(e) = 2f \cdot a$. A real solution has $0 \le a_i \le 1$. These small equations describe actual stresses because the quasiprimary and remaining Virasoro identities are controlled in this current-free commutant.

At $c(e) = 2$ and $t \neq 0$, the linear equations give $a_0 = 41/44 - 30a_1/11$ and $a_2 = 93/196 + 18a_1/49$. Writing $b = a_1$ leaves two necessary quadratics,

$$2500b^2 + 620b - 39 = 0, \qquad 10676b^2 - 10820b + 309 = 0.$$

The first factors as $(250b - 13)(10b + 3)$. Its roots are $13/250$ and $-3/10$. Substitution in the second polynomial gives, respectively, $-3512064/15625$ and $112896/25$, both nonzero. Thus no common root exists, even over $\mathbb{C}$. The $t = 0$ subset stresses also have no charge two. This is a complete exclusion in this four-dimensional slice, not in all of U_2.

The nonzero-t solutions do give four real irrational charges:

$$1.537523906\ldots, \quad 2.025004294\ldots, \quad 3.117852848\ldots, \quad 3.605333236\ldots.$$

They occur in complementary pairs summing to $36/7$, the charge of the diagonal-affine commutant. The one closest to two is

$$c_* = \frac{18}{7} - 4\sqrt{\frac{4166013889 - 141778560\sqrt{274}}{97483514569}}. \tag{18.9}$$

If $z = c_* - 18/7$, then

$$4776692213881z^4 - 6532309777952z^2 + 1524572989696 = 0.$$

Its displayed squared offset is irrational, so c_* is not exactly two. Rational C_2-cofinite VOAs have rational central charge; therefore a chiral algebra with this stress cannot be strongly rational, nor can a finite extension with the same stress become strongly rational.

18.8 What the irrational commutant actually contains

Choose the stress e of charge c_* and put

$$f = \omega_{E_8} - e, \qquad N_e = \mathrm{Com}(\mathrm{Vir}_f, V_{E_8}), \qquad M_f = \mathrm{Com}(N_e, V_{E_8}).$$

The vacuum-like-vector lemma proves

$$M_f = \mathrm{Com}(\mathrm{Vir}_e, V_{E_8}), \qquad N_e = \mathrm{Com}(M_f, V_{E_8}).$$

Indeed a vector killed by the translation $e_{(0)}$ is killed by all nonnegative modes of the whole N_e algebra, not only the Virasoro modes. This yields the first equality; the second follows from the defining commutant and the reverse containment.

The diagonal affine algebra belongs to M_f. Write

$$f = \omega_{A_{1,40}} + g, \qquad c_g = \frac{36}{7} - c_*.$$

The exact current decomposition under diagonal $SU(2)$ is

$$248 = 10(3) + 10(5) + 10(7) + 6(9) + 4(11).$$

There is no singlet. Positivity of the complementary affine energy therefore gives $(N_e)_1 = 0$.

Construction input 18.4 (Vacuum kernels of the irrational factor)**.** Complete exact low-grade calculations of the ambient complementary energy kernels give

$$(\dim(N_e)_0, \dots, \dim(N_e)_4) = (1, 0, 1, 1, 2). \tag{18.10}$$

Thus the vacuum algebra agrees with its Virasoro vacuum module through weight four. Its weight-four basis is $L^e_{-4}\mathbf{1}, (L^e_{-2})^2\mathbf{1}$. Their Gram determinant is $c_*^2(5c_* + 22)/2 > 0$, so the two states are independent. The opposite algebra has the exact finite kernel

$$(M_f)_2 = (A_{1,40})_2 \oplus \mathbb{C}g, \qquad \dim(M_f)_2 = 10.$$

There is no additional spin-three or spin-four vacuum generator in N_e. This rules out ordinary W_3 for this factor. It does not prove $N_e = \mathrm{Vir}_e$: a first additional generator may occur later. Similarly the opposite-algebra statement is not the all-grade equality $M_f = A_{1,40} \otimes \mathrm{Vir}_g$.

18.9 Three inequivalent proposals for the reduced state space

The smallest attempted reduction projects onto the complementary Virasoro vacuum on both chiralities. Its partition function would be $|\chi_{N_e}|^2$. It cannot be modular invariant. If it were, the ratio $\chi_{N_e}(-1/\tau)/\chi_{N_e}(\tau)$ would have constant modulus one and hence be constant; at $\tau = i$ positivity makes the constant one. The one-dimensional modular representation would have $S = 1$, $T = e^{-2\pi i c_*/24}$. The relation $(ST)^3 = S^2$ would force $c_* \in 8\mathbb{Z}$. Additional sectors are therefore mandatory.

A second proposal branches only over Vir_f,

$$V_{E_8} = \bigoplus_h B_h \otimes L(c_f, h), \qquad \mathcal{H}_{\mathrm{naive}} = \bigoplus_h B_h \otimes \overline{B_h}.$$

This produces infinitely many reduced vacua. Already the three diagonal affine currents are complementary Virasoro primaries of weight one and have e-energy zero. Pairing their multiplicity space with itself adds nine vacuum states. More generally $A_{1,40}$ has infinitely many Virasoro primaries: its high-temperature growth has effective charge $20/7$, whereas a finite sum of irreducible Virasoro characters at that charge grows with effective charge at most one. Every such affine primary still has zero e-energy.

The more careful third proposal branches over the full opposite algebra,

$$V_{E_8} = \bigoplus_\lambda B_\lambda \otimes (M_f)_\lambda, \qquad \mathcal{H}_{\mathrm{red}} = \bigoplus_\lambda B_\lambda \otimes \overline{B_\lambda}.$$

The unitary ordinary branching is an algebraic countable semisimple sum, even though the two factors need not be rational. To see why, the commuting product $N_e \otimes M_f$ has the full ambient stress. Its representation on V_{E_8} is therefore unitary with finite-dimensional nonnegative integer ambient grades. On the least remaining grade, split the finite-dimensional star-closed zero-mode algebra into irreducible top spaces. A simple top generates a simple unitary module: any invariant submodule has a grade-by-grade invariant orthogonal complement; if the submodule misses the simple top, that complement contains the top and hence everything it generates. Removing these summands and repeating grade by grade exhausts the representation.

The finite-dimensional simple tops for the commuting product factor into a simple top for each factor. Their generated modules are tensor products of ordinary simple modules. Each separate graded piece is finite: tensoring it with one fixed nonzero opposite ground vector embeds it in a single finite ambient grade. There are only countably many such summands. Grouping identical M_f modules produces the multiplicity spaces B_λ above; fixing their opposite ground weight again proves finite-dimensional grades and lower boundedness for each B_λ. All these

statements concern algebraic finite-energy vectors, so no direct integral or completion has been silently inserted.

This prescription has exactly one zero-energy vector, because all zero-e-energy states lie in the full opposite vacuum algebra. But deleting the complementary energies may bring infinitely many different labels into a bounded reduced-energy interval. Ambient trace convergence does not prove reduced trace convergence.

It also does not permit selecting attractive branches at will. Ambient currents have positive e-weights near

$$.0597642, \quad .0609687, \quad .0624674, \quad .0672163.$$

Their complementary weights are between zero and one, so they are tops for both factors. The full-opposite diagonal prescription therefore retains scalar dimensions twice all these values, including the first three below .134433. Ambient $SU(2)$ multiplicities are not automatically reduced bulk multiplicities, but the existence of these branches is unavoidable for this specified pairing.

18.10 What sewing must still establish

An actual reduced local CFT needs all of the following mathematical data: a complete branching with finite reduced degeneracies or a specified continuous measure; a convergent torus amplitude and its modular transformation; a positive local bulk OPE obeying sewing; and a transport of Haagerup defects that preserves junction compositions and associators. Integral spin from pairing matching complementary labels is only one necessary check.

The next two worked arguments turn the convergence warning into precise mathematical tests. Their additional hypotheses are part of their statements; the low-grade calculations do not establish them.

18.10.1 A conditional divergence theorem

Proposition 18.5. *Assume the all-grade identity $M_f = A_{1,40} \otimes \mathrm{Vir}_g$. For the complete full-$M_f$ branching, the diagonal reduced state space has divergent thermal trace at every positive inverse temperature. The conclusion does not require $N_e = \mathrm{Vir}_e$ or irreducibility of the multiplicity spaces B_λ.*

Proof. Let $c_A = 20/7$ and let $\beta, \epsilon > 0$ be dimensionless inverse temperatures conjugate to the two Virasoro energies. Thus their modular parameters are $\tau_e = i\beta/(2\pi)$ and $\tau_f = i\epsilon/(2\pi)$, with $q_e = e^{-\beta}$ and $q_f = e^{-\epsilon}$. For a physical inverse temperature b on a circle of length L and propagation speed v, the corresponding dimensionless parameter is $2\pi vb/L$, in units $\hbar = k_B = 1$. Here the two parameters can be varied independently as trace variables. Define the normalized two-temperature trace

$$F(\beta, \epsilon) = \mathrm{Tr}_{V_{E_8}} e^{-\beta(L_0^e - c_*/24) - \epsilon(L_0^f - c_f/24)}.$$

The two nonnegative commuting energies sum to the ambient energy, so convergence follows by comparison with an ordinary E_8 trace at $\min(\beta, \epsilon)$. The same estimate on compact subsets of two upper half-planes proves normal convergence of the vacuum two-variable trace. The Krauel–Miyamoto multivariable modular theorem applies to orthogonal conformal vectors summing to the stress of a regular ambient VOA, with this analyticity; its vacuum clause supplies the descendant analyticity needed in the theorem. Neither reduced factor is required to be rational. The ambient E_8 algebra is holomorphic and has $S = (1)$, so the theorem gives

$$F(\beta, \epsilon) = F(4\pi^2/\beta, 4\pi^2/\epsilon).$$

This is simultaneous inversion of two variables; it does not assert separate modular invariance in either variable. Retaining the positive vacuum term on the right gives

$$F(\beta,\epsilon) \geq \exp\left(\frac{\pi^2 c_*}{6\beta} + \frac{\pi^2 c_f}{6\epsilon}\right).$$

Under the stated product hypothesis, an opposite simple module is one of the forty-one affine integrable modules tensored with a unitary Virasoro module of lowest weight $h \geq 0$. A Verma spanning bound therefore gives, uniformly in λ,

$$\chi_{(M_f)_\lambda}(\epsilon) \leq U(\epsilon) := \max_j \chi_{A_j}(\epsilon)\, e^{\epsilon c_g/24} \prod_{n\geq 1} (1-e^{-n\epsilon})^{-1}.$$

Finite affine modular transformation and eta inversion imply

$$\limsup_{\epsilon\downarrow 0} \epsilon \log U(\epsilon) \leq \frac{\pi^2(c_A+1)}{6}.$$

The oscillator contribution is one. It is an upper bound on character growth, not a replacement of the actual central charge c_g by one.

Write $\chi_{B_\lambda}(\beta) = \mathrm{Tr}_{B_\lambda} e^{-\beta(L_0^e - c_*/24)}$, including the same chiral Casimir shift. If $R(\beta) = \sum_\lambda \chi_{B_\lambda}(\beta)$ were finite, positive branching would imply $F(\beta,\epsilon) \leq U(\epsilon)R(\beta)$. Multiply logarithms by ϵ and let ϵ decrease to zero. The lower and upper bounds force $c_f \leq c_A + 1$, contradicting $c_f = c_A + c_g$ and $c_g > 1$. Hence $R(\beta)$ diverges for every $\beta > 0$.

One must still justify divergence of the diagonal square sum; a divergent sum need not have divergent squares in general. Here the positive state-by-state character expansion gives

$$\chi_{B_\lambda}(\beta)^2 \geq \chi_{B_\lambda}(2\beta),$$

because the square contains every pair of a basis vector with itself. Summing yields a lower bound $R(2\beta) = \infty$ for the diagonal thermal trace. □

The proof isolates a general obstruction: a proposed complementary algebra whose simple characters all have a uniform growth rate below the complementary central charge cannot be discarded in this all-branches diagonal fashion. Additional opposite fields might change the uniform bound, or a different left/right pairing might change the construction. Neither possibility is settled by vacuum dimensions through weight four.

18.10.2 A continuous diagonal density can be solved exactly

An alternative is a noncompact theory with a positive continuum of weights. It is important to speak of a partition *density*: a continuum direct integral need not have a literal finite Hilbert-space trace. A direct integral is the continuum analogue of a direct sum of sectors, with a measure replacing the discrete sum over labels. Integrating their characters gives a partition density only after specifying that measure and its normalization. Suppose this density itself, not merely a regulator-dependent finite part, obeys exact modular inversion.

Proposition 18.6 (Strict diagonal continuum)**.** *Fix $c > 1$. Pair only identical positive-weight irreducible unitary Virasoro modules, with a positive measure μ whose Laplace integral is finite at every positive argument. Assume no vacuum atom and exact simultaneous modular inversion at independently positive left and right inverse temperatures. Then, up to a positive overall constant,*

$$d\mu(h) = \frac{\mathbf{1}_{h>(c-1)/24}}{\sqrt{h-(c-1)/24}}\, dh.$$

In particular its effective central charge is one.

Proof. Put $A = (c-1)/24$. A positive-weight unitary irreducible at $c > 1$ has character

$$\chi_h(\beta) = \frac{e^{-\beta(h-A)}}{\eta(\beta)}, \qquad \eta(\beta) = e^{-\beta/24}\prod_{n\geq 1}(1-e^{-n\beta}).$$

There are no positive real degenerate highest weights in this unitary range. For $1 < c < 25$, write $c = 1+6Q^2$ and $Q = b+b^{-1}$ with $|b| = 1$. The Kac values $[Q^2-(rb+s/b)^2]/4$ are nonreal when $r \neq s$ and nonpositive when $r = s$. For $c \geq 25$, take $b > 0$; then $rb+s/b \geq Q$ for positive integers r, s, again giving nonpositive values. Thus a positive-weight module has the Verma character above. The vacuum is the exception, with its $L_{-1}\mathbf{1} = 0$ relation, and is treated separately below. Define $G(s) = \int e^{-s(h-A)}d\mu(h)$. Strict diagonality gives

$$Z(\beta,\bar\beta) = \frac{G(\beta+\bar\beta)}{\eta(\beta)\eta(\bar\beta)}.$$

Here β and $\bar\beta$ are independently positive, dimensionless parameters in the same convention as above. Normal convergence gives the analytic continuation needed when deriving this identity from a physical modular amplitude.

Set $s = \beta+\bar\beta$, $p = \beta\bar\beta$, and $t = 4\pi^2 s/p$. Eta inversion, $\eta(\beta) = \sqrt{2\pi/\beta}\,\eta(4\pi^2/\beta)$, turns modular invariance into

$$G(t) = \frac{\sqrt{p}}{2\pi}G(s) = \sqrt{\frac{s}{t}}G(s).$$

For fixed s, positive temperatures realize every $0 < p \leq s^2/4$. Thus this equation holds whenever $st \geq 16\pi^2$. For arbitrary s_1, s_2, choose a common large u with $s_i u \geq 16\pi^2$. It follows that $\sqrt{s_1}G(s_1) = \sqrt{u}G(u) = \sqrt{s_2}G(s_2)$. Hence $G(s) = C/\sqrt{s}$.

The gamma integral identifies the positive measure:

$$\int_A^\infty \frac{e^{-sh}}{\sqrt{h-A}}\,dh = \sqrt{\pi}\frac{e^{-As}}{\sqrt{s}}.$$

Uniqueness of positive Laplace transforms rules out any other measure. For an elementary justification, weight it by $e^{-s_0 h}$ to make a finite measure and push it to $x = e^{-h} \in [0,1]$. The Laplace values at s_0+n fix all polynomial moments. Density of polynomials in continuous functions on $[0,1]$ fixes that measure uniquely. Undoing the positive weight fixes the original measure.

Its threshold is $h_{\min} = A$, so $c_{\text{eff}} = c - 24h_{\min} = 1$. □

Worked example 18.7 (The noncompact boson as a partition density)**.** The compact boson gives a familiar way to obtain an inverse-square-root measure. Use the radius convention of Chapter 2,

$$p_L = \frac{m}{R}+\frac{nR}{2}, \qquad p_R = \frac{m}{R}-\frac{nR}{2}, \qquad m,n \in \mathbb{Z}.$$

For independently positive $\beta, \bar\beta$, its partition function is

$$Z_R(\beta,\bar\beta) = \frac{1}{\eta(\beta)\eta(\bar\beta)}\sum_{m,n\in\mathbb{Z}} e^{-\beta p_L^2/2-\bar\beta p_R^2/2}.$$

The eta factors count oscillators and include the two chiral Casimir shifts. We use the same dimensionless inverse temperatures and eta normalization as in Proposition 18.6.

Take $R \to \infty$ at fixed positive temperatures. To check suppression of winding even when the two temperatures differ, put $s = \beta+\bar\beta$ and $a = \beta-\bar\beta$. Completing the square gives

$$\frac{\beta}{2}p_L^2+\frac{\bar\beta}{2}p_R^2 = \frac{s}{2}\left(\frac{m}{R}+\frac{anR}{2s}\right)^2+\frac{\beta\bar\beta}{2s}n^2R^2.$$

The momentum Gaussian sum is $O(R)$ uniformly in its shift, as follows by comparison with its Gaussian integral, or by Poisson summation. The second term therefore suppresses the entire $n \neq 0$ contribution after division by R. For $n = 0$, the spacing of $p = m/R$ is $1/R$, so the remaining sum is a Riemann sum:

$$\begin{aligned} \mathcal{Z}_\infty(\beta, \bar{\beta}) &:= \lim_{R\to\infty} \frac{Z_R(\beta, \bar{\beta})}{R} \\ &= \frac{\int_{\mathbb{R}} e^{-sp^2/2}\, dp}{\eta(\beta)\eta(\bar{\beta})} = \frac{\sqrt{2\pi}}{\eta(\beta)\eta(\bar{\beta})\sqrt{\beta+\bar{\beta}}}. \end{aligned} \tag{18.11}$$

Dividing instead by the target-space circumference $2\pi R$ changes only the overall constant. Eta inversion verifies exact simultaneous modular inversion of this density.

The surviving primary weights are diagonal: $h = \bar{h} = p^2/2$. Both signs of p give the same Virasoro weight, and changing variables gives

$$\int_{\mathbb{R}} f(p^2/2)\, dp = \sqrt{2} \int_0^\infty f(h) h^{-1/2}\, dh.$$

Thus our normalization has $d\mu(h) = \sqrt{2}\, h^{-1/2} dh$ for $h > 0$. The singularity at zero is integrable and carries no atom. At $c = 1$, the Heisenberg Fock character is $e^{-\beta h}/\eta(\beta)$. It agrees with the irreducible Virasoro character at generic momentum. The exceptional degenerate weights $h = j^2/4$, $j \in \mathbb{Z}_{\geq 0}$, form a countable set of measure zero for this continuous measure. Their distinct Virasoro decompositions, including the special vacuum relations at $p = 0$, do not change the density integral.

There is still no literal finite Hilbert-space trace in the noncompact limit. The zero-mode Hilbert space is $L^2(\mathbb{R})$, and the heat operator has an infinite spatial-volume trace. Equation (18.11) is a volume-normalized trace, equal to 2π times the trace per unit target-space circumference, obtained from the compact regulator. The finite-radius $m = n = 0$ sector contributes $1/R$ times its oscillator factor and disappears from this density. Its constant zero-mode wavefunction has no normalizable limit on the real line. A generalized zero-momentum state and the local identity operation remain meaningful, but there is no normalizable vacuum atom in the spectral measure.

The continuum threshold is zero, so $c_{\text{eff}} = 1 - 24 \cdot 0 = 1$. For example at equal temperatures, eta inversion gives

$$\mathcal{Z}_\infty(\beta, \beta) \sim \frac{\sqrt{\beta}}{2\sqrt{\pi}} e^{\pi^2/(3\beta)} \qquad (\beta \downarrow 0).$$

The polynomial prefactor records the continuum; the exponential has the common chiral effective charge one.

For $c > 1$, the proposition forces the same Gaussian measure shape after the shift $h = A + p^2/2$, where $A = (c-1)/24$. This comparison explains the threshold density and again gives $c - 24A = 1$. The compact-boson limit itself still has $c = 1$. Constructing a theory at $c = c_*$ would additionally require actual fields, OPEs, and compatible sewing and defect data. This Gaussian calculation supplies none of those additional data for a Liouville or Haagerup construction.

At $c = c_*$, effective charge two would instead require $h_{\min} = (c_* - 2)/24$, about .00104185. The diagonal modular density forces the larger threshold $(c_* - 1)/24$, about .0427085. Thus this precise continuum repair does not do the proposed job. It also does not identify a Liouville interaction theory: characters and a measure supply no OPE coefficients or defect junctions.

Adding a positive vacuum atom cannot fix the strict diagonal ansatz. The vacuum character has the extra numerator $1 - e^{-\beta}$. After stripping eta factors, fix $\bar{\beta} > 0$ and send β to zero. The original numerator stays bounded by Laplace finiteness. Modular inversion would therefore make

the transformed numerator tend to zero after its square-root prefactor. But the transformed positive vacuum term grows at least as $\exp(4\pi^2 A/\beta)$, a contradiction. This reasoning requires positivity; cancellations in a signed formal expression would not support it. Reducible branching blocks and asymmetric Virasoro pairings remain outside the strict diagonal hypothesis.

18.11 What a genuine defect descent would guarantee

The inherited extension action preserves the two commutant spaces because it intertwines the stress modes. This is weaker than a tensor action on a sewn reduced theory. Restricted normalized operations may cease to be extremal, and their compositions need not supply the required fusion junction spaces. There is nevertheless a useful positive criterion once a genuine tensor functor exists.

Proposition 18.8 (Haagerup tensor descent cannot collapse simples)**.** *Let F be a unitary strong tensor functor from $\mathcal{H}_3$ into a locally semisimple rigid category with simple unit and finite-dimensional Hom spaces on the image. Then F is fully faithful.*

Proof. Take the full category $\mathcal{D}$ of simple summands of the images of the six simples. It is a fusion category: products and duals stay inside the finite list of image summands. The induced functor to $\mathcal{D}$ is dominant, meaning every simple target is a summand of an image. Its right adjoint gives a connected central algebra $A = R(\mathbf{1})$ and

$$\operatorname{Dim}\mathcal{D} = \frac{\operatorname{Dim}\mathcal{H}_3}{\operatorname{FPdim} A}.$$

If A is not the unit, its central lift contains the unit once and at least one nonunit central simple. The smallest such dimension in $Z(\mathcal{H}_3)$ is $3d$, so $\operatorname{FPdim} A \geq 1 + 3d = d^2$. Consequently

$$\operatorname{Dim}\mathcal{D} \leq 3 + \frac{3}{d^2} < \frac{5+\sqrt{5}}{2}.$$

Every nonpointed fusion category has global dimension at least the last number. At rank two this is the Fibonacci minimum from $X^2 = 1 + nX$, $n \geq 1$. At rank at least three, a noninvertible has squared dimension at least two and two further simples contribute at least two, giving global dimension at least four. Thus $\mathcal{D}$ would be pointed. But $F(\rho)$ still has Frobenius–Perron dimension d, irrational, whereas every object of a pointed category has integer dimension. This is impossible. Therefore $A = \mathbf{1}$; the dominant functor is an equivalence onto $\mathcal{D}$, proving full faithfulness into the target. □

The premise is substantial. A BRST reduction, a branching construction, or a generalized quotient would have to build this tensor functor, including its unit and junction compatibility. Checking the Haagerup fusion polynomial for a few compressed operators does not meet that premise. The proposition says what a correct construction would preserve, rather than declaring that the missing construction has already been carried out.

18.12 A current test that survives changes of pairing

A unitary theory with unique normalizable vacuum and two independent commuting Hermitian holomorphic currents has a rank-two Heisenberg stress of charge two. Subtracting it at charge c_* would leave a unitary Virasoro vacuum with $0 < c_* - 2 < 1/2$, which is impossible. An explicit negative-norm witness at residual charge k is

$$-108L_{-6}\mathbf{1} - 264L_{-4}L_{-2}\mathbf{1} + 93L_{-3}^2\mathbf{1} + 64L_{-2}^3\mathbf{1},$$

whose squared norm is $12k(2k-1)(128k+6943) < 0$ on that interval. This is a finite Virasoro calculation, so it can be checked without invoking a classification of unitary minimal models. In the ordered vacuum basis $L_{-6}\mathbf{1}, L_{-4}L_{-2}\mathbf{1}, L_{-3}^2\mathbf{1}, L_{-2}^3\mathbf{1}$, the Gram matrix is

$$k\begin{pmatrix} 35/2 & 5 & 18 & 24 \\ 5 & (5k+16)/2 & 14 & (9k+48)/2 \\ 18 & 14 & 8k+36 & 30 \\ 24 & (9k+48)/2 & 30 & 3(k+8)(k+16)/4 \end{pmatrix}.$$

For example, $L_4L_{-2}^3\mathbf{1} = (48+9k)L_{-2}\mathbf{1}$ gives the second-row, fourth-column entry after applying L_2. Multiplying the matrix by $(-108, -264, 93, 64)$ on both sides gives $3072k^3+165096k^2-83316k$, which factors as the negative expression above. Quotienting null states cannot remove a vector of strictly negative norm in a unitary representation.

This is a conditional criterion for rejecting c_*. The lattice chain's nearly spin-one states have not been proved to be two such currents, so the criterion cannot yet be applied as an unconditional identification or exclusion.

The exact construction at charge eight, the rational charge-six coset, and the irrational chiral stress are three distinct achievements. The physical ρ-projector chain remains a fourth problem: one needs a controlled map from its lattice operators and sectors to the full continuum theory. Neither a matching central charge nor a matching pair of low dimensions supplies that map.

18.13 Exercises

Exercise 18.1. Verify the norm, index, and determinant calculation in (18.1). Explain why the squarefree determinant proves saturation of L_0 in its plane.

Solution on page 307.

Exercise 18.2. Derive the six-vacuum factor in (18.6). What additional information would be needed to interpret a multiplicity-six vacuum as a full theory with a nontrivial vacuum algebra?

Solution on page 307.

Exercise 18.3. Compute $c(A_{1,40})$ and the GKO charge $286/287$. Check that adding one Cartan stress gives $573/287$, and compare both exact distances from two with $c_* - 2$.

Solution on page 308.

Exercise 18.4. Derive (18.8) from the Griess multiplication table. Explain why the central charge is $2f \cdot a$ rather than an arbitrary quadratic expression once the idempotent equations hold.

Solution on page 308.

Exercise 18.5. Compute the Virasoro Gram matrix on $L_{-4}\mathbf{1}, L_{-2}^2\mathbf{1}$ and its determinant. Use it to justify the two-dimensional lower bound at weight four in (18.10).

Solution on page 308.

Exercise 18.6. Show directly from $\rho^2 = 1+\rho+\alpha\rho+\alpha^2\rho$ that any fusion-compatible group-valued assignment to simple objects is trivial. Explain which transmission assumption is needed to apply this algebraic observation to a current coset.

Solution on page 308.

Exercise 18.7. Explain why finiteness below every ambient-energy cutoff need not imply finiteness below every reduced-energy cutoff after one positive commuting energy is discarded. Illustrate the issue using projections of a rank-two integer lattice onto a line of irrational slope.

Solution on page 309.

Exercise 18.8. Give positive numbers a_n for which $\sum_n a_n$ diverges while $\sum_n a_n^2$ converges. Why does the stronger character inequality $\chi_\lambda(\beta)^2 \geq \chi_\lambda(2\beta)$ nevertheless complete the proof of Proposition 18.5? State where integrality of state multiplicities is used.

Solution on page 309.

Exercise 18.9. Starting from $\eta(\beta) = \sqrt{2\pi/\beta}\,\eta(4\pi^2/\beta)$, rederive the functional equation for G in Proposition 18.6. Explain why checking only equal temperatures would not justify the step that compares arbitrary s_1 and s_2 through one common large u.

Solution on page 309.

Exercise 18.10. Prove that a nonpointed fusion category of rank at least three has global dimension at least four. For rank two, show that $X^2 = \mathbf{1} + nX$ forces the nonpointed minimum $(5+\sqrt{5})/2$. Use these bounds to explain the role of $\operatorname{FPdim} A \geq d^2$ in the tensor-descent proposition.

Solution on page 309.

Chapter 19

From a fusion category to a quantum chain

We now return to the six-object $\mathcal{H}_3$ fusion category and construct a microscopic quantum chain from it. This is the lattice construction track introduced in the reading guide; the reduced CFTs of the preceding chapter will serve as comparison theories. A quantum Hamiltonian needs one further choice: which local fusion channels should cost or release energy? This chapter makes that choice explicit. We construct the Hilbert space of the physical Haagerup chain, derive its local projector Hamiltonian, and explain its translation and categorical symmetries. The required linear algebra is modest: orthogonal projections, eigenvectors of a small adjacency matrix, and Fourier sums. The important conceptual step is to distinguish a fusion label, a physical anyon, and an ordinary local spin. The fusion-space conventions of Chapter 1 and the explicit Haagerup associator in Chapter 4 are the categorical inputs.

19.1 The physical object and the fusion-path basis

Write the six simple objects as

$$I_j = \alpha^j, \qquad X_j = \alpha^j \rho, \qquad j \in \mathbb{Z}/3\mathbb{Z}.$$

Their dimensions are $\mathrm{qdim}(I_j) = 1$ and $\mathrm{qdim}(X_j) = d$, where

$$d^2 = 1 + 3d, \qquad d = \frac{3 + \sqrt{13}}{2}.$$

We put one physical object ρ at every site. The physical objects are all the same; it is their possible fusion histories that fluctuate.

An open fusion history starts with a label a, fuses successively with the physical ρ objects, and ends with a label b. At a trivalent vertex the outgoing label must occur in the fusion of the incoming label with ρ. The multiplicities here are zero or one, so no additional vertex index is needed. For example,

$$I_j \rho = X_j, \qquad X_j \rho = I_j + X_0 + X_1 + X_2.$$

An invertible height must therefore be followed by its corresponding noninvertible height. A noninvertible height can be followed by any noninvertible height or by its one matching invertible height.

Definition 19.1 (Regular periodic path space)**.** The length-L periodic Hilbert space $\mathcal{H}_{\mathrm{path},L}$ has an orthonormal basis

$$|x_0, x_1, \ldots, x_{L-1}\rangle$$

indexed by admissible closed paths, meaning that $x_{i+1} \subset x_i \otimes \rho$ for every i, with indices modulo L. The inner product is the ordinary orthonormal fusion-basis inner product.

The word "regular" records that all starting heights and compatible closed flux sectors are included. It is a choice of periodic Hilbert space, not a synonym for the sector obtained by fusing all L physical objects to the tensor unit. Open chains with fixed endpoint labels give other spaces.

One can embed $\mathcal{H}_{\mathrm{path},L}$ in $(\mathbb{C}^6)^{\otimes L}$ by identifying each height with a six-state basis vector and setting the coefficients of forbidden words to zero. This embedding is useful for ordinary Schmidt decompositions. Chapter 22 develops those decompositions and their tensor-network realization. Nevertheless, $\mathcal{H}_{\mathrm{path},L}$ is a constrained subspace: changing one height arbitrarily can violate two adjacent fusion constraints. A Hamiltonian or tensor ansatz must preserve these constraints.

Worked example 19.2 (Counting closed paths exactly)**.** In the ordered height basis $(I_0, I_1, I_2; X_0, X_1, X_2)$, right fusion by ρ has adjacency matrix

$$N_\rho = \begin{pmatrix} 0 & I_3 \\ I_3 & J_3 \end{pmatrix},$$

where J_3 is the all-ones matrix. Its Lth power counts length-L paths, and taking the trace closes them. On the color-independent subspace, N_ρ is the matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 3 \end{smallmatrix}\right)$. Its eigenvalues are d and $-d^{-1}$. On each of the two color-orthogonal directions it is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, with eigenvalues $1, -1$. Thus

$$\dim \mathcal{H}_{\mathrm{path},L} = d^L + (-d^{-1})^L + 2(1 + (-1)^L). \tag{19.1}$$

This apparently irrational expression is an integer. Indeed, if $t_L = d^L + (-d^{-1})^L$, then

$$t_0 = 2, \quad t_1 = 3, \quad t_{L+2} = 3t_{L+1} + t_L.$$

Consequently the path-space dimensions at $L = 3, 4, 6, 9$ are $36, 123, 1302, 46764$, respectively. The dimension grows as d^L, even though the local height alphabet has six letters.

This calculation already explains why symmetry reduction and matrix-free algorithms matter. A final energy table is small; the vector needed to obtain it can have exponentially many entries. Nothing about the file size of a reported spectrum estimates the memory needed for its calculation.

19.2 An F-move becomes a local projector

Fix three consecutive heights a, e, c. They represent a piece of fusion tree in which a first fuses with one physical ρ to e, and then with the next physical ρ to c. There is another fusion tree: first fuse the two physical objects into $f \subset \rho^2$, and then fuse a with f into c. The unitary matrix $F_c^{a\rho\rho}$ changes between these trees. We use

$$|((a\rho)e\rho); c\rangle = \sum_f F_c^{a\rho\rho}[e, f]\, |(a(\rho\rho)f); c\rangle. \tag{19.2}$$

The allowed e and f depend on the fixed endpoints. Every sum below is over allowed channels only.

In the second basis, asking whether the physical pair fuses to ρ is simply a diagonal yes-or-no measurement. Transforming that projection back to the path basis gives

$$(P^{(\rho)}_{a,c})_{e',e} = \overline{F_c^{a\rho\rho}[e', \rho]} F_c^{a\rho\rho}[e, \rho]. \tag{19.3}$$

If the ρ channel is absent for the given endpoints, this block is zero. Otherwise it is an outer product and has rank one.

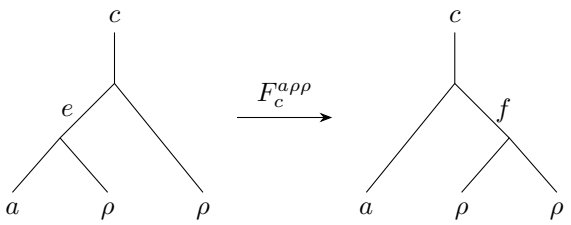


Figure 19.1: The local recoupling move. The neighboring heights a, c are fixed. In the left tree the middle path height e varies; in the right tree the physical pair's fusion channel f is measured.

Proposition 19.3. *For each endpoint block, $P^{(\rho)}$ is an orthogonal projection. On the periodic path space, the operator $P_i^{(\rho)}$ changes only the height x_i and preserves admissibility.*

Proof. Put $v_e = \overline{F[e,\rho]}$. Unitarity gives $v^\dagger v = 1$, and $P = vv^\dagger$. Hence $P^\dagger = P$ and $P^2 = v(v^\dagger v)v^\dagger = P$. Both the original and the updated middle height are allowed channels with the same endpoints, so the two adjacent fusion constraints remain satisfied. All other constraints are unchanged. □

The physical Hamiltonian studied here is

$$H_L = -\sum_{i=0}^{L-1} P_i^{(\rho)}. \tag{19.4}$$

The minus sign rewards the ρ channel. Changing the sign, rewarding the unit channel, or using another associator defines another dynamical problem. Neighboring terms usually do not commute, because the fusion of sites $i, i+1$ and that of sites $i+1, i+2$ are incompatible measurements. The pentagon controls the consistency of all reassociations; it does not make these overlapping measurements simultaneously diagonal.

The simple bounds $-L \le H_L \le 0$ follow from $0 \le P_i \le 1$. Saturating the lower bound requires a state in the $+1$ eigenspace of every projector. Such a state need not exist at a given length. A sum of local projections is not automatically frustration free.

Worked example 19.4 (An exact local imaginary-time gate)**.** Since $P^2 = P$, its exponential is

$$e^{\epsilon P} = 1 + (e^\epsilon - 1)P.$$

Thus the Trotter gate for $e^{-\epsilon H}$ on a single bond is an exact linear combination of 1 and P_ρ. Approximation enters when products of overlapping gates replace the exponential of their sum, not in the single-projector exponential itself. This gives a direct tensor-network route from the Hamiltonian to a two-dimensional Euclidean model.

19.3 Why the associator is physical data

A change of orthonormal basis at every trivalent vertex changes the printed F-symbols. If all operators and states are transformed consistently, it does not change the spectrum. This is a gauge change. In contrast, two inequivalent unitary associators with the same fusion rules need not give the same local projectors or spectra.

The categories commonly called $\mathcal{H}_2$ and $\mathcal{H}_3$ provide exactly this warning. Both have the six-object fusion ring used above. The physical chain in (19.4) uses the specified real unitary $\mathcal{H}_3$ associator. A uniform positive solution occurring in several algebraic constructions belongs to $\mathcal{H}_2$. Substituting it because the object names and fusion rules agree changes the chain.

Construction input 19.5 (A finite distinction between $\mathcal{H}_2$ and $\mathcal{H}_3$)**.** For the same rho-channel coupling and the regular periodic length-four path space, the repository's exact local-data calculation gives

$$\operatorname{Tr}(H_2^2) - \operatorname{Tr}(H_3^2) = \frac{16}{3d} > 0.$$

The space has dimension 123, as independently obtained from (19.1). This is a finite algebraic distinction between the two Hamiltonians, not a statement about their eventual infrared limits.

Unitary equivalence preserves every trace of a power. The displayed difference therefore excludes a unitary equivalence of these two finite Hamiltonians at the stated coupling and closure. It does not exclude a more elaborate equivalence involving module categories, boundary seams, changed couplings, or a common infrared fixed point. Such a proposal must specify those extra choices.

19.4 Translation and the ordinary C_3 charge

Let U_{tr} roll the height word one place to the right. Let A increase every color label by one, preserving whether the height is invertible or noninvertible. In a compatible gauge the Hamiltonian commutes with both, and $U_{\mathrm{tr}}^L = A^3 = 1$. They commute with each other. A simultaneous eigenspace is labeled by

$$U_{\mathrm{tr}} = e^{2\pi i m/L}, \qquad A = e^{2\pi i q/3}, \quad m \in \mathbb{Z}/L\mathbb{Z}, \quad q \in \mathbb{Z}/3\mathbb{Z}.$$

There are $3L$ possible ordinary symmetry sectors, though their dimensions need not be equal.

The formal sector projector is

$$\Pi_{m,q} = \frac{1}{3L} \sum_{r=0}^{L-1} \sum_{s=0}^{2} e^{-2\pi i(mr/L + qs/3)} U_{\mathrm{tr}}^r A^s.$$

It is often implemented through orbit representatives rather than as a dense matrix. Short orbits require care: if a combined translation and color shift fixes a path, the chosen character must be trivial on its stabilizer or its projected vector vanishes. Dividing indiscriminately by $\sqrt{3L}$ gives the wrong normalization on such orbits.

For $h_j = -P_j^{\rho}$, our translation convention is $U_{\mathrm{tr}} h_j U_{\mathrm{tr}}^{-1} = h_{j+1}$. Define

$$H_m = \sum_{j=0}^{L-1} e^{-2\pi i m j/L} h_j. \tag{19.5}$$

Changing the summation variable gives $U_{\mathrm{tr}} H_m U_{\mathrm{tr}}^{-1} = e^{2\pi i m/L} H_m$. Therefore H_m raises momentum by m. The sign can look reversed if one uses a left translation or the opposite Fourier exponential. There is no physical disagreement, but the state selector and Fourier operator must use one convention together.

Worked example 19.6 (A selection rule without diagonalization)**.** Suppose $|a\rangle$ has charge q_a and momentum m_a, and an operator O has charge q_O and momentum m_O. If $\langle b|O|a\rangle \neq 0$, then

$$q_b = q_a + q_O \pmod 3, \qquad m_b = m_a + m_O \pmod L.$$

This follows by inserting $A^{-1}A$ and $U_{\mathrm{tr}}^{-1}U_{\mathrm{tr}}$ on both sides of the matrix element. It is an exact lattice statement and needs no CFT limit.

19.5 Dragging an auxiliary line around the ring

Ordinary color rotation is only the invertible part of the symmetry. To construct a general categorical line, put an auxiliary object c around the periodic fusion tree and move it past each physical ρ by an F-move. Closing the auxiliary line gives an operator Y_c. In the orthonormal path convention its matrix elements are a cyclic product of the corresponding F coefficients:

$$\langle b|Y_c|a\rangle = \prod_i F(c, a_i, \rho, b_{i+1}; b_i, a_{i+1}). \tag{19.6}$$

The six-argument notation specifies the external objects and internal channels by

$$F(a, b, c, d; e, f) = F_d^{abc}[e, f].$$

It avoids suppressing the height indices that make the product meaningful. There is no extra quantum-dimension prefactor in this convention. In the selected gauge the invertible line Y_α is the color rotation A defined above.

Two auxiliary lines can be fused before or after being moved through the chain. Pentagon coherence equates the two operations, giving

$$Y_a Y_b = \sum_c N_{ab}^c Y_c.$$

The same local graphical move shows that these lines commute with the fusion-channel Hamiltonian. For self-dual ρ, unitarity makes Y_ρ self-adjoint. These statements use the associator, not merely the fusion matrices considered as abstract integer matrices.

Worked example 19.7 (Topological eigenvalues in the neutral sector)**.** From the fusion ring,

$$Y_\rho^2 = 1 + (1 + A + A^2)Y_\rho.$$

On $A = 1$ states this reduces to $Y_\rho^2 = 1 + 3Y_\rho$, so the only possible eigenvalues are d and $-d^{-1}$. The associated spectral projections are

$$\Pi_+ = \frac{Y_\rho + d^{-1}}{d + d^{-1}}, \qquad \Pi_- = \frac{d - Y_\rho}{d + d^{-1}}.$$

These formulas are useful for sector checks. They do not say that every neutral eigenstate belongs to the vacuum conformal family. Both sectors contain many lattice energies.

The operator Y_ρ is not a unitary symmetry. Its eigenvalues have different absolute values, and its fusion square is a sum of lines. Replacing it by its polar unitary preserves a Hilbert-space operation, but need not preserve locality or the multiplication of local fields. One cannot infer a new ordinary CFT symmetry just by taking the polar part of a commuting matrix.

In a conformal limit with a unique vacuum, a topological line acts on the vacuum by its quantum dimension. The stress family is then in the $Y_\rho = d$, $A = 1$ sector because topological lines preserve the stress tensor. Conversely, finding that an energy-density source has these labels does not identify it as pure stress: other invariant fields can have the same labels. The continuum interpretation of line eigenvalues and twisted sectors is developed in Chapter 16; the finite fusion equations alone do not supply that limiting dictionary.

19.6 The category supplies kinematics, the Hamiltonian dynamics

It is tempting to read a central charge directly from the categorical dimension d, the number of simple objects, or the modular data of the Drinfeld center. There is a simple obstruction to any such general rule.

Proposition 19.8 (Spectator underdetermination)**.** *Suppose a nonchiral CFT $\mathcal{X}$ realizes a specified Haagerup defect action. For any independent nonchiral CFT $\mathcal{B}$, the product $\mathcal{X} \otimes \mathcal{B}$ realizes the same specified action through $D_a \otimes 1$, while its central charge is $c_{\mathcal{X}} + c_{\mathcal{B}}$.*

Proof. Fusion and junction maps on the first factor are unchanged when tensored with the identity of the second. The stress tensors add, so their central terms add. A defect-sector partition function is multiplied by the modular-invariant spectator partition function, preserving its sector transformation law. □

This does not make the fixed Hamiltonian's infrared charge arbitrary. Adding a spectator changes the microscopic system. It shows what must be learned dynamically once the category has been chosen. The topological central charge of a modular tensor category and the coefficient of the nonchiral chain's Casimir energy answer different questions.

A similar distinction occurs for strange correlators. A topological bulk tensor network supplies a state $|\Psi_{\mathcal{C}}\rangle$; a boundary choice $\langle\Omega|$ supplies the amplitude $\langle\Omega|\Psi_{\mathcal{C}}\rangle$. The boundary tensor contains additional dynamics. In particular, a two-strand gate in the span $a1 + bP_1$ cannot distinguish ρ from the other nonunit channels, whereas the gate in the worked example above does. An entire decorated strange-correlator cell also contains algebra and module tensors. Its Hamiltonian limit must be established by an explicit local contraction, including seams and sectors, rather than by recognizing a familiar fusion category in one of its tensors.

19.7 Exercises

Exercise 19.1. Use the recurrence for t_L to compute $\dim \mathcal{H}_{\text{path},5}$ and $\dim \mathcal{H}_{\text{path},8}$. Explain why the leading exponential growth is d^L and why this differs from 6^L.

Solution on page 310.

Exercise 19.2. Let F be a unitary $n \times n$ matrix and let S be a subset of its columns. Show that the sum of the corresponding rank-one channel projectors is an orthogonal projector of rank $|S|$. What changes if the fusion multiplicity of a selected channel is greater than one?

Solution on page 310.

Exercise 19.3. For a finite abelian group G acting by permutations on an orthonormal basis, construct the character projection of one orbit vector. Prove that it vanishes precisely when the character is nontrivial on the orbit stabilizer, and normalize it when it is nonzero.

Solution on page 310.

Exercise 19.4. On the alpha-neutral space prove that $Y_\rho - 2$ is invertible and that

$$(Y_\rho - 2)^\dagger (Y_\rho - 2) = 5 - Y_\rho.$$

Give upper and lower bounds for its squared norm on a unit vector. Why is it incorrect to call $Y_\rho - 2$ a projection on this whole space?

Solution on page 310.

Exercise 19.5. Write the dimension of the open path space with fixed endpoints a, b as a matrix element of N_ρ^L. Explain why a periodic trace and a fixed-endpoint matrix element need not have the same low-energy spectrum.

Solution on page 310.

Chapter 20

What a finite spectrum can tell us about a CFT

A finite quantum chain has a discrete spectrum whether its infinite-volume limit is critical, massive, or separated from either regime by a long crossover. The task is therefore not simply to find low energies. It is to identify a scaling organization that persists as the circumference grows. This chapter connects the cylinder formulas of CFT to measured lattice quantities, derives several central-charge diagnostics, and explains the different errors attached to an eigenpair, a variational state, and a continuum extrapolation. The Haagerup chain provides useful examples because its finite calculations can be very accurate while their limiting interpretation remains unsettled. The state-operator correspondence and cylinder transformation used below were derived in Chapter 2.

20.1 The cylinder dictionary and its assumptions

For a nonchiral CFT with $c_L = c_R = c$, the Hamiltonian on a circle of circumference L is

$$H_{\mathrm{CFT}} = \frac{2\pi v}{L}\left(L_0 + \bar{L}_0 - \frac{c}{12}\right). \tag{20.1}$$

Here v converts spatial and temporal units. A lattice Hamiltonian also contains an extensive nonuniversal energy $e_\infty L$. Consequently a critical periodic chain is expected to have

$$E_0(L) = e_\infty L - \frac{\pi v c}{6L} + \cdots, \tag{20.2}$$

$$E_a(L) - E_0(L) = \frac{2\pi v}{L}\left(h_a + \bar{h}_a + n + \bar{n}\right) + \cdots. \tag{20.3}$$

The dots include corrections from irrelevant fields, running couplings, microscopic geometry, and the approach to a possible fixed point. Their form is extra information, not part of the leading formula.

An energy gap and a conformal scaling gap are different quantities. A gapped chain has a positive lower bound on excitation energies above its vacuum sector, independent of sufficiently large L along a size sequence compatible with the proposed order and boundary sector. With several pure vacua, that sector may be a cluster of nearly degenerate finite-volume states; their small internal splittings do not define the bulk gap. A CFT may have a positive lowest nonidentity scalar dimension $\Delta_{\min}$ while remaining gapless, since its corresponding energy difference is $2\pi v\Delta_{\min}/L$.

Equation (20.3) assumes that one common velocity describes the fields being compared. A system with decoupled sectors of different velocities need not admit this single-cylinder

interpretation before its Hamiltonian is rescaled sector by sector. We also assume that the chosen finite-volume state approaches the intended CFT vacuum.

The conformal dimension and spin are

$$\Delta = h + \bar{h}, \qquad s = h - \bar{h}.$$

For descendants the changes are $\Delta \mapsto \Delta + n + \bar{n}$ and $s \mapsto s + n - \bar{n}$. Near a microscopic carrier momentum $k_\star$, the lattice momentum has the form

$$k = k_\star + \frac{2\pi}{L}(s + n - \bar{n}) + \cdots . \tag{20.4}$$

The carrier matters in the Haagerup chain, whose low-energy states occur near 0 and the two commensurate momenta $\pm 2\pi/3$. The momentum relative to one of these cones is the quantity compared with conformal spin. A state at total lattice momentum $2\pi/3$ can still be a residual scalar.

For lengths not divisible by three, the carrier is incompatible with periodic momentum quantization. Those lengths probe a different residual translation boundary condition. They are valuable data, but their vacuum energies and momentum offsets should not be pooled with the divisible lengths without a twisted-sector dictionary.

Worked example 20.1 (Ratios eliminate a common velocity)**.** If two tracked primary gaps obey the same leading formula, then

$$\frac{E_a - E_0}{E_b - E_0} \longrightarrow \frac{\Delta_a}{\Delta_b}.$$

Neither the bulk energy nor v occurs. If a separately identified stress state has $\Delta_T = 2$, one may define the finite diagnostic

$$\Delta_a^{(T)}(L) = 2\frac{E_a(L) - E_0(L)}{E_T(L) - E_0(L)}.$$

This removes an independently fitted velocity, but replaces it by the assumption that the denominator tracks pure stress.

20.2 Canonical calibration: the periodic critical Ising chain

Before analyzing an unfamiliar chain, it helps to run the cylinder dictionary through a model whose finite spectrum can be found exactly.

Worked example 20.2 (From a periodic spin spectrum to Ising CFT)**.** Take Pauli matrices σ_j^a and

$$H_{\mathrm{I}} = -J\sum_{j=1}^{L}\left(\sigma_j^x\sigma_{j+1}^x + \sigma_j^z\right), \qquad J > 0, \qquad \sigma_{L+1}^a = \sigma_1^a. \tag{20.5}$$

The lattice spacing is one, so the circumference is L. We take L even; the displayed stress-pair construction below uses $L \geq 4$. The equality of the interaction and transverse-field coefficients is the critical point in this Pauli-matrix convention. Changing from Pauli matrices to spin operators $\sigma^a/2$ without changing the couplings would change the normalization of every energy.

A periodic spin ring has two fermion boundary conditions. The conserved spin-flip parity is $P = \prod_j \sigma_j^z$; here parity means internal spin flip, not spatial reflection. Introduce Jordan–Wigner fermions

$$c_j = \left(\prod_{m<j}\sigma_m^z\right)\sigma_j^+, \qquad \sigma_j^+ = \frac{\sigma_j^x + i\sigma_j^y}{2}.$$

The string makes fermions on different sites anticommute. At the same site,

$$n_j = c_j^\dagger c_j = \frac{1-\sigma_j^z}{2}, \qquad P = (-1)^{\sum_j n_j}.$$

For an interior bond the strings cancel to give $\sigma_j^x \sigma_{j+1}^x = (c_j^\dagger - c_j)(c_{j+1}^\dagger + c_{j+1})$. Across the last bond their uncancelled product is the total parity. The quadratic Hamiltonian therefore has the uniform form

$$H_{\mathrm{I}} = -J \sum_{j=1}^{L} \left[(c_j^\dagger - c_j)(c_{j+1}^\dagger + c_{j+1}) + 1 - 2n_j \right], \qquad c_{L+1} = -P c_1,$$

where the boundary sign is imposed within a fixed parity sector. Consequently the physical Hilbert space is

$$\mathcal{H}_{\mathrm{spin}} = \mathcal{H}_{\mathrm{NS,\ even}} \oplus \mathcal{H}_{\mathrm{R,\ odd}}. \tag{20.6}$$

Here NS means antiperiodic fermions, the Neveu–Schwarz sector, with momenta $k = (2n+1)\pi/L$. R means periodic fermions, the Ramond sector, with $k = 2n\pi/L$. These are two sectors of the same periodic spin Hamiltonian. Each fermion Fock space must still be projected to the indicated parity. Keeping both full Fock spaces would count twice as many states as the 2^L spin Hilbert space.

A Fourier transform leaves a two-mode problem for each pair $k, -k$. In the basis $(c_k, c_{-k}^\dagger)$ its single-particle matrix can be written

$$\mathsf{h}(k) = 2J \left[(1-\cos k)\tau_z + \sin k\, \tau_y \right],$$

where the τ matrices act on this particle-hole pair. Its positive eigenvalue is

$$\begin{aligned} \epsilon(k) &= 2J\sqrt{(1-\cos k)^2 + \sin^2 k} \\ &= 4J \left| \sin \frac{k}{2} \right|. \end{aligned} \tag{20.7}$$

The associated Bogoliubov transformation mixes creation and annihilation operators in opposite-momentum pairs and preserves fermion parity. It is canonical: the new operators γ_k obey the same fermion anticommutation relations. The diagonal form is

$$H_{\mathrm{I}} = \sum_k \epsilon(k) (\gamma_k^\dagger \gamma_k - \tfrac{1}{2}),$$

with each allowed quasiparticle momentum counted once. Its empty-mode energy is therefore minus one half of the dispersion sum. In the NS sector there are no unpaired $k = 0, \pi$ modes for even L. Its Bogoliubov vacuum has even parity, and each quasiparticle flips parity. Physical NS excitations must therefore contain an even number of quasiparticles.

The Ramond sector requires more care. It contains unpaired $k = 0$ and $k = \pi$ modes, with terms $2J(1-\cos k)(n_k - \frac{1}{2})$. At criticality the $k = 0$ term is exactly zero, whereas $k = \pi$ costs $4J$ to occupy. The paired-mode vacuum has even parity. The physical R ground state has the π mode empty and the zero mode occupied, supplying the required odd parity at no energy cost. Thus its energy is the usual zero-point sum. The two possible zero-mode occupations belong to different fermion parities; they are not two physical R ground states of the periodic spin problem.

Two finite trigonometric sums give the Casimir energy. Put $x = \pi/(2L)$. Choosing momenta in $[0, 2\pi)$ makes every sine in the zero-point sums nonnegative. The NS vacuum energy is

$$E_0(L) = -\frac{1}{2} \sum_{k \in \mathrm{NS}} \epsilon(k) = -2J \sum_{n=0}^{L-1} \sin[(2n+1)x].$$

The elementary identity $2\sin x \sin y = \cos(y-x) - \cos(y+x)$ telescopes:

$$2\sin x \sum_{n=0}^{L-1} \sin[(2n+1)x] = \cos 0 - \cos(2Lx) = 2.$$

Therefore

$$E_0(L) = -2J \csc \frac{\pi}{2L}. \tag{20.8}$$

The same argument in the physical R sector gives

$$2\sin x \sum_{n=0}^{L-1} \sin(2nx) = \cos(-x) - \cos[(2L-1)x] = 2\cos x,$$

and hence

$$E_{\mathrm{R}}(L) = -2J \cot \frac{\pi}{2L}. \tag{20.9}$$

The zero-mode occupation changes no term in this sum. Since $\csc x > \cot x$ for the present x, the NS state is the ground state of the full spin Hamiltonian.

The small-momentum slope of the exact dispersion gives an independent velocity:

$$\epsilon(k) = 2J|k| - \frac{J}{12}|k|^3 + O(|k|^5), \qquad v = 2J.$$

Using $\csc x = x^{-1} + x/6 + 7x^3/360 + O(x^5)$,

$$E_0(L) = -\frac{4J}{\pi}L - \frac{\pi J}{6L} - \frac{7\pi^3 J}{1440L^3} + O(L^{-5}). \tag{20.10}$$

Thus $e_\infty = -4J/\pi$. Comparing the L^{-1} coefficient with $-\pi vc/(6L)$, using the independently known $v = 2J$, gives $c = 1/2$. The coefficient alone would only have given $vc = J$. The long-wavelength modes form a massless Majorana fermion, providing an independent continuum description with the same central charge.

Even here a finite-size central-charge estimator is not exactly its limiting value. If the exact e_∞ and v are used,

$$-\frac{6L}{\pi v}\big[E_0(L) - e_\infty L\big] = \frac{1}{2} + \frac{7\pi^2}{480L^2} + O(L^{-4}).$$

The correction is a property of an exactly solved finite Hamiltonian, not an eigensolver error.

The spin, energy, and stress levels. The odd-parity ground lies above the NS ground by

$$g_\sigma(L) := E_{\mathrm{R}}(L) - E_0(L) = 2J\big(\csc x - \cot x\big) = 2J\tan\frac{x}{2}. \tag{20.11}$$

The last equality uses $(1-\cos x)/\sin x = \tan(x/2)$. In particular,

$$g_\sigma(L) = \frac{\pi J}{2L} + \frac{\pi^3 J}{96L^3} + O(L^{-5}), \qquad \frac{L g_\sigma(L)}{2\pi v} = \frac{1}{8} + \frac{\pi^2}{384L^2} + O(L^{-4}).$$

This is the Ising spin-field sector, with limiting $(h, \bar{h}) = (1/16, 1/16)$. The sector identification has more content than an energy fit: the lattice order operator σ_j^x changes P, and its Jordan–Wigner string changes the Majorana boundary condition between NS and R. In the continuum this boundary-condition change is the spin field. Both parity-sector ground states have physical momentum zero. One can see this directly in the spin basis: the off-diagonal matrix elements are nonpositive, adjacent pair flips connect each fixed-parity sector, and its positive ground vector is invariant under translation.

In the NS sector a single quasiparticle at $k = \pi/L$ is not a physical periodic-spin state: it has the wrong parity. The allowed opposite-momentum pair at $k = \pm\pi/L$ has zero total momentum and gap

$$g_\epsilon(L) = 2\epsilon(\pi/L) = 8J \sin x. \tag{20.12}$$

Its known-velocity dimension estimate is

$$\Delta_\epsilon^{(v)}(L) := \frac{Lg_\epsilon(L)}{2\pi v} = \frac{2L}{\pi} \sin\frac{\pi}{2L} = 1 - \frac{\pi^2}{24L^2} + O(L^{-4}). \tag{20.13}$$

In the Majorana continuum theory this pair is $\psi_{-1/2}\bar{\psi}_{-1/2}|0\rangle$, with weights $(1/2, 1/2)$: the energy field ϵ.

The same-direction pair at $k = \pi/L, 3\pi/L$ is also allowed. Its total momentum is $4\pi/L$ modulo 2π, and its gap is

$$\begin{aligned} g_T(L) &= 4J(\sin x + \sin 3x) = 16J \sin x \cos^2 x, \\ \frac{Lg_T(L)}{2\pi v} &= 2 - \frac{7\pi^2}{12L^2} + O(L^{-4}). \end{aligned} \tag{20.14}$$

The momenta correspond to continuum half-integer fermion modes $1/2$ and $3/2$. This identifies the limiting state, rather than merely matching its momentum. With $\{\psi_r, \psi_s\} = \delta_{r+s,0}$ and $T = -\frac{1}{2} :\psi\partial\psi:$,

$$L_{-2}|0\rangle = \frac{1}{2}\psi_{-3/2}\psi_{-1/2}|0\rangle.$$

The two-fermion vector has norm one, so the stress state has norm squared $1/4 = c/2$, as required by the Virasoro algebra. Its weights are $(2, 0)$ and its spin is $+2$. The negative-momentum pair gives the opposite chirality. At $L = 4$ the total lattice momenta of the two opposite-chirality pairs coincide modulo 2π; their distinct continuum chiral labels should not be inferred from that short-ring momentum alone.

The parity projection also explains the bosonic local continuum spectrum. Even-even chiral NS fermion parity gives the vacuum family, and odd-odd chiral NS parity gives the energy family. The projected R space gives the spin family. Thus its continuum Hilbert space is

$$\mathcal{H}_{\text{Ising}} = (\mathcal{V}_0 \otimes \overline{\mathcal{V}_0}) \oplus (\mathcal{V}_{1/2} \otimes \overline{\mathcal{V}_{1/2}}) \oplus (\mathcal{V}_{1/16} \otimes \overline{\mathcal{V}_{1/16}}),$$

where $\mathcal{V}_h$ is the irreducible Ising Virasoro module. An unprojected single chiral fermion of spin $1/2$ is not an additional local field in this periodic bosonic theory.

Two legitimate calibrations approach from opposite sides. The exact velocity $v = 2J$ is an infinite-volume slope. A stress-gap estimate uses a different quantity,

$$\widehat{v}_T(L) := \frac{Lg_T(L)}{4\pi} = \frac{4JL}{\pi} \sin x \cos^2 x = 2J\left[1 - \frac{7\pi^2}{24L^2} + O(L^{-4})\right].$$

It tends to v, but differs from it at every finite size. Using this estimate to normalize the energy-field gap gives

$$\Delta_\epsilon^{(T)}(L) := \frac{Lg_\epsilon(L)}{2\pi\widehat{v}_T(L)} = 2\frac{g_\epsilon(L)}{g_T(L)} = \sec^2\frac{\pi}{2L} = 1 + \frac{\pi^2}{4L^2} + O(L^{-4}). \tag{20.15}$$

The exact identity follows by cancelling $\sin x$ between $g_\epsilon = 8J \sin x$ and $g_T = 16J \sin x \cos^2 x$. The known-velocity estimate (20.13) approaches dimension one from below; the stress-calibrated estimate approaches it from above. They use the same exact eigenvalues and track the same

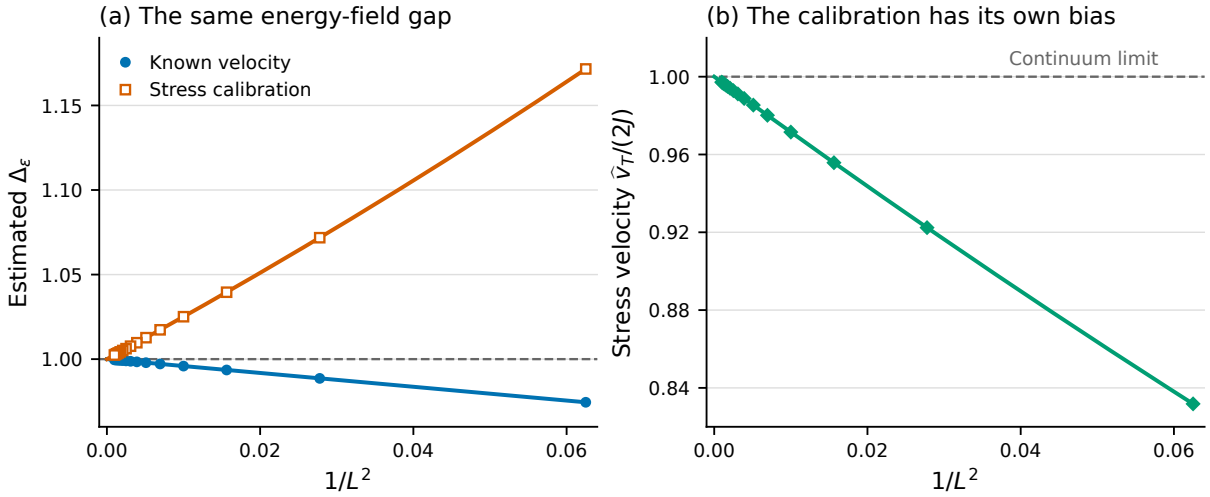


Figure 20.1: Two finite-size calibrations of the same critical Ising energy-field gap, using the exact formulas in Example 20.2. Symbols mark even $L = 4, 6, \ldots, 32$; curves show the smooth continuation of the formulas to real $L \geq 4$. Put $z = \pi/(2L)$. The known-velocity estimate is $\Delta_\epsilon^{(v)} = \sin z/z$, whereas the stress-calibrated estimate is $\Delta_\epsilon^{(T)} = \sec^2 z$. They approach the same dimension from opposite sides. The right panel shows the calibration bias itself: $\hat{v}_T/(2J) = (\sin z/z)\cos^2 z$. Dashed lines show the common limiting value 1 at $1/L^2 = 0$. These are exact Ising formulas, not fits or Haagerup-chain data.

limiting field. Removing the velocity by a ratio does not remove the unequal finite-size corrections of the two levels.

For this nearest-neighbor Ising Hamiltonian the corrections can be derived from the full dispersion and the finite sums: the cubic term in $\epsilon(k)$ produces L^{-3} corrections to gaps and hence L^{-2} corrections to dimension estimates. That power is consistent with dimension-four irrelevant contributions in a continuum expansion, but the power alone does not specify a unique irrelevant operator. Other Hamiltonians can have different coefficients, vanishing leading coefficients, or logarithmic running such as the current example in Chapter 14.

This calibration illustrates what becomes possible when the boundary conditions, parity projection, velocity, eigenstates, and continuum field dictionary are all independently controlled. For the Haagerup chain, exact finite eigenvalues or a small residual do not supply those missing identifications. Its finite-size analysis must still determine which sectors approach a common vacuum and velocity, which states approach the proposed fields, and what correction law governs that approach.

20.3 An accurate eigenpair is a finite statement

Suppose a normalized numerical vector u has Rayleigh value E and residual $r = (H - E)u$. In a finite Hermitian problem,

$$\operatorname{dist}(E, \operatorname{spec} H) \leq \|r\|. \tag{20.16}$$

To prove this, expand $u = \sum_n c_n |n\rangle$ in an exact eigenbasis. Then $\|r\|^2 = \sum_n |c_n|^2 (E_n - E)^2$. If every exact energy were farther than $\|r\|$, the weighted average would be larger than $\|r\|^2$.

This useful bound does not establish that the nearby eigenvalue is the lowest one in a sector, that all lower levels have been found, or that the state has a particular conformal identity. With an independent spectral gap separating a target eigenspace from the rest, one can turn a residual

into a bound on the unwanted component of u. Without such separation, nearly degenerate eigenvectors can rotate substantially while their energies remain accurate.

For a variational normalized state u, the energy is an upper bound on the ground energy in the allowed Hilbert space. A difference of two variational energies is generally not an upper or lower bound on an excitation gap: the two approximation errors can have either ordering. A small optimizer gradient tests stationarity within the chosen coordinates, and a small variance tests closeness to some energy eigenspace. Neither alone proves global minimality.

Remark 20.3 (Three separate questions). A useful report answers three questions separately: how accurately was a finite matrix problem solved; how completely were its relevant sectors covered; and how well is the infinite-size correction law controlled? Many significant mistakes arise by answering the third question with an impressive number from the first.

20.4 Casimir energy: why the velocity remains

Divide (20.2) by L and write $x = L^{-2}$. The leading relation is a line,

$$\frac{E_0(L)}{L} = e_\infty - \frac{\pi v c}{6} x + \cdots .$$

Its slope determines the product vc. The ground-state energy alone does not distinguish a larger central charge with a smaller velocity from a smaller central charge with a larger velocity.

The stored small-chain ground energies for the physical $\mathcal{H}_3$ model are

$$E_0(3) = -3, \quad E_0(6) = -5.271980047610, \quad E_0(9) = -7.737524752482.$$

The reported length-three value saturates the local lower bound at the displayed precision. A two-size fit using $6, 9$ gives the descriptive value $vc = 2.34379137903$. With two observations and two unknown coefficients the line interpolates exactly. An exact interpolation is not evidence that the omitted terms vanish.

One way to estimate v is to identify a conformal level with known dimension. Another is to compare independently normalized energy and current responses. Both approaches need a field dictionary. Inferring v by declaring the first convenient momentum-two state to be stress builds the desired answer into the normalization.

20.5 Deriving the stress-source estimator

Use the energy Fourier mode H_2 of (19.5). Under a specific lattice-to-CFT identification of the energy density,

$$H_2 \longrightarrow \frac{2\pi v}{L}(L_{-2} + \bar{L}_2). \tag{20.17}$$

Acting on the vacuum, the second term vanishes. The Virasoro commutator

$$[L_2, L_{-2}] = 4L_0 + \frac{c}{2}$$

therefore gives $\|L_{-2}|0\rangle\|^2 = c/2$. For a unit-normalized stress state $|T\rangle$, put $M = \langle T|H_2|0\rangle$ and $g_T = E_T - E_0$. The ideal cylinder formulas are

$$|M|^2 = \left(\frac{2\pi v}{L}\right)^2 \frac{c}{2}, \qquad g_T = \frac{4\pi v}{L}.$$

Dividing cancels the unknown velocity:

$$\widehat{c}_T(L) = \frac{8|M|^2}{g_T^2}. \tag{20.18}$$

There is no extra factor of two for the opposite chirality. A complex Fourier mode of the stated sign creates one chiral stress state from the vacuum under this identification. On a ring so short that the two Fourier labels coincide, such as $m = 2$ and $m = -2$ at $L = 4$, the source momentum alone cannot establish its chirality.

Numerical record 20.4 (Measured stress candidates)**.** The physical-chain calculation uses the neutral momentum-two sector and selects a low-energy state with large overlap with $H_2|0\rangle$. The following are saved finite results, not extrapolated central charges:

L	g_T	$\lvert M\rvert^2$	$\widehat{c}_T(L)$
6	1.271980047610	1.159949183395	5.735461284
9	1.153308959914	.516954434465	3.109216187
12	.950908727688	.293569361127	2.597305575
15	.793841246905	.189323099189	2.403401186

The later $L = 18$ continuation has $g_T = .6768446828683299$ and conditional $\widehat{c}_T(18) \simeq 2.30768$, with a full eigenpair residual of order 4×10^{-14}. Its stress-source fraction is approximately .95444.

The dominant state in the older calculations is not the lowest state at momentum two. This alone is an important lesson: lattice momentum does not identify a conformal family. If several states form a nearly degenerate candidate stress cluster, the basis-independent overlap is the sum of their squared amplitudes. But this sum equals stress weight only if other contributions to the cluster have been excluded or separated.

20.5.1 Energy-density improvements

The periodic sum is unchanged under

$$h_j \mapsto h_j + B_{j+1} - B_j.$$

For the Fourier mode with $k = 2\pi m/L$, the change is

$$\delta H_m = (e^{ik} - 1) \sum_j e^{-ikj} B_j.$$

If B has a continuum field of dimension Δ_B, its extensive Fourier mode scales as $L^{1-\Delta_B}$, and the finite difference contributes another L^{-1}. Thus $\Delta_B = 1$ gives the same L^{-1} scaling as the stress mode. A current descendant can contaminate a dimension-two source without changing the total Hamiltonian.

For a correctly separated pure stress eigenstate, an orthogonal current descendant does not change the stress matrix element. The difficulty is establishing that separation. Largest-overlap selection and high captured source weight are evidence about a finite source, not a proof of (20.17).

20.5.2 If the limiting ground is not the identity

The vacuum hypothesis is substantive. Suppose instead that the regulated ground states approach a spin-zero highest-weight threshold with $h = \bar{h} = h_0 > 0$. Its cylinder energy would involve

$$c_{\text{eff}} = c - 24h_0$$

in the Casimir term. The Virasoro algebra, however, still has central charge c. Its descendants obey

$$\|L_{-1}|h_0\rangle\|^2 = 2h_0, \qquad \|L_{-2}|h_0\rangle\|^2 = 4h_0 + c/2.$$

Thus the ground must have a nonzero same-symmetry momentum-one descendant at relative dimension one, provided the regulated low-energy stress action converges. A controlled sum of its level-two stress-source weight would return $c + 8h_0$ under the vacuum estimator normalization, not c_{eff}.

The sum matters: above a nonvacuum primary, the level-two space contains both $L_{-2}|h_0\rangle$ and $L_{-1}^2|h_0\rangle$, which are not orthogonal in general. Their Gram matrix is the one derived in Chapter 2. One selected lattice eigenstate need not carry all the L_{-2} weight.

For a concrete proposed mechanism, the bosonic cigar has

$$c = 2 + \frac{6}{k-2}, \qquad h_0 = \frac{1}{4(k-2)}, \qquad c_{\text{eff}} = 2.$$

Its identity is not normalizable; regulated radial levels approach a continuum threshold. Hence a Casimir estimate near two is compatible with a larger Virasoro charge in that theory. It also brings extra obligations: level-one spectral weight and a growing density of radial states. Under the simplest angular-charge assignment, a charged scalar of threshold-relative dimension $1/(2k)$ equal to $1/8$ would fix $k = 4$, giving $c = 5$, $h_0 = 1/8$, and $c + 8h_0 = 6$.

At $L = 9, 12, 15$, the saved first fully Haagerup-invariant momentum-one gaps have effective dimensions about $2.800, 2.846, 2.884$, using the saved epsilon-descendant velocity calibration. They do not display the required threshold descendant near dimension one in the observed regime. This is a conditional diagnostic, since logarithmic convergence can be slow and the low-energy stress map itself needs control. It does not exclude every noncompact CFT or license replacing c by c_{eff} in the vacuum sewing formulas.

The named calibration assumes that a selected momentum-one level is the first descendant of the neutral scalar ϵ: its gap from that scalar is set equal to $2\pi v/L$. This is a different finite-size prescription from setting the stress gap equal to $4\pi v/L$.

20.6 Entropy on a finite ring

For an interval of length ℓ in a periodic ground state, the CFT formula is

$$S_L(\ell) = \frac{c}{3} \log \left[\frac{L}{\pi} \sin \left(\frac{\pi \ell}{L} \right) \right] + s_0 + \cdots . \tag{20.19}$$

The logarithm contains the chord length, not simply ℓ. At small ℓ/L it approaches $\log \ell$; near the midpoint it reflects the finite circumference. An additive entropy constant can be eliminated by comparing two cuts or two sizes, but cutoff and boundary corrections remain.

For the stored ordinary height-embedding entropies,

$$S_6(3) = 2.258824289079, \qquad S_9(4) = 2.529902677312.$$

Since the nine-site ring has no integer cut at exactly $L/2$, the appropriate denominator in the two-size slope uses the actual chord lengths. This gives the effective value $c = 2.08438313923$. Replacing $\sin(4\pi/9)$ by one introduces a small but avoidable geometric error.

This attractive number is not by itself a central-charge determination. Within each size, fits over cuts $\ell \geq 2$ give approximately 3.3880 at $L = 6$ and 2.7302 at $L = 9$; the first fit has only two points. Different cut residues, short distances, and finite-size corrections clearly matter. A visually smooth curve and a close rational or integer slope do not bound them.

The partial trace convention must also be specified. The displayed entropies arise from the ordinary embedding in $(\mathbb{C}^6)^{\otimes L}$. An anyonic partial trace can add quantum-dimension boundary weights. These constructions may share a leading universal logarithm under appropriate hypotheses, but their finite entropies need not agree. One must not combine their constants or finite slopes as though they were the same observable. Chapter 22 explains the ordinary Schmidt spectrum and the extra entropy of a symmetry cat.

20.6.1 An open-chain caution

For an open critical chain the leading coefficient is $c/6$, rather than $c/3$. Boundary degrees of freedom also contribute entropy. In an open $\mathcal{H}_2$ representation related to a real-q spin-one chain, a low $S^z = 0$ state showed substantial entropy growth from length 48 to 96, while its low $S^z = 1$ partner changed its midpoint entropy by only about .00764. The nearly polarized edge state removes most of the apparent growth. This is evidence about boundary-sensitive finite-chain states in the $\mathcal{H}_2$ model, not a conclusion about $\mathcal{H}_3$. Its pedagogical point is general: a boundary cat or entangled edge pair can imitate bulk critical entropy over a long range.

20.7 A thermal modular test using the whole spectrum

Modular invariance supplies a different consistency test. Let $g_a = E_a - E_0$ run through every state in the untwisted periodic Hilbert space, with all charge and momentum multiplicities. At inverse temperature β, define

$$Z_{\mathrm{gap}}(\beta) = \sum_a e^{-\beta g_a}.$$

In a compact CFT with the vacuum identified and one velocity, restoring the cylinder vacuum energy gives

$$Z(\beta) = e^{\beta a} Z_{\mathrm{gap}}(\beta), \qquad a = \frac{\pi c v}{6L}.$$

The exchange of the two torus cycles maps $\beta \mapsto L^2/(v^2\beta)$. At its fixed point $\beta_* = L/v$, the function $Z(\beta_* e^t)$ is even in t. Its odd derivatives therefore vanish. This physical inverse temperature corresponds to the dimensionless $b = 2\pi$ of Chapter 14. The exchange is the same torus S operation introduced in Chapter 2; projecting onto one ordinary charge sector would generally transform to a different, spatially twisted trace.

Let $\mu = \langle g \rangle_{\beta_*}$ and $m_j = \langle (g-\mu)^j \rangle_{\beta_*}$ in the probability distribution proportional to $e^{-\beta_* g}$. The first derivative gives $a = \mu$, or

$$c = \frac{6\beta_* \mu}{\pi}. \tag{20.20}$$

The third derivative uses $(\beta\partial_\beta)^3 = \beta\partial_\beta + 3\beta^2\partial_\beta^2 + \beta^3\partial_\beta^3$. After imposing $a = \mu$, it becomes

$$\beta_* m_3 = 3m_2. \tag{20.21}$$

One can use this condition to seek β_* without first choosing a particular stress state. But the fifth derivative is an independent requirement:

$$15\beta_*^2 m_2 - 25\beta_*^3 m_3 + 10\beta_*^4 m_4 - \beta_*^5 m_5 = 0.$$

Solving the third condition does not enforce the fifth.

For the physical chain, the root near velocity one gives effective charges

$$c_6 = 2.31250, \qquad c_9 = 2.24215, \qquad c_{12} = 2.19799$$

in the stored diagnostic. Length six uses the complete spectrum; the larger rings use specified low-state cutoffs in every ordinary sector. The fifth condition remains unsatisfied. A compact-boson control reproduces $c = 1$ and its higher conditions, so the construction of the diagnostic itself has a useful independent check.

High moments amplify spectral tails. A tiny omitted contribution to Z_{gap} can have a much larger effect on m_5. Tail bounds require both a bound on how many states remain and a genuine lower bound on their energies. Small residuals of the retained eigenvectors supply neither completeness nor that lower threshold. These modular equations are continuum requirements tested on finite spectra; they are not exact identities of a generic finite lattice Hamiltonian.

20.8 Why fitting more decimals does not control the limit

For the stress values above, choose the illustrative variable $x = L^{-2}$. A linear fit through sizes $9, 12, 15$ gives intercept 1.989673; a line through $12, 15$ gives 2.058682; a quadratic through $9, 12, 15$ gives 2.125928. Two of these prescriptions interpolate their data exactly, with no remaining fit degrees of freedom.

Proposition 20.5 (Finite interpolation does not determine a limit)**.** *Let $f(x_i) = y_i$ be prescribed at finitely many distinct positive points x_i. If p interpolates those data, then*

$$p_\lambda(x) = p(x) + \lambda \prod_i (x - x_i)$$

has exactly the same data for every λ, while $p_\lambda(0)$ can take any value.

Proof. The added product vanishes at each observed point and is nonzero at zero. Varying its coefficient changes the unobserved intercept freely. □

The proposition does not say all extrapolation is useless. A known correction exponent, a controlled remainder, monotonicity, or a renormalization-group analysis can rule out much of this freedom. It says those are the ingredients that justify an extrapolation. Small eigensolver residuals cannot substitute for them. Marginally irrelevant operators can produce logarithmic corrections that look nearly polynomial over accessible sizes, while a large finite correlation length can produce a broad apparently critical window.

An honest comparison therefore retains several prescribed scales and fit windows, rather than choosing the one nearest a desired value. The spread is a sensitivity diagnostic. Without a justified model for the corrections it is not a statistical confidence interval.

20.9 Tracking light fields across sectors

The Greek letters in this inventory label tracked Haagerup-chain states. In particular, ϵ does not identify a state with the Ising energy field of Example 20.2.

The completed ordinary-sector inventory at lengths $9, 12, 15$ covers all $27, 36, 45$ intrinsic-charge and momentum sectors. It finds finite light-shell multiplicities 2 for ϕ, 2 for χ, 4 for ψ, η and their adjoints, and 1 for ϵ. The neutral cone pair $\chi, \chi^\dagger$ must not be confused with the higher invariant scalar accessed by a unit-channel source. Here ϕ labels the lowest intrinsic-charge pair, χ the intrinsically neutral pair at the nonzero carriers, and ϵ the low neutral scalar. Their full charge pattern and normalized matrix elements are introduced in Chapter 21.

Numerical record 20.6 (Velocity-independent gap ratios)**.** The saved ratios are

	$L = 12$	$L = 15$	$L = 18$
g_χ/g_ϕ	1.77474751	1.78423508	1.79085991
g_ϵ/g_ϕ	1.96087659	1.96657684	1.97006943
g_χ/g_ϵ	.90507863	.90727962	.90903391

At $L = 18$, $g_\phi = .048135824413041206$, $g_\chi = .08620451829960096$, and $g_\epsilon = .0948309161570915$.

These data make the fields' relative separation concrete without a velocity assumption. They also show continuing drift. Finite shell multiplicities are not a theorem about the complete continuum scalar spaces. An unseen level can descend, an accidental degeneracy can develop, and finite scalar assignments can mix with descendants. Nevertheless, a complete finite-sector census is stronger evidence than following only a favored charge line.

The source-visible invariant singlet gives another useful example. Its stress-normalized dimension decreases from 2.29254 to 2.22221 to 2.18339 at $L = 12, 15, 18$. Several formal L^{-2} intercepts lie near 2.095, while independent amplitude-based effective dimensions lie near 2.030, 2.053, and 2.082. This is a field near the marginal threshold in finite diagnostics. It does not establish an exactly marginal scalar, a limiting dimension above two, or a beta function.

20.10 Exercises

Exercise 20.1. Prove (20.16). Suppose in addition that every eigenvalue outside a specified target cluster is at least γ from E. Bound the squared norm of the component of u outside that cluster in terms of $\|r\|$ and γ.

Solution on page 311.

Exercise 20.2. Derive (20.18) directly from the Virasoro commutator. Repeat the calculation for a vacuum mode L_{-n} and derive its squared norm $cn(n^2 - 1)/12$.

Solution on page 311.

Exercise 20.3. Verify the Fourier transformation of $B_{j+1} - B_j$. For a continuum field of dimension Δ_B, explain the scaling of its contribution to H_m when m is fixed as $L \to \infty$.

Solution on page 311.

Exercise 20.4. Use two sizes L_1, L_2 to eliminate e_∞ in (20.2). Write the resulting expression for vc and identify the combination of omitted corrections that survives.

Solution on page 311.

Exercise 20.5. An observed state has momentum one and gap corresponding to $\Delta_\phi + 1$ in a chosen scale. Compare its putative weights as $L_{-1}\phi$ with those of a chiral current. Why does proximity to that gap fail to establish a current?

Solution on page 311.

Exercise 20.6. Derive the first, third, and fifth thermal modular conditions from the evenness of $Z(\beta_* e^t)$. Explain why a partition function restricted to the intrinsic neutral sector alone need not satisfy them.

Solution on page 311.

Chapter 21

Local fields, sewing overlaps, and Ward tests

Energy levels tell us which states might exist; matrix elements tell us how their fields interact. This chapter develops two ways to approach those interactions from a lattice chain. One uses fixed local operators and their matrix elements. The other glues normalized wavefunctions into a pair-of-pants geometry. Both are natural consequences of the state-operator correspondence, but they have different finite-size corrections. We derive phase-independent three-point invariants, the unavoidable descendant content of a spatially averaged primary, and a useful response test for a proposed running coupling. These calculations also explain why a large overlap with one low state is not a general error bound for an OPE coefficient. The primary and descendant conventions are those of Chapter 2; Chapter 17 explains the full-CFT consistency conditions that any limiting overlap interpretation must obey.

21.1 From a local lattice source to a normalized field

Let o_j be a fixed finite-range lattice density. Its momentum-filtered average is

$$O_k = \frac{1}{L}\sum_j e^{-ikj} o_j.$$

For a neutral source its vacuum mean should first be subtracted. For a field at a nonzero carrier momentum, the phase removes that carrier before comparison with a residual scalar zero mode. The support and coefficients of o_j should be kept fixed while L changes.

Suppose that in a critical limit

$$o_j \sim a_\phi \phi(x) + \sum_b a_b \mathcal{O}_b(x).$$

The coefficient a_ϕ is a nonuniversal microscopic normalization. If $|\phi\rangle$ is the corresponding unit-normalized cylinder state, then

$$A_\phi(L) = \langle \phi | O | 0 \rangle \sim a_\phi (2\pi/L)^{\Delta_\phi}$$

in a fixed coordinate convention. A ratio such as

$$c_O(L) = \frac{\langle \phi^\dagger | O | \phi \rangle}{\langle \phi | O | 0 \rangle} \tag{21.1}$$

can cancel a_ϕ and the common cylinder factor. Under convergence to the intended primary insertion and states, it approaches the normalized three-point coefficient $C(\phi, \phi, \phi)$.

The conditional phrase matters. The denominator tests O on the vacuum; the numerator tests it on an excited state. An unwanted operator can have a small vacuum-to-low-state matrix element and a substantial excited-state matrix element. No algebraic inequality turns a single large vacuum overlap into a uniform operator-norm estimate.

21.1.1 A finite-dimensional operator optimization

One can choose a span of local words $o^{(a)}$ with the desired charge and momentum and set $O = \sum_a c_a O^{(a)}$. The source Gram matrix is

$$G_{ab} = \langle 0|(O^{(a)})^\dagger O^{(b)}|0\rangle.$$

The source squared norm is $c^\dagger G c$. Linear constraints can normalize the overlap with a target and cancel overlaps with selected unwanted states. Minimizing the Gram norm subject to those constraints is a finite linear algebra problem. Null operator directions must be removed consistently; otherwise a formally large coefficient can represent a zero or nearly zero source.

The resulting recipe is useful when trained at one reference size and then transferred unchanged. Retraining independently at every size may produce a sequence of operators with no fixed local continuum limit. Equally, choosing the recipe that most closely matches a favorite CFT value makes subsequent agreement circular. Predetermined operator ranges, all successful recipes, and failures should remain visible.

Worked example 21.1 (Solving the source optimization)**.** Assume the independent source Gram matrix is positive definite, and define $b_a = \langle 0|(O^{(a)})^\dagger|\phi\rangle$. The target amplitude of $O = \sum_a c_a O^{(a)}$ is $b^\dagger c$. For $b \neq 0$, minimizing $c^\dagger G c$ subject to $b^\dagger c = 1$ gives

$$c_{\mathrm{opt}} = \frac{G^{-1}b}{b^\dagger G^{-1} b}, \qquad \|O_{\mathrm{opt}}|0\rangle\|^2 = \frac{1}{b^\dagger G^{-1} b}.$$

Indeed Cauchy–Schwarz applied to $G^{1/2}c$ and $G^{-1/2}b$ gives the lower bound, saturated by the displayed vector. The maximum target fraction is $b^\dagger G^{-1} b$, the squared projection of $|\phi\rangle$ onto the allowed source span.

For two normalized sources with real mutual overlap $1/5$ and target overlaps $3/5$ each, G has diagonal entries one and off-diagonal entries $1/5$, while $b = (3/5, 3/5)^T$. The optimum is $c = (5/6, 5/6)^T$ and its fraction is $3/5$. Each source alone has fraction $9/25$. The improvement is a finite linear-algebra fact about the chosen span, with no CFT assumption. Whether this optimized source has a stable local continuum interpretation is the separate transfer-in-size test.

Definition 21.2 (Vacuum-source fraction)**.** For a normalized target $|\phi\rangle$ and source $s = O|0\rangle$, define

$$f_\phi = \frac{|\langle \phi|s\rangle|^2}{\|s\|^2}.$$

The complement $1 - f_\phi$ is exactly the fraction of the finite source squared norm outside that state, provided the norm and normalization are known.

This is a strong finite statement, even if no other states have been saved. What it does not measure is $\|O|a\rangle\|$ for other $|a\rangle$ or the difference between O and a continuum operator in a norm valid on an entire low-energy space.

21.2 Complex phases and information that cannot be rephased away

Normalize charged fields by $\langle \phi(z)\phi^\dagger(0)\rangle = |z|^{-2\Delta_\phi}$. The transformation $\phi \mapsto e^{i\theta_\phi}\phi$ preserves this normalization and changes the cubic $c_\phi = C(\phi, \phi, \phi)$ by $e^{3i\theta_\phi}$. Thus an individual complex cubic has a conventional phase.

For the charge pattern

$$\phi : (1,0), \quad \chi : (0,1), \quad \psi : (1,1), \quad \eta : (1,-1)$$

under $C_3 \times C_3$, define

$$\lambda = C(\phi, \chi, \psi^\dagger), \quad \nu = C(\phi, \psi, \eta), \quad \kappa = C(\chi, \psi, \eta^\dagger).$$

The first charge is intrinsic alpha charge. The second records the proposed order-three internal part of the carrier translation, distinct from the residual conformal spin. Its survival as a faithful continuum symmetry is a premise tested in Chapter 24. The rephasing algebra below uses the displayed selection rules; it does not by itself prove that continuum symmetry interpretation. All three are allowed by charge conservation. Their phases transform as

$$\lambda \mapsto e^{i(\theta_\phi + \theta_\chi - \theta_\psi)}\lambda, \quad \nu \mapsto e^{i(\theta_\phi + \theta_\psi + \theta_\eta)}\nu,$$

and similarly for κ. Cubing and dividing by self cubics cancels the field phases:

$$I_\lambda = \frac{\lambda^3 c_\psi}{c_\phi c_\chi}, \qquad I_\nu = \frac{\nu^3}{c_\phi c_\psi c_\eta}, \qquad I_\kappa = \frac{\kappa^3 c_\eta}{c_\chi c_\psi}. \tag{21.2}$$

The ratio $J = \kappa\nu/(\lambda c_\psi)$ is also invariant and obeys $J^3 = I_\nu I_\kappa / I_\lambda$.

Worked example 21.3 (Parity need not make every invariant real)**.** Suppose spatial parity acts as $P\phi = \phi$, $P\chi = \chi^\dagger$, $P\psi = \eta$, $P\eta = \psi$, in a suitable phase convention. Then $c_\eta = c_\psi$ and parity can make κ and c_χ real, so I_κ is real. But parity sends λ to the different coupling $C(\phi, \chi^\dagger, \eta^\dagger)$. It does not send every field to its adjoint. It therefore does not force I_λ to equal its complex conjugate.

A unitary symmetry that simultaneously conjugates all three fields in I_λ would force its reality: its field phases cancel between numerator and denominator. A nonreal invariant can therefore obstruct such a symmetry. This is more informative than finding a complex entry in an arbitrary eigenvector basis. It is also different from the antiunitary conjugation associated with a real Hamiltonian.

Numerical record 21.4 (Finite mixed-tensor fingerprint)**.** At $L = 15$, prescribed range-four/five recipes give $|\lambda| \simeq 1.420$–1.432. Three different choices of which field supplies the inserted local operator agree within approximately .26–.68% for matched recipes. The direct chi insertion gives

$$\Re I_\lambda \in [1.114433, 1.119331], \qquad \Im I_\lambda \in [-.203273, -.199851].$$

These ranges describe deterministic operator variation. They are not confidence intervals for continuum data. The nonreal finite invariant is nevertheless unchanged by the saved state rephasings.

Chapter 24 applies this invariant to a specified canonical Cardy hypothesis, with the required spectral completeness and sewing assumptions kept explicit.

21.3 A local primary creates an entire translation family

The state-operator correspondence gives a precise benchmark for a source fraction. Let ϕ be a scalar primary with $h = \bar{h} = \Delta/2$. On the plane,

$$\phi(z, \bar{z})|0\rangle = e^{zL_{-1} + \bar{z}\bar{L}_{-1}}|\phi\rangle.$$

Radial quantization and the global conformal algebra imply

$$\|L_{-1}^n|h\rangle\|^2 = n!(2h)_n,$$

where $(a)_n = a(a+1)\cdots(a+n-1)$. This follows inductively from $[L_1, L_{-1}] = 2L_0$.

On the unit circle, spatial averaging keeps precisely the terms with equal left and right levels. The normalized scalar translation descendant is

$$|D_n\rangle = \frac{L_{-1}^n \bar{L}_{-1}^n|\phi\rangle}{n!(\Delta)_n}.$$

Its amplitude relative to the primary amplitude is $(\Delta)_n/n!$. Orthogonality of different energies then gives

$$F_{\mathrm{vac}}(\Delta) = \frac{\|O_0|0\rangle\|^2}{|\langle\phi|O_0|0\rangle|^2} = \sum_{n=0}^{\infty}\left[\frac{(\Delta)_n}{n!}\right]^2. \tag{21.3}$$

For $0 \le \Delta < 1/2$, Gauss summation yields

$$F_{\mathrm{vac}}(\Delta) = \frac{\Gamma(1-2\Delta)}{\Gamma(1-\Delta)^2}, \qquad p_{\mathrm{primary}} = \frac{\Gamma(1-\Delta)^2}{\Gamma(1-2\Delta)}. \tag{21.4}$$

The first daughter has relative amplitude Δ and squared weight Δ^2. A pure local primary therefore creates more than its primary state. Demanding a source fraction of exactly one would impose the wrong continuum target.

The convergence condition has a physical meaning. The equal-time two-point function behaves near coincidence as $|\theta|^{-2\Delta}$. Its spatial integral is finite precisely for $\Delta < 1/2$. For larger dimensions, a microscopic contact region can dominate the unfiltered source norm even though its low-energy matrix elements have a good scaling limit.

21.3.1 Why a finite fraction can exceed the CFT fraction

Let $O = L^{-1}\sum_j o_j$. A regular short-distance contribution to its variance is of order L^{-1}, whereas the long-distance primary contribution is of order $L^{-2\Delta}$. Thus a possible expansion is

$$\mathrm{Var}(O) = AL^{-2\Delta} + B/L + \cdots.$$

After normalization by the primary overlap squared,

$$F_L = F_{\mathrm{vac}} + B'L^{2\Delta-1} + \cdots. \tag{21.5}$$

The coefficient B' is not constrained to be positive: it measures the difference between a microscopic regularization and the ideal continuum contribution. At light dimensions near .13–.28, this correction decays much more slowly than L^{-2}.

For example, using the conditional $L = 18$ stress dimensions gives primary fractions approximately $.95865, .79759, .84008$ for ϕ, ϵ, χ. Measured fixed range-five source fractions are approximately $.985, .962, .969$. These numbers do not prove that the continuum fields are impure or that genuine descendants have been removed. They show why finite source purity cannot be turned into a percent-level continuum OPE error merely by subtracting from one.

21.3.2 Excited sources have a different descendant coefficient

There is a related but different calculation for $O_0|\phi\rangle$. In this calculation O_0 is the unit-circle average of the normalized continuum primary. A microscopic source must first have its common vacuum-to-primary amplitude divided out before making this comparison. Suppose a scalar

primary β of weights (h,h) occurs in $\phi\times\phi$, with coefficient C_β. For equal external weights, the chiral global OPE coefficient multiplying $L_{-1}^n\beta$ is $(h)_n/[n!(2h)_n]$. After normalizing left and right descendants, the scalar amplitude is

$$C_\beta\frac{(h)_n^2}{n!(2h)_n}.$$

Consequently

$$\|O_0|\phi\rangle\|^2\geq\sum_{\beta\ \text{scalar}}|C_\beta|^2\sum_{n\geq0}\frac{(h_\beta)_n^4}{(n!)^2(2h_\beta)_n^2}. \tag{21.6}$$

Other Virasoro directions and higher-spin primary families can add positive weight. The first scalar daughter amplitude here is $C_\beta\Delta_\beta/4$, whereas the vacuum-source amplitude above was Δ. Mixing these two external-state geometries produces a factor-of-four error.

If the identity is allowed in the channel, its global family contributes only the $n=0$ term. One should not substitute $h=0$ into the displayed ratio of Pochhammer symbols for $n>0$: the vacuum's translation descendants vanish. Other vacuum Virasoro directions, such as $T\bar{T}$, are separate positive contributions.

21.4 Ward vectors, phases, and nonorthogonal candidate spans

A candidate daughter should satisfy energy, momentum, norm, and matrix element tests together. A lattice approximation to a Virasoro mode can produce a Ward vector S. One may project it onto a fixed collection of residual-checked low states, but those states need not be exactly orthogonal after numerical reconstruction.

Let the columns of Y be the retained states, let $G=Y^\dagger Y$, and put $B=Y^\dagger S$. For positive invertible G, the actual orthogonal projector is

$$\Pi_Y=YG^{-1}Y^\dagger,\qquad\|\Pi_Y S\|^2=B^\dagger G^{-1}B.$$

If $A=Y^\dagger O|0\rangle$, the overlap with the coherently phased normalized Ward projection is

$$\frac{B^\dagger G^{-1}A}{\sqrt{B^\dagger G^{-1}B}\,\langle\phi|O|0\rangle}.$$

Independent rephasings of the numerical eigenvectors cancel in this formula. Replacing the complex amplitudes by their absolute values before forming the sum destroys that coherent test.

In an earlier calculation, a two-candidate span at $L=15$ has about .9870 of a conditional ideal daughter squared norm, yet only .2418 of the complete Ward-source squared norm. Its bare-source vacuum ratios have real parts near .098, compared with the conditional primary dimension .1464; trained range-five ratios are smaller. The attractive norm agreement therefore does not settle the family assignment. An energy near "primary plus two" is only one of several necessary checks.

Worked example 21.5 (A local improvement preserves an equal-energy numerator)**.** For a fixed local operator O and real t, define

$$O_t=O+t[H,[H,O]].$$

Between exact energy eigenstates,

$$\langle n|O_t|m\rangle=\big[1+t(E_n-E_m)^2\big]\langle n|O|m\rangle.$$

If ϕ and $\phi^\dagger$ are exactly degenerate, the cubic numerator in (21.1) is unchanged. With primary and daughter gaps g_0,g_1, their vacuum ratio changes by $(1+tg_1^2)/(1+tg_0^2)$. Choosing $t=-g_1(L_\star)^{-2}$

kills the daughter amplitude at one reference size while changing the primary denominator much less when $g_0 \ll g_1$.

The same fixed t must then be used at every size. If the gaps scale as L^{-1}, the improvement vanishes relatively as L^{-2} in the limit. Retuning t with L would be a different operator sequence. This example does not assert that the actual diagonal word recipes have this commutator form; it exhibits a concrete possible mechanism.

21.5 Heat filtering and a conditional omitted-tail bound

A positive-time filter reduces sensitivity to short-distance source weight. For exact finite eigenstates define

$$F_L(\tau) = \frac{\|e^{-\tau(H-E_\phi)}O|0\rangle\|^2}{|A_\phi|^2} = \sum_j \left|\frac{A_j}{A_\phi}\right|^2 e^{-2\tau(E_j-E_\phi)}.$$

The candidate primary contributes one. In an ideal CFT, with $\omega_L = 2\pi v/L$, the scalar descendants lie $2n\omega_L$ above it, giving

$$F_{\mathrm{CFT}}(\tau) = {}_2F_1(\Delta,\Delta;1;q), \qquad q = e^{-4\tau\omega_L}. \tag{21.7}$$

Keeping $\tau\omega_L$ fixed as L grows suppresses microscopic energies while preserving nontrivial descendant weights. Keeping physical τ fixed sends $q \to 1$ instead.

Suppose a set J of exact orthonormal eigenstates is retained and every omitted source-coupled state has energy at least E_{cut}. Positivity then proves

$$0 \le F_L(\tau) - F_J(\tau) \le e^{-2\tau(E_{\mathrm{cut}}-E_\phi)}\left[F_L(0) - \sum_{j\in J}|A_j/A_\phi|^2\right].$$

The full unfiltered norm is essential. So is the true omitted-energy cutoff: the largest energy among a few accurate Ritz vectors is not automatically such a cutoff. This distinction separates a rigorous positive bound from a useful conditional spectral diagnostic.

21.6 Sewing two circles into one

Let $L = L_1 + L_2$. In a common height embedding, define the literal wavefunction overlap

$$A_{abc}(L_1, L_2) = \langle c, L|(|a, L_1\rangle \otimes |b, L_2\rangle).$$

The outgoing parent is a bra. Seams, orientation, fusion-vertex gauge, and normalization of all three states must be held fixed. Exact microscopic gluing is a well-defined finite operation even when its continuum description has not been established.

21.6.1 What the finite joining map actually does

The tensor product in the overlap formula is taken in the ordinary height embedding of Chapter 19. It does not mean that two closed fusion paths automatically form a closed path of the combined length. Write

$$\iota_L : \mathcal{H}_{\mathrm{path},L} \longrightarrow (\mathbb{C}^6)^{\otimes L}$$

for the isometric embedding that sends each admissible path to its height word. Its adjoint deletes the forbidden-word components. After identifying the two ambient tensor products with the ambient length-L space, the actual joining map is

$$J_{L_1,L_2} = \iota_L^\dagger(\iota_{L_1} \otimes \iota_{L_2}), \qquad L = L_1 + L_2. \tag{21.8}$$

Thus

$$A_{abc} = \langle c, L | J_{L_1,L_2}(|a, L_1\rangle \otimes |b, L_2\rangle).$$

The children and parent are individually normalized before this contraction.

To see the seam constraint, let u and v be allowed child words. The concatenated word retains all their internal edges, but replaces their two closing edges by

$$u_{L_1-1} \longrightarrow v_0, \qquad v_{L_2-1} \longrightarrow u_0.$$

The image under J is the concatenated parent basis vector if both new edges are allowed, and zero otherwise. For example, the length-three words (I_0, X_0, X_0) and (I_1, X_1, X_1) are each valid closed paths. Their concatenation fails both new seam constraints: $X_0\rho$ does not contain I_1, and $X_1\rho$ does not contain I_0. Its image is zero. Concatenating (I_0, X_0, X_0) with itself instead gives a valid unit-norm parent word.

Distinct surviving child pairs give distinct parent words. Hence $J^\dagger J$ is the orthogonal projector onto the child pairs that survive the two seam tests; $JJ^\dagger$ projects onto the parent words whose two cut blocks separately close. In particular J is a partial isometry: its nonzero singular values are one, and

$$\|J\| \le 1, \qquad |A_{abc}| \le \big\|J(|a\rangle \otimes |b\rangle)\big\| \le 1$$

for normalized states. One should not renormalize the projected joined vector separately for each pair of children. That would change the multilinear overlap tensor being compared with the pants amplitude.

This elementary contraction property is also the source of the error bound used later. If u, v and $\widetilde{u}, \widetilde{v}$ are normalized child states with respective norm differences δ_1, δ_2, then

$$\|u \otimes v - \widetilde{u} \otimes \widetilde{v}\| \le \delta_1 + \delta_2.$$

Contracting with J and a fixed unit parent cannot increase this bound. Dividing both amplitudes by the same nonzero vacuum overlap therefore gives a difference at most $(\delta_1 + \delta_2)/|A_{000}|$. A small vacuum overlap can amplify an otherwise small absolute state error.

Categorical cups use a different normalization. There is no extra quantum-dimension factor in (21.8): its height basis is orthonormal. Square roots of quantum dimensions do occur when a graphical categorical contraction is converted to normalized states. For a simple object x, a standard unitary coevaluation $R_x : \mathbf{1} \to x^* \otimes x$ obeys

$$R_x^\dagger R_x = d_x \, 1_{\mathbf{1}}.$$

The cup vector of norm one is consequently $d_x^{-1/2} R_x$. For $x = \rho$, this uses the positive dimension $d = (3 + \sqrt{13})/2$. The unnormalized closed cup–cap diagram evaluates to d_x, while the norm of its normalized cup is one. These statements concern categorical duality maps; they do not insert a universal $\sqrt{d_x}$ into a literal height overlap. A different graphical sewing must specify its duality maps and their conversion to the chosen orthonormal basis.

The same distinction applies to gauge and phase choices. Under unitary changes of path coordinates U_L, representing the same finite map requires

$$J \longmapsto U_L J (U_{L_1}^\dagger \otimes U_{L_2}^\dagger).$$

Transforming the states while silently keeping the old numerical joining map need not preserve the overlap. Independent rephasings of the three states give $A_{abc} \mapsto e^{i\theta_a + i\theta_b - i\theta_c} A_{abc}$ because the parent is a bra. Coherent phase conventions, or the invariant combinations already constructed, are therefore needed for complex comparisons.

In the scaling picture, Euclidean time extends along three cylinders meeting at a pair of pants. Its conformal map to a punctured sphere relates ratios of overlaps to three-point functions. This geometric statement presumes that the chosen microscopic seams approach the specified continuum sewing.

21.6.2 The primary geometry factor

Put $x = L_1/L$ and define the positive entropy function

$$H(x) = -x \log x - (1-x)\log(1-x).$$

An explicit cylinder coordinate is

$$z(w) = \frac{L}{2\pi}\left[x \log(w-1) + (1-x)\log w + H(x)\right].$$

The incoming punctures are at $w = 1, 0$ and the outgoing one is at infinity. The derivative vanishes at $w = 1-x$; the constant $H(x)$ places this joining point at zero real cylinder time. Logarithm branches specify the seam phases.

The local disk coordinates of unit-normalized cylinder states are $\xi_1 = e^{2\pi z/L_1}$, $\xi_2 = e^{2\pi z/L_2}$, and $\xi_3 = e^{-2\pi z/L}$. Near the outgoing puncture use $u = 1/w$. The minus sign is essential: the incoming ends have $\Re z \to -\infty$, while the outgoing end has $\Re z \to +\infty$; all three local coordinates must vanish at their respective punctures. Normalizing states by these disk coordinates already includes the cylinder-to-plane state factors; one must not insert a second factor $(2\pi/L_i)^{\Delta_i}$. Their inverse Jacobians have magnitudes

$$\left|\frac{dw}{d\xi_1}\right| = e^{-H/x}, \qquad \left|\frac{dw}{d\xi_2}\right| = e^{-H/(1-x)}, \qquad \left|\frac{du}{d\xi_3}\right| = e^{H}.$$

A scalar primary of dimension Δ contributes the Δth power of the relevant magnitude. Dividing by the nonzero vacuum overlap with exactly the same lengths and seams cancels the overall conformal anomaly and its gluing normalization. Different vacuum shapes require a further shape factor, as in the current test below. The ideal CFT prediction for three scalar primaries in these coordinates, to which a correctly calibrated lattice ratio should converge, is

$$\frac{A_{abc}}{A_{000}} = e^{-H(x)[\Delta_a/x + \Delta_b/(1-x) - \Delta_c]} C_{ab}^{\ c}. \tag{21.9}$$

The coefficient is the plane correlator with the outgoing conjugate field at infinity. For nonzero spins the same calculation retains the holomorphic and antiholomorphic Jacobians separately; their coordinate phases must be fixed before comparing complex amplitudes.

At equal halves the factor becomes $2^{-2\Delta_a - 2\Delta_b + \Delta_c}$. For a normalized conjugate pair with outgoing vacuum, $C_{a\bar{a}}^{\ 0} = 1$, so the ratio is

$$\exp\left[-\frac{\Delta_a H(x)}{x(1-x)}\right].$$

This is the geometric function used in the later current test; for spinful conjugate fields it gives the magnitude, with the coordinate phase retained separately. At equal halves and dimension one its value is $1/16$. The exponential rescaling supplies an OPE estimator only if the lattice states actually converge to those primaries.

21.6.3 The vacuum size dependence

For the normalizable identity vacuum in this periodic pants geometry, the conformal-anomaly calculation gives

$$A_{000}(xL, (1-x)L) \sim C(x) L^{-c/8}, \qquad 0 < x < 1 \quad \text{fixed}.$$

The microscopic seam must approach the specified sewing, with no additional leading scaling field at the junction. The local twofold branch has scaling dimension $\Delta_2 = c/8$; Section 22.6 derives

that dimension from the Schwarzian. Applying it to the normalized three-cylinder geometry is the geometric input here. The squared overlap consequently has exponent $c/4$, and the logarithmic bipartite fidelity is $-\log|A_{000}|^2 = (c/4)\log L - 2\log|C(x)| + \cdots$. The primary ratios above had canceled this vacuum anomaly and could not determine it by themselves.

At two sizes with the same split and seam, the constant cancels:

$$c^{(\mathrm{vac})}_{L,L'} = -8\frac{\log|A_{000}(xL',(1-x)L')/A_{000}(xL,(1-x)L)|}{\log(L'/L)}.$$

This is a velocity-free finite-size diagnostic. Corrections to the leading geometric law remain in the ratio.

The saved equal-split vacuum overlap is .4306101916422253 for $9+9 \to 18$, compared with the earlier $6+6 \to 12$ value .47886168385663636. Their two-size exponent gives $c^{(\mathrm{vac})}_{12,18} = 2.0955399664$. This is an informative geometric observable; it does not control conical, bulk, or logarithmic corrections.

21.6.4 A stress overlap from the Schwarzian

Stress is not a primary, and this gives a particularly simple pants observable. With the same x and $H(x)$, put

$$B(x) = x(1-x)e^{2H(x)}.$$

For one normalized chiral stress state, the ideal pants map gives

$$\left|\frac{A_{00T}}{A_{000}}\right| = B(x)\sqrt{c/8}. \tag{21.10}$$

For example, an outgoing local coordinate can be written $\xi = e^{-H(x)}u(1-u)^{-x}$. Reversing its power series gives

$$u = e^H\xi - xe^{2H}\xi^2 + \frac{3x^2-x}{2}e^{3H}\xi^3 + O(\xi^4).$$

Thus $u'(0) = e^H$, $u''(0) = -2xe^{2H}$, and $u'''(0) = 3(3x^2-x)e^{3H}$. The Schwarzian $u'''/u' - \frac{3}{2}(u''/u')^2$ is $\{u,\xi\}_{\xi=0} = -3x(1-x)e^{2H(x)}$. Inserting the stress transformation law $T(\xi) = (du/d\xi)^2T(u) + (c/12)\{u,\xi\}$ and dividing by $\sqrt{c/2}$ gives the signed ratio $-B(x)\sqrt{c/8}$ in this coordinate convention, hence (21.10). A rephased stress state can change the sign; the modulus is invariant. At equal halves $B = 1$.

The value comes from the Schwarzian vacuum term, not from a primary coefficient $C(1,1,T)$. This explains why it survives even though an ordinary plane one-point function of stress vanishes. Agreement with the energy-Fourier estimator would be a useful common-stress test; it is not guaranteed at accessible sizes.

21.7 Connected overlap invariants

All eight overlaps with either vacuum or one selected field on each leg contain more information than a single three-field amplitude. Normalize by A_{000}, call the resulting quantities R, and set

$$\begin{aligned} m_1 &= R_{100}, \quad m_2 = R_{010}, \quad m_3 = R_{001}, \\ B_{12} &= R_{110} - m_1m_2, \quad B_{13} = R_{101} - m_1m_3, \quad B_{23} = R_{011} - m_2m_3, \\ K &= R_{111} - m_1R_{011} - m_2R_{101} - m_3R_{110} + 2m_1m_2m_3. \end{aligned}$$

These are the algebraic analogues of connected moments. Under an independent affine change of external field on each leg, $X_i \mapsto u_iX_i + v_i1$, direct expansion gives

$$B_{ij} \mapsto u_iu_jB_{ij}, \qquad K \mapsto u_1u_2u_3K.$$

Thus $K^2/(B_{12}B_{13}B_{23})$ is unchanged.

For identical normalized Hermitian scalar primaries in an ideal pants geometry, the external coordinate factors cancel between the triple coefficient and the square root of the three pair coefficients. With consistent real signs,

$$\frac{K}{\sqrt{B_{12}B_{13}B_{23}}} = C_{XXX}.$$

The pair factors must be nonzero. For a charged order-three field, C_{XXX} can be nonzero while $\langle XX\rangle = 0$; the identical-label quotient is then undefined. One must use an appropriate conjugate-label construction instead. For stress, its two- and three-point normalizations instead give

$$c = \frac{8B_{12}B_{13}B_{23}}{K^2}. \tag{21.11}$$

The complex products, rather than products of absolute values, are required. A wrong sign can turn an inconsistent finite tensor into an apparently positive central charge.

Affine cancellation has a definite scope. It removes identity mixing, leg rescalings, and the affine terms generated by stress coordinate changes. A physical nonidentity insertion at the branch point produces a four-point correlator with a connected part that need not be fixed by the lower moments. The scalar invariant therefore does not cancel every conical correction.

21.7.1 A stronger cancellation for three stresses

Stress has additional Ward identities that a general scalar lacks. Assume one common local branch expansion is used in all eight overlaps, with exact continuum vacuum or holomorphic stress on each external leg. A cover field of dimension w contributes with relative size $L^{-w/2}$ at a twofold branch. Correlators of any nonvacuum Virasoro primary with only external stresses vanish: Ward recursion reduces them to the primary's zero one-point function. The same holds for its descendants. Only the vacuum branch family can contribute.

An extra branch stress $T(b)$ acts on a plane stress correlator by

$$\sum_i \left[\frac{1}{b-z_i}\partial_{z_i} + \frac{2}{(b-z_i)^2} + \frac{c}{2(b-z_i)^4}\mathcal{R}_i\right],$$

where $\mathcal{R}_i$ removes the ith stress from that moment. These are respectively an infinitesimal displacement of a puncture, a leg rescaling, and an affine leg shift. The connected invariant (21.11) is invariant under all three. Its linear response to the branch stress therefore vanishes; differentiating in b gives the same result for every $\partial^n T(b)$.

The first new vacuum quasiprimary is $\Lambda =: TT : -(3/10)\partial^2 T$, with squared norm $N_\Lambda = c(5c+22)/10$. Choose plane punctures $0, 1, \infty$, where the unperturbed connected pairs are $c/2$ and the triple is c. Ward identities give, for a branch $\Lambda(b)$,

$$\delta B_{01} = \frac{N_\Lambda}{b^4(b-1)^4}, \qquad \delta B_{0\infty} = \frac{N_\Lambda}{b^4}, \qquad \delta B_{1\infty} = \frac{N_\Lambda}{(b-1)^4},$$
$$\delta K = \frac{2N_\Lambda(b^2-b+1)^2}{b^4(b-1)^4}.$$

In $\delta \log c = \sum \delta B_{ij}/B_{ij} - 2\delta K/K$, these cancel by the elementary identity

$$1 + (b-1)^4 + b^4 = 2(b^2-b+1)^2.$$

Thus even the linear weight-four correction, of nominal order L^{-2}, disappears. Differentiating also removes $\partial\Lambda$, of nominal order $L^{-5/2}$.

Because the invariant is nonlinear, this does not yet control quadratic terms. For the deliberately incomplete branch state $1 + tT(b)$, put $q = b^{-2}$, $r = (b-1)^{-2}$. Direct substitution of the stress Ward moments shows that the quadratic term in $\log c$ is

$$(c+2)\left[-\tfrac{1}{2}(q^2 + r^2 + q^2r^2) + qr(q + r + 1)\right] = 0.$$

The same polynomial identity proves the equality. Differentiating in the common branch location removes the mixed nonlinear $T, \partial T$ term. Vacuum fields through weight five are spanned by the derivatives just considered and Λ; the only nonlinear combinations below total weight six are T^2 and $T\partial T$. Hence every local branch correction through order $L^{-5/2}$ cancels under the stated hypotheses.

Order L^{-3} can survive. For example, the same bare branch at $b = 1/2$, with $c = 2$, gives $c_{\text{connected}}(t) = 2 + 6144t^3 + O(t^4)$. This is a calculation of the allowed local branch contribution, not an error estimate for lattice data. Bulk irrelevant fields, external-state contamination, and independently regulated branch points are not removed by it. In particular it does not justify fitting every finite three-stress estimator to $c + aL^{-3}$.

Numerical record 21.6 (A useful disagreement between observables)**.** The scalar connected overlap invariant is 1.12049321 at $6 + 6$ and 1.12397612 at $9 + 9$. Meanwhile the $L = 18$ fixed range-five local epsilon sources give absolute self cubics approximately .93242–.93755. Both finite diagnostics are fairly stable, but they disagree substantially if both are read as the same continuum OPE coefficient. This is evidence that observable-dependent corrections remain, not an invitation to average the two numbers.

The finite stress tensor supplies an even sharper consistency check. For equal halves, define the raw pair ratios

$$K_{13} = \frac{A_{T0T}A_{000}}{A_{T00}A_{00T}}, \qquad K_{12} = \frac{A_{TT0}A_{000}}{A_{T00}A_{0T0}}.$$

The ideal Ward values are $K_{13} = 1 - 16/c$ and $K_{12} = 1 + 2/c$. A saved $6 + 6$ input-output ratio gives the encouraging effective $c = 2.07127$. But its input-input ratio is $K_{12} = -4.21244$, which has the wrong sign relative to the positive-charge Ward prediction. The all-eight connected formula returns -6.43646 there and -43.18719 at $6 + 9$. These are failures of the selected finite tensor to obey the continuum Ward relations, not negative physical central charges. A separate Ising control gives .49938 and .49955 at nearby sizes, approaching its known $c = 1/2$.

Even a correction hypothesis can be overdetermined. In the equal-split model with only a branch insertion $1 + tT(1/2)$, retain the local-coordinate Schwarzian shifts:

$$X_0 = T(0) + c/2, \qquad X_1 = T(1) + c/2, \qquad X_\infty = T(\infty) - c/16.$$

Ward identities give

$$\langle X_0X_1X_\infty\rangle_t = \frac{c(64 + 30c - c^2)}{64}(1 + 24t).$$

For $c > 0$, $c \neq 32$, the nearly zero raw three-stress overlap would require $t = -1/24$. At that value, $K_{12} = -3 - 6/c$ and $K_{13} = 3/2 - 24/c$, hence

$$K_{13} = 4K_{12} + 27/2.$$

The measured K_{12} predicts about -3.34974, whereas the measured K_{13} is about -6.7247. One arbitrary stress-branch coefficient cannot reconcile the three measurements. Higher branch fields, bulk corrections, and external-state errors remain separate possibilities. The lesson is to test a correction model against more observables than it has adjustable parameters.

21.8 A current test independent of the mixing angle

A state with spin one need not be a holomorphic current. A current has $(h,\bar h)=(1,0)$ and must satisfy both chirality and normalized gluing tests. A conjugate holomorphic current has the same spin; conjugating its internal charge does not reverse its chirality. Pairing opposite momenta can therefore test the wrong overlap.

For normalized conjugate primaries of dimension Δ, the identity OPE coefficient is one. At split x, the ideal pants amplitude normalized by the same-shape vacuum overlap is

$$G(x;\Delta)=\exp\!\left[-\frac{\Delta H(x)}{x(1-x)}\right].$$

For a current and equal halves this is $1/16$. If the vacuum denominator is instead taken at a different shape y, the leading vacuum shape factor must also be included:

$$G(x;1)\exp[-\tfrac{c}{2}(f(x)-f(y))],\qquad f(x)=\frac{F(x)+F(1-x)}{12},$$

where $F(x)=(3-3x+2x^2)\log x/(1-x)$. Both children must grow when using this expression asymptotically. Keeping one child small cannot be repaired merely by inserting the correct leading shape factor.

To allow mixing, take orthonormal bases u_i, v_j of two proposed current subspaces and form

$$M_{ij}=\frac{\langle 0,L|(|u_i,L_1\rangle\otimes|v_j,L_2\rangle)}{A_{000}(L,y)}.$$

Unitary basis changes give $M\mapsto U^TMV$. Its singular values are invariant under state phases and arbitrary unitary mixing. If the current and its conjugate both converge inside the retained subspaces, the splits approach fixed fractions $x,y\in(0,1)$, and the gluing converges as assumed, then

$$\liminf_{L\to\infty}\|M_L\|_2\ge G(x;1)\exp[-\tfrac{c}{2}(f(x)-f(y))].$$

For asymptotically equal splits and a vacuum denominator of the same shape, this reduces to the bound $1/16$. This is simply the variational definition of the largest singular value: the amplitude of any normalized pair in the two spaces is at most $\|M\|_2$. Enlarging either space can increase this norm; it does not show that all added states are currents.

The stored $7+8\to15$ test compares with the fixed-point geometric value .06162. Its largest singular values for respectively $1,2,3,4,6$ states per child are

$$.01034,\quad .01977,\quad .03395,\quad .05437,\quad .05584.$$

Removing the projected lowest-scalar level-one descendant reduces the six-state value to about .03454. These finite deficits disfavor a current hidden in the specified small subspaces, but finite corrections and missing subspace weight are uncontrolled. At a smaller parent length the six-state norm exceeds the current reference. There is no monotonic no-current theorem in these data.

A norm inequality makes the missing information explicit. Suppose normalized current approximants lie within distances δ_1,δ_2 of normalized vectors in the selected spaces, and their full normalized gluing differs from the ideal value by at most η. Literal gluing against a unit parent is a contraction, so

$$\|M\|_2\ge G_{\mathrm{ideal}}-\eta-\frac{\delta_1+\delta_2}{|A_{000}(L,y)|}.$$

A parent-vector error adds another analogous term. A rigorous finite exclusion needs these errors; high state fidelity on a single short ring does not supply their asymptotic bounds.

Finally, for an actual spin-one primary, the energy-mode convention $H_{+1} \to \omega_L(L_{-1} + \bar{L}_1)$ gives

$$\|H_{-1}|J\rangle\|^2 = \omega_L^2(\Delta_J - 1), \qquad \|H_{+1}|J\rangle\|^2 = \omega_L^2(\Delta_J + 1).$$

Thus a current has vanishing lowering norm. A nonprimary has extra positive lowering contributions, and an unfiltered lattice norm can also contain ultraviolet weight. A projected low-energy Ward test, enlarged until stable, is the appropriate comparison to the continuum identity.

21.9 Perturbing the Hamiltonian: a velocity-free response test

Take $H(\theta) = H + \theta V$ and a differentiable normalized eigenbranch. The Hellmann–Feynman identity gives

$$\frac{dE_a}{d\theta} = \langle a|V|a\rangle.$$

At an exact degeneracy one first diagonalizes V in that eigenspace. For a gap $g_a = E_a - E_0$, define $d_a = \langle a|V|a\rangle - \langle 0|V|0\rangle$. Then

$$\frac{d}{d\theta}\left(\frac{g_a}{g_b}\right) = \frac{d_a g_b - g_a d_b}{g_b^2}. \tag{21.12}$$

Adding an identity to V cancels in d_a. Adding bH changes d_a by bg_a and cancels in (21.12). The response of a gap ratio is therefore insensitive to energy shifts and overall velocity rescalings.

For $V = +\sum_j P_j^1$, the completed $L = 18$ response in coordinates $(g_\chi/g_\phi, g_\epsilon/g_\phi)$ is approximately $(.37336, .00731)$, whereas the 15-to-18 logarithmic-size secant is $(.03634, .01916)$. They are not parallel. Thus those two finite drifts are not already described by a pure change of this one bare coupling with every other correction neglected. It would be stronger than the evidence to infer two infrared running couplings: the bare operator can mix several fields and need not coincide with the direction controlling size drift.

Similarly, extracting the sign of a marginal beta function from a finite scalar self-matrix element requires an identified dimension-two primary, a normalized scalar source, and controlled contamination by stress and other fields. Persistent finite signs can guide a theory; they cannot supply these missing premises.

21.10 Exercises

Exercise 21.1. Verify all three phase invariants in (21.2) and the relation for J^3. Show that a symmetry conjugating all three fields in I_λ forces that invariant to be real even when it attaches independent unit phases to the fields.

Solution on page 312.

Exercise 21.2. Prove by induction that $\|L_{-1}^n|h\rangle\|^2 = n!(2h)_n$. Use this to derive the first three terms of (21.3). Explain why the threshold $\Delta = 1/2$ is also visible from the equal-time two-point function.

Solution on page 312.

Exercise 21.3. Expand $X_i \mapsto u_i X_i + v_i 1$ in the definitions of B_{ij} and K and prove their stated transformation laws. Identify which assumption would fail if a fourth physical operator were inserted at the sewing branch point.

Solution on page 312.

Exercise 21.4. Prove the omitted-tail inequality for $F_L(\tau)$. Construct a three-level example showing why small residuals of two retained eigenvectors do not provide an energy cutoff for the omitted state.

Solution on page 313.

Exercise 21.5. Derive (21.12). For two gap ratios collected in a vector R, show why changing the normalization of θ rescales $\partial_\theta R$ without rotating it. What further hypothesis is needed to compare this vector with $\partial_{\log L} R$?

Solution on page 313.

Exercise 21.6. Show that unitary changes of the two incoming state bases act on the gluing matrix as $M \mapsto U^T M V$ and preserve its singular values. Prove the displayed finite error inequality from the contraction property of gluing. Why does the inequality become weaker when the vacuum overlap is small?

Solution on page 313.

Exercise 21.7. Verify $1 + (b-1)^4 + b^4 = 2(b^2 - b + 1)^2$ and use it to cancel the linear Λ correction in the connected stress invariant. For $q = b^{-2}$, $r = (b-1)^{-2}$, show that it also implies $q^2 + r^2 + q^2 r^2 = 2qr(q + r + 1)$. Explain why two distinct branch locations are not covered by differentiating this one-location identity.

Solution on page 313.

Chapter 22

Tensor networks, entanglement, and controlled computation

Exact diagonalization exposes the finite spectrum directly, but its state space grows exponentially. Matrix product states trade the full wavefunction for a structured variational family whose cost is governed by a bond dimension. To use them for the Haagerup problem, we must preserve fusion-path constraints and distinguish physical correlation lengths from symmetry-cat effects. This chapter introduces the essential tensor algebra, derives the transfer-matrix and entropy diagnostics, and explains what a converged finite-bond calculation can establish. The final section treats resource estimates as part of the mathematical design of an algorithm.

22.1 Compressing a wavefunction by successive Schmidt decompositions

For a bipartite pure state,

$$|\Psi\rangle = \sum_{a=1}^{D} \sqrt{p_a}\, |a\rangle_L |a\rangle_R, \qquad p_a \geq 0, \quad \sum_a p_a = 1.$$

The Schmidt rank counts the independent correlations crossing the cut. To obtain this decomposition, write the wavefunction coefficients in product bases as a matrix M_{ij}. Its singular-value decomposition $M = U\Sigma V^\dagger$ gives orthonormal left and right vectors and nonnegative singular values $\Sigma_{aa} = \sqrt{p_a}$. The right Schmidt vector has coefficients $\overline{V_{ja}}$ in this convention. Tracing over either side leaves eigenvalues p_a; the eigenvectors on the two sides need not resemble the original product bases.

Successive Schmidt decompositions along an open chain produce a matrix product representation

$$\Psi(s_1, \ldots, s_L) = A_1^{s_1} A_2^{s_2} \cdots A_L^{s_L},$$

with boundary vectors understood. Every finite state admits such a representation, but the exact bond dimensions can grow exponentially. An MPS approximation restricts these dimensions.

If one truncates a single Schmidt decomposition to the largest D probabilities and renormalizes, the squared overlap with the original state is $\sum_{a \leq D} p_a$. This is an exact one-cut statement. A long variational calculation involves many cuts, changing bases, and an optimization over tensors; a discarded weight reported at one update is not automatically the global ground-state error.

The one-cut truncation is optimal for a precise reason. Any normalized state of Schmidt rank at most D lies in $W \otimes \mathcal{H}_R$ for some left subspace W of dimension at most D. If P_W is its projector, Cauchy–Schwarz bounds its overlap with Ψ by $\|(P_W \otimes 1)\Psi\|$. The square of this norm

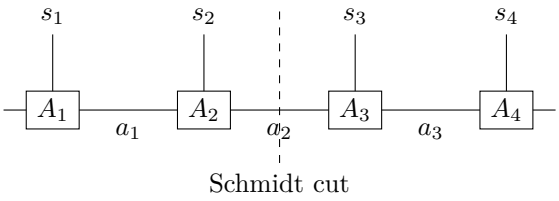


Figure 22.1: Physical indices point upward; horizontal virtual indices are contracted. The virtual dimension at a cut bounds its Schmidt rank. In a constrained edge MPS the virtual space is further organized by the fusion height.

is $\mathrm{Tr}(P_W \rho_L)$, maximized by the D largest eigenvalues of ρ_L. The normalized truncated Schmidt state attains the bound. Writing $\varepsilon = \sum_{a>D} p_a$, its fidelity is $1-\varepsilon$ and its phase-aligned squared vector error is $2-2\sqrt{1-\varepsilon}$. These are different error conventions.

Worked example 22.1 (A bond restriction changes the variational minimum)**.** For two qubits, let $|\Phi\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ and $H = -|\Phi\rangle\langle\Phi|$. Both Schmidt probabilities are $1/2$. Every normalized product state therefore has squared overlap at most $1/2$ with Φ, so its energy is at least $-1/2$. A bond-two state represents Φ exactly and reaches the true ground energy -1. A perfectly converged bond-one optimization would still miss half the binding energy. Convergence within a variational family and accuracy of that family are separate issues.

For a uniform periodic MPS,

$$\Psi(s_1, \ldots, s_L) = \mathrm{Tr}(A^{s_1} \cdots A^{s_L}), \tag{22.1}$$

the same tensor is used at every site. The trace enforces cyclic invariance of the amplitude. It does not require the state to be a unique ground state or to have finite correlation length.

There is a gauge freedom $A^s \mapsto X^{-1} A^s X$, which telescopes inside the trace. Overall rescaling changes the unnormalized state by a length-dependent scalar and leaves normalized expectation values unchanged. Optimizing raw coordinates without accounting for these directions can create ill-conditioned Hessians and misleading gradient norms.

22.2 An exact fusion-compatible ansatz

The path basis suggests a more economical object. Assign a virtual space V_a of dimension D_a to each height a, and a matrix

$$W_{ab} : V_b \longrightarrow V_a$$

to every allowed edge $a \to b$. Set $W_{ab} = 0$ on forbidden edges. The periodic amplitude is

$$\Psi(x_0, \ldots, x_{L-1}) = \mathrm{Tr}\big(W_{x_0 x_1} W_{x_1 x_2} \cdots W_{x_{L-1} x_0}\big). \tag{22.2}$$

Every nonzero amplitude is an admissible fusion path. No penalty term is needed to enforce the constraint. Heightwise gauges $W_{ab} \mapsto X_a^{-1} W_{ab} X_b$ again telescope.

The ansatz can be embedded in an ordinary MPS on the six-state height space. On $V = \bigoplus_a V_a$, define A^a to have only its ath block row nonzero, with (a,b) block W_{ab}. A product of such tensors forces the intermediate block indices to match the physical height word, and its trace is exactly (22.2). This bridge allows ordinary MPS results to be applied after the relevant support and redundant virtual directions are understood.

One can instead work in the full height-spin tensor product and add a positive energy penalty for forbidden edges. That is another legitimate variational scheme, but constraint leakage becomes

an observable that must be monitored. Small energy changes do not guarantee small leakage. For a constrained Hamiltonian, an exactly admissible ansatz has a particularly transparent variational interpretation.

Worked example 22.2 (Why a product of edge weights is already correlated)**.** Take $D_a = 1$, so every allowed W_{ab} is a scalar. The amplitude is then a product of nearest-neighbor factors, but it is generally not an ordinary product state: each physical height occurs in two neighboring factors. Its squared norm is the partition function of a one-dimensional classical transfer matrix $K_{ab} = |W_{ab}|^2$. This is the simplest nontrivial constrained MPS and will reappear in the exact module construction of Chapter 23.

22.3 What one variational update actually solves

The variational principle says that a normalized admissible state has energy at least the true ground energy. It does not say that the nonlinear energy function of all MPS tensors has a single minimum. A common strategy holds every tensor except one fixed. Collect that tensor's allowed entries into a vector a. The many-body state then depends linearly on a:

$$|\Psi(a)\rangle = \sum_{\mu} a_\mu |v_\mu\rangle.$$

The environment vectors v_μ need not be orthogonal or linearly independent. Their Gram and Hamiltonian matrices are

$$N_{\mu\nu} = \langle v_\mu | v_\nu \rangle, \qquad H^{\mathrm{eff}}_{\mu\nu} = \langle v_\mu | H | v_\nu \rangle.$$

Consequently the energy is a generalized Rayleigh quotient,

$$E(a) = \frac{a^\dagger H^{\mathrm{eff}} a}{a^\dagger N a}.$$

Varying with respect to $\overline{a}$ at fixed norm gives

$$H^{\mathrm{eff}} a = E N a. \tag{22.3}$$

The lowest solution minimizes the energy in this linear update space. If N has a kernel, those coordinates represent the zero physical state and must first be removed. On its positive support, $b = N^{1/2} a$ converts the problem to the ordinary Hermitian matrix $N^{-1/2} H^{\mathrm{eff}} N^{-1/2}$.

Worked example 22.3 (Why the norm environment cannot be dropped)**.** Use the preceding two-qubit Hamiltonian and the nonorthogonal update basis $v_1 = |00\rangle$, $v_2 = |\Phi\rangle$. Direct inner products give

$$N = \begin{pmatrix} 1 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1 \end{pmatrix}, \qquad H^{\mathrm{eff}} = -\begin{pmatrix} 1/2 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1 \end{pmatrix}.$$

The generalized eigenpair $a = (0,1)^T$, $E = -1$ is the exact ground state. The orthogonal physical state has coordinates $a = (\sqrt{2}, -1)^T$ and energy zero. Diagonalizing H^{eff} alone would instead give eigenvalues $-3/2$ and zero. The spurious value lies below the Hamiltonian's lowest eigenvalue: it is a coordinate-norm error, not an improved variational state.

On an open chain, successive Schmidt gauges can make the left and right environments orthonormal at the update site; then $N = 1$ on the retained support. Periodic and uniform formulations generally retain nontrivial environments or enforce their fixed-point equations. The edge ansatz additionally restricts updates to allowed height blocks, so every vector in the update space remains admissible.

The exact linear-update argument treats site tensors as independent variables. In a strictly uniform ansatz, changing the shared tensor changes every site simultaneously, so the global state is no longer linear in its entries. A uniform algorithm's local effective equation must be combined with consistent environments and the full uniform stationarity condition. It cannot be justified by silently treating all the other copies of the same variable as fixed.

A sweep repeats these local minimizations. Each exact unrestricted one-tensor update cannot raise the energy, but the final point can still be a local stationary point of the full nonlinear problem. A two-tensor update can change the intermediate Schmidt support; its subsequent truncation can raise the energy again. Neither a small local residual nor a small coordinate gradient bounds the missing directions outside the chosen bond dimensions. Independent initializations, increasing variational support, energy variance, and a stable set of physical observables answer different parts of that accuracy question.

The effective matrices in (22.3) need not be stored densely. Their action on a trial tensor can be contracted through environments. This is the link between the variational derivation and the resource discussion below: the mathematical Rayleigh quotient does not prescribe a dense allocation.

22.4 Transfer operators and correlation lengths

For the ordinary tensor (22.1), define the completely positive transfer map

$$\mathcal{E}(X) = \sum_s A^s X (A^s)^\dagger.$$

Completely positive means that $\mathrm{id}_{M_r(\mathbb{C})} \otimes \mathcal{E}$ preserves positive semidefinite matrices for every auxiliary dimension r. The displayed formula has this property because each summand acts as $Z \mapsto (1 \otimes A^s) Z (1 \otimes A^s)^\dagger$. A transfer map need not preserve the ordinary matrix trace. Suppose its leading eigenvalue is simple and positive. Rescale the tensors so this eigenvalue is one, and choose right and left fixed matrices $r, l \geq 0$ with

$$\mathcal{E}(r) = r, \qquad \mathcal{E}^*(l) = l, \qquad \mathrm{Tr}(lr) = 1.$$

An inserted physical operator O gives

$$\mathcal{E}_O(X) = \sum_{s,t} \langle s|O|t\rangle A^t X (A^s)^\dagger,$$

and its infinite-chain expectation is $\langle O\rangle = \mathrm{Tr}[l\mathcal{E}_O(r)]$. This specifies the stationary infinite-chain state associated with the chosen fixed metrics. A unique clustering limit of periodic rings additionally needs the primitive component discussed below; a simple leading eigenvalue alone does not exclude other peripheral roots.

For separated one-site insertions,

$$\langle O_0 P_n\rangle = (l|\mathcal{E}_O \mathcal{E}^{n-1} \mathcal{E}_P|r),$$

using Hilbert–Schmidt bra-ket notation. Subtracting the fixed-point projector $\mathcal{P} = |r)(l|$ removes the product of means. An eigenvalue $\lambda = |\lambda| e^{ik}$ below the unit circle contributes an exponentially decaying oscillation $|\lambda|^n e^{ikn}$. Its length and wave number are

$$\xi_\lambda = -\frac{1}{\log|\lambda|}, \qquad k_\lambda = \arg\lambda. \tag{22.4}$$

Polynomial factors can accompany Jordan blocks, but do not change the leading exponential rate.

The longest transfer length need not occur in every correlation function. A source can have zero overlap with that transfer eigenmode by symmetry. Thus a length extracted from a particular local observable and the largest length in a computed transfer sector need not coincide. A partial transfer eigensolve should be described as a spectrum among the retained modes, not as a complete periphery certificate.

22.4.1 Primitivity, periodicity, and cats

A primitive transfer map has an algebraically simple leading eigenvalue, no other eigenvalues of the same modulus, and strictly positive fixed metrics on its minimal support. Merely having one distinct peripheral eigenvalue is insufficient: the identity map on matrices has only the eigenvalue 1, with a large eigenspace. A primitive map describes a clustering pure phase in the usual injective MPS setting. Two different mechanisms can spoil the naive use of (22.4).

First, a direct sum of inequivalent injective tensors describes a superposition of distinct phases. The transfer has more than one stationary block. A symmetric cat of ordered phases has vanishing one-point order but nonzero long-distance order correlations. Second, a tensor can be irreducible but periodic, with peripheral roots of unity. A three-cycle produces roots $1, e^{2\pi i/3}, e^{-2\pi i/3}$. Neither phenomenon is evidence for a critical connected correlation length. After blocking p sites, a subleading blocked eigenvalue measures $\xi = -p/\log|\lambda_{\text{blocked}}|$ in original lattice sites.

Worked example 22.4 (A three-state cat)**.** Let $|o_j\rangle = |j\rangle^{\otimes L}$ for three orthonormal one-site states $|j\rangle$, with $L \geq 2$. These ordered product states are cycled by alpha. Let the local order operator M have values $m\omega^j$ on them, with $\omega = e^{2\pi i/3}$. In the normalized symmetric cat, $\langle M\rangle = 0$, but

$$\langle M_0^\dagger M_n\rangle = |m|^2$$

at every separation. The infinite apparent length measures mixing of ordered components. In any individual component the connected correlator is zero. A tiny finite-bond splitting of a cat transfer root can produce an enormous but physically inappropriate number from (22.4).

For the Haagerup variational sequence, primitive-component and intraphase transfer restrictions are therefore part of the observable definition. Raw near-unit roots cannot simply be fed into an entropy fit. Equivalent redundant tensor blocks are another possibility: they can multiply peripheral eigenvalues without representing different physical vacua. One must distinguish a minimal state decomposition from the displayed virtual dimension.

22.5 Schmidt probabilities from the fixed metrics

For an injective ordinary MPS in the thermodynamic limit, the nonzero Schmidt probabilities are the eigenvalues of

$$\sigma = \sqrt{l}\, r \sqrt{l}, \qquad \operatorname{Tr}\sigma = 1,$$

in a consistent virtual convention. A gauge transformation changes the metrics but not these probabilities. Replacing the expression by the diagonal entries of r, or by eigenvalues of a tensor block without its left metric, generally gives the wrong entropy.

The entropies are

$$S_1 = -\sum_a p_a \log p_a, \qquad S_n = \frac{1}{1-n}\log\sum_a p_a^n, \qquad S_\infty = -\log p_{\max}.$$

For a cat, one must specify whether the full cat entropy or the entropy of one primitive component is being used. For components with mutually orthogonal Schmidt supports on both sides of the cut

and classical weights w_j, the von Neumann entropy includes the mixing entropy $-\sum_j w_j \log w_j$ as well as the component entropies averaged with weights w_j. For general Rényi order the block-diagonal density matrix instead gives

$$S_n = \frac{1}{1-n} \log \left(\sum_j w_j^n \operatorname{Tr} \rho_j^n \right),$$

where ρ_j is the normalized reduced state of component j. The mixing contribution is real, but it is not the entanglement of a single clustering vacuum.

In a constrained height representation, a cut can retain coherent boundary-charge information. Pinching the reduced density matrix into height sectors, weighting those sectors by quantum dimensions, and taking the ordinary Schmidt spectrum are three different operations. The entropy convention must be fixed before comparing ranks or models.

22.6 Why entropy measures the central charge

The coefficient in an entropy fit comes from the stress-tensor anomaly. We can derive it using the Schwarzian transformation law of Chapter 2. Consider the vacuum of a unitary full CFT on the infinite line, with $c_L = c_R = c$, and an interval $A = [u, v]$ of length $\ell = v - u$. Keep a microscopic cutoff a so that its reduced density matrix is well defined. For an integer $n \geq 2$, multiplying n copies of this density matrix and tracing joins the copies cyclically along A. The resulting Euclidean surface has n sheets, with branch points at u and v.

Equivalently, use n copies of the CFT on one plane and insert fields $\mathcal{T}_n(u)$ and $\widetilde{\mathcal{T}}_n(v)$ that implement the cyclic permutation and its inverse. The stress tensor in this replica theory is the sum $T^{\mathrm{tot}} = \sum_{j=1}^n T_j$, with chiral central charge nc. The branch geometry is uniformized by

$$z(w) = \left(\frac{w-u}{w-v} \right)^{1/n}.$$

Different sheets choose different phases of this root. Those phases do not change its Schwarzian derivative. The vacuum stress in the z plane vanishes, so on each sheet

$$\langle T_j(w) \rangle = \frac{c}{12} \{z, w\}.$$

To evaluate it, write $g(w) = (w-u)/(w-v)$. A direct differentiation gives $\{g, w\} = 0$ and $\{g^{1/n}, g\} = (1-n^{-2})/(2g^2)$. The Schwarzian chain rule then gives

$$\langle T^{\mathrm{tot}}(w) \rangle = \frac{c}{24} \left(n - \frac{1}{n} \right) \frac{(u-v)^2}{(w-u)^2 (w-v)^2}.$$

On the other hand, the Ward identity for two conjugate primaries of weight h_n gives exactly the same rational function with coefficient h_n. We have therefore calculated the replica weights:

$$\begin{aligned} h_n = \bar{h}_n &= \frac{c}{24} \left(n - \frac{1}{n} \right), \\ \Delta_n = 2h_n &= \frac{c}{12} \left(n - \frac{1}{n} \right). \end{aligned} \tag{22.5}$$

The factor n in the total stress is essential. Using only one sheet's stress would miss it.

The twist two-point function now determines the interval moment:

$$\operatorname{Tr} \rho_A^n = C_n (\ell/a)^{-2\Delta_n}, \qquad C_1 = 1.$$

The constants C_n depend on the cutoff convention. Substitution in the definition of the Rényi entropy gives

$$S_n(A) = \frac{c}{6}\left(1+\frac{1}{n}\right)\log(\ell/a) + k_n. \tag{22.6}$$

Here $k_n = \log C_n/(1-n)$ records the nonuniversal constant. The physical continuation near $n=1$ is the one defined by powers of the regulated positive density matrix. It yields $S_1 = (c/3)\log(\ell/a) + k_1$. Integer replica values by themselves are not an abstract uniqueness theorem for analytic continuation. For a periodic spatial circle of circumference L, the plane-to-cylinder map replaces ℓ by the chord length:

$$S_n(\ell, L) = \frac{c}{6}\left(1+\frac{1}{n}\right)\log\left[\frac{L}{\pi a}\sin\frac{\pi\ell}{L}\right] + k_n.$$

This uses the conformal vacuum on the circle. A finite lattice has additional corrections to the scaling formula.

A half-infinite bipartition has one entangling endpoint. To obtain a finite entropy, consider a clustering massive vacuum near the critical point, with one large length ξ. Its replica moment is a single twist expectation. Scaling gives

$$\operatorname{Tr}\rho_{\text{half}}^n \sim A_n(a/\xi)^{\Delta_n}, \qquad S_n^{\text{half}} = \frac{c}{12}\left(1+\frac{1}{n}\right)\log(\xi/a) + s_n + o(1).$$

Here A_n depends on the massive theory and regulator. The power uses one twist dimension, whereas an interval two-point function uses two. This explains the factor of two between the two geometries. Replacing this physical correlation length by the length of a finite-bond MPS is a further scaling hypothesis, considered in the next section.

Worked example 22.5 (Ising entropies and an ordered cat)**.** For the Ising CFT, $c = 1/2$. The logarithmic coefficients of S_1, S_2, S_∞ on an interval are respectively $1/6, 1/8, 1/12$; the corresponding single-cut coefficients are $1/12, 1/16, 1/24$. The last values use the regime in which the scaling formula also controls the largest Schmidt probability. A common factor-of-two error would turn the correct Ising single-cut von Neumann slope $1/12$ into a spurious estimate $c = 1/4$ if one used the interval coefficient $c/3$.

The three-component product-state cat considered above instead has $\operatorname{Tr}\rho_A^n = 3^{1-n}$ across a nontrivial cut, hence $S_n = \log 3$ for every n. Its entropy has no logarithmic length slope. More generally, suppose three components have mutually orthogonal Schmidt supports on both sides of the cut, the same Schmidt probabilities p_a, and equal weights. The full probabilities are $p_a/3$, each repeated three times. Thus $S_n^{\text{cat}} = S_n^{\text{component}} + \log 3$. A fixed mixing constant does not change an asymptotic slope; changing component weights along a variational sequence can change a finite entropy secant.

Global orthogonality of the components alone would not suffice. For instance the three qutrit states $|\Phi_j\rangle = 3^{-1/2}\sum_{a=0}^{2}\zeta_3^{ja}|a,a\rangle$ are orthonormal and each has three equal Schmidt probabilities. Yet their normalized equal superposition is $|0,0\rangle$, a product state. Their Schmidt supports overlap; the preceding ordered-cat formula uses the stronger support condition explicitly.

22.7 Finite-entanglement scaling as a hypothesis to test

A finite-bond MPS approximation to a critical ground state often acts as an infrared regulator. Under the usual single-scale critical assumptions, the half-chain Rényi entropy behaves as

$$S_n = \frac{c}{12}(1+1/n)\log\xi + s_n + \cdots. \tag{22.7}$$

Thus adjacent entropy secants are converted to central-charge diagnostics with factors $6, 8, 12$ for $n = 1, 2, \infty$. All orders should use the same pair of tensors and the same physically specified correlation length. Mixing an entropy from one checkpoint with a length from a later optimization has no reason to obey the formula.

If a symmetry-breaking finite-bond state has a local order amplitude $|m| \sim \xi^{-\Delta}$ for one dominant scaling field, then

$$\Delta_{\mathrm{eff}} = -\frac{\log(|m_b|/|m_a|)}{\log(\xi_b/\xi_a)}.$$

There is no factor of one half unless the observable used is $|m|^2$. Several microscopic order sources can have different finite slopes because their subleading mixtures differ.

Numerical record 22.6 (The present stationary sequence)**.** In the completed October 5 record, the largest checkpoint qualified by the unchanged stationarity test has rank parameter 144, energy density approximately $-.84533646847$, and component transfer length approximately 154.8613 sites. Three consecutive accepted coordinate gradients and a fresh uncached same-point check pass the threshold 10^{-6}; the latter gives 7.67×10^{-7}. The rank-156 trial reaches the lower energy $-.84533724589$, but its gradient 6.27×10^{-5} fails the same criterion and it is excluded from the qualified sequence.

The newest adjacent conditional entropy estimates are

rank pair	c_1	c_2	c_∞
120–132	2.75608	2.65579	2.35197
132–144	1.55824	1.59321	1.47876

The comparison retains all 28 contiguous rank windows. A fit through the four checkpoints of ranks $108, 120, 132, 144$ gives $c_1 = 2.09890$. Its proximity to two does not make the strongly varying adjacent slopes a plateau. Stationarity qualifies the variational data for comparison; it does not establish the single-scale conformal assumptions of (22.7) or a continuum error bound.

Here the saved rank parameter r labels height-fiber dimensions $D_{I_j} = r$ and $D_{X_j} = 3r$. The cyclic implementation further splits them into three phase blocks of dimensions $r/3$ and r. Thus rank parameter 144 does not mean that the full ordinary embedding has bond dimension 144; its block structure is part of the tensor definition and of the resource estimate.

The lower variational energy at a larger rank is useful even when its entropy has not converged. It does not justify mixing that point into a qualified scaling sequence. Conversely, a strict coordinate-gradient gate is an optimization criterion, not a physical approximation-error bound. A high-quality sequence should retain energy, independent gradient checks, variance, transfer residuals, primitive-component tests, and entropies at the same tensors.

There is a second possible interpretation of growing ξ: a model with a very large but finite physical correlation length may look critical until the variational regulator exceeds that scale. Failure to see saturation at current ranks does not prove a divergent exact length. The gapped alternatives of the next chapter make this possibility especially relevant.

22.8 Energy variance and the transfer resolvent

For $H_L = \sum_j h_j$ in a clustering translation-invariant state,

$$\lim_{L\to\infty} \frac{\mathrm{Var}(H_L)}{L} = \langle h_0^2\rangle - e^2 + 2\sum_{r\ge 1} \Re(\langle h_0 h_r\rangle - e^2), \tag{22.8}$$

provided the connected series is summable. Here $e = \langle h\rangle$. For $h = -P_\rho$, $h^2 = -h$, so the coincident term is $-e - e^2$. Overlapping adjacent terms must be handled separately; their ordered product is not necessarily Hermitian, although the full variance is real.

The distant tail can be summed without enumerating separations. On the centered transfer subspace,

$$\sum_{n=0}^{\infty}(\mathcal{E}-\mathcal{P})^n=(1-\mathcal{E}+\mathcal{P})^{-1}.$$

Applying this reduced resolvent to a centered insertion vector turns the infinite sum into a linear equation. It can be solved through matrix-free Krylov methods, which repeatedly apply $\mathcal{E}$ but never store its dense matrix. The stationary direction is explicitly removed; otherwise the inverse is singular.

The variance density is an independent variational diagnostic. For an exact eigenstate on a finite ring it vanishes. A small density in an infinite ansatz does not by itself bound its fidelity with a particular ground state, because that requires spectral information. It also does not decide whether a decreasing variational energy sequence approaches a critical state or a massive one.

22.9 Local correlators and disconnected plateaus

An ordered MPS can be useful for testing proposed fields, but its long-distance raw moments include its mean. For a charged density O with $\langle O\rangle = m$, clustering gives

$$\langle O_0^\dagger O_r\rangle \to |m|^2, \qquad \langle O_0 O_r O_{2r}\rangle \to m^3.$$

The absolute normalized triple-to-pair ratio therefore approaches one, irrespective of an intermediate-distance CFT coefficient. Averaging over an ordered cat removes $\langle O\rangle$ but can retain these plateaus. Component centering and connected cumulants answer a different question from averaging over all symmetry-related vacua.

A possible scaling window must satisfy two competing inequalities: the separation is much larger than the operator support and microscopic scale, but much smaller than the variational or physical correlation length. At currently available bond dimensions that window can be narrow. Period-three oscillations also require demodulation or separate residue classes. A smooth ratio at a few separations is a diagnostic; it does not establish its universal limit.

The same issue arises for energy structure factors. If the local energy density really approaches $v(T+\bar{T})/(2\pi)$, the nonanalytic small-momentum part is proportional to $cv^2|q|^3$. Analytic contact terms, density improvements, finite transfer length, and higher powers can coexist. Cancelling a fitted analytic term does not remove every omitted correction, and an estimate of cv^2 still requires a velocity to become an estimate of c.

22.10 Designing a calculation whose resources are understood

A useful tensor identity can be computationally unusable in its literal form. An ordinary bond-D transfer matrix has D^2 rows and D^2 columns, hence D^4 entries. Complex double precision uses $16D^4$ bytes for that matrix alone. Its input tensor, copies, factorizations, and a solver workspace may coexist with it.

The danger is not hypothetical in this project: an older optimization path contained approximately $468D^4$ complex entries in one hidden workspace, about 47.18 GiB at $D = 51$, before other live objects. Inspecting only the allocating wrapper would have missed an in-place call that created the workspace. The correct object to audit is the actual execution path.

Matrix-free contractions often replace a D^4 storage cost by a small number of D^2 vectors and D^3 arithmetic steps, although the exact exponents depend on the tensor structure. Fusion-height blocks and cyclic phase blocks can reduce both constants and powers. Streaming one endpoint pair or one local contraction at a time avoids holding all intermediates together. A

reduced-resolvent solve should use a bounded Krylov restart rather than an unbounded stored basis.

Worked example 22.7 (Why "small bond rank" can be misleading)**.** A categorical line can enlarge each height fiber by summing over fusion-compatible incoming fibers. In the actual rho-line lift, a rank pair $(r, 3r)$ becomes $(3r, 10r)$. A coherent direct sum with the original tensor has ranks $(4r, 13r)$. Even if the original r looks small, forming a dense transfer matrix of the full ordinary embedding can already exceed desktop memory by orders of magnitude. The height-block representation, not the nickname of the original rank, determines the allocation.

A peak estimate should count simultaneously live basis arrays, operator tables, Krylov vectors, coordinate/gradient copies, library workspaces, decompression buffers, and cache charged to the job. A read-only checkpoint consumer can have small resident arrays but substantial file-cache use. Swap is not an additional compute budget: paging a memory-exhausted eigensolve can make a shared desktop unusable even without an immediate allocation failure.

For reproducing calculations on the project machine, the canonical machine policy and its central guarded launcher govern admission, memory reservations, CPU affinity, and thread counts. Historical commands in research notes are provenance, not current permission to reuse an old larger budget. A research method should degrade gracefully: reuse checkpoints, work in a smaller sector, stream a contraction, or prove an algebraic statement when safe numerical admission is unavailable.

22.11 Exercises

Exercise 22.1. Take the normalized two-qutrit state $|\Psi\rangle = \sqrt{1/2}|00\rangle + \sqrt{1/3}|11\rangle + \sqrt{1/6}|22\rangle$ and $H = -|\Psi\rangle\langle\Psi|$. Find the normalized optimal Schmidt-rank-two approximation, its fidelity, its vector-norm error, and its energy. Compare with the best rank-one energy. Explain why allowing arbitrary local basis changes cannot improve these optima.

Solution on page 313.

Exercise 22.2. Verify that the block-row ordinary tensor constructed from W_{ab} has periodic amplitude (22.2). Identify the virtual index constraints imposed by multiplying two consecutive tensors.

Solution on page 314.

Exercise 22.3. For the scalar-edge family with three colors, real t, and $W_t(i, j) = t$ when $i = j$ and 1 otherwise, compute the norm-transfer eigenvalues and the nontrivial correlation length. Analyze $t = 0$, $t = 1$, and large $|t|$.

Solution on page 314.

Exercise 22.4. Compute the ordinary bipartite entropy of the three-component product-state cat in the example. Explain why a nonzero entropy constant and long-range order do not imply a nonzero central charge.

Solution on page 314.

Exercise 22.5. Starting from the transfer expression for separated correlators, derive the centered geometric series and its reduced resolvent. State what fails when another eigenvalue has modulus one.

Solution on page 314.

Exercise 22.6. Derive the conversion factors $6, 8, 12$ from (22.7). Suppose $S_1 = a \log \xi + b + u/\log \xi$. Compute the effective secant between two lengths and explain why a slowly drifting slope need not have reached its limiting value.

Solution on page 314.

Chapter 23

Module categories, exact MPS, and gapped alternatives

An unusual noninvertible symmetry can restrict the possible vacuum structure without forcing a gapless theory. To see this concretely, we construct finite-bond states from a Haagerup algebra object and explain why a sufficiently long-range parent Hamiltonian is gapped and symmetric on the same physical-rho path space as the target chain. We then derive the corresponding finite-dimensional vacuum form factors. These are valuable alternatives against which to compare the chain data, but they are not assertions that the target Hamiltonian realizes that phase. The needed categorical ideas are algebra objects, their modules, and the compatibility of an action with reassociation. The matrix-product notation comes from Chapter 22; the relation between algebra modules and physical defects was introduced in Chapter 16.

23.1 Why several vacua can be required

For an ordinary finite symmetry group, a symmetry-broken gapped phase has a finite set of pure vacua permuted by the group. A noninvertible line can send one vacuum to a sum of vacua. Its multiplicities form nonnegative integer matrices satisfying the fusion rules. These matrices are often called a nonnegative integer matrix representation, or NIMrep.

The matrices are not all the data. There must also be consistent associativity maps describing how line junctions act on interfaces between vacua. Those maps make the vacua into a module category over the symmetry category. A representation of the abstract fusion ring on a Hilbert space need not lift to such a positive semisimple vacuum module. This distinction prevents a small matrix trick from proving the existence of a physical phase.

An invariant finite-volume ground state can be a cat of several pure vacua. Its one-point functions can vanish even when each component is ordered. In an ordinary gapped tunneling picture, several finite-volume levels collapse toward a common vacuum energy while a separate bulk gap remains positive. Their splittings may be exponentially small at large circumference. A broad crossover can make those splittings look like $1/L$ before the asymptotic regime is reached.

Worked example 23.1 (A gapped ordered chain on compatible and frustrated rings)**.** Consider an ordinary three-state clock at each site, with orthonormal basis $|s_j\rangle$, $s_j \in \mathbb{Z}/3\mathbb{Z}$. Define a diagonal Hamiltonian by its energy on each product configuration:

$$H_{\mathrm{clock}}|s_1,\dots,s_L\rangle = J\sum_{j=1}^{L}(1-\delta_{s_{j+1},\,s_j+1})|s_1,\dots,s_L\rangle, \qquad s_{L+1}=s_1, \quad J>0.$$

Take $L \geq 3$. All color arithmetic is modulo three. The local terms are commuting projections times J: a satisfied bond advances the color by one, while an unsatisfied bond costs J. The

three zero-defect configurations on the infinite chain are $s_j = a + j$, with $a \in \mathbb{Z}/3\mathbb{Z}$. These are period-three pure bulk vacua, cycled by one-site translation or a global color shift.

The finite spectrum follows from the defect residues

$$r_j = s_{j+1} - s_j - 1 \in \mathbb{Z}/3\mathbb{Z}, \qquad \sum_{j=1}^{L} r_j = -L \pmod 3.$$

Their nonzero entries count the energy in units of J. Conversely, any residue sequence satisfying this sum, together with one initial color, reconstructs a unique periodic configuration. When $3 \mid L$, all $r_j = 0$ is allowed and gives exactly three ground states of energy zero. One nonzero residue cannot have sum zero. Two residues $+1, -1$ can, so the first excited energy is exactly $2J$. With energies ordered from E_0 and counted with multiplicity,

$$E_0 = E_1 = E_2 = 0, \qquad E_3 - E_0 = 2J \quad (3 \mid L).$$

There is a vacuum cluster and a size-independent gap above it.

If $3 \nmid L$, at least one defect is forced. One residue $r_j = -L$ satisfies the constraint; there are L choices of its bond and three choices of initial color. Hence the ground energy is J with degeneracy $3L$. In particular $E_3 - E_0 = 0$ along these sizes. This does not diagnose a critical bulk: the incompatible ring contains a domain wall whose position costs no energy in this exactly diagonal model. The staggered variable $t_j = s_j - j$ makes the mechanism transparent. In the bulk the Hamiltonian is the ferromagnetic clock projector onto unequal neighboring t values; the ring closes with $t_{L+1} = t_1 - L$, a nontrivial color twist precisely when $3 \nmid L$.

The distinction between a pure vacuum and a cat is equally explicit. Let $Z_j|s_j\rangle = \omega^{s_j}|s_j\rangle$, with $\omega = e^{2\pi i/3}$. In each product vacuum connected correlations between distinct sites vanish. On a compatible ring, the equal superposition of the three product vacua instead has

$$\langle Z_j \rangle = 0, \qquad \langle Z_i^\dagger Z_j \rangle = \omega^{j-i}.$$

Its nondecaying order correlation coexists with the gap $2J$. This elementary clock model illustrates the finite-volume issues; it makes no assertion about the phase of the Haagerup Hamiltonian.

The physical $\mathcal{H}_3$ category has indecomposable vacuum module possibilities of ranks two, four, and six. Translation can also permute equivalent components. Thus the word "symmetric" need not mean a unique pure vacuum. A finite collection of vacua can carry the full symmetry consistently.

23.2 An algebra object is a multiplication with F-moves

An algebra object consists of an object Q, a unit $\eta : 1 \to Q$, and a multiplication $m : Q \otimes Q \to Q$ satisfying unit and associativity equations. Associativity compares $m(m \otimes 1)$ with $m(1 \otimes m)$ after applying the category's associator. In a fusion basis it is a system of polynomial equations whose coefficients are the F-symbols.

For a unitary separable Frobenius algebra, the adjoint $m^\dagger$ is compatible with multiplication, and a normalization can make

$$mm^\dagger = \operatorname{qdim}(Q)\, 1_Q.$$

The Frobenius identities are not decorative. They are what lets multiplication and comultiplication pass through one another in a positive tensor construction. A connected algebra has $\dim \operatorname{Hom}(1, Q) = 1$.

Worked example 23.2 (The associator can forbid an apparent algebra)**.** One might try $Q' = 1 + \rho$, since its summands look simple and its fusion products contain the needed output channels. Normalize the unit and write $u = m(\rho, \rho; 1)$, $v = m(\rho, \rho; \rho)$. In the physical $\mathcal{H}_3$ associator, the two forbidden noninvertible output channels in the three-rho associativity equation require

$$\frac{u}{\sqrt{d}} + y_1 v^2 = 0, \qquad \frac{u}{\sqrt{d}} + y_2 v^2 = 0,$$

where $y_1 \neq y_2$. Hence $u = v = 0$. The remaining multiplication is a square-zero extension; its pairing through the tensor-unit channel is degenerate, so it is not a connected separable unitary algebra. Fusion-ring closure alone therefore fails to construct $1 + \rho$ as a Q-system for this associator.

This local calculation also distinguishes the physical rank-four $\mathcal{H}_3$ module from a superficially similar $\mathcal{H}_2$ module. A category name or a shared fusion graph is not a substitute for checking its associativity equations.

Construction input 23.3 (The physical Haagerup algebra)**.** In the specified real $\mathcal{H}_3$ gauge, the repository constructs

$$Q = 1 + \rho + \alpha\rho$$

with unit, associativity, Frobenius, and specialness identities checked exactly in its algebraic coefficient field. It also constructs the four simple right modules and their action coefficients. The following MPS and parent arguments take these finite algebraic identities as input; they do not infer them from numerical optimization.

The finite input can be made quite explicit. Put $r = \rho$ and $s = \alpha\rho$, and define real core constants

$$x = \frac{2 - \sqrt{13}}{3}, \qquad y_{1,2} = \frac{5 - \sqrt{13} \mp \sqrt{6(1 + \sqrt{13})}}{12}, \qquad z = \frac{1 + \sqrt{13}}{6}.$$

Here the minus sign defines y_1. Set

$$D = \sqrt{z^2 - y_1^2}, \quad u = \sqrt{d}, \quad v = -y_1/D, \quad \zeta = z/D, \quad w = \sqrt{d - 1 - v^2}, \quad t = -\frac{y_2}{z} w,$$

with positive square roots. The unit coefficients are one, and the remaining nonzero multiplication coefficients are

$$\begin{aligned}
&m(r, r; 1) = m(s, s; 1) = u,\\
&m(r, r; r) = v,\\
&m(r, r; s) = m(r, s; r) = m(s, r; r) = w,\\
&m(s, s; r) = m(r, s; s) = m(s, r; s) = \zeta,\\
&m(s, s; s) = t.
\end{aligned}$$

These are coefficients on normalized trivalent fusion vertices, not the multiplication table of an ordinary three-dimensional vector space with a trivial associator. Every unlisted coefficient is zero.

Worked example 23.4 (Two specialness blocks)**.** On the tensor-unit output, $mm^\dagger$ is the sum of the squares of $m(1, 1; 1)$, $m(r, r; 1)$, and $m(s, s; 1)$, namely $1 + 2u^2 = 1 + 2d = \mathrm{qdim}(Q)$. On the r output, it is

$$2 + v^2 + 2w^2 + \zeta^2.$$

The radical definitions give $\zeta^2 - v^2 = 1$ and $w^2 = d - 1 - v^2$, reducing the expression to the same $1 + 2d$. These checks illustrate how the coefficients are normalized. They do not replace the associativity equations, including forbidden channels, or the separate Frobenius checks.

In the convention of (19.2), a full associativity component has the concrete form

$$\sum_e m(a,b;e)m(e,c;T)F_T^{abc}[e,f] = m(b,c;f)m(a,f;T).$$

All allowed f and all simple totals T, including those outside Q, enter the test. This is a small enough example to see exactly what an algebra-object construction claims while still using a nontrivial Haagerup associator.

A right module is an object M with action $\mu_M : M \otimes Q \to M$ satisfying the analogous unit and associativity equations. The category acts on modules from the left: $c \otimes M$ inherits a right action by reassociation. Decomposing this induced module into simples gives the vacuum fusion matrices. Here Q belongs to the nonbraided line category $\mathcal{H}_3$. Its right modules form the category of vacua on which the lines act; the argument requires no independent tensor product of two vacua.

23.3 The four right modules and their physical tensors

Three modules are $\alpha^j Q$, with underlying supports

$$\alpha^j + \alpha^j\rho + \alpha^{j+1}\rho.$$

The fourth, denoted M_{fix}, has underlying object $\rho + \alpha\rho + \alpha^2\rho$ and is fixed by the alpha action up to module isomorphism. Its existence can be seen from the free module $\rho \otimes Q$ after removing two known simple summands.

The multiplicity vector of ρQ in the ordered six-object list is $(1,0,1,3,2,2)$. It decomposes as

$$\rho Q \cong Q \oplus \alpha^2 Q \oplus M_{\mathrm{fix}}.$$

The exact action coefficients provide orthogonal coisometries onto the first two summands; their orthogonal complement gives the last one. The endomorphism dimension $\dim \mathrm{End}_Q(\rho Q) = \dim \mathrm{Hom}(\rho, \rho Q) = 3$ then shows that the nonzero complement is simple, once the first two distinct simple summands have been identified.

Restrict the physical algebra input of a module action to the ρ summand of Q. In the multiplicity-free modules here, this gives scalar edge weights

$$W_M(a,b) = \mu_M(a,\rho;b).$$

The periodic state is

$$\Psi_M(a_0,\dots,a_{L-1}) = \prod_i W_M(a_i,a_{i+1}). \tag{23.1}$$

More generally one takes an ordinary trace over multiplicity fibers. Forbidden fusion edges vanish, so this is an exactly admissible physical-rho state.

23.3.1 Why the categorical line has the expected action

Apply the F-product operator Y_c from (19.6) to (23.1). At each output height, summing the internal input height is precisely the induced right action on $c \otimes M$. A heightwise module decomposition turns this action into a direct sum of simple module actions. The changes of basis cancel on consecutive edges and inside the final cyclic trace. Therefore

$$Y_c\Psi_M = \sum_N N_{c,M}^N \Psi_N. \tag{23.2}$$

No quantum-dimension prefactor appears. The closure uses ordinary cyclic traces in the orthonormal fusion coordinates; a categorical pivotal trace would be a different contraction.

The finite constructor checked this statement literally on every periodic path at lengths three and four, as well as checking the module equations from which it follows generally. Those small-ring checks validate the implementation and normalization. The all-length argument is the telescoping module decomposition, not an extrapolation from two rings.

After a cyclic relabeling of the three ordered modules, the rho multiplicity matrix is

$$R = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}. \tag{23.3}$$

Alpha fixes the first state and cycles the last three. Its triplet block is $J - P$, where P fixes one color and exchanges the other two. This is the category-correct rank-four $\mathcal{H}_3$ action. In (23.2) it acts on the canonical, generally unnormalized finite-ring module tensors. Those tensors need not be orthogonal at finite circumference. In the gapped vacuum description below, the same symmetric matrix acts on an orthonormal basis of simple vacua. These are distinct normalization statements; independently normalizing four finite-ring vectors would insert ratios of their norms into the matrix elements.

23.4 A scalar-edge family that can be solved by hand

For the alpha-fixed module, a real height gauge makes all diagonal weights equal and all off-diagonal weights equal. After an overall normalization, consider the larger one-parameter family

$$W_t(i,j) = \begin{cases} t, & i = j, \\ 1, & i \neq j, \end{cases} \qquad i, j \in \mathbb{Z}/3\mathbb{Z}, \quad t \in \mathbb{R}.$$

Its norm transfer is

$$K_t = (t^2 - 1)I + J.$$

The leading eigenvalue is $t^2 + 2$ and the two other eigenvalues are $t^2 - 1$. The leading probability on colors is uniform. Even at $t = 0$ the transfer is primitive: a complete graph on three vertices without loops has both two-step and three-step returns and hence period one. For nonzero t, the positive diagonal makes aperiodicity immediate.

The module point is a particular negative solution

$$t_0^2 + t_0 = 1/d.$$

The nonzero amplitudes on all three colors show how an alpha-invariant primitive state can live entirely on noninvertible heights. There is no requirement that every height appear with nonzero probability in every allowed physical state.

The energy of this entire family is also a small local contraction. Write x, y_1, y_2, z for the relevant rho-projector column entries in the physical F gauge. For equal endpoint colors, the two-edge amplitude vector is $(t^2, 1, 1)$ and its rho-channel component is $xt^2 + y_1 + y_2$. For unequal endpoint colors it is a permutation of $(t, t, 1)$ and its component is $(y_1 + y_2)t + z$. The transfer Perron normalization gives

$$e(t) = -\frac{(xt^2 + y_1 + y_2)^2 + 2((y_1 + y_2)t + z)^2}{(t^2 + 2)^2}. \tag{23.4}$$

This is an exact infinite-state variational energy, not a finite-size fit. At the exact negative module point it is only about $-.065$, far above the optimized physical-chain energy density near

$-.845$. The construction supplies a controlled state and a symmetry example, not a competitive ground-state approximation by itself.

Module associativity gives another exact property. Recoupling two adjacent physical rho legs into a channel f produces

$$m(\rho,\rho;f)\,\mu_M(a,f;b).$$

Since $\alpha^2\rho$ is absent from Q, its coefficient vanishes. Every constructed module state is therefore annihilated by every $P_i^{(\alpha^2\rho)}$. It is a frustration-free state of the different Hamiltonian $+\sum_i P_i^{(\alpha^2\rho)}$. It has not thereby been shown that this short Hamiltonian has no other ground states or has a positive thermodynamic gap.

23.5 A gapped parent on the same constrained path space

We now explain the stronger long-range parent construction. Its input is the four exact tensors, and its conclusion concerns a particular sufficiently long interaction range. It does not change the physical object per site.

For an ordered block Q, in the support order $(1,\rho,\alpha\rho)$, the weight matrix has the form

$$W_Q=\begin{pmatrix}0&1&0\\u&v&w\\0&w&\zeta\end{pmatrix},\qquad u,v,w,\zeta>0.$$

It is invertible because $\det W_Q=-u\zeta$. Its norm graph has links $I\leftrightarrow R\leftrightarrow S$ and loops on R,S, so it is primitive. In the real height gauge, the fixed block is $W_f=B_f[(t_0-1)I+J]$, where $B_f>0$ is its common off-diagonal weight. It is also invertible:

$$\det W_f=B_f^3(t_0+2)(t_0-1)^2\neq 0.$$

Neither $t_0=-2$ nor $t_0=1$ solves $t_0^2+t_0=1/d$.

Embed each block as an ordinary three-dimensional virtual tensor

$$A_M^a=|a\rangle\,\mathrm{row}_a(W_M),$$

with zero tensor when a is outside the support. A physical word then yields

$$A_M^{a_0}\cdots A_M^{a_{\ell-1}}=\left[\prod_{j=0}^{\ell-2}W_M(a_j,a_{j+1})\right]|a_0\rangle\,\mathrm{row}_{a_{\ell-1}}(W_M).$$

The products of words span the virtual matrix algebra once enough words distinguish the blocks and vary their endpoints.

Proposition 23.5 (An explicit block-injectivity witness)**.** *Five-site words span the direct sum of the four* 3×3 *virtual matrix algebras of these module tensors.*

Proof. In ordered block j, choose a word a,R_j,I_j,R_j,b with arbitrary supported endpoints. Every edge has nonzero weight. The unique invertible height I_j excludes the other three modules. Varying a,b, and using invertibility of W_j, spans all nine matrix units in that block.

For the fixed block, choose an all-noninvertible five-site word with any prescribed endpoints that visits all three colors. Every edge is nonzero. No ordered module supports all three colors, so it isolates the fixed block. Again the endpoint matrices span all nine matrix units. □

Block injectivity is stronger than merely finding a primitive transfer spectrum separately in each block. It also separates the blocks by physical words. It is the appropriate input for a finite-family MPS parent theorem. Equivalently, after blocking, every virtual matrix can be detected by a suitable physical word within its chosen block. This is the linear-algebra meaning of *injectivity* being used here.

For an interval of length ℓ, define the local range

$$S_\ell = \mathrm{span}\{|v_{ab}^M\rangle : M, a, b\},$$

where $|v_{ab}^M\rangle$ sums all internal height words with fixed endpoints a, b, weighted by the product of the W_M edges. Because W_M is invertible, this is precisely the ordinary MPS range with arbitrary virtual boundary matrices. Different endpoint pairs have orthogonal physical support. Its orthogonal projector Π_ℓ therefore preserves both endpoint heights and maps admissible interiors to admissible interiors.

The parent term $h_\ell = 1 - \Pi_\ell$ preserves the global constrained path space. In the unconstrained six-state embedding, forbidden interior words are penalized. Restricting to the path space is a reducing restriction of the operator, not a change in its intended low-energy state family.

23.5.1 Why the local parent respects the categorical lines

Closure of the four periodic states under Y_c would not by itself prove that every local parent term commutes with Y_c. The needed statement is local. Fix the endpoints of an interval and all outside MPO indices. Acting with the open F-product line on an interval module vector is the induced module $c \otimes M$ on that interval. The same module decomposition used in (23.2) telescopes at every internal height, leaving only boundary intertwiners. Hence it sends the whole local range S_ℓ into itself.

Thus the subspace with this interval in S_ℓ and arbitrary compatible outside data is invariant under every closed Y_c and under $Y_c^\dagger = Y_{c^*}$. It is therefore reducing. Its orthogonal projector commutes with the closed line on the full compatible chain, giving $[h_\ell, Y_c] = 0$.

For a sufficiently long fixed interval, the standard block-injective MPS parent theorem gives a thermodynamic gap and four periodic ground states on sufficiently large rings. All four MPS lie in the admissible path space. Restricting the positive parent to this reducing subspace keeps those ground states and cannot create eigenvalues below its gap. The same local term can be placed at every site, preserving one-site translation. The five-site word argument is a proof device, not a requirement to break physical translation to a five-site cell.

Theorem 23.6 (Conditional on the exact module input)**.** *The physical one-rho-per-site path Hilbert space admits a finite-range, one-site-translation-invariant gapped Hamiltonian commuting with the actual $\mathcal{H}_3$ categorical lines and realizing the rank-four module of vacua.*

No minimal interaction range or numerical gap bound is asserted. In particular, this theorem does not prove a gap for the short missing-channel anchor. Its open kernels exceed the module ranges already at small lengths. Nor does it determine the phase of $-\sum P_\rho$. It rules out a proposed kinematic argument that these symmetries and this object per cell force every compatible Hamiltonian to be gapless.

Worked example 23.7 (A local kernel does not determine the longer parent)**.** The distinction can be checked by exact dimensions. On three height sites, the four-module range is in fact the entire kernel of the single $P_{\alpha^2\rho}$ term. At equal noninvertible endpoints, there are four middle heights and a rank-one constraint. The three available module vectors are independent: one has an invertible middle height unavailable to the others; after removing it, the fixed module has a third-color middle height unavailable to the remaining ordered module. At unequal noninvertible

endpoints, the middle space has dimension three and the two module vectors are independent, since only the fixed module reaches the third color.

The other endpoints contribute one dimension whenever they belong to an ordered module. Summing equal invertible, mixed, equal noninvertible, and unequal noninvertible endpoint sectors gives

$$\dim S_3 = 3 + 6 + 6 + 9 + 12 = 36.$$

Thus three-site words already separate the four virtual blocks; the five-site witness above was a simpler sufficient proof. The sum of the physical tensors is the invertible direct sum of their W_M matrices, so full spanning persists at every larger word length. In particular $\dim S_4 = 36$.

Now let D_ℓ count admissible open words of ℓ heights, including all endpoints. Unlike the periodic count, it is the sum of all entries of $N_\rho^{\ell-1}$. The color-independent two-by-two block gives $D_1 = 6$, $D_2 = 15$ and $D_{\ell+2} = 3D_{\ell+1} + D_\ell$, hence

$$D_3 = 51, \qquad D_4 = 168.$$

Each of the two missing-channel projectors on four heights has rank 51: fusing the selected pair into $\alpha^2\rho$ reduces the edge count by one, and its alpha permutation disappears when all endpoints are summed. Rank subadditivity therefore gives

$$\dim\bigl(\ker P_1 \cap \ker P_2\bigr) \geq 168 - 51 - 51 = 66 > 36 = \dim S_4.$$

No commutativity assumption is used. The equality of one local kernel with an MPS range has already failed to propagate at the next length. This is why a parent-gap theorem must use its actual intersection hypotheses, rather than only the existence of several annihilated MPS.

23.6 The rank-four vacuum algebra and its exact form factors

We can derive a sharp low-energy alternative if the target chain eventually approaches this gapped vacuum structure. Let $|f\rangle$ be the fixed pure vacuum and $|o_j\rangle$ the ordered three-cycle. Put

$$b = d - 2 = \frac{\sqrt{13} - 1}{2}, \qquad b^2 = 3 - b.$$

Choose the vacuum in the trivial categorical sector, $Y_\alpha = 1$ and $Y_\rho = d$. The four-dimensional ground space alone does not make this choice; alpha invariance still leaves a two-dimensional neutral space. The Perron eigenvector of (23.3) with eigenvalue d is

$$|0\rangle = \frac{b|f\rangle + |o_0\rangle + |o_1\rangle + |o_2\rangle}{\sqrt{b^2 + 3}}.$$

The ordered total probability is $p = 3/(b^2 + 3)$. The symmetric finite-volume vacuum is therefore not an equal mixture of four vacua.

There is a useful distinction between state normalization and idempotent normalization here. Let v_f, v_j be the local primitive idempotents, which multiply as $v_i v_j = \delta_{ij} v_i$, and let $\theta_i = \langle v_i, v_i\rangle$ in the positive Frobenius inner product. The line action on these idempotents is

$$Y_\rho v_j = (J - P)v_j + b^{-1} v_f, \qquad Y_\rho v_f = v_f + b\sum_j v_j.$$

Self-adjointness gives $\theta_f = b^2\theta_o$, with the three ordered norms equal. Passing to $|i\rangle = v_i/\sqrt{\theta_i}$ produces the symmetric integer matrix (23.3). Thus the unequal Perron probabilities are fixed by the positive vacuum metric; an equal-weight statistical mixture would calculate different form factors.

In the diagonal algebra of functions on these four vacua, define

$$\Phi(o_j) = p^{-1/2}\omega^j, \quad \Phi(f) = 0, \qquad E(o_j) = b/\sqrt{3}, \quad E(f) = -\sqrt{3}/b.$$

Direct averaging in $|0\rangle$ gives $\langle E\rangle = 0$ and $\langle E^2\rangle = \langle \Phi^\dagger \Phi\rangle = 1$. The cubic and mixed form factors are

$$C_{\Phi\Phi\Phi} = p^{-1/2} = \sqrt{\frac{13-\sqrt{13}}{6}}, \tag{23.5}$$

$$g = C_{\Phi\Phi^\dagger E} = b/\sqrt{3}, \qquad C_{EEE} = -1/\sqrt{3}. \tag{23.6}$$

For instance, Φ^3 is $p^{-3/2}$ on the ordered vacua, so its mean is $p^{-1/2}$. The neutral cubic follows by adding the ordered and fixed contributions with their unequal probabilities.

The algebra closes:

$$\Phi^\dagger\Phi = 1 + gE, \qquad E^2 = 1 + C_{EEE}E, \qquad |C_{\Phi\Phi\Phi}|^2 = 1 + g^2, \quad C_{EEE} = \frac{g^2-1}{g}. \tag{23.7}$$

These are ordinary multiplication identities in a finite-dimensional vacuum algebra. They do not use conformal invariance. A microscopic operator ratio approaches them only if its source and selected states converge to the stated vacuum projections.

The physical $\mathcal{H}_3$ predictions are approximately 1.25 for the charged cubic and .75 for the mixed neutral coefficient. The values about 1.66 and 1.33 arise from the different rank-four $\mathcal{H}_2$ action. The finite-chain values near 1.59 and 1.30 differ from the $\mathcal{H}_3$ targets, but do not alone exclude a late gapped crossover: vacuum projection and a finite bulk correlation length have not been established at these sizes.

23.7 Other vacuum modules and a symmetry filter

In the rank-two module, alpha is trivial and $R = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 3 \end{smallmatrix}\right)$. Its Perron amplitudes are proportional to $(1, d)$. The centered unit neutral field has values $d, -d^{-1}$ and cubic $d - d^{-1} = 3$. There is no charged order field in this vacuum algebra.

The regular rank-six module has two alpha triplets, with Perron amplitudes $(1, 1, 1, d, d, d)$. It has two charged pairs rather than one, and a neutral cubic of 3. Its charged cubic depends on the chosen linear combination of the two pairs. Under an additional closed-pair condition for the rho line, its upper possible magnitude is close to 1.5895. Matching that one number is not an identification: the extra charged pair and the distinct neutral predictions must also occur.

For the rank-four module, let $|s\rangle = (|o_0\rangle + |o_1\rangle + |o_2\rangle)/\sqrt{3}$. On (f, s),

$$Y_\rho = \begin{pmatrix} 1 & \sqrt{3} \\ \sqrt{3} & 2 \end{pmatrix}, \qquad (Y_\rho - 2)|s\rangle = \sqrt{3}|f\rangle.$$

This suggests extracting the missing fixed vacuum from an ordered cat. On the whole alpha-neutral Hilbert space, however, $B = Y_\rho - 2$ is invertible, not a projection, and $B^\dagger B = 5 - Y_\rho$. For a generic normalized state s,

$$\|Bs\|^2 = 5 - \langle Y_\rho\rangle.$$

Dividing by $\sqrt{3}$ is correct only for the exact module cat with $\langle Y_\rho\rangle = 2$.

Since $[H, Y_\rho] = 0$, the normalized filtered energy obeys

$$E_B - E = -\frac{\langle HY_\rho\rangle - E\langle Y_\rho\rangle}{5 - \langle Y_\rho\rangle}.$$

If $\mathrm{Var}(H_L) = O(L)$, its energy-density difference vanishes at least as $L^{-1/2}$ by Cauchy–Schwarz. Equal limiting energy densities after this symmetry filter are therefore a general variance consequence, not proof of a new pure vacuum.

A coherent tensor subtraction must retain mixed overlaps. Taking separate Perron limits of $Y_\rho s$ and s before subtracting can destroy the cancellation. At finite bond dimension, two close but inequivalent injective tensors can become orthogonal as the ring length tends to infinity. A failed finite-bond extraction therefore does not refute the rank-four phase, while small charged means do not prove primitivity.

23.8 Exercises

Exercise 23.1. Verify the fusion relation for (23.3) together with the alpha cycle. Find its eigenvalues on the neutral two-plane and its action on the charged complement.

Solution on page 315.

Exercise 23.2. Derive all three coefficients in (23.5)–(23.6) and the identities (23.7) by averaging functions on four points with the Perron probabilities.

Solution on page 315.

Exercise 23.3. Explain why invariance of the four periodic MPS under a closed line does not by itself imply symmetry of each local parent projector. Identify the stronger open-interval property used in the proof.

Solution on page 315.

Exercise 23.4. Derive the filtered-energy identity and its $O(L^{-1/2})$ density bound when $\mathrm{Var}(H_L) = O(L)$. Give an example of a cat of states with different energy densities for which the variance is instead of order L^2.

Solution on page 315.

Exercise 23.5. Suppose several MPS are annihilated by every term of a short frustration-free Hamiltonian. List the two additional conclusions needed before one can claim this Hamiltonian has exactly those ground states and a thermodynamic gap. Why does a gapped longer parent not supply them automatically?

Solution on page 315.

Chapter 24

Testing an identification without assuming one

Many theories can share an approximate central charge and a few low dimensions. A convincing identification must also explain multiplicities, charges, three-point tensors, locality, and the actual noninvertible junction data. This chapter develops several discriminating tests that are stronger than matching decimals. Some are exact theorems about candidate theories; their application to the chain remains conditional on a controlled continuum dictionary. Others are explicit ultraviolet constructions that organize the observed charges but have not yielded the required infrared theory. Keeping these roles separate allows a failed candidate to teach us something without being mistaken for a general no-go theorem.

24.1 What must match

A proposed identification has at least four layers. First, specify the microscopic model: the physical $\mathcal{H}_3$ associator, rho-channel Hamiltonian, regular periodic space, couplings, and twists. Second, specify the limiting kinematics: vacuum, velocity, central charge, complete low shells, and their charge and momentum assignments. Third, specify local dynamics: normalized OPE coefficients and a consistent crossing/sewing structure. Finally, identify the categorical lines with their fusion, associators, and action on the local fields and twisted sectors.

A chiral commutant with a certain central charge addresses only part of this list. A modular-invariant torus function addresses another part. Neither alone specifies a local full CFT. Conversely, a lattice model with the correct ordinary $C_3 \times C_3$ charges need not possess a Haagerup line. The irrational dimension $d = (3 + \sqrt{13})/2$ and its junctions are substantial extra data.

The central charge near two motivates comparison with two compact bosons, orbifolds, cosets, and interacting deformations. It does not justify assuming any one of those descriptions. The exact $c = 8$ research construction and the separate irrational chiral axis discussed in Chapter 18 should likewise not be relabeled as the infrared limit of the physical chain.

24.2 The simplest product-theory test

Suppose a proposal assigns $\phi = \sigma_A \otimes 1$, $\chi = 1 \otimes \sigma_B$, and $\psi = \sigma_A \otimes \sigma_B$ in a product CFT with normalized two-point functions. Factorization immediately gives

$$\Delta_\psi = \Delta_\phi + \Delta_\chi, \qquad C(\phi, \chi, \psi^\dagger) = 1.$$

The self cubics factorize as well, so $I_\lambda = 1$. These statements are insensitive to field phases once expressed in absolute coefficients and the invariant I_λ.

The physical chain's finite mixed coefficient near 1.42, nonreal I_λ, and nonadditive gaps disagree with that elementary dictionary at currently measured sizes. Turning this into a continuum exclusion requires limits of those quantities. It does already explain why recognizing two ordinary C_3 charges is insufficient to call the theory a product of two familiar charged CFTs.

24.3 Compact bosons and a sharp light-shell bound

A rank-two Narain theory is a unitary CFT of two compact bosons with a momentum-winding lattice of signature $(2,2)$. A vertex operator is labeled by (p_L, p_R), with

$$h = |p_L|^2/2, \qquad \bar{h} = |p_R|^2/2,$$

before adding nonnegative integer oscillator levels. The lattice pairing

$$B(p,q) = p_L \cdot q_L - p_R \cdot q_R$$

is integral. Normalized vertex three-point coefficients have modulus one when the charges sum to zero, and vanish otherwise. Cocycle phases affect their phases, not their magnitudes. This is the two-boson version of the circle construction in Chapter 2; the integral pairing encodes mutual locality between momentum-winding vertices.

A scalar vertex has $|p_L|^2 = |p_R|^2 = \Delta$. For scalar dimensions below $1/2$, oscillators are absent. Any two charges in light scalar shells obey

$$|B(p,q)| \le 2\sqrt{\Delta_p \Delta_q} < 1.$$

Integrality therefore forces $B(p,q) = 0$. All charges in the shells span a totally isotropic real subspace of dimension at most two. Projection to the left momenta is injective on this subspace, because a null vector with $p_L = 0$ also has $p_R = 0$. Charge addition can consequently be studied as addition of vectors in one Euclidean plane, with each dimension shell lying on a circle.

Theorem 24.1 (Unequal light scalar shells)**.** *Let A, B, C be normalized scalar fields in a compact rank-two Narain theory, with $0 < \Delta_A, \Delta_B, \Delta_C < 1/2$. If their dimensions are not all equal, then*

$$|C(A,B,C)| \le 1.$$

Arbitrary degeneracy and complex mixtures within each shell are allowed.

Proof. Choose the two unequal radii as the B, C inputs. Write their vertex coefficients as b_μ, c_ν, with squared sums one. For an output charge $-\lambda$ on the A circle, set

$$T_\lambda = \sum_{\mu+\nu=-\lambda} \epsilon(\mu,\nu) b_\mu c_\nu, \qquad |\epsilon| = 1.$$

Two circles in the plane have at most two intersection points, so each nonzero output has at most two input decompositions. If there are two, write $\mu + \nu = \mu' + \nu' = -\lambda$ and $\delta = \mu - \mu' = \nu' - \nu$. Equal radii within each input shell give

$$2\mu \cdot \delta = 2\nu' \cdot \delta = |\delta|^2, \qquad (\mu - \nu') \cdot \delta = 0.$$

The crossed input pairs have larger output radius:

$$|\mu + \nu'|^2 = |\lambda|^2 + |\delta|^2.$$

They are absent from the diagonal sum on the A circle. The interference term is bounded by

$$2|b_\mu c_\nu b_{\mu'} c_{\nu'}| \le |b_\mu c_{\nu'}|^2 + |b_{\mu'} c_\nu|^2.$$

Each crossed pair is used at most once. To see this, fix (μ, ν'). Their unequal radii imply $\mu \neq \nu'$, so δ lies on a fixed one-dimensional perpendicular. Writing $\delta = te$ gives $t^2 = 2t\mu \cdot e$, with at most one nonzero solution. Thus the original two decompositions are uniquely recovered.

Every interference term is therefore charged to distinct input pairs outside the allowed diagonal set. Summing proves $\sum_\lambda |T_\lambda|^2 \leq \sum_{\mu,\nu} |b_\mu|^2 |c_\nu|^2 = 1$. Finally, contracting T with the normalized coefficient vector of A and using Cauchy–Schwarz gives the result. □

This proof is useful beyond its numerical bound. It treats arbitrary shell degeneracy rather than a favored vertex basis, retains cocycle phases through absolute inequalities, and uses integrality before geometry. It is therefore not restricted to pure momentum states or a special torus radius.

For three equal light dimensions the sharp bound is instead $2/\sqrt{3}$. The unequal-radius uniqueness argument above no longer applies. A short separate argument gives the correct constant. Three nonzero equal-radius vectors adding to zero form an equilateral triad, and a charge belongs to at most one unordered such triad: the other two must be its rotations by ± 120 degrees. Thus the allowed triples partition into disjoint three-element sets.

On one triad, after taking absolute values, the output for normalized nonnegative vectors $b, c \in \mathbb{R}^3$ is $T_i = b_j c_k + b_k c_j$, with $\{i, j, k\} = \{1, 2, 3\}$. Direct expansion gives

$$\sum_i T_i^2 = 1 + \left(\sum_i b_i c_i \right)^2 - 2 \sum_i b_i^2 c_i^2 \leq 1 + \frac{1}{3} \left(\sum_i b_i c_i \right)^2 \leq \frac{4}{3}.$$

The first inequality uses Cauchy–Schwarz on the three products $b_i c_i$. A final contraction with a unit vector has magnitude at most $2/\sqrt{3}$. If several disjoint triads occur, their contributions are bounded by $(2/\sqrt{3}) \sum_T a_T b_T c_T$, where the restricted coefficient norms have squared sums at most one. Hölder's inequality bounds that last sum by one. Cocycle phases cannot increase this absolute bound.

For a Hermitian field, opposite triads have equal coefficient norms. Pairing them gives the stronger self-cubic bound $\sqrt{2/3}$: if the norm in each member of a pair is a_T, then $2 \sum_T a_T^2 \leq 1$ and the cubic is bounded by $(4/\sqrt{3}) \sum_T a_T^3 \leq \sqrt{2/3}$. The following example attains both constants when the corresponding vertex charges exist.

Worked example 24.2 (An equilateral vertex triad)**.** Choose three equal-length scalar charges p_1, p_2, p_3 with $p_1 + p_2 + p_3 = 0$, and normalize

$$\phi = (V_{p_1} + V_{p_2} + V_{p_3})/\sqrt{3}.$$

When the compatible vertex phases are chosen coherently, its cubic gets the six permutations of the charge-conserving triple, giving

$$C(\phi, \phi, \phi) = \frac{6}{3\sqrt{3}} = \frac{2}{\sqrt{3}}.$$

The Hermitian quadrature $X = (\phi + \phi^\dagger)/\sqrt{2}$ has cubic $\sqrt{2/3}$. A single normalized vertex has cubic magnitude at most one, but a degenerate-shell mixture can have a larger cubic.

The chain's finite $|\lambda|$ near 1.42 is thus a strong comparator if its limiting fields have distinct dimensions below $1/2$. The theorem applies to a compact rank-two parent and to untwisted fields that lift isometrically to it. It is not by itself a theorem about twisted fields of every orbifold or about every $c = 2$ CFT.

24.4 What an orbifold loophole requires

Gauging a finite symmetry introduces twisted sectors. Their local fields need not be ordinary sums of parent vertex operators, so one cannot apply a parent shell bound to them without an additional argument. Complete shell multiplicity provides one such argument in specified classes.

Suppose the entire real scalar space at a light dimension is one charged doublet $\phi, \phi^\dagger$, with nonzero cubic and $\langle \phi\phi^\dagger \rangle = 1$. Charge conservation gives $\langle \phi\phi \rangle = 0$. Its normalized real directions are

$$X_\theta = \frac{e^{i\theta}\phi + e^{-i\theta}\phi^\dagger}{\sqrt{2}}, \qquad C(X_\theta, X_\theta, X_\theta) = \frac{\Re(e^{3i\theta}c_\phi)}{\sqrt{2}}.$$

The factor $1/\sqrt{2}$ follows by expanding the cube: only the ϕ^3 and $(\phi^\dagger)^3$ terms survive. On this real two-plane the cubic is proportional to $\Re(z^3)$. Its orthogonal stabilizer is the six-element dihedral group D_3. Thus every symmetry preserving that isolated shell acts through D_3.

In an ordinary $\mathbb{Z}_2$ orbifold, the quantum involution must act on this plane either trivially or as a reflection. In either case there is an invariant real direction. The invariant direction lifts to an untwisted parent field, where the Hermitian Narain bound applies. At a suitable cubic extremum, its cubic magnitude is $|c_\phi|/\sqrt{2}$, giving

$$|c_\phi| \le 2/\sqrt{3}$$

for dimensions below $1/2$. This allows the original charged field to mix twisted and untwisted directions; the isolated full plane is the essential hypothesis.

For finite linear geometric point groups, the argument extends below dimension $2/9$: fields fixed by every quantum character lie in commutator sectors, whose nonidentity rotations have twist cost at least $2/9$. The following result handles a larger class of asymmetric actions, with a slightly smaller dimension interval.

24.4.1 Finite asymmetric actions and the shift subgroup

Proposition 24.3 (The current-preserving orbifold doublet bound)**.** *Gauge an effective finite group G of internal symmetries of a compact rank-two Narain theory. Assume the gauging is an ordinary consistent bosonic unitary orbifold and that G preserves a chosen two-dimensional Heisenberg current space on each chirality, setwise. Asymmetric lattice actions, momentum/winding phases, nonabelian groups, and consistent ordinary discrete torsion are allowed.*

If the entire real scalar eigenspace at $0 < \Delta < 5/36$ is one nontrivial order-three doublet, then its normalized charged cubic obeys

$$|c_\phi| \le 2/\sqrt{3}.$$

The premise preserves each two-dimensional current space, not each current separately. At an enhanced point, a symmetry can mix a chosen Cartan current with root vertex currents; such an action is outside this proposition unless an invariant pair of Cartans can be chosen. We first prove the stated result before addressing those enhanced cases.

Proof. Write the even unimodular charge lattice of signature $(2,2)$ as Γ. A current-preserving symmetry acts by

$$V_p \longmapsto u_g(p) V_{M_g p}, \qquad (J_L, J_R) \longmapsto (R_L(g)J_L, R_R(g)J_R),$$

where $R_L, R_R \in O(2)$ and M_g preserves the lattice and its chosen positive and negative planes. We will use three pieces of this action: its current-trivial kernel, the signs of its two determinants, and its finite rotations.

First gauge the current-trivial subgroup. Let $K = \ker(R_L \times R_R)$. The currents determine both charges, so $M_k = 1$ for $k \in K$. The vertex-operator OPE then gives $u_k(p+q) = u_k(p)u_k(q)$. Thus effectiveness makes K a finite abelian subgroup of $\mathrm{Hom}(\Gamma, U(1))$; it is normal as the kernel of a group action. Lattice self-duality identifies its characters with distinct rational shifts s_k mod Γ.

There are no twisted oscillators in a k sector: its charges are $\Gamma + s_k$ before projection. Define the invariant sublattice

$$\Gamma_0 = \{p \in \Gamma : u_k(p) = 1 \text{ for every } k \in K\}.$$

The pairing between Γ and K makes the map $\Gamma \to \widehat{K}$ onto. Indeed a proper image would have a nontrivial annihilating character, hence a nonidentity element of K acting trivially on every charge, contrary to effectiveness. Therefore $[\Gamma : \Gamma_0] = |K|$.

Projection in each k sector imposes a character condition on the original charge p. Discrete torsion changes which character occurs, but surjectivity still makes the solutions one nonempty coset of Γ_0. The $|K|$ sectors remain disjoint because their s_k mod Γ are distinct. Their union Γ' is additive and even integral by closure and locality of the gauged vertex OPE. Its covolume is

$$\mathrm{covol}(\Gamma') = \frac{\mathrm{covol}(\Gamma_0)}{|K|} = 1.$$

Hence Γ' is again even unimodular of signature $(2,2)$, with one oscillator Fock space per charge. The K orbifold is another rank-two Narain theory. This covolume argument does not require K to be a geometric target-space translation group.

Fields of the full G orbifold supported on K monodromies lift isometrically into this intermediate torus theory, subject to the remaining G equivariance. Their normalized sphere correlators agree. On a punctured sphere the local monodromies generate the holonomies; when these lie in K, the torsion restricts to K, and the gauge-volume factor cancels against vacuum normalization. No extra $\sqrt{|G|}$ multiplies a normalized cubic. We need this subspace and its correlators, without assuming that the remaining quotient acts geometrically.

Next bound the cost of a proper rotation. The two maps $\det R_L, \det R_R : G \to \{\pm 1\}$ are group characters. If both signs are positive, the charge action belongs to the identity component of real $SO(2,2)$. Its integral group is represented by

$$[SL(2,\mathbb{Z}) \times SL(2,\mathbb{Z})]/\{(I,I),(-I,-I)\}.$$

One can see the representation on integral two-by-two matrices with quadratic form minus the determinant: (A,B) acts by $X \mapsto AXB^{-1}$. A finite element has finite lifts A, B, since a suitable power is the same central sign in both factors. A finite-order integral $SL(2)$ matrix has trace $-2, -1, 0, 1$, or 2. At trace ± 2, finite order excludes a nonzero Jordan part, giving $\pm I$; the other traces give orders $3, 4, 6$. Its rotation fraction is thus among

$$0, \quad \frac{1}{2}, \quad \pm\frac{1}{3}, \quad \pm\frac{1}{4}, \quad \pm\frac{1}{6}.$$

For fractions a, b in the two factors, the physical chiral fractions are $a+b, a-b$, up to orientation and interchange. For instance, split the real two-by-two matrices into the planes commuting and anticommuting with the standard complex structure. Left and right rotations subtract their angles on one plane and add them on the other. Reducing each to $[0, 1/2]$ gives the complete nonidentity list:

Chiral rotation fractions	Scalar lower bound
(ν,ν), $\nu = 1/2, 1/3, 1/4, 1/6$	$\nu(1-\nu) \geq 5/36$
$(1/2, 0)$	$1/4$
$(1/3, 0)$	$2/9$
$(1/2, 1/6)$	$1/4$
$(1/12, 5/12)$	$35/144$

The two real chiral bosons rotated through fraction θ have oscillator ground weight $\theta(1-\theta)/2$. This normal-ordering shift can be checked with an exponential regulator, without assigning values to divergent sums. Put

$$S_a(\varepsilon) = \sum_{n\ge 0} (n+a) e^{-\varepsilon(n+a)} = -\partial_\varepsilon \frac{e^{-a\varepsilon}}{1-e^{-\varepsilon}}.$$

Its expansion is $S_a = \varepsilon^{-2} - (a^2 - a + 1/6)/2 + O(\varepsilon)$. The two twisted frequency lists start at θ and $1-\theta$, so their vacuum energy relative to two untwisted lists is

$$h_\theta = \frac{1}{2} \lim_{\varepsilon \downarrow 0} [S_\theta(\varepsilon) + S_{1-\theta}(\varepsilon) - 2S_1(\varepsilon)] = \frac{\theta(1-\theta)}{2}.$$

The divergent terms cancel before taking the limit. Zero-mode and excitation energies are nonnegative. A scalar obeys $h = \bar{h}$, so its dimension satisfies the stronger bound

$$\Delta \ge \max\{\theta_L(1-\theta_L), \theta_R(1-\theta_R)\}.$$

For example the last row cannot use the sum of two unequal oscillator ground weights as a scalar dimension: the smaller side must first rise to meet the larger. Level matching and projection may raise the actual scalar threshold further. Extra shift phases, or a lift of larger order, cannot lower the oscillator floor. Consequently a scalar below $5/36$ whose two determinant signs are positive must be supported in K sectors.

Finally let the cubic select those signs. If $c_\phi = 0$ the bound is immediate. Otherwise the complete real doublet has the nonzero harmonic cubic already described, with orthogonal stabilizer D_3. Every internal symmetry acts within this plane by the full-eigenspace hypothesis. In particular, the abelian quantum-character group $\widehat{G_{\mathrm{ab}}} = \mathrm{Hom}(G, U(1))$ has image $1, C_2$, or C_3 in D_3. A character χ acts on a g-twisted sector by $\chi(g)$, also with ordinary discrete torsion. Indeed rescaling a twisted group basis by $e_g \mapsto \chi(g) e_g$ preserves its torsion-modified multiplication because $\chi(g)\chi(h) = \chi(gh)$.

If the image is trivial, choose a unit real direction maximizing the cubic. If it is C_2, choose its reflection-fixed axis. In either case a unit Hermitian X is fixed by all quantum characters and satisfies $|C_{XXX}| = |c_\phi|/\sqrt{2}$. Characters separate the abelianization, so every supporting monodromy lies in $[G, G]$. Both determinant characters are then positive on it. The rotation bound puts the entire support in K.

If the image is C_3, the two complex lines are simultaneous quantum-character eigenlines. The group D_3 has a unique order-three subgroup, so these are also the physical charged lines spanned by ϕ and $\phi^\dagger$. Finite abelian duality identifies their characters with inverse elements of exact order three in G_{ab}. Every determinant character has order at most two and is therefore trivial on those elements. Both signs are again positive, and the support of ϕ lies in K. This is why low reflection-twisted sectors cannot escape the argument by mixing with untwisted fields. We never assumed that the physical order-three symmetry equals a quantum symmetry.

In the first two cases, lift X to the intermediate Narain theory and apply the Hermitian bound $|C_{XXX}| \le \sqrt{2/3}$. Its fixed-axis normalization gives $|c_\phi| \le \sqrt{2}\sqrt{2/3} = 2/\sqrt{3}$. In the C_3 case, lift ϕ itself and apply the complex bound $2/\sqrt{3}$ directly. All lifts preserve the dimension and normalized cubic, completing the proof. □

Worked example 24.4 (A shift orbifold changes the radius)**.** The kernel step has a familiar one-boson model. Use the circle convention of Chapter 2:

$$h = \frac{1}{2}(m/R + wR/2)^2, \qquad \bar{h} = \frac{1}{2}(m/R - wR/2)^2, \qquad m, w \in \mathbb{Z},$$

gauge the translation $X \mapsto X + 2\pi R/n$. It acts on a momentum vertex by $e^{2\pi i m/n}$, so projection requires $m = nm'$. The k-twisted sector has $w \in \mathbb{Z} + k/n$; combining all sectors gives $w = w'/n$

with $w' \in \mathbb{Z}$. Substitution yields the same weight formulas with radius $R' = R/n$ and integer charges m', w'. No oscillator has been twisted. The smaller invariant momentum lattice and the new fractional winding sectors restore an even self-dual charge lattice together. Keeping only the invariant untwisted states would omit the second half of this construction.

24.4.2 What happens when currents are enhanced

At a rank-two torus point, chiral charges of squared norm two form a simply laced root system of rank at most two. The maximal weight-one Lie algebra on either chirality is therefore one of

$$\mathfrak{u}(1)^2, \qquad \mathfrak{su}(2)_1 \oplus \mathfrak{u}(1), \qquad \mathfrak{su}(2)_1^{\oplus 2}, \qquad \mathfrak{su}(3)_1.$$

Their affine and Heisenberg stresses exhaust charge two. A finite symmetry preserves the full current algebra, but may move its chosen Cartan. This is the gap left by the preceding proposition, not a reason to apply it without checking its hypothesis.

Construction input 24.5 (Completed enhanced-parent inventory)**.** For an ordinary consistent effective finite orbifold of any unitary rank-two Narain theory, including enhanced parents, asymmetric chiral actions and permitted discrete torsion, an isolated order-three scalar doublet obeys

$$|c_\phi| \le 2/\sqrt{3} \quad \text{if} \quad \frac{13}{100} < \Delta < \frac{61}{450}.$$

The completed October 1 inventory combines the preceding current-preserving theorem with specialized finite-group classification and twisted-spectrum lemmas. Those additional lemmas are imported here; the following case split explains their role without claiming to reproduce their full proofs.

The additional inputs have a simple interpretation. A current line preserved up to sign can be squared to give a self-adjoint invariant operator on the isolated scalar plane. When the sign-fixing kernel has a complementary $SU(2)$ projection with element orders at most five, its spectral restrictions force the cubic to vanish for $1/8 < \Delta < 9/50$. On an A_2 chirality, bounded inner projective orders at most seven instead give no scalar at all for $13/100 < \Delta < 5/36$, including actual outer extensions. For two A_1 factors with bounded projections, the relevant scalars are odd under the quantum character of factor exchange, so their cubic vanishes in the required interval. These results control cases where no invariant Cartan is immediately visible.

The current-algebra inventory then proceeds as follows.

- For $\mathfrak{u}(1)^2$, the whole current space already is an invariant Heisenberg space.
- For $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$, the central line is characteristic. A finite $SO(3)$ projection either has orders at most five, invoking the current-line test, or is cyclic or dihedral with a preserved rotation axis. That axis and the central line give an invariant Cartan.
- For two $\mathfrak{su}(2)$ factors, first take the kernel of the factor permutation. Two bounded projections use the two-factor test. Two cyclic or dihedral projections admit axes compatible with any exchange: choose one axis and transport it by an exchange, whose square belongs to the kernel. If just one projection has large order, the factors cannot be exchanged; its axis is a preserved current line and the other projection is bounded.
- For $\mathfrak{su}(3)$, the finite inner image can be primitive irreducible, imprimitive, or reducible. The primitive exceptional groups have bounded projective orders and use the spectral test. Apart from bounded exceptions, an irreducible imprimitive group has a unique invariant triangle of complex lines; every normalizer preserves that triangle and hence its Cartan. Reducible $2 + 1$ blocks have a characteristic central current line. Three distinct

joint-character lines give a Cartan directly, while repeated joint characters fall under the current-line or bounded-order cases.

Outer $SU(3)$ automorphisms are semilinear in this discussion: they can conjugate the complex coordinates. Uniqueness of an invariant triangle makes its preservation by such a normalizer valid; merely choosing a convenient triangle would not suffice. The exceptional small cases are handled by the separate bounded spectral lemmas. Left and right current algebras may have different enhancements. Apply the inventory to each independently, then conjugate their invariant Cartans by the connected current groups. This is a change to an isomorphic toroidal presentation, after which Proposition 24.3 applies.

The interval in Construction input 24.5 lies inside every component interval. Its upper endpoint $61/450$ is excluded because it can occur in a bounded two-factor spectrum. Neither the inventory nor the simpler doublet theorem classifies all compact $c = 2$ CFTs, generalized categorical gaugings, ineffective topological factors, or noncompact theories.

For comparison, the illustrative continuum hypothesis $\Delta_\phi = .134$, together with a limiting cubic near the finite estimate $|c_\phi| \simeq 1.584$, would violate the bound in both stated intervals if established. They are not confidence intervals. Most importantly, isolation in one measured charge sector is not isolation of the entire continuum scalar shell. An additional exactly degenerate scalar removes a premise even when it has a different charge. These theorems supply conditional exclusions of construction classes, not an identification of the chain's continuum.

24.5 Ordinary crossing can demand an additional scalar

Crossing is useful without first guessing a named CFT. Choose the Hermitian field $X = (\phi + \phi^\dagger)/\sqrt{2}$ and make the charged self cubic positive by a field phase. Restrict its four-point function to a line. Unitarity gives a positive diagonal $SL(2,\mathbb{R})$ block expansion with

$$g_D(z) = z^D\, {}_2F_1(D, D; 2D; z), \qquad g_0 = 1.$$

Writing $\delta = \Delta_\phi$, define crossing vectors

$$F_D(z) = (1-z)^{2\delta} g_D(z) - z^{2\delta} g_D(1-z).$$

Crossing is $\sum_D a_D F_D = 0$ with $a_D \geq 0$. The known identity, charged-doublet, and epsilon terms have weights

$$a_0 = 1, \qquad a_\delta = |c_\phi|^2/2, \qquad a_e = g_\phi^2,$$

where $e = \Delta_\epsilon$ and $g_\phi = C(\phi, \phi^\dagger, \epsilon)$.

A linear functional can separate allowed spectra. At $z = 1/2$, put $C = D(D-1)$, $r_D = g'_D/g_D$, and $\kappa = 6/5$. The block Casimir equation gives $g''_D = 2g'_D + 8Cg_D$ and $g'''_D = 8(C+1)g'_D$. Direct differentiation therefore yields

$$\begin{aligned} \mathcal{L}(F) &= \frac{F''' - 8\kappa F'}{16\, 2^{-2\delta}}, \\ \frac{\mathcal{L}(F_D)}{g_D} &= (C + A - \kappa) r_D - 4\delta(3C + B - \kappa), \\ A &= 1 - 6\delta + 6\delta^2, \qquad B = 1 - 3\delta + 2\delta^2. \end{aligned}$$

Euler's integral for the hypergeometric function implies

$$2\sqrt{2}D \leq r_D \leq 3D, \qquad (2/3)^D \leq g_D(1/2) \leq (1 + 2^{-D})/2.$$

These are bounds at every positive dimension, not estimates at a finite sample of dimensions.

To derive them, let t have the symmetric beta distribution with density proportional to $t^{D-1}(1-t)^{D-1}$ on $[0,1]$. Euler's integral writes

$$g_D(1/2) = 2^{-D}\mathbb{E}[(1-t/2)^{-D}].$$

The weight $w(t) = (1-t/2)^{-D}$ is increasing. In the probability measure reweighted by w, its mean μ therefore satisfies $\mu \geq 1/2$: the covariance of two increasing real functions is nonnegative. Integrating the derivative of $t^D(1-t)^D(1-t/2)^{-D}$ gives

$$\mathbb{E}_w\left[(2-t) - \frac{2}{2-t}\right] = 0.$$

Jensen's inequality for $u \mapsto 1/u$ implies $(2-\mu)^2 \geq 2$, hence $\mu \leq 2-\sqrt{2}$. Differentiating Euler's representation of g_D and using the same identity yields $r_D = 4D - 2D\mu$. This proves the two derivative bounds. Finally, convexity bounds $w(\mathbb{E}t)$ below $\mathbb{E}w$, while the chord joining $w(0)$ and $w(1)$ bounds it above. Multiplication by 2^{-D} gives the two block bounds.

For $\delta \in [.13, .14]$, substituting the derivative bounds proves $\mathcal{L}(F_D) > 0$ for $0 < D \leq .16$ and for $D \geq 1.75$. For the latter range, $C \geq 21/16$ makes the coefficient of r_D positive; the lower derivative bound gives a positive polynomial lower bound. Explicitly, using $2\sqrt{2} > 113/40$ and $D \geq 7/4$,

$$\frac{\mathcal{L}(F_D)}{g_D} \geq \left(\frac{791}{160} - \frac{42}{25}\right)C - \frac{791}{160}\frac{1153}{1250} + \frac{36153}{125000} \geq .012780875 > 0.$$

The coefficient of C is positive, so its minimum occurs at $C = 21/16$; there is no untested high-dimension tail. For the former range, the coefficient of r_D is negative, and the upper derivative bound gives

$$3D^3 - 4.68D^2 - 1.0872D + .289224,$$

which decreases to the positive value .007752 at $D = .16$.

Worked example 24.6 (A hand crossing certificate)**.** Assume the illustrative continuum values

$$\delta = .134, \quad e = .263, \quad |c_\phi| = 1.584, \quad g_\phi = 1.3.$$

The same bounds give $\mathcal{L}(F_0) > .303$, $\mathcal{L}(F_\delta) > .9 \times .083$, and $\mathcal{L}(F_e) > -(37/40) \times .246$. Since $|c_\phi|^2/2 > 5/4$, the known contribution is greater than

$$.303 + \frac{5}{4}\frac{9}{10}(.083) - \frac{169}{100}\frac{37}{40}(.246) = .0118155 > 0.$$

All numbers in this inequality are terminating rationals. If every other exchange lay outside $(.16, 1.75)$, positivity would make the whole functional crossing sum positive, a contradiction. Hence an additional coupled exchange is required in that interval.

For identical bosonic scalars, nonzero-spin primaries have even integer spin and hence dimension at least two by unitarity. Below 1.75, the required exchange is therefore scalar. One must still check that it is not merely a new diagonal $SL(2,\mathbb{R})$ primary inside an already listed two-dimensional conformal family. For a scalar family, the left and right global weights agree. Their tensor product decomposes into diagonal primaries at relative levels $n = 0, 1, 2, \ldots$, with interchange parity $(-1)^n$. The scalar three-point tensor is symmetric in the two equal factors, so its odd-relative-level coefficients vanish. The first extra coupled diagonal primary is at $\Delta + 2$. A new global primary inside a Virasoro module also first occurs at level two, since level one is the ordinary translation descendant. Neither mechanism can produce the required exchange below 1.75. The conclusion is an additional two-dimensional scalar Virasoro primary with a nonzero $X \times X$ coupling, possibly degenerate with a supplied scalar.

The displayed numbers are hypotheses about continuum data, not confidence intervals extracted from the chain. Moreover, the finite chain already contains an appropriate higher charged candidate. Its saved $L = 15$ dimension in one prescribed scale is about .731, with a nonzero coefficient around .45. Including it changes the finite functional sum's sign. Thus the crossing demand is compatible with an observed extra channel. Partial sums neither solve crossing nor show a violation; descendants and the high spectrum remain.

24.6 A nonreal invariant and the scope of a Cardy test

The light three-point tensor can constrain more than individual OPE magnitudes. Assume the complete real scalar shells have dimensions $2, 2, 4$ for the phi, chi, and psi/eta groups, with the charge pattern above, nonzero self cubics, and the stated physical parity. The nonzero self cubics restrict the orthogonal transformations on the two light planes to D_3. On the four-plane, the cubic is proportional to $\Re(z^3 + w^3)$; its Hessian determinant is proportional to $|z|^2|w|^2$, forcing a symmetry to preserve or exchange the two coordinate planes. The nonzero mixed triangles then recover the normal $C_3 \times C_3$ subgroup.

If $\Im I_\lambda \neq 0$, simultaneous inversion of both charge axes is forbidden. With the given parity, the image of all scalar-tensor symmetries on these shells is $C_{3,\alpha} \times D_{3,S}$. This classifies an image on selected fields, not a possible kernel acting only on higher states, and it does not turn spatial parity into an orientation-preserving internal line.

One consequence concerns a specific rational full-CFT construction. Assume a unitary bosonic rational chiral algebra V with positive nonvacuum lowest weights and its canonical $A = 1$ Cardy sewing,

$$\mathcal{H}_{\text{bulk}} = \bigoplus_i M_i \otimes \overline{M_i^*}.$$

The canonical construction includes the OPE multiplication and sewing of Chapter 17, not only the displayed Hilbert-space sum. The positivity hypothesis ensures that V is the maximal holomorphic algebra of this full theory: only the right vacuum module can contain a state of right weight zero. Every internal symmetry preserves this maximal left algebra and its right analogue setwise, even if it acts nontrivially on their fields. It therefore permutes the irreducible module-pair summands. Without this step, counting summands of a chosen smaller chiral algebra would not constrain an arbitrary physical symmetry.

At a scalar weight $0 < h < 1$, all states in such a summand are chiral ground states. A module ground space of dimension n_i contributes n_i^2 bulk directions. An isolated real charged doublet must therefore arise from a nonselfdual pair with one-dimensional ground spaces: two isolated selfdual real lines cannot carry a nontrivial order-three rotation by permutation.

The canonical identity-algebra sewing has a spatial parity exchanging the two chiral sides. On each such doublet it conjugates the charge. Applying this to both light doublets gives simultaneous inversion and forces I_λ real under the full-shell and mixed-tensor hypotheses. A nonreal limiting value would exclude that canonical completion.

This is not a theorem that every unitary CFT has real three-point coefficients, nor an exclusion of every rational CFT. Other full algebras, other sewing data, additional degenerate fields, or a different limit lie outside its assumptions. Even a charge-conjugation torus partition function alone is weaker than the specified canonical sewing hypothesis.

24.7 A dimension obstruction for rational lattice orbifolds

The scalar-shell arguments above need detailed low-energy information. A different test uses only the finite category of defects, provided one specifies the chiral algebra that those defects preserve.

It extends the pointed-lattice argument of Chapter 18 to rational finite orbifolds, including their twisted sectors.

Let L be an even positive-definite lattice and G a finite faithful group of unitary automorphisms of V_L. Put $W = V_L^G$. Assume that W is rational and C_2-cofinite, and that every irreducible g-twisted V_L-module, except the untwisted vacuum module itself, has strictly positive lowest weight. These are the hypotheses of the orbifold global-dimension theorem being used; finiteness of G alone is not being offered as a proof of them. The theorem gives

$$\operatorname{FPdim}\operatorname{Rep}(W) = |G|^2 \operatorname{FPdim}\operatorname{Rep}(V_L) = |G|^2|L^*/L|.$$

The last equality follows because all lattice modules are invertible and have dimension one.

Proposition 24.7 (The retained rational-orbifold algebra matters)**.** *Assume the preceding orbifold hypotheses and a unitary rational full CFT described over $\mathcal{C} = \operatorname{Rep}(W)$ by a simple separable Frobenius algebra A, with the usual local sewing construction. Its W-preserving defect category cannot contain $\mathcal{H}_3$ as a full fusion subcategory.*

Proof. These defects form the fusion category $\mathcal{D} = {}_A\mathcal{C}_A$. Here simplicity of A means simplicity as an A–A bimodule: $\operatorname{End}_{A-A}(A) = \mathbb{C}$, so the defect unit is simple. The categories $\mathcal{D}$ and $\mathcal{C}$ are Morita equivalent, so their Frobenius–Perron global dimensions agree. Therefore $\operatorname{FPdim}\mathcal{D} = |G|^2|L^*/L|$ is an integer. A fusion category with integer global Frobenius–Perron dimension is called *weakly integral*; the corresponding dimension theorem says that every simple X satisfies $\operatorname{FPdim}(X)^2 \in \mathbb{Z}$. The Haagerup simple ρ violates this necessary condition:

$$\operatorname{FPdim}(\rho)^2 = d^2 = \frac{11 + 3\sqrt{13}}{2} \notin \mathbb{Z}.$$

A full fusion embedding preserves simplicity and these positive dimensions, giving a contradiction. □

The simplicity qualification is important: a sum of simple objects can have irrational squared dimension even in a weakly integral category. For a genuine unitary strong tensor action of $\mathcal{H}_3$, however, Proposition 18.8 already proves full faithfulness into a semisimple rigid target with simple unit. Thus sending ρ to a reducible defect does not evade this particular Haagerup obstruction.

Worked example 24.8 (A finite-group center has noninvertible objects)**.** Consider the untwisted group category $Z(\mathrm{Vec}_{S_3})$. Its simples are labeled by a conjugacy class $[g]$ and an irreducible representation π of its centralizer; their dimensions are $|[g]| \dim \pi$. The identity class gives dimensions $1, 1, 2$. The transposition class has size three and centralizer $\mathbb{Z}_2$, giving $3, 3$. The three-cycle class has size two and centralizer $\mathbb{Z}_3$, giving $2, 2, 2$. Consequently

$$\operatorname{FPdim} Z(\mathrm{Vec}_{S_3}) = 1^2 + 1^2 + 2^2 + 3^2 + 3^2 + 2^2 + 2^2 + 2^2 = 36 = |S_3|^2.$$

Finite-group constructions can therefore produce noninvertible sectors while still obeying strong arithmetic restrictions. Weak integrality does not require every simple dimension to be an integer: the Ising dimensions $1, 1, \sqrt{2}$ illustrate the distinction. Their squares remain integers, unlike the Haagerup value above.

The algebra A may encode a non-diagonal pairing or a local extension; the proof is not restricted to the Cardy choice $A = \mathbf{1}$. It does require the proposed defects to preserve W on both chiralities. Preserving only the total stress is weaker, and does not place a defect in ${}_A\mathcal{C}_A$. Nonrational radii or orbifolds, a new action breaking W, and asymmetric constructions without this specified rational category remain outside the argument. Conversely, within its stated framework the exclusion needs no extrapolated scalar gap, isolated doublet, or numerical OPE coefficient.

24.8 Nearby rational theories: test the retained algebra

A central charge is a coarse invariant. The rational numbers 21/10, 25/12, 13/6, and 44/21 all lie near the finite Haagerup diagnostics, but a theory at one of those charges must still have the required symmetries and scalar spectrum. We now describe a second family of tests, complementary to the canonical Cardy phase test: retain a specified rational chiral algebra, allow its bosonic local extensions and physical non-diagonal pairings, and ask whether any such full theory has enough internal symmetry. Throughout this test assume a bosonic unitary rational full CFT with a unique vacuum, retaining the stated algebra in both chiralities and satisfying the usual completeness and sewing assumptions.

The distinction between a retained algebra and a number is essential. Excluding full theories containing one particular W-algebra does not exclude every theory with its central charge. A deformation can also destroy the proposed chiral subalgebra, even when its ultraviolet description used it.

24.8.1 Why two order-three actions can be compulsory

Suppose the ordinary internal part S of translation survives faithfully, has order three, and commutes with the full Haagerup action. It cannot equal α or α^2, because

$$\alpha^k \rho = \rho \alpha^{-k}$$

and the two objects $\rho\alpha^k, \rho\alpha^{-k}$ are distinct for $k = 1, 2$. Under the stated faithfulness assumptions, α, S generate $\mathbb{Z}_3^2$. If one then gauges a suitable order-three subgroup, a mixed anomaly can alter the extension law and produce $\mathbb{Z}_9$ rather than a direct product. An obstruction to every subgroup of order nine covers both possibilities. This discussion assumes a precise original-versus-gauged field dictionary; a scalar in an imposed spatially twisted sector is not automatically a local scalar of the original theory.

24.8.2 The extension and symmetry strategy

There are several logically separate steps. Integer conformal weight is necessary for a module to occur in a bosonic chiral extension, because the algebra object must have trivial twist. It is not sufficient: multiplication, associativity, commutativity, and positivity still constrain the support. A useful additional necessary condition is total positivity of each constituent's quantum dimension under all number-field embeddings. Once the actual extension algebras are classified, their chiral automorphisms and their invertible representation objects can be studied.

For a physical rational full CFT, pass to the maximal left and right chiral algebras that actually occur. The full theory then has a multiplicity-one graph pairing

$$\mathcal{H} = \bigoplus_X X \otimes \overline{F(X)}$$

for a braided equivalence of the corresponding representation categories, with its associative OPE multiplication. A symmetry fixing both chiral algebras pointwise acts by a phase λ_X on each summand. Nonzero OPE channels require

$$\lambda_X \lambda_Y = \lambda_Z \quad \text{whenever } N_{XY}^Z > 0.$$

Thus these phases are tensor automorphisms of the identity. In a modular category their group is identified with the invertible objects through the nondegenerate braiding. Bounding that group and the chiral automorphism images bounds the full internal symmetry. The OPE assumption is important: an arbitrary formal matrix commuting with S, T need not provide this graph algebra.

Worked example 24.9 (All minimal-model products at $c = 21/10$)**.** The unitary Virasoro minimal charges are $c_m = 1 - 6/[m(m+1)]$, $m \geq 3$; write $M_m = M(m, m+1)$. A product at $21/10$ must have exactly three nontrivial factors. Two have charge below two; five have charge at least $5/2$. Four Ising factors give two, and replacing any by the next model raises the charge by at least $1/5$, overshooting $21/10$.

Order the three labels. If the smallest is at least five, the total is at least $12/5$, which is too large. If the smallest is four, all three must be four. If it is three, the other charges sum to $8/5$. Taking the next label to be three, four, or five would require the last charge to be respectively $11/10$, $9/10$, or $4/5$. Only the last occurs, since $c_m = 9/10$ would require $m(m+1) = 60$. Larger next labels already overshoot. Hence the complete product list is

$$M_4^{\otimes 3}, \qquad M_3 \otimes M_5 \otimes M_5.$$

This elementary classification is the beginning of the test, not its conclusion; non-diagonal pairings and extensions must still be included.

For three tricritical Ising factors, the only possible local extension supports lie in the even products of their three weight-$3/2$ fermion currents, a bosonic $\mathbb{Z}_2^2$. These add no weight-two fields. Chiral automorphisms can therefore permute the three primitive stress idempotents, with a two-group kernel. Their order-three part is at most one three-cycle. That cycle acts faithfully on suitable module classes, so a multiplicity-one left-right graph pairing cannot admit independent left and right three-cycles.

There is also a short arithmetic check on the remaining symmetry kernel. Every symmetry in that kernel is an invertible defect preserving the original product algebra $V = M_4^{\otimes 3}$. All V-preserving defects form a bimodule fusion category Morita equivalent to $\mathrm{Rep}(V)$, of global dimension $64(\varphi+2)^3$, where $\varphi = (1+\sqrt{5})/2$. The smaller symmetry kernel need not be that whole category. An invertible order-three defect would give a fusion subcategory of dimension three. Fusion-subcategory divisibility would force $64(\varphi+2)^3/3$ to be an algebraic integer. Its field norm is

$$\frac{64^2\, 5^3}{9},$$

which is not an integer, a contradiction. Combining the chiral and kernel arguments excludes an internal subgroup of order nine throughout this retained-algebra family. The norm calculation is not a statement that every category of nonintegral dimension has no group symmetry; it uses this particular algebraic integer.

For $M_3 \otimes M_5^{\otimes 2}$, the only integer-weight supports are the two weight-three M_5 currents and their product. Thus the possible extensions are indexed by the five subgroups of $\mathbb{Z}_2^2$. Resolving the local modules shows that order-nine symmetry forces both M_5 factors to extend to Potts. The resulting retained algebra is Ising times two Potts algebras. It necessarily contains a unique scalar

$$\sigma_{\mathrm{I}} \otimes 1 \otimes 1, \qquad \Delta = 1/8,$$

in every physical pairing. To see the last assertion directly, Potts-pair weight residues have denominator fifteen, whereas differences of distinct Ising weights $0, 1/2, 1/16$ do not. T-commutation separates three Ising blocks N_i of the modular invariant. The vacuum diagonal block of S-commutation says that N_0 commutes with the Potts-pair matrix S_{P}. The off-diagonal vacuum-row equation is

$$(S_{\mathrm{I}})_{0i} S_{\mathrm{P}} N_i = N_0 S_{\mathrm{P}} (S_{\mathrm{I}})_{0i}.$$

The nonzero Ising vacuum row and invertibility of S_{P} therefore give $N_i = N_0$: the invariant is $I_{\mathrm{I}} \otimes N_0$. This retains the Ising spin field with the vacuum Potts pair. No other scalar occurs at $1/8$, since the smallest nonvacuum Potts chiral weight is $1/15 > 1/16$. A unique Hermitian scalar is fixed by any odd-order symmetry: its real one-dimensional orthogonal action is only a sign.

Consequently this second family conflicts with a limiting jointly neutral scalar gap above 1/8, if both order-three actions survive. The chain's finite neutral gap near .26 motivates the test; its limit remains an additional premise.

24.8.3 Why a nearby ordinary W_3 model is not enough

The unitary minimal algebras $W_3(p, p+1)$, $p \geq 4$, have

$$c_p = 2 - \frac{24}{p(p+1)}.$$

They approach two from below. Approximate central-charge agreement near two therefore motivates testing this family, while an established charge strictly above two would already distinguish it. Here we ask whether they can support the required defects while those defects preserve this rational W_3 algebra. Throughout this subsection, a full theory means the unitary rational sewing construction from a separable Frobenius algebra A that is simple as an A–A bimodule.

Here is a first arithmetic filter that can be understood without the classification of full theories. Put

$$K_p = \mathbb{Q}(\zeta_{2p}, \zeta_{2(p+1)}), \qquad \zeta_n = e^{2\pi i/n}.$$

The positive quantum dimensions of the chiral simple objects belong to K_p. In the rational sewing description by a simple separable Frobenius algebra A, a defect X has normalized dimension

$$\mathrm{FPdim}_{{}_A\mathcal{C}_A} X = \frac{\mathrm{FPdim}_{\mathcal{C}} X}{\mathrm{FPdim}_{\mathcal{C}} A} \in K_p, \qquad \mathcal{C} = \mathrm{Rep}(W_3(p, p+1)).$$

The numerator and denominator use the underlying objects in $\mathcal{C}$. This also applies to a reducible defect, since sums stay in the same field.

Only primes dividing $2p(p+1)$ can ramify in this cyclotomic field. On the other hand, $\mathbb{Q}(\sqrt{13})$ has discriminant thirteen, so thirteen ramifies in it. The fact used here is that a subfield cannot introduce a ramified prime absent from its containing field. Since $\sqrt{13} = 2d - 3$, a defect of Haagerup dimension $d = (3 + \sqrt{13})/2$ would therefore require

$$13 \mid p(p+1).$$

The first surviving height is $p = 12$, with $c_p = 24/13$. This is a necessary test, not a proof that this height works; it still leaves infinitely many heights, with both adjacent heights at least twelve.

An elementary gap controls ordinary affine dimensions. For affine $SU(3)$, let the *height* n mean level $n - 3$. Shifted labels are positive integers a, b with $a + b < n$, and

$$D_n(a, b) = \frac{\sin(\pi a/n)\sin(\pi b/n)\sin(\pi(a+b)/n)}{\sin^2(\pi/n)\sin(2\pi/n)}.$$

The W_3 dimensions are products from its two adjacent heights; simultaneous simple-current field identification leaves them unchanged. The vacuum label is $(1, 1)$. The expression is symmetric in the positive triple $(a, b, n-a-b)$, so sort it with $a \leq b \leq n-a-b$. The invertible orbit has $(a, b) = (1, 1)$ and dimension one. The fundamental and its conjugate orbits have

$$f_n = D_n(1, 2) = 1 + 2\cos(2\pi/n), \qquad 1 + \sqrt{3} \leq f_n < 3 \quad (n \geq 12).$$

Every other sorted label has $a = 1, b \geq 3$, or $a, b \geq 2$. Reducing to $(1, 3)$ or $(2, 2)$ can only lower the dimension: the derivative signs of the numerator with respect to a, b are those

of $\sin((2a+b)\pi/n)$ and $\sin((a+2b)\pi/n)$, nonnegative in the sorted region. Moreover, with $[m] = \sin(m\pi/n)/\sin(\pi/n)$,

$$\frac{D_n(2,2)}{D_n(1,3)} = \frac{[2]^2}{[3]} > 1, \qquad D_n(1,3) = 2\cos(2\pi/n) + 4\cos^2(2\pi/n) \geq 3+\sqrt{3}.$$

Since $3 < d < 3+\sqrt{3}$, no affine simple at these heights has dimension d. Two noninvertible factors have product at least $(1+\sqrt{3})^2 > d$. This handles the diagonal Cardy theory, but not yet every sewing algebra.

Construction input 24.10 (Categorical classification inputs for this test)**.** The unitary diagonal affine coset has its standard full branching rule and extension/local-module correspondence. The complete W_3 modular invariant classification has exceptional affine heights $8, 12, 24$. At heights at least twelve away from these exceptions, the indecomposable module categories of the ordinary affine $SU(3)$ fusion category are the simple-current cases and their products with the invertible charge-conjugation module category. Tensoring by the latter preserves the dual fusion category up to equivalence. These are classification inputs; the reduction below explains why they apply to the actual sewing algebras here.

Pass from torus matrices to actual sewing algebras. Choose the two adjacent heights as n, q, with $3 \nmid q$; their order need not be increasing. Let $\mathcal{A}_n$ be the affine $SU(3)$ category at height n, and $\mathcal{D}_q$ the triality-zero subcategory at height q. Triality is the central $\mathbb{Z}_3$ charge, equal to $a+2b$ modulo three in shifted labels. Since $3 \nmid q$, the pointed $\mathbb{Z}_3$ subcategory $\mathcal{P}_q$ is nondegenerate and $\mathcal{A}_q \simeq \mathcal{P}_q \boxtimes \mathcal{D}_q$.

The coset/extension theorem realizes the W_3 category as local modules for the diagonal $\mathbb{Z}_3$ graph algebra in

$$\mathcal{A}_p \boxtimes \mathcal{A}_4 \boxtimes \mathcal{A}_{p+1}^{\mathrm{rev}}.$$

This is an algebraic realization, not an inference from matching S, T. Its reconstruction premise holds because the denominator-vacuum branch of the numerator vacuum contains only the W_3 vacuum. The branching rule reconstructs exactly the three simultaneous simple-current translates of the vacuum. Here charge conjugation on the reversed factor lets us write the same current exponent in all three graph labels. Their graph multiplication is unique up to isomorphism because $H^2(\mathbb{Z}_3, \mathbb{C}^*) = 0$.

After extracting $\mathcal{D}_q$, take objects with trivial $\mathcal{P}_q$ component. Locality uniquely determines their $\mathcal{A}_4$ component from their triality. This correction changes the braiding but not the affine associator: the pointed height-four category has a realization with trivial associator and braiding $c(a,b) = e^{2\pi i ab/3}$ on $\mathbb{Z}_3$. Induction is fully faithful on these objects, because its Hom formula sums over the three graph translates and only the trivial translate has trivial $\mathcal{P}_q$ component. The local-category dimension is $\mathrm{FPdim}(\mathcal{A}_n)\, 3\, 3/3^2 = \mathrm{FPdim}(\mathcal{A}_n)$, so no additional objects remain. Consequently

$$\mathcal{C} \simeq \widetilde{\mathcal{A}}_n \boxtimes \mathcal{D}_q^{\pm},$$

where $\widetilde{\mathcal{A}}_n$ is monoidally equivalent to ordinary $\mathcal{A}_n$, with a triality correction to its braiding; the sign allows reversal of braiding. Neither change alters dimensions.

Let $\mathcal{L}$ be the Lagrangian full center of any sewing algebra. Among the conductor-compatible heights, the classified coefficient matrices have the form $M_n \otimes I_{\mathcal{D}_q}$, up to label conjugation. Intersect $\mathcal{L}$ with $\mathcal{D}_q \boxtimes \mathcal{D}_q^{\mathrm{rev}}$, interchanging these two factors if the braiding was reversed. The intersection theorem makes this a commutative separable algebra, also called an *étale algebra*, not just a subobject. Its graph underlying object has dimension $\mathrm{FPdim}(\mathcal{D}_q)$, so it is already Lagrangian in that factor. After condensing it, the factor has local category Vec: no nontrivial object remains for a mixed gluing. The classification of connected étale algebras in a product therefore factors $\mathcal{L}$ as an algebra. A possible graph autoequivalence gives an invertible module

category and does not change dimensions. Thus every simple defect has dimension uv, where u is a simple dimension in a Morita dual of ordinary $\mathcal{A}_n$, and v is an ordinary dimension in $\mathcal{D}_q$. This is the step that a factorized partition function alone would miss.

All nonexceptional heights now follow from the gap. In the classified simple-current cases, the algebra has support 1 or $1 + J + J^2$. A length-three orbit gives defect dimension $D_n(a, b)$. At a fixed point the possible dimensions are $D_n(a, b)/3$ or $D_n(a, b)$. The stabilizer of the combined left and right action is $\mathbb{Z}_3$ on a free orbit, and $\mathbb{Z}_3^2$ at a fixed point. Its projective irreducible representations have dimension one, or dimension three when the latter group's commutator pairing is nondegenerate. Dividing the underlying-object dimension by $\mathrm{FPdim}\, A = 3$ gives the stated possibilities. A fixed point requires $3 \mid n$ and $(a, b) = (n/3, n/3)$, and

$$\frac{D_n(n/3, n/3)}{3} = \frac{\sqrt{3}}{8 \sin^2(\pi/n) \sin(2\pi/n)} \geq 3 + 2\sqrt{3} > d.$$

Every allowed u or v is therefore one, lies in $[1 + \sqrt{3}, 3)$, or exceeds d; two noninvertible factors are too large. The affine classification includes all module categories at these heights, including the charge-conjugation cases, so no sewing algebra is omitted.

Only the pair $(12, 13)$ *requires a separate finite input.* The other exceptional heights, eight and twenty-four, have no adjacent height divisible by thirteen. At height twelve, all affine dimensions lie in $\mathbb{Q}(\sqrt{3})$: the fundamental and its dual generate the fusion ring and both have dimension $1 + \sqrt{3}$. Every Morita-dual simple dimension u also lies there, by the dimension-ratio formula. At height thirteen, $v = D_{13}(a, b) \in \mathbb{Q}(\zeta_{13})$, whereas

$$\mathbb{Q}(\sqrt{3}) \cap \mathbb{Q}(\zeta_{13}) = \mathbb{Q}.$$

If $uv = d$, then $u = d/v$ belongs to both fields, so u is rational and v must be a rational multiple of d.

Construction input 24.11 (The exact height-thirteen comparison)**.** For all sixty-six pairs $a, b > 0$, $a + b < 13$, the three invertible labels $(1, 1), (1, 11), (11, 1)$ have dimension one; all other sixty-three dimensions have degree six over $\mathbb{Q}$. None is therefore a rational multiple of the quadratic number d.

The finite verification takes place exactly in $\mathbb{Q}[z]/(1 + z + \cdots + z^{12})$, with $z = \zeta_{13}$ and $q_{13} = -z^7 = e^{\pi i/13}$. It evaluates $[m] = (q_{13}^m - q_{13}^{-m})/(q_{13} - q_{13}^{-1})$ and $D_{13}(a, b) = [a][b][a + b]/[2]$. Galois substitution gives exact degrees; reality and invariance under $z \mapsto z^3$ test membership in $\mathbb{Q}(\sqrt{13})$. This is an exhaustive finite algebraic input, not a comparison of approximate decimal dimensions. The textbook imports its archived exact certificate without replaying that calculation; Appendix C explains this scope.

For example, the fundamental value is $D_{13}(1, 2) = 1 + z + z^{-1}$. It generates the real subfield of $\mathbb{Q}(\zeta_{13})$, which has degree six; the complete finite input extends this check to every other label, including labels outside the triality-zero subcategory.

This disposes of every sewing algebra at the exceptional pair, including both exceptional modular-invariant types at height twelve. The argument even excludes a reducible defect of dimension d. In the nonexceptional cases every noninvertible simple has dimension at least $1 + \sqrt{3} > d - 1$, so such an object could only be a single noninvertible simple, already excluded, or a sum of invertibles of integer dimension. In the exceptional case a summand with noninvertible v again exceeds $d - 1$ and would have to be the whole object; if every v is invertible, the sum stays in $\mathbb{Q}(\sqrt{3})$.

Thus a unitary strong tensor Haagerup action preserving the stated rational W_3 algebra is excluded for every $p \geq 4$ and every rational sewing algebra in the stated framework. A tensor functor preserves the positive dimension, so its image of ρ would have supplied the forbidden

dimension d. This says nothing about Virasoro-only defects of a W_3 theory or another CFT at the same central charge. It is also different from Chapter 18, where an explicitly constructed irrational factor lacked an independent spin-three vacuum generator.

24.8.4 Principal W algebras and an arithmetic filter

The unitary principal type-A charges are

$$c(N,k) = (N-1)\left[1 - \frac{N(N+1)}{(N+k)(N+k+1)}\right], \qquad k \geq 1.$$

At $k = 1$ they lie below two. For $N = 2, 3$ their charge is at most two. For $N \geq 7, k \geq 2$, $c(N,k) \geq c(7,2) = 34/15 > 2.2$. Checking $N = 4, 5, 6$ therefore leaves exactly $W_4(8,9)$ at $13/6$ and $W_6(8,9)$ at $25/12$ in $2 < c < 2.2$.

The extension problem cannot be solved by looking only at the original simple currents. In the W_4 coset $(SU(4)_4 \otimes SU(4)_1)/SU(4)_5$, some noncurrent integer-weight labels initially appear possible. A typical denominator quantum dimension among these is

$$d_\mu = \frac{\sin^2(4\pi/9)}{\sin(\pi/9)\sin(2\pi/9)}.$$

Its Galois image multiplying sine arguments by five is negative: the numerator remains positive, while $\sin(5\pi/9)\sin(10\pi/9) < 0$. Choosing the common-field automorphism $113 \equiv 1 \pmod{16}$ and $113 \equiv 5 \pmod{18}$ fixes the numerator $SU(4)_4$ dimensions while producing that negative denominator image. It is an actual automorphism of the common cyclotomic field because $\gcd(113, 288) = 1$. The corresponding coset objects fail total positivity and cannot occur in an extension algebra. This is an exact sign argument, without a floating-point test near zero.

After that filter, the surviving extension algebras transfer to the known $SU(4)_4$ list, including its exceptional $SO(15)_1$ extension. All their added fields begin at weight four, leaving one real weight-three generator. An odd-order chiral automorphism fixes it, as it fixes the stress; these generate the original principal algebra as a vertex algebra. The remaining extension automorphisms and invertible modules are two-groups. Thus every physical full theory retaining this principal W_4 algebra lacks order-three internal symmetry.

Construction input 24.12 (Completed rational-algebra tests)**.** The completed exact label and extension audits establish the following restrictions, including bosonic local chiral extensions and physical non-diagonal pairings:

Retained algebra	Restriction
$M_4^{\otimes 3}$, $c = 21/10$	No internal subgroup of order nine.
$M_3 \otimes M_5^{\otimes 2}$, $c = 21/10$	Order-nine symmetry forces the two Potts extensions and a jointly neutral scalar of dimension $1/8$; $\mathbb{Z}_9$ is absent.
$W_6(8,9)$, $c = 25/12$	No internal subgroup of order nine.
$W_4(8,9)$, $c = 13/6$	No internal symmetry of order three.
$W(E_8)(35,36)$, $c = 44/21$	No internal symmetry of order three; every proper bosonic chiral extension is excluded by the exact Galois sign test.

The finite exhaustive label calculations and the quoted extension classifications are inputs to this inventory. The preceding examples explain the main methods; they do not independently reproduce every label in those audits.

For W_6, the only nontrivial bosonic extension is by a $\mathbb{Z}_3$ of integer-spin currents. Its fixed-point resolution produces no new invertibles beyond a $\mathbb{Z}_2$; its possible chiral three-cycle acts faithfully on resolved module classes. This again prevents two independent left-right three-cycles. For the E_8 principal algebra, all five nonvacuum integer-weight supports have a negative Galois conjugate. Its distinct real strong-generator weights $2, 8, 12, 14, 18, 20, 24, 30$ let one inductively fix every generator under an odd-order automorphism, and its representation category has no nontrivial invertible objects. These are different mechanisms leading to the stated symmetry restrictions.

24.8.5 Why the exceptional E_6 branch stays separate

The principal coset

$$V = \frac{(E_6)_3 \otimes (E_6)_1}{(E_6)_4}$$

also has $c = 21/10$, but the previous Galois filter does not dispose of it. Its six integer-weight supports are the vacuum, two currents of weight eight, a conjugate pair X, X^* of weight two, and a field Y of weight three. All pass the necessary total-positivity test. Moreover a genuine exceptional affine $(E_6)_4$ extension supplies a commutant algebra for a proposed coset extension. Under the specified categorical transfer, the following partial-category calculation applies. This does not claim an independent construction of every intermediate VOA. The branch cannot be dismissed by assuming that all extensions are simple-current extensions.

The unextended diagonal theory and the diagonal ordinary current extension have a scalar at $\Delta = 1/10$. That is a useful test of those pairings, not every intermediate algebra. Under the specified exceptional commutant-algebra transfer, a partial extension has category

$$\mathrm{Rep}(V_Y) \simeq \mathcal{C}((E_6)_3) \boxtimes \mathrm{Ising}_{5/2}.$$

The subscript specifies the Ising modular category with fermion weight residue $1/2$ and spin-field residue $5/16$; it is not a claim that an ordinary critical Ising factor of charge $1/2$ appears as a tensor factor of this VOA. Its exact twist residues imply no positive local scalar below $7/45$ and none in $(9/40, 2/5)$ in any physical pairing retaining that extension. Under the further fully condensed branch, the category is

$$(\mathrm{Fib}^{\mathrm{rev}})^{\boxtimes 3} \boxtimes \mathrm{Ising}_{5/2},$$

and its allowed positive scalar dimensions below one are contained in

$$\{1/5,\ 9/40,\ 2/5,\ 3/5,\ 5/8\}.$$

These constraints need only twists, once the braided category of the retained algebra is established. A local scalar has $h = \bar{h} = \Delta/2$; if $0 < \Delta < 1$, its actual chiral weight lies in $(0, 1/2)$ and is determined there by its residue modulo one. Summing the Fibonacci residues $0, 3/5, 1/5, 4/5$ with the Ising residues $0, 1/2, 5/16$ gives the last list. A pairing may omit entries; it cannot add a residue absent from the category.

As a consistency check, the partial category has topological central charge $78/5 + 5/2 = 181/10 \equiv 21/10 \pmod 8$. The fully condensed category gives $3(-14/5) + 5/2 = -59/10 \equiv 21/10 \pmod 8$. Both agree with the coset charge modulo eight. This necessary check does not turn either categorical calculation into a uniquely identified VOA or full theory.

The charged scalar near .13 and neutral scalar near .26 would each test these exceptional branches if their limits are established. Yet a full CFT can retain an algebra that is mathematically extendible without containing its largest possible extension. One cannot impose the fully condensed twist list on every intermediate theory. Nor does equality of representation categories identify VOAs, characters, OPEs, or chiral automorphisms. A complete analysis of every intermediate algebra and full pairing remains outside the completed result.

24.9 A constructive ultraviolet family: three Potts order fields

The charge pattern also suggests an explicit organizing model. Start with three critical three-state Potts CFTs, total $c = 12/5$, and their normalized order fields σ_i of dimension $2/15$. Preserve

$$\alpha : (\sigma_1, \sigma_2, \sigma_3) \mapsto \omega(\sigma_1, \sigma_2, \sigma_3),$$

and $S : (\sigma_1, \sigma_2, \sigma_3) \mapsto (\sigma_1, \omega\sigma_2, \omega^2\sigma_3)$. Then $\phi = \sigma_1$, $\psi = \sigma_2$, $\eta = \sigma_3$ have the desired ordinary charges. Reflection combined with exchanging copies two and three gives the desired parity.

The neutral scalar $W = \sigma_1\sigma_2\sigma_3$ has dimension $2/5$. A Hermitian perturbation $uW + \bar{u}W^\dagger$ can have a genuine relative phase once the three self-cubic phases are fixed. Thermal anisotropy preserving the exchange of copies two and three is needed to split phi from psi/eta. The relevant couplings generated by products of W must also be retained; a one-complex- parameter action is not closed under renormalization.

The missing charge line has a simple composite basis

$$w_1 = \sigma_2\sigma_1^\dagger, \quad w_2 = \sigma_3\sigma_2^\dagger, \quad w_3 = \sigma_1\sigma_3^\dagger,$$

all of dimension $4/15$. Choose the common Potts-field phase so that the self cubic c_σ is real and positive. If $\chi = \sum_i a_i w_i$ with $\sum_i |a_i|^2 = 1$, factorization gives

$$c_\chi = c_\sigma^2 \sum_i a_i^3 + 6a_1a_2a_3, \qquad \lambda = a_1, \quad \kappa = a_2, \quad \nu = 0.$$

The six mixed permutations explain how a composite field can have a large self cubic. At the ultraviolet point, however, $|\lambda| \le 1$, and the scalar shells have extra degeneracies. The observed finite chain tensor would require substantial interaction effects and a demonstrated infrared splitting.

A strong-coupling clock reduction of this family has an ordinary C_3^2 duality whose line dimension, if topological at a critical point, is $\sqrt{9} = 3$. It is not the Haagerup rho dimension, and its fusion square is the sum of nine invertible group lines. Ordinary self-duality therefore does not supply the missing Haagerup junctions or prove a continuous transition. The ultraviolet family is a concrete place to look, not an identified CFT.

24.10 The remaining identification problem

In the completed October 5 research record, the physical regular $\mathcal{H}_3$ chain $H = -\sum P_\rho$ has no identified continuum CFT. Its data suggest a long nearly critical regime, with several central-charge diagnostics near two but substantial drift, separated light charge shells, and a detailed nontrivial mixed three-point tensor. They do not yet establish ultimate masslessness, a controlled exact central charge, or continuum OPE coefficients with error bounds.

Numerical record 24.13 (What the latest finite comparisons add)**.** The 64-state Ward runs empirically cover all four predetermined descendant-energy windows at $L = 12$ and the first three at $L = 15$. The widest $L = 15$ window remains truncated. Agreement between independent starts and recovery of known states support this coverage, but do not certify the entire low spectrum. Intermediate filtering improves several energy and norm comparisons while leaving operator-dependent amplitude discrepancies. A vector can align closely with the proposed Ward direction and still have the wrong normalized source amplitude; the descendant assignment must satisfy both tests.

Two invariant local directions,

$$V_1 = \sum_j P_{\mathbf{1},j}, \qquad V_{\text{diff}} = \sum_j (P_{\alpha\rho,j} - P_{\alpha^2\rho,j}),$$

both couple strongly to the same chosen invariant-state candidate. Their stress-subtracted diagonal responses on four fixed states give six nonzero pairwise minors at $L = 12, 15$ in the stored floating-point data, indicating two finite response directions. This tests a concrete common-operator model at those sizes. It does not establish two infrared primaries or two marginal couplings; subleading fields and finite-size corrections can produce two finite response directions even when only one field dominates in the infrared. Separate equal-energy comparisons in the two charged families show small decreasing catalog-wide deviations from proportionality over $L = 9, 12, 15$. These aggregate diagnostics do not bound each individual pair; nonzero individual discrepancies are retained. They support further common-field tests but prove neither an exact operator identity nor a limiting OPE normalization.

Two stationary module-based variational trials with height-fiber dimensions $18/54$ have visible charged height correlations through separation 513. One has a pronounced period-three pattern; the other is nearly uniform at the longest measured distances. These are direct physical insertions, so they add information beyond auxiliary transfer eigenvalues. Both trials are finite-capacity local optimization endpoints. The observations do not establish an infinite-distance plateau, a ground-state vacuum count, or a bulk gap. An ordered finite-bond approximation can also occur when the exact chain is critical.

The constructive algebraic results are valuable on their own. The exact module tensors and their gapped parent show that symmetry alone does not settle the phase. The $c = 8$ construction has its own finite and analytic hypotheses. The irrational chiral axis near 2.025004294 has a definite stress tensor but still needs the full local completion and defect descent required for a chain identification. Gaussian bounds, crossing functionals, and Cardy tests can reject specified dictionaries when their premises are established; they cannot select a unique theory by elimination without an exhaustive classification.

To make a limiting primary-field dictionary concrete, choose a candidate full CFT and a fixed scaling-dimension cutoff. Track the entire low-energy space below that cutoff along a specified size sequence, including the sectors required by the proposal. A single assignment of these spaces must match rescaled energies, residual momenta, and symmetry actions. Near-degenerate multiplets permit a change of basis; they do not permit a different field identification for each observable. The vacuum and velocity must likewise be shared.

For each proposed continuum field, choose a controlled sequence of lattice operators and its normalization and mixing coefficients. Use that same choice for vacuum-to-field amplitudes, matrix elements between primary states, and descendant matrix elements. The first group fixes the two-point normalization; the second tests OPE coefficients; the third tests the Ward identities and descendant assignment. Fitting an independent scale to each matrix element would discard precisely the consistency that makes this a field dictionary. Several microscopic recipes for the same field provide additional checks when their separately fixed limits agree.

A correction model should then explain several observables using the same irrelevant or running couplings and predict levels or amplitudes not used in the fit. Example 20.2 shows that two valid finite-size calibrations can approach the same limit from opposite sides. Example 14.3 shows that one running coupling can produce different logarithms in different fields. These are lessons about consistency, not correction exponents to assign to the Haagerup chain.

A finite, mutually consistent table can support a strong numerical identification. A proof that the chain has that scaling limit requires further control of convergence of the state and operator data, compatibility as the cutoff increases, and the resulting continuum correlations. The existence of the candidate CFT is a separate question from this convergence. Finally, the lattice lines must have the claimed limiting action and junction spaces, with associators equivalent to the physical $\mathcal{H}_3$ data. Agreement of loop dimensions or fusion multiplicities alone does not establish that part of the dictionary.

For a gapped interpretation with q thermodynamic vacua, choose periodic sizes compatible

with the proposed spatial period and boundary or flux sector. Along that size sequence the corresponding target is a complete collapsing cluster, $E_a(L)-E_0(L)\to 0$ for $0\le a<q$, separated from a positive bulk scale

$$\liminf_{L\to\infty}(E_q(L)-E_0(L))>0.$$

The energies are ordered with multiplicity. A period-three ordered phase, for example, can force a domain wall on rings whose length is not divisible by three; those rings need a separate spectral analysis. Example 23.1 shows this effect exactly in a commuting clock Hamiltonian. Connected lengths within individual pure vacua must stay finite; a symmetric cat can retain long-range order across the whole vacuum cluster. The exact rank-four comparison module does not predetermine the number of vacua of the physical projector Hamiltonian. The critical and gapped alternatives both call for consistent state, operator, and symmetry data.

24.11 Exercises

Exercise 24.1. Derive the product-theory predictions for Δ_ψ, λ, and I_λ. Explain which of them survive an arbitrary rephasing of the three fields.

Solution on page 316.

Exercise 24.2. Prove that a totally isotropic subspace of a real vector space of signature $(2,2)$ has dimension at most two by projecting to one positive definite factor. Explain why the strict light-dimension inequality is needed to turn the integral pairing into zero.

Solution on page 316.

Exercise 24.3. Count the charge-conserving triples in the equilateral vertex example and recover $2/\sqrt{3}$. Why does this not contradict the unequal-shell theorem?

Solution on page 316.

Exercise 24.4. Derive the functional expression from the block Casimir identities. Check the displayed rational lower bound .0118155. State exactly which unmeasured continuum assumptions are needed to apply it to the lattice chain.

Solution on page 316.

Exercise 24.5. For asymmetric rotation fractions $(1/12, 5/12)$, compute the two oscillator ground weights. Compare their sum with the lower bound on a scalar dimension and explain the difference. Why is an order-two determinant character trivial on an order-three element of the abelianization? Finally recover the charged bound $2/\sqrt{3}$ from the fixed real axis and the Hermitian parent bound, keeping the normalization factors.

Solution on page 317.

Exercise 24.6. Derive the cubic of $\chi=\sum_i a_i w_i$ in the three-Potts example by independent charge conservation in the three factors. Show why a large self cubic does not imply a large mixed coefficient λ.

Solution on page 317.

Exercise 24.7. Let $\varphi=(1+\sqrt{5})/2$. Compute the field norms of $\varphi+2$ and $64(\varphi+2)^3/3$. Explain the algebraic-integer obstruction to an order-three fusion subcategory in the tricritical Ising defect kernel. Which extra argument is needed to control symmetries that act nontrivially on the chiral algebra?

Solution on page 317.

Exercise 24.8. Combine the twist residues $0, 3/5, 1/5, 4/5$ and $0, 1/2, 5/16$ and recover the displayed possible scalar dimensions strictly between zero and one. Explain why the result is a necessary list and why it applies only when the specified extended algebra is actually retained by the full CFT.

Solution on page 317.

Part V

Further mathematical tools

Chapter 25

Algebraic constructions and exact certificates

The order-three Haagerup category belongs to a much larger algebraic problem. Can one construct associators for similar fusion rules at other group orders? How can special functions produce exact finite categorical data? And what does a computer certificate establish when the desired conclusion concerns an infinite-dimensional CFT? This chapter gives a guided answer. Its purpose is to explain the logic and several small exact calculations, rather than reproduce the full contour analysis or a catalogue of number fields.

25.1 The cyclic Haagerup–Izumi family

Replace $\mathbb{Z}_3$ by $G = \mathbb{Z}/n\mathbb{Z}$, with n odd. There are $2n$ simple objects A_g and R_g, with fusion

$$\begin{aligned} A_g A_h &= A_{g+h}, & A_g R_h &= R_{g+h}, \\ R_g A_h &= R_{g-h}, & R_g R_h &= A_{g-h} + \sum_{k\in G} R_k. \end{aligned} \tag{25.1}$$

The dimension equation is

$$z^2 = nz + 1.$$

To avoid changing the meaning of the fixed order-three dimension d elsewhere in the book, write the two general roots as

$$d_n = \frac{n + \sqrt{n^2+4}}{2}, \qquad \delta_n = \frac{n - \sqrt{n^2+4}}{2} = -d_n^{-1}. \tag{25.2}$$

The positive Frobenius–Perron dimension is d_n. The algebraic construction described below naturally begins at the negative spherical dimension δ_n.

At $n = 1$, the rule reduces to Fibonacci fusion. The positive and negative spherical categorifications are the Fibonacci and Yang–Lee categories. At $n = 3$, the rule becomes the Haagerup ring, but the associator still has to be identified: a construction of the ring's categorification does not automatically select the $\mathcal{H}_3$ category used for the physical chain.

25.2 The finite equations before the special function

For a useful untwisted presentation, the nontrivial information is encoded in an $n \times n$ matrix $\mathsf{A}_{g,h}$. The group labels are taken modulo n. Set $\delta = \delta_n$ for this section. Among the required

equations are rotation symmetry, an axis sum, and the quadratic and cubic identities

$$\mathsf{A}_{g,h} = \mathsf{A}_{-h,g-h} = \mathsf{A}_{h-g,-g}, \tag{25.3}$$

$$\sum_h \mathsf{A}_{h,0} = -\delta^{-1}, \tag{25.4}$$

$$\sum_m \mathsf{A}_{m+g,h}\mathsf{A}_{h,m} = \mathbf{1}_{g=0} - \delta^{-1}\mathbf{1}_{h=0}, \tag{25.5}$$

$$\sum_m \mathsf{A}_{m,g+h}\mathsf{A}_{g,m+k}\mathsf{A}_{h,m+l} = \mathsf{A}_{g+l,k}\mathsf{A}_{h+k,l} - \delta^{-1}\mathbf{1}_{g=0}\mathbf{1}_{h=0}. \tag{25.6}$$

The symbol $\mathbf{1}_P$ is the indicator of a condition P. It is not an object of the category.

The quadratic identity expresses cancellation of inverse recouplings. The cubic identity compares the three-recoupling and two-recoupling routes around a pentagon. Its correction term is important: it accounts for an additional invertible intermediate channel when the relevant labels vanish. Deleting that term would give a superficially cleaner equation for the wrong fusion space.

The reconstruction theorem is usually stated with an additional quartic identity. In this setup the quartic equations follow by contracting the quadratic and cubic identities. Thus one can first solve the conceptually simpler pentagon-type system and then match the hypotheses of the reconstruction theorem. The conclusion is a complex spherical fusion category with the specified fusion rules and dimension parameter.

These equations are polynomial identities over a field. They do not contain complex conjugates. Consequently their algebraic meaning is different from a unitary identity such as $F^\dagger F = I$.

Worked example 25.1 (The entire order-one coefficient problem)**.** When $n = 1$, there is a single matrix entry $\mathsf{A} = -\delta^{-1}$, with $\delta^2 = \delta + 1$. The quadratic equation becomes

$$\delta^{-2} = 1 - \delta^{-1},$$

which follows after dividing the dimension equation by δ^2. The cubic becomes

$$-\delta^{-3} = \delta^{-2} - \delta^{-1}.$$

Multiplying by δ^3 gives $-1 = \delta - \delta^2$, again exactly the dimension equation. Both dimension roots solve the polynomial conditions. Positivity has not entered this calculation.

25.3 A factorized ansatz and its exceptional point

Put $\tau_0 = \delta - 1$ and choose nonzero auxiliary coefficients W_g satisfying

$$W_0 = -1, \qquad W_g W_{-g} = \delta \quad (g \neq 0).$$

A useful core is

$$\mathsf{A}_{a,b} = \begin{cases} \dfrac{W_a W_{b-a}}{\tau_0 W_b}, & b \neq 0, \\ \dfrac{-1 + \tau_0 \mathbf{1}_{a=0}}{\tau_0}, & b = 0. \end{cases} \tag{25.7}$$

For nonzero b, the reciprocal relation permits the equivalent product $W_a W_{b-a} W_{-b}/(\delta\tau_0)$. Its three indices sum to zero, making the cyclic symmetry of (25.3) visible. The point $a = b = 0$ requires an extra correction and cannot be recovered by applying the nonzero reciprocal relation to W_0.

For example,

$$\mathsf{A}_{0,0} = \frac{\delta - 2}{\delta - 1}, \qquad \mathsf{A}_{a,0} = \mathsf{A}_{0,a} = \mathsf{A}_{a,a} = -\frac{1}{\delta - 1} \quad (a \neq 0).$$

The ansatz therefore imposes a particular fixed-axis condition. It selects a family of categories; it is not a consequence of the Haagerup–Izumi fusion ring alone.

Worked example 25.2 (An order-three invariant distinguishes coefficient families)**.** Move the dimension parameter to its positive value $d = (3 + \sqrt{13})/2$. The fixed-axis entry becomes

$$\mathsf{A}_{0,0} = \frac{d-2}{d-1} = \frac{\sqrt{13}-1}{\sqrt{13}+1} = \frac{7-\sqrt{13}}{6}.$$

This equals the $\mathcal{H}_2$ invariant entry from Chapter 4. It differs from the $\mathcal{H}_3$ entry $(2-\sqrt{13})/3$ by $d/3$. The factorized ansatz has therefore imposed a restrictive associator condition even at the smallest nontrivial group order. Equality of this one entry does not identify the entire category with $\mathcal{H}_2$ or supply a positive adjoint. Its success in solving the Haagerup fusion ring does not justify substituting its coefficient for the $\mathcal{H}_3$ block in the physical chain.

The auxiliary numbers W_g are not automatically physical OPE coefficients. They are coordinates in a factorization used to construct an associator. A separate dictionary would be needed to interpret one as a normalized observable of a CFT.

25.4 Why two gamma periods differ by the group order

Choose positive periods

$$\omega_1 = d_n + 1, \qquad \omega_2 = 1 + d_n^{-1}. \tag{25.8}$$

The dimension equation gives three useful identities:

$$\omega_1 - \omega_2 = n, \qquad \omega_1\omega_2 = \omega_1 + \omega_2, \qquad \frac{1}{\omega_1} + \frac{1}{\omega_2} = 1. \tag{25.9}$$

These are the reason for the choice. A shift by the cyclic order is the difference of the two special-function shifts.

25.4.1 A convergent definition, including the subtraction

The special function can be introduced with ordinary complex analysis. For positive periods write $\Omega = \omega_1 + \omega_2$ and first restrict to the open strip $0 < \Re z < \Omega$. Define

$$\gamma(z) = \exp\left\{ -\int_0^\infty \left[\frac{\sinh((2z-\Omega)t)}{2\sinh(\omega_1 t)\sinh(\omega_2 t)} - \frac{2z-\Omega}{2\omega_1\omega_2 t} \right] \frac{dt}{t} \right\}. \tag{25.10}$$

This is a definition by an absolutely convergent integral, not a prescription to subtract two divergent answers. Keep the terms inside the same integrand.

To see the cancellation, put $\lambda = 2z - \Omega$. Taylor expansion at the integration endpoint gives

$$\frac{\sinh(\lambda t)}{2\sinh(\omega_1 t)\sinh(\omega_2 t)} = \frac{\lambda}{2\omega_1\omega_2 t} + \frac{\lambda(\lambda^2 - \omega_1^2 - \omega_2^2)}{12\omega_1\omega_2}\, t + O(t^3).$$

After subtraction and division by t, the integrand is bounded as $t \to 0$. At infinity its hyperbolic-function part decays exponentially because $|\Re(2z-\Omega)| < \Omega$, while the subtracted part is $O(t^{-2})$ after division by t. These bounds are uniform on compact subsets of the strip, giving holomorphy there by the standard parameter-integral theorem. Exponentiation makes γ nonzero on this strip.

Two identities are immediate. At $z = \Omega/2$ the entire integrand vanishes, so $\gamma(\Omega/2) = 1$. Replacing z by $\Omega - z$ negates the integrand, giving $\gamma(z)\gamma(\Omega - z) = 1$. For real z in the strip the logarithm is real, hence $\gamma(z) > 0$.

The remaining special-function work establishes the two shifts and meromorphic continuation:

$$\begin{aligned}\gamma(z+\omega_1) &= 2\sin(\pi z/\omega_2)\gamma(z),\\ \gamma(z+\omega_2) &= 2\sin(\pi z/\omega_1)\gamma(z),\\ \gamma(z)\gamma(\omega_1+\omega_2-z) &= 1.\end{aligned} \tag{25.11}$$

Their relation to the integral is visible already on the overlap $0 < \Re z < \omega_2$. Subtract the logarithms at $z+\omega_1$ and z, using the hyperbolic addition identity, to obtain

$$\log\gamma(z+\omega_1) - \log\gamma(z) = -\int_0^\infty \left[\frac{\cosh((2z-\omega_2)t)}{\sinh(\omega_2 t)} - \frac{1}{\omega_2 t}\right]\frac{dt}{t}.$$

This is the convergent integral for $\log(2\sin(\pi z/\omega_2))$ on that strip, with its branch real on the real interval. Its derivative is the usual cotangent integral, and the value at $z = \omega_2/2$ fixes the constant $\log 2$. Interchanging the periods gives the other shift. The integral evaluations and continuation are part of the analytic theorem quoted here; the display shows why the sine appears rather than treating the shift as an arbitrary functional equation.

The continued function has poles at $-r\omega_1 - s\omega_2$ and zeros at $\Omega + r\omega_1 + s\omega_2$, for $r, s \geq 0$. At our periods $\omega_1/\omega_2 = d_n$ is irrational, so these are simple. There is a useful normalization check that needs no numerical sample. Write $R = \operatorname{Res}_{z=0}\gamma(z)$. Taking the limit in both shifts gives

$$\gamma(\omega_1) = \frac{2\pi R}{\omega_2}, \qquad \gamma(\omega_2) = \frac{2\pi R}{\omega_1}.$$

Their product is one by reflection. Positivity as $z \to 0^+$ chooses the positive residue, so

$$R = \frac{\sqrt{\omega_1\omega_2}}{2\pi}, \qquad \gamma(\omega_1) = \sqrt{d_n}, \qquad \gamma(\omega_2) = d_n^{-1/2}.$$

The integral normalization has therefore fixed the normalization needed by the finite coefficient construction.

25.4.2 From the strip function to cyclic samples

Set $F(z) = 1/\gamma(\omega_2 + z)$, so $F(0) = \sqrt{d_n}$. Reflection gives

$$F(z)F(n-z) = 1. \tag{25.12}$$

Taking the ratio of the two shift equations gives

$$\frac{F(z+n)}{F(z)} = \frac{\sin(\pi z/\omega_1)}{\sin(\pi z/\omega_2)}.$$

For a nonzero integer a, the reciprocal-period relation in (25.9) therefore implies

$$F(a+n) = (-1)^{a+1}F(a). \tag{25.13}$$

At $a = 0$ both sines vanish; their ratio must be replaced by its limit $\omega_2/\omega_1 = d_n^{-1}$. Thus the origin is exceptional. Many errors in finite sampling arguments come from using a generic shift formula at precisely such a removable singularity without taking the correct limit.

With representatives $1 \leq a \leq n-1$, define

$$W_0 = -1, \qquad W_a = -(-1)^{a(a+n)/2}\frac{F(a)}{\sqrt{d_n}}. \tag{25.14}$$

The exponent is an integer because n is odd. Reflection gives $F(a)F(n-a)=1$, and the total sign exponent in W_aW_{n-a} is

$$\frac{a(a+n)+(n-a)(2n-a)}{2}=a(a-n)+n^2.$$

Its first term is even and its second odd. Therefore

$$W_aW_{-a}=-d_n^{-1}=\delta_n, \tag{25.15}$$

which proves exactly the reciprocal identity needed in (25.7).

This is a useful example of an analytic expression producing exact finite algebraic relations. It does not yet prove all quadratics or cubics: those involve sums over the group rather than only pairs of reflected samples.

25.5 Fourier transform, contour identities, and reconstruction

On $\mathbb{Z}/n\mathbb{Z}$, finite Fourier transform turns convolution into pointwise multiplication. For a function f put

$$\widehat{f}(k)=\sum_{a=0}^{n-1}f(a)e^{-2\pi ika/n},\qquad f(a)=\frac{1}{n}\sum_{k=0}^{n-1}\widehat{f}(k)e^{2\pi ika/n}.$$

The quadratic coefficient equations contain translated convolution sums. Computing the Fourier transform of a column of $\mathsf{B}=\tau_0\mathsf{A}$ makes their cancellation accessible to the gamma reflection identity. Some recoupling blocks are anti-circulant, so it would be inaccurate to say that the transform simply diagonalizes every F-matrix. It diagonalizes the relevant cyclic translations and organizes their products.

25.5.1 The analytic column identity and its domain

Write $u_h(a)=\mathsf{B}_{a,h}$, with $1\le h<n$, and choose the principal frequency

$$\theta_h(k)\equiv\frac{2\pi k}{n}+\pi(h-1)\pmod{2\pi},\qquad -\pi\le\theta_h(k)\le\pi.$$

The supplied analytic column identity is

$$\widehat{u}_h(k)=-\omega_2e^{-ih\theta_h(k)/2}\gamma\left(\frac{\Omega+h}{2}-\frac{\Omega\theta_h(k)}{2\pi}\right)\gamma\left(\frac{\Omega+h}{2}+\frac{\Omega\theta_h(k)}{2\pi}\right). \tag{25.16}$$

The right side involves only two gamma values instead of an n-term sum. The frequency representative matters for the continuous contour integral used to prove it, even though the original finite transform is periodic in k. At an endpoint $\theta=\pm\pi$, the congruence forces h even, and the two displayed values agree.

Here is the connection with residues. The kernel

$$K_h(z)=\frac{\pi e^{-i\theta z}}{\sin\pi z}\frac{F(z)}{F(z-h)}$$

has residue $(-1)^ae^{-i\theta a}F(a)/F(a-h)$ at an integer a. The choice of θ reproduces the signs of the finite samples. There is one exceptional sample: the actual column entry differs from its generic interpolation at $a=h$. Accounting for it gives

$$\sum_{a=0}^{n-1}\operatorname{Res}_{z=a}K_h(z)=(-1)^h\sqrt{d_n}F(h)\left(\widehat{u}_h(k)+\omega_2e^{-2\pi ikh/n}\right).$$

The rectangle residue theorem converts this sum to a two-gamma integral and the remaining crossed-pole residues. Those residues cancel the displayed exceptional-sample correction in the final answer. Omitting the correction would give a wrong Fourier formula.

The continuation used here has a precise domain. For fixed $|\theta| < \pi$, allow a complex column parameter h' in

$$\{\Re h' > \Omega(|\theta|/\pi - 1)\} \setminus (\omega_1\mathbb{Z}_{\geq 0} + \omega_2\mathbb{Z}_{\geq 0}).$$

Choose first a negative real h_0 in this domain and a vertical line $h_0 < \Re z < 0$, which separates the two pole families. Locally fixed contour deformations, their crossed-pole corrections, and uniform tail estimates continue the integral through the connected domain. The final positive integer $0 < h < n$ avoids the excluded pinch set. At this final value the integral still decays at $|\theta| = \pi$, so endpoint limits are valid. This is the restricted continuation argument in the expanded and formal development; an unqualified appeal to meromorphic continuation would omit essential work.

25.5.2 A quadratic system becomes one Fourier product

Reflection and a period shift in (25.16) give

$$\widehat{u}_h(k)\widehat{u}_{n-h}(-k) = \omega_2^2 e^{-2\pi i k h/n}. \tag{25.17}$$

One can check the signs explicitly. By swapping $h, n-h$ and complex conjugating real columns, it suffices to use odd h and $0 \leq k \leq (n-1)/2$. Put $\lambda_k = \Omega k/n$. The four gamma arguments in the product are

$$x_\pm = (\Omega + h)/2 \pm \lambda_k, \quad y_1 = \Omega + (n-h)/2 - \lambda_k, \quad y_2 = (n-h)/2 + \lambda_k.$$

They obey $x_+ + y_1 = \Omega + \omega_1$ and $x_- + y_2 = \omega_1$. Reflection and the period shifts reduce their product to

$$\frac{\sin(\pi[(h-n)/(2\omega_2) + \omega_1 k/n])}{\sin(\pi[(\Omega+h)/(2\omega_1) - \omega_2 k/n])} = (-1)^{(h-n)/2+k}.$$

Indeed the two arguments divided by π sum to the integer $(h-n+2)/2+k$, and the denominator's argument divided by π lies strictly between zero and one. The exponential prefactors contribute the same sign times $e^{-2\pi i k h/n}$, proving (25.17).

Now define the unscaled quadratic convolution

$$C_h(g) = \sum_a \mathsf{B}_{g+a,h}\mathsf{B}_{h,a}.$$

Rotation symmetry gives $\mathsf{B}_{h,a} = u_{-h}(a-h)$. Transforming in g and substituting $b = g + a$, then $c = a - h$, yields

$$\begin{aligned}\widehat{C}_h(k) &= \widehat{u}_h(k)\sum_a u_{-h}(a-h)e^{2\pi i k a/n} \\ &= \widehat{u}_h(k)e^{2\pi i k h/n}\widehat{u}_{-h}(-k) = \omega_2^2 = \tau_0^2.\end{aligned}$$

The inverse transform of the constant function is $\tau_0^2\mathbf{1}_{g=0}$. Dividing by τ_0^2 proves the quadratic equation for every $h \neq 0$ simultaneously. At $h = 0$ the exceptional axis is elementary:

$$C_0(g) = n - 2\tau_0 + \tau_0^2\mathbf{1}_{g=0} = \tau_0^2(\mathbf{1}_{g=0} - \delta^{-1}).$$

The last equality follows from $\delta^2 = n\delta + 1$ and $\tau_0 = \delta - 1$. This completes all quadratic equations. The analytic input has been isolated in one column formula; the remaining proof is finite Fourier algebra.

Worked example 25.3 (An exact order-three matrix without numerical samples)**.** Take $n = 3$ and $\delta = (3 - \sqrt{13})/2 < 0$. The sample signs give $W_1 < 0$ and $W_2 > 0$. Put $p = W_1^2/W_2 > 0$, $q = W_2^2/W_1 < 0$. Then

$$\mathsf{B} = \begin{pmatrix} \delta - 2 & -1 & -1 \\ -1 & -1 & p \\ -1 & q & -1 \end{pmatrix}, \qquad \mathsf{A} = \frac{\mathsf{B}}{\delta - 1}. \tag{25.18}$$

The reciprocal sample identity gives $pq = \delta$. The now-proved quadratic equation at $(g, h) = (1, 1)$ reads $1 - p - q = 0$. Thus

$$p = \frac{1 + \sqrt{2\sqrt{13} - 5}}{2}, \qquad q = \frac{1 - \sqrt{2\sqrt{13} - 5}}{2}.$$

For a direct check, at $h = 1$ the three unscaled quadratic sums are $2 + pq$, $1 - p - q$, $1 - p - q$. They equal $\tau_0^2, 0, 0$, since $(\delta - 1)^2 = \delta + 2$ at $n = 3$. The $h = 2$ column has the same checks with p, q exchanged, and the axis case was computed above.

This also gives an exact special-function sample. Since $F(1)F(2) = 1$ and $p = F(1)^3/\sqrt{d}$,

$$F(1) = (p\sqrt{d})^{1/3}, \qquad F(2) = F(1)^{-1},$$

with the positive real cube root. This small-order identity was deduced from the analytic quadratic theorem; it is not an independent proof of that theorem at every order. The matrix still has negative categorical dimension. Its quadratics alone do not prove the cubic reconstruction identities; the next example supplies that extra finite argument at order three.

Worked example 25.4 (The complete order-three cubic system)**.** Once the preceding matrix is given algebraically, its coefficient identities can be verified without evaluating a gamma function. Put $a = \delta - 2$, $\tau_0 = \delta - 1$, and $\kappa = 7 - \delta$. The three relations $\delta^2 = 3\delta + 1$, $p + q = 1$, and $pq = \delta$ imply

$$p^2 = p - \delta, \quad q^2 = q - \delta, \quad \tau_0^2 = \delta + 2, \quad a\tau_0 = 3, \quad a^2 = 5 - \delta, \quad \tau_0^3/\delta = \kappa.$$

Multiplying the cubic equation by τ_0^3 clears every denominator. With all indices reduced modulo three, it becomes

$$\begin{aligned} C_{g,h}(k, l) &:= \sum_{m=0}^{2} \mathsf{B}_{m,g+h}\mathsf{B}_{g,m+k}\mathsf{B}_{h,m+l} \\ &= \tau_0 \mathsf{B}_{g+l,k}\mathsf{B}_{h+k,l} - \kappa \mathbf{1}_{g=0}\mathbf{1}_{h=0}. \end{aligned} \tag{25.19}$$

The exceptional correction is present for *every* k, l when $g = h = 0$.

Three entries explain the cancellations. At $(g, h, k, l) = (1, 1, 0, 0)$, the sum is $-1 + p - p^2 = \tau_0$. At $(1, 1, 1, 1)$ it is $p^3 - 2$, and

$$p^3 - 2 = -\tau_0 p - \tau_0^2 = \tau_0(q - \delta) = \tau_0 q^2,$$

as the right side requires. At $(0, 0, 0, 0)$ the residual is

$$a^3 - 2 - \tau_0 a^2 + \kappa = -a^2 + 5 - \delta = 0.$$

The last cancellation would fail if the invertible-channel correction were omitted.

Here is a complete hand check beyond those representatives. For each fixed g, h, arrange the left side into a matrix with rows $k = 0, 1, 2$ and columns $l = 0, 1, 2$. Expanding its three

summands and using the displayed relations gives

$$C_{0,0} = \begin{pmatrix} a^3-2 & -a^2+a-1 & -a^2+a-1 \\ -a^2+a-1 & -a^2+a-1 & 3a \\ -a^2+a-1 & 3a & -a^2+a-1 \end{pmatrix},$$

$$C_{0,1} = \begin{pmatrix} -3 & \tau_0 & -\tau_0 p \\ \tau_0 & -\tau_0 q & -\tau_0 q \\ -3 & -\tau_0 p & \tau_0 \end{pmatrix},$$

$$C_{1,1} = \begin{pmatrix} \tau_0 & \tau_0 & 3p \\ \tau_0 & p^3-2 & \tau_0 \\ 3p & \tau_0 & \tau_0 \end{pmatrix}, \qquad C_{1,2} = \begin{pmatrix} \tau_0 & -\tau_0 q & -3 \\ -3 & -\tau_0 q & \tau_0 \\ -\tau_0 p & \tau_0 & -\tau_0 p \end{pmatrix}.$$

Substitution into the right side of (25.19) gives the same matrices. In the exceptional block, besides the already checked corner, the only identities needed are

$$-a^2+a-1 = \tau_0 - \kappa = 2\delta - 8, \qquad 3a = \tau_0\delta - \kappa.$$

The other blocks use $a\tau_0 = 3$ and $p^3 - 2 = \tau_0 q^2$.

Finally, exchanging (g,k) with (h,l) preserves the equation. Simultaneously negating all four indices exchanges p and q in the matrix, and the three scalar relations are invariant under that exchange. The four tables therefore cover the classes $\{(0,0)\}$, $\{(0,1),(1,0),(0,2),(2,0)\}$, $\{(1,1),(2,2)\}$, and $\{(1,2),(2,1)\}$ of ordered pairs (g,h). Each table includes all nine k,l choices, proving all 81 cubic equations.

This is an independent finite coefficient verification after the explicit matrix is specified. Its identification with the gamma samples still uses the analytic argument above, and the all-order theorem requires more work. At the positive dimension embedding $\delta = d$, one can separately choose $p, q = (1 \pm i\sqrt{4d-1})/2$, making A Hermitian. This matrix condition is not by itself the construction of the full unitary associator; that passage uses the appropriate reconstruction theorem.

25.5.3 The further work needed for reconstruction

The cubic residual, the difference between the two sides of (25.6), is treated similarly. Residue calculations convert finite sums into integrals built from four gamma factors. An Euler-type integral transformation and a four-term contour identity establish the required cancellation in a range where the contours and convergence are controlled. A finite-degree interpolation argument, together with index symmetries of the factorized ansatz, then recovers the remaining coefficients. Finally, algebraic contraction gives the quartics required by reconstruction.

Each step answers a different question. The contour argument proves an analytic identity on a justified domain. Interpolation extends the finite set of coefficient identities. Reconstruction turns the coefficient solution into a category. None of these is replaced by finding small numerical residuals for several values of n.

Construction input 25.5 (Recorded formal coefficient theorem)**.** For every odd positive integer n, the specified gamma samples give complex coefficient data satisfying the rotation and axis equations, all quadratics, all cubics, and all reconstruction quartics. The repository's completed Lean development constructs the hyperbolic gamma function from its regularized integral and proves the required analytic identities. Its final coefficient theorem does not assume those identities as additional hypotheses.

The recorded verification is an all-order proof, not a collection of finite residual checks. Its source supplies the analytic interface by a concrete constructor and then passes that constructor

to the general coefficient theorem. The recorded build and axiom audit report only Lean's standard logical axioms, with no admitted proofs or additional axioms. Preparing this textbook included reading the final declarations and their stated scope; it did not include an independent rebuild of the entire development.

The short analytic manuscript and this formal development should also be distinguished. A later source review identifies missing continuation-domain justifications and an overly broad initial parameter range in the short manuscript. The expanded analytic and formal developments supply the more precise domain arguments. We therefore quote the coefficient result with its recorded formal scope, rather than treating the short contour sketch in this chapter as an independent proof.

Theorem 25.6 (Categorical consequence quoted from the uniform construction)**.** *For every odd order $n \geq 3$, the explicit gamma-sample construction gives a complex spherical Haagerup–Izumi category with noninvertible categorical dimension δ_n. A simultaneous change of field embedding gives a pseudounitary category with the same fusion rules and dimension d_n.*

The order-one Fibonacci and Yang–Lee cases are treated separately by the elementary calculation above. The displayed categorical consequence additionally uses the Evans–Gannon reconstruction theorem and the change of dimension embedding explained next. These categorical steps are mathematical arguments outside the formal coefficient theorem: that theorem has no conclusion asserting the existence of a tensor category, a VOA, or a full CFT. The finite audits discussed below are a third kind of verification, with their own finite scopes. None of these results is a uniform construction of unitary CFTs or an identification of an anyon-chain limit. The order-three unitary categories used elsewhere in the book are known separately and have explicit positive associator presentations.

25.6 Why Galois conjugation does not carry an adjoint automatically

The automorphism of the quadratic dimension field interchanging δ_n and d_n extends to an automorphism σ of $\mathbb{C}$. One first extends it to the algebraic closure of $\mathbb{Q}$, fixes a transcendence basis, and then extends to the algebraic closure of the resulting field. Apply this same σ to every coefficient of the solution. After adjoining the dimension variable and clearing nonzero denominators, the reconstruction equations are polynomial over $\mathbb{Q}$. The transformed coefficients therefore satisfy those same equations with dimension d_n. This argument does not require a prior proof that all gamma samples belong to a specified number field.

It need not preserve complex conjugation:

$$\sigma(\overline{z}) \quad \text{need not equal} \quad \overline{\sigma(z)}.$$

Thus a statement involving adjoints and positive norms is not transported merely because its polynomial part is. In particular, the following are progressively different assertions:

1. The fusion ring has a positive dimension character.
2. A spherical category has dimensions equal to its Frobenius–Perron dimensions; it is pseudounitary.
3. The category admits a compatible positive adjoint, making it unitary.
4. A vacuum VOA has a positive invariant Hermitian form.
5. Its irreducible modules and their intertwining operators have the positivity needed for a unitary tensor category and a local full theory.

The final two assertions concern additional CFT structures and cannot be deduced from the first three without the relevant construction and analytic theorems.

There is also a useful obstruction on the negative side. A spherical structure on these fusion rules, when it exists, is unique: the ratio of two pivotal structures acts by scalars t_x on simples. Because ρ^2 contains both $\mathbf{1}$ and ρ, one has $t_\rho^2 = 1 = t_\rho$, so $t_\rho = 1$. The other summands and multiplication by invertibles then force every $t_x = 1$. A category with negative $\dim \rho$ cannot be made positive merely by choosing a different pivotal structure.

25.7 What an exact finite certificate can prove

The chiral construction uses finite-dimensional spaces of lattice states, exact OPE identities, and rank statements. A certificate should state the finite assertion, the coefficient field, and the argument converting the recorded data into that assertion.

For example, a matrix M over a number field has rank at least r if a specified $r \times r$ minor has nonzero determinant. After clearing denominators, one can reduce modulo a prime avoiding those denominators. A nonzero reduced determinant proves that the original determinant was nonzero. This is an exact argument, even if a floating calculation was used earlier to discover which minor to choose.

The upper bound is a separate matter. It can come from exact linear relations, a known kernel, or a representation-theoretic dimension bound. A nonzero $r \times r$ minor alone proves $\operatorname{rank} M \geq r$, not equality.

Worked example 25.7 (Why every component of a coefficient algebra can matter)**.** Consider $R = \mathbb{Q}[t]/(t^2 - 1)$. This is a product of two fields, not a field. The element $1 + t$ is nonzero at $t = 1$ but vanishes at $t = -1$. Thus testing a determinant only at the first component does not prove invertibility in all components.

If a coefficient presentation is known to be one number field, a single good specialization showing nonzero determinant already proves the determinant is a nonzero field element. If the presentation is an algebra with several possible embeddings or branches and the claim concerns all of them, one must control the appropriate components. A convenient sufficient certificate is a nonzero norm, obtained as the product of determinant values at all split points modulo a good prime. It is sufficient, not necessary: a zero reduced norm may simply mean the prime divides a nonzero characteristic-zero norm.

A second useful certificate uses positivity. If a residual vector v lies in a space with an established positive-definite Hermitian form and exact arithmetic gives $\langle v, v\rangle = 0$, then $v = 0$. Without positive definiteness, a nonzero isotropic vector can have zero norm. The metric is part of the certificate's mathematical hypotheses.

Finite verification must then be connected to an all-weight theorem. Checking the first several graded pieces of a VOA does not exclude a new current generated by higher fields. The strong-generation argument in Chapter 8 supplies the implication from specified finite products and their provenance to closure in every weight. Similarly, a formal proof of a rank lemma does not amount to a formalization of the surrounding VOA, tensor-category, or conformal-net theory.

25.8 The role of arithmetic special values

The coefficient fields, their Galois actions, and special-function values lead naturally to arithmetic questions. One may ask whether the samples can be expressed through units in explicitly described number fields, whether class-field or Stark-type constructions organize their conjugates,

and which embeddings admit positive categorical structures. Such questions can make exact coefficients more intelligible and can provide new ways to construct or certify them.

They do not, by themselves, specify a conformal spectrum or an OPE. In particular, a cubic number field has degree three; this use of "cubic" is unrelated to a three-point coefficient in a CFT. An identity between arithmetic special values becomes relevant to the physics only after an explicit chain of identifications links those values to the categorical or chiral data being used.

For the order-three center, the auxiliary quadratic data involve the prime thirteen, and their isometry class is tightly constrained. At larger odd orders the corresponding discriminant n^2+4 may be composite. The general metric-group description of the center does not automatically say that every associator selects the same cyclic group and quadratic form. Structural form, evaluation of a selected family, categorical unitarity, and CFT realization remain separate questions.

25.9 Exercises

Exercise 25.1. Derive every identity in (25.9) from $d_n^2 = nd_n+1$. Then derive the nonzero-integer sample shift (25.13) and compute the exceptional limit at the origin.

Solution on page 318.

Exercise 25.2. Verify the parity calculation following (25.14) for arbitrary odd n. Explain why the chosen integer representatives must be stated when expressions such as $(-1)^{a(a+n)/2}$ appear.

Solution on page 318.

Exercise 25.3. Give a 2×2 rational matrix that is invertible but becomes singular after rounding its entries to six decimal places. Prove its invertibility exactly. Explain the asymmetry between a nonzero determinant modulo a good prime and a zero determinant modulo that prime.

Solution on page 318.

Exercise 25.4. In $\mathbb{Q}[t]/(t^2-1)$ compute the product of the two evaluations of $1+t$ and of $2+t$. Which element is invertible? Reduce modulo five and compare with the exact answer.

Solution on page 318.

Exercise 25.5. Explain why applying a field automorphism to polynomial pentagon equations is legitimate, while applying it to a positivity claim requires additional work. Give an example of an algebraic number whose being positive in one real embedding is not preserved in another real embedding.

Solution on page 319.

Source note. Evans–Gannon supply the categorical reconstruction framework. For the uniform coefficient theorem, the recorded formal source is `lean/HaagerupIzumi/Analytic/GammaConstruction.lean`, at revision `698b66a`; its main result is `EGEquations_ruijsenaars`. The recorded build scope is described in `lean/README.md`, and the short manuscript's analytic exposition is assessed in the September 14 source review `proof_revised/formalization_review.md`. The formal result covers every odd order, whereas the later finite audits certify their specified finite statements. Neither should be read as formal verification of every structure in the CFT realization.

Chapter 26

Noncyclic constructions and extended Haagerup

The name Haagerup appears in several related mathematical problems. Replacing the order-three group by another finite abelian group is one operation. Extending the initial string of a principal graph is another. Taking an orbifold or a Morita partner is a third. These operations change different parts of the data, and their outputs need not belong to one family of fusion rules.

This chapter develops two useful comparisons with the preceding construction. Noncyclic Haagerup–Izumi coefficients illustrate how signs, integer lifts, and exact polynomial certificates work together. Extended Haagerup illustrates how a planar algebra can specify an entire tensor category without printing all its associator matrices. The small calculations will be explicit; the large finite certificates will be identified as supplied inputs. Neither comparison supplies a new conformal field theory.

26.1 Keeping the fusion rules in view

For a finite abelian group G, put $n = |G|$. The generalized Haagerup rules considered here have simples A_g, R_g, $g \in G$, and the same formulas as (25.1):

$$A_g A_h = A_{g+h}, \qquad A_g R_h = R_{g+h}, \qquad R_g A_h = R_{g-h}, \qquad R_g R_h = A_{g-h} + \sum_{k \in G} R_k.$$

There are n invertible and n noninvertible simples. The positive dimension of every R_g is

$$d_n = \frac{n + \sqrt{n^2 + 4}}{2}, \qquad \mathrm{FPdim}(\mathcal{C}) = n(1 + d_n^2).$$

The group law matters even when two groups have the same order. For example, the invertibles for $\mathbb{Z}_8$ contain an element of order eight, whereas those for $\mathbb{Z}_4 \times \mathbb{Z}_2$ do not. Their fusion rings are already different before choosing an associator.

For $|G| > 1$, these also differ from *near-group* rules, which have only one noninvertible simple together with a group of invertibles. The phrase “related to Haagerup” should therefore always be followed by a statement of the simple objects and their fusion.

Even group order introduces an additional issue. The coefficient presentation generally needs a family of matrices $\mathsf{A}_g(p, q)$ and signs $\epsilon_h(g)$, rather than the single untwisted matrix used in Chapter 25. The index g labels a *sheet* of the coefficient data. It is not an extra simple-object label beyond the displayed fusion ring. Transport equations relate the sheets. The following construction makes their signs concrete for certain rank-two group quotients.

26.2 Why a determinant phase appears

Take

$$\Lambda = \mathbb{Z}^2, \qquad \Gamma = 2\mathbb{Z} \times 2m\mathbb{Z}, \qquad G = \Lambda/\Gamma, \qquad n = 4m,$$

and choose either root δ of $\delta^2 = n\delta + 1$. Write $\tau = \delta - 1$ and

$$M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \epsilon_h(g) = (-1)^{h^T M g}.$$

Integer representatives are used until all signs have been evaluated. The relation $M + M^T \equiv J \pmod 2$ will make the phase cancellations work.

An auxiliary weight is a nonzero function on integer lifts satisfying

$$W(0) = -1,$$
$$W(a + 2z) = (-1)^{a^T M z} W(a) \qquad (2z \in \Gamma), \tag{26.1}$$
$$W(a)W(-a) = \delta \qquad (a \notin \Gamma). \tag{26.2}$$

Thus W is not quite a function on G: changing a lift can change its sign. This prescribed change is called its monodromy here. It is compatible with consecutive shifts because the extra cross term in the exponent is even. At a lift of the zero class, the weight is -1 and its reciprocal product is 1, not δ.

Define the parity correction

$$\sigma(a) = (-1)^{\lfloor a/2 \rfloor^T M (a \bmod 2)}, \qquad D_g(p, q) = \sigma(g)\sigma(g + p)\sigma(g + q)\sigma(g + p + q),$$

where division and remainder are coordinatewise. The useful identity is

$$\sigma(g + 2h)/\sigma(g) = \epsilon_h(g).$$

Now set

$$B(p, q) = i^{\det(p,q)} \frac{W(p)W(q - p)}{W(q)} + \tau \mathbf{1}_{q \in \Gamma} (-1)^{(q/2)^T M p}, \tag{26.3}$$
$$\mathsf{A}_g(p, q) = \tau^{-1} D_g(p, q) B(p, q). \tag{26.4}$$

The expression involving $q/2$ is used only when $q \in \Gamma$, so its exponent is integral. The determinant couples the two coordinates. Omitting it would change the coefficient identities.

Worked example 26.1 (Descent from integer lifts)**.** Shift p to $p + 2z$ with $2z \in \Gamma$. The determinant and weight factors in the first term of B contribute the sign

$$(-1)^{\det(z,q) + p^T M z + (q-p)^T M z} = (-1)^{z^T M q} = \epsilon_z(q).$$

The equality uses $M + M^T \equiv J \pmod 2$. The exceptional term has the same transformation. Meanwhile $D_g(p, q)$ acquires exactly the same sign, so their product is unchanged. The q shift works similarly, and changing the representative of g leaves the whole expression unchanged. Thus $\mathsf{A}_g(p, q)$ is well defined on the finite group even though W itself has nontrivial monodromy.

Reciprocity gives the three axis values

$$B(p, 0) = B(0, p) = B(p, p) = \tau \mathbf{1}_{p \in \Gamma} - 1.$$

For example, at $q = 0$ and nonzero residue p, the weight quotient is $-\delta$; adding τ changes it to -1. This explains the exceptional correction in (26.3). In particular

$$\mathsf{A}_g(p, 0) = \mathbf{1}_{p=0} - \frac{1}{\delta - 1}, \qquad \sum_{p \in G} \mathsf{A}_g(p, 0) = -\frac{1}{\delta}. \tag{26.5}$$

The last equality follows from $\delta^2 = n\delta + 1$.

The determinant ansatz solves the linear transport and rotation equations for every admissible weight. It also gives the pointwise identity

$$\mathsf{A}_g(p,q)\mathsf{A}_g(q,p) = \frac{\delta}{(\delta-1)^2} \quad (p,q \neq 0,\ p \neq q). \tag{26.6}$$

Here the determinant phases cancel and the remaining product is $W(q-p)W(p-q) = \delta$. However, the summed quadratic and cubic equations remain to be solved. Pointwise reciprocity is much less information than a full recoupling identity.

Worked example 26.2 (The Klein-four coefficient matrix)**.** For $m = 1$, choose $t^2 = -\delta$ and set $W(a) = t$ at the three nonzero representatives. One solution has

$$(\delta-1)\mathsf{A}_0 = \begin{pmatrix} \delta-2 & -1 & -1 & -1 \\ -1 & -1 & -it & -it \\ -1 & -it & -1 & -it \\ -1 & -it & -it & -1 \end{pmatrix}, \qquad \delta^2 = 4\delta + 1,$$

in the order $(0,0),(0,1),(1,0),(1,1)$. To see one phase, take $p = (0,1)$ and $q = (1,0)$. The determinant is -1, so its phase is $-i$; the monodromy rule and quotient of weights give the remaining factor t.

At $\delta = 2+\sqrt{5}$, choose $t = i\sqrt{\delta}$. Then $-it = \sqrt{\delta}$ and the displayed matrix is real symmetric. At the negative root $\delta = 2-\sqrt{5}$, choose $t = \sqrt{-\delta} > 0$; its off-diagonal lower entries are imaginary and do not satisfy the same Hermitian condition. Polynomial transport to another embedding has preserved the algebraic equations, not the original positive adjoint.

One quadratic normalization can be checked immediately:

$$\sum_{a\in G} \mathsf{A}_0(a,0)^2 = \frac{(\delta-2)^2+3}{(\delta-1)^2} = 1-\delta^{-1}.$$

Indeed the numerator is 8, the denominator is $2(\delta+1)$, and $(5-\delta)(\delta+1) = 4$. This one calculation illustrates the dimension equation's role; the complete known Klein-four solution requires all the remaining coefficient equations as well.

For odd m, the three nonzero elements of $G[2]$ have weights whose squares equal $-\delta$. Multiplication of W by a group character changes no coefficient quotient. It can fix two of those three signs, leaving one discrete sign. The other $4m-4$ elements form $2m-2$ inverse pairs, giving that many independent nonzero weights. At $m = 3$, only four continuous parameters remain.

The same descent proof works for any finite-index $\Gamma \subseteq 2\mathbb{Z}^2$. This is a general theorem about the *linear ansatz*. It does not prove existence of solutions to the nonlinear equations for every such quotient.

26.3 Two different achievements at orders eight and twelve

Before using cyclic order eight, one extension of the previous chapter must be stated. The even-cyclic coefficient problem has signed sheets. After removing their parity factors, a column of the scalar kernel obeys

$$B(a+n,h) = (-1)^h B(a,h).$$

Even columns are periodic and odd columns are antiperiodic. Their finite Fourier bases therefore use frequencies k or $k+\frac{1}{2}$, respectively: translating $e^{2\pi i(k+1/2)a/n}$ by n multiplies it by -1. The same gamma function supplies the samples, but the frequency bookkeeping has changed. The supplied even-cyclic analytic manuscript proves the signed quadratic and cubic identities. The order-eight calculation below imports its quadratic identities; it does not reprove that analytic extension or claim that it belongs to the all-odd Lean theorem described in the previous chapter.

26.3.1 An exact special-function identification at order eight

The group $\mathbb{Z}_4 \times \mathbb{Z}_2$ has a published generalized Haagerup construction. Its radical coefficients can be expressed using one degree-eight scalar. A separate calculation in the supplied work identifies its negative-embedding weights with products of the same double-sine samples that occur in cyclic order eight. The category's existence and this special-function identification are different statements.

Put

$$D_8 = 4 + \sqrt{17}, \qquad t = D_8^{-1/2}, \qquad F_k = \mathrm{Sin}_2\left(1 + \frac{kD_8}{D_8 + 1}; D_8, 1\right).$$

To fix the normalization without a naming ambiguity, use the gamma of Chapter 25 with periods $\omega_1 = D_8 + 1$, $\omega_2 = 1 + D_8^{-1}$. Then

$$F_k = \gamma(\omega_2 + k)^{-1}.$$

Thus this double sine is the rescaled *reciprocal* gamma; its period shifts divide by the corresponding sine factors. All samples used below lie in its positive real strip. Set $y = F_2$ and $r = F_3/F_1$. Four of the proved cyclic quadratic equations have the following exact elimination consequences:

$$y^2 r^2 = tr^2 + 2, \qquad 2tr^4 + (3 - t^2)r^2 - 2t = 0, \qquad t^4 + 8t^2 - 1 = 0. \tag{26.7}$$

The finite elimination is a supplied rational ideal-membership certificate. Once these two equations are available, the branch identification can be done by hand.

Write $s = t^2 = \sqrt{17} - 4$. The second equation gives

$$r^{-2} - r^2 = \frac{3 - s}{2t} > 0.$$

Positivity of the gamma samples implies $0 < r < 1$. Using $s^2 + 8s - 1 = 0$ and squaring yields

$$r^2 + r^{-2} = \frac{9 + s}{2}, \qquad r + r^{-1} = \frac{5 + s}{2}.$$

The positive square roots are forced at both steps. Thus

$$r = \frac{1 + \sqrt{17} - \sqrt{2 + 2\sqrt{17}}}{4} =: \lambda_8^2. \tag{26.8}$$

The quartic $r^4 - r^3 - 2r^2 - r + 1 = 0$, together with the second eliminant, also gives $tr^3 = 1 - 2r$. The solution to Exercise 26.2 derives this identity explicitly on the positive branch. Multiply the first eliminant by r: $y^2 r^3 = tr^3 + 2r = 1$. Since $y > 0$,

$$F_2 = \lambda_8^{-3}, \qquad F_3/F_1 = \lambda_8^2. \tag{26.9}$$

This is an exact equality, not recognition from a decimal fit. It also shows why an eliminant alone is insufficient: without positivity one has not selected the correct reciprocal root or the sign of y.

26.3.2 An algebraic existence certificate at order twelve

For $G = \mathbb{Z}_2 \times \mathbb{Z}_6$, use

$$t = \sqrt{\sqrt{37} - 6}, \qquad \delta = -t^2 = 6 - \sqrt{37}, \qquad t^4 + 12t^2 - 1 = 0.$$

The four unknown weights can be specified algebraically. A convenient auxiliary variable u is the unique root in $(3, 4)$ of

$$u^3 + t(t^2 + 13)u^2 - (26 + 2t^2)u - t(48 + 4t^2) = 0.$$

For example, $2/7 < t < 3/10$ proves that the polynomial changes sign between 3 and 4 and has positive derivative there. Set $C = (u + \sqrt{u^2+4})/2$, so $u = C - C^{-1}$. Clearing its inverse powers gives a monic sextic

$$\begin{aligned} P_t(C) = C^6 + t(t^2+13)C^5 - (29+2t^2)C^4 - t(74+6t^2)C^3 \\ + (29+2t^2)C^2 + t(t^2+13)C - 1. \end{aligned} \tag{26.10}$$

The supplied construction expresses the four normalized weights z_j as rational functions of t, C. The coefficient equations can therefore be checked in the finite rational algebra

$$E = (\mathbb{Q}[t]/(t^4 + 12t^2 - 1))[C]/(P_t(C)). \tag{26.11}$$

Its basis consists of the 24 monomials $t^i C^j$, with $0 \le i < 4$ and $0 \le j < 6$. We call this a finite algebra; irreducibility is not needed for the verification method. An inverse is accepted only with a polynomial Bézout identity proving its denominator is a unit. After that, every asserted identity has 24 rational coefficients to check.

Construction input 26.3 (The recorded order-twelve certificate)**.** The specified algebraic weights solve every signed transpose coefficient equation for $\mathbb{Z}_2 \times \mathbb{Z}_6$. Their $1,728$ indexed quadratics reduce to 16 nonzero Laurent-polynomial types, and their $248,832$ indexed cubics to 246 types; all remainders in (26.11) vanish. The recorded certificate includes a second extraction of the indexed equations, nonvanishing denominators, and exact real root isolation. These recorded finite checks are inputs to this chapter; they have not been rerun while preparing the textbook.

There is a further exact step to unitarity. At positive dimension $\delta_+ = 6 + \sqrt{37}$, put $t = i\sqrt{\delta_+}$. The certificate proves that all six roots C of the sextic lie on the unit circle. Hence ordinary conjugation acts by

$$t \longmapsto -t, \qquad C \longmapsto C^{-1}.$$

It also proves $z_j(t, C) z_j(-t, C^{-1}) = 1$. Consequently every normalized z_j has modulus one. The nonzero representative weights are t, tz_j, or signed t/z_j, and real sign monodromy determines their other lifts. Thus every nonzero-residue weight has squared modulus δ_+. Equation (26.6) then upgrades to

$$\mathsf{A}_g(q,p) = \overline{\mathsf{A}_g(p,q)}.$$

Indeed both the product and $|\mathsf{A}_g(p,q)|^2$ equal $\delta_+/(\delta_+ - 1)^2$; divide by the nonzero first factor. The axis entries are already real. This explains how a separate conjugation certificate supplies information absent from the polynomial equations alone.

With the corresponding reconstruction theorem, the recorded result is a unitary category with 24 simples and noninvertible dimension $6 + \sqrt{37}$. The fixed axis condition supplies the Q-system $\mathbf{1} \oplus R_0$, giving subfactor index $7 + \sqrt{37}$. Six positive coefficient embeddings do not by themselves mean six inequivalent categories: gauge changes and group automorphisms must first be considered.

Remark 26.4 (What remains open in the gamma formula). Put $D_{12} = 6 + \sqrt{37}$ and use the corresponding double-sine samples F_k as above. The proposed negative-embedding identities are

$$(z_0, z_1, z_2, z_3) = \left(\frac{F_2 F_5}{F_1}, \frac{F_2}{F_4}, \frac{F_2}{F_4^2}, -\frac{F_5}{F_1} \right).$$

They match the independently refined algebraic branch to high precision, but their exact identification remains open in the current source. The algebraic existence and unitarity certificate does not depend on this conjecture. Conversely, a tiny residual from these products does not prove their equality to the exact weights. One relation follows immediately from the proposed products, $z_0 z_2 + z_1^2 z_3 = 0$; obtaining one such relation is far short of the full coefficient system.

26.4 The same quadratic unit, two quotient groups

There is a useful arithmetic reason to compare cyclic and noncyclic examples at the same order. Let

$$D = 2m + \sqrt{4m^2+1}, \qquad \omega = \frac{1+\sqrt{4m^2+1}}{2}.$$

Then $D = 2\omega + 2m - 1$ and $\omega^2 = \omega + m^2$. The quadratic order $\mathbb{Z}[D]$ has index two in $\mathbb{Z}[\omega]$. Imposing $D = 1$ in $D^2 - 4mD - 1 = 0$ gives

$$\mathbb{Z}[D]/(D-1)\mathbb{Z}[D] \cong \mathbb{Z}_{4m}.$$

In the basis $(1, \omega)$, multiplication by $D - 1$ instead has matrix

$$2\begin{pmatrix} m-1 & m^2 \\ 1 & m \end{pmatrix}.$$

Its entries have gcd 2 and its determinant is $-4m$, so its Smith invariant factors are 2 and $2m$. Therefore

$$\mathbb{Z}[\omega]/(D-1)\mathbb{Z}[\omega] \cong \mathbb{Z}_2 \times \mathbb{Z}_{2m} \quad \text{as additive groups.} \tag{26.12}$$

At $m = 1$ this compares $\mathbb{Z}_4$ with the Klein group; at $m = 3$ it compares $\mathbb{Z}_{12}$ with $\mathbb{Z}_2 \times \mathbb{Z}_6$. The dimension parameter is the same while the group changes. This arithmetic observation motivates the common special function. It does not derive the weight formulas or solve the nonlinear equations.

26.4.1 Why an auxiliary action's kernel need not shrink the group

Reconstruction must recover exactly the $2|G|$ promised simples. A subtlety arises when the signed group action on the first auxiliary algebra has a kernel. One cannot simply identify the corresponding group elements and continue to claim the original fusion ring. The supplied algebraic reconstruction removes that kernel by enlarging the auxiliary algebra.

Here is the mechanism. Start with the Leavitt algebra L with generators s, t_g, s', t'_g, whose relations say that the column (s, t_g) and row (s', t'_g) are inverse rectangular matrices:

$$s's = 1, \quad t'_g t_h = \delta_{g,h}, \quad s't_g = t'_g s = 0, \qquad ss' + \sum_g t_g t'_g = 1.$$

The primes are algebraic partners; they do not denote an adjoint. The coefficient equations produce endomorphisms α_g, ρ obeying covariance and the splitting of ρ^2 on L.

Let $D = \mathbb{C}^G$ be functions on the finite group, with translations $(\theta_g f)(x) = f(x-g)$. Form the unital algebraic free product

$$\mathcal{B} = L *_{\mathbb{C}} D_1 *_{\mathbb{C}} D_2.$$

This consists of finite linear combinations of alternating words from the three factors, subject only to their own relations and the common scalar unit. In particular every factor embeds. Extend the action by

$$\widetilde{\alpha}_g(f_1) = (\theta_g f)_1, \qquad \widetilde{\alpha}_g(f_2) = (\theta_{-g} f)_2,$$

and extend ρ by

$$\widetilde{\rho}(f_1) = f_2, \qquad \widetilde{\rho}(f_2) = s f_1 s' + \sum_g t_g (\theta_{-g} f)_2 t'_g.$$

The second expression preserves multiplication because distinct Leavitt corners annihilate each other; it preserves the unit because the corner projections sum to one. The free-product universal

property therefore gives an endomorphism. The extended group action is faithful on D_1, regardless of its behavior on L.

The full reconstruction proof checks covariance and

$$\widetilde{\rho}^2(b) = sbs' + \sum_g t_g \widetilde{\alpha}_g \widetilde{\rho}(b) t'_g \qquad (b \in \mathcal{B}).$$

Checking this on each free factor suffices. On D_1 it is the displayed definition; on D_2 the coefficient quadratics cancel the mixed terms. A further intertwiner lemma shows that this enlargement creates no unwanted intertwiners between the proposed simple endomorphisms. Its useful idea is that an equation $ux = xv$, with u homogeneous of positive degree and left invertible in L, cannot be satisfied by a nonzero component of x containing new free-product letters: it would produce a finite-dimensional subspace invariant under multiplication by u, contradicting the grading. These checks supply all the group labels and their distinct simple objects. The relevant grading is concrete: $\deg s = 2$, $\deg t_g = 1$, $\deg s' = -2$, and $\deg t'_g = -1$; every Leavitt relation is homogeneous.

This is an algebraic construction for the full signed transpose system with $\eta_g = 1$ and finite abelian G. It needs neither an analytic completion nor a positive functional. A separate positive Hermitian reconstruction is used for a unitary conclusion, as in the order-twelve example. A repair of faithfulness is not itself a proof of unitarity.

26.5 Extended Haagerup changes the construction problem

Extended Haagerup comes from a different principal graph. Its first branching occurs later: its planar algebra agrees with the Temperley–Lieb diagrams through box degree seven, and an extra generator first appears in degree eight. This property is called *seven-supertransitivity*. The corresponding Haagerup example is three-supertransitive. The word "extended" refers to the additional initial graph string.

Write x for its subfactor index, reserving the fixed symbol d for the order-three Haagerup dimension elsewhere in this book:

$$x = 3 + 2\cos\frac{4\pi}{13} + 2\cos\frac{6\pi}{13}, \qquad x^3 - 8x^2 + 17x - 5 = 0, \qquad x \approx 4.37720. \tag{26.13}$$

The two original unitary even parts have ranks six and eight. Their common global dimension is

$$D_{\mathrm{EH}} = 125x^2 - 450x + 145, \qquad D_{\mathrm{EH}}^3 - 585D_{\mathrm{EH}}^2 + 8450D_{\mathrm{EH}} - 21125 = 0. \tag{26.14}$$

The published Morita classification has four unitary categories, of ranks $6, 8, 8, 8$. The rank-six category is conventionally EH_1 and the original rank-eight category EH_2.

Proposition 26.5 (A dimension obstruction to identifying the families)**.** *No category with the generalized Haagerup fusion rules above is Morita equivalent to an extended Haagerup category.*

Proof. For $n \geq 1$ the generalized Haagerup global Frobenius–Perron dimension is

$$n(1 + d_n^2) = 2n + n^2 d_n.$$

It has degree two over $\mathbb{Q}$, since $n^2 + 4$ is not a square. The cubic for D_{EH} reduces modulo two to $u^3 + u^2 + 1$, which has no root in $\mathbb{F}_2$ and is irreducible. Thus D_{EH} has degree three. Morita equivalence preserves global Frobenius–Perron dimension, so these numbers cannot be equal. The argument uses the positive Frobenius–Perron dimension and remains valid for a nonunitary categorification of the generalized Haagerup ring. □

The transferable idea is to construct finite exact tensor data and then reconstruct a category. The specific group-indexed coefficient system does not transfer to extended Haagerup.

26.6 A planar algebra as a tensor presentation

A shaded planar algebra assigns a vector space $P_{k,+}$ or $P_{k,-}$ to a box with $2k$ boundary points. A diagram connecting several boxes by nonintersecting strings defines a multilinear operation on their labels. The signs specify the two checkerboard shadings. Substitution of diagrams agrees with composition of the corresponding operations. Vertical stacking gives multiplication, horizontal placement gives tensor product, and closing strings gives a trace. Reflection supplies the adjoint in a planar $*$-algebra.

For a subfactor planar algebra the empty diagram is the scalar identity in each zero-box space, and the trace pairing is positive. In an ambient graph planar algebra there are several zero-box coordinates, one for each vertex of the corresponding parity. An identity asserted to equal a scalar there must hold with the same scalar at every such vertex. Checking only one coordinate would be insufficient.

Put $\beta = \sqrt{x} > 2$ and define quantum integers by

$$[0] = 0, \quad [1] = 1, \quad [k+1] = \beta[k] - [k-1].$$

A contractible string loop evaluates to β. The Jones–Wenzl projection $f^{(k)}$ is the part of k parallel strands annihilated by every Temperley–Lieb turnback e_i, which joins neighboring strands on the same side of the box. Its trace is $[k+1]$. This turnback condition is different from annihilation by every boundary cap: a Jones–Wenzl projection has a nonzero partial trace. The first nontrivial case can be derived without any machinery. If e joins two neighboring strands by a cup and cap, then $e^2 = \beta e$. Therefore

$$f^{(2)} = 1 - \beta^{-1}e, \qquad (f^{(2)})^2 = f^{(2)}, \qquad ef^{(2)} = 0.$$

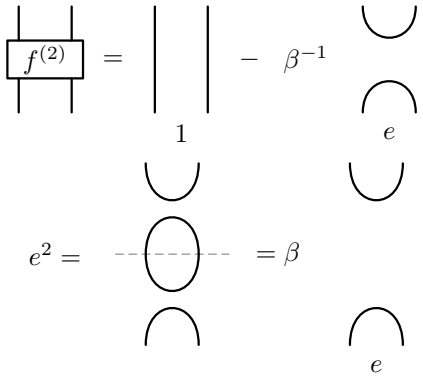


Figure 26.1: The two-strand Jones–Wenzl projection, with checkerboard shading suppressed. Morphisms are read from bottom to top, and $\beta = \sqrt{[M : N]}$. The normalized Jones projection is e/β. In the lower row the dashed line marks the glued boundary of two copies of e. The resulting internal loop evaluates to β, leaving the cup–cap operator and proving $e^2 = \beta e$.

Closing the two diagrams gives $\mathrm{tr}(1) = \beta^2$ and $\mathrm{tr}(e) = \beta$, so $\mathrm{tr}(f^{(2)}) = x - 1 = [3]$.

The extended Haagerup generator is an element $S \in P_{8,+}$, an eight-box with sixteen boundary points. It is self-adjoint, annihilated by all caps, and has eigenvalue -1 under two-click rotation R. One-click rotation H changes the shading; its direction is fixed by the sign of the last moment below. Write tr_0 for diagram closure into the zero-box space, and tr for its composition with the faithful positive scalar Perron–Frobenius trace.

Theorem 26.6 (Published extended Haagerup reconstruction criterion)**.** *In the graph planar algebra of a finite connected bipartite graph of norm β, suppose S has the preceding adjoint, cap, and rotation properties and*

$$\operatorname{tr}_0(S^2) = [9], \qquad \operatorname{tr}_0(S^3) = 0, \qquad \operatorname{tr}_0(S^4) = [9],$$
$$\operatorname{tr}_0(H(S)^3) = i\frac{[18]}{\sqrt{[8][10]}}. \tag{26.15}$$

Then the generated positive planar algebra is the extended Haagerup planar algebra. Taking direct sums and splitting projections in its two even parts gives EH_2 and EH_1.

The finite moments carry more information than their short list suggests because the ambient trace is positive and the planar operations impose strong compatibility. Here is the first step. Uncappability implies $f^{(8)}S = S = Sf^{(8)}$. Write $f = f^{(8)}$. The second and fourth moments give

$$\operatorname{tr}((S^2 - f)^*(S^2 - f)) = [9] - 2[9] + [9] = 0.$$

Faithfulness and positivity force

$$S^2 = f. \tag{26.16}$$

An indefinite pairing would not justify this inference.

The remaining part of the published theorem derives diagram relations from positive Gram matrices. They allow generators to be moved toward the outside of a diagram and adjacent pairs to be reduced using (26.16). This is the *jellyfish evaluation* method. It evaluates every closed diagram and identifies the generated planar algebra. The full diagram reduction is an input here; four numerical moment matches alone would not replace its theorem.

Worked example 26.7 (Finding objects by splitting one operator)**.** Equation (26.16) gives two orthogonal projections

$$p_+ = \frac{f + S}{2}, \qquad p_- = \frac{f - S}{2}.$$

Indeed $p_\pm^2 = p_\pm$, $p_+p_- = 0$, and $p_+ + p_- = f$. The trace of S vanishes by uncappability, so both projections have dimension $[9]/2$. The principal graph identifies them as the two depth-eight simple objects. The algebraic formula proves they are projections; simplicity and the completeness of the simple list use the reconstructed principal graph.

This illustrates how a category is extracted from a diagram presentation. Objects are projections, morphisms are planar maps cut down by the relevant source and target projections, and tensor product places projections side by side. It is the same idempotent-completion operation encountered in Chapter 5, now with explicit planar coordinates.

26.7 Twenty-one seeds and a ternary word tensor

The generator has many graph-loop coordinates, but its essential coefficient data can be compressed. Set

$$\lambda_{\mathrm{EH}} = i\sqrt{x - 2}, \qquad \lambda_{\mathrm{EH}}^6 + 2\lambda_{\mathrm{EH}}^4 - 3\lambda_{\mathrm{EH}}^2 - 5 = 0. \tag{26.17}$$

For this section abbreviate this scalar by λ. Substituting $x = 2 - \lambda^2$ into the index cubic proves its sextic equation. Thus exact arithmetic uses six rational coordinates, with reduction $\lambda^6 = -2\lambda^4 + 3\lambda^2 + 5$. Conjugation at the chosen embedding sends λ to $-\lambda$.

Collapse each relevant loop to a word of length eight in three arm labels $0, 1, 2$. The resulting coefficients p_w satisfy signed cyclic and reflection rules

$$p_{Lw} = -p_w, \qquad p_{Jw} = \overline{p_w}, \tag{26.18}$$

where

$$L(w_0 \cdots w_7) = w_1 \cdots w_7 w_0, \qquad J(w_0 \cdots w_7) = w_0 w_7 \cdots w_1.$$

There is also a linear relation in each slot:

$$p_{a0b} + p_{a1b} + p_{a2b} = 0. \tag{26.19}$$

Consequently the binary words determine all $3^8 = 6,561$ ternary coefficients. Up to the signed dihedral symmetries and vanishing orbits, only 21 seed values are needed. The seed values are published data; the following tensor packaging is a derived reorganization of them.

Introduce vectors

$$u_0 = (1, 0), \qquad u_1 = (0, 1), \qquad u_2 = (-1, -1).$$

Let T be the eight-linear form whose values on the binary basis vectors are the prescribed binary coefficients. Then

$$p_w = T(u_{w_0}, \ldots, u_{w_7}). \tag{26.20}$$

Because $u_0 + u_1 + u_2 = 0$, linearity proves (26.19) automatically. Repeatedly replacing letter 2 by $-0 - 1$ gives the unique extension from binary words. This is an ordinary finite tensor identity, not an approximate compression.

Worked example 26.8 (Computing a new word coefficient)**.** Two seeds are

$$c_1 = p_{00000001} = -2\lambda^4 - \lambda^2 + 9, \qquad c_2 = p_{00000011} = -\lambda^5 - \lambda^3 + 3\lambda.$$

Rotation gives $p_{00000010} = -c_1$. Applying the slot relation at the last letter gives

$$\begin{aligned} p_{00000012} &= -p_{00000010} - p_{00000011} = c_1 - c_2 \\ &= \lambda^5 - 2\lambda^4 + \lambda^3 - \lambda^2 - 3\lambda + 9. \end{aligned}$$

Every step is an exact signed linear operation. The coefficient vector in the basis $1, \lambda, \ldots, \lambda^5$ is $(9, -3, -1, 1, -2, 1)$.

One can package the entire dihedral extension without recursion. For a binary word b, define $M_b(w) = \prod_j (u_{w_j})_{b_j}$. Let $\mathcal{R}$ be the 21 seed representatives, let c_r be their polynomials, and let s_r be the size of the ordinary dihedral stabilizer of r. Then

$$p_w = \sum_{r \in \mathcal{R}} \frac{1}{s_r} \sum_{k=0}^{7} (-1)^k \left[c_r(\lambda) M_{L^k r}(w) + c_r(-\lambda) M_{L^k J r}(w) \right]. \tag{26.21}$$

For binary w, $M_b(w)$ is simply $\mathbf{1}_{b=w}$. Every matching dihedral image is counted s_r times, explaining the denominator. Compatibility of the seeds with their stabilizers makes those contributions equal. The formula has integer polynomial coefficients despite the displayed division. For ternary w, multilinearity completes the proof.

To recover actual graph-loop coefficients, multiply these word coefficients by the prescribed loop signs and ratios of positive Perron–Frobenius vertex weights. The graph planar algebra then turns multiplication into finite path-block matrix multiplication. The recorded exact audit checks all 96 nonempty blocks needed for the moment conditions. This finite calculation is distinct from the hand proof of the word symmetries. Neither has been rerun for this textbook.

26.7.1 The coefficient field and the gauge field

The field $\mathbb{Q}(\lambda)$ is useful for exact word arithmetic. Positive square roots introduced by a particular orthonormal choice of path or trivalent basis may enlarge the displayed field of matrix entries. This does not mean that the underlying category requires every such square root in every gauge.

A *split rational form* over a field K means a tensor category defined over K in which the simple objects and their scalar endomorphism algebras are already split, and whose extension of scalars gives the complex category. This is stronger than merely writing some unsplit algebra over K. The published field result distinguishes the original even parts: EH_1 has a split form over $\mathbb{Q}(x)$, whereas EH_2 requires the noncyclotomic degree-six field represented here by $\mathbb{Q}(\lambda)$. The parameter often used for the latter is $-1/\lambda$, which generates the same field. These statements concern tensor structure, not just the numerical dimensions.

26.8 From projections to associators

Choose a representative projection p_a for every simple object. For each fusion space choose a basis $x_i \in \mathrm{Hom}(p_c, p_a \otimes p_b)$ and form its Gram matrix

$$H_{ij} = d_c^{-1} \operatorname{tr}(x_i^* x_j), \qquad d_c = \operatorname{tr}(p_c).$$

Simplicity makes $x_i^* x_j$ a scalar times p_c. Positivity therefore makes H positive definite, and

$$v_\mu = \sum_i x_i (H^{-1/2})_{i\mu}$$

is an isometric basis: $v_\mu^* v_\nu = \delta_{\mu\nu} p_c$. Completeness over all channels says $\sum_{c,\mu} v_\mu v_\mu^* = p_a \otimes p_b$.

For three objects, construct the two tree bases

$$L_{e,\alpha,\beta} = (v_{e,\alpha}^{ab} \otimes p_c) v_{d,\beta}^{ec}, \qquad R_{f,\mu,\nu} = (p_a \otimes v_{f,\mu}^{bc}) v_{d,\nu}^{af}.$$

In the planar category horizontal placement is strictly associative. The skeletal associator records only the change between these two decomposition bases. With the left-channel row and right-channel column convention of (1.5), it is

$$(F_d^{abc})_{(e,\alpha,\beta),(f,\mu,\nu)} = d_d^{-1} \operatorname{tr}(R_{f,\mu,\nu}^* L_{e,\alpha,\beta}). \tag{26.22}$$

Both bases are orthonormal, so this matrix is unitary. For four objects the two pentagon routes insert different intermediate complete tree bases. Their completeness projectors are identities, and both routes reduce to the same initial-to-final overlap. This proves the pentagon, including all multiplicity labels.

Thus the exact planar generator determines the full associator up to gauge. Printing a complete table is still substantial additional work: one must explicitly split every simple, choose all fusion bases, and evaluate all overlaps. The existence of a complete presentation should not be confused with possession of a fully enumerated table.

Worked example 26.9 (A three-by-three extended Haagerup recoupling)**.** In either even part take $X = f^{(2)}$ and $Y = f^{(4)}$. At these small depths the category agrees with Temperley–Lieb, so

$$X^2 = \mathbf{1} + X + Y, \qquad \mathrm{qdim}(X) = y = x - 1, \qquad \mathrm{qdim}(Y) = z = x^2 - 3x + 1.$$

The dimension identity is $y^2 = 1 + y + z$. In a conventional Temperley–Lieb trivalent gauge, with

both channel orders $(\mathbf{1}, X, Y)$, the recoupling is

$$F_X^{XXX} = \begin{pmatrix} \frac{1}{y} & -\frac{1}{\sqrt{y}} & \frac{\sqrt{z}}{y} \\ -\frac{1}{\sqrt{y}} & \frac{x-3}{x-2} & \frac{\sqrt{z/y}}{x-2} \\ \frac{\sqrt{z}}{y} & \frac{\sqrt{z/y}}{x-2} & \frac{1}{(x-2)y} \end{pmatrix}. \tag{26.23}$$

Its identification as an actual recoupling uses the Temperley–Lieb diagram evaluation; its orthogonality has a short independent proof. Put $a = x - 2$, so $y = a + 1$ and $z = a^2 + a - 1$. The vector

$$v = \left(\sqrt{\frac{a}{2y}}, \frac{1}{\sqrt{2a}}, -\sqrt{\frac{z}{2ay}} \right)^T$$

has norm one because $a^2 + y + z = 2ay$. Direct entry comparison gives

$$F_X^{XXX} = I - 2vv^T.$$

It is an orthogonal reflection, with eigenvalue -1 along v and $+1$ on its orthogonal plane. In particular its square is I without expanding nine matrix products.

This block uses only six strands, below the new degree-eight generator. It is therefore an especially accessible example, but does not display all the distinctive extended Haagerup recouplings. The recorded local checks of 77 low-depth blocks and 90 scalar pentagon equations cover a stated Temperley–Lieb range. The general pentagon follows from the reconstruction and basis-overlap argument, not from extrapolating those finite numerical tests.

26.9 What this teaches about a CFT construction

Both methods produce tensor categories with exact finite presentations. The noncyclic method uses signed endomorphism coefficients; the extended Haagerup method uses planar diagrams and positive trace relations. A Drinfeld center can then organize braided sectors, and its modular data provide further invariants. None of these operations alone constructs a vacuum VOA, a positive-energy Hilbert space of local fields, or correlators satisfying analytic sewing. Those additional steps are the subject of the CFT construction chapters.

In particular, the existence of a noncyclic unitary category does not identify the continuum limit of the order-three Haagerup chain. A resemblance between modular representations likewise does not identify two categories or produce an extended Haagerup CFT. Each proposed transfer must specify which tensor category, which chiral algebra, and which analytic realization are being transferred.

26.10 Exercises

Exercise 26.1. Suppose $G = H \times K$ with both factors nontrivial and $W(h,k) = U(h)V(k)$, $W(0,0) = -1$. Assume $W(g)W(-g) = \delta \neq 0$ for every nonzero g. Show that $\delta = 1$. Explain why independent scalar weights for the two factors cannot directly give the generalized Haagerup reciprocal ansatz.

Solution on page 319.

Exercise 26.2. Starting from (26.7) with $t, y, r > 0$, derive $r + r^{-1} = (5 + t^2)/2$ and select its root $r < 1$. Identify exactly where positivity is used. Given $tr^3 = 1 - 2r$, derive $y = \lambda_8^{-3}$.

Solution on page 319.

Exercise 26.3. At $m = 3$, write the multiplication matrix for $D-1$ on $\mathbb{Z}[\omega]$, find its determinant and entry gcd, and recover $\mathbb{Z}_2 \times \mathbb{Z}_6$. Why does knowing only the order of this quotient fail to distinguish it from $\mathbb{Z}_{12}$?

Solution on page 320.

Exercise 26.4. Let $S = S^*$, $f = f^* = f^2$, and $fS = S = Sf$ in a finite dimensional $*$-algebra with faithful positive trace. Suppose $\mathrm{tr}(S^2) = \mathrm{tr}(S^4) = \mathrm{tr}(f)$ and $\mathrm{tr}(S) = 0$. Derive $S^2 = f$ and verify the projection and trace formulas for $(f \pm S)/2$. Which inference would fail for an indefinite trace pairing?

Solution on page 320.

Exercise 26.5. Using (26.18), show that a word fixed by an odd power of L has zero coefficient. Explain why the multilinear rule determines all ternary coefficients uniquely from the binary ones. Compute $p_{00000002}$ from the seed $p_{00000001} = c_1$ and $p_{00000000} = 0$.

Solution on page 320.

Exercise 26.6. Verify the norm of v and the $(1,2)$ and $(2,3)$ entries of $I - 2vv^T$ in (26.23). Prove that the matrix is unitary and find its determinant. Does this calculation alone prove that the matrix is an extended Haagerup associator?

Solution on page 320.

Source note. The published noncyclic order-eight example is Grossman–Izumi–Snyder, *The Asaeda–Haagerup fusion categories*, Theorem 4.5. The determinant construction, order-eight gamma elimination, and recorded order-twelve certificates are taught from `hi_noncyclic_analytic_note.tex`, `hi_noncyclic_gis_gamma_proof.tex`, and their exact certificate artifacts. The current open gamma-identification status is recorded in `hi_noncyclic_gamma_identification_status.md`. The free-product reconstruction mechanism comes from `hi_noncyclic_reconstruction_extension.tex`. The imported even-cyclic identities are supplied in `hi_even_cyclic_analytic_proof.tex`.

The planar algebra construction and moment criterion are published results of Bigelow–Morrison–Peters–Snyder, *Constructing the extended Haagerup planar algebra*, and Grossman–Morrison–Penneys–Peters–Snyder, *The Extended Haagerup fusion categories*. The rational-form distinction is due to Morrison–Snyder, *Non-cyclotomic fusion categories*. The local report `extended_haagerup_reconstruction_and_coefficients.tex` reorganizes the published seeds into the closed tensor formula and records independent exact checks; it does not claim a new category or a complete printed associator table.

Appendix A

Notation and a working glossary

The subject joins several mathematical languages. The same letter can have a conventional meaning in more than one of them. This appendix records the principal conventions and identifies the distinctions that matter when moving between chapters.

A.1 Fields and states

Symbol	Meaning
$\mathbf{1}$	Vacuum state, tensor unit, or identity object according to context. The vacuum simple-module label is also written 0.
$Y(a,z)$	Field associated to state a; $Y(a,z)=\sum_n a_{(n)}z^{-n-1}$.
$a_{(n)}b$	The coefficient of $(z-w)^{-n-1}$ in the OPE of a and b; it has weight $\mathrm{wt}(a)+\mathrm{wt}(b)-n-1$.
T, D	Translation operator on a VOA, equal to L_{-1}. The construction chapters often use D. It is distinct from a stress field $T(z)$ and a modular matrix T.
ω, $T(z)$	Conformal vector and corresponding stress field; $Y(\omega,z)=T(z)=\sum_n L_n z^{-n-2}$.
H_L, H_R	In the construction chapters, Cartan currents of the internal affine factors at levels one and thirty-nine. Both are holomorphic; the subscripts do not denote the two physical chiralities. In the chain chapter, H_L instead denotes the Hamiltonian of a chain of length L.
ω_L, ω_R	Sugawara conformal vectors of those internal factors, with charges $1, 117/41$ and fields T_L, T_R.
$a_{(-1)}b$	Normal-product state, corresponding to $:Y(a,z)Y(b,z):$.
$h, \bar{h}$	Holomorphic and antiholomorphic conformal weights.
Δ	Full scaling dimension $h+\bar{h}$.
s_{spin}	Conformal spin $h-\bar{h}$. Some local physics formulas abbreviate it to s; this is distinct from the categorical abbreviation $1+d^2$.
c	One chiral central charge. For the nonchiral theories here, $c_L = c_R = c$; their sum is $2c$.
v	The nonuniversal velocity in a lattice-to-cylinder energy conversion; elsewhere a generic state vector when no velocity is involved.
$\mathcal{H}_{\mathrm{path},L}$	The length-L periodic chain Hilbert space, with orthonormal basis the admissible closed fusion paths.

Symbol	Meaning
U_{tr}	The chain's unitary right cyclic shift, with $U_{\mathrm{tr}}^{L} = 1$ and $U_{\mathrm{tr}} h_j U_{\mathrm{tr}}^{-1} = h_{j+1}$. It is distinct from the chiral algebra U and VOA translation T.
$\widehat{a}$	A field divided by its positive norm. It does not fix the remaining phase of a complex field.
$a^{\dagger}$	Radial adjoint or Hilbert-space adjoint, according to context; not an instruction to add a right-moving field.
θ	An antilinear PCT operation in VOA sections; θ_i can also denote a categorical twist, distinguished by its indexed scalar use.

A.2 Categories and modular data

Symbol	Meaning
$\mathcal{H}_3$	The specified unitary six-object Haagerup fusion category used for the physical chain.
$\mathcal{H}_2$	A different unitary category with the same six-object fusion ring, Morita equivalent to $\mathcal{H}_3$.
α	The invertible order-three Haagerup object. A lattice root is also conventionally denoted by α; its lattice context or subscript distinguishes it.
ρ	A noninvertible Haagerup simple with $\rho^2 = \mathbf{1} + \rho + \alpha\rho + \alpha^2\rho$.
d	The positive quantum dimension $(3+\sqrt{13})/2$, satisfying $d^2 = 1+3d$.
s	In the category and extension formulas, $s = 1 + d^2 = 2 + 3d$.
$\mathrm{Dim}(\mathcal{C})$	Global dimension, $\sum_i d_i^2$, for a positive spherical fusion category.
$\mathcal{D}$	Total quantum dimension $\sqrt{\mathrm{Dim}(\mathcal{C})}$ of the modular category under discussion. For $\mathcal{C} = Z(\mathcal{H}_3)$, $\mathcal{D} = 3s = \mathrm{Dim}(\mathcal{H}_3)$.
$Z(\mathcal{C})$	Drinfeld center, whose objects have compatible half-braidings with every object of $\mathcal{C}$.
F_d^{abc}	Associator matrix between two bases of a fusion space. Left channels index rows and right channels index columns; see (1.5). Elsewhere F can denote a lowering current or a forgetful functor; its arguments distinguish these uses.
X^*	Categorical dual object. Dualizing a morphism is linear; it differs from taking a Hermitian adjoint, which conjugates scalars.
S	Modular matrix, normalized by $S_{00} = 1/\mathcal{D}$. For a pointed lattice example we use the negative Fourier sign.
Θ	Categorical twist matrix, with entries $e^{2\pi i h_i}$.
T_{ch}	Character transformation matrix, $e^{-2\pi i c/24}\Theta$. At $c = 8$ its common prefactor is $e^{-2\pi i/3}$.
a	One of the twelve central simple labels. It is not the invertible Haagerup object α.
$\mathcal{C}_A$	All right modules of an algebra object A. For the connected commutative separable algebras in the braided setting used here, the braiding supplies a left action and defines the balanced tensor product $\otimes_A$.

Symbol	Meaning
$\mathcal{C}_A^0$	Local modules: if $r : X \otimes A \to X$ is the action, then $r\, c_{A,X} c_{X,A} = r$. This is an equation for the action, not necessarily identity of the double braiding on the entire underlying tensor product.
${}_A\mathcal{C}_A$	Algebra bimodules, describing the relevant full-theory defects in the rational construction.
$\mathcal{C}^{\mathrm{op}}$	The convention in this book reverses the order of the tensor product, not the direction of morphisms.
$\mathcal{C}^{\mathrm{rev}}$	For a braided category, reverse the braiding and twist. This differs from reversing tensor order.
$\mathcal{C} \boxtimes \mathcal{D}$, $M \boxtimes_V N$	The first is the external product of two categories, with pairs of simple labels. The second fuses two modules of one VOA V. Neither is the plain tensor product of the underlying module vector spaces.

A.3 The main algebras and their charges

Object	Role
V_L	Positive even lattice VOA, with central charge rank L.
$B = V_{E_6 \oplus A_2}$	The immediate ambient algebra of the six-current construction; $c = 8$ and $\dim B_1 = 86$.
$H = V_{E_8}$	The original holomorphic lattice extension of B; $c = 8$ and $\dim H_1 = 248$.
U	The actual vertex algebra generated by the two affine current triples, the ambient conformal vector, and the field u. Under the finite closure inputs, $\dim U_1 = 6$.
H'	The holomorphic extension selected through the canonical $\mathcal{H}_3$ algebra. It is isomorphic to V_{E_8} but its embedding of U is not automatically the original one.
$A_{1,k}$	The simple integrable affine $L_k(\mathfrak{sl}_2)$ VOA, of charge $3k/(k+2)$.
$K(\mathfrak{sl}_2, k)$	The parafermion commutant of the Cartan Heisenberg algebra, of charge $2(k-1)/(k+2)$.
Potts	The $k = 3$ parafermion algebra, of charge $4/5$.
Level-thirteen parafermion	The $k = 13$ factor, of charge $8/5$.
$\mathcal{T}$	The rank-two lattice–Potts core, of charge $14/5$. Its detailed lattice extension is defined in Chapter 9.
A_L	The larger order-thirteen core extension, of charge $22/5$ and fourteen simple modules. The subscript is part of the core notation, not a declaration that it is an affine level.
C_6	The rational central-charge-six commutant constructed from the saturated plane lattice under the accepted $c = 8$ construction.
c_*	The irrational chiral charge $2.025004294\ldots$ arising from a stress axis. It is not an identified central charge of the physical chain.

A.4 A glossary of the main logical distinctions

Abelian current. A current whose OPE with the other currents in the selected abelian family has only a central double pole and no Lie-bracket simple pole. The double pole need not vanish.

Anomaly. An obstruction to a specified symmetry operation, such as gauging a group action with nontrivial associator class. A chiral gravitational anomaly and a finite-group associator anomaly are different data; the context must specify which is meant.

Associator. A coherent change between the two ways of parenthesizing a tensor product. Its pentagon relation is more information than associativity of the fusion multiplicities.

Boundary category. The category of boundary conditions or all extension modules in the specified construction. It can be nonbraided even though the bulk chiral category is braided.

Bounded localized vertex operators. A property combining bounded annulus-regularized insertion maps, with suitable energy damping on the inserted state, and locality of the bounded algebras they generate. It is stronger than boundedness of an insertion for each fixed state separately.

Branching. Decomposition of a representation after restricting to a subalgebra. Multiplicity spaces can carry a commuting algebra.

Canonical. A construction determined naturally after its input data and conventions have been specified. The word does not remove the need to state which category, algebra, or normalization is being used.

Cauchy complete category. A category in which every idempotent splits: if $p^2 = p$ on X, there is an object representing its image, with inclusion and projection composing to p. This is the categorical version of retaining the range of a projection as an object.

CFT type. A nonnegatively integer-graded VOA with finite-dimensional homogeneous spaces and $V_0 = \mathbb{C}\mathbf{1}$. This does not assert rationality.

Character. A trace counting states by conformal energy. It forgets most operator products and can coincide for distinct simple modules.

Commutant. Fields whose singular OPE with a subalgebra vanishes. Constructing a commutant is a chiral operation.

Completely positive map. A linear map preserving positive semidefinite matrices even after tensoring with the identity on any finite auxiliary matrix space. The MPS transfer formula $\mathcal{E}(X) = \sum_s A^s X (A^s)^\dagger$ makes this property explicit; trace preservation is a separate condition.

Complete unitarity. Positivity of module forms and canonical fusion forms, beyond positivity of the vacuum space.

Completely rational net. A chiral conformal net satisfying the split and strong-additivity properties and having finite two-interval index. These hypotheses give a finite unitary modular sector theory.

Conformal block. A chiral solution of the symmetry Ward identities for specified external representations. A factorization basis labels blocks by intermediate representations and junction multiplicities, summing their descendants. A general block may be a linear combination of that basis. Full correlators additionally specify how holomorphic and antiholomorphic blocks are paired.

Conformal net. An assignment of a von Neumann algebra of bounded observables to each interval, satisfying inclusion, locality, conformal covariance, and the vacuum and positive-energy conditions used in the text.

Construction input. A model-specific finite identity, membership, rank, or structural datum accepted from the exact construction record. It is a mathematical premise with a stated verification scope.

Core of an operator. A subspace of an operator's domain dense in its graph norm. Approximating a vector there also approximates its image. Ordinary Hilbert-space density alone does not supply this property.

Effective central charge. The parameter governing a leading asymptotic trace or entropy under its stated spectral assumptions; it need not equal the Virasoro central charge without a normalizable vacuum of lowest weight zero.

Energy bound. An estimate controlling field modes by a polynomial in their mode number and in the conformal Hamiltonian. Such estimates make smearing and operator closure possible on controlled domains.

Étale algebra. A commutative separable algebra object in the braided category under discussion. Connectedness is an additional condition, $\dim \mathrm{Hom}(\mathbf{1}, A) = 1$.

Exact finite result. A statement about the specified finite algebraic calculation or finite chain, proved with appropriate arithmetic or certification. It does not alone control an infinite-size limit.

Fiber functor. A faithful exact strong monoidal functor from a fusion category to finite-dimensional vector spaces. It is much more structure than a representation of the fusion ring by ordinary matrices.

Frobenius–Schur indicator. The trace of pivotal cyclic rotation on $\mathrm{Hom}(\mathbf{1}, X^{\otimes n})$. It detects information about the associator and duality beyond the fusion rules. In a spherical fusion category it also equals the categorical twist trace of center induction divided by the global dimension.

Fusion category. A rigid semisimple complex linear tensor category with finitely many simple isomorphism classes, finite-dimensional morphism spaces, and simple tensor unit. It need not have a braiding or a positive adjoint.

Fusion space. A junction multiplicity space such as $\mathrm{Hom}(c, a \otimes b)$. Its dimension is a fusion coefficient, not the dimension of a physical lattice Hilbert space.

Gap. An energy gap is an excitation-energy separation above the vacuum sector. A scalar scaling gap is the lowest nonidentity scalar dimension Δ. A positive scaling gap in a CFT corresponds to energies of order $1/L$.

Haag duality. On the conformal vacuum Hilbert space, the algebra assigned to an interval equals the commutant of the algebra assigned to its complementary interval. The representation and cyclic-vacuum hypotheses matter.

Holomorphic. A rational chiral algebra with only its vacuum module. The corresponding full theory still has infinitely many local fields.

Invariant field. A field on which each simple symmetry line acts by its vacuum eigenvalue, equal to that line's quantum dimension.

Ishibashi state. A formal solution of chiral boundary gluing equations within a compatible representation pair. Its propagated overlap is a character; the unregularized state is not generally normalizable.

Isotropic subgroup. A subgroup of a finite discriminant form on which the quadratic function vanishes modulo one. It determines an even lattice extension in the lattice setting.

Jones–Wenzl projection. The Temperley–Lieb idempotent on a fixed number of parallel strands killed by the adjacent turnback operators. It removes the lower-through-strand summands and makes simple projections explicit.

Lagrangian algebra. A connected commutative separable algebra whose squared dimension equals the global dimension of the ambient modular category; its local module category is trivial.

Marginal by dimension. A scalar of dimension two. Exact marginality additionally requires vanishing beta functions and compatibility among deformations.

Modular category. A braided ribbon fusion category with nondegenerate braiding; equivalently in this setting, its normalized S matrix is invertible. Positivity is a further property when a unitary structure is specified.

Module category. A category on which a fusion category acts, with coherent associativity maps. It encodes boundary and gapped vacuum patterns unavailable from the fusion ring alone.

Morita equivalence. Equivalence through an invertible bimodule-category relation. Morita-equivalent fusion categories have equivalent centers but need not have the same simple objects or the same chain Hamiltonian.

Near-group rules. Fusion rules with a group of invertible simples and only one noninvertible simple. Generalized Haagerup rules instead have one noninvertible simple for each group element.

Normal convergence. Uniform absolute convergence on compact subsets of the relevant complex parameter domain. It permits the analytic operations used with the trace functions in the stated domains.

Ordinary module. A module with diagonalizable energy, finite-dimensional eigenspaces, and lower truncation along each integer-shift weight string. A simple ordinary module here has one lowest-weight grading; a direct sum can have several.

Perron eigenvector. A positive eigenvector of a nonnegative matrix at its spectral radius. For a matrix with strongly connected directed graph it is unique up to scale; primitivity additionally excludes other eigenvalues of the same modulus. Such vectors supply positive weights in fusion graphs and classical transfer matrices.

Planar algebra. Vector spaces of labeled boxes equipped with multilinear operations specified by planar string diagrams. Substituting diagrams agrees with composition of operations. A positive subfactor planar algebra reconstructs tensor categories by taking sums and splitting projections.

Primary versus quasiprimary. A Virasoro primary is killed by all positive Virasoro modes; a quasiprimary need only be killed by L_1 and have definite weight.

Primitive transfer operator. A finite-dimensional completely positive transfer map for which, after normalization and restriction to its supported algebra, the eigenvalue one is simple, all other eigenvalues have modulus less than one, and the left and right fixed metrics are strictly positive. Equivalently in this setting, a suitable positive power sends every nonzero positive operator to a strictly positive operator. A symmetry cat can fail this property even when its pure components are well behaved.

Projective object. An object P for which any morphism $P \to Y$ lifts through every surjection $X \to Y$. Projectivity of the vacuum is a central step in the regularity argument; right exactness of a functor is a different property.

Pseudounitary category. A fusion category whose categorical and Frobenius–Perron global dimensions agree, with the corresponding canonical positive spherical dimensions. The definition does not supply a positive adjoint on morphisms.

Q-system. A positive separable Frobenius algebra object in a unitary tensor category, in the appropriate normalization. It is the algebraic datum of a finite-index extension or a module construction.

Ramified prime. For a number field, a rational prime dividing its field discriminant. Ramification persists in a larger number field, which lets it obstruct a proposed field inclusion. For $\mathbb{Q}(\zeta_N)$, only primes dividing N can ramify.

Regularity. Complete decomposition of weak modules into ordinary simple modules, in the VOA sense used in the construction.

Right exact functor. A functor preserving cokernels, and therefore sequences $X \to Y \to Z \to 0$ in the relevant abelian setting. It need not preserve kernels or split extensions.

Schmidt decomposition. The singular-value decomposition of a pure-state coefficient matrix across a bipartition. Its squared singular values are the reduced density-matrix eigenvalues and determine the bipartite entropy.

Sewing. Compatibility of correlation functions under cutting, inserting a complete set of states, and gluing surfaces. Sphere crossing and torus modular invariance are related parts of this larger requirement.

Smith normal form. An integer matrix diagonalized using invertible integer row and column operations, with successive nonzero diagonal entries dividing the next. Those entries describe the cyclic factors of its presented abelian group.

Spherical structure. A coherent choice of duality traces for which left and right categorical traces agree. It permits isotopy-invariant loop evaluations and categorical dimensions.

Split rational form. A tensor category defined over a smaller field with its simple objects and scalar endomorphism algebras already split there. Extending scalars gives the complex category. A field of dimensions alone need not be such a field of definition.

Strong generation. Spanning by derivatives and normal products of a set of fields. It is stronger than arbitrary-mode vertex generation.

Strongly rational VOA. A simple, self-contragredient, CFT-type VOA that is rational and C_2-cofinite, with the module-theoretic conventions stated in the regularity chapter.

Tensorator. The coherent isomorphisms comparing the image of a tensor product with the tensor product of images under a monoidal functor. A linear equivalence does not supply them.

Tube algebra. A finite algebra of annular line networks whose simple representations encode center charge sectors of a fusion category. Its finite representation dimension is not a quantum dimension or the dimension of the dynamical CFT sector. It counts copies of that dynamical sector and therefore multiplies its state degeneracies.

Twined versus twisted. A twined trace inserts a line operator in an ordinary trace; a twisted trace counts states in a Hilbert space with a defect crossing the spatial circle.

Unitary fusion category. A fusion category with compatible positive adjoints on morphisms and unitary structural maps. Positive Frobenius–Perron dimensions alone do not imply this structure.

Weakly integral. A fusion category whose global Frobenius–Perron dimension is an integer. Its simple dimensions have integer squares but need not be integers: the Ising dimensions $1, 1, \sqrt{2}$ give global dimension four. The square condition concerns simple objects, not their arbitrary sums.

Weak module. A VOA module satisfying the field and Jacobi identities without a preassigned ordinary energy grading. Regularity supplies a decomposition into ordinary simple modules; it is not part of the weak-module definition.

Zhu algebra. An associative algebra built from a VOA whose representations describe the degree-zero action on module top spaces. Its use here relates complete component tops to the spectrum of partial conformal energies.

Appendix B

Exercise solutions

B.1 Solutions for Chapter 1

Exercise 1.1. Tensoring $m_L\mathbf{1} \oplus n_L\tau$ with τ gives

$$m_{L+1} = n_L, \qquad n_{L+1} = m_L + n_L.$$

At $L = 0$ the tensor product is the unit, so the pairs for $L = 0, 1, 2, 3, 4, 5$ are

$$(1, 0), (0, 1), (1, 1), (1, 2), (2, 3), (3, 5).$$

The recursion matrix has eigenvalues φ and $-\varphi^{-1}$. Its powers have integer entries, but their dominant asymptotic growth is controlled by the irrational eigenvalue φ. This is no different in principle from an integer recursion whose closed form involves cancelling irrational terms.

Exercise 1.2. The left fusion-tree vector is multiplied by $u^e_{ab}u^d_{ec}$, while the right tree with channel f is multiplied by $u^f_{bc}u^d_{af}$. Expressing the rescaled left vector in the rescaled right basis gives their ratio in (1.7). When all six labels equal ρ, both products are $(u^\rho_{\rho\rho})^2$, so the ratio is one. For generic labels the ratio need not be one; changing a junction phase can change the entry without changing the category.

Exercise 1.3. For two channels f, g,

$$\begin{aligned}(P^{(f)}P^{(g)})_{ee'} &= \sum_k \overline{F_{ef}}F_{kf}\overline{F_{kg}}F_{e'g} \\ &= \overline{F_{ef}}F_{e'g}\,\delta_{fg} = \delta_{fg}(P^{(f)})_{ee'}.\end{aligned}$$

The sum over k is column orthogonality. Similarly, $\sum_f \overline{F_{ef}}F_{e'f} = \delta_{ee'}$ is row orthogonality. Omitting the complex conjugates would give the wrong projector in a general unitary gauge, even though it would go unnoticed for the real Fibonacci and $\mathcal{H}_3$ blocks displayed in the text.

Exercise 1.4. The identities $v^2 = u$ and $u^2 = 1 - u$ give

$$u^2 - uv^2 = 0, \qquad uv + u^2v = v(u + u^2) = v, \qquad v^2 - u^3 = u - u(1 - u) = u^2.$$

The four remaining off-diagonal entries use either $v^2 = u$ or the already identical value $-uv$; the middle diagonal entries also use $v^2 = u$. Both matrix products therefore equal

$$\begin{pmatrix} 0 & u & v \\ u & u & -uv \\ v & -uv & u^2 \end{pmatrix}.$$

Dividing the $(1,3)$ equation by v uses $v \neq 0$. The other root of $u^2 + u = 1$ is negative, so the condition $v^2 = u$ cannot hold with v real. This excludes that root from the stated positive real ansatz; it is not an argument that the algebraic fusion ring has only one categorification.

Exercise 1.5. The roots are $\varphi = (1+\sqrt{5})/2$ and $-\varphi^{-1} = (1-\sqrt{5})/2$. Since the complexified ring is $\mathbb{C}[t]/(t^2-t-1)$, evaluation at either root is a unital algebra homomorphism to $\mathbb{C}$, hence a one-dimensional representation of that algebra. The word "one-dimensional" describes the space on which the algebra acts, not the value of its generator. An ordinary finite-dimensional vector space has a nonnegative integer dimension. The positive root is irrational and the other root is negative, so neither is such a dimension. In particular, a tensor functor sending τ to an ordinary vector space would require an integer N with $N^2 = 1 + N$, which is impossible. The positive categorical dimension instead measures asymptotic fusion-space growth.

B.2 Solutions for Chapter 2

Exercise 2.1. The commutator $[L_1, L_{-1}] = 2L_0$ gives, by induction,

$$L_1 L_{-1}^n |h\rangle = n(2h+n-1) L_{-1}^{n-1} |h\rangle.$$

Taking an inner product and using $L_{-1}^\dagger = L_1$ therefore multiplies the previous norm by $n(2h+n-1)$. Starting with a unit primary gives $n!(2h)_n$. For equal external weights, the primary three-point function has $\langle \phi(\infty)\phi(1)\mathcal{O}_p(t)\rangle = C(1-t)^{-p}$. Differentiating n times at $t=0$ inserts $L_{-1}^n \mathcal{O}_p$ and gives $C(p)_n$. Each three-point descendant coefficient is therefore $(p)_n$ relative to the primary coupling. Writing x for the cross-ratio as in the chapter, the block in the reduced correlator $x^{2h}G(x)$ is consequently

$$x^p \sum_{n\geq 0} \frac{(p)_n^2}{n!(2p)_n} x^n = x^p \left(1 + \frac{p}{2}x + \frac{p(p+1)^2}{4(2p+1)}x^2 + \cdots\right).$$

Multiplying by x^{-2h} restores the block of the unreduced correlator displayed in the chapter. The identity block is treated separately at $p = 0$.

Exercise 2.2. Let $v = (L_{-2} + aL_{-1}^2)|h\rangle$. Acting with L_1 gives $[3 + a(4h+2)]L_{-1}|h\rangle$, so at $h = 1/16$ we need $a = -4/3$. The L_2 action is $4h + c/2 + 6ah$. At $c = 1/2, h = 1/16$ it equals $1/4 + 1/4 - 1/2 = 0$. Thus

$$\left(L_{-2} - \frac{4}{3}L_{-1}^2\right)|1/16\rangle$$

is the null vector. Higher positive modes annihilate it by grading. Its vanishing is the input for the Ising differential equation in the sewing chapter.

Exercise 2.3. For $(m,n) = (1,1)$, $\bar{h} = 0$ requires $R^{-1} - R/2 = 0$, hence $R = \sqrt{2}$. Then $p_L = \sqrt{2}$ and $h = p_L^2/2 = 1$. Under $(R,m,n) \mapsto (2/R, n, m)$, p_L is unchanged and p_R changes sign. Both conformal weights are unchanged. The holomorphic current at the fixed radius is the exponential current of the A_1 lattice example.

Exercise 2.4. The fourth-order pole of $(T_1+T_2)(z)(c_2T_1 - c_1T_2)(0)$ is $(c_2c_1 - c_1c_2)/2 = 0$, and the remaining singular poles are $2A/z^2 + \partial A/z$. Thus A is primary for the total stress. Orthogonality of the two stress states gives

$$\|A\|^2 = \frac{1}{2}c_1c_2(c_1+c_2) > 0.$$

Its self-OPE has second-order coefficient $2(c_2^2 T_1 + c_1^2 T_2)$. Equality with $2A$ would require $c_2^2 = c_2$ and $c_1^2 = -c_1$. The second condition is impossible for $c_1 > 0$, so A is not a stress tensor in this normalization. A primary of weight two is therefore not automatically a Virasoro stress.

Exercise 2.5. Put $t = v\beta/L$. The vacuum contribution to $Z(it)$ at large t is $e^{\pi ct/6}$, because the cylinder vacuum energy uses $c_L + c_R = 2c$. Modular invariance replaces t by $1/t$. As $\beta \to 0$,

$$\log Z \sim \frac{\pi c}{6t} = \frac{\pi c L}{6v\beta}.$$

If the vacuum is not normalizable, the low-temperature trace is controlled by the actual spectral threshold and its measure. One cannot insert a unit-degeneracy vacuum term; this is where effective central charge and possible power or logarithmic prefactors enter.

Exercise 2.6. One needs normalized OPE coefficients controlling quadratic beta functions; higher-order integrated correlators, including local contributions when insertion points coincide, controlling further obstructions; and an analysis of which couplings describe inequivalent theories while preserving the specified symmetry. The coincident-point contributions are called *contact terms*. Couplings removable by a field redefinition or a symmetry transformation are *redundant directions* and must not be counted as independent parameters. One must also check whether a putative marginal field becomes a descendant when a current ceases to be conserved; this is one mechanism of *multiplet recombination*. These are additional tasks, not data supplied by the character count. Vanishing beta functions must hold on a locus of couplings after these identifications. A count of thirty-six dimension-two primaries alone supplies none of those conclusions.

B.3 Solutions for Chapter 3

Exercise 3.1. The singular products have $n = 0, \dots, 6$ and respective weights $6, 5, 4, 3, 2, 1, 0$. Products with $n > 6$ vanish because their weights would be negative. CFT type gives $V_0 = \mathbb{C}\mathbf{1}$, so $a_{(6)}b$ is a scalar multiple of the vacuum; $a_{(4)}b$ is the weight-two candidate. This list permits a product to be zero; weight counting does not establish its nonvanishing.

Exercise 3.2. Write the exponent of the sign as a bilinear form $B(a, b)$ over $\mathbb{Z}/2\mathbb{Z}$. Bilinearity gives

$$B(a, b) + B(a + b, c) = B(b, c) + B(a, b + c),$$

which is exactly the two-cocycle identity. For the triangular root-basis choice, $B(a, b) - B(b, a)$ is $\sum_{i>j} G_{ij}(a_i b_j + a_j b_i)$ modulo two. The diagonal terms in (a, b) are even, so this is (a, b) modulo two. Hence the commutator ratio is $(-1)^{(a,b)}$. A coboundary has the form $f(a)f(b)/f(a+b)$ and is symmetric in a, b; it cancels out of that ratio.

Exercise 3.3. The oscillator series begins $1 + q + 2q^2 + \cdots$. For the vacuum lattice, $\Theta_L = 1 + 2q + O(q^4)$, giving

$$\Theta_L \prod_{n \geq 1} (1 - q^n)^{-1} = 1 + 3q + 4q^2 + \cdots.$$

For the shifted lattice, the momentum weights are $(m + 1/2)^2$. Thus

$$\Theta_{L+\alpha/2} = q^{1/4}(2 + 2q^2 + \cdots), \qquad \frac{\Theta_{L+\alpha/2}}{\prod_{n\geq 1}(1 - q^n)} = q^{1/4}(2 + 2q + 6q^2 + \cdots).$$

The two shortest momenta explain the two-dimensional top. The module is nevertheless invertible: adding its discriminant coset twice gives L. An invertible simple has categorical dimension one, independently of its top multiplicity.

Exercise 3.4. Every oscillator of index $n \geq 2$ can be removed modulo C_2:

$$J_{-n}b = \frac{1}{(n-1)!}(\mathsf{T}^{n-2}J)_{(-2)}b \in C_2(V).$$

Commuting the creation oscillators shows that only powers of $J_{-1}\mathbf{1}$ remain. To see that these powers have no relation, count oscillators. A monomial of weight w has at most w oscillators, with equality only when every index is one. For homogeneous Fock monomials a, b, the state $a_{(-2)}b$ has weight $\mathrm{wt}(a) + \mathrm{wt}(b) + 1$, but Wick products have at most $\mathrm{wt}(a) + \mathrm{wt}(b)$ oscillators; contractions only lower their number. Thus every resulting monomial contains an index at least two. No C_2 term has a pure J_{-1} component in the Fock basis. The quotient is $\mathbb{C}[x]$, an infinite-dimensional algebra. The VOA is strongly generated by J, so this is the desired counterexample to an implication from finite strong generation.

Exercise 3.5. Subtract the charge-one Cartan boson:

$$\frac{3k}{k+2} - 1 = \frac{2(k-1)}{k+2}.$$

At $k = 3$ and 13 this is $4/5$ and $8/5$. Identification with the parafermion factors additionally needs their generating fields, OPEs, null relations, and the proof that the relevant simple quotient is realized. Central charge is only one invariant of a VOA.

Exercise 3.6. The transported basis squared norms are $4, 40, 40$, since g^{-1} rescales each vector by the inverse of the stated factor. Each of the last two coefficients has squared modulus $(13 + 27)/400 = 1/10$. Therefore

$$\|v\|_g^2 = 4 + 40/10 + 40/10 = 12.$$

Adding coefficient magnitudes alone would give $6/5$, which uses the old orthonormal form on the new coordinates.

B.4 Solutions for Chapter 4

Exercise 4.1. The invertible factors give $A_g A_h = \alpha^{g+h} = A_{g+h}$ and $A_g R_h = \alpha^{g+h}\rho = R_{g+h}$. The relation $\rho\alpha^h = \alpha^{-h}\rho$ gives $R_g A_h = \alpha^{g-h}\rho$. It also gives

$$R_g R_h = \alpha^{g-h}\rho^2 = A_{g-h} + \sum_k R_{g-h+k} = A_{g-h} + \sum_k R_k.$$

The product of A_g with an invertible object contains the unit exactly for $A_g A_{-g}$, and $R_g R_h$ contains the unit exactly when $g = h$, proving the duality rule. Finally, $\alpha\rho = R_1$ and $\rho\alpha = R_2$ are distinct simples. A braiding would make them isomorphic, so no braiding exists on this six-object category.

Exercise 4.2. Put $r = \sqrt{13}$. From the radicals,

$$y_- + y_+ = \frac{5-r}{6}, \qquad y_-y_+ = \frac{(5-r)^2 - 6(1+r)}{144} = \frac{2-r}{9} = \frac{x}{3}.$$

Consequently

$$xy_- + y_-y_+ + y_+x = \frac{(2-r)(7-r)}{18} = \frac{3-r}{2} = -d^{-1}.$$

Together with $x + y_- + y_+ = -d^{-1}$, this gives $x^2 + y_-^2 + y_+^2 = d^{-2} + 2d^{-1} = 1 - d^{-1}$. The first row of the full matrix has norm squared $d^{-2} + 3d^{-1} = 1$. Its inner product with any lower row is $d^{-1/2}(d^{-1} + x + y_- + y_+) = 0$. Each lower row has norm squared $d^{-1} + x^2 + y_-^2 + y_+^2 = 1$, and two distinct lower rows have inner product $d^{-1} + xy_- + y_-y_+ + y_+x = 0$. Thus all rows are orthonormal. This checks the inverse of one recoupling map. The pentagon also compares compositions of different blocks, for every admissible choice of external labels; those equations have not been checked by this calculation.

Exercise 4.3. Let $S_0 = \sum_g Y_{A_g}$ and $S_1 = \sum_g Y_{R_g}$. Summing the fusion rules gives

$$S_0^2 = 3S_0, \quad S_0S_1 = S_1S_0 = 3S_1, \quad S_1^2 = 3S_0 + 9S_1.$$

For $N = S_0 + dS_1$, it follows that

$$N^2 = 3(1+d^2)S_0 + (6d + 9d^2)S_1 = 3sN,$$

where $6d + 9d^2 = 3sd$ follows from $d^2 = 1 + 3d$. Therefore $P_+ = N/(3s)$ is idempotent. Also

$$Y_\rho S_0 = S_1, \qquad Y_\rho S_1 = S_0 + 3S_1,$$

so $Y_\rho N = dS_0 + (1+3d)S_1 = dN$. No eigenvectors, associator entries, or numerical approximations are needed for this ring identity.

Exercise 4.4. The block-constant subspace gives characteristic polynomial $x^2 - 3x - 1$. On the two copies of the sum-zero subspace, the matrix exchanges the blocks, giving $+1$ and -1 each twice. Thus

$$\det(xI - N_\rho) = (x^2 - 3x - 1)(x^2 - 1)^2.$$

Taking traces of powers gives (4.22). The recursion yields $C_2 = 11, C_3 = 36, C_4 = 119, C_5 = 393, C_6 = 1298$; adding the even-length contribution four gives 1302 paths at $L = 6$.

Exercise 4.5. On $\mathbb{C}^3 \oplus \mathbb{C}^2 \oplus (\mathbb{C}^2)^{\oplus 4}$ with its standard positive form, take

$$Y_\alpha = I_3 \oplus I_2 \oplus \begin{pmatrix} \omega & 0 \\ 0 & \omega^2 \end{pmatrix}^{\oplus 4},$$

$$Y_\rho = dI_3 \oplus (-d^{-1}I_2) \oplus \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^{\oplus 4}.$$

Each block obeys the fusion rules, and $Y_\alpha^\dagger = Y_{\alpha^2}$, $Y_\rho^\dagger = Y_\rho$. This is a unitary $*$-representation of dimension $3 + 2 + 4 \cdot 2 = 13$; it does not require the noninvertible operator Y_ρ itself to be a unitary matrix. The neutral projector E has rank five, since it is the identity on both kinds of scalar representation and zero on every charged doublet. The ranks of P_+, P_-, and P_{ch} are $3, 2, 8$, respectively. There are five α-neutral states but only three fully Haagerup-invariant states. Neutrality under the invertible subgroup is therefore a strictly weaker condition.

B.5 Solutions for Chapter 5

Exercise 5.1. The scalar σ_1 must obey $\sigma_1^3 = 1$, and $\sigma_h = \sigma_1^h$. Its three choices are ω^k, giving the labels (g, k) for all nine pairs. The twist is ω^{gk}. Five pairs have $gk = 0$: the three with $g = 0$ and the two further pairs with $k = 0$. Two pairs have product one, and two have product two. Hence the twist multiplicities are $5, 2, 2$ for $1, \omega, \omega^2$.

Exercise 5.2. If $g \neq g'$, multiplication vanishes. Otherwise set $t = h + h'$ in the double sum:

$$\begin{aligned} p_{g,k}p_{g,l} &= \frac{1}{9}\sum_{t,h'} \omega^{-k(t-h')-lh'} e_{g,t} \\ &= \frac{1}{9}\sum_{t} \omega^{-kt} e_{g,t} \sum_{h'} \omega^{(k-l)h'} = \delta_{kl} p_{g,k}. \end{aligned}$$

Summing over k kills every $h \neq 0$, so $\sum_k p_{g,k} = e_{g,0}$. Summing over g gives the algebra unit. Only $p_{0,k}$ occur in the untwisted sector.

Exercise 5.3. For a fixed invertible c, tensoring on either side permutes the six simples. For each a there is exactly one b giving a one-dimensional morphism space, so these choices contribute $3 \cdot 6 = 18$ in total.

Fix now $c = R_l$. If $a = A_i, b = A_j$, then $ca = R_{l-i}$ and $bc = R_{j+l}$, so a morphism exists exactly when $j = -i$. There are three such one-dimensional spaces. If exactly one of a, b is noninvertible, each of the nine pairs contributes one, giving nine in each orientation. If both are noninvertible, the two products share all three noninvertible summands, contributing $9 \cdot 3 = 27$. For $a = R_i, b = R_j$, their invertible summands are A_{l-i} and A_{j-l}; these agree exactly when $j = 2l - i$. There are three such pairs, adding three. Thus each noninvertible c contributes $3 + 9 + 9 + 27 + 3 = 51$, and

$$\dim \mathrm{Tube}(\mathcal{H}_3) = 18 + 3 \cdot 51 = 171.$$

This independently checks the matrix-block sum in (5.16).

Exercise 5.4. The forgetful functor preserves the positive categorical dimension. The images have dimensions 1, $1 + 3d = d^2$, $2 + 3d = s$, and $3d$, respectively. For the center object a, its ordinary tube module has dimension $2 + 1 + 1 + 1 = 5$, while its categorical dimension is $2 + 3d$. The former counts summand multiplicities; the latter weights each noninvertible summand by its categorical dimension d. They are different trace constructions and need not agree.

Exercise 5.5. For a right action $r_M : M \otimes A \to M$, locality is the extra equation

$$r_M \circ c_{A,M} \circ c_{M,A} = r_M.$$

An ordinary module satisfies the associative action equations; it need not satisfy this double-braiding equation. If A is Lagrangian in a positive modular category $\mathcal{C}$, as here, then

$$\mathrm{Dim}(\mathcal{C}_A^0) = \frac{\mathrm{Dim}(\mathcal{C})}{\mathrm{qdim}(A)^2} = 1, \qquad \mathrm{Dim}(\mathcal{C}_A) = \frac{\mathrm{Dim}(\mathcal{C})}{\mathrm{qdim}(A)} = \mathrm{qdim}(A).$$

Thus the local category is Vec, while the all-module category can be nontrivial. For the canonical algebra in $Z(\mathcal{H}_3)$ these categories are respectively Vec and $\mathcal{H}_3$. Holomorphicity concerns the first category. The decomposition $A = \bigoplus_i m_i X_i$ gives an object and its multiplicities. Multiplication still requires actual maps $\mu : A \otimes A \to A$ and a unit, satisfying associativity and commutativity with the specified associator and braiding. Those maps are not contained in the list of summands.

B.6 Solutions for Chapter 6

Exercise 6.1. Summing over the intermediate label (a, b) gives

$$\begin{aligned}(S^2)_{(g,k),(h,l)} &= \frac{1}{9}\sum_{a,b}\omega^{-ka-bg-bh-la} \\ &= \left(\frac{1}{3}\sum_{a}\omega^{-(k+l)a}\right)\left(\frac{1}{3}\sum_{b}\omega^{-(g+h)b}\right) \\ &= \delta_{l,-k}\delta_{h,-g}.\end{aligned}$$

For example $(1, 0)$ is conjugate to $(2, 0)$, rather than to itself. Therefore charge conjugation is not the identity matrix.

Exercise 6.2. Let $\mathbf{1}_m$ be the length-m all-ones column and $J_{m,n}$ the all-ones rectangular matrix. Direct arithmetic gives

$$AB = \frac{1}{36} - \frac{1}{52} = \frac{1}{117}, \qquad A^2 + B^2 = \frac{1}{18} + \frac{1}{26} = \frac{11}{117}.$$

Using $u^2 = 1/9$ and $v^2 = 1/13$, the upper-left diagonal entries of S^2 are $11/117 + 4/9 + 6/13 = 1$, and its off-diagonal entries are $2/117 + 4/9 - 6/13 = 0$. The row sums $C_4\mathbf{1}_4 = -u\mathbf{1}_4$ and $D_6\mathbf{1}_6 = v\mathbf{1}_6$, together with $A + B = u$ and $A - B = -v$, give the first two rows' mixed blocks:

$$u(A + B - u)\mathbf{1}_4^{\mathsf{t}} = 0, \quad v(A - B + v)\mathbf{1}_6^{\mathsf{t}} = 0, \quad v(B - A - v)\mathbf{1}_6^{\mathsf{t}} = 0.$$

The 4×6 block is $uvJ_{4,6} - uvJ_{4,6} = 0$. Finally,

$$C_4^2 + 2u^2 J_4 = I_4, \qquad D_6^2 + 2v^2 J_6 = I_6$$

by the Fourier sums in the chapter. Symmetry supplies the transposed blocks, completing the entire identity $S^2 = I$.

Exercise 6.3. The squares are $\{1, 3, 4, 9, 10, 12\}$ and the nonsquares are $\{2, 5, 6, 7, 8, 11\}$. The list $6j^2$ for $j = 1, \ldots, 6$ is $6, 11, 2, 5, 7, 8$ modulo thirteen. For the Gauss–Milgram calculation, $1 + \omega + \omega^2 = 0$ combines the four dimension-s terms to s^2. Since $s = 1 + d^2$ and $d^4 = 11d^2 - 1$,

$$1 + d^4 + s^2 = 2 + 2d^2 + 2d^4 = 24d^2.$$

The six residual twists contribute $9d^2(-1 - \sqrt{13})/2$, so

$$p^+ = \frac{d^2}{2}(39 - 9\sqrt{13}).$$

Using $d^2 = (11 + 3\sqrt{13})/2$ and multiplying out gives $p^+ = (39 + 9\sqrt{13})/2 = 3s$.

Exercise 6.4. For $i = 1$, $[2i] = 2$; for $i = 4$, $[8] = 5$ because $8 \equiv -5$ modulo thirteen. Hence

$$\begin{aligned}d_1^2 &= \mathbf{1} + a + c_0 + c_1 + c_2 + d_1 + d_3 + d_4 + d_5 + d_6, \\ d_4^2 &= \mathbf{1} + a + c_0 + c_1 + c_2 + d_1 + d_2 + d_3 + d_4 + d_6.\end{aligned}$$

The absent residual objects are d_2 and d_5, respectively. Both dimension sums equal $1 + 4s + 15d = 9 + 27d = 9d^2$.

Exercise 6.5. The first two components of Sm are each $A + B + 2u = 1$. The a component is $2u + 4/3 = 2$; each of the three c_j components is $2u - 2/3 = 0$. Every final component is $v - v = 0$. Thus $Sm = m$. Each nonzero entry of m lies at twist one, so $\Theta m = m$. Symmetry of S gives $Smm^{\mathsf{t}} = mm^{\mathsf{t}} = mm^{\mathsf{t}}S$, and the analogous statement for Θ is immediate. To build a local CFT one still needs, among other things, actual chiral modules and vertex operators, a compatible algebra multiplication on the extension, and the appropriate positivity and sewing results. A modular-invariant integer matrix contains none of those multiplication maps by itself.

Exercise 6.6. In the first statement, the *braided chiral representation category* $\mathrm{Rep}(U)$ is a Drinfeld center. Its Gauss–Milgram phase is one, so the chiral central charge obeys $e^{2\pi i c_U/8} = 1$. The value $c_U = 8$ satisfies this constraint. In the second statement, $\mathcal{H}_3$ is a fusion category of full-CFT symmetry lines. That does not identify the theory's entire chiral representation category with $Z(\mathcal{H}_3)$. Its center organizes symmetry sectors, which is a different assertion. Hence a proposed nonchiral theory with $c_L = c_R = 2$ has no contradiction with the first construction: its left-right gravitational anomaly vanishes, and the symmetry claim alone imposes no condition $c_L = 0 \bmod 8$. This explains compatibility; it does not establish the proposed $c = 2$ theory or an anyon-chain scaling limit.

B.7 Solutions for Chapter 7

Exercise 7.1: cocycle signs. Let $B(a, b)$ be the exponent in the cocycle, considered modulo two. The diagonal terms cancel in $B(a, b) - B(b, a)$, while the remaining terms give the off-diagonal part of (a, b). The omitted diagonal part is $2\sum_i a_i b_i$, which is zero modulo two. Hence the cocycle commutator is $(-1)^{(a,b)}$. For the second identity,

$$B(a, -a) \equiv \sum_i a_i^2 + \sum_{i>j} a_i a_j G_{ij} = \frac{(a, a)}{2} \pmod 2.$$

Evenness makes the last expression integral. In this root basis the diagonal entries are two; an analogous formula in a general even basis uses $G_{ii}/2$ in its diagonal exponent.

Exercise 7.2: a weight-two exponential. The opposite-momentum cocycle is $+1$. Expanding the exponential gives the terms through degree three in $\exp(za(-1)+z^2a(-2)/2+z^3a(-3)/3+\cdots)$:

$$\begin{aligned} Y(e^a, z)e^{-a} = {} & z^{-4}\mathbf{1} + z^{-3}a(-1)\mathbf{1} \\ & + \frac{z^{-2}}{2}(a(-2) + a(-1)^2)\mathbf{1} \\ & + z^{-1}\left(\frac{a(-3)}{3} + \frac{a(-1)a(-2)}{2} + \frac{a(-1)^3}{6}\right)\mathbf{1} + O(1). \end{aligned}$$

The coefficient of z^{-n-1} has weight $3 - n$, since both input states have weight two. Thus the four displayed coefficient weights are $0, 1, 2, 3$, exactly as required. There is no need to enumerate any other lattice momentum because the output momentum is fixed at zero.

Exercise 7.3: the affine share of u. The two highest weights are 1 and 13, so

$$h_K = \frac{1 \cdot 3}{4(1+2)} + \frac{13 \cdot 15}{4(39+2)} = \frac{1}{4} + \frac{195}{164} = \frac{59}{41}.$$

The complementary weight is $2 - 59/41 = 23/41$. It is positive, as it must be in the unitary complementary-stress decomposition. This also checks the recorded pair $180/1271 + 13/31 =$

$(180 + 533)/1271 = 713/1271 = 23/41$ for the two neutral stresses. The affine formula applies without an additional grade because u is an affine highest vector; an arbitrary descendant with the same horizontal charges need not have this energy.

Exercise 7.4: transport preserves unitarity. Applying θ_g twice cancels the two oscillator signs, complex conjugates the coefficients twice, and multiplies opposite momentum factors $\prod_i \widetilde{c}_i^{\beta_i}$ and $\prod_i \widetilde{c}_i^{-\beta_i}$. The result is the identity. For $v \neq 0$, invertibility of g gives $\langle v, v\rangle_g = \|g^{-1}v\|^2 > 0$. More explicitly, the torus factors are positive real numbers, so antilinearity leaves them unchanged on the second application: $\theta_g^2(e^\beta) = (\prod_i \widetilde{c}_i^{\beta_i})(\prod_i \widetilde{c}_i^{-\beta_i})e^\beta = e^\beta$. For the simple roots in E_R, the transported operation gives

$$\theta_g(E_R) = \sum_i c_i e^{-\alpha_i} = -F_R.$$

For weight-one primary fields the mode-adjoint formula has a further minus sign, so $(E_R)^\dagger_{(n)} = (F_R)_{(-n)}$. Ordinary coefficient conjugation would omit the unequal c_i and would therefore fail to give this compact adjoint relation.

Exercise 7.5: determinant and index. If $L \subset \widetilde{L}$ has index r, a fundamental region of $\widetilde{L}$ has volume $1/r$ times that of L. The Gram determinant is the square of that volume, giving $\det \widetilde{L} = \det L/r^2$. Here $r = 3$ and $\det L = 9$, so the extended determinant is one. Isotropy is the condition that the adjoined discriminant classes have integral conformal weights, and hence that the overlattice remains even. It also ensures integral pairings: for classes x, y in the isotropic subgroup, $q(x + y) - q(x) - q(y) = (x, y)$ vanishes modulo $\mathbb{Z}$. For $E_6 \oplus A_2$, the glue generator has quadratic value $2/3 + 1/3 = 1$ and order three. Its multiples therefore form the required isotropic subgroup of size three. The determinant formula uses this subgroup's size, not the size nine of the whole discriminant group.

B.8 Solutions for Chapter 8

Exercise 8.1: all negative modes. Translation covariance gives $(Da)_{(n)} = -na_{(n-1)}$. For example $(D^2a)_{(-1)} = (Da)_{(-2)} = 2a_{(-3)}$. More generally each differentiation supplies the next factor $1, 2, \ldots, r - 1$, giving $(D^{r-1}a)_{(-1)} = (r - 1)!a_{(-r)}$. Thus negative modes are normal products with translated left inputs. Once translation, normal products, and nonnegative modes preserve a space, every mode preserves it, which is exactly vertex-subalgebra closure.

Exercise 8.2: the missing lowest line. Choose weight bases $v_0, \ldots, v_{m-1}$ and $w_0, \ldots, w_{n-1}$ ordered from lowest to highest. The raising image in the first factor contains every v_i with $i > 0$, and the second contains every w_j with $j > 0$. Their sum therefore contains all tensor basis vectors except $v_0 \otimes w_0$. That vector cannot be in the image because neither raising operator can reach a joint-lowest weight. The quotient is one dimensional. Equivalently, normalize the basis so that $E_L v_i = v_{i+1}$ below the top and $E_R w_j = w_{j+1}$. Every basis tensor except $v_0 \otimes w_0$ then has an explicit preimage under one of the two operators. This is a quotient of vector spaces; it need not be a representation of the lowering operators.

Exercise 8.3: counting the raising quotient. The summands whose left representation has dimension one contribute

$$3 + 1 + 5 + 2 + 3 + 1 + 2 + 1 = 18$$

irreducible constituents and

$$9 + 5 + 35 + 18 + 33 + 13 + 30 + 19 = 162$$

dimensions. Those whose left representation has dimension two contribute $1+1+2+2+2+1+1 = 10$ constituents and $4+8+24+32+40+24+28 = 160$ dimensions. Thus P has dimension 322 and 28 constituents, with multiplicity. Each constituent contributes precisely one joint-lowest vector to the quotient by $E_L P + E_R P$, by Exercise 8.2. The image therefore has dimension $322 - 28 = 294$. Multiplicity copies require separate seeds even when their representation types agree.

Exercise 8.4: specialization can lose rank. The matrix $\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)$ has rank two over $\mathbb{Q}$ and rank one over $\mathbb{F}_p$. A nonzero minor modulo p was nonzero over $\mathbb{Q}$, but a vanishing minor modulo p may merely have been divisible by p. The scalar residual p is the simplest example of a nonzero integer residual that reduces to zero. Thus lower rank certificates and exact zero-residual certificates require different evidence. For rational matrices, first clear denominators and choose a prime which does not divide them. Without that condition, the proposed specialization might not even be defined.

Exercise 8.5: the positive-mode bootstrap. For a weight-two state, the Virasoro commutator formula gives

$$[L_1, Y(a,z)] = z^2 \partial_z Y(a,z) + 4zY(a,z) + Y(L_1 a, z).$$

Comparing the coefficient of z^{-n-1} yields

$$[L_1, a_{(n)}] = (2-n)a_{(n+1)} + (L_1 a)_{(n)}.$$

Quasiprimary means $L_1 a = 0$, so the last term vanishes. Apply the $n = 0$ identity to p and rearrange:

$$2a_{(1)}p = L_1(a_{(0)}p) - a_{(0)}L_1 p.$$

Thus control of the zero product, together with the already proved L_1 stability of its output space, controls the next positive mode. Quasiprimary does not mean that $L_2 a$ vanishes; this is why the subsequent L_2 argument separately keeps track of scalar terms.

Exercise 8.6: why one induction index fails. In the term $b_{(n)}(a_{(0)}t)$, a normal expression for $a_{(0)}t$ may contain a single generator of weight $\mathrm{wt}(t)+1$. The left input b has weight $\mathrm{wt}(s)-1$, so the total input weight is unchanged. Induction on total weight alone has made no progress. The left weight has decreased, however, so lexicographic induction applies. Composite and derivative terms are already dealt with by filtered Wick and do not need this second index.

B.9 Solutions for Chapter 9

Exercise 9.1: parafermion charges. Subtracting the charge-one Cartan stress gives $c_{\mathrm{PF}_k} = 3k/(k+2) - 1 = 2(k-1)/(k+2)$. Thus the three substitutions are $2/4 = 1/2$, $4/5$, and $24/15 = 8/5$, respectively. The first is the Ising parafermion algebra. Notice that the parafermion central charges approach two from below as k tends to infinity; proximity to two by itself therefore selects no particular level or full theory.

Exercise 9.2: three opposite momentum pairs. Using $r^2 = 13$, $t^2 = -3$, and $\bar{t} = -t$, the three products are

$$p_{\beta_1}p_{-\beta_1} = p_{\beta_3}p_{-\beta_3} = \frac{r^2 - 9t^2}{1000} = \frac{40}{1000} = \frac{1}{25},$$

and

$$p_{\beta_2}p_{-\beta_2} = \frac{49 - 9r^2t^2}{10000} = \frac{49 + 351}{10000} = \frac{1}{25}.$$

The transported PCT multiplies an opposite exponential by $10, 1, 10$ for the three positive momenta. Directly,

$$\begin{aligned} 10\overline{p_{\beta_1}} &= (-r + 3t)/10 = p_{-\beta_1}, \\ \overline{p_{\beta_2}} &= (7 - 3rt)/100 = p_{-\beta_2}, \\ 10\overline{p_{\beta_3}} &= -(r + 3t)/10 = p_{-\beta_3}. \end{aligned}$$

The opposite directions use the reciprocal factors, so each two-term exponential pair is fixed. Each Heisenberg oscillator changes sign under PCT; its quadratic products are fixed as well. Consequently the entire Potts stress is PCT real in the transported positive form.

Exercise 9.3: the spin-three simple pole. Skew symmetry gives $a_{(0)}a = -a_{(0)}a + D(a_{(1)}a) - \frac{1}{2}D^2(a_{(2)}a) + \frac{1}{6}D^3(a_{(3)}a) - \cdots$ for an even field a. Here "even" means bosonic parity, despite the odd conformal weight three; no fermionic minus sign is present. Use $a_{(1)}a = b_4$, $a_{(2)}a = DL$, $a_{(3)}a = 2L$. Derivatives of the scalar higher pole vanish. Thus

$$2a_{(0)}a = Db_4 - \frac{1}{2}D^3L + \frac{1}{3}D^3L = Db_4 - \frac{1}{6}D^3L.$$

Dividing by two gives the displayed self-row.

Exercise 9.4: the two isotropic lines. If $j = 0$, isotropy forces $k = 0$. Otherwise divide by j^2 to obtain $(k/j)^2 = -1$ in $\mathbb{F}_{13}$. The two roots are 5 and -5. Adjoining both lines would adjoin their span, which is the entire two-dimensional discriminant group. It contains $(1, 0)$, whose quadratic value is $-1/13$, so such an extension would not have integral vacuum grading. Explicitly $5^2 = 25 \equiv -1 \pmod{13}$, and $(1, 5), (1, -5)$ are independent because their determinant -10 is nonzero modulo thirteen. A quadratic polynomial over a field has at most two roots, so there are no further isotropic lines.

Exercise 9.5: coefficients of the stress identity. In the basis $(\omega_L, \omega_R, e, f)$, the four coefficients on the proposed right side are

$$\begin{aligned} \omega_L &: 1, \\ \omega_R &: \frac{3403}{4563} + \frac{5}{27} + \frac{35}{507} = \frac{3403 + 845 + 315}{4563} = 1, \\ e &: \frac{3}{8} + \frac{5}{8} = 1, \\ f &: \frac{13}{14} + \frac{1}{14} = 1. \end{aligned}$$

They therefore reproduce the sum of the four orthogonal stresses. The norm of a vector gives only one quadratic constraint; it does not determine these four projection coefficients. More fundamentally, a correct central term is only one pole in a self-OPE. Establishing the Potts or parafermion subalgebra also requires the remaining singular products, the requisite higher fields, and the stated translation and PCT properties.

Exercise 9.6: positivity without commutativity. Take

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Their spectra are $\{1,2\}$ and $\{1,3\}$, respectively, so both are positive definite and self-adjoint. But $[A,B] = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. For $\alpha, \beta > 0$ and every vector v,

$$(\alpha+\beta)\|v\|^2 \le \langle(\alpha A + \beta B)v, v\rangle \le (2\alpha+3\beta)\|v\|^2.$$

This follows by bounding each expectation separately and adding. No common eigenbasis was used. The C_2 proof does the same thing with the partial energies on a fixed ambient grade: it bounds each expectation on the same vector in $C_2(V)^\perp$, then uses the positive stress identity to bound the total energy.

B.10 Solutions for Chapter 10

Exercise 10.1: the graded quotient. For homogeneous a, b, the vector $a_{(-2)}b$ has weight $\mathrm{wt}(a) + \mathrm{wt}(b) + 1$. Hence $C_2(V)$ is a direct sum of homogeneous subspaces. In finite-dimensional V_n every subspace has an orthogonal complement, and $V_n/(C_2(V) \cap V_n)$ is represented by that complement. This is finite-dimensional linear algebra grade by grade; no assertion about Hilbert-space closure of the infinite direct sum is needed.

Exercise 10.2: the trace step. A diagonalizable operator with nonnegative eigenvalues has trace the sum of those eigenvalues. Zero trace forces each to vanish, so the operator vanishes. Without diagonalizability, $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ has zero spectrum and trace but is nonzero. This is precisely why the regular component top decomposition is used on a generalized weight-zero space: nilpotence of the total energy alone is insufficient. The argument does not require a positive form on the module carrying these operators. It uses diagonalizability and the signs of their eigenvalues, both supplied separately by the component theory.

Exercise 10.3: downward induction to the vacuum condition. For each fixed a, lower truncation supplies an integer $N \ge 3$ with $a_{(n)}v = 0$ for all $n \ge N$. Since $L_{-1}v = 0$, translation covariance implies $(n+1)a_{(n)}v = -L_{-1}a_{(n+1)}v$. Descending from N first reaches $a_{(3)}v = 0$, and then gives

$$3a_{(2)}v = -L_{-1}a_{(3)}v = 0, \qquad 2a_{(1)}v = -L_{-1}a_{(2)}v = 0, \qquad a_{(0)}v = -L_{-1}a_{(1)}v = 0.$$

The coefficient is nonzero for every $n \ge 0$. At $n = -1$ it would vanish, exactly where a creation mode may act nontrivially. In particular the argument imposes no erroneous vanishing of $a_{(-1)}v$, which is the desired copy of the state a.

Exercise 10.4: a nonsplit extension. Let $A = \mathbb{C}[\varepsilon]/(\varepsilon^2)$. Its regular module fits into $0 \to \mathbb{C}\varepsilon \to A \to A/(\varepsilon) \to 0$, with both endpoints the simple module on which ε acts as zero. A section would send 1 to $1 + b\varepsilon$, but $\varepsilon(1+b\varepsilon) = \varepsilon \ne 0$, so no section is A-linear. The simple module is therefore not projective. In the VOA proof, the positive stress identity supplies the missing vacuum-like lift and thereby rules out the analogous nonsplit extension onto the unit.

Exercise 10.5: factorial versus exponential. For n even, the final $n/2$ factors of $n!$ are at least $n/2$, so $n! \ge (n/2)^{n/2}$. Once $n/2 > A^2$, this is greater than A^n. A fixed prefactor can be absorbed by choosing n larger. To see this uniformly for even and odd n, put $r_n = n!/A^n$. For $n+1 \ge 2A$ one has $r_{n+1}/r_n = (n+1)/A \ge 2$; hence r_n eventually exceeds every fixed constant. In the moderate-growth argument take $A = C_X^2$ and the fixed constant $d_0(\mathbf{1})^2$. Then, for all sufficiently large n,

$$\dim \mathrm{End}(X^{\otimes n}) \le d_0(\mathbf{1})^2 C_X^{2n} < n!,$$

which is exactly the hypothesis needed there.

Exercise 10.6: the logical uses of the hypotheses. Positive coefficients let separate upper bounds on partial-energy expectations add to an upper bound on ambient energy. They also make a zero sum of nonnegative partial traces force each trace to vanish. Component regularity supplies ordinary semisimple restrictions, finitely many simple types, and, together with its C_2 consequences, the finite component quotient bounds. Grading compatibility makes component modes have their expected ambient degrees. It places a lowest ambient space in complete component tops and makes each partial energy preserve finite ambient grades.

The unique simple component module at weight zero identifies those zero-energy tops with vacuum tops. Their translation operators therefore annihilate the selected vectors, giving the vacuum-like lift needed for projectivity. If another simple had weight zero, the trace argument would still force its energy to vanish, but would not force its translation mode to vanish. Both uniqueness of the V module of weight zero and vacuum projectivity would lose their stated proof. The C_2 bound itself would remain valid if the other component hypotheses remained true.

Braiding is a premise of the categorical rigidity theorem used after projectivity. Before rigidity, right exactness turns a surjection $P^{\oplus d_0(Z)} \twoheadrightarrow Z$ into one onto $Z \otimes X$; this gives the exponential bound on tensor-power endomorphisms. Only after rigidity has been established is tensoring known to be exact, allowing projectivity of the unit to imply projectivity of every object. Interchanging these stages would be circular.

B.11 Solutions for Chapter 11

Exercise 11.1: the x_4 top. At $m = 6$, the lattice–Potts contribution is $36/156 = 3/13$. The paired parafermion charge is $k = -5 \cdot 6 = -4$ modulo thirteen, and its energy is $4 \cdot 5/15 - 16/13 = 4/3 - 16/13 = (52 - 48)/39 = 4/39$. The sum is $9/39 + 4/39 = 1/3$. The opposite charge supplies the conjugate minimizer. Since these are distinct lattice charge sectors and each has one top state here, their contributions add to a two-dimensional top. They are not two copies produced by left/right bulk pairing.

Exercise 11.2: checking the top certificate. The coefficient $4680(17/10)$ equals 7956. For the requested three variables the left-side coefficients are

$$x_0 : 7956, \qquad x_5 : 14(7956) - 148560 = -37176, \qquad y_4 : 14(7956) - 81480 = 29904.$$

The right side gives respectively 7956, $-8460 - 28716 = -37176$, and $28716 + 1188 = 29904$. In particular the negative x_5 coefficient is absorbed by the actual representation constraints $y_2 - x_5 \geq 0$ and $y_4 - x_5 - x_6 \geq 0$.

Ignoring those constraints and taking a weighted average of individual row ratios gives only the bound by their maximum. The x_5 row alone has ratio $619/273 > 2$. Such a bound could not exclude an additional integer-graded module beginning at weight two. The current character inequalities force the high Cartan charges of that row to come with lower-charge companions; their combined contribution yields the stronger bound $17/10$. The individual ratio is bookkeeping, not a separate ambient energy eigenvalue.

Exercise 11.3: multiplicities are squared. Schur's lemma gives $\mathrm{Hom}(X_i, X_j) = 0$ for $i \neq j$ and $\mathrm{End}(X_i) = \mathbb{C}$. Hence the endomorphism algebra is $\bigoplus_i M_{m_i}(\mathbb{C})$ and has dimension $\sum_i m_i^2$. The branching $1 + X + Y + Y^*$ therefore gives endomorphism dimension $1 + 1 + 1 + 1 = 4$ if the four simples differ, and dimension $1 + 1 + 2^2 = 6$ when it becomes $1 + X + 2Y$. The boundary-rank formula established in the chapter equates these dimensions with the two possible boundary ranks. This latter identification is an extension-theory result, not part of Schur's lemma for an arbitrary semisimple object.

Exercise 11.4: the complex fusion algebra. In a one-dimensional representation, write the images as scalars a, r. The quadratic relation rules out $r = 0$, since that would give $0 = 1$. The cross relation now gives $ra = a^2 r$, hence $a = a^2$; together with $a^3 = 1$, this forces $a = 1$. The remaining quadratic is $r^2 - 3mr - 1 = 0$, whose roots are d_m and $-d_m^{-1}$. For the two-dimensional representation take

$$\alpha = \begin{pmatrix} \zeta_3 & 0 \\ 0 & \zeta_3^2 \end{pmatrix}, \qquad \rho = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The cross relation is immediate, and $(1+\alpha+\alpha^2)\rho = 0$, so the quadratic relation reduces to $\rho^2 = 1$. Also $\alpha^3 = 1$ entry by entry. The two distinct eigenlines of α are exchanged by ρ, so neither is a common invariant line; the representation is irreducible. These inequivalent irreducibles have squared dimensions $1 + 1 + 4 = 6$, exhausting the six-dimensional semisimple complex fusion algebra.

Exercise 11.5: the arithmetic exclusion. Expanding the difference gives

$$\begin{aligned} 36m^2(9m^2 + 4) - 13(4m^2 + 2)^2 &= 116m^4 - 64m^2 - 52 \\ &= 4(m^2 - 1)(29m^2 + 13). \end{aligned}$$

For $m > 1$ it is strictly positive. Both sides of the preceding root-sum inequality are positive, so squaring would give the opposite weak inequality $13(4m^2 + 2)^2 \geq 36m^2(9m^2 + 4)$; this is the contradiction outside the exceptional squareclass.

For $m = 2$, $\sqrt{9m^2 + 4} = 2\sqrt{10}$, and the second moment is

$$\frac{-4 + (2\varepsilon - 2)2\sqrt{10}}{2} = \begin{cases} -2, & \varepsilon = 1, \\ -2 - 4\sqrt{10}, & \varepsilon = -1. \end{cases}$$

The first sum is $-2 - 2\sqrt{10}$, expressed with $4m^2 + 2 = 18$ roots. The conductor-forty bound $a + 8b$ is exactly $2 + 8(2) = 18$, so its equality condition requires the squared-root sum to be $a = 2$. The second-moment formulas do lie in $\mathbb{Q}(\sqrt{10})$, as required for that equality condition. Neither permitted indicator sign gives the required value. Thus the exceptional case does not survive.

Exercise 11.6: center global dimension. The sum is

$$1 + d^4 + 4(1 + d^2)^2 + 6(3d)^2.$$

Squaring $d^2 - 1 = 3d$ gives $d^4 - 2d^2 + 1 = 9d^2$, hence $d^4 - 11d^2 + 1 = 0$. Subtracting the displayed sum from $9(1 + d^2)^2$ gives $4(d^4 - 11d^2 + 1) = 0$. Thus the center global dimension is the square of the boundary global dimension $3(1 + d^2)$. Its total quantum dimension is the positive square root, namely $3(1 + d^2)$. Its rank is twelve; these are three different invariants.

B.12 Solutions for Chapter 12

Exercise 12.1. Set $a = -2 \log r > 0$. The continuous function $f(x) = (x+1)e^{-ax}$ has derivative $e^{-ax}[1 - a(x + 1)]$. Thus

$$\sup_{n \geq 0} (n + 1)r^{2n} \leq \begin{cases} 1, & a \geq 1, \\ e^{a-1}/a, & 0 < a < 1. \end{cases}$$

Choosing an integer nearest $1/a - 1$ also shows that the discrete supremum diverges like $1/(ea)$ as $a \downarrow 0$. The estimate is finite for every proper annulus and loses uniformity as the damping annulus degenerates.

Exercise 12.2. Because P preserves the common core, $1-P$ preserves it too, and $T_0(1-P) = (1-P)T_0$. Apply the same graph-limit argument to both projections. Their ranges then give a direct-sum decomposition of the *closed operator and its domain.* The adjoint of this direct sum is the direct sum of the adjoints, proving reduction of T^*. Only proving invariance of one subspace would not by itself give this direct-sum statement.

Exercise 12.3. Full faithfulness identifies each image endomorphism algebra with $\mathbb{C}$, so the image of a simple is simple in the target semisimple category. It also preserves the vanishing of Hom spaces between distinct simples, making the twelve images pairwise nonisomorphic. They exhaust the twelve target simples, and additivity exhausts finite direct sums. A tensorator would be a coherent natural isomorphism $F(X)\otimes F(Y)\to F(X\otimes Y)$. Counting objects and morphism dimensions neither supplies these maps nor verifies their pentagon and braiding compatibilities.

Exercise 12.4. A full rotation on the $h=1/3$ module is $e^{2\pi i/3}$, while on the integer-graded vacuum VOA it is one. If $v\in U$ is homogeneous, every mode changes energy by the integer $\mathrm{wt}(v)-n-1$. Hence every mode commutes with $Z=e^{2\pi iL_0}$, and $BZ^mY(v,z)Z^{-m}A = BY(v,z)A$ on the test domain. Bounded extension preserves this equality. The paired cancellation does not make the individual charged-sector phase equal to one.

Exercise 12.5. For four mutually inequivalent summands, Schur's lemma gives $\mathbb{C}^4$, of dimension four. If $Y\cong Y^*$, the Y multiplicity space is two-dimensional, giving $\mathbb{C}\oplus\mathbb{C}\oplus M_2(\mathbb{C})$, of dimension six and noncommutative. The canonical extension argument translates these dimensions into the rank-four and rank-six alternatives. It is the repeated simple, rather than four displayed summands, that produces a matrix block.

B.13 Solutions for Chapter 13

Exercise 13.1. The five partitions give, respectively,

$$8,\quad 8^2,\quad \binom{9}{2},\quad 8\binom{9}{2},\quad \binom{11}{4}.$$

Their sum is $8+64+36+288+330=726$. The E_8 theta coefficients through weight four are $1, 240, 2160, 6720, 17520$. Hence the VOA coefficient is

$$17520+6720(8)+2160(44)+240(192)+726=213126.$$

This separates momentum degeneracy from oscillator degeneracy.

Exercise 13.2. Through grade three, multiplication by $P(q)=1-q-q^2+\cdots$ gives

	$v(0)$	$v(1)$	$v(2)$	$v(3)$
b	80	1170	9300	53162
d_4	5	224	2504	16975

For example, $v_b(3)=65042-10630-1250=53162$. Convolution with $1/P(q)=1+q+2q^2+3q^3+\cdots$ returns $a(1)=v(1)+v(0)$, $a(2)=v(2)+v(1)+2v(0)$, and $a(3)=v(3)+v(2)+2v(1)+3v(0)$. These recover all the displayed character entries.

Exercise 13.3. A twist determines h modulo one, so the three stated weights have the same twist. The possible $2/13$ top would produce a pole with coefficient γ. Existence of the specified fundamental modular solutions, together with $\dim U_1 = 6 + 13\gamma$ and the independent six-current theorem, forces $\gamma = 0$. The uniquely selected fundamental solution then has its first nonzero d_3 coefficient at $15/13$. Excluding $2/13$ by itself would not exclude $28/13$; that final selection uses the known fundamental solution.

Exercise 13.4. The Grothendieck group is free on the isomorphism classes of simples. Taking a character is a linear map from that group to a space of functions, and the map need not be injective. Defect insertions, including tube projectors, can distinguish equal unflavoured characters. Flavoured traces or matrix elements of specified current zero modes can also distinguish modules when the associated actions differ.

Exercise 13.5. If v is a modular-invariant constant vector, adding tv does not change any negative-power coefficient. The compact modular-curve argument then proves uniqueness only modulo such invariant constants. One must prescribe the projection of the constant coefficient onto the invariant subspace, or an equivalent value normalization, to remove the ambiguity.

B.14 Solutions for Chapter 14

Exercise 14.1. Squaring the seven top dimensions gives

$$3^2 + 5^2 + 7^2 + 13^2 + 14^2 + 27^2 + 42^2 = 9 + 25 + 49 + 169 + 196 + 729 + 1764 = 2941.$$

These entries follow the increasing dimensions in the chapter's table. A scalar in a diagonal sector has equal grades on the two sides, so any positive grade adds at least two to its dimension. This explains why the relevant scalar list uses only top states.

For the spinning levels, each positive-weight top vector supplies one grade-one descendant. It cannot vanish in a unitary module, because $\|L_{-1}|h\rangle\|^2 = 2h\langle h|h\rangle > 0$. Thus $v_i(1) = a_i(1) - a_i(0)$. In the d_4 sector,

$$(h, \bar{h}) = \left(\frac{5}{13} + 1, \frac{5}{13}\right), \qquad \Delta = \frac{23}{13}, \quad s_{\text{spin}} = 1,$$

and the multiplicity is $5(229 - 5) = 1120$. Interchanging the two grades gives another 1120 states of spin -1. Similarly, the d_1 top has weight $6/13$, giving $\Delta = 2(6/13) + 1 = 25/13$ and $7(292 - 7) = 1995$ primary components for each spin sign.

Exercise 14.2. Integer grading leaves exactly three pairs of weights at total dimension two: $(1,1)$, $(2,0)$, and $(0,2)$. At $(1,1)$ there are $248^2 = 61504$ scalar primaries. The chiral character has 4124 states at weight two. Of these, 248 are the independent derivatives $L_{-1}J^A$ of the currents, and one is $L_{-2}\mathbf{1}$, the stress state. There is no $L_{-1}\mathbf{1}$ descendant. Subtracting leaves $4124 - 248 - 1 = 3875$ chiral primaries. Each of the pairs $(2,0)$ and $(0,2)$ contains one copy of this decomposition. Thus the total is

$$61504 + 2(3875) + 2(248) + 2 = 69752.$$

The last 498 states are not Virasoro primaries.

Exercise 14.3. For each omitted state, factor $E^k e^{-bE} = e^{-b_0 E}[E^k e^{-(b-b_0)E}]$ and bound the bracket by its supremum for $E \geq E_D$. Positivity permits summation. With $a = b - b_0 > 0$, the suprema are

$$\sup_{E \geq E_D} Ee^{-aE} = \begin{cases} E_D e^{-aE_D}, & aE_D \geq 1, \\ (ae)^{-1}, & aE_D < 1, \end{cases}$$

and

$$\sup_{E\ge E_D} E^2 e^{-aE} = \begin{cases} E_D^2 e^{-aE_D}, & aE_D \ge 2, \\ 4/(a^2e^2), & aE_D < 2. \end{cases}$$

Heat capacity uses the variance $\langle E^2\rangle - \langle E\rangle^2$, so one needs control of the zeroth, first, and second weighted sums, including their common denominator. A small unweighted tail can still carry a substantial energy moment.

Exercise 14.4. There are six dimension-zero states, which describes a multiple-vacuum or disconnected situation instead of the specified connected theory with a unique vacuum. Division by six can destroy integral multiplicities and does not construct identity fields, OPEs, or sewing maps. A numerical rescaling of a partition function cannot repair missing local-theory data.

Exercise 14.5. The $36 = 6^2$ count is the diagonal theory's scalar dimension-two subspace invariant under all its Verlinde lines. The $1488 = 6 \cdot 248$ count is invariant under the selected first-factor $\mathcal{H}_3$ action in the extended full theory. The $61504 = 248^2$ count is the entire scalar dimension-two space of that extended theory, with no symmetry restriction. Different ambient Hilbert spaces and line actions are being used.

The invariant spaces themselves can also be checked. In the diagonal theory, modular orthogonality makes the projector in Chapter 16 select precisely $U \otimes \overline{U}$. In the extension, use the line matrices on $H' = U \oplus b \oplus (\mathbb{C}^2 \otimes a)$. The invariance conditions are $\widehat{\alpha}\psi = \psi$ and $\widehat{\rho}\psi = d\psi$. The a multiplicity block has α eigenvalues ω, ω^2, so it contributes no invariant vector. The b block survives α but has ρ eigenvalue $-d^{-1} \neq d$, so it also contributes none. The entire U block survives. Since the chosen first factor does not act on the right movers, the full invariant space is $U \otimes \overline{H'}$. Its scalar weight pair $(1,1)$ has $6 \cdot 248$ components. Imposing the independent right action repeats the same argument and reduces this to 6^2.

Exercise 14.6. The two block products are

$$X^{\mathsf{t}}\eta = \begin{pmatrix} 0 & -\varepsilon \\ \varepsilon^{\mathsf{t}} & 0 \end{pmatrix}, \qquad \eta X = \begin{pmatrix} 0 & \varepsilon \\ -\varepsilon^{\mathsf{t}} & 0 \end{pmatrix}.$$

They sum to zero. Thus e^{tX} preserves the quadratic form $p^{\mathsf{t}}\eta p$, and its infinitesimal action is the stated charge boost: differentiate $p(t) = e^{tX}p(0)$ at $t = 0$ and read its two blocks. In particular,

$$\delta(p_L^2 - p_R^2) = 2p_L^{\mathsf{t}}\varepsilon p_R - 2p_R^{\mathsf{t}}\varepsilon^{\mathsf{t}} p_L = 0, \qquad \delta\Delta = 2p_L^{\mathsf{t}}\varepsilon p_R.$$

For $r = 1$ and $\varepsilon = -1$, $X^2 = I$ and $e^{tX} = I\cosh t + X\sinh t$. Starting at $(p_L, p_R) = (\sqrt{2}, 0)$ yields $(\sqrt{2}\cosh t, -\sqrt{2}\sinh t)$. The spin remains $(p_L^2 - p_R^2)/2 = 1$, although $\bar{h} = \sinh^2 t > 0$ for $t \neq 0$. This is a statement about the charge contribution, with oscillator and neutral labels held fixed. Establishing a local conformal family uses the abelian current-current theorem in the chapter; the matrix identity alone would not construct one or transport the Haagerup defect junctions.

B.15 Solutions for Chapter 15

Exercise 15.1. Write the momenta at $(\infty, 1, x, 0)$ as $(-a_i, a_j, -a_k, a_\ell)$. Neutrality requires the multisets $\{i,k\}$ and $\{j,\ell\}$ to agree. There are three choices with all labels equal. For $i \neq k$, the choices $(j,\ell) = (i,k)$ and (k,i) give six each. The finite-distance factors are $x^{-(a_k,a_\ell)}(1-x)^{-(a_j,a_k)}$. They are, respectively, $x^{-4}(1-x)^{-4}$, $x^{-4}(1-x)^{-2}$, and $x^{-2}(1-x)^{-4}$. Each assignment has coefficient $1/9$, giving the three terms in the chapter.

Exercise 15.2. At order four the blocks of exchanged weights one, two, and three contribute $2/3$, $21/5$, and $29/3$. Subtracting them from the reduced-correlator coefficient gives

$$A_4 = \frac{65}{3} - \frac{2}{3} - \frac{21}{5} - \frac{29}{3} = \frac{107}{15}.$$

In the charged ordering the reduced correlator is $x^4 G_+(x) = (4/3)x^6 + \cdots$, so the first exchanged global primary has weight six and squared coefficient $4/3$. The corresponding unit-normalized three-point coefficient has magnitude $2/\sqrt{3}$.

Exercise 15.3. Only pure-factor contractions survive, and $\langle T_i(\infty) T_i(1) T_i(0) \rangle = c_i$. For $A = c_2 T_1 - c_1 T_2$ the unnormalized three-point coefficient is $c_1 c_2 (c_2^2 - c_1^2)$. Divide its square by $\|A\|^6 = [c_1 c_2 (c_1 + c_2)/2]^3$ to obtain

$$C^2_{\widehat{A}\widehat{A}\widehat{A}} = \frac{8(c_2 - c_1)^2}{c_1 c_2 (c_1 + c_2)}.$$

When $c_1 = c_2$, exchanging the two Virasoro factors within their two-stress vacuum subalgebra sends A to $-A$, so its odd correlator vanishes. Its positive norm does not vanish; it remains a nonzero field.

Exercise 15.4. At the Potts point $5c_0 + 22 = 26$ and $\lambda(5c_0 + 22) = (4/13)26 = 8$, so the weight-four primary norm is zero. For the simple pole in the four-point function, $\frac{1}{2}\partial b$ contributes $\frac{1}{2}(4B) = 2B$ and $-\frac{1}{12}\partial^3 T_0$ contributes $-72N/12 = -6N = -2c_0$. The total is $2B - 2c_0$.

Exercise 15.5. Using $q = (2, 0, -2)$ gives

$$K = \frac{1}{39} \begin{pmatrix} 24 & -13 & -11 \\ -13 & 26 & -13 \\ -11 & -13 & 24 \end{pmatrix}.$$

The vectors $(1, 1, 1)$, $(1, 0, -1)$, and $(1, -2, 1)$ are eigenvectors with eigenvalues 0, $35/39$, and 1. They are mutually orthogonal and span the space, so K is positive semidefinite. Positivity concerns the quadratic form, not the signs of its individual matrix entries.

B.16 Solutions for Chapter 16

Exercise 16.1. On the orthogonal summand $M_i \otimes \overline{M_i}$, the projector candidate has eigenvalue

$$p_i = \frac{1}{\mathcal{D}^2} \sum_a \frac{S_{a0}}{S_{00}} \frac{S_{ai}}{S_{0i}} = \frac{\delta_{0i}}{\mathcal{D}^2 S_{00} S_{0i}} = \delta_{0i}.$$

Here the middle equality is orthogonality of the real S columns, and the last uses $S_{00} = 1/\mathcal{D}$ when $i = 0$. Therefore Π_{inv} squares to itself and is Hermitian on the orthogonal sector decomposition. Its vacuum eigenvalue before normalization is $\sum_a d_a^2 = \mathcal{D}^2$, which explains the denominator. Dividing by $\mathcal{D}$ would leave the vacuum eigenvalue $\mathcal{D}$ and would fail idempotence.

Exercise 16.2. On the scalar blocks, $\alpha = 1$ and the two possible ρ eigenvalues solve $x^2 = 1 + 3x$. On the two-dimensional block, $\alpha^3 = 1$, $\rho^2 = 1$, and conjugation by the swap matrix sends α to α^{-1}. Moreover $1 + \alpha + \alpha^2 = 0$, so $\rho + \alpha\rho + \alpha^2\rho = 0$ there. These checks verify the generating relations and hence all products. To make the noncommutative products explicit, put $r_i = \alpha^i \rho$ with indices modulo three. Then

$$\alpha^j r_i = r_{i+j}, \qquad r_i \alpha^j = r_{i-j}, \qquad r_i r_j = \alpha^{i-j} \rho^2 = \alpha^{i-j} + r_0 + r_1 + r_2.$$

The middle equality in the last formula follows by moving α^j past ρ with $\rho\alpha^j = \alpha^{-j}\rho$. The traces of α on the three multiplicity blocks are $1, 1, -1$; those of ρ are $d, -d^{-1}, 0$. Multiplying by the respective characters and the unchanged antiholomorphic factor gives the two twined traces. For example, on $\mathbb{C}^2 \otimes a$ the first-factor operator is $\widehat{\alpha}|_{\mathbb{C}^2} \otimes I_a$, so its graded trace is $(\omega + \omega^2)\,\mathrm{Tr}_a\, q^{L_0 - c/24} = -\chi_a$. The matrix acts on the two-copy multiplicity space, not on a two-dimensional approximation to the infinite module a.

Exercise 16.3. The forgetful table gives $R\alpha = c_0 \oplus c_1 \oplus c_2$. The three summands have lowest weights 1, 1/3, and 2/3, respectively. Thus the ground comes only from c_1. Its restriction multiplicity is $n_{\alpha,c_1} = 1$, whereas the ordinary dimension of its chiral top is three. In the first-factor twisted Hilbert space the antiholomorphic E_8 vacuum has multiplicity one, so the lowest full level has

$$(h, \bar{h}) = \left(\frac{1}{3}, 0\right), \qquad \Delta = s_{\mathrm{spin}} = \frac{1}{3}, \qquad d_{\mathrm{states}} = 1 \cdot 3 \cdot 1 = 3.$$

There is no second contribution with the chiralities exchanged for this specified first-factor twist. A twisted state sits at a defect endpoint or in a defect boundary condition; the ordinary untwisted locality requirement of integer spin does not apply to it unchanged.

Exercise 16.4. Put $x = \alpha^i\rho$ and $y = \alpha^j\rho$. Duality reverses the order, and the fusion relation then gives $x^* = \rho\alpha^{-i} = \alpha^i\rho = x$. An oriented boundary-changing field from x to y is counted by x^*y, not by deleting the dual in a general formula. For $i \neq j$, $(\alpha^i\rho)^*(\alpha^j\rho) = \alpha^{i-j} + \rho + \alpha\rho + \alpha^2\rho$. Indeed, $x^*y = \alpha^{i-j}\rho^2$, and multiplying the three noninvertible summands of ρ^2 by α^{i-j} only permutes them. The nontrivial invertible term restricts to $c_0 + c_1 + c_2$, contributing three states of weight 1/3. Each noninvertible contributes another three. The total is $3 + 3 \cdot 3 = 12$, compared with nine in the self-boundary case, where the invertible term is the vacuum boundary. Equivalently, the character χ_{c_1} occurs four times between distinct noninvertible boundaries and three times in a self-annulus; each copy has three top states. The distinct-boundary annulus has no identity state: the coefficient of the tensor unit in x^*y is zero when $x \neq y$. These are boundary weights $h = 1/3$ in a single chiral open-channel trace, rather than bulk scalar dimensions $2h = 2/3$.

Exercise 16.5. Six counts the simples of the Haagerup boundary fusion category; twelve counts those of its braided Drinfeld center; thirty-six counts the simples of the relevant bimodule defect category equivalent, up to the stated opposite convention, to a product of the boundary category with its opposite. The center is braided; the boundary and bimodule categories are used as fusion categories. A Lagrangian algebra has only the trivial local module category, although its all-module category has six simples. Replacing all modules by local ones would discard precisely the nonlocal boundary sectors.

B.17 Solutions for Chapter 17

Exercise 17.1. For $0 < z < 1$, put $u = \sqrt{1 + \sqrt{1 - z}}$ and $v = \sqrt{1 - \sqrt{1 - z}}$. Then $u^2 + v^2 = 2$ and $uv = \sqrt{z}$. Consequently $(u \pm v)/\sqrt{2} = \sqrt{1 \pm \sqrt{z}}$. The common factor $[z(1-z)]^{-1/8}/\sqrt{2}$ is unchanged by $z \mapsto 1 - z$, proving the Hadamard fusing relation. It mixes both blocks: the continued identity block is $(\mathcal{F}_1 + \mathcal{F}_\epsilon)/\sqrt{2}$. Its absolute square therefore cannot equal $|\mathcal{F}_1|^2$ for all z.

Exercise 17.2. Write $D = \left(\begin{smallmatrix} a & b \\ \bar{b} & d \end{smallmatrix}\right)$. The relative phase of the two monodromy eigenvalues is -1. With the convention $G = \mathcal{F}^{\mathsf{t}} D \overline{\mathcal{F}}$, the condition $M^{\mathsf{t}} D \overline{M} = D$ forces $b = 0$. For $D = \mathrm{diag}(a, d)$, the condition $F^{\mathsf{t}} D \overline{F} = D$ with the real Hadamard F forces $a = d$. Thus $D = aI$. Positivity gives $a \geq 0$, and a nontrivial theory with unit vacuum coefficient has $a = 1$. Because $\mathcal{F}_\epsilon \sim \frac{1}{2} z^{3/8}$, the energy-channel bulk coefficient is $1/4$, giving $|C_{\sigma\sigma\epsilon}| = 1/2$.

Exercise 17.3. An equivalence of full local theories preserves all correlators and their sewing structures. An algebra-object description specifies the chiral multiplication, unit, and compatibilities that enter the rational full-theory construction. Forgetting multiplication leaves an underlying module, which still remembers how the chiral algebra acts. Taking its character retains only energy multiplicities. Thus OPE products are lost on passing to the module, and most representation matrices are lost on passing to the character. An equality at a weaker level does not reconstruct the stronger one. In the rational construction, Morita equivalence of appropriate algebra objects can describe equivalent full theories even when the algebra objects are not isomorphic.

Exercise 17.4. If infinitely many orthogonal states have energy $E \leq E_*$, each contributes at least $e^{-\beta E_*} > 0$ to the trace. The partial sums are unbounded for every $\beta > 0$. A common vacuum-energy shift only multiplies this divergence by a finite positive constant. Positivity rules out cancellations; an alternating formal series is not a positive thermal trace.

Exercise 17.5. One needs the chosen reduced Hilbert space and its left-right pairing, bounded or otherwise controlled line operators on that space, junction maps with the specified associator, compatibility with the local OPE, and sewing consistency. For a nontrivial branching extension, one also needs coherence with the complementary-sector multiplicities and the algebra object defining the reduction. A torus trace can detect spectra and certain line eigenvalue sums. It does not record the junction associator or all the local and higher-surface sewing identities.

B.18 Solutions for Chapter 18

Exercise 18.1: the perpendicular lattice. The orthogonal summands make the glue norm the sum $2/3 + 2/3 + 1/2 + 78/36 = 4$. Its components have orders $3, 3, 2, 6$ modulo their respective lattices, so the full glue class has order $\mathrm{lcm}(3, 3, 2, 6) = 6$. Its pairings with the original lattice are integral: the A_2 weights are dual-lattice vectors, while $(\alpha/2, \alpha) = 1$ and $(\gamma_{78}/6, \gamma_{78}) = 13$. The even norm therefore gives an even integral index-six extension. Its determinant is

$$\frac{3 \cdot 3 \cdot 2 \cdot 78}{6^2} = \frac{1404}{36} = 39.$$

If the plane lattice L_0 had a larger saturation of index $r > 1$ in E_8, that integral lattice would have determinant $39/r^2$. Integrality of its Gram matrix makes this an integer, so r^2 would divide the squarefree integer 39, impossible. This argument uses the actual integral ambient lattice; an arbitrary real enlargement has no such determinant restriction.

Exercise 18.2: why branching multiplicities square. For each lattice charge n, the three nonzero branching functions are $\chi_n, \chi_n, 2\chi_n$. Pairing each full multiplicity space with its complex conjugate produces multiplicities $1^2, 1^2, 2^2$. Summing gives $6Z_{L_0}$. The lattice vacuum occurs at $n = 0$, so the zero-energy multiplicity is six. These are the multiplicities of the original embedded extension $A = \mathbf{1} \oplus X \oplus 2Y$. The calculation does not choose between its possible $\mathcal{H}_2$ and $\mathcal{H}_3$ boundary categories, nor replace it by a separately selected canonical algebra.

A partition function alone supplies no product on that six-dimensional vacuum space. A full interpretation requires a unital associative commutative vacuum product, its positive pairing and adjoint, and compatible local OPE and sewing data for all other sectors. In a unitary finite vacuum algebra the mutually orthogonal primitive idempotents describe the separate vacuum components; defect junctions must act consistently on this algebra. Six disconnected identical lattice theories give one possible partition function with this multiplicity, but do not automatically supply the intended Haagerup action. A unique-vacuum, clustering theory cannot simply keep the six-dimensional zero-energy space unchanged.

Exercise 18.3: exact near-two charges. The affine charge is $3(40)/(40+2) = 20/7$. Hence the diagonal GKO charge is

$$1 + \frac{117}{41} - \frac{20}{7} = \frac{286}{287} = 1 - \frac{1}{287}.$$

Adding one Cartan stress gives $573/287 = 2 - 1/287$. Its distance below two is $1/287$, approximately .00348432, whereas $c_* - 2$ is approximately .025004294 and is positive. For completeness, the GKO stress by itself is $288/287$ below two, and the diagonal affine stress is $6/7$ above two. These exact rational stresses and the irrational axis are different vectors and different chiral algebras; closeness of their charges does not identify their full theories.

Exercise 18.4: solving for a Virasoro axis. Expand $e * e$ using the table. The coefficient of e_i is $2a_i^2 + t^2\lambda_i/f_i$ and that of x is $2t\lambda\cdot a$. Equating them to $2a_i$ and $2t$ gives $a_i(1-a_i) = \lambda_i t^2/(2f_i)$ and the second equation. The three ratios are

$$\frac{11/41}{286/287} = \frac{7}{26}, \qquad \frac{192/1271}{2400/1271} = \frac{2}{25}, \qquad \frac{49/31}{70/31} = \frac{7}{10},$$

giving the stated q_i. Let $\omega' = e_0 + e_1 + e_2$. Invariance and $\omega' * e = 2e$ give $2\langle e, e\rangle = \langle \omega', e * e\rangle = 2\langle \omega', e\rangle$. Therefore $c(e) = 2\langle e, e\rangle = 2\sum_i f_i a_i$ on the idempotent locus.

Exercise 18.5: the vacuum grade-four Gram matrix. Using $[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}$ and $L_{-1}\mathbf{1} = 0$, the matrix in the stated basis is

$$\begin{pmatrix} 5c & 3c \\ 3c & c(c+8)/2 \end{pmatrix}.$$

For example the mixed entry follows from $L_4 L_{-2}^2\mathbf{1} = 6L_2L_{-2}\mathbf{1} = 3c\mathbf{1}$. The first diagonal entry is $\langle \mathbf{1}, [L_4, L_{-4}]\mathbf{1}\rangle = 5c$. For the other one, commute L_2 past both factors to obtain $L_2L_{-2}^2\mathbf{1} = (8+c)L_{-2}\mathbf{1}$; a second L_2 gives $(8+c)c/2$. These are inner products because the unitary adjoint is $L_n^\dagger = L_{-n}$ and $\|\mathbf{1}\| = 1$. Its determinant is $5c^2(c+8)/2 - 9c^2 = c^2(5c+22)/2$, positive for $c > 0$. The two vacuum descendants are therefore independent at c_*.

Exercise 18.6: the trivial grading. Let g_X be the group element assigned to a simple X, with $g_Z = g_X g_Y$ whenever Z occurs in $X \otimes Y$. The tensor unit has $g_{\mathbf{1}} = 1$, either by the unital assignment or by $g_{\mathbf{1}}^2 = g_{\mathbf{1}}$ in a group. Since both $\mathbf{1}$ and ρ occur in ρ^2, $g_\rho^2 = 1 = g_\rho$. Thus $g_\rho = 1$. The constituent $\alpha\rho$ then implies $g_\alpha g_\rho = g_{\alpha\rho} = g_\rho^2 = 1$, so $g_\alpha = 1$; the same holds for every remaining simple. The group need not be abelian for this argument.

To apply this to currents, each simple line must transmit the chosen current algebra by a definite invertible rotation, and every nonzero fusion junction must identify the rotation of each output constituent with the product of the two input rotations. Those rotations then give the required group-valued assignment. Preserving only the stress, or retaining arbitrary matrix elements of a line after a projection, does not establish this transmission property.

Exercise 18.7: bounded projected energy. Project $(m, n) \in \mathbb{Z}^2$ to $m + \theta n$ with irrational θ. There are infinitely many pairs whose projected value lies in any fixed sufficiently large interval: for each n, choose m nearest to $-\theta n$, giving absolute value at most $1/2$. Their full Euclidean lengths grow, so a finite ambient-energy cutoff would remove all but finitely many. Discarding the perpendicular energy removes that control. To put this in the form of positive commuting energies, use

$$E_{\parallel} = \frac{(m+\theta n)^2}{1+\theta^2}, \qquad E_{\perp} = \frac{(n-\theta m)^2}{1+\theta^2}, \qquad E_{\parallel} + E_{\perp} = m^2 + n^2.$$

The selected states have $E_{\parallel} \le 1/[4(1+\theta^2)]$, so their reduced thermal sum already diverges at any positive temperature. Irrationality makes the linear projection injective: the issue is infinitely many states in a bounded energy interval, not necessarily infinite multiplicity at one exact eigenvalue. This example illustrates the convergence problem; it does not identify the actual irrational commutant branching with a freely projected lattice.

Exercise 18.8: a sum and its diagonal square. The example $a_n = 1/n$ has a divergent sum and a convergent square sum. Therefore the divergence of $R(\beta)$ alone would be insufficient. A character is more special: write it as a sum over an actual basis, $\chi_\lambda(\beta) = \sum_j e^{-\beta E_j}$, including repetitions for multiplicities. Its square is a sum over ordered pairs of basis vectors. Keeping only the pairs (j, j) gives $\chi_\lambda(\beta)^2 \ge \sum_j e^{-2\beta E_j} = \chi_\lambda(2\beta)$. All terms are positive, and integer multiplicities permit this basis interpretation: a multiplicity d contributes d^2 ordered pairs and at least d diagonal pairs. The shifted energies E_j may be negative; positivity of their exponentials is all that is used. Since $R(2\beta)$ diverges for every $\beta > 0$, the diagonal thermal sum diverges too.

Exercise 18.9: independent temperatures. Write $s = \beta + \bar\beta$, $p = \beta\bar\beta$, and $t = 4\pi^2 s/p$. The inverted temperatures sum to t, while eta inversion gives

$$\eta(4\pi^2/\beta)\eta(4\pi^2/\bar\beta) = \frac{\sqrt{p}}{2\pi}\eta(\beta)\eta(\bar\beta).$$

Equating the original and inverted amplitudes therefore gives $G(t) = \sqrt{p}\,G(s)/(2\pi) = \sqrt{s/t}\,G(s)$. For fixed $s > 0$, independently positive temperatures allow all $0 < p \le s^2/4$, equivalently all t with $st \ge 16\pi^2$. Given $s_1, s_2 > 0$, choose $u \ge 16\pi^2 \max(s_1^{-1}, s_2^{-1})$. Both pairs (s_i, u) are allowed, forcing $\sqrt{s_1}G(s_1) = \sqrt{s_2}G(s_2)$.

If only equal temperatures were checked, one would have $p = s^2/4$ and only the involution $t = 16\pi^2/s$. It relates each argument to one partner, with no freedom to connect two arbitrary arguments through a common u. An invariance under this involution alone does not force $\sqrt{s}G(s)$ to be constant. The stronger two-variable modularity hypothesis is essential to the uniqueness proof.

Exercise 18.10: the dimension threshold. For a noninvertible simple X, the product $X \otimes X^*$ contains the unit and at least one further simple. All simple Frobenius–Perron dimensions are at least one, so $\mathrm{FPdim}(X)^2 \ge 2$. If there are at least three simples, two others each contribute at least one to global dimension, proving the bound four.

At rank two, the nonunit simple is self-dual and $X^2 = \mathbf{1} + nX$ for an integer $n \ge 1$. Its positive dimension is $(n + \sqrt{n^2+4})/2$, an increasing function of n. At $n = 1$ the global dimension is $1 + ((1+\sqrt{5})/2)^2 = (5+\sqrt{5})/2$. In the descent argument, a nontrivial central adjoint algebra would have dimension at least d^2, forcing the image category below both nonpointed thresholds:

$$\mathrm{Dim}\,\mathcal{D} \le \frac{3(1+d^2)}{d^2} = 3 + \frac{3}{d^2} < \frac{10}{3} < \frac{5+\sqrt{5}}{2},$$

where $d > 3$ gives the strict bound. Its being pointed would contradict the irrational dimension d of the image of ρ, since any object of a pointed category is a direct sum of dimension-one simples. The adjoint algebra in this argument is $R(\mathbf{1})$ for the hypothesized descent functor; it is not the original E_8/U extension algebra, despite the chapter's reused local letter A.

B.19 Solutions for Chapter 19

Exercise 19.1. The recurrence gives

$$t_2 = 11,\ t_3 = 36,\ t_4 = 119,\ t_5 = 393,\ t_6 = 1298,\ t_7 = 4287,\ t_8 = 14159.$$

For odd length the extra term in (19.1) vanishes, so $\dim \mathcal{H}_{\text{path},5} = 393$. For even length it is four, giving $\dim \mathcal{H}_{\text{path},8} = 14163$. The eigenvalue $d > 1$ strictly dominates the other five eigenvalues in modulus. The alphabet bound 6^L counts all words, most of which violate at least one fusion constraint.

Exercise 19.2. Let v_f be the conjugated fth column in the convention of (19.3). Unitarity says $v_f^\dagger v_g = \delta_{fg}$. For $P_S = \sum_{f\in S} v_f v_f^\dagger$, multiplication gives $P_S^2 = P_S = P_S^\dagger$, and its range is the $|S|$-dimensional span of those columns. With fusion multiplicities, a channel has an additional orthonormal multiplicity label, and one sums over all selected copies. It is the subspace, not a single scalar label, that defines the projection.

Exercise 19.3. For a basis vector $|x\rangle$ with stabilizer $K \subset G$, form

$$v_\chi = \sum_{g\in G} \overline{\chi(g)}\, |gx\rangle.$$

Every repeated orbit entry is multiplied by $\sum_{k\in K} \overline{\chi(k)}$. Character orthogonality makes this zero unless $\chi|_K = 1$. In the surviving case, choose one representative per coset and divide its character sum by $\sqrt{|G|/|K|}$. Distinct cosets give orthogonal basis vectors, so the normalized vector is

$$|\chi; x\rangle = \frac{1}{\sqrt{|G|/|K|}} \sum_{g\in G/K} \overline{\chi(g)}\, |gx\rangle.$$

It is independent of the coset representatives precisely because $\chi|_K = 1$. Equivalently, the original full-group sum has norm $\sqrt{|G||K|}$, not $\sqrt{|G|}$. Replacing g by $h^{-1}g$ in the sum shows that $h|\chi; x\rangle = \chi(h)|\chi; x\rangle$.

Exercise 19.4. Using self-adjointness and $Y_\rho^2 = 1 + 3Y_\rho$,

$$(Y_\rho - 2)^2 = Y_\rho^2 - 4Y_\rho + 4 = 5 - Y_\rho.$$

Its eigenvalues are $d - 2$ and $-d^{-1} - 2$, both nonzero. On a unit vector, $5 - d \le \|(Y_\rho - 2)s\|^2 \le 5 + d^{-1}$. An orthogonal projection has only eigenvalues zero and one; these values plainly do not have that property. The special extraction formula in the rank-four vacuum module uses one particular input vector, not a global projection identity.

Exercise 19.5. For L physical rho objects between fixed external heights a, b, the dimension is $(N_\rho^L)_{ab}$: matrix multiplication sums all intermediate fusion histories. Summing the diagonal entries instead identifies the two ends and permits every compatible starting height. It changes both the state space and the closing Hamiltonian terms. Even when the infinite bulk theory agrees, fixed endpoints can select boundary conformal families or carry edge states absent from a given periodic sector. The equal leading growth d^L does not equate their finite spectra.

B.20 Solutions for Chapter 20

Exercise 20.1. Expand $u = \sum_n c_n |n\rangle$. Then $\|r\|^2 = \sum_n |c_n|^2 (E_n - E)^2$. This is at least the squared distance from E to the spectrum, since the weights sum to one. If Π projects onto the target cluster and all other eigenvalues are at least γ from E, the same sum is at least $\gamma^2 \|(1-\Pi)u\|^2$. Hence the unwanted squared weight is at most $\|r\|^2/\gamma^2$. The value of γ is a separate spectral input.

Exercise 20.2. For the vacuum, $L_n|0\rangle = 0$ for $n \geq -1$ and $L_0|0\rangle = 0$. Therefore

$$\|L_{-n}|0\rangle\|^2 = \langle 0|[L_n, L_{-n}]|0\rangle = \frac{c}{12} n(n^2-1).$$

At $n = 2$ this is $c/2$. The source amplitude squared is $(2\pi v/L)^2 c/2$, whereas the stress gap squared is $(4\pi v/L)^2$. Their ratio is $c/8$, proving the estimator. The calculation assumes the source has the displayed Virasoro normalization; it does not establish the lattice-to-CFT map.

Exercise 20.3. Put $k = 2\pi m/L$ and use periodic indices. In the first sum below, relabel $i = j + 1$:

$$\sum_j e^{-ikj}(B_{j+1} - B_j) = e^{ik} \sum_i e^{-iki} B_i - \sum_j e^{-ikj} B_j.$$

The phase is periodic because $e^{ikL} = 1$, so the difference is $(e^{ik} - 1)B_m$. It vanishes identically for $m = 0$, as required for the unchanged total Hamiltonian.

For fixed nonzero m, matrix elements of a scaling field B_j of dimension Δ_B between normalized low-energy states scale as $L^{-\Delta_B}$. Summing its Fourier component over L sites gives $O(L^{1-\Delta_B})$ when the relevant continuum matrix element is nonzero. Since $e^{ik} - 1 = 2\pi i m/L + O(L^{-2})$, the improvement contributes $O(L^{-\Delta_B})$. A dimension-one current can therefore contribute at the same L^{-1} order as the extensive stress Fourier mode. A field of dimension greater than one is subleading under this scaling assumption. Selection rules may make a particular matrix element vanish; dimension counting alone does not guarantee a nonzero coefficient.

Exercise 20.4. Write $E_0(L)/L = e_\infty - A/L^2 + r(L)$, with $A = \pi v c/6$. Subtracting the two sizes eliminates e_∞ and gives

$$\begin{aligned}(vc)_{\text{two sizes}} &= -\frac{6}{\pi} \frac{E_0(L_1)/L_1 - E_0(L_2)/L_2}{L_1^{-2} - L_2^{-2}}, \\ &= vc - \frac{6}{\pi} \frac{r(L_1) - r(L_2)}{L_1^{-2} - L_2^{-2}}.\end{aligned}$$

Thus even small per-site remainders can matter when divided by the difference of two small inverse-square lengths. Two exact energies solve for two leading coefficients exactly, but do not estimate this remainder.

Exercise 20.5. If ϕ is scalar of dimension Δ_ϕ, then $L_{-1}\phi$ has weights $(\Delta_\phi/2+1, \Delta_\phi/2)$ and spin one. A holomorphic current has weights $(1, 0)$. The former has nonzero right weight whenever $\Delta_\phi > 0$. A momentum-one level near $\Delta_\phi + 1$ is therefore consistent with an ordinary descendant and is not evidence for a current. It could also mix with an unrelated finite-size primary.

Exercise 20.6. Let $D = \beta\partial_\beta$, $a = \pi c/(6\beta_*)$, and $\mu_\beta = \langle g \rangle_\beta$. Directly differentiating $Z = e^{\beta a} \sum_j e^{-\beta g_j}$ gives

$$\frac{DZ}{Z} = \beta(a - \mu_\beta).$$

Its vanishing at $\beta = \beta_*$ gives $a = \mu_{\beta_*} = \mu$, or $c = 6\beta_*\mu/\pi$. At that point, $Z^{(j)}/Z = (-1)^j m_j$ for $j \geq 2$, where the derivatives are in β and the moments are centered. Repeated use of $D\beta^j = \beta^j(D + j)$ gives

$$\begin{aligned} D^3 &= \beta\partial_\beta + 3\beta^2\partial_\beta^2 + \beta^3\partial_\beta^3, \\ D^5 &= \beta\partial_\beta + 15\beta^2\partial_\beta^2 + 25\beta^3\partial_\beta^3 \\ &\quad + 10\beta^4\partial_\beta^4 + \beta^5\partial_\beta^5. \end{aligned}$$

The first-derivative term vanishes. The third condition is $3\beta_*^2 m_2 - \beta_*^3 m_3 = 0$, hence $\beta_* m_3 = 3m_2$. The fifth condition is

$$15\beta_*^2 m_2 - 25\beta_*^3 m_3 + 10\beta_*^4 m_4 - \beta_*^5 m_5 = 0.$$

These are derivatives of Z itself, so the m_j are central moments; replacing them by cumulants would require differentiating $\log Z$ instead. The odd derivatives vanish because modular S acts by $t \mapsto -t$ in $\beta = \beta_* e^t$. A charge projection inserts a symmetry operator into the torus trace. Exchanging cycles generally turns that insertion into a spatial twist; the projected partition function alone is therefore not S-invariant.

B.21 Solutions for Chapter 21

Exercise 21.1. The phase of λ^3 is $3\theta_\phi + 3\theta_\chi - 3\theta_\psi$. Multiplication by c_ψ and division by $c_\phi c_\chi$ cancel it. The same bookkeeping for ν^3 gives $3\theta_\phi + 3\theta_\psi + 3\theta_\eta$, exactly the denominator phase. For $\kappa^3 c_\eta$ it gives $3\theta_\chi + 3\theta_\psi - 3\theta_\eta + 3\theta_\eta$, again exactly the denominator phase. The definitions also give

$$\frac{I_\nu I_\kappa}{I_\lambda} = \frac{\nu^3\kappa^3}{\lambda^3 c_\psi^3} = J^3.$$

All these ratios require their displayed denominators to be nonzero. If a symmetry maps each field to a unit phase times its adjoint, it conjugates each three-point coefficient. All unit phases cancel in I_λ, so invariance says $I_\lambda = \overline{I_\lambda}$. No claim that individual coefficients are real was required.

Exercise 21.2. The commutator identity

$$L_1 L_{-1}^n |h\rangle = n(2h + n - 1) L_{-1}^{n-1} |h\rangle$$

follows by commuting L_1 past the n lowering operators. The sum of their $2L_0$ terms is $2\sum_{j=0}^{n-1}(h + j) = n(2h + n - 1)$. Writing $N_n = \|L_{-1}^n|h\rangle\|^2$ therefore gives $N_n = n(2h + n - 1)N_{n-1}$, with $N_0 = 1$, and hence $N_n = n!(2h)_n$. The first three terms of the scalar vacuum-source series are

$$1 + \Delta^2 + \left[\frac{\Delta(\Delta + 1)}{2}\right]^2 + \cdots .$$

The equal-time integral near coincidence contains $\int_0^\varepsilon \theta^{-2\Delta}\, d\theta$, finite precisely for $2\Delta < 1$. At the endpoint it diverges logarithmically.

Exercise 21.3. Treat the normalized overlaps as a linear moment functional with $\langle 1\rangle = 1$, and set $\widetilde{X}_i = X_i - m_i$. Then

$$B_{ij} = \langle \widetilde{X}_i \widetilde{X}_j \rangle, \qquad K = \langle \widetilde{X}_1 \widetilde{X}_2 \widetilde{X}_3 \rangle.$$

After $X_i \mapsto u_i X_i + v_i$, the new centered variable is $u_i \widetilde{X}_i$. Multilinearity proves the factors $u_i u_j$ and $u_1 u_2 u_3$, including for complex data. The third leg's u_3 denotes its amplitude factor after converting the outgoing ket change into a bra change.

A physical branch insertion does not invalidate these affine identities. It changes the moment functional itself, replacing three-point moments by normalized four-point correlators. Their connected part contains extra OPE data. Thus it invalidates the inference that the invariant equals the original three-field coefficient, unless an additional Ward or selection argument removes that branch contribution.

Exercise 21.4. Each retained-state summand is nonnegative. For every omitted state, its exponential factor is at most $e^{-2\tau(E_{\mathrm{cut}}-E_\phi)}$. Factoring this out of the omitted sum proves the bound. To see why residuals do not provide the cutoff, take a diagonal Hamiltonian with energies $0, 1, 2$ and retain the exact eigenvectors at 0 and 2. Both residuals vanish, but the omitted state has energy $1 < 2$. A source with weight on all three detects the failure of using the largest retained energy as an omitted-state threshold.

Exercise 21.5. For $H(\theta) = H + \theta V$, Hellmann–Feynman gives $d_a = \partial_\theta g_a = \langle a|V|a\rangle - \langle 0|V|0\rangle$ along the chosen differentiable eigenstate branch. The quotient rule gives

$$\partial_\theta\left(\frac{g_a}{g_b}\right) = \frac{d_a g_b - g_a d_b}{g_b^2}.$$

In a degenerate space one must first choose the branches by diagonalizing the perturbation there. Under $V \mapsto V + a1 + bH$, the gap derivatives become $d_a + bg_a$, so the added numerator is $bg_a g_b - bg_a g_b = 0$. A coupling reparametrization multiplies every coordinate derivative by the same Jacobian. Comparing its direction with size drift assumes the perturbation probes the same running continuum coupling and that other corrections are negligible. The algebraic cancellation of energy rescaling does not prove either premise.

Exercise 21.6. Write $u'_i = \sum_k u_k U_{ki}$ and $v'_j = \sum_l v_l V_{lj}$. The gluing amplitude is linear in both incoming kets, giving $M' = U^T M V$. Both U^T and V are unitary, so $M'^\dagger M'$ is unitarily conjugate to $M^\dagger M$ and has the same eigenvalues.

For normalized pairs, the tensor-product difference satisfies

$$\|u \otimes v - u' \otimes v'\| \le \|(u - u') \otimes v\| + \|u' \otimes (v - v')\| \le \delta_1 + \delta_2.$$

Taking an inner product with a unit parent cannot increase that norm. Divide by $|A_{000}|$, subtract the continuum gluing error η, and use $|x^T M y| \le \|M\|_2$ for normalized coefficient vectors. This proves the stated bound. A small denominator amplifies an absolute vector error; cancellation of a parent's overall normalization is not cancellation of its directional state error.

Exercise 21.7. Expanding either side gives $2b^4 - 4b^3 + 6b^2 - 4b + 2$. Consequently

$$\sum_{i<j} \frac{\delta B_{ij}}{B_{ij}} = \frac{2N_\Lambda}{c} \frac{1 + (b-1)^4 + b^4}{b^4(b-1)^4} = 2\frac{\delta K}{K}.$$

This is exactly the cancellation in the logarithmic variation of the estimator. Multiplying the identity in q, r by $b^4(b-1)^4$ reduces it to the same polynomial identity. The quadratic variation is a bilinear Hessian $H(v_b, v_b)$; differentiation controls $H(v_b, \partial_b v_b)$. It supplies no identity for $H(v_b, v_{b'})$ at independent b, b', so two separately regulated singularities require another calculation.

B.22 Solutions for Chapter 22

Exercise 22.1. The largest two Schmidt probabilities sum to $5/6$. Their normalized truncation is

$$|\Psi_2\rangle = \sqrt{3/5}|00\rangle + \sqrt{2/5}|11\rangle.$$

Its squared overlap with Ψ is $5/6$, its phase-aligned vector-norm error is $\sqrt{2-2\sqrt{5/6}}$, and its energy is $-5/6$. The best rank-one energy is $-1/2$. The proof by the projector onto the left Schmidt support already optimizes over every subspace of the prescribed dimension. It therefore includes arbitrary local basis changes; it is not restricted to coordinate truncation in the original product basis.

Exercise 22.2. The only nonzero block of A^a in block row a and column b is W_{ab}. In $A^{x_0}A^{x_1}$, the first tensor's column must therefore be the second tensor's row x_1, leaving $W_{x_0x_1}$ times the next block row. Induction produces $W_{x_0x_1}\cdots W_{x_{L-2}x_{L-1}}$ followed by the last row. The ordinary trace forces its last column back to x_0, giving $W_{x_{L-1}x_0}$ and the desired cyclic fiber trace. Any forbidden edge makes one factor zero. This proof works for unequal fiber dimensions, because consecutive rectangular maps have matching intermediate spaces.

Exercise 22.3. The norm matrix is $(t^2-1)I+J$. Its eigenvalues are t^2+2 on the uniform vector and t^2-1 twice on the color-orthogonal plane. The nontrivial length is

$$\xi = -\left[\log\left|\frac{t^2-1}{t^2+2}\right|\right]^{-1}.$$

At $t=0$ the ratio is $-1/2$, giving alternating color correlations and finite length $1/\log 2$. At $t=1$ it is zero; the uniform edge amplitudes give an ordinary product state on the supported three colors, with no nonzero-distance connected tail. At large $|t|$, $|\lambda| = 1-3/(t^2+2)$ and $-\log|\lambda| \sim 3/t^2$, so $\xi \sim t^2/3$ in lattice sites. This growth reflects long domains of equal colors. No conformal limit follows merely from this parameter limit.

Exercise 22.4. Across any nontrivial cut the three distinct product branches are orthogonal on both sides. Their Schmidt probabilities are $(1/3,1/3,1/3)$, so the entropy is $\log 3$. It is independent of interval length. The state has three ordered product components and a nondecaying order correlator; its standard symmetry-breaking parent has a finite ground-state degeneracy. Neither feature gives the logarithmic interval growth associated with a CFT central charge.

Exercise 22.5. Normalize $(l|r)=1$ and put $\mathcal{P}=|r)(l|$. The fixed-point relations imply $\mathcal{E}\mathcal{P}=\mathcal{P}\mathcal{E}=\mathcal{P}$. Thus $\mathcal{E}^n=\mathcal{P}+(\mathcal{E}-\mathcal{P})^n$ for $n\geq 1$. Subtracting the product of one-point functions in a separated correlator removes the $\mathcal{P}$ term. On the complementary space the spectral radius is less than one for an injective clustering tensor, so the geometric series converges even if the map is not diagonalizable:

$$\sum_{n\geq 0}(\mathcal{E}-\mathcal{P})^n = (1-\mathcal{E}+\mathcal{P})^{-1}.$$

The additional stationary direction of a cat remains after subtracting only one rank-one projector. A periodic peripheral root likewise has modulus one. The series then fails to converge on its coupled direction, and a connected primitive component must be specified before using this formula.

Exercise 22.6. At $n=1,2,\infty$, the logarithmic coefficients in (22.7) are respectively $c/6, c/8, c/12$. Let $x_a=\log\xi_a$ and $x_b=\log\xi_b$. For $S_1=ax+b+u/x$, the secant is

$$\frac{S_1(x_b)-S_1(x_a)}{x_b-x_a} = a-\frac{u}{x_ax_b}.$$

The corresponding central-charge diagnostic is six times this quantity. Increasing ξ by a large factor may still change the logarithms modestly, so the correction can drift slowly. This illustrative correction is not a derived law for the Haagerup sequence; it shows why apparent slope stability needs an independently justified remainder model.

B.23 Solutions for Chapter 23

Exercise 23.1. On the neutral basis (f, s), with $s = (o_0 + o_1 + o_2)/\sqrt{3}$, the rho matrix is $R_0 = \left(\begin{smallmatrix} 1 & \sqrt{3} \\ \sqrt{3} & 2 \end{smallmatrix}\right)$. Its characteristic polynomial is $x^2 - 3x - 1$, giving d and $-d^{-1}$. On the color-orthogonal plane, J vanishes and rho is $-P$, where P is the reflection of two colors. Hence rho squares to one on that plane. Alpha is the order-three rotation and $P\alpha P = \alpha^{-1}$.

On the neutral plane, $1+\alpha+\alpha^2 = 3$ and $R_0^2 = 1+3R_0$. On the charged plane that alpha sum is zero and $(-P)^2 = 1$. These two checks prove the full fusion relation $R^2 = 1 + (1 + \alpha + \alpha^2)R$, as well as $R\alpha = \alpha^{-1}R$. They identify the representation as the two scalar fusion representations plus the two-dimensional charged representation.

Exercise 23.2. The ordered total probability is $p = 3/(b^2 + 3)$ and the fixed probability is $1 - p = b^2/(b^2 + 3)$. Each ordered point has probability $1/(b^2 + 3)$. The charges satisfy $\omega^{3j} = 1$, so

$$\langle \Phi^3 \rangle = p\, p^{-3/2} = p^{-1/2}, \qquad \langle \Phi^\dagger \Phi E \rangle = p\, p^{-1} b/\sqrt{3} = b/\sqrt{3}.$$

For the neutral cubic,

$$\langle E^3 \rangle = \frac{b^3/\sqrt{3} - 3\sqrt{3}/b}{b^2 + 3} = \frac{b^2 - 3}{\sqrt{3}b} = -\frac{1}{\sqrt{3}},$$

using $b^2 + b = 3$. Checking the multiplication identities at the ordered and fixed points separately proves them as identities of functions. For example, $1 + gE$ equals $1 + b^2/3 = p^{-1}$ on an ordered point and zero on the fixed point, exactly the values of $\Phi^\dagger\Phi$. Similarly $1 - E/\sqrt{3}$ has values $b^2/3$ and $3/b^2$, equal to E^2. Substituting $g = b/\sqrt{3}$ yields $C_{EEE} = (g^2-1)/g$ and $p^{-1} = 1+g^2$. Finally $b^2 = (7 - \sqrt{13})/2$ gives $p^{-1} = (13 - \sqrt{13})/6$, recovering the charged cubic's radical expression.

Exercises 23.3 and 23.5. Invariance of a few global vectors constrains the operator only on their span. It says nothing about its action on the local range with arbitrary boundary data and arbitrary outside environment. The stronger statement is that the open line maps every interval module vector, with all allowed endpoint and MPO boundary indices, into the complete local module range. Closure under adjoints makes this range reducing, so its orthogonal projector commutes with the closed line.

For the short Hamiltonian, one still needs equality of its global zero space with the proposed MPS family and a positive gap above that zero space uniformly in ring size. A different longer parent can penalize directions left unpenalized by the short terms. Its gap does not provide an operator comparison bounding the short Hamiltonian from below by a positive multiple of that parent. The four-site kernel count supplies a concrete obstruction to the simplest such comparison.

Exercise 23.4. With $B = Y_\rho - 2$ and $B^\dagger B = 5 - Y_\rho$, commutation with H gives

$$E_B = \frac{5E - \langle HY_\rho \rangle}{5 - \langle Y_\rho \rangle}.$$

Subtracting E gives the covariance formula. The denominator is at least $5 - d > 0$, and $\mathrm{Var}(Y_\rho) \le (d + d^{-1})^2/4 = 13/4$. Cauchy–Schwarz therefore bounds the total energy difference by

$$|E_B - E| \le \frac{\sqrt{13}}{2(5-d)} \sqrt{\mathrm{Var}(H_L)}.$$

For variance $O(L)$, division by L gives $O(L^{-1/2})$. A cat with nonzero weights on branches of densities $e_1 \neq e_2$ has a contribution $w_1 w_2 (e_1 - e_2)^2 L^2$ to the variance. For an explicit example, take $H_L = \sum_j Z_j$ on qubits and $(|0\rangle^{\otimes L} + |1\rangle^{\otimes L})/\sqrt{2}$. The branches have energies $L, -L$, so $\langle H_L \rangle = 0$ and $\mathrm{Var}(H_L) = L^2$.

B.24 Solutions for Chapter 24

Exercise 24.1. Independent stress tensors make the dimension of a product field the sum of its factor dimensions. The mixed three-point function factors into two normalized two-point functions, giving $\lambda = 1$ in the displayed product convention. Moreover $c_\psi = c_\phi c_\chi$, hence $I_\lambda = 1$ whenever its denominator cubics are nonzero. Under independent rephasings, λ itself can acquire a phase, while $|\lambda|$, all dimensions, and I_λ remain unchanged. Thus the invariant comparison is stronger than comparing a printed phase of one matrix element.

Exercise 24.2. On a totally isotropic subspace, projection to the positive two-dimensional factor has zero kernel: a vector with positive part zero must also have negative part zero by nullness. The projection is injective, so the subspace has dimension at most two. For light scalar charges the integral pairing has absolute value strictly below one and must vanish. At a non-strict endpoint the value ± 1 is not excluded, so the planar reduction requires a separate argument.

Exercise 24.3. The only nonzero charge-neutral ordered triples are the six permutations of (p_1, p_2, p_3). Each coefficient product is $1/(\sqrt{3})^3$, giving $2/\sqrt{3}$. All three external dimensions are equal, exactly the case excluded by the unequal-shell hypothesis. Arbitrary degeneracy within unequal shells was already allowed by that theorem; it is equality of the three radii that changes the geometry.

Exercise 24.4. Write $q(z) = (1-z)^{2\delta}$ and $P(z) = q(z) g_D(z)$. Then $F_D(z) = P(z) - P(1-z)$, so the first and third derivatives at $1/2$ are respectively $2P'$ and $2P'''$. At that point,

$$\frac{q'}{q} = -4\delta, \qquad \frac{q''}{q} = 8\delta(2\delta - 1), \qquad \frac{q'''}{q} = -32\delta(2\delta - 1)(\delta - 1).$$

Insert $g_D'' = 2g_D' + 8C g_D$ and $g_D''' = 8(C+1) g_D'$, where $C = D(D-1)$, into the product rule. With the chapter's $A = 1 - 6\delta + 6\delta^2$ and $B = 1 - 3\delta + 2\delta^2$, this gives

$$\frac{P'''}{q} = 8(C + A) g_D' - 32\delta(3C + B) g_D, \qquad \frac{P'}{q} = g_D' - 4\delta g_D.$$

Substitution into the defined functional yields

$$\frac{\mathcal{L}(F_D)}{g_D} = (C + A - \kappa) r_D - 4\delta(3C + B - \kappa), \qquad r_D = g_D'/g_D.$$

For the identity $D = 0$, use $g_0 = 1$ directly rather than the positive-dimension integral representation.

The arithmetic in the certificate is exact:

$$\frac{5}{4}\frac{9}{10}(.083) = .093375, \qquad \frac{169}{100}\frac{37}{40}(.246) = .3845595,$$

and hence $.303 + .093375 - .3845595 = .0118155 > 0$. The identity, charged, and epsilon contributions have a positive total under the stated lower bounds. Every additional exchange outside $(.16, 1.75)$ has nonnegative coefficient and positive functional value, so crossing requires another coupled exchange inside that interval.

Application to the chain requires a unitary local continuum CFT with the specified normalized Hermitian scalar and a positive, convergent block expansion. The proposed dimensions and OPE magnitudes must be limiting continuum values, or controlled intervals over which the strict inequality survives. Finite eigenvector overlaps alone do not establish those premises. The additional interpretation as a new two-dimensional scalar uses integer locality, even spin in the identical-scalar OPE, and the descendant analysis in the chapter. It does not require assuming a particular central charge or a named Haagerup CFT. It also does not exclude the continuum when the required extra scalar is present.

Exercise 24.5. The oscillator formula gives $h_L = 11/288$ and $h_R = 35/288$. Their sum is $23/144$, but that pair is not scalar: its spin is $-1/12$. Nonnegative excitations and zero modes imply that a scalar must have both weights at least $35/288$, hence $\Delta \geq 35/144$. Level matching or projection can increase this threshold; the bound need not be attained.

If $a^3 = 1$ in the abelianization and $\chi^2 = 1$, then $\chi(a)^3 = \chi(a)^2 = 1$, so $\chi(a) = 1$. This applies to each determinant character independently. For the fixed Hermitian axis, the unit normalization $X = (e^{i\theta}\phi + e^{-i\theta}\phi^\dagger)/\sqrt{2}$ gives $|C_{XXX}| = |c_\phi|/\sqrt{2}$ at a cubic extremum. The isometric lift contributes no group-order factor. Thus $|c_\phi|/\sqrt{2} \leq \sqrt{2/3}$ gives the stated $2/\sqrt{3}$, with no further rescaling.

Exercise 24.6. Choose the common Potts field phase so that c_σ is real and positive. The triples w_i^3 then contribute $c_\sigma^2 a_i^3$: two factors contribute conjugate order cubics, whose product is $|c_\sigma|^2 = c_\sigma^2$, and the third contributes the identity. The six permutations of $w_1 w_2 w_3$ pair each Potts order field with its adjoint and contribute $6a_1a_2a_3$. Every other triple violates an independent ultraviolet charge. In $C(\sigma_1, \chi, \sigma_2^\dagger)$ only w_1 survives, giving $\lambda = a_1$ and hence $|\lambda| \leq 1$. A large cubic can arise from several charge-conserving permutations without increasing this one mixed matrix element.

Exercise 24.7. The conjugate of φ is $(1 - \sqrt{5})/2$. Therefore

$$N(\varphi + 2) = \frac{(5 + \sqrt{5})(5 - \sqrt{5})}{4} = 5, \qquad N\left(\frac{64(\varphi + 2)^3}{3}\right) = \frac{64^2 5^3}{9}.$$

An algebraic integer has an integer field norm, so the second number proves that the proposed quotient is not an algebraic integer. Fusion-subcategory divisibility then excludes an order-three pointed subcategory from the ambient category of defects preserving the original retained algebra, and hence from its smaller maximal-chiral-trivial symmetry kernel. It does not control a symmetry that permutes stress tensors and acts nontrivially on the chiral algebra. The separate three-cycle argument, together with the maximal graph pairing, is what bounds those chiral images and prevents independent left-right cycles.

Exercise 24.8. Reduce the sums modulo one, keep values strictly between zero and $1/2$, and then double them. Adding zero to the Fibonacci residues retains only $1/5$, yielding $\Delta = 2/5$.

Adding 1/2 retains 1/10 and 3/10, yielding 1/5 and 3/5. Adding 5/16 retains 5/16 and 9/80, yielding 5/8 and 9/40. This gives the chapter's five-element list.

For a scalar below dimension one, positive-energy integer grading leaves no freedom to add a nonzero integer to its chiral weight while remaining below 1/2. Therefore twist residues impose this necessary condition. The full pairing can remove possible entries, and the category does not fix their actual multiplicities or lowest characters. An intermediate smaller algebra has a different representation category, so this residue list cannot be imposed on it by performing a further mathematical condensation absent from the physical theory.

B.25 Solutions for Chapter 25

Exercise 25.1. Dividing $d_n^2 = nd_n + 1$ by d_n gives $d_n - d_n^{-1} = n$, hence the period difference. Expanding $(d_n + 1)(1 + d_n^{-1})$ gives $d_n + 2 + d_n^{-1} = \omega_1 + \omega_2$; division by their product proves the reciprocal identity. For nonzero integer a,

$$\sin(\pi a/\omega_1) = \sin(\pi a - \pi a/\omega_2) = (-1)^{a+1} \sin(\pi a/\omega_2).$$

The denominator is nonzero because ω_2 is irrational. Substituting this sine ratio into $F(a + n)/F(a) = \sin(\pi a/\omega_1)/\sin(\pi a/\omega_2)$ gives $F(a + n) = (-1)^{a+1}F(a)$. At the origin the sine ratio tends instead to $\omega_2/\omega_1 = d_n^{-1}$.

Exercise 25.2. For $1 \le a < n$, multiplication of the two sample formulas gives the sign exponent

$$\frac{a(a + n) + (n - a)(2n - a)}{2} = a(a - n) + n^2.$$

Since n is odd, a and $a - n$ have opposite parity, so their product is even; n^2 is odd. The two initial minus signs cancel, the remaining sign is minus, and reflection gives $W_a W_{n-a} = -F(a)F(n - a)/d_n = -d_n^{-1}$. The integer exponent is not itself a function of the residue class: replacing a by $a + n$ changes $a(a + n)/2$ by $n(a + n)$, which can be odd. For example at $n = 3$, representatives 2 and 5 give exponents 5 and 20. One must state the chosen representatives and include the accompanying gamma-shift factor when changing them. The separately defined origin $W_0 = -1$ also cannot be obtained by the generic nonzero-sample rule.

Exercise 25.3. Take

$$M = \begin{pmatrix} 1 & 1 \\ 1 & 1 + 10^{-8} \end{pmatrix}.$$

Its exact determinant is 10^{-8}, while rounding to six decimal places produces two identical rows. After clearing denominators, a nonzero determinant modulo a prime proves that the original integer determinant was nonzero. A zero residue only says that the prime divides that determinant; the integer could still be nonzero. Modular rank methods therefore give decisive lower bounds when they succeed and potentially inconclusive results when they fail. For an explicit failure at a prime with no denominator problem, $\mathrm{diag}(1, 7)$ has exact determinant seven but determinant zero modulo seven. A nonzero $r \times r$ minor modulo a permitted prime proves rank at least r over $\mathbb{Q}$; proving an upper bound still requires showing the larger minors vanish exactly.

Exercise 25.4. The two evaluations of $1 + t$ are 2 and 0, so their product is zero and the element is not invertible. The two evaluations of $2 + t$ are 3 and 1, so its product is three. It is invertible, with inverse $(2 - t)/3$. Modulo five both conclusions remain valid: the relevant products are zero and three. The distinction is visible in the algebra itself: $(1 + t)(1 - t) = 0$

with $1-t \neq 0$, whereas $(2+t)(2-t) = 4-t^2 = 3$. Since two and three remain invertible modulo five, the algebra still splits into its $t=1$ and $t=-1$ components, and the second inverse reduces to $4-2t$.

Exercise 25.5. A field automorphism preserves a polynomial equality by preserving addition and multiplication. Positivity is an order condition, and an adjoint involves complex conjugation, neither of which is part of that algebraic guarantee. For example the two real embeddings of $\mathbb{Q}(\sqrt{13})$ send the positive number $\sqrt{13}$ to opposite signs. Thus a transported algebraic solution must have its Hermitian and positivity properties checked separately.

B.26 Solutions for Chapter 26

Exercise 26.1. Choose nonzero $h \in H$ and $k \in K$. The reciprocal conditions on $(h,0)$ and $(0,k)$ read

$$U(h)U(-h)V(0)^2 = \delta, \qquad U(0)^2V(k)V(-k) = \delta.$$

Since $U(0)^2V(0)^2 = W(0,0)^2 = 1$, multiplying gives $U(h)U(-h)V(k)V(-k) = \delta^2$. The condition at (h,k) says that the same product equals δ. Nonvanishing forces $\delta = 1$. But $\delta^2 = n\delta + 1$ with $n > 0$ does not permit $\delta = 1$. Coupling the coordinates or changing the origin structure is therefore necessary; the direct product of independent reciprocal scalar functions cannot solve this ansatz.

Exercise 26.2. Put $s = t^2$. Since $s > 0$ and $s^2 + 8s - 1 = 0$, $s = \sqrt{17} - 4$ lies strictly between zero and one. Dividing the second eliminant by $2tr^2$ gives $r^{-2} - r^2 = (3-s)/(2t) > 0$, hence $0 < r < 1$. Squaring and adding four yields

$$(r^2 + r^{-2})^2 = \frac{(3-s)^2 + 16s}{4s} = \frac{9 + 10s + s^2}{4s} = \left(\frac{9+s}{2}\right)^2,$$

where $s^2 + 8s - 1 = 0$ reduces the last equality. Since $r^2 + r^{-2} > 0$ and $9 + s > 0$, the positive square root gives $r^2 + r^{-2} = (9+s)/2$. Thus

$$(r + r^{-1})^2 = (13+s)/2 = ((5+s)/2)^2,$$

and positivity again gives $r + r^{-1} = (5+s)/2$. Solving the resulting quadratic and choosing the root below one gives

$$r = \frac{5 + s - \sqrt{(5+s)^2 - 16}}{4} = \frac{1 + \sqrt{17} - \sqrt{2 + 2\sqrt{17}}}{4} = \lambda_8^2.$$

For completeness, the supplied identity $tr^3 + 2r = 1$ can also be derived from this positive branch. The relation $s^2 + 8s = 1$ gives

$$(5+s)^2 - 16 = s(9+s)^2.$$

Since $t > 0$, the selected radical is $t(9+s)$, and hence

$$r = \frac{5 + s - t(9+s)}{4} = \frac{t(5+s) + 2}{t(11+s) + 4}.$$

The second equality follows by cross-multiplication and reduction using $t^2 = s$ and $s^2 + 8s = 1$; its denominator is positive. Writing $C = (5+s)/2$, the quadratic $r^2 - Cr + 1 = 0$ gives

$$r^3 = (C^2 - 1)r - C = \frac{11+s}{2}r - \frac{5+s}{2}.$$

Consequently

$$tr^3 + 2r = \frac{t(11+s)+4}{2} r - \frac{t(5+s)}{2} = 1.$$

Multiplication of the first eliminant by r now gives $y^2 r^3 = tr^3 + 2r = 1$. The positive sample y is therefore $r^{-3/2} = \lambda_8^{-3}$. The signs of r, y and the positive square-root choices are analytic branch information that the polynomial equations alone do not supply.

Exercise 26.3. At $m = 3$, the relations are $D - 1 = 4 + 2\omega$ and $\omega^2 = \omega + 9$. Multiplication of the basis vectors $1, \omega$ gives $4 + 2\omega$ and $18 + 6\omega$, so the matrix is

$$\begin{pmatrix} 4 & 18 \\ 2 & 6 \end{pmatrix}.$$

Its determinant is $24 - 36 = -12$ and the gcd of its entries is 2. For a full-rank two-by-two integer matrix, the first Smith invariant is that gcd and the product of the two invariants is the absolute determinant. They are therefore $2, 6$. The quotient has 12 elements, but every element has order dividing six. The cyclic group of order twelve has an element of order twelve, so the cardinality alone loses essential group data.

Exercise 26.4. Let $Y = S^2 - f$. The support condition gives

$$\operatorname{tr}(Y^* Y) = \operatorname{tr}(S^4) - 2\operatorname{tr}(S^2) + \operatorname{tr}(f) = 0.$$

Faithful positivity implies $Y = 0$. Expanding with $S^2 = f$ and $fS = S$ gives

$$\left(\frac{f \pm S}{2}\right)^2 = \frac{2f \pm 2S}{4} = \frac{f \pm S}{2}, \qquad \frac{f+S}{2}\frac{f-S}{2} = 0.$$

They are self-adjoint and sum to f, and their traces are $\operatorname{tr}(f)/2$ because $\operatorname{tr}(S) = 0$. For an indefinite pairing a nonzero Y may have zero squared pairing, so the first inference to $Y = 0$ fails. For example, on $\mathbb{C} \oplus \mathbb{C}$ take $\operatorname{tr}(a, b) = a - b$, $f = (1, 1)$ and $S = 0$. The stated trace equalities and support conditions hold, but $S^2 \neq f$. The trace pairing is nondegenerate and indefinite; it cannot turn a zero squared pairing into a zero vector.

Exercise 26.5. If $L^k w = w$ with odd k, the rotation rule gives $p_w = p_{L^k w} = (-1)^k p_w = -p_w$, hence $p_w = 0$ over $\mathbb{C}$. Every letter 2 may be replaced by the sum of letters 0 and 1 with a minus sign, decreasing the number of 2's. This determines all coefficients from binary ones. Equivalently, the uniquely specified multilinear tensor evaluated at $u_2 = -u_0 - u_1$ performs every replacement at once; independence of the replacement order follows from multilinearity. In the given example,

$$p_{00000002} = -p_{00000000} - p_{00000001} = -c_1 = 2\lambda^4 + \lambda^2 - 9.$$

Exercise 26.6. The defining cubic is negative at $x = 4$ and tends to $+\infty$, so its selected largest root satisfies $x > 4$. Consequently $a = x - 2 > 2$, $y = a + 1 > 0$, and $z = a^2 + a - 1 > 0$; the displayed square roots make v a real vector. With $y = a + 1$ and $z = a^2 + a - 1$,

$$v^T v = \frac{a^2 + y + z}{2ay} = 1.$$

The two requested entries are

$$-2v_1 v_2 = -\frac{1}{\sqrt{y}}, \qquad -2v_2 v_3 = \frac{\sqrt{z/y}}{a}.$$

The other entries follow in the same way. Since $v^T v = 1$, $(I - 2vv^T)^2 = I - 4vv^T + 4v(v^T v)v^T = I$. The matrix is real symmetric, hence orthogonal and unitary. Its eigenvalues are $-1, 1, 1$, so its determinant is -1. This proves a matrix identity. Identifying it as a particular category's associator additionally requires the specified trivalent bases and their planar evaluations; many unrelated orthogonal matrices satisfy the same unitarity equation.

Appendix C

Proofs, finite certificates, and numerical evidence

A construction that uses exact algebra, analytic theorems, finite computations, formalization, and numerical experiments needs more than one notion of verification. This appendix explains those notions without asking the reader to reconstruct the entire research workspace.

C.1 An exact finite certificate has a mathematical target

A finite lattice calculation can certify a vector equality $a_{(n)}b = \sum_j c_j v_j$, a membership in a specified span, the rank of a matrix, positivity of a Gram matrix, or the cyclicity of a finite-dimensional top. The target must name the actual vectors, mode conventions, coefficient field, and basis. Otherwise the phrase "the check passed" leaves the mathematical claim unclear.

The exact construction uses the lattice oscillator formulas and algebraic coefficient fields such as $\mathbb{Q}(\sqrt{13}, \sqrt{-3})$. An equality in that field is stronger than a small floating residual. A saved coordinate relation can be checked by a small independent program without recomputing the original mode product. That verifies the relation among the supplied vectors. It does not independently verify that those vectors were correctly computed as the claimed lattice products. Both checks are useful, and their scopes should be distinguished.

The finite-closure theorem then converts an explicitly stated family of finite identities into an all-weight strong-generation conclusion. This implication is a mathematical proof, not an extrapolation from a list of low-weight dimensions. In particular, the provenance of recursively defined higher fields is part of the hypotheses: a vector with the right weight is not enough if it has not been shown to belong to the algebra generated by the original fields.

C.2 A ledger of implications for the charge-eight construction

Inputs apply row by row; the diagonal net route branches before the final fusion-positivity step. Finite construction inputs retain the status explained above; the structural implications do not independently replay them. Write $H = V_{E_8}$ for the original ambient extension and H' for the later selected extension.

Step	Needed input	Output and logical boundary
Finite closure	Specified PCT-stable lattice fields and ambient stress; actual memberships and product identities; recursive provenance of higher generators.	Theorem 8.4 gives all-weight strong generation. The stated finite identities give properness and $\dim U_1 = 6$, (8.9). A few ranks alone do not.

Step	Needed input	Output and logical boundary
Positive stresses	Actual PCT-stable embedded regular affine, Potts and parafermion algebras; their verified projection identities.	(9.13) expresses ω_U as a positive finite sum of their stresses. This is an energy identity, not a commuting tensor decomposition.
Strong rationality	That stress family, compatible component tops, vacuum splitting, and the finite-category rigidity argument.	C_2-cofiniteness and vacuum projectivity (Propositions 10.4, 10.8) lead to semisimplicity and strong rationality (Proposition 10.10; Chapter 10). Positivity of every U-module has not been assumed.
Algebraic branching	Strong rationality; finite core-top bounds, extension branching, rank-four exclusion and indicator moments.	$H\downarrow_U = \mathbf{1} \oplus X \oplus 2Y$, (11.7); the boundary ring is Haagerup and there are twelve chiral simples. This does not identify $Y = a$ or determine the unitary associator (Chapter 11).
Complete unitarity	Strong locality; independent algebraic and net sector counts; common domains and paired-annulus comparison; fusion positivity and unitary category classification.	Positive modules and canonical fusion forms; then $\mathrm{Rep}(U) \simeq Z(\mathcal{H}_3)$ as unitary ribbon categories. The original boundary category remains $\mathcal{H}_2$ or $\mathcal{H}_3$ (Chapter 12, especially (12.5)).
Selected extension	The center equivalence and the positive canonical Lagrangian algebra belonging specifically to $\mathcal{H}_3$; positive extension and charge-eight holomorphic classification theorems.	A selected extension $H'\downarrow_U = \mathbf{1} \oplus b \oplus 2a$, with $H' \cong V_{E_8}$. That ambient isomorphism does not identify the two embeddings of U (Chapter 14).
Full local theories	Diagonal: complete rationality of the net and Longo–Rehren. Extension: the chosen positive algebra and extension/sewing theorems.	The diagonal theory and $H' \otimes \overline{H'}$ exist, with different Hilbert spaces, (14.1) and (14.2). Modular invariance alone would not establish either sewing construction (Chapter 17).

Characters follow from the modular representation, admissible principal parts and six-current coefficient, with uniqueness supplied by Proposition 13.3. They count states in the constructed sectors; they do not replace the earlier domain or positivity arguments. None of the rows identifies the spin-chain infrared limit.

C.3 Reduction modulo a prime has a direction

Suppose a matrix has entries in an algebraic number field and all denominators remain invertible under a chosen reduction to a finite field. If a minor is nonzero after reduction, it was nonzero before reduction. Therefore

$$\operatorname{rank}_{\text{number field}} M \geq \operatorname{rank}_{\text{finite field}} \bar{M}.$$

A nonzero modular minor proves a characteristic-zero rank lower bound. A zero modular determinant does not prove a zero original determinant: reduction can lower rank. Nor does a large modular rank by itself prove that a target belongs to a span. A membership needs an exact expression or an independent upper-bound argument that closes the dimension count.

The choice of reduction matters. Algebraic generators must satisfy their defining polynomials in the finite field, and denominators must be checked. For example a map sending $\sqrt{13}$ to an element whose square is not thirteen is not a valid specialization of the intended coefficient ring, regardless of how plausible the resulting rank looks.

Worked example C.1 (A modular dependence that disappears in characteristic zero)**.** Let $R = \mathbb{Z}[1/2, \sqrt{13}] \subset K = \mathbb{Q}(\sqrt{13})$ and take

$$A = \begin{pmatrix} 1 & \sqrt{13} \\ \sqrt{13} & 16 \end{pmatrix}, \qquad \det A = 3.$$

Specializing the coefficient ring to $\mathbb{F}_3$ by $\sqrt{13} \mapsto 1$ is valid: the defining square relation is preserved and the denominator two is invertible. It sends A to the all-ones matrix, of rank one. Its two columns coincide modulo three. Over K, expressing the second column as a multiple of the first would instead force the coefficient to be $\sqrt{13}$. The lower coordinate would then be 13, while the required coordinate is 16. Thus this modular span membership does not lift to characteristic zero.

The specialization $R \to \mathbb{F}_{17}$ sending $\sqrt{13}$ to 8 is also valid, since $8^2 = 64 \equiv 13$. Now the determinant remains $16 - 64 \equiv 3 \neq 0$, so the specialized rank is two. The nonzero minor proves rank two over K as well. The prime three was admissible for the coefficients but lost rank for this particular matrix. These are maps from R, not field homomorphisms $K \to \mathbb{F}_p$.

Finally, positive definiteness is a separate real statement. The leading principal minors of A are 1 and 3, so Sylvester's criterion proves positivity under either real embedding of K. Neither finite-field rank calculation establishes that positivity.

C.4 What a proof assistant theorem actually assumes

A kernel-checked theorem can be highly informative while still being conditional. Consider the abstract theorem

$$(H : P) \longmapsto H.$$

It proves P from a hypothesis P using no unproved axiom in its body. Inspecting only the theorem's global axiom list would not tell us that an instance of P is still needed. Likewise, a formal dependency theorem can correctly compose finite certificate assumptions, external results, and manuscript lemmas without constructing every assumption for the intended lattice VOA.

The repository's formalization contains real concrete mathematics: mode identities, finite current and field relations, parts of the properness argument, character and category arithmetic, and logical composition of the construction. It also retains explicit external, certificate, and analytic hypotheses. Some conclusion nodes are concrete propositions about constructed objects; others encode the intended theorem as a proposition parameter. These distinctions are recorded in its trust-boundary document.

Accordingly, this textbook does not call the complete Haagerup CFT construction machine verified. Formalizing a finite identity, formalizing the implication from that identity, and supplying a complete formal instance of the whole theory are different tasks. The available formal results help locate the remaining work precisely.

There is also a separate, stronger recorded formal result about the uniform coefficient construction in Chapter 25. The theorem `HaagerupIzumi.exists_EGEquations_odd` constructs complex coefficient data satisfying the complete Evans–Gannon equations for every odd natural number. The underlying hyperbolic gamma interface is constructed from its regularized integral; the final theorem does not assume an unproved special-function interface. The recorded build and transitive axiom audit in `lean/README.md` report only Lean's standard logical axioms. The corresponding coefficient source revision is `698b66ac2c667e68b1e19c196fb91e84a0c82235`.

This all-order coefficient theorem is not a finite numerical spot check. It is also not a formal theorem that reconstructs a fusion category, proves a positive adjoint structure, or constructs a CFT. Those later steps have separate mathematical inputs. For this book the final declarations, constructors, source status, and recorded audit were inspected without rebuilding the Lean

development. The short revised analytic exposition has documented domain gaps that the expanded/formal development addresses; one should use the latter when following the complete analytic proof. Keeping this result separate from the conditional $c = 8$ formalization avoids both understating the coefficient proof and overstating its categorical conclusions.

C.5 Numerical precision and continuum control

For a normalized approximate eigenvector v of a Hermitian finite matrix H, a residual $r = \|Hv - \lambda v\|$ implies that some exact eigenvalue lies within r of λ. It does not identify which ordered eigenvalue it is without further isolation information, and it does not give a finite-size error bound relative to a continuum dimension.

Similarly, a variational MPS energy is an upper bound within the physical variational setting. A small gradient establishes stationarity in the stated coordinate chart and tolerance; it does not establish a global minimum. A transfer eigenvalue close to the unit circle can arise from critical correlations or from multiple symmetry-related components. A high overlap of a source with one vector controls that source action on the vacuum, not its matrix elements on every excited state.

The chain chapters retain the completed finite results and state their continuum assumptions locally. They do not convert decimal precision into asymptotic certainty. The open infrared problem requires both a candidate full theory and controlled comparisons of dimensions, symmetry sectors, and normalized local data.

C.6 Reproducing examples responsibly

Most examples in this book need only hand algebra, short exact polynomial manipulations, or existing finite tables. Reproducing them does not require rerunning the largest lattice calculations. The accompanying `check_examples.py` reproduces $S^2 = I$ and all $1,728$ Verlinde coefficients of the displayed Haagerup modular matrix in exact cyclotomic arithmetic. It also checks two twisted-character vectors and selected descendant, Gram-matrix, and correlator examples. Its recorded successful run has this finite scope; it does not certify the lattice construction. The separate `check_order_three.py` verifies the nine quadratics and all 81 cubics of Example 25.4 using four-integer normal forms. The hand reduction in that example explains why the cancellations occur, independently of the program's finite enumeration. The third checker, `check_core_tops.py`, reconstructs all fourteen core tops in Chapter 11. It keeps every tied charge and its multiplicity, checks the finite search cutoff, and verifies the trace certificate for the $17/10$ bound coefficient by coefficient. This confirms the displayed arithmetic under the core-realization and compatible-top hypotheses; it does not establish those representation-theoretic hypotheses by running a program. The Ising calibration figure uses only the closed formulas proved in its worked example, with no numerical diagonalization or fit.

Before any substantial numerical calculation, estimate all simultaneously live allocations: basis vectors, sparse assembly, eigensolver workspaces, temporary copies, and tensor contractions. The size of the final result is not a bound on peak memory.

For the shared research workspace accompanying this text, the canonical machine policy and central research launcher govern numerical admission. Historical scripts and resource exceptions are provenance, not current authorization. The textbook's PDF builder is a bounded typesetting tool with shell escape disabled; it does not execute scientific calculations or replay certificates.

The source files for the book are separated from the research sources. Document compilation checks references, layout, and syntax. A mathematical review checks equations and implications. Neither activity silently upgrades an unverified scientific input. Keeping those responsibilities explicit makes the book a reliable starting point for both learning and further research.

A short source guide

This book emphasizes explanations and calculations. The following guide is deliberately selective: it gives a route to the main constructions and to the strongest external inputs without turning each page into a literature survey. The repository manuscripts and the public literature have different roles. In particular, the detailed lattice realization developed in Part II is presented as the repository's research construction with the finite inputs and qualifications stated in the text.

For the elementary theta and eta transformations, the NIST theta-function formulas and eta modular transformations fix the analytic conventions used in the rank-one lattice example. The discriminant and cyclotomic ramification facts used in Example 18.3 are explained in J. S. Milne's *Algebraic Number Theory*, Sections 3 and 6.

Fusion categories, centers, and modular data

D. E. Evans and T. Gannon, *The exoticness and realisability of twisted Haagerup–Izumi modular data*, arXiv:1006.1326, is the principal source for the modular data and the vector-valued character construction used here. It is especially useful after Chapters 6 and 13, when the difference between modular data and a realized VOA is clear.

M. Izumi, *The classification of 3^n subfactors and related fusion categories*, arXiv:1609.07604, develops the operator-algebraic reconstruction and classification framework. P. Grossman and N. Snyder, *Quantum subgroups of the Haagerup fusion categories*, arXiv:1102.2631, is a source for the relations among the Haagerup categories and their module categories.

For the center and extension mechanisms, useful primary sources are M. Müger, *From subfactors to categories and topology II*, arXiv:math/0111205, and A. Davydov, M. Müger, D. Nikshych, and V. Ostrik, *The Witt group of non-degenerate braided fusion categories*, arXiv:1009.2117. The small pointed-center and algebra-object examples in this book are intended to make those structural results easier to enter.

The cyclic-rotation definition and center-trace formula for Frobenius–Schur indicators used in Chapter 11 are due to S.-H. Ng and P. Schauenburg; see *Frobenius–Schur indicators and exponents of spherical categories*, Theorem 4.1. The group and pointed-center calculations in that chapter fix the normalization needed for the Haagerup twist moments.

D. E. Evans and T. Gannon, *Non-unitary fusion categories and their doubles via endomorphisms*, arXiv:1506.03546, provides reconstruction machinery relevant to the arithmetic chapter. The uniform arithmetic existence argument discussed there comes from the repository research manuscript; it should not be attributed wholesale to this paper.

For Chapter 26, Grossman–Izumi–Snyder, *The Asaeda–Haagerup fusion categories*, supplies the published order-eight generalized Haagerup example. Its exact hyperbolic-gamma identification is a local result recorded in `hi_noncyclic_gis_gamma_proof.tex`. The distinction between the completed order-twelve algebraic construction and its open gamma identification is recorded in `hi_noncyclic_gamma_identification_status.md`, with the algebraic certificate explained in `hi_noncyclic_ansatz_z2xz6_exact.tex`.

For extended Haagerup, the primary construction is Bigelow–Morrison–Peters–Snyder, *Constructing the extended Haagerup planar algebra*.
Grossman–Morrison–Penneys–Peters–Snyder, *The Extended Haagerup fusion categories*, develops its fusion-category and Morita theory; Morrison–Snyder, *Non-cyclotomic fusion categories*, explains the field-of-definition distinction. The local `extended_haagerup_reconstruction_and_coefficients.tex` reorganizes the published coefficient seeds and explains their reconstruction into categorical data. These category constructions do not by themselves supply a new conformal field theory.

Rationality and the analytic bridge

T. Abe, G. Buhl, and C. Dong, *Rationality, regularity, and C_2-cofiniteness*, arXiv:math/0204021, is a basic source for the finiteness implications. R. McRae, *On rationality for C_2-cofinite vertex operator algebras*, arXiv:2108.01898, and P. Etingof and D. Penneys, *Rigidity of non-negligible objects of moderate growth in braided categories*, arXiv:2412.17681v3, are the deeper inputs in the regularity route. Their hypotheses are part of the argument; quoting their conclusions without checking the relevant module category would skip the main work of Chapter 10.

T. Arakawa, C. H. Lam, and H. Yamada, *Parafermion vertex operator algebras and W-algebras*, arXiv:1701.06229, is useful for the parafermion identification in Chapter 9. C. Dong and Q. Wang, *Quantum dimensions and fusion rules for parafermion vertex operator algebras*, arXiv:1411.7429, supplies the affine and parafermion branching, identification, and fusion formulas used in that chapter's Potts-module example.

S. Carpi, Y. Kawahigashi, R. Longo, and M. Weiner, *From vertex operator algebras to conformal nets and back*, arXiv:1503.01260, provides the locality and net framework. A. G. Henriques and J. E. Tener, *Every conformal net has an associated unitary VOA*, arXiv:2507.20735v3, supplies the reconstruction input in the version used by the research manuscript. J. E. Tener, *Fusion and positivity in chiral conformal field theory*, arXiv:1910.08257, is the source for the final fusion-positivity implication. Chapter 12 separates the module-counting, domain, and annulus-comparison steps needed to apply these results.

M. Krauel and M. Miyamoto, *A modular invariance property of multivariable trace functions for regular vertex operator algebras*, Theorem 1, supplies the simultaneous two-temperature transformation used in Chapter 18. The regular ambient algebra and trace analyticity are the relevant hypotheses there; the two smaller Virasoro algebras need not be rational.

B. Gui, *Q-systems and extensions of completely unitary vertex operator algebras*, arXiv:1908.03484, and Y.-Z. Huang, A. Kirillov, Jr., and J. Lepowsky, *Braided tensor categories and extensions of vertex operator algebras*, arXiv:1406.3420, explain the relation between categorical algebra objects and actual extensions. These are the relevant sources when moving from the center equivalence to the selected holomorphic theory.

For the actual diagonal full conformal net, K.-H. Rehren, *Canonical tensor product subfactors*, gives the Longo–Rehren route used here (Theorem 1.4, Proposition 1.5, and Corollary 1.6). For categorical sewing, the paper by J. Fjelstad, J. Fuchs, I. Runkel, and C. Schweigert, *TFT construction of RCFT correlators V*, proves mapping-class invariance and bulk and boundary factorization. The paper by M. Bischoff, Y. Kawahigashi, and R. Longo, *Characterization of 2D rational local conformal nets and its boundary conditions*, identifies the categorical full center with the corresponding net construction. Chapter 17 distinguishes these theorems from the analytic sewing statements used to interpret surface amplitudes.

The physical chain and the central-charge-eight proposal

For the controlled marginal directions, S. Förste and D. Roggenkamp, *Current-current deformations of conformal field theories, and WZW models*, arXiv:hep-th/0304234, gives the abelian current-deformation framework used in Chapter 14.

T.-C. Huang, Y.-H. Lin, K. Ohmori, Y. Tachikawa, and M. Tezuka, *Numerical evidence for a Haagerup conformal field theory*, arXiv:2110.03008, and R. Vanhove and collaborators, *A critical lattice model for a Haagerup conformal field theory*, arXiv:2110.03532, are two complementary starting points for the physical motivation. Their lattice constructions and evidentiary roles should be kept distinct.

The overlap approach developed in Part IV is related to Y. Liu, Y. Zou, and S. Ryu, *Operator fusion from wavefunction overlaps: Universal finite-size corrections and application to Haagerup model*, arXiv:2203.14992. The book derives the normalization and finite-size tests it uses, so the reader need not take an overlap value on trust as a direct measurement of central charge.

P. Calabrese and J. Cardy, *Entanglement Entropy and Quantum Field Theory*, arXiv:hep-th/0405152, supplies the interval and massive single-cut entropy framework. Chapter 22 derives the replica weights and separates those geometries from the finite-entanglement scaling hypothesis used for variational states.

For the rational-candidate tests in Chapter 24, Dong–Ren–Xu, *On Orbifold Theory*, Theorem 4.4 and Lemma 4.7, supplies the orbifold dimension formula under the stated hypotheses. Etingof–Nikshych–Ostrik, *On fusion categories*, Corollary 8.14 and Proposition 8.27, supplies Morita invariance and the weak-integrality dimension bound. For W_3, Beltaos–Gannon, *The W_N minimal model classification*, Theorem 3.3, classifies modular-invariant matrices; Edie-Michell–Gannon, *Type II quantum subgroups for quantum $\mathfrak{sl}_N$. II: Classification*, Theorem 1.1, Remark 1.2 and Section 5.1, provides the affine module-category input. The chapter explains the additional category and full-center transfers rather than identifying matrix classification with OPE sewing. The height-thirteen comparison is a separate finite exact repository input, and all these exclusions retain their stated chiral-algebra-preservation premises.

J. Albert, Y. Honda, J. Kaidi, and Y. Zheng, *Haagerup symmetry in $(E_8)_1$?*, arXiv:2512.08225v2, gives the public central-charge-eight proposal relevant to this project. The question mark is part of its title. A proposal based on modular and spectral structure, the repository's explicit lattice construction, and an identification of a microscopic chain are three different achievements.

The construction record used for this textbook

The main algebraic record is `hi_cft_c8_proof.tex`, with the structural and analytic arguments in `hi_cft_c8_manuscript_structure.tex` and `hi_cft_c8_manuscript_analytic.tex`. These files give the finite lattice gates and the global deductions on which Part II is based. The review note `hi_cft_c8_explicitness_review_20260928.md` and the expanded material in `c8_expanded_20260929/` help expose intermediate steps that are compressed in the shorter manuscript.

The spectrum, coupling, correlator, and defect records are in `c8_physics_20260927/`. They are the immediate research inputs for the detailed observables in Part III. This textbook rederives the small examples and explains their normalizations; it does not substitute a few successful low-weight calculations for a reconstruction of every bulk nonvacuum OPE.

The chain discussion uses the completed research summaries and their linked evidence, especially `hi_cft_20261001_status.md` and `hi_cft_20261001_theory_handoff.md`. The completed October 5 continuation is recorded in `hi_cft_20261005_status.md`, `hi_cft_20261005_chain_handoff.md`, and `hi_cft_20261005_theory_handoff.md`. Its

qualified variational sequence and completed operator and correlation comparisons are included in the later chain chapters. Earlier calculations retain their original finite scope; a later qualified tensor does not retroactively qualify an earlier endpoint.

Formal verification has a separate scope. The authoritative description of the current conditional development is `lean/HaagerupIzumi/C8/TRUST_BOUNDARY.md`. Appendix C explains how to read a trust boundary and why a theorem proved from named finite inputs is not the same as a machine verification of those inputs. The local audit notes in this textbook's `editorial/` directory record independent mathematical reviews of the exposition itself.

These paths identify the source snapshot in the accompanying workspace. The textbook source bundle is self-contained for typesetting, but it does not duplicate the entire research repository or all of its numerical checkpoints.

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